Topics on econometrics and causal inference

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Preface

This is a working notebook rather than a textbook. The chapters are posts I wrote while reading, teaching, and consulting on applied econometrics, often because a colleague or a student asked a question that did not have a tidy answer in the books I had. Each one stands on its own. The order did not follow a plan, and the same topic sometimes appears more than once as my understanding changed.

The common question is the one Angrist and Pischke (2009) emphasize: what is being identified, by what variation, under what assumptions. The posts ask that question in different settings: interactions and marginal effects, fixed and correlated random effects, matching and weights, difference-in-differences, synthetic control, count models, TMLE and AIPW, mediation, proximal causal inference, partial interference, conjoint and list experiments, uplift modeling, and policy trees.

Most examples are in R. I use Stata where it is more natural, and sometimes Python or Julia when the package is better there. The two companion books, *Introduction to Causal Econometrics* (Ao 2026b) and *Causal Econometrics with Julia* (Ao 2026a), organize the same material more systematically. This collection is the messier version: useful when I want to see how a specific method behaves on a specific problem.

Where to start

Forty-nine chapters in no particular order is not a reading order. Three routes through the material, for readers who want one:

Foundations. What a coefficient means before any causal language is attached. Interpreting interaction in a regression model → Marginal effects in models with fixed effects → Fixed or random effect, or both? → Matching and weighting, part 1 → part 2.

Modern causal estimation. Nuisance functions estimated by machine learning, with the estimand held fixed. Machine learning for causal effects → Recent causal inference tools, which covers AIPW and double machine learning → Longitudinal modified treatment policies → Intro to proximal causal inference → the proximal simulation.

Panel and policy methods. Designs where the variation comes from timing. Extended TWFE and other DiD estimators → Causal panel, and why DDDiD → Synthetic control → Causal forest in panel data → Policy learning by policytree.

Chapters outside these routes are self-contained and can be read on their own.

How to cite

Ao, Xiang. *Topics on Econometrics and Causal Inference*. https://xiangao.github.io/blog_book/.

```
@book{ao_topics_econometrics,
  author = {Ao, Xiang},
  title  = {Topics on Econometrics and Causal Inference},
  year   = {2026},
  url    = {https://xiangao.github.io/blog_book/}
}
```

Part I.

Regression Foundations

1. Interpreting interaction in a regression model

1.1. Interaction with two binary variables

Start with the simplest case: two binary variables x_1 and x_2 coded 0/1.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 \tag{1.1}$$

The model is saturated — three coefficients for three contrasts from the baseline $(x_1, x_2) = (0, 0)$, one for each of the other three cells. Often only the interaction term gets attention. When the two variables are treatment and group, the interaction is the difference in treatment effects between groups, and that may be all that matters. But in other designs all four cells are substantively interesting, and then β_1 and β_2 matter too.

Consider the model:

$$E(y) = \beta_0 + \beta_1 female + \beta_2 treatment + \beta_{12} female * treatment \tag{1.2}$$

The four cell means are:

	male	female
control	β_0	$\beta_0 + \beta_1$
treatment	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \beta_{12}$

Each β reads off the table: β_0 is the baseline cell mean, β_1 the female premium among controls, β_2 the treatment effect among males. The interaction

$$\beta_{12} = (E(Y|(1,1)) - E(Y|(1,0))) - (E(Y|(0,1)) - E(Y|(0,0))) \tag{1.3}$$

where $E(Y|(f,d))$ is the mean outcome for $female = f$ and $treatment = d$. It is the difference in differences — the treatment effect for females minus the treatment effect for males. When the research question is "does the treatment effect differ by group?", the interaction term is the answer. Replace female/male with before/after and the same expression is the standard DiD estimator.

But sometimes the interest is in all four cells. Union membership crossed with race, for example — we may want all six pairwise comparisons, not just the interaction. Stata's `margins` command computes these directly:

```
webuse union3
reg ln_wage i.union##i.black, r
margins union#black
margins union#black, pwcompare
```

```
. webuse union3
(NLS Women 14-24 in 1968)

. reg ln_wage i.union##i.black, r

Linear regression                               Number of obs     =      1,244
                                                F(3, 1240)        =      34.76
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0762
                                                Root MSE          =     .37699

------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     1.union |   .2045053   .0291682     7.01   0.000     .1472808    .2617298
     1.black |  -.1709034   .0308067    -5.55   0.000    -.2313425   -.1104644
             |
 union#black |
       1 1   |   .0386275   .0516609     0.75   0.455     -.062725      .13998
             |
       _cons |   1.657525   .0138278   119.87   0.000     1.630396    1.684653
------------------------------------------------------------------------------

. margins union#black

Adjusted predictions                                  Number of obs = 1,244
Model VCE: Robust

Expression: Linear prediction, predict()

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 union#black |
       0 0   |   1.657525   .0138278   119.87   0.000     1.630396    1.684653
       0 1   |   1.486621    .027529    54.00   0.000     1.432613     1.54063
       1 0   |    1.86203   .0256822    72.50   0.000     1.811644    1.912415
       1 1   |   1.729754   .0325611    53.12   0.000     1.665873    1.793635
------------------------------------------------------------------------------
```

```
. margins union#black, pwcompare

Pairwise comparisons of adjusted predictions             Number of obs = 1,244
Model VCE: Robust

Expression: Linear prediction, predict()

------------------------------------------------------------------
                |            Delta-method         Unadjusted
                |   Contrast   std. err.     [95% conf. interval]
----------------+-------------------------------------------------
    union#black |
(0 1) vs (0 0)  |  -.1709034   .0308067     -.2313425   -.1104644
(1 0) vs (0 0)  |   .2045053   .0291682      .1472808    .2617298
(1 1) vs (0 0)  |   .0722294   .0353756      .0028268     .141632
(1 0) vs (0 1)  |   .3754087   .0376487      .3015466    .4492709
(1 1) vs (0 1)  |   .2431328   .0426388      .1594807     .326785
(1 1) vs (1 0)  |  -.1322759   .0414705     -.2136359   -.0509159
------------------------------------------------------------------

.
```

The `margins` call reports the four cell means; `pwcompare` gives all six pairwise comparisons. In this example, all six pairwise contrasts are significant at 95% — yet the interaction term is insignificant ($p = 0.455$). The two facts are compatible. Each pairwise comparison asks whether two cells differ from each other. The interaction asks whether the *difference between two of those differences* is non-zero — a second-order quantity with its own, larger standard error. Here the union premium is clearly positive for both groups, but the gap between the two premiums (0.039) is not distinguishable from zero.

1.2. Interaction with continuous variables

With continuous x_1 and x_2,

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 \tag{1.4}$$

centering both variables — subtracting each mean before forming the product — is a linear transformation that leaves fitted values, R^2, and everything about β_{12} unchanged. What it changes is β_1 and β_2.

First, it reduces collinearity. When x_1 and x_2 are positive, $x_1 x_2$ is mechanically correlated with both, inflating the standard errors of β_1 and β_2. Centering breaks most of that correlation (all of it under normality), stabilizing the main-effect estimates without altering the interaction.

Second, it fixes the interpretation. In the uncentered model, β_1 is the effect of x_1 when $x_2 = 0$, and zero may be outside the support of x_2. After centering, β_1 is the effect of x_1 evaluated at the

mean of x_2 — a quantity that is always interpretable. When a dummy interacts with a continuous variable, only the continuous variable should be centered.

Stata's `margins` command computes the conditional predictions directly. The data are the built-in `auto` file: 74 1978 model cars, with `price` in dollars, `mpg` in miles per gallon, and `foreign` marking the 22 imports. We centre `mpg` at its mean, regress price on `foreign` interacted with centred mpg, and then ask for predicted prices by origin across a range of mpg values.

```
sysuse auto
sum mpg
gen mpg_centered=mpg-r(mean)
sum mpg_centered
reg price i.foreign##c.mpg_centered
margins foreign, at(mpg_centered=(-3 (1) 3))
marginsplot
graph export "marginsplot-interaction.svg", as(svg) replace
```

```
. sysuse auto
(1978 automobile data)

. sum mpg

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
         mpg |         74     21.2973    5.785503         12         41

. gen mpg_centered=mpg-r(mean)

. sum mpg_centered

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
mpg_centered |         74   -4.03e-08    5.785503  -9.297297    19.7027

. reg price i.foreign##c.mpg_centered

      Source |       SS           df       MS      Number of obs   =        74
-------------+----------------------------------   F(3, 70)        =      9.48
       Model |   183435285         3  61145094.9   Prob > F        =    0.0000
    Residual |   451630112        70  6451858.74   R-squared       =    0.2888
-------------+----------------------------------   Adj R-squared   =    0.2584
       Total |   635065396        73  8699525.97   Root MSE        =    2540.1

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     foreign |
```

```
    Foreign  |   1666.519    717.217     2.32   0.023     236.0751    3096.963
mpg_centered |  -329.2551   74.98545    -4.39   0.000    -478.8088   -179.7013
             |
     foreign#|
          c. |
mpg_centered |
    Foreign  |   78.88826   112.4812     0.70   0.485    -145.4485     303.225
             |
       _cons |   5588.295   369.0945    15.14   0.000     4852.159    6324.431
------------------------------------------------------------------------------

. margins foreign, at(mpg_centered=(-3 (1) 3))

Adjusted predictions                                      Number of obs = 74
Model VCE: OLS

Expression: Linear prediction, predict()
1._at: mpg_centered = -3
2._at: mpg_centered = -2
3._at: mpg_centered = -1
4._at: mpg_centered =  0
5._at: mpg_centered =  1
6._at: mpg_centered =  2
7._at: mpg_centered =  3

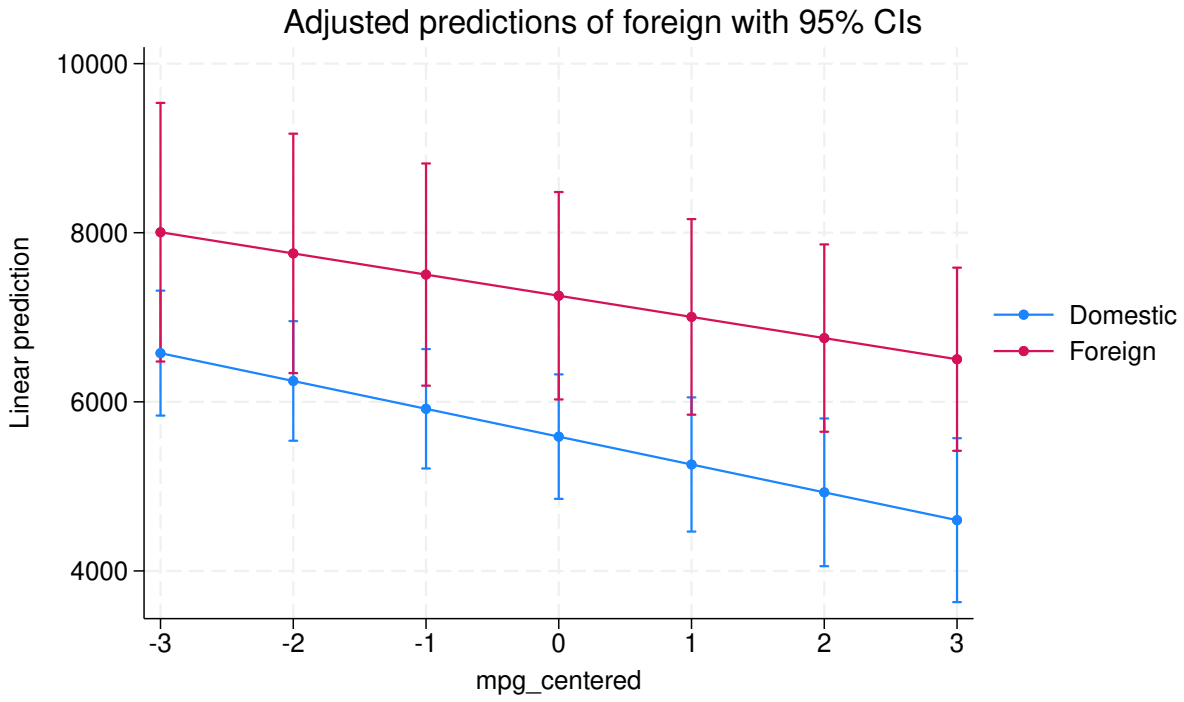
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 _at#foreign |
 1#Domestic  |    6576.06    370.446    17.75   0.000     5837.229    7314.891
  1#Foreign  |   8005.915   766.8178    10.44   0.000     6476.545    9535.284
 2#Domestic  |   6246.805   354.4734    17.62   0.000      5539.83     6953.78
  2#Foreign  |   7755.548   709.9327    10.92   0.000     6339.632    9171.464
 3#Domestic  |    5917.55   354.0032    16.72   0.000     5211.513    6623.587
  3#Foreign  |   7505.181   658.8306    11.39   0.000     6191.185    8819.177
 4#Domestic  |   5588.295   369.0945    15.14   0.000     4852.159    6324.431
  4#Foreign  |   7254.814   614.9548    11.80   0.000     6028.325    8481.303
 5#Domestic  |    5259.04    397.981    13.21   0.000     4465.292    6052.788
  5#Foreign  |   7004.447   579.9479    12.08   0.000     5847.778    8161.117
 6#Domestic  |   4929.785   437.9413    11.26   0.000     4056.338    5803.231
  6#Foreign  |   6754.081   555.4891    12.16   0.000     5646.192    7861.969
 7#Domestic  |    4600.53    486.253     9.46   0.000     3630.729    5570.331
  7#Foreign  |   6503.714   543.0057    11.98   0.000     5420.723    7586.704
------------------------------------------------------------------------------

. marginsplot
```

```
Variables that uniquely identify margins: mpg_centered foreign

. graph export "marginsplot-interaction.svg", as(svg) replace
file marginsplot-interaction.svg saved as SVG format

.
```



Mean mpg is 21.3, so the centred variable runs from −9.3 to 19.7 and has mean zero to seven decimals. That is what makes the coefficients readable. The constant of $5,588 is the predicted price of a domestic car at *average* fuel economy. In the uncentred model it would instead be the predicted price at `mpg = 0`, which is nine standard deviations below the smallest car in the data and has no meaning.

The `foreign` coefficient is 1666.5 with a standard error of 717.2 and $p = 0.023$: at average mpg, imports cost about $1,667 more than domestic cars. The slope on centred mpg is −329.3 with a standard error of 75.0, so among domestic cars each extra mile per gallon is worth about $329 less in price — efficient cars in 1978 were the cheap ones. The interaction is 78.9 with a standard error of 112.5 and $p = 0.485$, so the mpg slope for imports ($-329.3 + 78.9 = -250.4$) is not distinguishable from the domestic slope.

The graph shows predicted price for foreign and domestic cars across the range of mpg. Both lines slope down, the import line sits above the domestic one, and because the interaction is small and insignificant the two are close to parallel. The visible gap between them is the $1,667 main effect, not evidence of a differing slope.

1.3. Summary

With two binary variables the interaction model is saturated. Each coefficient is a contrast among the four cell means, and the interaction is a difference in differences: the treatment effect in one group minus that in the other. When all cell comparisons matter, `margins` and `pwcompare` give them directly. All six pairwise contrasts can be significant while the interaction is not, because they answer different questions.

With continuous variables, centering before forming the product leaves the fit and the interaction unchanged. It changes the main effects into effects at the mean of the other variable and reduces their collinearity with the product. In the `auto` data, centered at mean mpg, imports cost about \$1,667 more. Each extra mile per gallon lowers the price of a domestic car by about \$329. The interaction of 78.9 ($p = 0.485$) gives no evidence that the slope differs for imports.

2. Interaction term in a non-linear model

2.1. The issue

In a linear model with interaction,

$$E(y) = \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2, \tag{2.1}$$

the cross partial derivative is simply the coefficient:

$$\frac{\partial^2 E(y)}{\partial x_1 \partial x_2} = \beta_{12}. \tag{2.2}$$

In a nonlinear model — logit, Poisson, or any link function F — the same coefficient β_{12} sits on the *link* scale:

$$F(E(y)) = \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2. \tag{2.3}$$

The cross partial of $F(E(y))$ is still β_{12}. But the cross partial of $E(y)$ itself — the quantity most researchers want — is not:

$$\frac{\partial^2 E(y)}{\partial x_1 \partial x_2} = \beta_{12} G'(\eta) + (\beta_1 + \beta_{12} x_2)(\beta_2 + \beta_{12} x_1) G''(\eta), \tag{2.4}$$

where $G = F^{-1}$ and η is the linear predictor. This is the point of Ai and Norton (2003): the probability-scale interaction effect depends on every coefficient, varies across the covariate space, and can differ in sign and significance from β_{12}.

The deeper issue is that nonlinearity alone generates a nonzero cross partial — even without an interaction term. Drop β_{12} and the cross partial on $E(y)$ is still $\beta_1 \beta_2 G''(\eta) \neq 0$ whenever the link is curved. The marginal effect of x_1 depends on x_2 in any nonlinear model, interaction or not.

Greene (2018, sec. 17.3) and Buis (2010) draw the same conclusion: the model is specified on the link scale, and that is where coefficients and their tests are interpretable. In a logit, β_{12} is the interaction on log-odds; exponentiating gives an odds-ratio interaction that is multiplicative. Partial effects on the probability scale are *implications* of the model, not parameters of it — they should be computed from the fitted model (via `margins` or `marginaleffects`), not read off the coefficient table.

2.2. A worked example

Let's illustrate the Ai-Norton point directly: in a logit model with an interaction term, the cross partial derivative $\frac{\partial^2 E(y)}{\partial x_1 \partial x_2}$ (what we actually care about) is not the same object as the estimated coefficient β_{12} on $x_1 x_2$ in the linear index. We simulate data with a true interaction effect on the log-odds scale, fit a logit model, and then use the `marginaleffects` package for two things: the marginal effect of x_1 on the probability scale at several values of x_2, and the contrast of that slope between two values of x_2. We compute the local cross partial derivative itself by hand, from the true coefficients.

We draw $n = 2000$ observations with $x_1, x_2 \sim N(0, 1)$, independent of each other. The linear index is $\eta = -0.3 + 0.8x_1 + 0.5x_2 + 0.6\, x_1 x_2$, the success probability is $p = \Lambda(\eta)$, and $y \sim \text{Bernoulli}(p)$. So the true interaction on the log-odds scale is $\beta_{12} = 0.6$, a single constant — which is exactly the object the coefficient table estimates, and exactly the object that does *not* transfer to the probability scale.

```
library(dplyr)
library(marginaleffects)

set.seed(42)
n <- 2000
x1 <- rnorm(n)
x2 <- rnorm(n)
eta <- -0.3 + 0.8 * x1 + 0.5 * x2 + 0.6 * x1 * x2
p <- plogis(eta)
y <- rbinom(n, 1, p)
dat <- data.frame(y, x1, x2)

m <- glm(y ~ x1 * x2, data = dat, family = binomial)
summary(m)$coefficients
```

```
             Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -0.2348658 0.05017155 -4.681255 2.851237e-06
x1           0.7792602 0.05930028 13.140920 1.918795e-39
x2           0.5296149 0.05537644  9.563903 1.133977e-21
x1:x2        0.6624955 0.06403580 10.345705 4.376826e-25
```

The coefficient on `x1:x2` is 0.662 with a standard error of 0.064, against a true 0.6 — recovered to within one standard error, as it should be, since the logit is correctly specified. The other three are 0.779, 0.530 and −0.235 against true values of 0.8, 0.5 and −0.3.

That coefficient is $\hat{\beta}_{12}$, the interaction effect on the log-odds (linear index) scale — constant by construction. On the *probability* scale nothing is constant. Start with the marginal effect of x_1 itself, evaluated at three values of x_2:

```
slopes(m, variables = "x1", by = "x2", newdata = datagrid(x2 = c(-1, 0, 1)))
```

```
 x2 Estimate Std. Error     z Pr(>|z|)     S    2.5 % 97.5 %
 -1   0.0253     0.0159  1.59    0.112   3.2 -0.00591 0.0565
  0   0.1919     0.0147 13.09   <0.001 127.6  0.16314 0.2206
  1   0.3538     0.0243 14.59   <0.001 157.7  0.30630 0.4014

Term: x1
Type: response
Comparison: dY/dX
```

The marginal effect of x_1 on $P(y = 1)$ is 0.025 at $x_2 = -1$, 0.192 at $x_2 = 0$, and 0.354 at $x_2 = 1$ — a fourteen-fold change across the three values, where the log-odds interaction is one fixed number. Notice too that at $x_2 = -1$ the effect is not significant ($p = 0.112$) even though $\hat{\beta}_{12}$ is significant at better than 10^{-24}. Already the probability-scale answer and the coefficient tell different stories.

The next step is not the effect of x_1 at a given x_2 but how much that effect changes as x_2 moves. Over a finite interval that is a difference of slopes, a *slope contrast*, so it needs a hypothesis test on top of the `by` argument rather than a second call to `slopes()`. It is not the cross partial derivative $\partial^2 P/\partial x_1 \partial x_2$, which is the local rate at a single point:

```
slopes(m, variables = "x1", by = "x2",
       newdata = datagrid(x2 = c(-1, 1)),
       hypothesis = "b2 - b1 = 0")
```

```
 Hypothesis Estimate Std. Error  z Pr(>|z|)    S 2.5 % 97.5 %
    b2-b1=0    0.329     0.0299 11   <0.001 90.7  0.27  0.387

Type: response
```

The slope contrast is 0.329 with a standard error of 0.030, $z = 11$ — the difference between the slope at $x_2 = 1$ and the slope at $x_2 = -1$, $0.354 - 0.025$, with a proper standard error attached. It is a finite difference across two units of x_2, not the local derivative. At the true coefficients and $x_1 = 0$, the population slope contrast is 0.304, while the cross partial at $x_2 = 0$ is

$$\Lambda''(-0.3)(0.5)(0.8) + \Lambda'(-0.3)(0.6) = 0.161,$$

where Λ' and Λ'' are the first and second derivatives of the logistic function, -0.3 is the index at $x_1 = x_2 = 0$, and 0.5, 0.8 and 0.6 are the true coefficients on x_2, x_1 and $x_1 x_2$. It is large and highly significant, which is worth being honest about: in *this* example it agrees with $\hat{\beta}_{12}$ in significance, differing only in scale and magnitude. Note that it is nowhere near 0.662; the two numbers live on different scales and there is no reason to expect them to match. Ai and Norton's warning is that the two quantities *need not* agree — not that they never do. The dependable lesson is the one visible in the table above: on the probability scale the answer depends on where you evaluate it, so a single coefficient cannot summarise it.

2.3. Summary

In a nonlinear model the interaction coefficient lives on the link scale. On the scale of the expected outcome, the interaction effect is a cross partial derivative. It depends on every coefficient and on where it is evaluated, and the curvature of the link produces it even with no interaction term. Ai and Norton's point is that the coefficient and the probability-scale interaction need not agree in size, sign or significance. Following Greene and Buis, we test the coefficient on the scale where the model is specified, and we report probability-scale effects at stated covariate values.

In the logit example the interaction coefficient is 0.662 against a true 0.6. The marginal effect of x_1 on the probability is 0.025, 0.192 and 0.354 at $x_2 = -1, 0, 1$. The slope contrast, the change in that slope from $x_2 = -1$ to $x_2 = 1$, is 0.329; the local cross partial at the true coefficients is 0.161 at $x_2 = 0$. Here it agrees with the coefficient in significance, but no single number summarizes the probability scale.

3. Chow test and more

3.1. Chow test

To compare a coefficient across two subsamples, interact the subsample indicator with the covariates and run one pooled regression. If only the treatment is interacted, all other coefficients are constrained equal across subsamples; interacting everything relaxes that. This is the Chow test. (The overlapping-samples extension below follows Austin Nichols's Statalist code.)

The data are Stata's `nlsw88`, an extract from the 1988 National Longitudinal Survey of Young Women. With non-missing wage and hours, 938 of the women live in the South and 1,304 elsewhere, 2,242 in all. We regress hourly `wage` on weekly `hours`, and ask whether the slope differs by region. The chunk runs that four ways — two separate regressions, `suest`, and the pooled interacted regression — so the equivalences are visible side by side.

```
est clear
sysuse nlsw88, clear
reg wage hours if south
est sto south
reg wage hours if !south
est sto nonsouth
suest south nonsouth
est sto suest
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
reg wage south hours?
est sto chow
test _b[hours1]-_b[hours2]=0
esttab south nonsouth suest chow, nogaps mti
```

```
(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =       938
-------------+----------------------------------   F(1, 936)       =     12.47
       Model |  344.732583         1  344.732583   Prob > F        =    0.0004
    Residual |  25866.3404       936   27.634979   R-squared       =    0.0132
-------------+----------------------------------   Adj R-squared   =    0.0121
       Total |  26211.0729       937   27.973397   Root MSE        =    5.2569

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
       hours |   .0623497   .0176532     3.53   0.000     .0277053    .0969941
       _cons |   4.520583   .6957145     6.50   0.000     3.155242    5.885923
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     1,304
-------------+----------------------------------   F(1, 1302)      =     55.93
       Model |  1929.41943         1  1929.41943   Prob > F        =    0.0000
    Residual |  44919.1023     1,302  34.5000785   R-squared       =    0.0412
-------------+----------------------------------   Adj R-squared   =    0.0404
       Total |  46848.5217     1,303  35.9543528   Root MSE        =    5.8737

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .1107536     .01481     7.48   0.000     .0816995    .1398076
       _cons |   4.357811    .564756     7.72   0.000      3.24988    5.465743
------------------------------------------------------------------------------



Simultaneous results for south, nonsouth                Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
       hours |   .0623497   .0174426     3.57   0.000     .0281629    .0965366
       _cons |   4.520583   .6599379     6.85   0.000     3.227128    5.814037
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   3.319082   .1435305    23.12   0.000     3.037768    3.600397
-------------+----------------------------------------------------------------
nonsouth_m~n |
       hours |   .1107536   .0134936     8.21   0.000     .0843066    .1372006
       _cons |   4.357811   .4633326     9.41   0.000     3.449696    5.265927
-------------+----------------------------------------------------------------
nonsouth_l~r |
       _cons |   3.540962   .1006119    35.19   0.000     3.343766    3.738157
------------------------------------------------------------------------------


(4 missing values generated)

(4 missing values generated)

      Source |       SS           df       MS      Number of obs   =     2,242
```

```
-------------+----------------------------------   F(3, 2238)      =     36.91
       Model |  3502.37892         3  1167.45964   Prob > F        =    0.0000
    Residual |  70785.4426     2,238  31.6288841   R-squared       =    0.0471
-------------+----------------------------------   Adj R-squared   =    0.0459
       Total |  74287.8215     2,241  33.1494072   Root MSE        =     5.624

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       south |   .1627711   .9199871     0.18   0.860    -1.641346    1.966888
      hours1 |   .0623497   .0188858     3.30   0.001     .0253142    .0993852
      hours2 |   .1107536   .0141803     7.81   0.000     .0829456    .1385616
       _cons |   4.357811   .5407453     8.06   0.000     3.297397    5.418226
------------------------------------------------------------------------------



 ( 1)  hours1 - hours2 = 0

       F(  1,  2238) =    4.20
            Prob > F =    0.0405
```

	(1)	(2)	(3)	(4)
	south	nonsouth	suest	chow
main				
hours	0.0623***	0.111***	0.0623***	
	(3.53)	(7.48)	(3.57)	
south				0.163
				(0.18)
hours1				0.0623***
				(3.30)
hours2				0.111***
				(7.81)
_cons	4.521***	4.358***	4.521***	4.358***
	(6.50)	(7.72)	(6.85)	(8.06)
south_lnvar				
_cons			3.319***	
			(23.12)	
nonsouth_m~n				
hours			0.111***	
			(8.21)	
_cons			4.358***	
			(9.41)	

```
------------------------------------------------------------------------------
nonsouth_l~r
_cons                                              3.541***
                                                  (35.19)
------------------------------------------------------------------------------
N                     938            1304            2242            2242
------------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The separate regressions give an hours slope of 0.0623 in the South and 0.1108 elsewhere. The pooled regression with `hours1` and `hours2` returns 0.0623497 and 0.1107536 — the *same numbers to seven digits*, not merely similar ones. That is algebra, not luck: a fully interacted regression is the two subsample regressions written as one.

What does differ is the standard errors. The South slope carries 0.0177 in its own regression and 0.0189 in the pooled one, because the pooled model estimates a single error variance across both groups while the separate regressions each get their own (root MSE 5.26 in the South, 5.87 elsewhere). If that pooling is unattractive, `suest` avoids it: it stacks the two stored estimates with their joint covariance and reports robust standard errors, 0.0174 and 0.0135 here.

`test _b[hours1]-_b[hours2]=0` is the Chow test, and it gives $F(1, 2238) = 4.20$ with $p = 0.0405$. The returns to hours are steeper outside the South, and the gap of 0.048 dollars per hour is just significant at 5%. Note that the `south` dummy itself is 0.163 with $p = 0.860$ — the two regions differ in slope, not in intercept, which is precisely the pattern a treatment-only interaction would have missed. Stata's `suest` does the same job by stacking the two stored estimates and their covariance structure; the interaction approach is more flexible since it works with any estimator.

3.2. A comparison with two different outcomes

The same idea compares a coefficient across two different outcomes. Stack the two outcome variables into a long format, interact the covariates with the stacking indicator, and run one regression.

The chunk below does that in seven steps, and it is worth reading them in order because the payoff is the last two columns of the table agreeing.

1. `reg south wage hours` and `reg smsa wage hours`, each stored with `est sto`. These are the two equations we want to compare, fitted separately.
2. `suest south smsa` refits them jointly and stacks the two coefficient vectors into one, with a robust covariance matrix across equations. This is the benchmark: it is the standard way to test a cross-equation hypothesis.
3. `preserve`, because the next steps reshape the data and we want it back.
4. `gen Y1=south`, `gen Y2=smsa`, `gen id=_n`. Copy the two outcomes into a numbered pair and give every woman an id.
5. `reshape long Y, i(id) j(subsample)` turns each woman's one row into two: one carrying `Y = south` with `subsample == 1`, one carrying `Y = smsa` with `subsample == 2`. The stacked outcome is now a single variable `Y`.

6. `gen wage1=wage*(subsample==1)` and its three siblings. Each covariate is split into the part that acts in equation 1 and the part that acts in equation 2. A single regression on `wage1 wage2 hours1 hours2` therefore estimates a separate slope per equation, which is what the two separate regressions did.
7. `reg Y wage? hours? subsample, cluster(id)` fits it, and `test _b[wage1]-_b[wage2]=0` is the cross-equation test. The `?` is a Stata wildcard, so `wage?` expands to `wage1 wage2`.

The whole construction exists so that a hypothesis *across* two equations becomes a hypothesis *within* one, where `test` can reach both coefficients.

```
est clear
sysuse nlsw88, clear
reg south wage hours
est sto south
reg smsa wage hours
est sto smsa
suest south smsa
est sto suest
preserve
gen Y1=south
gen Y2=smsa
gen id=_n
reshape long Y, i(id) j(subsample)
gen wage1=wage*(subsample==1)
gen wage2=wage*(subsample==2)
gen hours1=hours*(subsample==1)
gen hours2=hours*(subsample==2)
* Each individual contributes TWO rows here (one per stacked outcome), so the
* errors are correlated within id.  Clustering on the stacking id is what makes
* the cross-equation test comparable to suest -- see the note below.
reg Y wage? hours? subsample, cluster(id)
test _b[wage1]-_b[wage2]=0
est sto stacked
* Match the preserve above, so the reshape does not leak into later chunks.
restore
* Note: suest stores coefficients under equation-qualified names (e.g.
* south:hours), while the single-sample models store the same
* coefficient simply as "hours"; esttab matches by exact coefficient
* name, so these land on separate rows instead of being aligned. Use
* esttab's coeflabel()/eqlabel() options, or rename coefficients on the
* stored estimates first, to get a directly comparable table.
esttab south smsa suest stacked, nogaps mti
```

```
(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =     2,242
```

```
-------------+----------------------------------   F(2, 2239)      =     30.60
       Model |   14.513685         2  7.25684251   Prob > F        =    0.0000
    Residual |  531.049205     2,239  .237181423   R-squared       =    0.0266
-------------+----------------------------------   Adj R-squared   =    0.0257
       Total |   545.56289     2,241   .24344618   Root MSE        =    .48701

------------------------------------------------------------------------------
       south | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |  -.0124053   .0018099    -6.85   0.000    -.0159545    -.008856
       hours |   .0047722   .0009916     4.81   0.000     .0028277    .0067167
       _cons |   .3372108   .0387354     8.71   0.000     .2612497    .4131718
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     2,242
-------------+----------------------------------   F(2, 2239)      =     36.17
       Model |  14.6274602         2  7.31373012   Prob > F        =    0.0000
    Residual |  452.719551     2,239  .202197209   R-squared       =    0.0313
-------------+----------------------------------   Adj R-squared   =    0.0304
       Total |  467.347012     2,241  .208543959   Root MSE        =    .44966

------------------------------------------------------------------------------
        smsa | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |   .0139103   .0016711     8.32   0.000     .0106332    .0171873
       hours |   .0003663   .0009155     0.40   0.689    -.0014291    .0021616
       _cons |   .5820585   .0357648    16.27   0.000      .511923    .6521941
------------------------------------------------------------------------------



Simultaneous results for south, smsa                     Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
        wage |  -.0124053   .0019287    -6.43   0.000    -.0161854   -.0086251
       hours |   .0047722   .0009825     4.86   0.000     .0028465    .0066978
       _cons |   .3372108   .0379918     8.88   0.000     .2627483    .4116733
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   -1.43893   .0096645  -148.89   0.000    -1.457872   -1.419988
-------------+----------------------------------------------------------------
smsa_mean    |
        wage |   .0139103   .0018313     7.60   0.000      .010321    .0174996
       hours |   .0003663   .0009099     0.40   0.687    -.0014172    .0021497
```

```
       _cons |   .5820585   .0361037    16.12   0.000     .5112965    .6528205
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |  -1.598512   .0195103   -81.93   0.000    -1.636751   -1.560272
------------------------------------------------------------------------------
```

```
(j = 1 2)

Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations            2,246   ->   4,492
Number of variables                  23   ->   23
j variable (2 values)                     ->   subsample
xij variables:
                                  Y1 Y2   ->   Y
-----------------------------------------------------------------------------
```

```
(8 missing values generated)

(8 missing values generated)
```

```
Linear regression                               Number of obs     =      4,484
                                                F(5, 2241)        =      96.34
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1091
                                                Root MSE          =     .46871

                                (Std. err. adjusted for 2,242 clusters in id)
------------------------------------------------------------------------------
             |               Robust
           Y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       wage1 |  -.0124053   .0019298    -6.43   0.000    -.0161896    -.008621
       wage2 |   .0139103   .0018323     7.59   0.000      .010317    .0175035
      hours1 |   .0047722    .000983     4.85   0.000     .0028444    .0066999
      hours2 |   .0003663   .0009104     0.40   0.688    -.0014191    .0021517
   subsample |   .2448477   .0548754     4.46   0.000     .1372358    .3524597
       _cons |    .092363   .0872219     1.06   0.290    -.0786812    .2634072
------------------------------------------------------------------------------
```

```
 ( 1)  wage1 - wage2 = 0

       F(  1,  2241) =   72.14
            Prob > F =    0.0000
```

	(1) south	(2) smsa	(3) suest	(4) stacked
main				
wage	-0.0124***	0.0139***	-0.0124***	
	(-6.85)	(8.32)	(-6.43)	
hours	0.00477***	0.000366	0.00477***	
	(4.81)	(0.40)	(4.86)	
wage1				-0.0124***
				(-6.43)
wage2				0.0139***
				(7.59)
hours1				0.00477***
				(4.85)
hours2				0.000366
				(0.40)
subsample				0.245***
				(4.46)
_cons	0.337***	0.582***	0.337***	0.0924
	(8.71)	(16.27)	(8.88)	(1.06)
south_lnvar				
_cons			-1.439***	
			(-148.89)	
smsa_mean				
wage			0.0139***	
			(7.60)	
hours			0.000366	
			(0.40)	
_cons			0.582***	
			(16.12)	
smsa_lnvar				
_cons			-1.599***	
			(-81.93)	
N	2242	2242	2242	4484

```
------------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

Read the table by columns. The first two are the separate regressions, the third is `suest`, and the fourth is the stacked regression. The `wage1` coefficient in column four equals the `wage` coefficient in column one, and `wage2` equals the `wage` coefficient in column two: stacking has reproduced both single-equation fits exactly, in one regression, with both coefficients now addressable by `test`. Note the sample size, 4,484 in the stacked column against 2,242 in the others — the reshape doubled the rows, which is the next point.

One detail matters a great deal here, and it is easy to skip. Stacking puts **two rows per person** in the data, so the two error terms belonging to the same person are correlated by construction. The cross-equation test is precisely a comparison *across* those two rows, so its standard error has to account for that correlation. Clustering on the stacking id does it, and it is not a cosmetic adjustment:

	test of the wage coefficient across equations
`suest`	$\chi^2(1) = 72.22$
stacked, **no** clustering	$F(1, 4478) = 114.12$
stacked, `cluster(id)`	$F(1, 2241) = 72.14$

Without clustering the statistic is inflated by more than half. With it, the stacked test reproduces `suest` almost exactly – the small remaining gap is just the F-versus-χ^2 finite-sample adjustment. Note that `robust` alone will *not* fix this; the cluster variable has to be the id you stacked on.

3.3. Overlapping samples

The same method extends to overlapping subsamples. Here south and smsa overlap — some observations belong to both — so a simple split is not possible.

The reshape trick from the previous section will not work here, because a woman who is both southern and in an SMSA has to appear in both subsamples rather than be assigned to one. `expand` handles that instead:

1. `ta south smsa` shows the overlap, which is the reason for what follows.
2. `preserve`, then `expand 2` duplicates every row, so each woman now has two copies.
3. `bys idcode: g n=_n` numbers a woman's two copies 1 and 2.
4. `keep if (n==1&south)|(n==2&smsa)` keeps copy 1 only if she is southern and copy 2 only if she is in an SMSA. A woman in both keeps both copies; a woman in neither drops out; a woman in exactly one keeps one. Each surviving row now represents one woman's membership in one subsample.
5. `g hours1=hours*(n==1)` and `g hours2=hours*(n==2)` split the covariate by subsample, as before. Within this restricted sample `n==1` means southern and `n==2` means SMSA, so the copy number *is* the subsample indicator.

6. `reg wage hours? n, cl(idcode)` fits both slopes at once, clustering on the woman rather than on a stacking id, since a woman can now contribute one row or two.

```
est clear
sysuse nlsw88, clear
ta south smsa
reg wage hours if south
est sto south
reg wage hours if smsa
est sto smsa
suest south smsa
est sto suest
preserve
expand 2
bys idcode: g n=_n
keep if (n==1&south)|(n==2&smsa)
* hours1 is hours in the south subsample (n==1), hours2 is hours in the
* smsa subsample (n==2); within this restricted sample n==1 <=> south==1
* and n==2 <=> smsa==1, so this is equivalent to but clearer than negating
* the complementary condition.
g hours1=hours*(n==1)
g hours2=hours*(n==2)
reg wage hours? n, cl(idcode)
est sto stacked
restore
esttab south smsa suest stacked, nogaps mti
```

```
(NLSW, 1988 extract)


  Lives in |     Lives in SMSA
 the south |  Not SMSA       SMSA |     Total
-----------+----------------------+----------
 Not south |       308        996 |     1,304
     South |       357        585 |       942
-----------+----------------------+----------
     Total |       665      1,581 |     2,246

      Source |       SS           df       MS      Number of obs   =       938
-------------+----------------------------------   F(1, 936)       =     12.47
       Model |  344.732583         1  344.732583   Prob > F        =    0.0004
    Residual |  25866.3404       936   27.634979   R-squared       =    0.0132
-------------+----------------------------------   Adj R-squared   =    0.0121
       Total |  26211.0729       937   27.973397   Root MSE        =    5.2569

------------------------------------------------------------------------------
```

```
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .0623497   .0176532     3.53   0.000     .0277053    .0969941
       _cons |   4.520583   .6957145     6.50   0.000     3.155242    5.885923
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     1,578
-------------+----------------------------------   F(1, 1576)      =     46.48
       Model |  1594.14881         1  1594.14881   Prob > F        =    0.0000
    Residual |  54048.2539     1,576  34.2945773   R-squared       =    0.0286
-------------+----------------------------------   Adj R-squared   =    0.0280
       Total |  55642.4027     1,577  35.2837049   Root MSE        =    5.8562

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .0953636   .0139872     6.82   0.000     .0679281    .1227991
       _cons |   4.861519   .5443826     8.93   0.000     3.793729    5.929309
------------------------------------------------------------------------------



Simultaneous results for south, smsa                  Number of obs = 1,934

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
       hours |   .0623497   .0174432     3.57   0.000     .0281617    .0965378
       _cons |   4.520583   .6599613     6.85   0.000     3.227082    5.814083
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |   3.319082   .1435356    23.12   0.000     3.037758    3.600407
-------------+----------------------------------------------------------------
smsa_mean    |
       hours |   .0953636   .0132806     7.18   0.000      .069334    .1213931
       _cons |   4.861519   .4842914    10.04   0.000     3.912325    5.810713
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |   3.534987   .0910825    38.81   0.000     3.356469    3.713506
------------------------------------------------------------------------------



(2,246 observations created)
```

```
(1,969 observations deleted)

(7 missing values generated)

(7 missing values generated)


Linear regression                               Number of obs     =      2,516
                                                F(3, 1933)        =      40.81
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0399
                                                Root MSE          =     5.6403

                            (Std. err. adjusted for 1,934 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   .0623497   .0174536     3.57   0.000     .0281198    .0965796
      hours2 |   .0953636   .0132886     7.18   0.000     .0693022     .121425
           n |   .3409364   .6124145     0.56   0.578    -.8601261    1.541999
       _cons |   4.179646   1.177889     3.55   0.000     1.869579    6.489713
------------------------------------------------------------------------------



----------------------------------------------------------------------------
                     (1)             (2)             (3)             (4)
                   south            smsa           suest         stacked
----------------------------------------------------------------------------
main
hours              0.0623***       0.0954***       0.0623***
                   (3.53)          (6.82)          (3.57)
hours1                                                            0.0623***
                                                                  (3.57)
hours2                                                            0.0954***
                                                                  (7.18)
n                                                                  0.341
                                                                  (0.56)
_cons               4.521***        4.862***        4.521***        4.180***
                   (6.50)          (8.93)          (6.85)          (3.55)
----------------------------------------------------------------------------
south_lnvar
_cons                                               3.319***
                                                   (23.12)
----------------------------------------------------------------------------
smsa_mean
```

```
hours                                                 0.0954***
                                                     (7.18)
_cons                                                  4.862***
                                                    (10.04)
------------------------------------------------------------------------------
smsa_lnvar
_cons                                                  3.535***
                                                    (38.81)
------------------------------------------------------------------------------
N                     938            1578            1934            2516
------------------------------------------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The cross-tab shows the overlap: of the 1,304 non-South women, 996 live in an SMSA, so the two subsamples share members and cannot be produced by a simple split. Expanding the data and keeping each observation under whichever subsample it belongs to handles that. The pooled `hours1` is 0.0623497, again reproducing the South regression exactly, and `hours2` is 0.0954 for the SMSA subsample. The `n` indicator is 0.341 with $p = 0.578$.

3.4. IV regression

The interaction approach works with IV. Interact both the endogenous variable and the instrument with the subsample indicator. Here `hours` is instrumented by `union` within each region.

The point is not to get a different estimate. The two interacted 2SLS coefficients reproduce the two subsample IV regressions exactly, just as they did for OLS. The point is that running them separately leaves the two estimates in different `ivregress` results with no joint covariance between them, so there is nothing to test the difference *with*. Interacting puts both in one estimation, so `test _b[hours1]-_b[hours2]=0` can reach both coefficients and their covariance. That is the whole reason for the construction, and it is the same reason as in the OLS case.

Two details specific to IV. The instrument must be split alongside the endogenous variable — `union1` and `union2`, not just `hours1` and `hours2` — because an instrument that is not subsample-specific would identify a single pooled first stage and defeat the purpose. And the number of instruments must still match the number of endogenous regressors: two of each here, so the system is exactly identified, as each subsample regression was on its own.

```
sysuse nlsw88, clear
ivregress 2sls wage (hours=union) if south
ivregress 2sls wage (hours=union) if !south
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
```

```
ivregress 2sls wage south (hours1 hours2 = union1 union2)
test _b[hours1]-_b[hours2]=0
```

```
(NLSW, 1988 extract)


Instrumental-variables 2SLS regression            Number of obs   =        798
                                                  Wald chi2(1)    =       2.61
                                                  Prob > chi2     =     0.1060
                                                  Root MSE        =     9.5807

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |     .97819   .6050822     1.62   0.106    -.2077492    2.164129
       _cons |  -30.96026   23.32846    -1.33   0.184     -76.6832    14.76268
------------------------------------------------------------------------------
Endogenous: hours
Exogenous:  union


Instrumental-variables 2SLS regression            Number of obs   =      1,079
                                                  Wald chi2(1)    =       6.03
                                                  Prob > chi2     =     0.0140
                                                  Root MSE        =     7.0372

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .6255843   .2546731     2.46   0.014     .1264342    1.124735
       _cons |   -14.9159   9.401508    -1.59   0.113    -33.34252    3.510714
------------------------------------------------------------------------------
Endogenous: hours
Exogenous:  union

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)


Instrumental-variables 2SLS regression            Number of obs   =      1,877
                                                  Wald chi2(3)    =      21.75
                                                  Prob > chi2     =     0.0001
```

```
                                                Root MSE        =     8.2154

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |     .97819   .5188511     1.89   0.059    -.0387394    1.995119
      hours2 |   .6255843    .297311     2.10   0.035     .0428654    1.208303
       south |  -16.04435   22.81705    -0.70   0.482    -60.76495    28.67625
       _cons |   -14.9159   10.97553    -1.36   0.174    -36.42754    6.595735
------------------------------------------------------------------------------
Endogenous: hours1 hours2
Exogenous:  south union1 union2


 ( 1)  hours1 - hours2 = 0

           chi2(  1) =    0.35
         Prob > chi2 =    0.5554
```

The IV estimates are 0.978 in the South and 0.626 elsewhere, against OLS values of 0.062 and 0.111 — an order of magnitude larger, with standard errors to match (0.519 and 0.297). And the Chow test now gives $\chi^2(1) = 0.35$ with $p = 0.5554$, where the OLS version rejected at $p = 0.0405$. Instrumenting has not overturned the difference between regions; it has made the data too imprecise to detect it. The lesson is about the test, not the subsamples: a Chow test inherits the precision of whatever estimator it is built on.

3.5. IV with fixed effects

With fixed effects the pooled regression needs the fixed effects themselves interacted with the subsample indicator. The first attempt below omits this and fails to reproduce the separate estimates:

> **Warning**
>
> The examples below cluster standard errors on `race`, which in `nlsw88` has only three categories (White, Black, Other). Cluster-robust standard errors rely on asymptotics in the number of clusters $G \to \infty$; with $G = 3$ they are severely downward-biased, inflating t-statistics and false-positive rates. These chunks use `race` purely to illustrate the fixed-effect/IV mechanics — in an actual analysis, cluster on a variable with many more categories (e.g., `industry` or `occupation`), or on the true level of treatment assignment/sampling design.

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
```

```
gen union2=union*(south==0)
ivreghdfe wage (hours=union) if south, a(race) cluster(race)
ivreghdfe wage (hours=union) if !south, a(race) cluster(race)
ivreghdfe wage south (hours1 hours2 = union1 union2) , a(race) cluster(race)
```

```
(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =       798
                                                      F(  1,     2) =     69.28
                                                      Prob > F      =    0.0141
Total (centered) SS     =   12186.8806                Centered R2   =   -6.4527
Total (uncentered) SS   =   12186.8806                Uncentered R2 =   -6.4527
Residual SS             =  90825.44631                Root MSE      =     10.68

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   1.107099    .133012     8.32   0.014     .5347947    1.679404
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.756
                                                    Chi-sq(1) P-val =    0.1852
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.233
                         (Kleibergen-Paap rk Wald F statistic):         13.977
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
```

```
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3                Number of obs =      1079
                                                      F(  1,     2) =      1.41
                                                      Prob > F      =    0.3564
Total (centered) SS     =  19086.22115                Centered R2   =   -2.3302
Total (uncentered) SS   =  19086.22115                Uncentered R2 =   -2.3302
Residual SS             =  63560.97528                Root MSE      =     7.682

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .7023322   .5905772     1.19   0.356    -1.838717    3.243381
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.789
                                                    Chi-sq(1) P-val =    0.3745
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                5.501
                         (Kleibergen-Paap rk Wald F statistic):          3.772
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
```

```
                                        20% maximal IV size              6.66
                                        25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                 (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3               Number of obs =      1877
                                                     F(  3,     2) =   1036.24
                                                     Prob > F      =    0.0010
Total (centered) SS     =    32142.319               Centered R2   =   -3.9041
Total (uncentered) SS   =    32142.319               Uncentered R2 =   -3.9041
Residual SS             =  157627.5718               Root MSE      =     9.174

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.142036   .0650507    17.56   0.003     .8621454    1.421927
      hours2 |   .6880691   .5265185     1.31   0.321    -1.577357    2.953495
       south |  -19.74773   22.10811    -0.89   0.466    -114.8712    75.37577
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.793
```

```
                                                    Chi-sq(1) P-val =    0.1806
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.584
                         (Kleibergen-Paap rk Wald F statistic):          8.484
Stock-Yogo weak ID test critical values: 10% maximal IV size             7.03
                                         15% maximal IV size             4.58
                                         20% maximal IV size             3.95
                                         25% maximal IV size             3.63
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.
------------------------------------------------------------------------------
Instrumented:         hours1 hours2
Included instruments: south
Excluded instruments: union1 union2
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

The separate regressions give 1.1071 and 0.7023. The pooled regression gives 1.1420 and 0.6881 — close, but not equal, and that is the point. The pooled coefficients do not match the separate regressions because the fixed effects are shared: one set of race effects is estimated off both subsamples at once, so neither subsample gets the within-group demeaning its own regression used. Interacting them with the subsample indicator fixes it:

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
```

```
gen race1=race*(south==1)
gen race2=race*(south==0)


ivreghdfe wage (hours=union) if south, a(race) cluster(race)
ivreghdfe wage (hours=union) if !south, a(race) cluster(race)
ivregress 2sls wage south (hours1 hours2 = union1 union2) i.race1 i.race2, cluster(race)
```

```
(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)

(368 missing values generated)



(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3               Number of obs =      798
                                                     F(  1,     2) =    69.28
                                                     Prob > F      =   0.0141
Total (centered) SS     =   12186.8806               Centered R2   =  -6.4527
Total (uncentered) SS   =   12186.8806               Uncentered R2 =  -6.4527
Residual SS             =  90825.44631               Root MSE      =    10.68

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   1.107099    .133012     8.32   0.014     .5347947    1.679404
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              1.756
                                                   Chi-sq(1) P-val =    0.1852
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.233
                         (Kleibergen-Paap rk Wald F statistic):         13.977
```

```
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                    (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

(MWFE estimator converged in 1 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3               Number of obs =      1079
                                                     F(  1,     2) =      1.41
                                                     Prob > F      =    0.3564
Total (centered) SS     =  19086.22115               Centered R2   =   -2.3302
Total (uncentered) SS   =  19086.22115               Uncentered R2 =   -2.3302
Residual SS             =  63560.97528               Root MSE      =     7.682

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hours |   .7023322   .5905772     1.19   0.356    -1.838717    3.243381
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.789
```

```
                                                    Chi-sq(1) P-val =    0.3745
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                5.501
                         (Kleibergen-Paap rk Wald F statistic):          3.772
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Hansen J statistic (overidentification test of all instruments):         0.000
                                                  (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         hours
Excluded instruments: union
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
        race |         3           3           0    *|
-----------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation

note: 3.race1 omitted because of collinearity.
note: 3.race2 omitted because of collinearity.

Instrumental-variables 2SLS regression            Number of obs   =      1,877
                                                  Wald chi2(7)    =  101696.08
                                                  Prob > chi2     =     0.0000
                                                  Root MSE        =     9.0693

                                 (Std. err. adjusted for 3 clusters in race)
------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.107099   .1085357    10.20   0.000     .8943732    1.319825
      hours2 |   .7023322   .4819806     1.46   0.145    -.2423324    1.646997
       south |  -21.52299   22.35225    -0.96   0.336    -65.33259    22.28662
             |
       race1 |
```

```
          1  |   3.054296   .1451752    21.04   0.000     2.769758    3.338834
          2  |   1.987268    .176538    11.26   0.000      1.64126    2.333277
          3  |          0  (omitted)
             |
       race2 |
          1  |  -.4816127    .434723    -1.11   0.268    -1.333654    .3704287
          2  |  -2.065527   .7694573    -2.68   0.007    -3.573635   -.5574181
          3  |          0  (omitted)
             |
       _cons |  -17.01633   18.01689    -0.94   0.345    -52.32879    18.29614
------------------------------------------------------------------------------
Endogenous: hours1 hours2
Exogenous:  south 1.race1 2.race1 1.race2 2.race2 union1 union2
```

This reproduces the separate estimates exactly: 1.107099 and 0.7023322 in the pooled regression against 1.107099 and 0.7023322 separately. The approach builds explicit dummies from the interaction of the fixed-effect variable and the subsample indicator, which becomes impractical when the number of fixed-effect levels is large.

One caution about how those dummies were built. `race1 = race*(south==1)` multiplies a *categorical code* by an indicator, so non-south observations are assigned `race1 = 0` and then `i.race1` treats 0 as its own category. That is safe here only because `race` has no zero level (it is coded 1/2/3). If the absorbed variable did have a 0 category, those observations and all the non-south ones would be silently collapsed into the same dummy. For a general recipe, build the interactions from the category dummies directly (`i.race#i.south`) rather than from the code.

Alternatively, pass the two interaction variables to `reghdfe`'s absorb option as a two-way fixed effect:

```
sysuse nlsw88, clear
gen hours1=hours*(south==1)
gen hours2=hours*(south==0)
gen union1=union*(south==1)
gen union2=union*(south==0)
gen race1=race*(south==1)
gen race2=race*(south==0)

ivreghdfe wage south (hours1 hours2 = union1 union2) , a(race1 race2) cluster(race)
test _b[hours1]-_b[hours2]=0
```

```
(NLSW, 1988 extract)

(4 missing values generated)

(4 missing values generated)

(368 missing values generated)
```

```
(368 missing values generated)



(MWFE estimator converged in 2 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics robust to heteroskedasticity and clustering on race

Number of clusters (race) =          3               Number of obs =     1877
                                                     F(  2,     2) = 13010.62
                                                     Prob > F      =   0.0001
Total (centered) SS     =  31273.10174               Centered R2   =  -3.9367
Total (uncentered) SS   =  31273.10174               Uncentered R2 =  -3.9367
Residual SS             =  154386.4216               Root MSE      =    9.089

------------------------------------------------------------------------------
             |               Robust
        wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      hours1 |   1.107099   .1331773     8.31   0.014     .5340838    1.680115
      hours2 |   .7023322   .5914076     1.19   0.357     -1.84229    3.246954
       south |          0  (omitted)
------------------------------------------------------------------------------
Underidentification test (Kleibergen-Paap rk LM statistic):              0.000
                                                     Chi-sq(1) P-val =    1.0000
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):                3.793
                         (Kleibergen-Paap rk Wald F statistic):          0.000
Stock-Yogo weak ID test critical values: 10% maximal IV size             7.03
                                         15% maximal IV size             4.58
                                         20% maximal IV size             3.95
                                         25% maximal IV size             3.63
Source: Stock-Yogo (2005).  Reproduced by permission.
NB: Critical values are for Cragg-Donald F statistic and i.i.d. errors.
------------------------------------------------------------------------------
Warning: estimated covariance matrix of moment conditions not of full rank.
         overidentification statistic not reported, and standard errors and
         model tests should be interpreted with caution.
Possible causes:
         number of clusters insufficient to calculate robust covariance matrix
         singleton dummy variable (dummy with one 1 and N-1 0s or vice versa)
partial option may address problem.
------------------------------------------------------------------------------
```

```
Collinearities detected among instruments: 1 instrument(s) dropped
Instrumented:         hours1 hours2
Included instruments: south
Excluded instruments: union1 union2
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
       race1 |         4           0           4     |
       race2 |         4           2           2     |
-----------------------------------------------------+


 ( 1)  hours1 - hours2 = 0

       F(  1,     2) =    0.31
            Prob > F =    0.6325
```

Absorbing `race1` and `race2` as a two-way fixed effect gives the same 1.107099 and 0.7023322 without building the dummies by hand, and scales to many fixed-effect levels. The Chow test is $F(1, 2) = 0.31$ with $p = 0.6325$ — but read that denominator: 2 degrees of freedom, because the standard errors are clustered on `race`, which has three categories. This is the problem the callout above flags, and it is why these IV/FE numbers illustrate mechanics rather than support any conclusion about the South.

3.6. Chow test with different covariates

The equations need not share the same covariates. If equation 1 includes X_2 and equation 2 does not,

$$Y_1 = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon_1, \qquad Y_2 = \gamma_0 + \gamma_1 X_1 + \epsilon_2, \tag{3.1}$$

the second equation implicitly sets the coefficient on X_2 to zero. In the stacked regression, interact X_2 with the first subsample only, so that the second equation gets no X_2 term:

```
sysuse nlsw88, clear
reg south wage hours tenure
est sto south
reg smsa wage hours
```

```
est sto smsa
suest south smsa
est sto suest
preserve
gen Y1=south
gen Y2=smsa
gen id=_n
reshape long Y, i(id) j(subsample)
gen wage1=wage*(subsample==1)
gen wage2=wage*(subsample==2)
gen hours1=hours*(subsample==1)
gen hours2=hours*(subsample==2)
replace tenure=0 if subsample==2
gen tenure1= tenure*(subsample==1)
reg Y wage? hours? tenure1 subsample, cluster(id)
est sto chow
test _b[hours1]-_b[hours2]=0
esttab  suest chow, nogaps mti
restore
```

```
(NLSW, 1988 extract)

      Source |       SS           df       MS      Number of obs   =     2,227
-------------+----------------------------------   F(3, 2223)      =     20.30
       Model |  14.4507388         3  4.81691294   Prob > F        =    0.0000
    Residual |  527.507052     2,223   .23729512   R-squared       =    0.0267
-------------+----------------------------------   Adj R-squared   =    0.0254
       Total |  541.957791     2,226  .243467112   Root MSE        =    .48713

------------------------------------------------------------------------------
       south | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        wage |  -.0119584   .0018359    -6.51   0.000    -.0155586   -.0083583
       hours |    .004828   .0010072     4.79   0.000     .0028528    .0068031
      tenure |  -.0024032   .0019221    -1.25   0.211    -.0061726    .0013661
       _cons |    .346361   .0392121     8.83   0.000     .2694648    .4232573
------------------------------------------------------------------------------

      Source |       SS           df       MS      Number of obs   =     2,242
-------------+----------------------------------   F(2, 2239)      =     36.17
       Model |  14.6274602         2  7.31373012   Prob > F        =    0.0000
    Residual |  452.719551     2,239  .202197209   R-squared       =    0.0313
-------------+----------------------------------   Adj R-squared   =    0.0304
       Total |  467.347012     2,241  .208543959   Root MSE        =    .44966

------------------------------------------------------------------------------
        smsa | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
        wage |   .0139103   .0016711     8.32   0.000     .0106332    .0171873
       hours |   .0003663   .0009155     0.40   0.689    -.0014291    .0021616
       _cons |   .5820585   .0357648    16.27   0.000      .511923    .6521941
------------------------------------------------------------------------------
```

```
Simultaneous results for south, smsa                  Number of obs = 2,242

------------------------------------------------------------------------------
             |               Robust
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
south_mean   |
        wage |  -.0119584    .001953    -6.12   0.000    -.0157862   -.0081307
       hours |    .004828   .0009971     4.84   0.000     .0028737    .0067822
      tenure |  -.0024032   .0018928    -1.27   0.204    -.0061131    .0013066
       _cons |    .346361    .038473     9.00   0.000     .2709554    .4217667
-------------+----------------------------------------------------------------
south_lnvar  |
       _cons |  -1.438451   .0096359  -149.28   0.000    -1.457337   -1.419565
-------------+----------------------------------------------------------------
smsa_mean    |
        wage |   .0139103   .0018313     7.60   0.000      .010321    .0174996
       hours |   .0003663   .0009099     0.40   0.687    -.0014172    .0021497
       _cons |   .5820585   .0361037    16.12   0.000     .5112965    .6528205
-------------+----------------------------------------------------------------
smsa_lnvar   |
       _cons |  -1.598512   .0195103   -81.93   0.000    -1.636751   -1.560272
------------------------------------------------------------------------------
```

```
(j = 1 2)

Data                               Wide   ->   Long
-----------------------------------------------------------------------------
Number of observations           2,246   ->   4,492
Number of variables                 23   ->   23
j variable (2 values)                    ->   subsample
xij variables:
                                 Y1 Y2   ->   Y
-----------------------------------------------------------------------------
```

```
(8 missing values generated)

(8 missing values generated)

(2,195 real changes made)

(15 missing values generated)


Linear regression                               Number of obs     =      4,469
                                                F(6, 2241)        =      81.10
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1091
                                                Root MSE          =      .4687

                                (Std. err. adjusted for 2,242 clusters in id)
------------------------------------------------------------------------------
             |               Robust
           Y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       wage1 |  -.0119584   .0019543    -6.12   0.000    -.0157909    -.008126
       wage2 |   .0139103   .0018325     7.59   0.000     .0103166    .0175039
      hours1 |    .004828   .0009978     4.84   0.000     .0028713    .0067846
      hours2 |   .0003663   .0009105     0.40   0.688    -.0014193    .0021519
     tenure1 |  -.0024032   .0018941    -1.27   0.205    -.0061176    .0013111
   subsample |   .2356975   .0550137     4.28   0.000     .1278144    .3435806
       _cons |   .1106636   .0878185     1.26   0.208    -.0615506    .2828778
------------------------------------------------------------------------------


 ( 1)  hours1 - hours2 = 0

       F(  1,  2241) =   10.03
            Prob > F =    0.0016


--------------------------------------------
                     (1)             (2)
                   suest            chow
--------------------------------------------
main
wage              -0.0120***
                  (-6.12)
hours             0.00483***
                   (4.84)
```

```
tenure           -0.00240
                  (-1.27)
wage1                          -0.0120***
                               (-6.12)
wage2                           0.0139***
                                (7.59)
hours1                         0.00483***
                                (4.84)
hours2                        0.000366
                                (0.40)
tenure1                       -0.00240
                               (-1.27)
subsample                        0.236***
                                (4.28)
_cons               0.346***     0.111
                   (9.00)       (1.26)
--------------------------------------------
south_lnvar
_cons               -1.438***
                 (-149.28)
--------------------------------------------
smsa_mean
wage                0.0139***
                    (7.60)
hours              0.000366
                    (0.40)
_cons                0.582***
                    (16.12)
--------------------------------------------
smsa_lnvar
_cons               -1.599***
                  (-81.93)
--------------------------------------------
N                     2242           4469
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The stacked model reproduces `suest`'s point estimates exactly: the hours coefficient is 0.004828 in the `south` equation and 0.0003663 in the `smsa` one, in both. Here the covariates differ — `tenure` appears in the first equation only. We include `tenure1` but no `tenure2`, which imposes the zero restriction in the `smsa` equation, and `subsample` stays the intercept shift. Setting `tenure = 0` in the second subsample first keeps the 15 women with missing `tenure` in the `smsa` equation, which does not use it. The Chow test on the hours coefficient gives $F(1, 2241) = 10.03$ with $p = 0.0016$: hours predicts living in the South quite differently from how it predicts living in an SMSA. (The `esttab` columns do not line up because `suest` stores equation-qualified names like `south:wage` while the stacked model stores plain `wage`.)

3.7. Conclusion

`suest` and the stacking method both test cross-equation hypotheses with robust standard errors. `sureg` does the same under a GLS framework, but it assumes homoscedastic errors across equations; under misspecification it remains consistent but its standard errors are invalid. When the equations share covariates `sureg` and OLS give the same point estimates; with different covariates they diverge.

The stacking approach is the most portable. It works in any language, extends to IV and fixed-effect models, and requires only that the single-equation estimator works — no special post-estimation infrastructure.

One pattern runs through every example in this chapter, and it is worth stating once on its own. A Chow test is a hypothesis about two coefficients that live in two different regressions. Stata's `test` can only reach coefficients inside a single stored estimation result, and, more importantly, testing a difference requires the covariance between the two estimates, which separate regressions never produce. Every construction here — the reshape for two outcomes, the `expand` for overlapping samples, the split instruments under IV, the absorbed fixed effects — exists to move both coefficients into one estimation so that the covariance exists and `test` can use it. `suest` does the same thing by a different route, refitting the stored estimates jointly with a robust cross-equation covariance matrix.

So the choice among them is practical rather than statistical. Use `suest` when the estimator supports it and the samples are the same. Stack when the samples overlap, when the covariates differ across equations, or when the estimator is one `suest` will not take — which, in practice, means most fixed-effect and IV commands.

4. Weights in OLS in causal inference

4.1. Unit level weights for OLS

This chapter follows Shinkre and Hazlett (2024).

Under unconfoundedness, the identifying assumption holds at each level of X. When X is discrete and low-dimensional, we can stratify and compute the ATE by the g-formula. In practice we usually run a linear regression

$$Y = \alpha + \beta D + \gamma X + \epsilon \tag{4.1}$$

and treat $\hat{\beta}$ as the treatment effect, relying on linearity to span the covariate space. The question is what this estimator is actually weighting.

Suppose we are interested in the ATE of D on Y. Say we have discrete variables X. Under unconfoundedness, the ATE is

$$\begin{aligned} \tau_{ATE} &= \sum E[Y(1) - Y(0)|X = x]P(X = x) \\ &= \sum (E[Y|D = 1, X = x] - E[Y|D = 0, X = x])P(X = x) \\ &= \sum DIM_x P(X = x) \end{aligned} \tag{4.2}$$

This is basically the g-formula, or regression adjustment formula. We compute the DIM (difference in means) between treated and control for each strata $X = x$, then the weighted average is the ATE, with $\hat{P}(X = x)$ as weights.

If we have high dimensional X, we usually do

$$Y = \beta_0 + \beta_1 X + \tau_{reg} D + \epsilon \tag{4.3}$$

We would hope $\hat{\tau}_{reg}$ is close to τ_{ATE}, but we are not sure.

OLS coefficients are:

$$\hat{\beta} = (Z'Z)^{-1} Z'y \tag{4.4}$$

Here I use Z to represent all regressors. Each coefficient is a weighted sum of the outcome,

$$\hat{\tau}_{reg} = \sum \omega_i Y_i \tag{4.5}$$

By FWL theorem,

$$\hat{\tau}_{reg} = \frac{\sum Y_i(D_i - \hat{d}(X_i))}{\sum(D_i - \hat{d}(X_i))^2} \tag{4.6}$$

Here $\hat{d}(X_i) = \hat{p}(D_i = 1|X_i)$ which is the predicted value of linear regression of D on X. The numerator is the covariance between Y and residualized D, the denominator is the variance of the residualized D. Note that the original FWL would have residualized Y here, but it turns out it works the same with the original Y.

Therefore the weights are

$$\omega_i = \frac{D_i - \hat{d}(X_i)}{\sum(D_i - \hat{d}(X_i))^2} \tag{4.7}$$

Note for the denominator, it's a constant. It does *not* scale the weights to sum to one — they sum to **zero**, because $D_i - \hat{d}(X_i)$ is an OLS residual from a regression that includes an intercept. What the denominator normalizes is the contrast: since $\sum D_i(D_i - \hat{d}(X_i)) = \sum(D_i - \hat{d}(X_i))^2$ (the residual is orthogonal to the fitted values), we get $\sum_i \omega_i D_i = 1$, i.e. the weights on the treated sum to $+1$ and those on the control sum to -1. That is what makes $\sum_i \omega_i Y_i$ a weighted *difference* of means. The numerator for treated units are $1 - \hat{d}(X_i)$, for control units it's $-\hat{d}(X_i)$ (negative). It is similar to IPW weights, which are $1/\hat{d}(X_i)$ for treated and $1/(1 - \hat{d}(X_i))$ for control. These two sets of weights go the same direction. For the treated, the higher probability of being treated, the lower weight; for the control, the higher probability of being treated, the larger the magnitude of the (negative) weight.

If these weights are the same as the IPW weights, then we'll get $\tau_{reg} = \tau_{ATE}$. However, they are usually different.

Another way to write this as weighted difference in means:

$$\hat{\tau}_{wdim} = \sum \omega_i Y_i = \sum_{i:D=1} \tilde{\omega}_i Y_i - \sum_{i:D=0} \tilde{\omega}_i Y_i \tag{4.8}$$

where, with $c = \sum(D_i - \hat{d}(X_i))^2$, the per-arm weights are $\tilde{\omega}_i = (1 - \hat{d}(X_i))/c$ for treated units ($D_i = 1$) and $\tilde{\omega}_i = \hat{d}(X_i)/c$ for control units ($D_i = 0$), so each arm's weights sum to one

Because $\hat{d}(X_i)$ is from a linear probability model, it's possible that it's not between 0 and 1. In that case, we can have negative weights. The negative weights are a sign that the fitted linear comparison extrapolates. For example, $\hat{d}(X_i) > 1$ will cause the weight $(1 - \hat{d}(X_i))/c$ to flip sign and become negative for a treated unit. The linear probability model then predicts this unit to be treated with probability above one, which suggests it may have no close match among the controls. It does not prove that: sampling noise and a misspecified linear model can also push $\hat{d}(X_i)$ outside $[0, 1]$, and conversely an in-range prediction does not establish common support. The mirror image happens for a control unit with $\hat{d}(X_i) < 0$: its weight $-\hat{d}(X_i)/c$ flips to *positive*, so it enters the contrast on the same side as the treated. Either way the unit contributes with the wrong sign.

The negative weights are invisible unless we examine $\hat{d}(X_i)$ directly — which is to say, unless we look at the propensity score. A linear regression of Y on D and X hides the extrapolation. So $\hat{\tau}_{\text{reg}}$ is a weighted average of outcomes with weights that differ from the ATE weights, and some of those weights may carry the wrong sign without the researcher knowing.

4.2. Strata level weights

Angrist (1998) has the strata-wise weights for OLS:

$$\hat{\tau}_{reg} = \frac{\sum_x \hat{\tau}_x \hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)}{\sum_x \hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)} \tag{4.9}$$

this shows that different from ATE, these weights are $\hat{d}(x)(1-\hat{d}(x))\hat{P}(X=x)$. The strata are weighted not only by $P(X_i = x)$, but also by the variance of treatment status. When $\hat{d}(X_i)$ is close to .5, the weight is the largest.

If there is no treatment effect heterogeneity, then it does not matter how you weight. When there is treatment effect heterogeneity, this estimator could be very different from ATE.

Shinkre and Hazlett show that Angrist's formula is a special case: it is exact when the probability of treatment is linear in X (as in a saturated model with discrete X). They derive a more general formula, but the Angrist version suffices under linearity of $d(X)$.

4.3. Recommended estimation approaches

Under constant effects and linearity, $E[Y \mid X, D] = \beta_0 + \beta_1 X + \beta_2 D$, a regression of Y on D and X recovers the ATE without further adjustment. When the linearity or constant-effects assumption fails, Shinkre and Hazlett suggest relaxing to *separate* linearity — allowing different intercepts for $Y(0)$ and $Y(1)$. Under that weaker assumption, several estimators are unbiased for the ATE.

The interaction (Winston Lin 2013) estimator regresses Y on D, $\tilde{X} = X - \bar{X}$, and $D\tilde{X}$; the coefficient on D is the ATE. The imputation estimator fits separate regressions for treated and control, predicts counterfactual outcomes for the other group, and takes the average difference — algebraically, it returns the same estimate. Stratification and g-computation do the same for discrete X: compute the stratum-level difference in means and weight by the stratum probability.

All of these still rely on linearity within each arm. When that fails, matching (e.g. mean-balancing) or nonparametric estimation is needed.

4.4. A worked example

We make the regression weights ω_i concrete and compare them to IPW weights $1/\hat{d}(X_i)$ (treated) and $1/(1-\hat{d}(X_i))$ (control).

We draw $n = 500$ observations with a single covariate $X \sim N(0,1)$, treatment $D \sim$ Bernoulli($\Lambda(0.5X)$) — so 238 units are treated — and outcome $Y = 1 + 2D + 1.5X + \varepsilon$ with $\varepsilon \sim N(0,1)$. The effect is constant at 2, so there is no heterogeneity for the weights to mis-average; the point of this first example is only to show that the weights reproduce the coefficient.

```r
library(dplyr)

set.seed(1)
n <- 500
X <- rnorm(n)
d_prob <- plogis(0.5 * X)
D <- rbinom(n, 1, d_prob)
Y <- 1 + 2 * D + 1.5 * X + rnorm(n)
dat <- data.frame(Y, D, X)

# Regression-implied weights via the FWL residualization of D on X.
d_hat <- fitted(lm(D ~ X, data = dat))
resid_D <- dat$D - d_hat
omega_reg <- resid_D / sum(resid_D^2)

# IPW weights (Horvitz-Thompson), using the same linear-probability d_hat
# purely for comparability with the regression weights above.
ipw <- ifelse(dat$D == 1, 1 / d_hat, 1 / (1 - d_hat))
# rescale by a single constant so the treated weights average 1; this only
# fixes the axis scale of the plot below and does not change any comparison
ipw <- ipw / sum(ipw[dat$D == 1]) * sum(dat$D == 1)

tau_reg <- sum(omega_reg * dat$Y)
tau_lm  <- coef(lm(Y ~ D + X, data = dat))["D"]

data.frame(tau_reg_from_weights = tau_reg, tau_from_lm_coef = tau_lm)
```

```
  tau_reg_from_weights tau_from_lm_coef
D             2.084644         2.084644
```

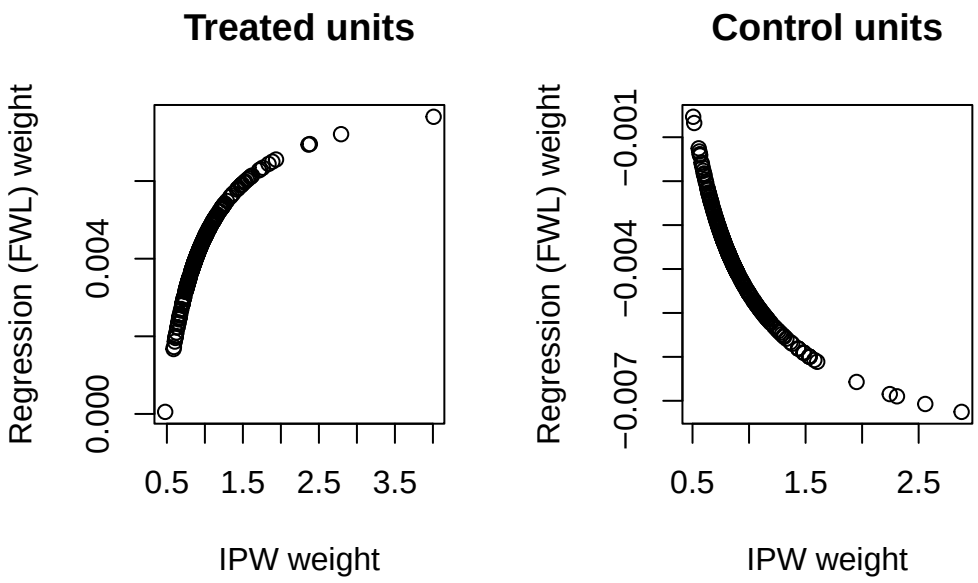
Both give 2.084644, identical to seven digits, confirming $\hat{\tau}_{reg} = \sum \omega_i Y_i$. (Both sit a little above the true 2; with $n = 500$ and a constant effect that is sampling noise, not weighting.)

Plotting `omega_reg` against `ipw` for treated and control units separately shows the two weighting schemes move in the same direction but are not proportional.

Be careful about what that does and does not establish. Non-proportional weights are *necessary* for $\hat{\tau}_{reg}$ to differ from the ATE, not sufficient. In this example the effect is constant at 2, so every unit's effect is the same number and any weights that form a difference of means return that number: $\hat{\tau}_{reg}$ and the ATE coincide here no matter how oddly the weights are distributed. A weighting problem needs two ingredients, and this design has only one. The section after next supplies the other.

```r
par(mfrow = c(1, 2))
plot(ipw[dat$D == 1], omega_reg[dat$D == 1],
     xlab = "IPW weight", ylab = "Regression (FWL) weight",
     main = "Treated units")
plot(ipw[dat$D == 0], omega_reg[dat$D == 0],
```

```
     xlab = "IPW weight", ylab = "Regression (FWL) weight",
     main = "Control units")
```




```
par(mfrow = c(1, 1))
```

Both panels slope upward and neither is a straight line through the origin: the two schemes rank units the same way but do not scale together, which is the whole content of $\hat{\tau}_{reg} \neq \hat{\tau}_{ATE}$.

Note also what this example does *not* show. Here $\hat{d}(X_i)$ stays inside $[0.062, 0.994]$, so every weight keeps its expected sign and the negative-weight problem discussed above never materializes. That is a property of this particular design, not a reassurance. We now strengthen the selection from $\Lambda(0.5X)$ to $\Lambda(3X)$, holding everything else fixed, and the linear probability model starts predicting outside the unit interval:

```
set.seed(2)
X2 <- rnorm(n)
D2 <- rbinom(n, 1, plogis(3 * X2))       # much stronger selection on X
dat2 <- data.frame(D = D2, X = X2)
d_hat2 <- fitted(lm(D ~ X, data = dat2))
omega2 <- (dat2$D - d_hat2) / sum((dat2$D - d_hat2)^2)

data.frame(
  dhat_below_0        = sum(d_hat2 < 0),
  dhat_above_1        = sum(d_hat2 > 1),
  treated_wrong_sign  = sum(dat2$D == 1 & omega2 < 0),
  control_wrong_sign  = sum(dat2$D == 0 & omega2 > 0)
)
```

```
  dhat_below_0 dhat_above_1 treated_wrong_sign control_wrong_sign
1           28           42                 42                 28
```

The fitted probabilities now run from -0.367 to 1.445. Twenty-eight units fall below zero and 42 rise above one, and the wrong-sign counts match those two numbers exactly: 42 treated units carry a negative weight and 28 control units carry a positive one. That is not a coincidence but the mechanism described above — every unit with $\hat{d} > 1$ here is treated, every unit with $\hat{d} < 0$ is control, and each flips the sign of its own contribution. In total 70 of 500 observations, 14% of the sample, enter the contrast on the wrong side.

Those units are the ones with no counterpart in the other arm, and a plain regression of Y on D and X gives you no indication that they are there. Nothing in the coefficient table changes appearance; only looking at $\hat{d}(X_i)$ reveals it.

This is a property of the *regression* weights specifically, and the reason is worth being explicit about. $\hat{d}(X_i)$ here is not a propensity score we chose to estimate with a linear probability model; it is the object OLS implicitly uses, because the FWL residualisation of D on X *is* a linear regression. Linear fitted values are not confined to $[0, 1]$, so out-of-range values, and with them wrong-sign weights, are always available to OLS. Genuine IPW weights built from a logit or probit propensity score cannot do this: $\hat{\pi}(X)$ lies strictly inside $(0, 1)$ by construction, so $1/\hat{\pi}$ and $1/(1-\hat{\pi})$ stay positive. IPW has its own failure mode when $\hat{\pi}$ approaches 0 or 1 — weights explode, and the estimate rests on a handful of observations — but it is a visible one. The `ipw` vector plotted above is deliberately built from the same linear $\hat{d}$ so that the two weighting schemes are comparable unit by unit; it is a diagnostic device, not the IPW estimator one would actually run.

4.5. When the weights actually bite

Neither example so far has shown $\hat{\tau}_{reg}$ miss the ATE, because neither had heterogeneous effects. This one does, and it is built on discrete X so that every quantity in this chapter can be computed exactly rather than approximated.

Let $X \in \{0, 1, 2\}$ with probabilities $0.4, 0.3, 0.3$; let the treatment probability be $d(x) = 0.5, 0.1, 0.9$; and let the effect be $\tau(x) = 1, 4, 4$. The outcome is $Y = 1 + \tau(X)D + 2X + \varepsilon$ with $\varepsilon \sim N(0, 1)$, and we draw $n = 20{,}000$. The design is chosen so that the stratum where treatment is most balanced, $x = 0$ with $d = 0.5$, is also the stratum with the small effect.

The population ATE follows from the definition: $0.4(1) + 0.3(4) + 0.3(4) = 2.8$. Angrist's strata-level formula above weights each stratum by $d(x)\{1-d(x)\}P(X = x)$ instead — that is $0.10, 0.027, 0.027$, or $0.649, 0.175, 0.175$ once normalised, against the true shares $0.4, 0.3, 0.3$. The $x = 0$ stratum carries 65% of the regression's weight and 40% of the ATE's, so the regression should be pulled toward that stratum's effect of 1.

```
set.seed(11)
n  <- 20000
px <- c(0.4, 0.3, 0.3)          # P(X = 0, 1, 2)
dx <- c(0.5, 0.1, 0.9)          # P(D = 1 | X)
tx <- c(1,   4,   4)            # tau(X)
```

```r
X   <- sample(0:2, n, TRUE, prob = px)
D   <- rbinom(n, 1, dx[X + 1])
Y   <- 1 + tx[X + 1] * D + 2 * X + rnorm(n)
dat3 <- data.frame(Y, D, X = factor(X))

# What OLS returns
tau_ols <- coef(lm(Y ~ D + X, data = dat3))["D"]

# Angrist's variance-weighted average of the stratum effects, computed from the
# population quantities rather than estimated
w   <- dx * (1 - dx) * px
ang <- sum(w * tx) / sum(w)

# g-computation on a saturated arm-specific model
m   <- lm(Y ~ D * X, data = dat3)
tau_g <- mean(predict(m, transform(dat3, D = 1)) -
              predict(m, transform(dat3, D = 0)))

# Lin's interaction estimator: D, centred covariates, and their interaction
Xc  <- scale(model.matrix(~ X, dat3)[, -1, drop = FALSE], scale = FALSE)
tau_lin <- coef(lm(dat3$Y ~ dat3$D + Xc + dat3$D:Xc))[2]

data.frame(population_ATE = sum(px * tx),
           angrist_formula = ang,
           ols_coefficient = tau_ols,
           g_computation = tau_g,
           lin_interaction = tau_lin, row.names = NULL)
```

```
  population_ATE angrist_formula ols_coefficient g_computation lin_interaction
1            2.8        2.051948        2.082158      2.839965        2.839965
```

The OLS coefficient is 2.08 against a population ATE of 2.80. It is not a noisy estimate of the ATE; it is a precise estimate of something else. Angrist's formula predicts 2.052 from the population quantities alone, and the regression delivers 2.082 — the gap between those two is sampling noise, while the gap between either of them and 2.80 is the weighting problem. The regression is answering a question about the stratum where treatment happens to be most balanced, and nothing in its output says so.

The example also bounds what can be said about ATT and ATU. Here the ATT and the ATU are 2.8 as well: $P(D=1)=0.5$, and the ATT is $(0.2\times 1+0.03\times 4+0.27\times 4)/0.5=2.8$, with the ATU the same by symmetry. So the OLS coefficient of 2.05 is not a weighted average of the ATT and ATU either. Unconfoundedness holds by construction. What fails is the linearity the ATT/ATU representation of Słoczyński (2022) needs: the stratum effects 1, 4, 4 at propensities 0.5, 0.1, 0.9 are not linear in the propensity score.

The two recommended estimators recover the ATE, and recover it identically: g-computation and Lin's interaction estimator both return 2.840, against a realised sample ATE of 2.814. That

they agree to every printed digit is the algebraic equivalence noted in the previous section, not a coincidence of this draw. Both work here because the saturated model in a discrete X satisfies separate linearity exactly — each arm's conditional mean is linear in the stratum dummies, since it is *saturated* in them.

The practical reading is that the weighting problem needs heterogeneity and non-proportional weights together. The first two examples had the weights; this one adds the heterogeneity, and the cost appears immediately. Since one rarely knows in advance whether effects are heterogeneous, the cheap insurance is to run the interaction or imputation estimator by default: under constant effects it returns what OLS would have returned, and under heterogeneity it returns the ATE that OLS would have missed.

4.6. Summary

The OLS coefficient on a treatment dummy, with covariates entered linearly, is a weighted sum of outcomes. The weights come from residualizing treatment on the covariates. In the language of strata, OLS weights each stratum's effect by the variance of treatment in it, $d(x)(1 - d(x))$, as well as by its size. That gives most weight to strata where treatment is close to 50-50. Because the implicit propensity is linear, it can leave $[0, 1]$, and then units with no counterpart in the other group enter with the wrong sign, invisibly. The OLS coefficient differs from the ATE when these weights are not proportional to the ATE's weights and effects vary across strata.

In the constant-effect example the weights reproduce the OLS coefficient of 2.08. Stronger selection puts 14% of the sample on the wrong side. In the heterogeneous discrete example the ATE is 2.80, and OLS gives 2.08, as Angrist's formula predicts, because it overweights the balanced stratum with the small effect. G-computation and Lin's interaction estimator both give 2.840. Running those by default costs nothing when effects are constant.

Part II.

Marginal Effects & Fixed Effects

5. Marginal effects in models with fixed effects

5.1. Marginal effects in a linear model

Stata's `margins` command computes predicted means and marginal effects, but care is needed when fixed effects are present. In a linear model the issue is benign: demeaning and dummy-variable approaches return the same coefficients and the same marginal effects.

We check that on the `auto` data, treating the 5-category repair record `rep78` as the panel identifier and regressing `price` on `mpg` interacted with `trunk` — once with `xtreg, fe` (within-demeaning) and once with `reg` plus `i.rep78` dummies.

```
clear
sysuse auto
xtset rep78
xtreg price c.mpg##c.trunk, fe
margins , dydx(mpg)
reg price c.mpg##c.trunk i.rep78
margins , dydx(mpg)
```

```
. clear

. sysuse auto
(1978 automobile data)

. xtset rep78

Panel variable: rep78 (unbalanced)

. xtreg price c.mpg##c.trunk, fe

Fixed-effects (within) regression               Number of obs     =         69
Group variable: rep78                           Number of groups  =          5

R-squared:                                      Obs per group:
     Within  = 0.2570                                         min =          2
     Between = 0.0653                                         avg =       13.8
     Overall = 0.2237                                         max =         30

                                                F(3, 61)          =       7.03
```

```
corr(u_i, Xb) = -0.4133                         Prob > F          =     0.0004

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -98.12003   226.8708    -0.43   0.667    -551.7763    355.5362
       trunk |   295.0544   343.3934     0.86   0.394    -391.6032     981.712
             |
       c.mpg#|
     c.trunk |  -12.23318   15.94713    -0.77   0.446    -44.12143    19.65506
             |
       _cons |    7574.85   5321.325     1.42   0.160    -3065.797     18215.5
-------------+----------------------------------------------------------------
     sigma_u |   992.2156
     sigma_e |  2631.2869
         rho |  .12449059   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(4, 61) = 0.86                     Prob > F = 0.4948

. margins , dydx(mpg)

Average marginal effects                                  Number of obs = 69
Model VCE: Conventional

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -268.4981   74.12513    -3.62   0.000    -413.7807   -123.2156
------------------------------------------------------------------------------

. reg price c.mpg##c.trunk i.rep78

      Source |       SS           df       MS      Number of obs   =        69
-------------+----------------------------------   F(7, 61)        =      3.19
       Model |   154453046         7  22064720.8   Prob > F        =    0.0061
    Residual |   422343913        61  6923670.71   R-squared       =    0.2678
-------------+----------------------------------   Adj R-squared   =    0.1838
       Total |   576796959        68  8482308.22   Root MSE        =    2631.3

------------------------------------------------------------------------------
       price | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -98.12003   226.8708    -0.43   0.667    -551.7763    355.5362
       trunk |   295.0544   343.3934     0.86   0.394    -391.6032     981.712
```

```
             |
       c.mpg#|
     c.trunk |  -12.23318   15.94713    -0.77   0.446    -44.12143    19.65506
             |
       rep78 |
          2  |   438.0002   2161.922     0.20   0.840    -3885.031    4761.031
          3  |   987.1363   2022.606     0.49   0.627    -3057.315    5031.587
          4  |   1240.944   2046.417     0.61   0.547     -2851.12    5333.008
          5  |    2605.83   2161.837     1.21   0.233    -1717.031    6928.691
             |
       _cons |   6355.731   5209.899     1.22   0.227    -4062.105    16773.57
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                    Number of obs = 69
Model VCE: OLS

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |  -268.4981   74.12513    -3.62   0.001    -416.7205   -120.2758
------------------------------------------------------------------------------

.
```

The two agree exactly. Both give -98.12003 on `mpg`, 295.0544 on `trunk` and -12.23318 on the interaction, and both average marginal effects are -268.4981 with a standard error of 74.12513. The only difference is that `xtreg` reports a z statistic and `reg` a t, so the confidence intervals differ in the fourth digit. Nothing here needs care.

5.2. Marginal effects in a non-linear model

In a nonlinear model the two approaches diverge. We now need a genuine panel and a count outcome, so we simulate one: 250 observations, 50 units observed over 5 periods. Within each unit, $\text{mpg} \sim N(20, 4^2)$ and $\text{trunk} \sim N(15, 3^2)$ vary period to period, while the unit effect $\alpha_i \sim N(0, 1)$ is drawn once and held fixed. The conditional mean is $\lambda_{it} = \exp(0.5 + 0.05\,\text{mpg}_{it} - 0.03\,\text{trunk}_{it} + \alpha_i)$ and $\text{accidents}_{it} \sim \text{Poisson}(\lambda_{it})$. The true coefficients are 0.05 and -0.03.

```
clear
set seed 42
set obs 250
```

```
gen id = ceil(_n/5)
bys id: gen t = _n
gen mpg = rnormal(20,4)
gen trunk = rnormal(15,3)
gen alpha = rnormal(0,1)
bys id: replace alpha = alpha[1]
gen lambda = exp(0.5 + 0.05*mpg - 0.03*trunk + alpha)
gen accidents = rpoisson(lambda)
xtset id
xtpoisson accidents mpg trunk, fe
margins , dydx(mpg)
margins , dydx(mpg) predict(nu0)
poisson accidents mpg trunk i.id
margins , dydx(mpg)
```

```
. clear

. set seed 42

. set obs 250
Number of observations (_N) was 0, now 250.

. gen id = ceil(_n/5)

. bys id: gen t = _n

. gen mpg = rnormal(20,4)

. gen trunk = rnormal(15,3)

. gen alpha = rnormal(0,1)

. bys id: replace alpha = alpha[1]
(200 real changes made)

. gen lambda = exp(0.5 + 0.05*mpg - 0.03*trunk + alpha)

. gen accidents = rpoisson(lambda)

. xtset id

Panel variable: id (balanced)

. xtpoisson accidents mpg trunk, fe
note: 1 group (5 obs) dropped because of all zero outcomes
```

```
Iteration 0:  Log likelihood = -358.34297
Iteration 1:  Log likelihood = -340.16596
Iteration 2:  Log likelihood = -340.16413
Iteration 3:  Log likelihood = -340.16413

Conditional fixed-effects Poisson regression         Number of obs    =    245
Group variable: id                                   Number of groups =     49

                                                     Obs per group:
                                                                  min =      5
                                                                  avg =    5.0
                                                                  max =      5

                                                     Wald chi2(2)     =  35.90
Log likelihood = -340.16413                          Prob > chi2      = 0.0000

------------------------------------------------------------------------------
   accidents | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090524     5.39   0.000     .0310521    .0665371
       trunk |  -.0316066   .0108931    -2.90   0.004    -.0529566   -.0102566
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                  Number of obs = 245
Model VCE: OIM

Expression: Linear prediction, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090524     5.39   0.000     .0310521    .0665371
------------------------------------------------------------------------------

. margins , dydx(mpg) predict(nu0)

Average marginal effects                                  Number of obs = 245
Model VCE: OIM

Expression: Predicted number of events (assuming u_i=0), predict(nu0)
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
```

```
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+------------------------------------------------------------------
         mpg |   .0850287   .0340547     2.50   0.013     .0182826    .1517747
------------------------------------------------------------------------------

. poisson accidents mpg trunk i.id

Iteration 0:  Log likelihood = -485.81671
Iteration 1:  Log likelihood = -453.45562
Iteration 2:  Log likelihood = -453.02481
Iteration 3:  Log likelihood = -452.93985
Iteration 4:  Log likelihood = -452.92127
Iteration 5:  Log likelihood = -452.91701
Iteration 6:  Log likelihood =   -452.916
Iteration 7:  Log likelihood = -452.91579
Iteration 8:  Log likelihood = -452.91575

Poisson regression                                      Number of obs =    250
                                                        LR chi2(51)   = 934.93
                                                        Prob > chi2   = 0.0000
Log likelihood = -452.91575                             Pseudo R2     = 0.5079

------------------------------------------------------------------------------
   accidents | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .0487946   .0090525     5.39   0.000      .031052    .0665371
       trunk |  -.0316067   .0108931    -2.90   0.004    -.0529568   -.0102565
             |
          id |
           2 |   .3464221   .5592835     0.62   0.536    -.7497533    1.442598
           3 |   1.123341   .5131078     2.19   0.029     .1176679    2.129014
           4 |   .1098041   .6057603     0.18   0.856    -1.077464    1.297073
           5 |   .4069152   .5702496     0.71   0.475    -.7107536    1.524584
           6 |   1.791221   .4776303     3.75   0.000     .8550829    2.727359
           7 |  -14.00991   475.5637    -0.03   0.976    -946.0976    918.0778
           8 |   .9632705   .5216941     1.85   0.065    -.0592311    1.985772
           9 |   .1039008   .6061094     0.17   0.864    -1.084052    1.291853
          10 |   .7748186   .5273812     1.47   0.142    -.2588296    1.808467
          11 |   .7331684   .5395719     1.36   0.174    -.3243732     1.79071
          12 |   1.114943   .5007908     2.23   0.026      .133411    2.096475
          13 |  -.2054342   .6343396    -0.32   0.746    -1.448717    1.037848
          14 |   1.248116   .4945786     2.52   0.012     .2787602    2.217472
          15 |   1.395876   .4914484     2.84   0.005     .4326552    2.359098
          16 |   1.689054   .4845687     3.49   0.000     .7393167    2.638791
          17 |   .9941942   .5171224     1.92   0.055    -.0193471    2.007735
          18 |   1.742704   .4844911     3.60   0.000      .793119    2.692289
          19 |   .9912213   .5168563     1.92   0.055    -.0217985    2.004241
          20 |  -1.134862   .8375742    -1.35   0.175    -2.776477    .5067532
```

```
          21  |   1.484731   .4879013     3.04   0.002     .5284622       2.441
          22  |   2.442088   .4643803     5.26   0.000     1.531919    3.352256
          23  |   .9732572   .5265941     1.85   0.065    -.0588483    2.005363
          24  |    .280425   .5859507     0.48   0.632    -.8680172    1.428867
          25  |   2.472329   .4641555     5.33   0.000     1.562601    3.382057
          26  |   .5602592   .5485592     1.02   0.307    -.5148971    1.635416
          27  |   1.537242   .4861325     3.16   0.002     .5844401    2.490044
          28  |    1.34391   .5026286     2.67   0.008     .3587763    2.329044
          29  |   2.370956   .4660218     5.09   0.000      1.45757    3.284342
          30  |   2.453329   .4642702     5.28   0.000     1.543377    3.363282
          31  |   .9383342   .5171502     1.81   0.070    -.0752616     1.95193
          32  |   .1850002   .5864331     0.32   0.752    -.9643876    1.334388
          33  |  -.2686233   .6711829    -0.40   0.689    -1.584118    1.046871
          34  |   1.547864    .485423     3.19   0.001     .5964525    2.499276
          35  |   1.557431   .4878123     3.19   0.001     .6013366    2.513526
          36  |   1.295147   .4965837     2.61   0.009     .3218611    2.268433
          37  |   2.668556   .4616456     5.78   0.000     1.763748    3.573365
          38  |   .9287953   .5326269     1.74   0.081    -.1151343    1.972725
          39  |  -.9247308   .8366882    -1.11   0.269     -2.56461     .715148
          40  |   1.358213   .4979867     2.73   0.006     .3821769    2.334249
          41  |  -.0472585   .6070096    -0.08   0.938    -1.236975    1.142458
          42  |   1.394504   .4958685     2.81   0.005     .4226201    2.366389
          43  |  -.5414444   .7303533    -0.74   0.458    -1.972911    .8900219
          44  |   .9881123    .513744     1.92   0.054    -.0188073    1.995032
          45  |   1.171458   .5165503     2.27   0.023     .1590378    2.183878
          46  |    .817007   .5269245     1.55   0.121    -.2157461     1.84976
          47  |   2.800173   .4596719     6.09   0.000     1.899233    3.701113
          48  |    .365911   .5704157     0.64   0.521    -.7520831    1.483905
          49  |   2.445633   .4649624     5.26   0.000     1.534324    3.356943
          50  |   .7177863   .5279692     1.36   0.174    -.3170143    1.752587
              |
       _cons  |  -.4227135   .5046249    -0.84   0.402     -1.41176     .566333
------------------------------------------------------------------------------

. margins , dydx(mpg)

Average marginal effects                                  Number of obs = 250
Model VCE: OIM

Expression: Predicted number of events, predict()
dy/dx wrt:  mpg

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         mpg |   .2238673   .0420551     5.32   0.000     .1414408    .3062937
------------------------------------------------------------------------------
```

.

(We use a simulated count outcome `accidents` and a proper panel with 50 units of 5 periods each, rather than `auto`'s continuous `price` and its 5-category `rep78` — a true panel needs many units, and a fixed-effect count model needs an actual count dependent variable.)

In this example, "xtpoisson, fe" and "poisson … i.id" return the same coefficients (both give `mpg` = .0487946 with a standard error of .0090524). Fixed effect Poisson model (sometimes called conditional fixed effect Poisson) is the same model as a Poisson model with dummies, just like a linear model (OLS with dummies is the same as fixed effect OLS). Poisson model and OLS are unique in this sense that there is no "incidental parameter" problem.

Both recover the truth: 0.0488 against a true 0.05 for `mpg`, and −0.0316 against −0.03 for `trunk`. Note the sample sizes differ, 245 against 250 — `xtpoisson, fe` drops the one group whose outcome is zero in every period, because such a group contributes nothing to the conditional likelihood. It changes neither coefficient here.

The `margins` calls return different marginal effects even though the models are the same, and the spread is not small:

call	AME of `mpg`	std. err.
`margins, dydx(mpg)` after `xtpoisson, fe`	0.0488	0.0091
`margins, dydx(mpg) predict(nu0)`	0.0850	0.0341
`margins, dydx(mpg)` after `poisson ... i.id`	0.2239	0.0421

Three answers from one coefficient, the largest 4.6 times the smallest. In the conditional fixed-effect Poisson the fixed effects are conditioned out — they are not estimated. The first row is not really a marginal effect at all. After `xtpoisson, fe`, `margins` defaults to the linear prediction `xb` (the output says `Expression: Linear prediction`), so it returns $\partial(X\beta)/\partial x = \beta$, the coefficient itself. After `poisson`, the default is the expected count. `margins, predict(nu0)` sets the fixed effect to zero, which is arbitrary, and note that the conditional model has no intercept either — it was swept out with the α_i. So `nu0` exponentiates $X\beta$ alone, giving a baseline count no unit in the data actually has. The Poisson with dummies estimates the fixed effects, so `margins` uses actual estimates and gets the largest value.

The arithmetic is easy to follow, since for a Poisson $\partial\lambda/\partial x = \beta\lambda$ and therefore every one of these AMEs is β times an average λ. Divide each row by 0.0488 and the implied baselines are 1.00, 1.74 and 4.59; the 1.00 only restates that the first row is β. The last matches $E[\lambda] = e^{1.05}E[e^{\alpha}] = 4.71$ from the DGP; the middle matches $E[e^{X\beta}]$ without intercept or fixed effect, $E[e^{0.05\,\mathrm{mpg}-0.03\,\mathrm{trunk}}] = e^{0.55+(0.05^2\cdot 16+0.03^2\cdot 9)/2} = 1.78$ from the DGP. The three rows differ only in what they assume about α_i, and none of them is a modelling subtlety — it is a choice of baseline. The third is the defensible one, but for the conditional model the safest route is to report the original coefficients and avoid `margins` for marginal effects entirely.

The conditional logit is worse. The fixed effects are not estimated, and unlike Poisson there is an incidental-parameter problem, so logit with dummies is inconsistent unless the panel is deep (roughly 20+ observations per unit). With conditional logit the predicted probability depends on

the unestimated α_i; `margins` after `clogit` or `xtlogit, fe` offers options like `pu0` (set all fixed effects to zero), but none is defensible.

In a fixed effect logit model,

$$log(P(y=1)/(1-P(y=1))) = \alpha_i + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 * x_2 \tag{5.1}$$

Here α_i is the unit fixed effect. Therefore

$$P(y=1) = F(\alpha_i + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 * x_2) \tag{5.2}$$

Without estimating α_i there is no way to predict P in a meaningful way. But the log-odds are linear in the covariates, and the marginal effect of x_1 or x_2 on the log-odds does not involve α_i. So `margins` with the `predict(xb)` option — marginal effects on the log-odds scale — is interpretable. Note that `xb` here is $X_{it}\beta$ only — the individual fixed effect α_i is not estimated in conditional logit and so is not part of the prediction. The logged-odds *effect* of a covariate ($\partial(X\beta)/\partial x$) therefore matches the original coefficient exactly and does not depend on α_i, but `xb` itself is a log-odds relative to each individual's (unidentified) baseline, not the individual's absolute log-odds. This lets you make linear extrapolations of relative log-odds across values of the covariates, but not of absolute predicted probabilities, since those would require α_i.

```
clear
webuse union
clogit union c.age##i.south not_smsa grade, group(idcode)
margins, at( age=(15 20 25 30 35 40) south=(0 1)) predict(xb)
marginsplot
graph export "marginsplot-clogit.svg", as(svg) replace
```

```
. clear

. webuse union
(NLS Women 14-24 in 1968)

. clogit union c.age##i.south not_smsa grade, group(idcode)
note: multiple positive outcomes within groups encountered.
note: 2,744 groups (14,165 obs) omitted because of all positive or
      all negative outcomes.

Iteration 0:  Log likelihood = -4518.8815
Iteration 1:  Log likelihood = -4512.8224
Iteration 2:  Log likelihood = -4512.8192
Iteration 3:  Log likelihood = -4512.8192

Conditional (fixed-effects) logistic regression         Number of obs = 12,035
                                                        LR chi2(5)    =  74.73
                                                        Prob > chi2   = 0.0000
```

```
Log likelihood = -4512.8192                              Pseudo R2     = 0.0082

------------------------------------------------------------------------------
       union | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0096842   .0050265     1.93   0.054    -.0001676     .019536
     1.south |  -1.382178    .276966    -4.99   0.000    -1.925022   -.8393346
             |
 south#c.age |
           1 |   .0208997   .0081247     2.57   0.010     .0049756    .0368238
             |
    not_smsa |   .0195233   .1131292     0.17   0.863    -.2022058    .2412523
       grade |   .0822276   .0419062     1.96   0.050      .000093    .1643622
------------------------------------------------------------------------------

. margins, at( age=(15 20 25 30 35 40) south=(0 1)) predict(xb)

Predictive margins                                      Number of obs = 12,035
Model VCE: OIM

Expression: Linear prediction, predict(xb)
1._at:  age   = 15
        south =  0
2._at:  age   = 15
        south =  1
3._at:  age   = 20
        south =  0
4._at:  age   = 20
        south =  1
5._at:  age   = 25
        south =  0
6._at:  age   = 25
        south =  1
7._at:  age   = 30
        south =  0
8._at:  age   = 30
        south =  1
9._at:  age   = 35
        south =  0
10._at: age   = 35
        south =  1
11._at: age   = 40
        south =  0
12._at: age   = 40
        south =  1

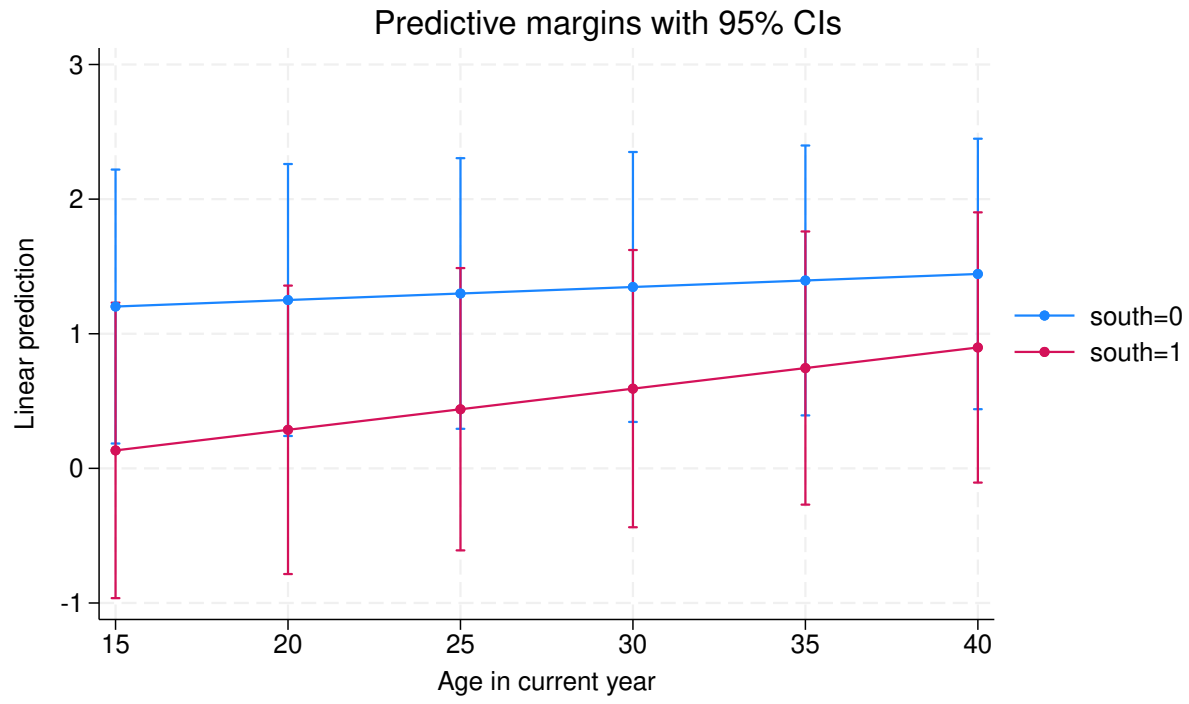
------------------------------------------------------------------------------
             |            Delta-method
```

```
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   1.202147   .5190753     2.32   0.021      .184778    2.219516
          2  |    .133464   .5599015     0.24   0.812    -.9639228    1.230851
          3  |   1.250568   .5153257     2.43   0.015      .240548    2.260588
          4  |   .2863834   .5465398     0.52   0.600    -.7848148    1.357582
          5  |   1.298989   .5127819     2.53   0.011     .2939548    2.304023
          6  |   .4393029   .5349589     0.82   0.412    -.6091973    1.487803
          7  |    1.34741   .5114619     2.63   0.008     .3449629    2.349857
          8  |   .5922223   .5252767     1.13   0.260    -.4373011    1.621746
          9  |   1.395831   .5113752     2.73   0.006     .3935538    2.398108
         10  |   .7451418   .5175997     1.44   0.150    -.2693351    1.759619
         11  |   1.444252   .5125224     2.82   0.005     .4397264    2.448777
         12  |   .8980612   .5120182     1.75   0.079    -.1054761    1.901598
------------------------------------------------------------------------------

. marginsplot

Variables that uniquely identify margins: age south

. graph export "marginsplot-clogit.svg", as(svg) replace
file marginsplot-clogit.svg saved as SVG format
```



This fits a conditional logit of union status on age, south, and their interaction, on 12,035 person-years from the `union` panel. The age coefficient is 0.0097 with a standard error of 0.0050, and the

south indicator is −1.382 with a standard error of 0.277. `margins` with `predict(xb)` then returns predicted log-odds at six ages crossed with south — twelve combinations — but not predicted probabilities, since those require α_i.

Read the vertical axis of that plot as a *relative* scale. Because α_i is not estimated, `xb` is each individual's log-odds measured from an unidentified baseline, so the level of any single point is arbitrary; what is interpretable is how the curves move across age and how far apart the south and non-south curves sit. Differences within the plot are the estimable quantities, not heights.

5.3. Summary

In a linear model, within-demeaning and unit dummies give the same coefficients and the same marginal effects, so fixed effects need no special care. In nonlinear models they diverge. Fixed-effects Poisson and Poisson with dummies give identical coefficients, because Poisson has no incidental-parameters problem. But the conditional model does not estimate the unit effects, so `margins` after it effectively sets the unit effect to zero. Since a Poisson marginal effect is the coefficient times an average predicted mean, the three `margins` calls give three different answers.

In the simulated panel both fits recover the true coefficient, 0.0488 against 0.05. The average marginal effects of the same coefficient are 0.0488, 0.0850 and 0.2239, depending on the baseline each prediction assumes. Conditional logit is worse, since the unit effects are not estimated and dummies would be inconsistent in short panels. Only effects on the log-odds, read relative to each unit's unknown baseline, are meaningful there.

6. Comparing Marginal effects with margins command

6.1. Comparing marginal effects

A common practice is to plot marginal effects with `margins` and `marginsplot`, then conclude two effects differ whenever their confidence intervals do not overlap. This can be wrong — overlapping intervals do not imply no significant difference, because the test depends on the joint distribution of the two estimates, not on their marginal distributions separately.

```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins female,  at(age=(20 30 40 50 60 70))
marginsplot
graph export "marginsplot1.svg", as(svg) replace
margins r.female,  at(age=(20 30 40 50 60 70))
marginsplot
graph export "marginsplot2.svg", as(svg) replace
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Adjusted predictions                                    Number of obs = 10,335
Model VCE: OIM
```

```
Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
  _at#female |
     1#Male  |   .0036348   .0009895     3.67   0.000     .0016953    .0055743
   1#Female  |   .0097316   .0018436     5.28   0.000     .0061184    .0133449
     2#Male  |    .007396   .0015622     4.73   0.000     .0043341    .0104579
   2#Female  |   .0160637   .0023434     6.85   0.000     .0114706    .0206568
     3#Male  |   .0149906   .0022831     6.57   0.000     .0105159    .0194653
   3#Female  |    .026406   .0027757     9.51   0.000     .0209657    .0318462
     4#Male  |   .0301469   .0029996    10.05   0.000     .0242677    .0360261
   4#Female  |    .043115   .0030923    13.94   0.000     .0370543    .0491758
     5#Male  |   .0596983   .0040248    14.83   0.000     .0518099    .0675867
   5#Female  |    .069641   .0040284    17.29   0.000     .0617454    .0775366
     6#Male  |   .1147889   .0089467    12.83   0.000     .0972536    .1323242
   6#Female  |   .1106004   .0078699    14.05   0.000     .0951757     .126025
------------------------------------------------------------------------------


Variables that uniquely identify margins: age female

file marginsplot1.svg saved as SVG format


Contrasts of adjusted predictions                     Number of obs = 10,335
Model VCE: OIM

Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

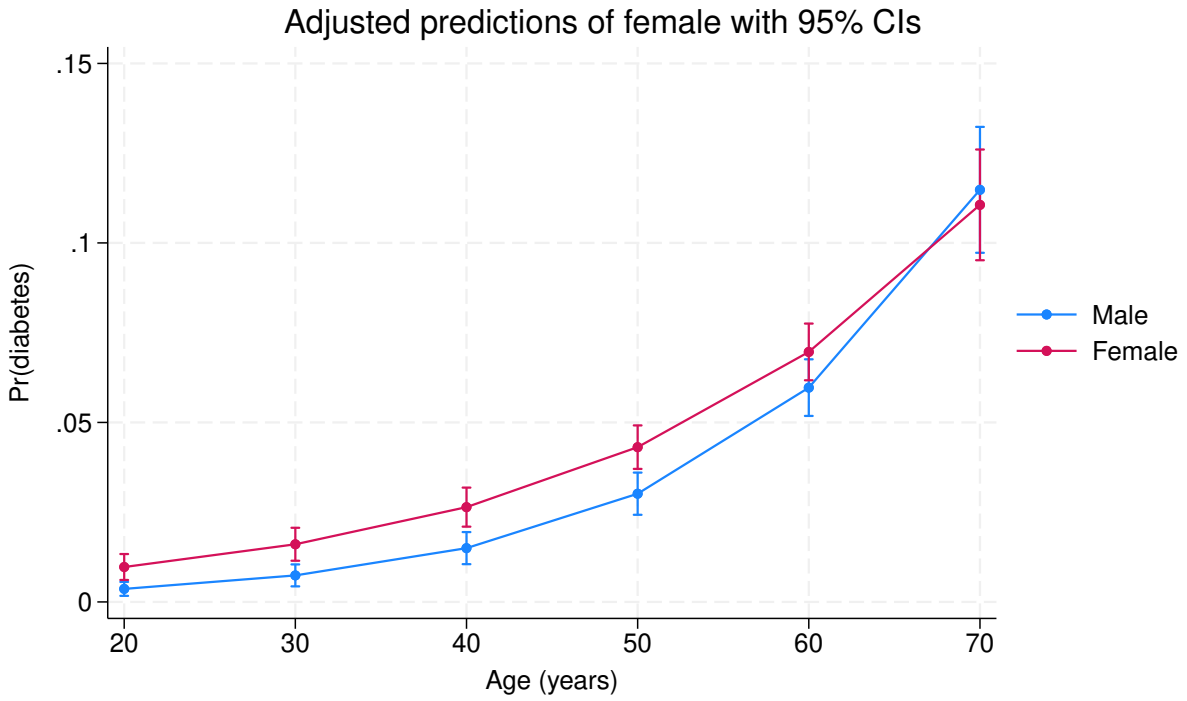
------------------------------------------------------
                   |         df        chi2     P>chi2
-------------------+----------------------------------
```

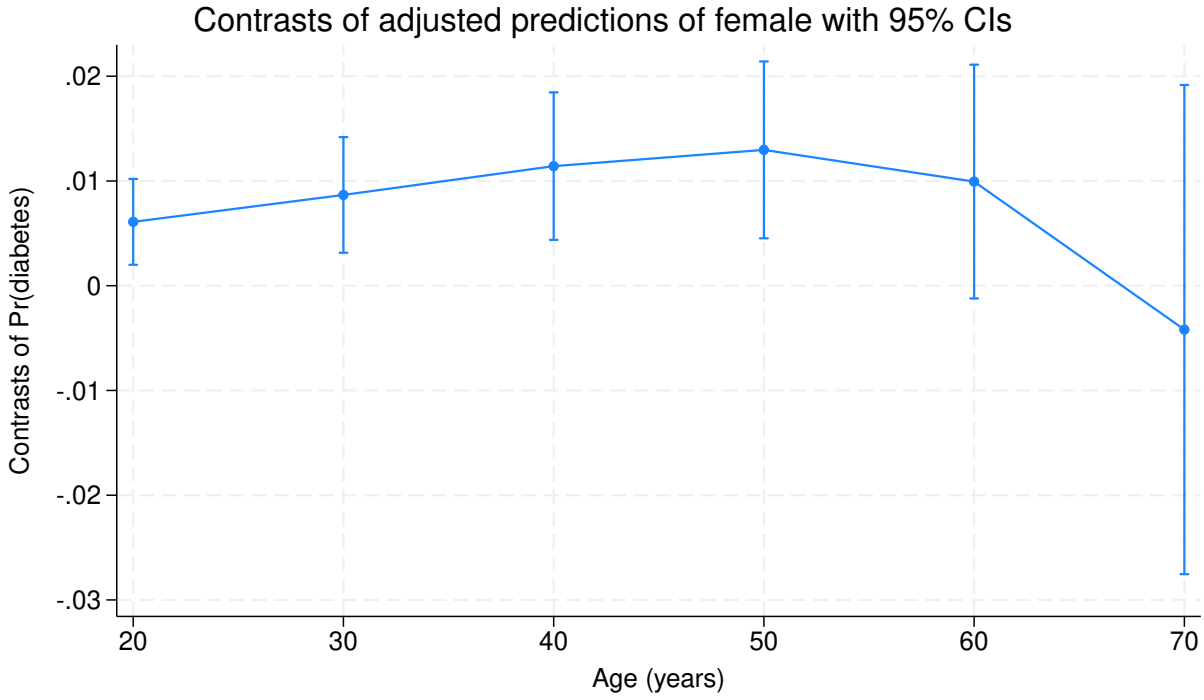
```
         female@_at |
(Female vs Male) 1  |          1        8.49     0.0036
(Female vs Male) 2  |          1        9.47     0.0021
(Female vs Male) 3  |          1       10.09     0.0015
(Female vs Male) 4  |          1        9.06     0.0026
(Female vs Male) 5  |          1        3.05     0.0808
(Female vs Male) 6  |          1        0.12     0.7252
             Joint  |          4       40.48     0.0000
----------------------------------------------------------

------------------------------------------------------------------------
                    |            Delta-method
                    |   Contrast   std. err.     [95% conf. interval]
--------------------+---------------------------------------------------
         female@_at |
(Female vs Male) 1  |   .0060969   .0020923      .0019959    .0101978
(Female vs Male) 2  |   .0086677   .0028164      .0031476    .0141878
(Female vs Male) 3  |   .0114154    .003594      .0043712    .0184595
(Female vs Male) 4  |   .0129682   .0043081      .0045244    .0214119
(Female vs Male) 5  |   .0099427   .0056945     -.0012183    .0211037
(Female vs Male) 6  |  -.0041885   .0119155     -.0275425    .0191654
------------------------------------------------------------------------

Variables that uniquely identify margins: age

file marginsplot2.svg saved as SVG format
```





We fit a logit of diabetes on female, age, and their interaction, on 10,335 NHANES II respondents. The interaction is −0.0207 with a standard error of 0.0078: on the log-odds scale the female-male gap narrows with age. On the probability scale it need not, and below it first widens. The first `margins` call gives predicted probabilities by gender; the second (with `r.female`) gives the female-vs-male contrast at each age. (On the general question of marginal effects in nonlinear models, see the

previous chapter — the coefficient on the link scale is usually more interpretable.)

Those contrasts run .0061, .0087, .0114, .0130, .0099 and −.0042 at ages 20 through 70, significant through age 50 and not thereafter — the gap widens, peaks in middle age, then closes and reverses sign.

Now the point of the chapter. The female effect at age 20 is .0061 with a 95% interval of [.0020, .0102], and at age 30 it is .0087 with an interval of [.0031, .0142]. Those intervals overlap across most of their length. It is tempting to conclude the two effects are not significantly different, but that conclusion does not follow. For two effects x and y from the same model,

$$\mathrm{Var}(x - y) = \mathrm{Var}(x) + \mathrm{Var}(y) - 2\,\mathrm{Cov}(x, y). \tag{6.1}$$

When x and y are positively correlated, as they are here, the variance of their difference can be much smaller than either individual variance. Estimates from the same fit are often positively correlated, but not always; the covariance has to be computed. So the difference can be significant even when the individual intervals overlap.

```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins r.female,  at(age=(20 30 40 50 60 70)) post
lincom _b[r1vs0.female@2._at] - _b[r1vs0.female@1._at]
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Contrasts of adjusted predictions                       Number of obs = 10,335
Model VCE: OIM

Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
```

```
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70

------------------------------------------------------
                    |         df        chi2     P>chi2
--------------------+---------------------------------
         female@_at |
(Female vs Male) 1  |          1        8.49     0.0036
(Female vs Male) 2  |          1        9.47     0.0021
(Female vs Male) 3  |          1       10.09     0.0015
(Female vs Male) 4  |          1        9.06     0.0026
(Female vs Male) 5  |          1        3.05     0.0808
(Female vs Male) 6  |          1        0.12     0.7252
             Joint  |          4       40.48     0.0000
------------------------------------------------------

------------------------------------------------------------------------------
                    |            Delta-method
                    |   Contrast   std. err.     [95% conf. interval]
--------------------+---------------------------------------------------------
         female@_at |
(Female vs Male) 1  |   .0060969   .0020923      .0019959    .0101978
(Female vs Male) 2  |   .0086677   .0028164      .0031476    .0141878
(Female vs Male) 3  |   .0114154    .003594      .0043712    .0184595
(Female vs Male) 4  |   .0129682   .0043081      .0045244    .0214119
(Female vs Male) 5  |   .0099427   .0056945     -.0012183    .0211037
(Female vs Male) 6  |  -.0041885   .0119155     -.0275425    .0191654
------------------------------------------------------------------------------


 ( 1)  - r1vs0.female@1bn._at + r1vs0.female@2._at = 0

------------------------------------------------------------------------------
             | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0025709   .0007946     3.24   0.001     .0010134    .0041283
------------------------------------------------------------------------------
```

The test gives a difference of .0025709 with a standard error of .0007946, $z = 3.24$ and $p = 0.001$. The two effects are different, despite intervals that overlap heavily.

The standard error is what makes the point. Each individual contrast carries a standard error of .0021 and .0028, but their *difference* carries only .00079 — smaller than either. Solving $\text{Var}(x-y) = \text{Var}(x) + \text{Var}(y) - 2\text{Cov}(x,y)$ for the correlation gives about 0.99: the two contrasts come from the same fitted model evaluated ten years apart, so they move together almost perfectly, and nearly all of their variance cancels in the difference.

Had we ignored the covariance and used $\sqrt{\text{Var}(x) + \text{Var}(y)} = .0035$, we would have got $z = 0.73$ and $p = 0.46$. Same data, same two numbers, opposite conclusion. That is the cost of reading overlap off a graph.

We can do this with native Stata syntax: `margins ..., post` posts the estimated contrasts as if they were regular coefficients (named `r1vs0.female@1._at`, `r1vs0.female@2._at`, etc. — check `matrix list e(b)` to see the exact names), so `lincom` can test any linear combination of them natively, without depending on a user-written command.

`margins` can compute these differences directly with `contrast(atcontrast(r))`, which compares each evaluation point to the reference:

```
webuse nhanes2f, clear
logit diabetes i.female##c.age, nolog
margins r.female,  at(age=(20 30 40 50 60 70)) contrast(atcontrast(r))
marginsplot
graph export "marginsplot3.svg", as(svg) replace
```

```
Logistic regression                                     Number of obs = 10,335
                                                        LR chi2(3)    = 353.06
                                                        Prob > chi2   = 0.0000
Log likelihood = -1822.5393                             Pseudo R2     = 0.0883

------------------------------------------------------------------------------
    diabetes | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      female |
     Female  |   1.404111   .4858776     2.89   0.004     .4518088    2.356414
         age |   .0714166   .0063231    11.29   0.000     .0590235    .0838097
             |
female#c.age |
     Female  |  -.0206572   .0078369    -2.64   0.008    -.0360172   -.0052972
             |
       _cons |  -7.041892   .3967299   -17.75   0.000    -7.819468   -6.264315
------------------------------------------------------------------------------


Contrasts of adjusted predictions                       Number of obs = 10,335
Model VCE: OIM

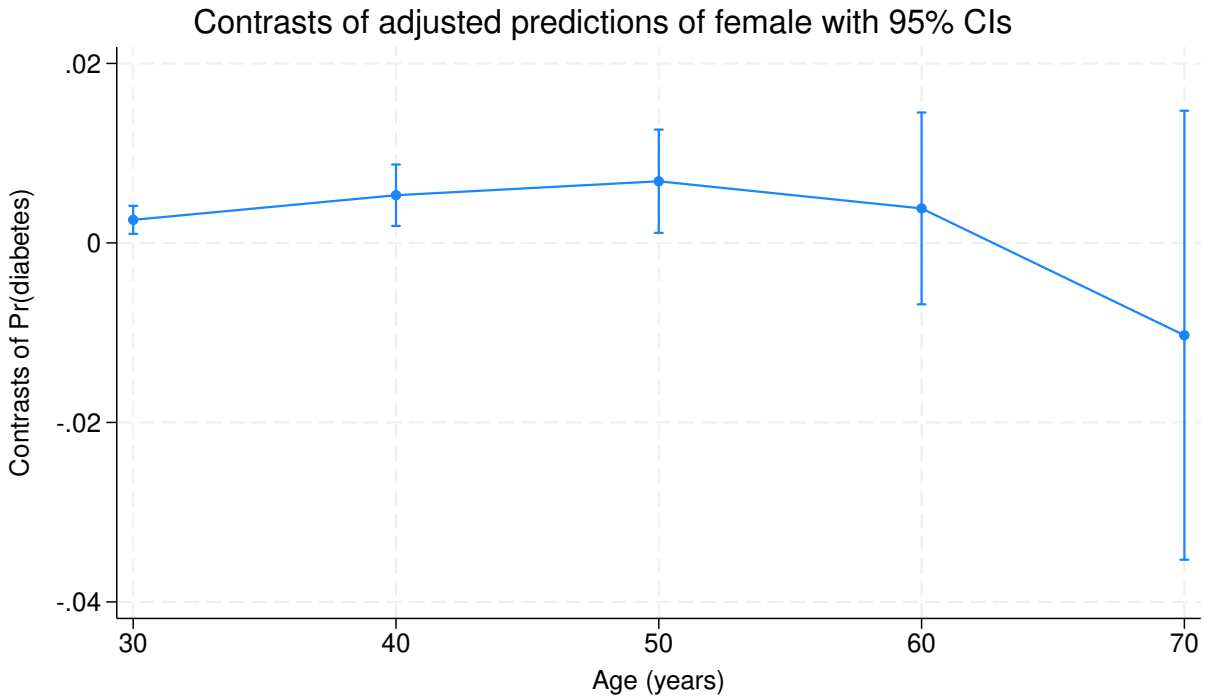
Expression: Pr(diabetes), predict()
1._at: age = 20
2._at: age = 30
3._at: age = 40
4._at: age = 50
5._at: age = 60
6._at: age = 70
```

```
------------------------------------------------------------
                           |         df        chi2     P>chi2
---------------------------+--------------------------------
               _at#female |
(2 vs 1) (Female vs Male)  |          1       10.47     0.0012
(3 vs 1) (Female vs Male)  |          1        9.28     0.0023
(4 vs 1) (Female vs Male)  |          1        5.48     0.0192
(5 vs 1) (Female vs Male)  |          1        0.50     0.4807
(6 vs 1) (Female vs Male)  |          1        0.65     0.4203
                     Joint |          4       20.04     0.0005
------------------------------------------------------------

------------------------------------------------------------------------------
                           |            Delta-method
                           |   Contrast   std. err.     [95% conf. interval]
---------------------------+--------------------------------------------------
               _at#female |
(2 vs 1) (Female vs Male)  |   .0025709   .0007946      .0010134    .0041283
(3 vs 1) (Female vs Male)  |   .0053185    .001746      .0018964    .0087407
(4 vs 1) (Female vs Male)  |   .0068713   .0029351      .0011187    .0126239
(5 vs 1) (Female vs Male)  |   .0038458   .0054532     -.0068422    .0145339
(6 vs 1) (Female vs Male)  |  -.0102854   .0127614     -.0352972    .0147264
------------------------------------------------------------------------------


Variables that uniquely identify margins: age

file marginsplot3.svg saved as SVG format
```



From this we see the difference of effects between point 2 and 1 is .0026, with $\chi^2(1) = 10.47$ and $p = 0.0012$ — the same conclusion the `lincom` test gave above, and indeed the same test: $3.24^2 = 10.5$. The remaining rows compare each later age to age 20, and the widening gap is significant through age 50 ((4 vs 1): $\chi^2 = 5.48$, $p = 0.019$) before losing precision.

6.2. Summary

Overlapping confidence intervals do not mean two estimates are statistically indistinguishable. A test of the difference depends on the variance of the difference, $\mathrm{Var}(x-y) = \mathrm{Var}(x)+\mathrm{Var}(y)-2\,\mathrm{Cov}(x,y)$, and the covariance is nowhere to be seen in a plot of the two intervals separately. Estimates read off the same fitted model at nearby covariate values are strongly positively correlated — 0.99 in this example — so most of their variance cancels in the difference, and the difference is pinned down far more precisely than either level is. The eyeball test is therefore conservative in a specific and misleading way: it reports no difference where a proper test finds one.

The chapter's numbers make the size of the error concrete. The female-vs-male contrasts at ages 20 and 30 have intervals overlapping across most of their length, yet their difference is 0.0026 with a standard error of 0.00079, $z = 3.24$, $p = 0.001$. Ignoring the covariance and combining the two standard errors as $\sqrt{\mathrm{Var}(x) + \mathrm{Var}(y)}$ would give $z = 0.73$ and $p = 0.46$ — same data, opposite conclusion.

So do not read differences off a `marginsplot`. Ask for them directly:

- `margins r.varname` returns a contrast rather than two levels, with the covariance already handled.

- `margins ..., contrast(atcontrast(r))` compares every evaluation point to the reference in one call, which is what produced the last table above.
- `margins ..., post` followed by `lincom` or `test` handles any single contrast you can write down, using only built-in commands.

The plot is still the right tool for showing how an effect moves across a covariate. It is the wrong tool for judging whether two points on it differ.

7. Fixed or Random Effect, or Both?

7.1. Panel data

With panel data — repeated observations over time or observations clustered at a higher level — the standard choice is between fixed and random effects. Consider

$$y_{it} = \beta_0 + \beta_1 x_{it} + c_i + \epsilon_{it} \tag{7.1}$$

for individuals $i = 1, \ldots, n$ measured at $t = 1, \ldots, T$, where c_i is an unobserved time-invariant individual effect. The two approaches differ in how they handle c_i.

Fixed effects eliminate c_i by demeaning or by including individual dummies (the two are equivalent in a linear model). In nonlinear models only Poisson avoids the incidental-parameter problem; the standard solution for other models is conditional likelihood (conditional logit, for example), which absorbs the fixed effects in the likelihood. When no conditional likelihood exists, the incidental-parameter bias shrinks with panel depth but does not vanish.

Random effects treat c_i as part of the error term. This is more efficient but requires x_{it} to be uncorrelated with c_i — an assumption economists are generally skeptical of.

7.2. Time-invariant variables

Sometimes we want the effect of a time-invariant variable, so the model becomes

$$y_{it} = \beta_0 + \beta_1 x_{it} + c_i + \gamma z_i + \epsilon_{it} \tag{7.2}$$

Fixed effects cannot identify γ because z_i is perfectly collinear with c_i. Random effects can still estimate both, treating z_i as another covariate.

7.3. Between-within model

The textbook advice is a Hausman test: random effects are more efficient when the uncorrelatedness assumption holds; if not, fixed effects are still consistent. The between-within model (BW) sidesteps the choice. Neuhaus and Kalbfleisch (1998) introduced the BW estimator,

$$y_{it} = \beta_0 + \beta_1 (x_{it} - \bar{x}_i) + \beta_2 \bar{x}_i + c_i + \gamma z_i + \epsilon_{it} \tag{7.3}$$

Here β_1 is the within effect — identical to the fixed-effect coefficient. β_2 is the between effect (mean of x on mean of y). γ is the effect of the time-invariant variable.

The other specification of BW estimator is

$$y_{it} = \beta_0 + \beta_1 x_{it} + \beta_2 \bar{x}_i + c_i + \gamma z_i + \epsilon_{it} \tag{7.4}$$

This is the same model reparametrized. β_1 is unchanged, and β_2 is now the between-minus-within difference (substitute $\beta_1(x_{it} - \bar{x}_i) + \beta_2 \bar{x}_i = \beta_1 x_{it} + (\beta_2 - \beta_1)\bar{x}_i$ to see this). This is the "contextual model"; β_2 is the contextual effect and acts as an embedded Hausman test.

The BW model has two advantages. It recovers the fixed-effect coefficient while estimating time-invariant covariates. And it extends naturally to cross-level interactions, random slopes, and other multilevel structures. Implementation is straightforward: run a random-effects model on either equation above.

7.3.1. Is this just Mundlak?

Yes. Equation 7.4 *is* the Mundlak device: Mundlak's regression puts x_{it} and $\bar{x}_i$ on the right-hand side of a random-effects model, which is exactly what that equation does. So the between-within model and the Mundlak device are one model in two parameterisations, fitted by the same random-effects command. It is worth doing the algebra, because the identity is easy to state loosely and easy to get wrong on the inference side.

Write $d_{it} = x_{it} - \bar{x}_i$ for the within deviation, and label the two specifications by their coefficients: b_1, b_2 for the hybrid form Equation 7.3 and p_1, p_2 for the contextual form Equation 7.4. Expanding the hybrid regressor block,

$$b_1(x_{it} - \bar{x}_i) + b_2 \bar{x}_i = b_1 x_{it} + (b_2 - b_1)\bar{x}_i, \tag{7.5}$$

so $p_1 = b_1$ and $p_2 = b_2 - b_1$. Substitution shows the map, but the reason behind it is worth stating more strongly: the two regressor matrices are related by an invertible linear transformation. With $X = [\, x_{it}, \bar{x}_i \,]$ and $W = [\, d_{it}, \bar{x}_i \,]$,

$$W = XA, \qquad A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \tag{7.6}$$

and $\det A = 1$, so the two designs span the same column space. Everything that depends only on that space is numerically identical: fitted values, residuals, the likelihood, the estimated variance components, and the GLS quasi-demeaning weight θ. Only the coordinates change, by $b = A^{-1}p$, and the covariance matrix by $\mathrm{Var}(b) = A^{-1}\mathrm{Var}(p)A^{-1\prime}$. Componentwise that is

$$\mathrm{Var}(p_1) = \mathrm{Var}(b_1), \qquad \mathrm{Var}(p_2) = \mathrm{Var}(b_1) + \mathrm{Var}(b_2) - 2\,\mathrm{Cov}(b_1, b_2), \tag{7.7}$$

which already says something useful: the standard error printed next to the between effect is not the standard error you want for the contextual effect.

7.3.1.1. Why the within coefficient is the fixed-effect estimator

This part needs a proof rather than a substitution. The relevant fact is that $\sum_t d_{it} = 0$ within every unit by construction, so d is orthogonal to any group-invariant column: for v_i constant within i,

$$\sum_{i,t} d_{it} v_i = \sum_i v_i \sum_t d_{it} = 0, \tag{7.8}$$

which covers the intercept, $\bar{x}_i$ and z_i alike.

Random effects is OLS after quasi-demeaning, $\tilde{w}_{it} = w_{it} - \theta_i \bar{w}_i$ with $\theta_i = 1 - \sqrt{\sigma_\epsilon^2 / (\sigma_\epsilon^2 + T_i \sigma_c^2)}$, where T_i is the number of periods observed for unit i. Apply it to the hybrid regressors. Because $\bar{d}_i = 0$ we get $\tilde{d}_{it} = d_{it}$: the within-deviation regressor passes through the GLS transform untouched, whatever θ_i is. The remaining regressors are group-invariant and stay so after scaling by $1 - \theta_i$. Nothing here needs T_i to be the same for every unit, so the one-way result also holds in an unbalanced panel; the two-way result in the Mundlak chapter does not. Orthogonality therefore survives the transform, and by Frisch-Waugh-Lovell

$$b_1 = \frac{\sum d_{it} \tilde{y}_{it}}{\sum d_{it}^2} = \frac{\sum d_{it} y_{it}}{\sum d_{it}^2} = \frac{\sum (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum (x_{it} - \bar{x}_i)^2}, \tag{7.9}$$

the second and third equalities both by Equation 7.8. That last expression is the within estimator. Note that θ_i has dropped out, so the identity holds for *any* θ_i — for random effects, for pooled OLS ($\theta = 0$) and for fixed effects ($\theta = 1$). That is why running a random-effects model on either equation returns the fixed-effect coefficient. For the conventional random-effects covariance $\hat{\sigma}^2 (\tilde{W}' \tilde{W})^{-1}$, where $\tilde{W}$ stacks the quasi-demeaned regressors $\tilde{w}_{it}$, it also gives $\mathrm{Cov}(b_1, b_2) = 0$ exactly. A cluster-robust covariance has no such block structure, so there the covariance term stays. In the conventional case, combined with Equation 7.7 yields $\mathrm{Var}(p_2) = \mathrm{Var}(b_1) + \mathrm{Var}(b_2)$ and $\mathrm{Cov}(p_1, p_2) = -\mathrm{Var}(b_1)$.

All of this is checkable on the chapter's own data. We take a single regressor, weeks worked, fit both parameterisations by `plm`, and compare against the within estimator and against the variance map:

```
library(plm); library(panelr); data("WageData")
d <- WageData
d$xbar <- ave(d$wks, d$id)          # unit mean
d$xdev <- d$wks - d$xbar            # within deviation

fe  <- plm(lwage ~ wks,          data = d, index = c("id", "t"), model = "within")
hyb <- plm(lwage ~ xdev + xbar,  data = d, index = c("id", "t"), model = "random")
mun <- plm(lwage ~ wks  + xbar,  data = d, index = c("id", "t"), model = "random")

b <- coef(hyb); p <- coef(mun); Vh <- vcov(hyb); Vm <- vcov(mun)

data.frame(
  check = c("p1 - b1", "p2 - (b2 - b1)", "b1 - FE within",
            "sum(xdev * xbar)", "Cov(b1, b2)",
            "Var(p2) - [Var(b1) + Var(b2)]", "Cov(p1, p2) + Var(b1)",
```

```
           "theta(hybrid) - theta(Mundlak)"),
  value = c(p["wks"] - b["xdev"],
           p["xbar"] - (b["xbar"] - b["xdev"]),
           b["xdev"] - coef(fe)["wks"],
           sum(d$xdev * d$xbar),
           Vh["xdev", "xbar"],
           Vm["xbar", "xbar"] - (Vh["xdev", "xdev"] + Vh["xbar", "xbar"]),
           Vm["wks", "xbar"] + Vh["xdev", "xdev"],
           hyb$ercomp$theta - mun$ercomp$theta),
  row.names = NULL)
```

```
                          check         value
1                      p1 - b1  2.385245e-18
2                p2 - (b2 - b1)  7.459311e-17
3                b1 - FE within  1.886512e-17
4               sum(xdev * xbar)  3.534950e-12
5                   Cov(b1, b2) -5.676184e-21
6  Var(p2) - [Var(b1) + Var(b2)] -4.404571e-20
7         Cov(p1, p2) + Var(b1)  2.117582e-21
8 theta(hybrid) - theta(Mundlak)  1.110223e-16
```

Every entry is zero to machine precision. The two parameterisations are the same fit, the within coefficient is the fixed-effect estimate, the hybrid regressors are orthogonal, and the variance map holds as derived.

7.3.1.2. So why keep both

The fit is one object, but the parameterisation settles three practical things.

Which quantity is printed. b_2 is the between effect and p_2 is between minus within. They are different parameters with non-interchangeable standard errors, and whichever form you fit, the other costs a `lincom`.

Which hypothesis is one line. $p_2 = 0$ says between equals within, which is exactly the random-effects assumption that the regressors are uncorrelated with c_i. The Mundlak form prints that as a single t; in hybrid form it is the contrast $b_1 = b_2$. The two tests agree numerically, but only one is printed. This is the Mundlak form of the Hausman test, asymptotically equivalent to the classical version and with the practical advantage of working directly with cluster-robust standard errors. The next chapter develops it and its two-way extension for difference-in-differences.

What the second coefficient is for. In multilevel work p_2 is not a diagnostic but the estimand: holding a pupil's own status fixed, what does the school mean do? The hybrid's b_2 answers a different question, the total between-unit association. Two literatures ask two questions of one regression, and each named the parameterisation that printed its answer. That is the honest reason both names survive.

There is also a reason to prefer the hybrid form in extensions. Its two regressors are exactly orthogonal, where x_{it} and $\bar{x}_i$ are correlated. That does not change the fit, but if you want a random

slope on the within component you need the deviation as its own column; a random slope on x_{it} would mix within and between variation.

7.3.1.3. Where the equivalence stops

The reparameterisation is exact for anything with a linear index, logit and Poisson mixed models included, since XA spans the same space and the index is unchanged. The two names stay interchangeable there.

The fixed-effect identity does not carry over. The proof above used two linear-model facts: that random effects is OLS on quasi-demeaned data, and that d is orthogonal to group constants. A logit or Poisson mixed model has no such transform, the unit effects do not sweep out, and the hybrid within coefficient is not the conditional-logit estimate but an approximation to it. Keep the two claims separate: the parameterisations are always the same model, while "within coefficient equals fixed effects" is a linear-model result. The last section of this chapter returns to that point.

7.4. BW model in R

R's `panelr` package implements BW models directly. The data are `WageData`, a balanced panel of 595 men observed over 7 years. We model log wage on weeks worked, union membership, marital status and occupation, all time-varying, plus two time-invariant covariates — black and female — that a fixed-effect model could not touch.

```
library(panelr)
data("WageData")
wages <- panel_data(WageData, id = id, wave = t)
model1 <- wbm(lwage ~ wks + union + ms + occ | blk + fem, data = wages)
summary(model1)
```

```
MODEL INFO:
Entities: 595
Time periods: 1-7
Dependent variable: lwage
Model type: Linear mixed effects
Specification: within-between

MODEL FIT:
AIC = 2036.78, BIC = 2119.13
Pseudo-R² (fixed effects) = 0.27
Pseudo-R² (total) = 0.69
Entity ICC = 0.57

WITHIN EFFECTS:
-----------------------------------------
               Est.   S.E.   t val.      p
---------- ------- ------ -------- ------
```

```
wks            0.00   0.00     1.06   0.29
union          0.06   0.03     2.53   0.01
ms            -0.08   0.03    -2.57   0.01
occ           -0.08   0.02    -3.32   0.00
-----------------------------------------

BETWEEN EFFECTS:
-------------------------------------------------
                      Est.   S.E.   t val.      p
------------------ ------- ------ -------- ------
(Intercept)           6.30   0.20    30.85   0.00
imean(wks)            0.01   0.00     2.25   0.02
imean(union)          0.15   0.03     4.67   0.00
imean(ms)             0.17   0.05     3.07   0.00
imean(occ)           -0.41   0.03   -13.31   0.00
blk                  -0.15   0.05    -2.81   0.00
fem                  -0.32   0.06    -4.96   0.00
-------------------------------------------------

p values calculated using df = 4153

RANDOM EFFECTS:
-----------------------------------
  Group      Parameter    Std. Dev.
---------- ------------- -----------
    id      (Intercept)    0.2992
 Residual                  0.2589
-----------------------------------
```

The within estimates are 0.00 for weeks, 0.06 for union, −0.08 for married and −0.08 for occupation. The between estimates for the same four variables are 0.01, 0.15, 0.17 and −0.41 — two to five times larger, and for `ms` the opposite sign. That gap is the reason the fixed-versus-random choice matters here: a random-effects model without the decomposition would blend the two and report neither. The time-invariant covariates come out at −0.15 for black and −0.32 for female, quantities a fixed-effect model cannot produce at all.

Compare with `lfe`.

```
library(lfe)
model2 <- felm(lwage ~ wks + union + ms + occ | id, data = wages)
summary(model2)
```

```
Call:
   felm(formula = lwage ~ wks + union + ms + occ | id, data = wages)

Residuals:
```

```
     Min       1Q   Median       3Q      Max
-1.89500 -0.16174  0.00652  0.17060  1.94521

Coefficients:
       Estimate Std. Error t value Pr(>|t|)
wks    0.001083   0.001019   1.063 0.287816
union  0.064320   0.025378   2.534 0.011305 *
ms    -0.082905   0.032226  -2.573 0.010132 *
occ   -0.077507   0.023359  -3.318 0.000916 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2589 on 3566 degrees of freedom
Multiple R-squared(full model): 0.7304   Adjusted R-squared: 0.6852
Multiple R-squared(proj model): 0.006509   Adjusted R-squared: -0.1601
F-statistic(full model):16.16 on 598 and 3566 DF, p-value: < 2.2e-16
F-statistic(proj model): 5.841 on 4 and 3566 DF, p-value: 0.0001106
```

`felm` gives 0.001083, 0.064320, −0.082905 and −0.077507 — the same four numbers `panelr` reported as within effects, identical to machine precision (`WageData` is a balanced panel of 7 waves and 595 individuals, but the one-way identity does not need balance). That is the BW model's selling point: it reproduces the fixed-effect estimates exactly while also estimating the time-invariant covariates `blk` and `fem`, which `felm` cannot. `lfe` supports clustered standard errors, which `panelr` does not.

```
model3 <- felm(lwage ~ wks + union + ms + occ | id | 0 | id, data = wages)
summary(model3)
```

```
Call:
   felm(formula = lwage ~ wks + union + ms + occ | id | 0 | id,      data = wages)

Residuals:
     Min       1Q   Median       3Q      Max
-1.89500 -0.16174  0.00652  0.17060  1.94521

Coefficients:
       Estimate Cluster s.e. t value Pr(>|t|)
wks    0.001083     0.001331   0.814   0.4160
union  0.064320     0.040936   1.571   0.1167
ms    -0.082905     0.047399  -1.749   0.0808 .
occ   -0.077507     0.031320  -2.475   0.0136 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2589 on 3566 degrees of freedom
Multiple R-squared(full model): 0.7304   Adjusted R-squared: 0.6852
Multiple R-squared(proj model): 0.006509   Adjusted R-squared: -0.1601
```

```
F-statistic(full model, *iid*):16.16 on 598 and 3566 DF, p-value: < 2.2e-16
F-statistic(proj model): 3.456 on 4 and 594 DF, p-value: 0.008358
```

Clustering on `id` leaves every coefficient untouched and changes the standard errors substantially: `union` goes from 0.0254 to 0.0409, `ms` from 0.0322 to 0.0474, `wks` from 0.00102 to 0.00133. Two of the four results do not survive. `union` falls from $p = 0.011$ to $p = 0.117$ and `ms` from 0.010 to 0.081; only `occ` stays significant at 5%. With 7 observations per man the residuals are serially correlated, and the unclustered standard errors were treating 4,165 observations as though they were independent draws.

7.5. BW model in Stata

Stata has no dedicated BW command, but `xtreg` suffices. We switch to `nlswork`, an unbalanced panel of 4,710 women over 28,510 person-years, and regress log wage on age.

```
webuse nlswork
xtset idcode
xtreg ln_w age, fe cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)


Fixed-effects (within) regression               Number of obs     =     28,510
Group variable: idcode                          Number of groups  =      4,710

R-squared:                                      Obs per group:
     Within  = 0.1026                                         min =          1
     Between = 0.0877                                         avg =        6.1
     Overall = 0.0774                                         max =         15

                                                F(1, 4709)        =     884.05
corr(u_i, Xb) = 0.0314                          Prob > F          =     0.0000

                            (Std. err. adjusted for 4,710 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0181349   .0006099    29.73   0.000     .0169392    .0193306
       _cons |   1.148214   .0177153    64.81   0.000     1.113483    1.182944
-------------+----------------------------------------------------------------
     sigma_u |  .40635023
```

```
     sigma_e |  .30349389
         rho |  .64192015   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

The fixed-effect estimate is .0181349 with a clustered standard error of .0006099: within a woman, each additional year of age raises log wage by about 1.8%. Keep that number in view — the BW model has to reproduce it.

Generate the group mean and run the BW estimation.

```
webuse nlswork
xtset idcode
bysort idcode: center age, prefix(d) mean(m)
xtreg ln_w dage mage i.race, re cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)

(generated variables: dage mage)


Random-effects GLS regression                   Number of obs     =     28,510
Group variable: idcode                          Number of groups  =      4,710

R-squared:                                      Obs per group:
     Within  = 0.1026                                         min =          1
     Between = 0.1040                                         avg =        6.1
     Overall = 0.0950                                         max =         15

                                                Wald chi2(4)      =    1335.89
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

                            (Std. err. adjusted for 4,710 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        dage |   .0181349     .00061    29.73   0.000     .0169394    .0193304
        mage |    .022558   .0011405    19.78   0.000     .0203226    .0247933
             |
        race |
       Black |  -.1190246   .0127418    -9.34   0.000    -.1439982   -.0940511
       Other |   .0974996   .0617364     1.58   0.114    -.0235016    .2185008
             |
       _cons |   1.037566   .0323185    32.10   0.000     .9742233     1.10091
```

```
-------------+----------------------------------------------------------------
     sigma_u |  .36581005
     sigma_e |  .30349389
         rho |  .59230575   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

In this BW model, we use the centered (within-group-demeaned) `dage` together with the group mean `mage`, which gives the standard within-between (Mundlak) decomposition: the coefficient on `dage`, .0181349, is exactly the fixed-effect ("within") coefficient on age reported above — the same seven digits, with a standard error of .00061 against .0006099 — and the coefficient on `mage`, .022558, is directly the between effect. Women who are older on average earn about 2.3% more per year of average age, against a within-woman return of 1.8%; the between effect is larger because it also carries cohort and selection differences that the within estimate sweeps out. And we have the effect of time-invariant covariate race estimated: Black women earn 11.9% less (−.1190246, SE .0127), while the "Other" category is not distinguishable from white (.0975, $p = 0.114$). The advantage of using xtreg is that we have clustered standard errors implemented.

Note: if we instead ran `xtreg ln_w age mage i.race, re cluster(idcode)` — using the raw (uncentered) `age` together with `mage` — we would get the *contextual model* instead: the coefficient on `age` stays .0181, but the coefficient on `mage` becomes .0044, the "contextual effect" (the *additional* between-effect on top of the within effect). The two models are algebraically related: contextual-model mage = Mundlak-model mage − Mundlak-model dage (here, .0226 − .0181 ≈ .0044).

7.6. BW model in non-linear models

Allison (2009) applies the BW model to binary outcomes (see also his blog post). The bias relative to conditional logit is an open question, but if the linear probability model is a reasonable approximation the BW approach should perform similarly for binary panel data.

7.7. Summary

Fixed effects is consistent when the unit effect is correlated with the regressors, but it cannot estimate time-invariant covariates. Random effects can, but only if that correlation is zero. The between-within model enters the within deviation and the unit mean of each time-varying regressor in a random-effects regression. Its within coefficient equals the fixed-effects estimate exactly, for any random-effects weight, and time-invariant covariates are still estimated. The Mundlak device is the same model in another parameterization. There the coefficient on the unit mean is between minus within, so its test is the Hausman test.

On `WageData` the within estimates match `felm` to machine precision, while the between estimates are two to five times larger. The model also estimates the race and sex gaps that fixed effects cannot. Clustering by person leaves the coefficients alone, but two of the four lose significance. In `nlswork`, Stata's `xtreg` with the centered age and its mean reproduces the fixed-effect estimate of .0181. In nonlinear models the parameterizations still coincide, but the within coefficient only approximates the conditional-logit estimate.

8. The Mundlak device in panel data and difference-in-differences

8.1. The basic idea

Suppose we have a panel-data model with an individual effect c_i:

$$y_{it} = X_{it}\beta + c_i + u_{it}$$

The fixed-effects estimator removes c_i by demeaning, so c_i can be correlated with X_{it}. But the same transformation also removes time-invariant variables such as gender, race, or baseline schooling.

Mundlak (1978) instead models c_i through the unit means of the time-varying covariates:

$$c_i = \bar{X}_i\theta + \eta_i, \qquad E[\eta_i \mid X_i] = 0$$

Here X_i is the unit's full covariate history. The conditional-mean restriction is an assumption. A linear projection of c_i on $\bar{X}_i$ always exists, but its residual is only uncorrelated with $\bar{X}_i$; it need not have mean zero given the whole history. For example, let X_{i1} and X_{i2} be independent and equal to -1 or 1 with equal probability, and let $c_i = X_{i1}X_{i2}$. Then c_i is uncorrelated with $\bar{X}_i$, so the projection coefficient is zero, yet $E[c_i \mid X_{i1}, X_{i2}] = X_{i1}X_{i2}$.

Substituting into the regression yields:

$$y_{it} = X_{it}\beta + \bar{X}_i\theta + \eta_i + u_{it}$$

We can estimate this equation by pooled OLS or random effects. By the Frisch–Waugh–Lovell theorem, the coefficient on X_{it} is the same as the fixed-effects coefficient. This algebraic equivalence holds whether or not the conditional-mean restriction is true; the restriction matters when we interpret θ or the coefficients on time-invariant variables, and in nonlinear models.

Testing $H_0 : \theta = 0$ gives a regression-based version of the Hausman test and works with cluster-robust standard errors. We can also add time-invariant covariates directly. They do not change $\hat{\beta}$ because within-unit deviations are orthogonal to every time-invariant variable.

8.2. A one-way panel example

I start with Stata's `nlswork` data, using 19,229 person-year observations:

```
clear
webuse nlswork
xtset idcode year
drop if union==. | age==.
bysort idcode: egen mean_age = mean(age)
bysort idcode: egen mean_union = mean(union)

* 1. Standard FE
xtreg ln_wage age union, fe cluster(idcode)

* 2. Pooled OLS with Mundlak means
reg ln_wage age union mean_age mean_union, cluster(idcode)

* 3. Random Effects with Mundlak means
xtreg ln_wage age union mean_age mean_union, re cluster(idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)


Panel variable: idcode (unbalanced)
 Time variable: year, 68 to 88, but with gaps
         Delta: 1 unit

(9,305 observations deleted)




Fixed-effects (within) regression               Number of obs     =     19,229
Group variable: idcode                          Number of groups  =      4,150

R-squared:                                      Obs per group:
     Within  = 0.0963                                         min =          1
     Between = 0.0433                                         avg =        4.6
     Overall = 0.0562                                         max =         12

                                                F(2, 4149)        =     331.52
corr(u_i, Xb) = 0.0127                          Prob > F          =     0.0000

                            (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139956    .0167058
       union |   .1055274   .0098582    10.70   0.000     .0862001    .1248546
       _cons |   1.248435   .0215661    57.89   0.000     1.206154    1.290716
-------------+----------------------------------------------------------------
     sigma_u |  .42353003
     sigma_e |  .26213464
         rho |  .72302816   (fraction of variance due to u_i)
------------------------------------------------------------------------------


Linear regression                               Number of obs     =     19,229
                                                F(4, 4149)        =     234.77
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0769
                                                Root MSE          =     .44953

                              (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139956    .0167059
       union |   .1055274   .0098587    10.70   0.000     .0861991    .1248556
    mean_age |  -.0064208    .001548    -4.15   0.000    -.0094556   -.0033859
  mean_union |   .1900785   .0223172     8.52   0.000     .1463249    .2338321
       _cons |   1.405264    .042734    32.88   0.000     1.321482    1.489046
------------------------------------------------------------------------------


Random-effects GLS regression                   Number of obs     =     19,229
Group variable: idcode                          Number of groups  =      4,150

R-squared:                                      Obs per group:
     Within  = 0.0963                                         min =          1
     Between = 0.0750                                         avg =        4.6
     Overall = 0.0768                                         max =         12

                                                Wald chi2(4)      =    1023.99
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000

                              (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006912    22.21   0.000     .0139959    .0167055
       union |   .1055274   .0098587    10.70   0.000     .0862047     .12485
```

```
    mean_age |  -.0061464   .0014254    -4.31   0.000    -.0089401   -.0033527
  mean_union |    .212183   .0210098    10.10   0.000     .1710046    .2533614
       _cons |   1.362101   .0383784    35.49   0.000     1.286881    1.437322
-------------+----------------------------------------------------------------
     sigma_u |  .38505558
     sigma_e |  .26213464
         rho |  .68331727   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

All three models yield identical coefficients: $\hat{\beta}_{\text{age}} = 0.0153507$ (SE $= 0.00069$) and $\hat{\beta}_{\text{union}} = 0.1055274$ (SE $= 0.00986$).

8.2.1. Adding a time-invariant covariate

We add `race` (constant within individual) to the Mundlak Pooled OLS model:

```
reg ln_wage age union mean_age mean_union i.race, cluster(idcode)
```

```
Linear regression                               Number of obs     =     19,229
                                                F(6, 4149)        =     189.66
                                                Prob > F          =     0.0000
                                                R-squared         =     0.1069
                                                Root MSE          =      .4422

                            (Std. err. adjusted for 4,150 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |   .0153507   .0006913    22.21   0.000     .0139955    .0167059
       union |   .1055274   .0098592    10.70   0.000     .0861981    .1248566
    mean_age |  -.0070345   .0015342    -4.59   0.000    -.0100424   -.0040266
  mean_union |   .2227511   .0220026    10.12   0.000     .1796142     .265888
             |
        race |
      Black  |  -.1790995   .0143624   -12.47   0.000    -.2072575   -.1509415
      Other  |   .0882461   .0715315     1.23   0.217    -.0519939    .2284861
             |
       _cons |   1.466452   .0426087    34.42   0.000     1.382916    1.549988
------------------------------------------------------------------------------
```

The coefficients on `age` and `union` remain unchanged to seven digits, while the coefficient on `race` is identified (Black women earn -17.9% relative to white women, SE $= 0.0144$).

8.3. Two-way Mundlak and balanced panels

Wooldridge (2025) extends Mundlak's idea to Two-Way Fixed Effects (TWFE):

$$y_{it} = X_{it}\beta + c_i + f_t + u_{it}$$

The Two-Way Mundlak (TWM) estimator adds both unit means ($\bar{X}_{i\cdot}$) and time means ($\bar{X}_{\cdot t}$):

$$y_{it} = X_{it}\beta + \bar{X}_{i\cdot}\lambda + \bar{X}_{\cdot t}\theta + u_{it}$$

8.3.1. What changes in an unbalanced panel

The simple equivalence between TWM and TWFE requires the unit and time factors to be orthogonal. This holds automatically in a balanced panel, where every unit is observed in every period.

In an unbalanced panel, separate unit and time means do not span the two-way dummy-variable space. To recover the equivalence, we keep the year dummies as regressors and add their unit means ($\bar{d}_{it}$), which describe each unit's observation pattern, s_{it}/T_i. This is no longer a two-way Mundlak regression. It is the one-way Mundlak device over units applied to the enlarged regressor set (X_{it}, d_t), the same construction as B in the next section.

```
preserve
webuse nlswork, clear
keep if !missing(ln_wage, age, union)
xtset idcode year

bysort idcode: egen double m_age   = mean(age)
bysort idcode: egen double m_union = mean(union)
egen double t_age   = mean(age),   by(year)
egen double t_union = mean(union), by(year)

* Unit means of time dummies
levelsof year, local(yrs)
foreach y of local yrs {
    bysort idcode: egen double my_`y' = mean(year==`y')
}

quietly xtreg ln_wage age union i.year, fe
scalar b_twfe = _b[age]

* Naive TWM with unit and time means only
quietly regress ln_wage age union m_age m_union t_age t_union
scalar b_twm = _b[age]

* One-way Mundlak over units with explicit year dummies and their unit means
quietly regress ln_wage age union m_age m_union i.year my_*
```

```
scalar b_fix = _b[age]

di "TWFE                           age = " %9.7f b_twfe
di "Naive TWM (unit + time means)  age = " %9.7f b_twm
di "TWM + unit means of dummies    age = " %9.7f b_fix

restore
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

(9,305 observations deleted)


Panel variable: idcode (unbalanced)
 Time variable: year, 70 to 88, but with gaps
         Delta: 1 unit




70 71 72 73 77 78 80 82 83 85 87 88




TWFE                           age = 0.0276645

Naive TWM (unit + time means)  age = 0.0117968

TWM + unit means of dummies    age = 0.0276645
```

TWFE gives $\hat{\beta}_{\text{age}} = 0.0276645$. The simple TWM regression gives only 0.0117968. Once we keep the year dummies and add their unit means, the estimate is 0.0276645, equal to TWFE. (All the means are generated with `egen double`; in single precision the last digit differs.)

8.4. More than two fixed effects

With three or more fixed-effect factors, such as individual i, year t, and group g, one set of means for each dimension is not generally enough. It works only when the factors are mutually orthogonal, as in a Latin-square design. A safer approach is to eliminate the high-dimensional factor, usually

the individual, keep dummies for the other factors, and add the within-individual means of every time-varying regressor, including those dummies.

```
preserve
capture program drop twm3
program define twm3
    args label
    quietly {
        capture drop x_i x_t x_g mt_* mg_* x y ft hg
        sort t
        by t: gen double ft = rnormal() if _n==1
        by t: replace ft = ft[1]
        sort g
        by g: gen double hg = rnormal() if _n==1
        by g: replace hg = hg[1]
        gen double x = 0.6*ci + 0.5*ft + 0.4*hg + rnormal()
        gen double y = 1*x + ci + ft + hg + rnormal()
        bysort i: egen double x_i = mean(x)
        bysort t: egen double x_t = mean(x)
        bysort g: egen double x_g = mean(x)
        levelsof t, local(ts)
        foreach k of local ts {
            bysort i: egen double mt_`k' = mean(t==`k')
        }
        levelsof g, local(gs)
        foreach k of local gs {
            bysort i: egen double mg_`k' = mean(g==`k')
        }
        areg y x i.t i.g, absorb(i)
        scalar b_fe = _b[x]
        regress y x x_i x_t x_g
        scalar b_A = _b[x]
        regress y x x_i i.t i.g mt_* mg_*
        scalar b_B = _b[x]
    }
    di "`label'"
    di "   Three-way FE                       = " %9.6f b_fe
    di "   A: One set of means per factor     = " %9.6f b_A  "   gap: " %9.6f (b_A-b_fe)
    di "   B: Means over i, dummies included  = " %9.6f b_B  "   gap: " %9.6f (b_B-b_fe)
end

clear
set seed 20260731
set obs 300
gen i = _n
gen double ci = rnormal()
expand 10
sort i, stable
```

```
by i: gen t = _n

* Non-orthogonal design (random assignment of g)
gen g = 1 + int(runiform()*5)
twm3 "Non-orthogonal factor assignment:"

* Orthogonal design (Latin square)
replace g = mod(i+t,5)+1
twm3 "Orthogonal factor assignment (Latin square):"

capture program drop twm3
restore
```

```
Number of observations (_N) was 0, now 300.
```

```
(2,700 observations created)
```

```
Non-orthogonal factor assignment:
   Three-way FE                      =  0.994424
   A: One set of means per factor     =  1.007648   gap:  0.013224
   B: Means over i, dummies included  =  0.994424   gap:  0.000000

(2,435 real changes made)

Orthogonal factor assignment (Latin square):
   Three-way FE                      =  0.996471
   A: One set of means per factor     =  0.996471   gap:  0.000000
   B: Means over i, dummies included  =  0.996471   gap:  0.000000
```

Construction B (means over i including factor dummies) is exact in both orthogonal and non-orthogonal designs.

8.5. Difference-in-differences with common timing

In standard DiD with common treatment timing:

$$y_{it} = \beta w_{it} + c_i + f_t + u_{it}$$

where $w_{it} = d_i \cdot \text{post}_t$. Because d_i is time-invariant and post_t is unit-invariant: - Unit mean: $\bar{w}_{i\cdot} = d_i \cdot \bar{p}$ - Time mean: $\bar{w}_{\cdot t} = \bar{d} \cdot \text{post}_t$

So we can span the Mundlak projection with d_i and the post-treatment indicator, or with the full set of time dummies. We can also interact treatment with time to allow the treatment effect to vary:

```
clear
set obs 100
set seed 1234567
gen id = _n
gen w1 = rnormal()
gen w2 = rnormal()
gen x1 = rnormal() + w1 + 2*w2
expand 4
sort id, stable
by id: gen year=2011+_n
gen post=(year>2013)
gen d=(id>50)
gen w=d*post
gen u = rnormal()

gen f2014=(year==2014)
gen f2015=(year==2015)

* DGP with growing treatment effect: 2.5 in 2014, 3.5 in 2015 (ATT = 3.0)
gen y1 = 1 + 2*x1 + 3*w1 + u
gen logy = y1 + 2.5*d*f2014 + 3.5*d*f2015

xtset id year
```

```
Number of observations (_N) was 0, now 100.
```

```
(300 observations created)
```

```
Panel variable: id (strongly balanced)
 Time variable: year, 2012 to 2015
         Delta: 1 unit
```

```
* 1. TWFE with year-specific treatment effects
xtreg logy c.w#c.f2014 c.w#c.f2015 i.year, fe vce(cluster id)

* 2. Pooled OLS with Mundlak controls (d and year dummies)
reg logy c.w#c.f2014 c.w#c.f2015 d f2014 f2015, vce(cluster id)
lincom (c.w#c.f2014 + c.w#c.f2015)/2

* 3. Adding time-invariant controls (x1 and interaction with d)
reg logy c.w#c.f2014 c.w#c.f2015 d f2014 f2015 x1 c.d#c.x1, vce(cluster id)
```

```
Fixed-effects (within) regression               Number of obs     =        400
Group variable: id                              Number of groups  =        100

R-squared:                                      Obs per group:
     Within  = 0.6025                                         min =          4
     Between = 0.0099                                         avg =        4.0
     Overall = 0.0322                                         max =          4

                                                F(5, 99)          =      97.08
corr(u_i, Xb) = -0.0024                         Prob > F          =     0.0000

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.570823   .2569637    10.00   0.000     2.060951    3.080695
             |
 c.w#c.f2015 |   3.334939   .2746487    12.14   0.000     2.789977    3.879902
             |
        year |
       2013  |  -.1351205    .137473    -0.98   0.328    -.4078966    .1376557
       2014  |  -.2320952   .2022437    -1.15   0.254    -.6333905    .1692002
       2015  |    .147704   .2269835     0.65   0.517    -.3026806    .5980885
             |
       _cons |    .364572   .0861151     4.23   0.000     .1937011     .535443
-------------+----------------------------------------------------------------
     sigma_u |   7.132491
     sigma_e |  1.0368898
         rho |  .97930335   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

```
Linear regression                               Number of obs     =        400
                                                F(5, 99)          =      98.53
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0322
                                                Root MSE          =     7.2067

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.570823   .2569637    10.00   0.000     2.060951    3.080695
             |
 c.w#c.f2015 |   3.334939   .2746487    12.14   0.000     2.789977    3.879902
             |
           d |   -.059725   1.442034    -0.04   0.967    -2.921033    2.801583
      f2014 |  -.1645349   .1912728    -0.86   0.392    -.5440616    .2149918
      f2015 |   .2152642   .2173966     0.99   0.324    -.2160979    .6466263
      _cons |   .3268743   1.022832     0.32   0.750    -1.702645    2.356394
------------------------------------------------------------------------------


 ( 1)  .5*c.w#c.f2014 + .5*c.w#c.f2015 = 0

------------------------------------------------------------------------------
        logy | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   2.952881   .2224535    13.27   0.000     2.511485    3.394277
------------------------------------------------------------------------------


Linear regression                               Number of obs     =        400
                                                F(7, 99)          =     166.99
                                                Prob > F          =     0.0000
                                                R-squared         =     0.8485
                                                Root MSE          =     2.8589

                                  (Std. err. adjusted for 100 clusters in id)
------------------------------------------------------------------------------
             |               Robust
        logy | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 c.w#c.f2014 |   2.570823   .2576184     9.98   0.000     2.059652    3.081994
             |
 c.w#c.f2015 |   3.334939   .2753484    12.11   0.000     2.788588     3.88129
             |
           d |    .704106   .5693191     1.24   0.219    -.4255466    1.833759
```

```
       f2014 |  -.1645349   .1917601    -0.86   0.393    -.5450286    .2159587
       f2015 |   .2152642   .2179505     0.99   0.326    -.2171969    .6477253
          x1 |   2.500374   .1451124    17.23   0.000     2.212439    2.788308
             |
    c.d#c.x1 |   .0290171   .2018959     0.14   0.886    -.3715881    .4296224
             |
       _cons |    .203081   .3958344     0.51   0.609    -.5823403    .9885023
------------------------------------------------------------------------------
```

The DGP sets the effects at 2.5 in 2014 and 3.5 in 2015, an average of 3.0. TWFE estimates 2.571 and 3.335 in this sample, and the pooled Mundlak regression reproduces both exactly; `lincom` gives their average, 2.953. Adding the time-invariant control x_1 and its interaction with d leaves the two estimates unchanged at 2.571 and 3.335: both controls are constant within a unit, and the treatment effects are identified from changes within units.

8.6. What to remember

For a one-way model, adding unit means to pooled OLS or random effects recovers the within coefficient and lets us include time-invariant variables. For two-way models, the simple result holds in a balanced panel; an unbalanced panel also needs unit means of the time dummies. With more fixed-effect dimensions, I would eliminate the high-dimensional factor and retain explicit dummies for the others. The same formulation is useful in DiD because it accommodates time-varying effects and time-invariant moderators in an ordinary regression.

9. Correlated Random Effect

9.1. Panel data

The standard panel-data setup is:

$$y_{it} = \alpha + X_{it}\beta + v_i + \epsilon_{it} \tag{9.1}$$

9.2. Fixed effect

The fixed-effect estimator runs OLS on demeaned data:

$$(y_{it} - \bar{y}_i) = (X_{it} - \bar{X}_i)\beta + (\epsilon_{it} - \bar{\epsilon}_i) \tag{9.2}$$

Both α and v_i drop out after demeaning by unit.

9.3. Random effect

Random effects treat $v_i + \epsilon_{it}$ as a composite error with two variance components: $\hat{\sigma}_e^2$ (idiosyncratic) and $\hat{\sigma}_u^2$ (individual). The GLS transformation is

$z_{it}^* = z_{it} - \hat{\theta}_i \bar{z}_i$

with $\hat{\theta}_i = 1 - \sqrt{\hat{\sigma}_e^2/(T_i\hat{\sigma}_u^2 + \hat{\sigma}_e^2)}$, where T_i is the number of observations for unit i. Given estimated variance components, we run OLS on the transformed variables and iterate.

9.4. Correlated random effect

The correlated random-effect (CRE) model runs a random-effects regression of y_{it} on X_{it}, $\bar{X}_i$, time-invariant covariates w_i, and a constant.

RE requires v_i and X_{it} to be uncorrelated — an assumption most economists reject. FE is consistent regardless, but it cannot estimate time-invariant covariates (race, gender, etc.). CRE includes time-invariant variables, and Mundlak and Wooldridge show that its coefficients on X_{it} are identical to the FE estimates. Its coefficients on w_i are another matter. Any part of the unit effect v_i that moves with w_i is indistinguishable from the coefficient on w_i, so that coefficient is causal only under an extra restriction: that v_i is mean-independent of w_i once $\bar{X}_i$ is controlled. Testing whether the coefficients on $\bar{X}_i$ are jointly zero gives the Mundlak test for choosing between RE and CRE.

9.5. Example

Stata 18 added a `cre` option to `xtreg`. The following uses `nlswork`, 28,101 person-years from the National Longitudinal Survey of Young Women with complete data on the model's variables. We regress log wage on `tenure` and `age`, which vary within a person, plus `race`, which does not — the covariate a fixed-effect model has to drop.

```
webuse nlswork

xtreg ln_wage tenure age i.race, cre vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

note: 2.race omitted from xt_means because of collinearity.
note: 3.race omitted from xt_means because of collinearity.

Correlated random-effects regression            Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.2346                                         avg =        6.0
     Overall = 0.1890                                         max =         15

                                                Wald chi2(4)      =    1685.18
corr(xit_vars*b, xt_means*γ) = 0.5474           Prob > chi2       =     0.0000

                           (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
xit_vars     |
      tenure |   .0211313   .0012113    17.44   0.000     .0187572    .0235055
         age |   .0121949   .0007414    16.45   0.000     .0107417     .013648
             |
        race |
      Black  |  -.1312068   .0117856   -11.13   0.000    -.1543061   -.1081075
      Other  |   .1059379   .0593177     1.79   0.074    -.0103225    .2221984
             |
       _cons |     1.2159   .0306965    39.61   0.000     1.155736    1.276064
-------------+----------------------------------------------------------------
xt_means     |
      tenure |   .0376991    .002281    16.53   0.000     .0332283    .0421698
         age |  -.0011984   .0013313    -0.90   0.368    -.0038077    .0014109
             |
        race |
```

```
      Black  |          0  (omitted)
      Other  |          0  (omitted)
-------------+----------------------------------------------------------------
     sigma_u |  .33334407
     sigma_e |  .29808194
         rho |  .55567161   (fraction of variance due to u_i)
------------------------------------------------------------------------------
Mundlak test (xt_means = 0): chi2(2) = 331.5144          Prob > chi2 = 0.0000
```

CRE gives `tenure` = .0211313 and `age` = .0121949, and estimates race as well: Black women earn 13.1% less (−.1312068, SE .0118), while the "Other" category is marginal (.1059, $p = 0.074$). The coefficients on the group means are .0376991 for mean tenure and −.0011984 for mean age.

Stata prints the Mundlak test at the foot of the table: $\chi^2(2) = 331.51$, $p < 0.0001$. The group means are jointly far from zero, so plain random effects is rejected — v_i and the regressors are correlated, as the reported corr = 0.5474 between the two blocks says directly.

To compare with a fixed effect model:

```
webuse nlswork

xtreg ln_wage tenure age i.race, fe vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

note: 2.race omitted because of collinearity.
note: 3.race omitted because of collinearity.

Fixed-effects (within) regression               Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.1916                                         avg =        6.0
     Overall = 0.1456                                         max =         15

                                                F(2, 4698)        =     766.79
corr(u_i, Xb) = 0.1302                          Prob > F          =     0.0000

                            (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |   .0211313   .0012112    17.45   0.000     .0187568    .0235059
         age |   .0121949   .0007414    16.45   0.000     .0107414    .0136483
             |
        race |
```

```
      Black  |          0  (omitted)
      Other  |          0  (omitted)
             |
       _cons |   1.256467   .0194187    64.70   0.000     1.218397    1.294537
-------------+----------------------------------------------------------------
     sigma_u |  .39034493
     sigma_e |  .29808194
         rho |  .63165531   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

The coefficients on `tenure` and `age` are identical: .0211313 and .0121949 in both, with standard errors agreeing to the seventh decimal (.0012112 against .0012113). Race is reported as `0 (omitted)` here, being collinear with the unit effects — the one thing FE cannot do and CRE can. This is the Mundlak result in practice: CRE gives you the fixed-effect estimates *and* the time-invariant coefficients from a single random-effects fit.

We can replicate this manually by including the group means in a RE model:

```
webuse nlswork
* Restrict the mean calculation to the estimation sample (listwise-deleted
* on the model's variables), matching what xtreg's native cre option does
* internally. Both tenure and age have missing values in nlswork, so
* computing age_mean/tenure_mean over all rows for an idcode (rather than
* only rows with complete data on ln_wage, tenure, age, and race) would
* make the manual means diverge from cre's.
egen age_mean = mean(age) if !missing(ln_wage, tenure, age, race), by(idcode)
egen tenure_mean = mean(tenure) if !missing(ln_wage, tenure, age, race), by(idcode)
xtreg ln_wage tenure tenure_mean age age_mean i.race, vce(cluster idcode)
```

```
(National Longitudinal Survey of Young Women, 14-24 years old in 1968)

(433 missing values generated)

(433 missing values generated)


Random-effects GLS regression                   Number of obs     =     28,101
Group variable: idcode                          Number of groups  =      4,699

R-squared:                                      Obs per group:
     Within  = 0.1296                                         min =          1
     Between = 0.2346                                         avg =        6.0
     Overall = 0.1890                                         max =         15

                                                Wald chi2(6)      =    2688.54
corr(u_i, X) = 0 (assumed)                      Prob > chi2       =     0.0000
```

```
                           (Std. err. adjusted for 4,699 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |   .0211313   .0012113    17.44   0.000     .0187572    .0235055
 tenure_mean |   .0376991    .002281    16.53   0.000     .0332283    .0421698
         age |   .0121949   .0007414    16.45   0.000     .0107417     .013648
    age_mean |  -.0011984   .0013313    -0.90   0.368    -.0038077    .0014109
             |
        race |
       Black |  -.1312068   .0117856   -11.13   0.000    -.1543061   -.1081075
       Other |   .1059379   .0593177     1.79   0.074    -.0103225    .2221984
             |
       _cons |     1.2159   .0306965    39.61   0.000     1.155736    1.276064
-------------+----------------------------------------------------------------
     sigma_u |  .33334407
     sigma_e |  .29808194
         rho |  .55567161   (fraction of variance due to u_i)
------------------------------------------------------------------------------
```

This is what Stata's `cre` option does internally. Every number matches the first table: `tenure` .0211313, `age` .0121949, Black −.1312068, Other .1059379, constant 1.2159, and the same variance components (σ_u .3333, σ_e .2981, ρ .5557). The `cre` option is a convenience, not a different estimator.

The one detail worth care is in the code above: the group means must be computed on the estimation sample. Both `tenure` and `age` have missing values in `nlswork`, so averaging over all of a woman's rows rather than only her complete ones would give different means and break the equivalence.

9.6. Summary

Random effects is efficient but assumes the unit effect is uncorrelated with the regressors. Fixed effects drops that assumption but cannot estimate time-invariant covariates. The correlated random effects model adds the unit means of the time-varying regressors to a random-effects regression. That gives the fixed-effects slopes and estimates for time-invariant covariates in one fit. The joint test on the means is the Mundlak version of the Hausman test.

On the `nlswork` data, CRE and fixed effects give identical coefficients on tenure and age, .0211 and .0122. CRE also estimates the race gap, with Black women earning about 13% less. The Mundlak test strongly rejects plain random effects. Adding the means by hand reproduces Stata's `cre` option exactly, provided the means are computed on the estimation sample.

10. Correlated random effects in linear and nonlinear models

```
library(dplyr)
library(fixest)
library(bacondecomp)
library(lme4)
library(marginaleffects)
```

10.1. The CRE idea

Suppose a panel-data model has an unobserved individual effect v_i:

$$y_{it} = \alpha + X_{it}\beta + v_i + \epsilon_{it}$$

The usual random-effects model assumes that v_i is independent of the covariates, so $E[v_i \mid X_i] = 0$. This is often hard to defend. Fixed effects allows v_i to be correlated with X_{it} by removing v_i through demeaning or first-differencing, but then we cannot estimate coefficients on time-invariant covariates. The fixed-effects approach is also less convenient in nonlinear models.

Correlated random effects (CRE) provides another way to allow this correlation (Mundlak 1978; Chamberlain 1982; Wooldridge 2010). We write the projection of v_i on the history of X_i as

$$v_i = \psi + \bar{X}_i\xi + a_i, \qquad E[a_i \mid X_i] = 0$$

where $\bar{X}_i = \frac{1}{T}\sum_{t=1}^{T} X_{it}$ is the unit mean of the time-varying covariates. Substituting it into the outcome model gives

$$y_{it} = (\alpha + \psi) + X_{it}\beta + \bar{X}_i\xi + a_i + \epsilon_{it}$$

So the practical step is simple: add the unit means $\bar{X}_i$ to the regression.

10.2. Linear models

In a balanced linear panel, pooled OLS or random-effects GLS with $\bar{X}_i$ gives the same coefficient on X_{it} as the within estimator.

10.2.1. An example using the `castle` data

The `castle` data form a balanced panel of 50 states observed for 11 years ($N = 550$). I use log homicide as the outcome and poverty and log police employment as the covariates.

```
data("castle")

# 1. Two-way Fixed Effects
fe_model <- feols(l_homicide ~ poverty + l_police | state + year, data = castle)

# 2. Standard Random Effects (assumes Cov(X, v) = 0)
re_model <- lmer(l_homicide ~ poverty + l_police + factor(year) + (1 | state), data = castle)

# 3. CRE Pooled OLS (Mundlak device)
castle2 <- castle |>
  group_by(state) |>
  mutate(poverty_mean = mean(poverty, na.rm = TRUE),
         l_police_mean = mean(l_police, na.rm = TRUE))

cre_pols <- feols(l_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year),
                  data = castle2, cluster = ~state)

# 4. CRE Random Effects
cre_re <- lmer(l_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year) + (1 | state
               data = castle2)

# Compare estimates on poverty and l_police
res_table <- data.frame(
  Estimator = c("Two-way Fixed Effects", "Standard Random Effects", "CRE Pooled OLS", "CRE Random Effects
  beta_poverty = c(coef(fe_model)["poverty"], fixef(re_model)["poverty"],
                   coef(cre_pols)["poverty"], fixef(cre_re)["poverty"]),
  beta_police  = c(coef(fe_model)["l_police"], fixef(re_model)["l_police"],
                   coef(cre_pols)["l_police"], fixef(cre_re)["l_police"])
)
res_table[, 2:3] <- round(res_table[, 2:3], 6)
knitr::kable(res_table, caption = "Comparison of linear panel estimators.")
```

Table 10.1.: Comparison of linear panel estimators.

Estimator	beta_poverty	beta_police
Two-way Fixed Effects	-0.027068	0.066033
Standard Random Effects	-0.000799	0.166591
CRE Pooled OLS	-0.027068	0.066033
CRE Random Effects	-0.027068	0.066033

10.2.2. Comparing the estimates

The two-way fixed-effects model, CRE pooled OLS, and CRE random-effects model give the same coefficients: $\beta_{\text{poverty}} = -0.027068$ and $\beta_{\text{police}} = 0.066033$. Standard random effects, which omits the unit means, gives $\beta_{\text{poverty}} = -0.000799$. The difference is large because the standard model rules out correlation with the state effect. For pooled OLS, we should cluster the standard errors by state because observations within a state are correlated.

10.3. Binary outcomes

For a binary outcome $y_{it} \in \{0, 1\}$, the latent model is:

$$P(y_{it} = 1 \mid X_{it}, v_i) = \Phi(X_{it}\beta + v_i)$$

The exact equivalence between fixed effects and CRE no longer holds in nonlinear models. Conditional logit requires conditional independence of y_{it} over time given X_i and v_i; serial correlation in the idiosyncratic errors violates this assumption. It also conditions v_i out of the likelihood, so we cannot recover partial effects that depend on v_i.

An alternative is pooled probit with the unit means included. With standard errors clustered by unit, this approach allows arbitrary serial correlation (Wei Lin and Wooldridge 2019), and average marginal effects are easy to calculate.

10.3.1. An example with a binary outcome

For an illustration, I define `high_homicide` to equal one when the homicide rate is above the sample mean:

```
castle2 <- castle2 |>
  mutate(high_homicide = ifelse(l_homicide > mean(l_homicide, na.rm = TRUE), 1, 0))

# 1. CRE Pooled Probit
cre_probit <- glm(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year),
                  data = castle2, family = binomial(link = "probit"))

# 2. Linear Probability Model with CRE
cre_lpm <- lm(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year),
              data = castle2)

# 3. CRE Random Effects Probit
cre_re_probit <- glmer(high_homicide ~ poverty + poverty_mean + l_police + l_police_mean + factor(year) +
                       data = castle2, family = binomial(link = "probit"))

# Average marginal effects; state-clustered SEs for the pooled fits
ame_probit <- avg_slopes(cre_probit, variables = c("poverty", "l_police"), vcov = ~state)
```

```r
ame_lpm    <- avg_slopes(cre_lpm, variables = c("poverty", "l_police"), vcov = ~state)
ame_re     <- avg_slopes(cre_re_probit, variables = c("poverty", "l_police"))

ame_comp <- data.frame(
  Model = c("CRE Pooled Probit", "CRE Linear Probability Model", "CRE Random Effects Probit"),
  AME_poverty = c(ame_probit$estimate[ame_probit$term == "poverty"],
                  ame_lpm$estimate[ame_lpm$term == "poverty"],
                  ame_re$estimate[ame_re$term == "poverty"]),
  SE_poverty  = c(ame_probit$std.error[ame_probit$term == "poverty"],
                  ame_lpm$std.error[ame_lpm$term == "poverty"],
                  ame_re$std.error[ame_re$term == "poverty"])
)
ame_comp[, 2:3] <- round(ame_comp[, 2:3], 5)
knitr::kable(ame_comp, caption = "Average Marginal Effects across binary CRE models.")
```

Table 10.2.: Average Marginal Effects across binary CRE models.

Model	AME_poverty	SE_poverty
CRE Pooled Probit	-0.06477	0.04300
CRE Linear Probability Model	-0.06406	0.04325
CRE Random Effects Probit	-0.06477	0.03336

All three specifications give an average marginal effect of poverty close to -0.064. For the pooled probit and the linear probability model, the standard errors are clustered by state (`vcov = ~state`) and are about 0.043. The random-effects probit reports a model-based standard error of 0.033. Its estimated state-effect variance is essentially zero (the fit is singular), so here it reduces to pooled probit with conventional standard errors, which ignore serial correlation within a state.

In this example, CRE pooled probit with clustered standard errors gives us the partial effect without imposing the serial-independence assumption of the random-effects probit. The unit means and the clustering solve different problems. The means address correlation between the state effect and the covariates. The clustering addresses dependence across years within a state.

10.4. What we learn

For a balanced linear panel, adding the unit averages $\bar{X}_i$ to pooled OLS or random effects recovers the fixed-effects coefficients. In binary and count models, CRE is useful because we can calculate average marginal effects and avoid estimating a fixed effect for every unit. Pooled estimation with unit-clustered standard errors is especially attractive when we do not want to assume serial independence.

Part III.

Treatment Effects & Matching

11. Treatment effects and matching

11.1. Treatment effects in observational studies

Most economic data are observational. With unobserved confounders we need instrumental variables, but instruments are hard to find and harder to justify. Under conditional independence — no unobserved confounders given the covariates — we can proceed without them.

Stata's `teffects` suite implements the standard estimators under conditional independence. (The companion `eteffects` handles endogenous treatment via a control-function approach with instruments.) This chapter walks through how `teffects` works.

11.1.1. Regression adjustment (RA)

RA allows all covariate effects to differ between treatment and control — equivalent to an outcome model with treatment interacted with all covariates.

```
clear
webuse bweightex
teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke)
reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)
margins r.mbsmoke
```

```
. clear

. webuse bweightex
(Hypothetical birthweight data)

. teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke)

Iteration 0:  EE criterion = 1.223e-24
Iteration 1:  EE criterion = 1.792e-25

Treatment-effects estimation                    Number of obs     =         60
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
```

```
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -389.3099   37.00882   -10.52   0.000    -461.8458   -316.7739
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3613.769   29.97438   120.56   0.000     3555.021    3672.518
------------------------------------------------------------------------------

. reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)

      Source |       SS           df       MS      Number of obs   =        60
-------------+----------------------------------   F(9, 50)        =     18.11
       Model |  1376306.34         9  152922.927   Prob > F        =    0.0000
    Residual |  422241.592        50  8444.83184   R-squared       =    0.7652
-------------+----------------------------------   Adj R-squared   =    0.7230
       Total |  1798547.93        59  30483.8633   Root MSE        =    91.896

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |
     Smoker  |  -32.87702   193.2827    -0.17   0.866    -421.0967    355.3426
   prenatal1 |   -23.8431   38.31014    -0.62   0.537    -100.7913    53.10507
    mmarried |  -40.39753   50.21942    -0.80   0.425    -141.2662    60.47114
        mage |    27.9537   7.671423     3.64   0.001     12.54519    43.36221
       fbaby |  -2.030497   40.76029    -0.05   0.960    -83.89995    79.83896
             |
     mbsmoke#|
 c.prenatal1 |
     Smoker  |   8.436558   56.38855     0.15   0.882    -104.8232    121.6963
             |
     mbsmoke#|
  c.mmarried |
     Smoker  |   33.92362   61.34015     0.55   0.583     -89.2817    157.1289
             |
     mbsmoke#|
      c.mage |
     Smoker  |  -15.98608   9.006494    -1.77   0.082    -34.07616    2.103995
             |
     mbsmoke#|
     c.fbaby |
     Smoker  |   14.86782   57.58771     0.26   0.797    -100.8005    130.5362
             |
```

```
       _cons |   2976.807   151.0117    19.71   0.000     2673.491    3280.123
------------------------------------------------------------------------------

. margins r.mbsmoke

Contrasts of predictive margins                               Number of obs = 60
Model VCE: OLS

Expression: Linear prediction, predict()

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
     mbsmoke |          1      116.06     0.0000
             |
 Denominator |         50
------------------------------------------------

------------------------------------------------------------------------------
                      |            Delta-method
                      |   Contrast   std. err.     [95% conf. interval]
----------------------+-------------------------------------------------------
              mbsmoke |
(Smoker vs Nonsmoker) |  -389.3099   36.13675     -461.8927   -316.7271
------------------------------------------------------------------------------
```

The data here are `bweightex`, a 60-observation extract with birthweight as the outcome and maternal smoking as the treatment.

`teffects ra` returns the same ATE as `margins r.mbsmoke` after the interacted regression: both give −389.3099 grams. The standard errors differ slightly, 37.01 against 36.14, because `teffects` accounts for the estimation of the outcome model differently from `margins`' delta method. The point estimate is the same object computed two ways.

For the ATET (average treatment effect on the treated):

```
clear
webuse bweightex
teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke), atet
reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)
margins r.mbsmoke, subpop(mbsmoke)
```

```
. clear

. webuse bweightex
(Hypothetical birthweight data)
```

```
. teffects ra (bweight prenatal1 mmarried mage fbaby) (mbsmoke), atet

Iteration 0:  EE criterion = 1.159e-24
Iteration 1:  EE criterion = 5.133e-26

Treatment-effects estimation                    Number of obs     =         60
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -437.4721   53.43513    -8.19   0.000     -542.203   -332.7412
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3693.639   53.80537    68.65   0.000     3588.182    3799.095
------------------------------------------------------------------------------

. reg bweight i.mbsmoke##c.(prenatal1 mmarried mage fbaby)

      Source |       SS           df       MS      Number of obs   =        60
-------------+----------------------------------   F(9, 50)        =     18.11
       Model |  1376306.34         9  152922.927   Prob > F        =    0.0000
    Residual |  422241.592        50  8444.83184   R-squared       =    0.7652
-------------+----------------------------------   Adj R-squared   =    0.7230
       Total |  1798547.93        59  30483.8633   Root MSE        =    91.896

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |
     Smoker  |  -32.87702   193.2827    -0.17   0.866    -421.0967    355.3426
   prenatal1 |   -23.8431   38.31014    -0.62   0.537    -100.7913    53.10507
    mmarried |  -40.39753   50.21942    -0.80   0.425    -141.2662    60.47114
        mage |    27.9537   7.671423     3.64   0.001     12.54519    43.36221
       fbaby |  -2.030497   40.76029    -0.05   0.960    -83.89995    79.83896
             |
     mbsmoke#|
 c.prenatal1 |
     Smoker  |   8.436558   56.38855     0.15   0.882    -104.8232    121.6963
             |
     mbsmoke#|
```

```
  c.mmarried |
     Smoker  |   33.92362   61.34015     0.55   0.583     -89.2817    157.1289
             |
     mbsmoke#|
      c.mage |
     Smoker  |  -15.98608   9.006494    -1.77   0.082    -34.07616    2.103995
             |
     mbsmoke#|
     c.fbaby |
     Smoker  |   14.86782   57.58771     0.26   0.797    -100.8005    130.5362
             |
       _cons |   2976.807   151.0117    19.71   0.000     2673.491    3280.123
------------------------------------------------------------------------------

. margins r.mbsmoke, subpop(mbsmoke)

Contrasts of predictive margins                           Number of obs   = 60
Model VCE: OLS                                            Subpop. no. obs = 30

Expression: Linear prediction, predict()

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
     mbsmoke |          1       73.30     0.0000
             |
 Denominator |         50
------------------------------------------------

------------------------------------------------------------------------------
                     |            Delta-method
                     |   Contrast   std. err.     [95% conf. interval]
---------------------+--------------------------------------------------------
               mbsmoke |
(Smoker vs Nonsmoker)  |  -437.4721   51.09606     -540.1015   -334.8426
------------------------------------------------------------------------------
```

The ATET compares potential outcomes for the treated subpopulation, hence the `subpop(mbsmoke)` option in `margins`. Both routes give -437.4721, again identical, and again with slightly different standard errors (53.44 against 51.10). Note the ATET is 48 grams larger in magnitude than the ATE of -389.31: smoking mothers are drawn disproportionately from the groups where smoking does most damage.

11.1.2. Inverse Probability Weighting (IPW)

IPW has two steps. First, estimate the treatment model (typically logit) to obtain the propensity score. Second, use inverse-probability weights to compute the outcome difference between treated

and control units.

These steps produce consistent estimates of the effect parameters because the treatment is assumed to be independent of the potential outcomes after conditioning on the covariates.

We can manually conduct the two steps, but the nice thing about using Stata's teffects is that it takes account of the noise of estimating probability in the first step when calculating standard errors in the second step.

11.1.2.1. example

```
clear
webuse cattaneo2
teffects ipw (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu, probit)
probit mbsmoke mmarried c.mage##c.mage fbaby medu
predict ps
replace ps = 1/ps if mbsmoke==1
replace ps = 1/(1-ps) if mbsmoke==0
reg bweight mbsmoke [pweight=ps]
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. teffects ipw (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu, probit)

Iteration 0:  EE criterion = 4.622e-21
Iteration 1:  EE criterion = 8.453e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : inverse-probability weights
Outcome model  : weighted mean
Treatment model: probit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -230.6886   25.81524    -8.94   0.000    -281.2856   -180.0917
-------------+----------------------------------------------------------------
POmean       |
```

```
     mbsmoke |
  Nonsmoker  |   3403.463   9.571369   355.59   0.000     3384.703    3422.222
-------------------------------------------------------------------------------

. probit mbsmoke mmarried c.mage##c.mage fbaby medu

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2042.6734
Iteration 2:  Log likelihood = -2040.5088
Iteration 3:  Log likelihood = -2040.5061
Iteration 4:  Log likelihood = -2040.5061

Probit regression                                       Number of obs =  4,642
                                                        LR chi2(5)    = 380.48
                                                        Prob > chi2   = 0.0000
Log likelihood = -2040.5061                             Pseudo R2     = 0.0853

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |  -.6484821   .0526991   -12.31   0.000    -.7517705   -.5451938
        mage |   .1744327   .0352437     4.95   0.000     .1053562    .2435092
             |
      c.mage#|
      c.mage |  -.0032559   .0006462    -5.04   0.000    -.0045224   -.0019894
             |
       fbaby |  -.2175962   .0491066    -4.43   0.000    -.3138433    -.121349
        medu |  -.0863631   .0098692    -8.75   0.000    -.1057064   -.0670198
       _cons |  -1.558255   .4511589    -3.45   0.001     -2.44251       -.674
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

. replace ps = 1/ps if mbsmoke==1
(864 real changes made)

. replace ps = 1/(1-ps) if mbsmoke==0
(3,778 real changes made)

. reg bweight mbsmoke [pweight=ps]
(sum of wgt is 9,193.96990537643)

Linear regression                               Number of obs     =      4,642
                                                F(1, 4640)        =      79.08
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0389
                                                Root MSE          =     573.36
```

```
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -230.6886   25.94182    -8.89   0.000    -281.5469   -179.8303
       _cons |   3403.463   9.616992   353.90   0.000     3384.609    3422.317
------------------------------------------------------------------------------
```

We switch here to `cattaneo2`, the full 4,642-birth dataset, and a richer propensity model (a probit in marital status, a quadratic in maternal age, first-baby status and education).

`teffects ipw` gives an ATE of −230.6886 with a standard error of 25.82. The manual two-step replication matches the point estimate exactly, −230.6886; the only difference is the standard error (the manual version does not account for first-stage estimation noise). Note this is well short of the −389 found by RA above — but on different data and a different covariate set, so the two are not comparable.

11.1.3. Doubly robust estimators

RA models the outcome; IPW models the treatment assignment. Doubly-robust estimators model both, requiring only one to be correctly specified.

Stata implements the AIPW combination of Robins and Rotnitzky (1995) and the IPWRA combination of Wooldridge (2010).

The AIPW estimator augments the IPW estimator with a correction term built from the outcome model. Double robustness is best stated as a **consistency** property: AIPW is consistent if *either* the treatment model *or* the outcome model is correctly specified (not necessarily both). It is *not* that the correction term literally "goes to zero" when the treatment model is right and the outcome model is wrong — with a misspecified outcome model the correction term does not generally vanish. Rather, the two pieces are constructed so that the estimator's bias cancels in the limit as long as one of the two models is right.

The IPWRA estimator uses IPW probability weights when performing RA. The weights do not affect the accuracy of the RA estimator if the treatment model is wrong and the outcome model is correct. The weights correct the RA estimator if the treatment model is correct and the outcome model is wrong.

Here we run the two commands, then manually replicate both as a two-step ("plug-in") procedure. `teffects` estimates the treatment and outcome models jointly by GMM/ML, using the joint estimation to get standard errors that account for first-stage estimation noise; the manual two-step version below gets essentially the same point estimate but not the correct standard errors (for the same reason noted above for IPW).

```
clear
webuse cattaneo2
teffects ipwra (bweight mmarried mage prenatal1 fbaby)  (mbsmoke mmarried mage prenatal1 fbaby)
teffects aipw (bweight mmarried mage prenatal1 fbaby) (mbsmoke mmarried mage prenatal1 fbaby)
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. teffects ipwra (bweight mmarried mage prenatal1 fbaby)  (mbsmoke mmarried mag
> e prenatal1 fbaby)

Iteration 0:  EE criterion = 9.066e-22
Iteration 1:  EE criterion = 2.902e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : IPW regression adjustment
Outcome model  : linear
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -238.7679   24.38353    -9.79   0.000    -286.5587    -190.977
-------------+----------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3402.851   9.538741   356.74   0.000     3384.155    3421.546
------------------------------------------------------------------------------

. teffects aipw (bweight mmarried mage prenatal1 fbaby) (mbsmoke mmarried mage
> prenatal1 fbaby)

Iteration 0:  EE criterion = 2.115e-22
Iteration 1:  EE criterion = 1.343e-26

Treatment-effects estimation                    Number of obs     =      4,642
Estimator      : augmented IPW
Outcome model  : linear by ML
Treatment model: logit
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
     mbsmoke |
    (Smoker  |
```

```
         vs  |
 Nonsmoker)  |  -239.0294    24.2524    -9.86   0.000    -286.5632   -191.4955
-------------+------------------------------------------------------------------
POmean       |
     mbsmoke |
  Nonsmoker  |   3402.839   9.538926   356.73   0.000     3384.143    3421.535
------------------------------------------------------------------------------
```

Manual IPWRA: first estimate the propensity score, then run *weighted* regression adjustment within each treatment arm, using the inverse probability of the *observed* treatment status as weights:

```
clear
webuse cattaneo2
logit mbsmoke mmarried mage prenatal1 fbaby
predict ps

gen wgt1 = mbsmoke/ps
gen wgt0 = (1-mbsmoke)/(1-ps)
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1 [pweight=wgt1]
predict mu1_ipwra
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0 [pweight=wgt0]
predict mu0_ipwra
gen tau_ipwra_i = mu1_ipwra - mu0_ipwra
sum tau_ipwra_i
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. logit mbsmoke mmarried mage prenatal1 fbaby

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2089.7838
Iteration 2:  Log likelihood = -2082.3958
Iteration 3:  Log likelihood = -2082.3879
Iteration 4:  Log likelihood = -2082.3879

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(4)    = 296.72
                                                        Prob > chi2   = 0.0000
Log likelihood = -2082.3879                             Pseudo R2     = 0.0665

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
    mmarried |   -1.10666    .091279   -12.12   0.000    -1.285564   -.9277569
        mage |  -.0187894   .0082217    -2.29   0.022    -.0349036   -.0026751
   prenatal1 |  -.2721103   .0943067    -2.89   0.004    -.4569481   -.0872725
       fbaby |  -.5428134   .0864561    -6.28   0.000    -.7122642   -.3733625
       _cons |   .1297324   .2095812     0.62   0.536    -.2810391     .540504
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

.
. gen wgt1 = mbsmoke/ps

. gen wgt0 = (1-mbsmoke)/(1-ps)

. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1 [pweight=wgt1]
(sum of wgt is 4,620.9067401886)

Linear regression                               Number of obs     =        864
                                                F(4, 859)         =       3.12
                                                Prob > F          =     0.0145
                                                R-squared         =     0.0142
                                                Root MSE          =     563.01

------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   130.5978   39.90558     3.27   0.001     52.27399    208.9217
        mage |  -6.495147    4.95637    -1.31   0.190    -16.22316    3.232866
   prenatal1 |   23.29089   40.04843     0.58   0.561    -55.31335    101.8951
       fbaby |   45.19338   48.48447     0.93   0.352    -49.96852    140.3553
       _cons |   3206.397   126.7646    25.29   0.000     2957.592    3455.202
------------------------------------------------------------------------------

. predict mu1_ipwra
(option xb assumed; fitted values)

. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0 [pweight=wgt0]
(sum of wgt is 4,643.79376959801)

Linear regression                               Number of obs     =      3,778
                                                F(4, 3773)        =      26.89
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0336
                                                Root MSE          =     567.52
```

```
------------------------------------------------------------------------------
             |               Robust
     bweight | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   160.9954   26.87191     5.99   0.000     108.3105    213.6803
        mage |   3.061463   2.181356     1.40   0.161    -1.215289    7.338214
   prenatal1 |    62.4539   28.28456     2.21   0.027      6.99938    117.9084
       fbaby |  -67.38469   20.29791    -3.32   0.001    -107.1806   -27.58876
       _cons |   3188.522   56.19956    56.74   0.000     3078.338    3298.707
------------------------------------------------------------------------------

. predict mu0_ipwra
(option xb assumed; fitted values)

. gen tau_ipwra_i = mu1_ipwra - mu0_ipwra

. sum tau_ipwra_i

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
 tau_ipwra_i |      4,642   -238.7679    97.94189  -481.7334   6.216553
```

The mean of `tau_ipwra_i` matches `teffects ipwra`'s ATE exactly: -238.7679 in both.

Manual AIPW: estimate outcome models $\hat{\mu}_1(X)$, $\hat{\mu}_0(X)$ (unweighted, one per arm) and the propensity score $\hat{p}(X)$, then combine via the augmented-IPW formula

$$\hat{\tau}_i^{aipw} = D_i \frac{Y_i - \hat{\mu}_1(X_i)}{\hat{p}(X_i)} + \hat{\mu}_1(X_i) - (1 - D_i)\frac{Y_i - \hat{\mu}_0(X_i)}{1 - \hat{p}(X_i)} - \hat{\mu}_0(X_i) \tag{11.1}$$

```
clear
webuse cattaneo2
logit mbsmoke mmarried mage prenatal1 fbaby
predict ps

reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1
predict mu1_aipw
reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0
predict mu0_aipw
gen aipw_i = mbsmoke*(bweight-mu1_aipw)/ps + mu1_aipw - (1-mbsmoke)*(bweight-mu0_aipw)/(1-ps) - mu0_aipw
sum aipw_i
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)
```

```
. logit mbsmoke mmarried mage prenatal1 fbaby

Iteration 0:  Log likelihood = -2230.7484
Iteration 1:  Log likelihood = -2089.7838
Iteration 2:  Log likelihood = -2082.3958
Iteration 3:  Log likelihood = -2082.3879
Iteration 4:  Log likelihood = -2082.3879

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(4)    = 296.72
                                                        Prob > chi2   = 0.0000
Log likelihood = -2082.3879                             Pseudo R2     = 0.0665

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   -1.10666    .091279   -12.12   0.000    -1.285564   -.9277569
        mage |  -.0187894   .0082217    -2.29   0.022    -.0349036   -.0026751
   prenatal1 |  -.2721103   .0943067    -2.89   0.004    -.4569481   -.0872725
       fbaby |  -.5428134   .0864561    -6.28   0.000    -.7122642   -.3733625
       _cons |   .1297324   .2095812     0.62   0.536    -.2810391     .540504
------------------------------------------------------------------------------

. predict ps
(option pr assumed; Pr(mbsmoke))

.
. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==1

      Source |       SS           df       MS      Number of obs   =       864
-------------+----------------------------------   F(4, 859)       =      3.69
       Model |  4586519.21         4   1146629.8   Prob > F        =    0.0055
    Residual |   266914157       859  310726.609   R-squared       =    0.0169
-------------+----------------------------------   Adj R-squared   =    0.0123
       Total |   271500676       863  314601.015   Root MSE        =    557.43

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   133.6617   41.33023     3.23   0.001     52.54164    214.7818
        mage |  -7.370881   3.983409    -1.85   0.065    -15.18924    .4474735
   prenatal1 |   25.11133   43.64639     0.58   0.565    -60.55473    110.7774
       fbaby |   41.43991   41.82146     0.99   0.322     -40.6443    123.5241
       _cons |   3227.169   102.3392    31.53   0.000     3026.305    3428.033
------------------------------------------------------------------------------

. predict mu1_aipw
(option xb assumed; fitted values)
```

```
. reg bweight mmarried mage prenatal1 fbaby if mbsmoke==0

      Source |       SS           df       MS      Number of obs   =     3,778
-------------+----------------------------------   F(4, 3773)      =     30.00
       Model |    37916015         4  9479003.74   Prob > F        =    0.0000
    Residual |  1.1922e+09     3,773  315979.752   R-squared       =    0.0308
-------------+----------------------------------   Adj R-squared   =    0.0298
       Total |  1.2301e+09     3,777  325683.776   Root MSE        =    562.12

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |   160.9513   24.55483     6.55   0.000     112.8092    209.0933
        mage |   2.546828   1.935504     1.32   0.188    -1.247908    6.341564
   prenatal1 |   64.40859   25.83668     2.49   0.013     13.75338    115.0638
       fbaby |   -71.3286    19.4895    -3.66   0.000    -109.5396   -33.11763
       _cons |   3202.746   51.20076    62.55   0.000     3102.362     3303.13
------------------------------------------------------------------------------

. predict mu0_aipw
(option xb assumed; fitted values)

. gen aipw_i = mbsmoke*(bweight-mu1_aipw)/ps + mu1_aipw - (1-mbsmoke)*(bweight-
> mu0_aipw)/(1-ps) - mu0_aipw

. sum aipw_i

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
      aipw_i |      4,642   -239.0294     1618.64  -29389.38   16313.36
```

The mean of `aipw_i` is −239.0294, matching `teffects aipw`'s ATE to four decimals. (One might expect a small gap, since `teffects` estimates the treatment and outcome models jointly while this is a two-step plug-in; here the point estimates coincide, and only the standard errors would differ.)

The two doubly-robust estimators land within 0.3 grams of each other, −238.77 for IPWRA and −239.03 for AIPW, both about 8 grams away from plain IPW's −230.69.

The unit-level `aipw_i` values are worth a glance before trusting the mean. They have a standard deviation of 1,618.6 and run from −29,389 to +16,313 — two orders of magnitude wider than the estimate itself. That spread comes from dividing residuals by $\hat{p}$ or $1 - \hat{p}$ for units with extreme propensity scores. The average is well behaved, but it is an average over some very large individual terms, which is exactly why AIPW can be unstable when overlap is poor.

11.2. Matching

Matching does not solve the endogeneity problem. It addresses selection on observables only; with unobserved confounders it does not help.

What matching does is balance the covariate distributions across treatment and control groups. Regression does this too, but matching makes the lack of overlap visible — a plain regression can extrapolate far outside the support without warning. Running regression on a matched sample combines the strengths of both approaches.

Below we cover three matching methods in Stata: propensity score matching, nearest-neighbor matching, and coarsened exact matching.

11.2.1. Propensity score matching

Exact matching on all covariates is ideal but infeasible in high dimensions. Rosenbaum and Rubin (1983) showed that the propensity score provides a scalar sufficient statistic, reducing the matching problem to one dimension.

Stata's `teffects psmatch` and `psmatch2` both estimate the propensity score, match, and compute the treatment effect with correct standard errors in one step.

```
clear
webuse cattaneo2
teffects psmatch (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu), atet
psmatch2 mbsmoke mmarried c.mage##c.mage fbaby medu, logit ties
reg bweight mbsmoke   [aweight=_weight]
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. teffects psmatch (bweight) (mbsmoke mmarried c.mage##c.mage fbaby medu), atet

Treatment-effects estimation                   Number of obs      =      4,642
Estimator      : propensity-score matching     Matches: requested =          1
Outcome model  : matching                                  min =          1
Treatment model: logit                                     max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
```

```
 Nonsmoker)  |  -236.7848   26.57789    -8.91   0.000    -288.8765    -184.693
------------------------------------------------------------------------------

. psmatch2 mbsmoke mmarried c.mage##c.mage fbaby medu, logit ties

Logistic regression                                     Number of obs =  4,642
                                                        LR chi2(5)    = 375.00
                                                        Prob > chi2   = 0.0000
Log likelihood = -2043.2504                             Pseudo R2     = 0.0841

------------------------------------------------------------------------------
     mbsmoke | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
    mmarried |  -1.145706   .0918962   -12.47   0.000     -1.32582    -.965593
        mage |    .321518   .0638472     5.04   0.000     .1963798    .4466563
             |
      c.mage#|
      c.mage |  -.0060368   .0011849    -5.09   0.000    -.0083592   -.0037144
             |
       fbaby |  -.3864258   .0880445    -4.39   0.000    -.5589898   -.2138618
        medu |  -.1420833   .0173215    -8.20   0.000    -.1760328   -.1081338
       _cons |  -2.950915   .8102504    -3.64   0.000    -4.538976   -1.362853
------------------------------------------------------------------------------

. reg bweight mbsmoke   [aweight=_weight]
(sum of wgt is 1,728)

      Source |       SS           df       MS      Number of obs   =     3,671
-------------+----------------------------------   F(1, 3669)      =    151.87
       Model |    51455506         1    51455506   Prob > F        =    0.0000
    Residual |  1.2431e+09     3,669  338808.237   R-squared       =    0.0397
-------------+----------------------------------   Adj R-squared   =    0.0395
       Total |  1.2945e+09     3,670  352736.492   Root MSE        =    582.07

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -236.7848   19.21387   -12.32   0.000    -274.4557   -199.1138
       _cons |   3374.444   13.58626   248.37   0.000     3347.807    3401.082
------------------------------------------------------------------------------

.
```

`teffects psmatch` gives the ATT of smoking on birthweight, −236.7848 with a standard error of 26.58. The manual replication uses `psmatch2` with `logit` and `ties` (by default `teffects` includes all tied matches, while `psmatch2` keeps only one), then runs a weighted regression, and gives −236.7848 — the same to four decimals. Point estimates match; standard errors should be taken from `teffects`.

The nearest-neighbour ATT sits close to the two doubly-robust ATEs above, −238.8 and −239.0, though it is a different estimand on a different subpopulation.

11.2.2. Nearest neighbor matching

`teffects nnmatch` uses Mahalanobis distance by default and can also enforce exact matching on discrete covariates. It is nonparametric.

```
clear
webuse cattaneo2
teffects nnmatch  (bweight mmarried mage fbaby medu) (mbsmoke) , atet
teffects psmatch (bweight) (mbsmoke mmarried mage fbaby medu), atet
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. teffects nnmatch  (bweight mmarried mage fbaby medu) (mbsmoke) , atet

Treatment-effects estimation                   Number of obs      =      4,642
Estimator      : nearest-neighbor matching     Matches: requested =          1
Outcome model  : matching                                  min =          1
Distance metric: Mahalanobis                               max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |  -239.2433   25.68524    -9.31   0.000    -289.5854   -188.9011
------------------------------------------------------------------------------

. teffects psmatch (bweight) (mbsmoke mmarried mage fbaby medu), atet

Treatment-effects estimation                   Number of obs      =      4,642
Estimator      : propensity-score matching     Matches: requested =          1
Outcome model  : matching                                  min =          1
Treatment model: logit                                     max =         74
------------------------------------------------------------------------------
             |              AI robust
     bweight | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

```
ATET         |
     mbsmoke |
    (Smoker  |
         vs  |
 Nonsmoker)  |   -245.711   26.38675    -9.31   0.000    -297.4281   -193.9939
-------------------------------------------------------------------------------

.
```

Note the specification difference: `nnmatch` lists covariates in the outcome parentheses, `psmatch` lists them in the treatment parentheses.

11.2.3. Coarsened Exact Matching (CEM)

CEM is available as a separate Stata package.

```
clear
webuse cattaneo2
cem  mmarried mage fbaby medu, treatment(mbsmoke)
reg bweight mbsmoke   [aweight=cem_weights]
* Note: cem creates the variable "cem_weights" (plural). We spell it out
* in full here rather than relying on Stata's automatic variable-name
* abbreviation matching, to keep the reference unambiguous.
```

```
. clear

. webuse cattaneo2
(Excerpt from Cattaneo (2010) Journal of Econometrics 155: 138–154)

. cem  mmarried mage fbaby medu, treatment(mbsmoke)

Matching Summary:
-----------------
Number of strata: 274
Number of matched strata: 135

              0     1
      All  3778   864
  Matched  3421   827
Unmatched   357    37


Multivariate L1 distance: .25424705

Univariate imbalance:
```

```
              L1     mean      min      25%      50%      75%      max
mmarried  6.4e-15  8.4e-15        0        0        0        0        0
    mage   .04955  -.00887        1        0        1        0       -2
   fbaby  6.1e-15  6.1e-15        0        0        0        0        0
    medu   .03809  -.03316        0        0        0        0        0

. reg bweight mbsmoke   [aweight=cem_weights]
(sum of wgt is 4,248.00000000005)

      Source |       SS           df       MS      Number of obs   =     4,248
-------------+----------------------------------   F(1, 4246)      =    121.12
       Model |  40566772.9         1  40566772.9   Prob > F        =    0.0000
    Residual |  1.4221e+09     4,246  334928.257   R-squared       =    0.0277
-------------+----------------------------------   Adj R-squared   =    0.0275
       Total |  1.4627e+09     4,247   344401.26   Root MSE        =    578.73

------------------------------------------------------------------------------
     bweight | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     mbsmoke |  -246.8017   22.42533   -11.01   0.000    -290.7671   -202.8364
       _cons |   3383.394   9.894625   341.94   0.000     3363.996    3402.793
------------------------------------------------------------------------------

. * Note: cem creates the variable "cem_weights" (plural). We spell it out
. * in full here rather than relying on Stata's automatic variable-name
. * abbreviation matching, to keep the reference unambiguous.
.
```

CEM creates exact-match strata on coarsened covariates and generates weights for the final regression. The standard errors do not account for the matching step.

11.3. Modern matching in R: balance diagnostics and entropy weighting

The 2019 vintage of this chapter used Stata's `teffects` throughout. R's `MatchIt` and `WeightIt` packages have since become the standard for matching workflows, and `cobalt` provides publication-quality balance tables and plots that are now expected in applied papers.

11.3.1. Balance checking with cobalt

After any matching or weighting step, always inspect covariate balance. `cobalt`'s `bal.tab()` and `love.plot()` are the go-to tools:

```
library(MatchIt)
library(cobalt)

# Propensity score matching on the lalonde data
data("lalonde", package = "MatchIt")

m_out <- matchit(treat ~ age + educ + race + married + nodegree +
                   re74 + re75,
                 data = lalonde, method = "nearest", distance = "glm",
                 ratio = 1)

# Balance table: standardized mean differences before and after matching
bal.tab(m_out, stats = c("mean.diffs", "variance.ratios"), thresholds = c(m = 0.1))
```

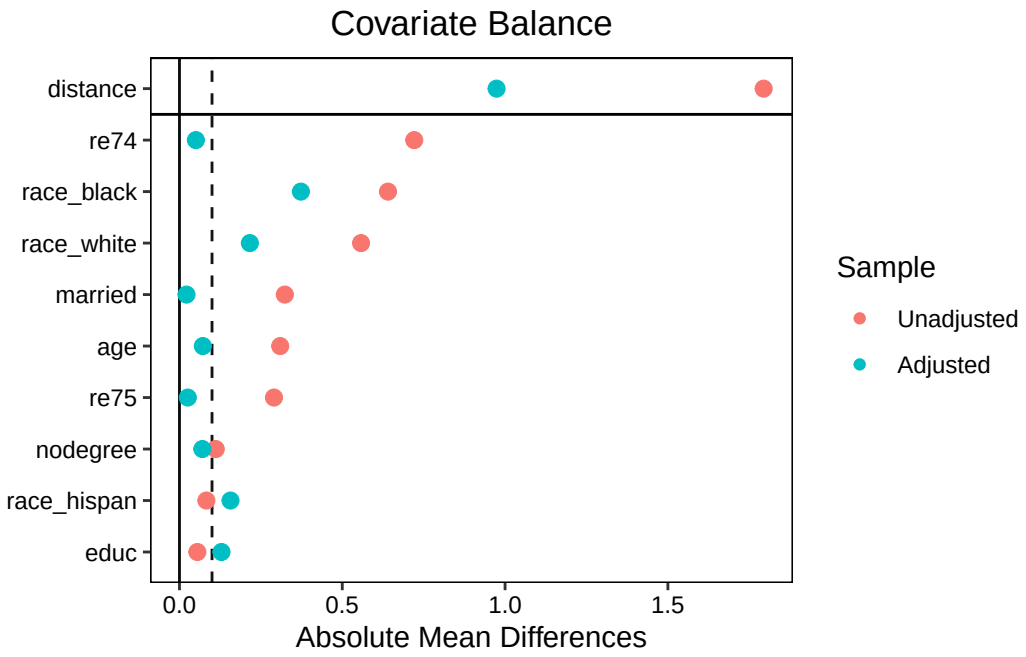
```
Balance Measures
                Type Diff.Adj        M.Threshold V.Ratio.Adj
distance    Distance   0.9739                          0.7566
age          Contin.   0.0718     Balanced, <0.1      0.4568
educ         Contin.  -0.1290 Not Balanced, >0.1      0.5721
race_black    Binary   0.3730 Not Balanced, >0.1           .
race_hispan   Binary  -0.1568 Not Balanced, >0.1           .
race_white    Binary  -0.2162 Not Balanced, >0.1           .
married       Binary  -0.0216     Balanced, <0.1           .
nodegree      Binary   0.0703     Balanced, <0.1           .
re74         Contin.  -0.0505     Balanced, <0.1      1.3289
re75         Contin.  -0.0257     Balanced, <0.1      1.4956

Balance tally for mean differences
                   count
Balanced, <0.1         5
Not Balanced, >0.1     4

Variable with the greatest mean difference
   Variable Diff.Adj        M.Threshold
 race_black    0.373 Not Balanced, >0.1

Sample sizes
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
```

```
# Love plot: visualize balance improvement
love.plot(m_out, stats = "mean.diffs", threshold = 0.1,
          abs = TRUE, var.order = "unadjusted")
```



The love plot shows each covariate's standardized mean difference (SMD) before matching (open circles) and after (filled circles). The vertical line at SMD = 0.1 is the conventional threshold for "balanced." If post-matching points stay to the right of the line, the matching has not achieved adequate balance on those covariates, even if it reduced the imbalance, and a different specification is needed.

That is what happens here, which is why it is worth running the diagnostic rather than assuming. Matching does reduce the largest imbalances: `re74` falls to −0.05 and the black indicator to 0.37, from unadjusted SMDs of −0.72 and 0.64 (printed in the matching chapter's balance table). It does not bring them all below 0.1. 1:1 nearest-neighbour matching on the propensity score leaves four of the nine covariates *above* the 0.1 threshold, `race` worst of all (an adjusted SMD of 0.37 for the black indicator), while discarding 244 of the 429 controls. Matching on a propensity score balances the score, not necessarily the covariates that went into it.

11.3.2. Entropy balancing

Propensity score matching discards unmatched units and can reduce effective sample size substantially. **Entropy balancing** (Hainmueller 2012) keeps all units and reweights the control group so that weighted covariate moments exactly match the treatment group — without discarding anyone.

```
library(WeightIt)
library(cobalt)

# Entropy balancing: exact balance on means, variances, and covariances
w_out <- weightit(treat ~ age + educ + race + married + nodegree +
                    re74 + re75,
                  data = lalonde, method = "ebal", estimand = "ATT",
                  moments = 1)   # moments=1: balance means only
```

```
                                # moments=2: balance means + variances

summary(w_out)
```

```
                 Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.           ||                     1.
control 0.008 |--------------------------| 4.062

- Units with the 5 most extreme weights by group:

            NSW5    NSW4    NSW3    NSW2    NSW1
 treated       1       1       1       1       1
         PSID423 PSID196 PSID412 PSID118 PSID226
 control   3.073   3.235    3.45   3.897   4.062

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.834 1.287   1.101       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     98.46     185
```

```
# Check balance
bal.tab(w_out, stats = "mean.diffs", thresholds = c(m = 0.1))
```

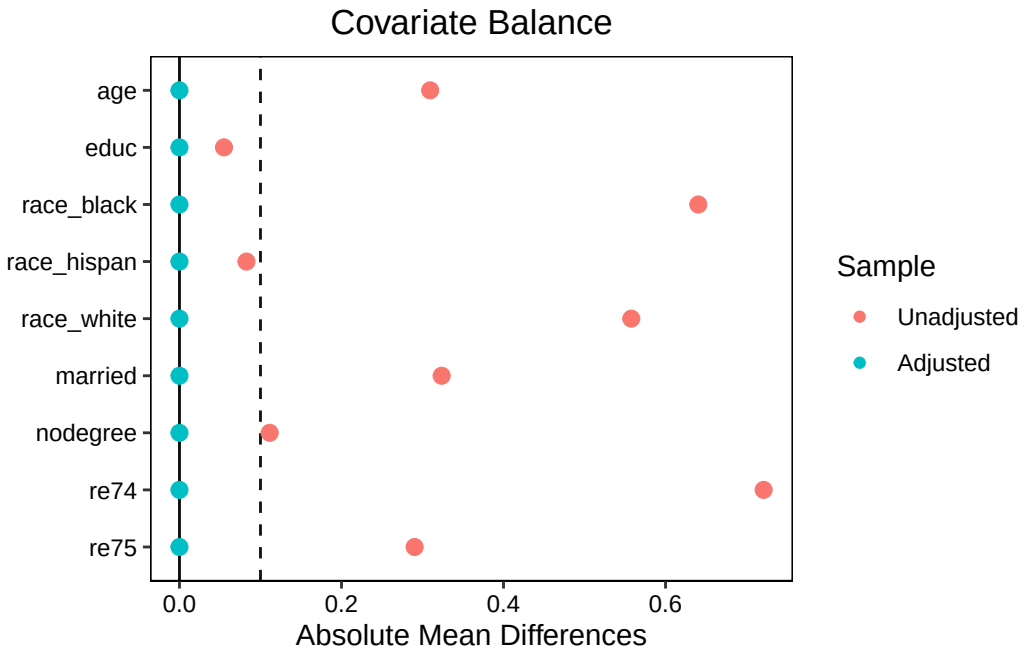
```
Balance Measures
               Type Diff.Adj    M.Threshold
age         Contin.       -0 Balanced, <0.1
educ        Contin.       -0 Balanced, <0.1
race_black   Binary        0 Balanced, <0.1
race_hispan  Binary        0 Balanced, <0.1
race_white   Binary       -0 Balanced, <0.1
married      Binary       -0 Balanced, <0.1
nodegree     Binary        0 Balanced, <0.1
re74        Contin.       -0 Balanced, <0.1
re75        Contin.       -0 Balanced, <0.1
```

```
Balance tally for mean differences
                    count
Balanced, <0.1          9
Not Balanced, >0.1      0

Variable with the greatest mean difference
 Variable Diff.Adj    M.Threshold
     re74       -0 Balanced, <0.1

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     98.46     185
```

```
love.plot(w_out, threshold = 0.1, abs = TRUE)
```




```
# Weighted outcome regression
library(lmtest); library(sandwich)
m_eb <- lm(re78 ~ treat + age + educ + race + married + nodegree +
             re74 + re75,
           data = lalonde, weights = w_out$weights)
coeftest(m_eb, vcov = vcovHC(m_eb, type = "HC3"))["treat", ]
```

```
    Estimate   Std. Error      t value     Pr(>|t|)
1273.2618139  820.1848528    1.5524083    0.1210883
```

Entropy balancing delivers on its promise here: the largest absolute standardized mean difference after weighting is **exactly zero**, since the weights are chosen to satisfy the mean-balance constraints as equalities. All 429 controls are retained.

Method	Pros	Cons
PSM (nearest neighbor)	Intuitive; respects overlap	Discards units; balance not guaranteed
CEM	Exact balance in bins	Requires discretizing continuous vars
Entropy balancing	Exact mean balance; keeps all units	Weights can be extreme; no variance balance by default
IPW (logistic)	Simple; relies only on the treatment model	Not doubly robust on its own; extreme weights if overlap is poor
AIPW / IPWRA	Doubly robust: consistent if *either* the treatment or outcome model is correct	Needs both models specified; more moving parts

For most applications, entropy balancing or IPW with trimmed weights (via `WeightIt`'s `trim()`) is preferred over PSM: better effective sample size, guaranteed balance, and easy integration with doubly-robust estimators.

See also the weighting chapter for IPW estimators and the OLS-ATE chapter for regression-based approaches.

11.4. Summary

Under conditional independence, Stata's `teffects` estimates treatment effects by regression adjustment, inverse probability weighting, and the doubly robust AIPW and IPWRA, which are consistent if either model is right. It also offers propensity-score and nearest-neighbour matching. Each estimator can be reproduced by hand. The point estimates match exactly, and `teffects` adds standard errors that account for the first-stage estimation. Matching only handles selection on observables. Its value is to make the lack of overlap visible.

In the birth-weight data, IPW gives −230.7 g, IPWRA −238.8 g and AIPW −239.0 g. The individual AIPW terms range over tens of thousands of grams, a warning about poor overlap. In R, 1:1 nearest-neighbour matching on the propensity score leaves four of nine covariates above the 0.1 balance threshold. Entropy balancing hits mean balance exactly and keeps every control.

Systematic treatment: R · Julia.

12. When is OLS coefficient the ATE

12.1. OLS cannot be ATE (most of the time)

Słoczyński (2022) clarifies when the OLS coefficient on a treatment dummy can and cannot be read as the ATE.

Consider the standard setup:

$$y = \alpha + \tau d + X\beta \tag{12.1}$$

Here d is the treatment dummy and X the covariates, under unconfoundedness. If the treatment effect is homogeneous, τ is the ATE. Under heterogeneous effects, Słoczyński shows that τ is a convex combination of ATT and ATU under an extra linearity condition, and that its weights tend to run opposite to the treatment shares: the more treated units there are, the less weight falls on ATT.

By definition,

$$\tau_{ATE} = \rho\tau_{ATT} + (1-\rho)\tau_{ATU} \tag{12.2}$$

Let $p(X)$ be the linear projection of d on X, the "linear propensity score", and let v_1 and v_0 be its variance among the treated and the untreated. The paper's main result writes the OLS coefficient as

$$\tau = w_1\tau_{ATT} + w_0\tau_{ATU}, \qquad w_1 = \frac{(1-\rho)\,v_0}{\rho\,v_1 + (1-\rho)\,v_0}, \quad w_0 = 1 - w_1. \tag{12.3}$$

Strictly, the theorem is about linear-projection analogues of ATT and ATU, built by regressing y on $p(X)$ within each group. They equal τ_{ATT} and τ_{ATU} when $E[y(1) \mid X]$ and $E[y(0) \mid X]$ are linear in $p(X)$. Unconfoundedness alone does not give that, and without it the OLS coefficient need not be any convex combination of ATT and ATU. When $v_1 \approx v_0$, $w_1 \approx 1-\rho$, which gives the reversed-share approximation

$$\tau \approx (1-\rho)\tau_{ATT} + \rho\tau_{ATU}.$$

where $\rho = P(d=1)$, the proportion of treated; $\tau_{ATT} = E(y(1) - y(0)|d=1)$, $\tau_{ATU} = E(y(1) - y(0)|d=0)$, and $\tau_{ATE} = E(y(1) - y(0))$.

Comparing Equation 12.2 with Equation 12.3, τ and τ_{ATE} differ unless w_1 happens to equal ρ or the effect does not vary. The weights depend on how the linear propensity score varies within each

group, not on the raw treated share alone, so a 50/50 treated share does not by itself make OLS recover the ATE. The reversed-share form is the case of equal variances; the more imbalanced the treatment probabilities, the further OLS can be from the ATE.

Given the true propensity score, ATT, ATU, and ATE can be separated. In practice we use the estimated propensity score. Słoczyński provides the `hettreatreg` package in R and Stata.

The data are `nswcps`: the NSW experimental treatment group combined with a large CPS comparison sample, so the treated are a tiny fraction of the whole. The outcome is 1978 earnings and the covariates are age, its square, education, race, marital status, degree status, and earnings in 1974 and 1975. This is the classic LaLonde setting, and its extreme treatment imbalance is exactly what makes the weighting result bite.

```
library(hettreatreg)
data("nswcps")

summary(lm(re78 ~ treated + age + age2 + educ + black + hispanic + married + nodegree  + re74 + re75, da
```

```
Call:
lm(formula = re78 ~ treated + age + age2 + educ + black + hispanic +
    married + nodegree + re74 + re75, data = nswcps)

Residuals:
   Min     1Q Median     3Q    Max
-25130  -3601   1274   3668  55040

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 7634.34415  736.67074  10.363  < 2e-16 ***
treated      793.58704  548.25433   1.447  0.14778
age         -233.67749   41.18067  -5.674 1.42e-08 ***
age2           1.81437    0.56099   3.234  0.00122 **
educ         166.84923   28.65984   5.822 5.94e-09 ***
black       -790.60856  213.24523  -3.708  0.00021 ***
hispanic    -175.97512  218.99126  -0.804  0.42166
married      224.26599  149.84542   1.497  0.13450
nodegree     311.84453  178.51743   1.747  0.08068 .
re74           0.29534    0.01222  24.175  < 2e-16 ***
re75           0.47064    0.01216  38.700  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7002 on 16166 degrees of freedom
Multiple R-squared:  0.4762,    Adjusted R-squared:  0.4758
F-statistic:  1469 on 10 and 16166 DF,  p-value: < 2.2e-16
```

```r
outcome <- nswcps$re78
treated <- nswcps$treated
our_vars <- c("age", "age2", "educ", "black", "hispanic", "married", "nodegree", "re74", "re75")
covariates <- subset(nswcps, select = our_vars)

hettreatreg(outcome, treated, covariates, verbose = TRUE)
```

```
"OLS" is the estimated regression coefficient on treated.

   OLS = 793.6

P(d=1) = 0.011
P(d=0) = 0.989

    w1 = 0.983
    w0 = 0.017
 delta = -0.971

   ATE = -6751
   ATT = 928.4
   ATU = -6840

OLS = w1*ATT + w0*ATU = 793.6
```

The output is the theorem in numbers. OLS puts 793.6 on `treated`. Only 1.1% of the sample is treated, $P(d = 1) = 0.011$, and the implied OLS weights are $w_1 = 0.983$ on ATT and $w_0 = 0.017$ on ATU. Compare those with the ATE's own weights, which are $\rho = 0.011$ on ATT and 0.989 on ATU: they are very nearly reversed. The package prints the identity directly, OLS $= w_1$ ATT $+ w_0$ ATU $= 793.6$, which closes exactly.

The consequences are not subtle. ATT is 928.4, ATU is -6840, and ATE is -6751. So the OLS coefficient of 793.6 sits within 135 of the ATT and roughly 7,500 away from the ATE — and on the *opposite side of zero*. Reading 793.6 as "the effect of training on earnings" would get the sign wrong for the population. The `delta` of -0.971 that the package reports is the gap between the OLS weight on ATT and ρ; it is close to -1, its extreme value.

Słoczyński's paper itself is an algebraic decomposition of the OLS coefficient τ in terms of ATT/ATU weights; it does not propose or endorse the propensity-score-as-covariate estimator below as a general-purpose way to recover ATT/ATU/ATE. What follows is our own illustrative exercise, done in three steps, to see how such an estimator would behave in practice: First, estimate the propensity score equation, basically

$$d = X\gamma \tag{12.4}$$

Second, include the predicted propensity score as a covariate and estimate the model with treatment and control group separately. Note that regressing Y linearly on the scalar propensity score $p(X)$ assumes $E[Y(d) \mid p(X)]$ is linear in $p(X)$ — a strong and generally untested assumption, unlike matching/weighting directly on the propensity score, or regression adjustment on the covariates X themselves, which are the standard approaches. Treat the numbers below as an exploratory illustration rather than a recommended estimator.

Third, predict the counterfactuals for the full sample. Then we get the ATE, ATT and ATU.

One more reason to treat this as illustration rather than method: the propensity score here comes from a linear probability model, and on these data it runs from -0.034 to 0.154 — some fitted "probabilities" are negative. That is harmless when the score is used merely as a regressor, as it is here, but it would be fatal if the same score were used as an inverse-probability weight.

Let's see with the same data:

```
m_propensity <- lm(treated ~ age + age2 + educ + black + hispanic + married + nodegree  + re74 + re75, da
ps <- predict(m_propensity)
df_combined <- data.frame(nswcps, ps)
df.1 <- df_combined[which(treated == 1),]
df.0 <- df_combined[which(treated == 0),]
m_ot <- lm(re78 ~ ps, data = df.1)
ot <- suppressWarnings(predict(m_ot, newdata = df_combined)) # add newdata option to predict for full sar

m_oc <- lm(re78 ~ ps, data = df.0)
oc <- suppressWarnings(predict(m_oc, newdata = df_combined)) # add newdata option to predict for full sar

te <- ot - oc
ate <- mean(te)
df_combined$te <- te
att <- as.numeric(mean(df_combined[which(treated == 1),'te']))
atu <- as.numeric(mean(df_combined[which(treated == 0),'te']))

print(paste("ate=", signif(ate, 4)))
```

```
[1] "ate= -6751"
```

```
print(paste("att=", signif(att, 4)))
```

```
[1] "att= 928.4"
```

```
print(paste("atu=", signif(atu, 4)))
```

```
[1] "atu= -6840"
```

This reproduces the package's decomposition exactly: ATE −6751, ATT 928.4, ATU −6840, the same three numbers to four significant figures.

We see in this case OLS coefficient is hugely different from ATE. This is because $P(d = 1)$ is only .01, about 1 percent; we have much larger control group than treatment group. In this case, the OLS coefficient is far different from ATE; in fact, it's pretty close to ATT.

Now, what's the intuition that OLS coefficient is just the opposite as the ATE in terms of weighting ATT and ATU? The reason is that OLS is to predict actual outcome, not counterfactual outcome. For calculating ATE, we need counterfactual outcome predicted. Now suppose we have a lot of control observations, we'd much better predicting $(y(0)|d == 0)$ rather than $(y(1)|d == 1)$. That is, we are better getting the slope of $(y(0)|d == 0, X = x)$; therefore predicting $(y(0)|d == 1, X = x)$. Therefore although treatment group has less observations, but because the $(y(0)|d == 1)$ is better predicted, the ATT is better estimated, thus more weight. If the opposite happens, then the ATU should be more heavily weighted. Thus the counter-intuitive weighting.

12.2. Regression adjustment

Słoczyński's method is to include propensity score as a covariate. This propensity score serves as a proxy for all covariates. As Rosenbaum and Rubin (1983) showed, the propensity score is a balancing score (treatment is independent of the covariates given the propensity score). Another way to allow treatment effect to differ across all covariates is to allow interaction of treatment dummy with all covariates, then I am doing "Regression Adjustment" (see Stata's `teffects ra`, used in the treatment effects and matching chapter; note this is not `treatreg`, which was the old endogenous-treatment command, superseded by `etregress`). We can calculate ATE, ATT, ATU after the linear model.

```
library(tidyverse)
lm1 <- lm(re78 ~ treated*(age + age2 +  educ + black + hispanic + married + nodegree  + re74 + re75), da
y1 <- predict(lm1,newdata=nswcps %>% mutate(treated=1))
y0 <- predict(lm1,newdata=nswcps %>% mutate(treated=0))
ate=mean(y1-y0) #ate
print(paste("ate=",  signif(ate, 4)))
```

```
[1] "ate= -4930"
```

```
y1.att <- predict(lm1,newdata=nswcps %>% filter(treated==1))
y0.att <- predict(lm1,newdata=nswcps %>% filter(treated==1) %>% mutate(treated=0))
att=mean(y1.att)-mean(y0.att) #att
print(paste('att=',  signif(att, 4)))
```

```
[1] "att= 796"
```

```
y1.atu <- predict(lm1,newdata=nswcps %>% filter(treated==0) %>% mutate(treated=1))
y0.atu <- predict(lm1,newdata=nswcps %>% filter(treated==0))
atu=mean(y1.atu)-mean(y0.atu) #atu
print(paste('atu=', signif(atu, 4)))
```

```
[1] "atu= -4996"
```

Regression adjustment gives ATE −4930, ATT 796 and ATU −4996.

We see this is somewhat different from Słoczyński's method, which gave −6751, 928.4 and −6840: the two agree on sign and order of magnitude but not on level, unsurprisingly, since one regresses on a scalar propensity score and the other interacts the treatment with all nine covariates. But the point is the original OLS coefficient 794 is nowhere close to ATE under either approach — −4930 or −6751, both far away and both negative. In this case, it's very close to ATT (796 by regression adjustment, 928 by the propensity-score route), as the treatment group is very small.

12.3. Summary

Under unconfoundedness with heterogeneous effects, and when the potential-outcome means are linear in the linear propensity score, the OLS coefficient on a treatment dummy is a weighted average of the ATT and the ATU. Słoczyński shows that the weights are roughly the reverse of the ones that define the ATE: OLS puts weight near the control share on the ATT. So the OLS coefficient is close to the ATE only when the groups are about equal in size, or when the effect does not vary.

In the NSW-CPS data only 1.1% of the sample is treated. OLS gives 793.6, with 98% of its weight on the ATT of 928.4. The ATE is −6751, on the other side of zero. Regression adjustment with full interactions gives a different level, an ATE of −4930 and an ATT of 796, but the same conclusion: the OLS coefficient is close to the ATT and nowhere near the ATE.

Systematic treatment: R · Julia.

13. Matching and Weighting Part 1: matching

This chapter follows R's `MatchIt` and `WeightIt` packages (Ho et al. 2011; Greifer, n.d.).

13.1. Assumptions

Matching and weighting require three assumptions for causal identification in observational studies: (1) SUTVA, (2) ignorability (unconfoundedness), and (3) overlap (positivity). Under ignorability, conditioning on X makes treatment assignment D independent of the potential outcomes $Y(0)$ and $Y(1)$. In practice, the distribution of X is rarely balanced across treatment and control groups — matching and weighting restore that balance.

13.2. Distance

Matching finds control units close to treated units in X-space. The first question is how to define "close."

13.2.1. Propensity score

The most common distance measure is the propensity score (PS), which is the probability of being treated given X. The benefit is that when there is high dimension of X, we can reduce the dimension to a single number, the PS.

There are many ways to estimate the propensity score, such as logit, or more flexible parametric "gam", or other machine learning methods, such as "gbm", "lasso", "rpart", "randomforest", "cbps", etc. Here CBPS has seen more popularity. The propensity scores are estimated using the covariate balancing propensity score (CBPS) algorithm, which is a form of logistic regression where balance constraints are incorporated into a generalized method of moments estimation of the model coefficients.

13.2.2. Distance from covariates

Or we can compute distance from covariates directly, not using propensity score. The distance can be computed using Mahalanobis distance, or Euclidean distance, or other distance measures.

13.3. Matching methods

With a distance, we then decide which method to use. The first few matching methods all need to specify a distance measure. They are matching based on distance. Then there are stratum matching methods, which do not need a distance measure. The last two methods are subsetting methods.

13.3.1. Nearest neighbor matching

NNM is widely used; it is also known as greedy matching. It matches each treated unit to the closest control unit in terms of distance. The closest control unit is defined as the one with the smallest distance to the treated unit. If there are multiple control units with the same distance and we want a 1:1 match, one is randomly selected.

13.3.2. Optimal pair matching

NNM is not optimal in the sense that it does not necessarily minimize the total distance between matched pairs. Optimal pair matching is a method that minimizes the total distance (or some other overall criterion) between matched pairs. It is more computationally intensive than NNM, but it can lead to better balance.

13.3.3. Optimal full matching

OFM assigns each unit (treated or control) to a subclass. Within each subclass it computes the sum of distances, and it chooses the subclasses to minimize the total distance across all of them. It is more computationally intensive than NNM and optimal pair matching. Weights are used based on subclass membership. Because all units are included, it can be used to estimate ATE.

13.3.4. Generalized Full matching

GFM is a variant of OFM. It uses a faster algorithm for large datasets.

13.3.5. Genetic matching

Genetic matching (`method = "genetic"`, Diamond and Sekhon (2013)) runs nearest neighbor matching on a *scaled* generalized Mahalanobis distance, where the scaling weights on the covariates are not fixed in advance but chosen by a genetic algorithm that searches for the weighting giving the best covariate balance. That search is the point of the method — and also why it is far slower than the alternatives above.

13.3.6. Exact matching

It's a stratum matching. First subclasses are defined, by combinations of covariate values. Then units are assigned to subclasses. When a subclass has only either control or treated units, it is discarded.

Exact matching is nonparametric, but it works only when covariates are discrete and it happens that there are enough treated and control units in each subclass.

13.3.7. Coarsened exact matching

CEM is a more popular stratum matching. First bins are created after coarsening the covariates. then use exact matching on the covariate bins. CEM is nonparametric. User needs to decide how many bins to create for each covariate.

13.3.8. Subclassification

Subclassification is another kind of stratum matching. It uses the propensity score to define bins, usually quantiles of PS in the treated group, or control group or overall, depending on target estimand.

13.3.9. Cardinality and profile matching

Cardinality matching involves finding the largest sample that satisfies user-supplied balance constraints and constraints on the ratio of matched treated to matched control units. Profile matching involves identifying a target distribution (e.g., the full sample for the ATE or the treated units for the ATT) and finding the largest subset of the treated and control groups that satisfy user-supplied balance constraints with respect to that target.

13.4. Example

Here is an example with Lalonde data, we are interested in the effect of treatment on "re78" (1978 real earnings). There are 614 observations, 185 treated and 429 control, with covariates age, education, race, marital status, degree status, and earnings in 1974 and 1975. The first `matchit` call uses `method = NULL`, which fits the propensity score but performs no matching — a way of getting the pre-match balance table.

```
library("MatchIt")
data("lalonde")

head(lalonde)
```

```
     treat age educ   race married nodegree re74 re75       re78
NSW1     1  37   11  black       1        1    0    0  9930.0460
NSW2     1  22    9 hispan       0        1    0    0  3595.8940
NSW3     1  30   12  black       0        0    0    0 24909.4500
NSW4     1  27   11  black       0        1    0    0  7506.1460
NSW5     1  33    8  black       0        1    0    0   289.7899
NSW6     1  22    9  black       0        1    0    0  4056.4940
```

```
# No matching; constructing a pre-match matchit object
m.out0 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  method = NULL,
                  distance = "glm")
summary(m.out0)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, method = NULL, distance = "glm")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5774        0.1822          1.7941     0.9211    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6444
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Sample Sizes:
          Control Treated
```

```
All           429     185
Matched       429     185
Unmatched       0       0
Discarded       0       0
```

The imbalance is severe. Standardized mean differences are −0.72 for 1974 earnings, 1.76 for the black indicator, −1.88 for white, −0.83 for married and −0.31 for age, and the propensity score itself differs by 1.79 standard deviations. The treated average $2,096 in 1974 earnings against $5,619 for controls, and 84% are black against 20%. These are not two samples that differ at the margin.

Or use "cobalt" to see the balance:

```
library("cobalt")
bal.tab(treat ~ age + educ + race + married + nodegree + re74 + re75,
        data = lalonde, estimand = "ATT", thresholds = c(m = .05))
```

```
Balance Measures
               Type Diff.Un      M.Threshold.Un
age         Contin. -0.3094 Not Balanced, >0.05
educ        Contin.  0.0550 Not Balanced, >0.05
race_black   Binary  0.6404 Not Balanced, >0.05
race_hispan  Binary -0.0827 Not Balanced, >0.05
race_white   Binary -0.5577 Not Balanced, >0.05
married      Binary -0.3236 Not Balanced, >0.05
nodegree     Binary  0.1114 Not Balanced, >0.05
re74        Contin. -0.7211 Not Balanced, >0.05
re75        Contin. -0.2903 Not Balanced, >0.05

Balance tally for mean differences
                    count
Balanced, <0.05         0
Not Balanced, >0.05     9

Variable with the greatest mean difference
 Variable Diff.Un      M.Threshold.Un
     re74 -0.7211 Not Balanced, >0.05

Sample sizes
    Control Treated
All     429     185
```

`bal.tab` says the same thing against a stricter 0.05 threshold: all nine covariate contrasts fail, with `re74` the worst at −0.7211.

First do a PS match. The chunks below run several distance measures in turn — the logit of a logistic propensity score, a GAM score with smooth terms in age and education, CBPS targeting the ATC with replacement, Mahalanobis distance on the raw covariates, and full matching on a

probit score. Each prints its own balance table; what changes across them is the `distance` row and how much of the covariate imbalance survives.

```
# Matching on logit of a PS estimated with logistic
# regression:
m.out1 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "glm",
                  link = "linear.logit")
summary(m.out1)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "glm", link = "linear.logit")

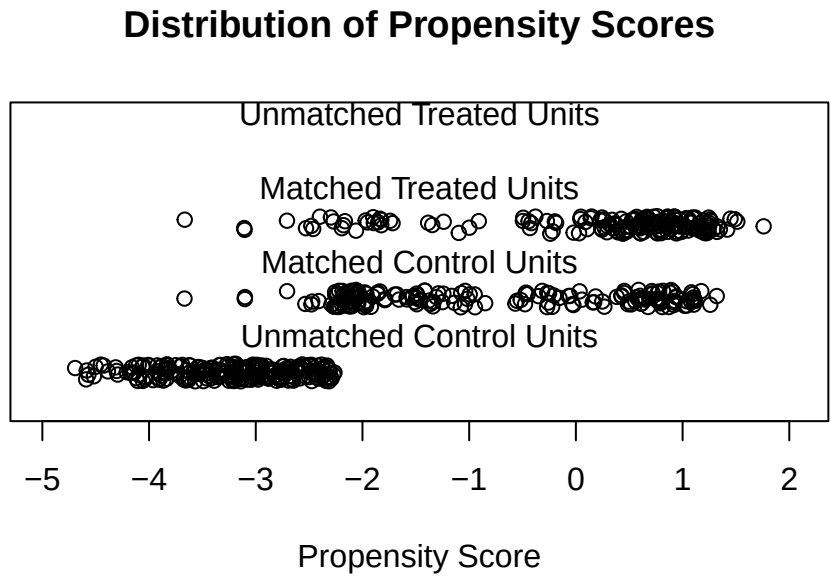
Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2657       -2.1500          2.1426     0.5398    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6444
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2657       -0.7706          0.9192     0.7712    0.1321
age              25.8162       25.3027          0.0718     0.4568    0.0847
educ             10.3459       10.6054         -0.1290     0.5721    0.0239
raceblack         0.8432        0.4703          1.0259          .    0.3730
```

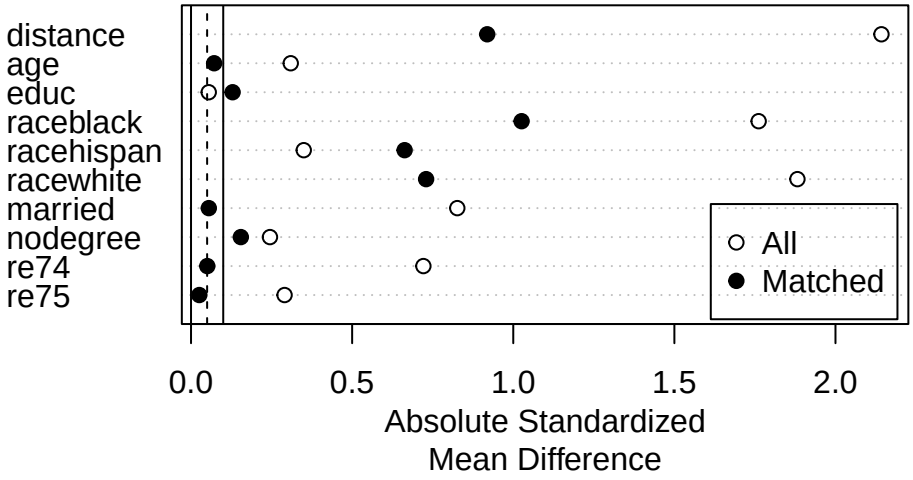
```
racehispan        0.0595        0.2162         -0.6629          .    0.1568
racewhite         0.0973        0.3135         -0.7296          .    0.2162
married           0.1892        0.2108         -0.0552          .    0.0216
nodegree          0.7081        0.6378          0.1546          .    0.0703
re74           2095.5737     2342.1076         -0.0505     1.3289    0.0469
re75           1532.0553     1614.7451         -0.0257     1.4956    0.0452
           eCDF Max Std. Pair Dist.
distance     0.4216          0.9193
age          0.2541          1.3938
educ         0.0757          1.2474
raceblack    0.3730          1.0259
racehispan   0.1568          1.0743
racewhite    0.2162          0.8390
married      0.0216          0.8281
nodegree     0.0703          1.0106
re74         0.2757          0.7965
re75         0.2054          0.7381

Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

```
plot(m.out1, type = "jitter", interactive = FALSE)
```



```
plot(summary(m.out1))
```



We see there are still some imbalance on variables such as "raceblack". We can use more flexible models for the PS.

```
# GAM logistic PS with smoothing splines (s()):
m.out2 <- matchit(treat ~ s(age) + s(educ) +
                    race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "gam")
summary(m.out2)
```

```
Call:
matchit(formula = treat ~ s(age) + s(educ) + race + married +
    nodegree + re74 + re75, data = lalonde, distance = "gam")

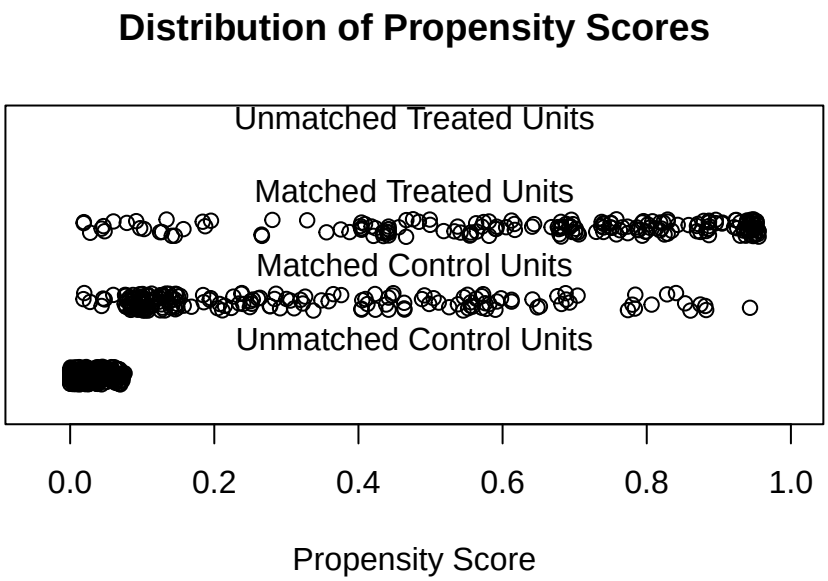
Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.6566        0.1481          1.9123     1.6161    0.4155
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
```

```
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
           eCDF Max
distance     0.7172
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876
age          0.1577
educ         0.1114

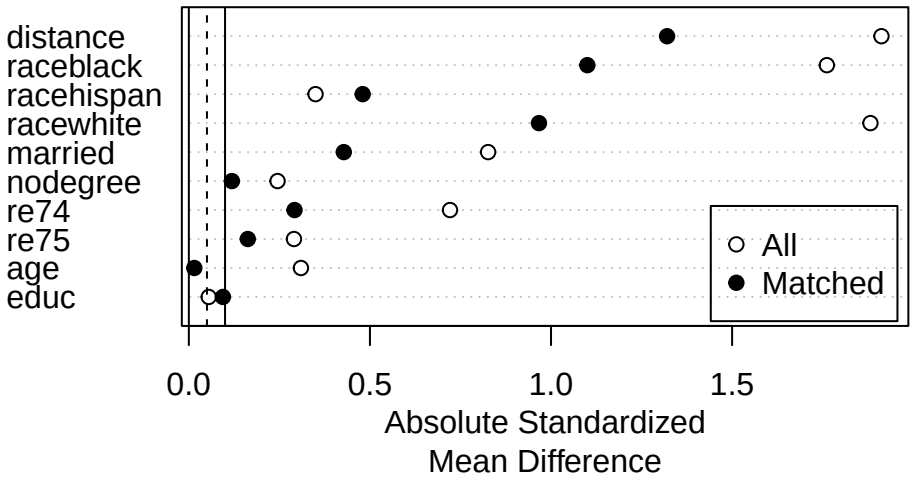
Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.6566        0.3055          1.3204     1.2324    0.1853
raceblack         0.8432        0.4432          1.1002          .    0.4000
racehispan        0.0595        0.1730         -0.4800          .    0.1135
racewhite         0.0973        0.3838         -0.9667          .    0.2865
married           0.1892        0.3568         -0.4278          .    0.1676
nodegree          0.7081        0.6541          0.1189          .    0.0541
re74           2095.5737     3521.7556         -0.2919     1.0512    0.1346
re75           1532.0553     2056.2000         -0.1628     1.2120    0.0917
age              25.8162       25.7081          0.0151     0.7958    0.0270
educ             10.3459       10.1568          0.0941     0.6186    0.0270
           eCDF Max Std. Pair Dist.
distance     0.5189          1.3204
raceblack    0.4000          1.1002
racehispan   0.1135          0.8914
racewhite    0.2865          1.1491
married      0.1676          1.0627
nodegree     0.0541          0.9749
re74         0.4270          0.8894
re75         0.2378          0.8453
age          0.0865          1.1619
educ         0.0919          1.2447

Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

```
plot(m.out2, type = "jitter", interactive = FALSE)
```




```
plot(summary(m.out2))
```



Or use CBPS.

```
# CBPS for ATC matching w/replacement, using the just-
# identified version of CBPS (setting method = "exact"):
m.out3 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "cbps",
                  estimand = "ATC",
                  distance.options = list(method = "exact"),
                  replace = TRUE)
summary(m.out3)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "cbps", distance.options = list(method = "exact"),
    estimand = "ATC", replace = TRUE)

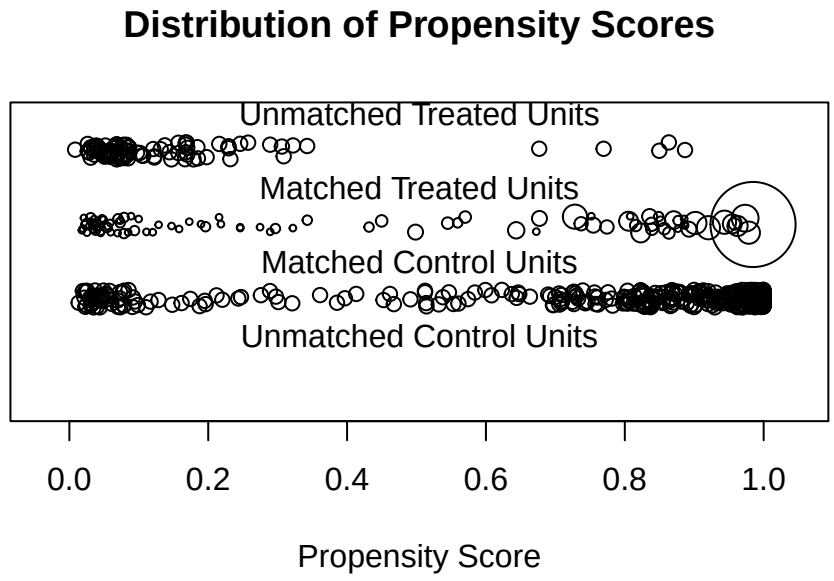
Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.2801        0.7758         -1.5464     0.9710    0.3525
age              25.8162       28.0303         -0.2053     0.4400    0.0813
educ             10.3459       10.2354          0.0387     0.4959    0.0347
raceblack         0.8432        0.2028          1.5928          .    0.6404
racehispan        0.0595        0.1422         -0.2369          .    0.0827
racewhite         0.0973        0.6550         -1.1732          .    0.5577
married           0.1892        0.5128         -0.6475          .    0.3236
nodegree          0.7081        0.5967          0.2270          .    0.1114
re74           2095.5737     5619.2365         -0.5190     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2838     0.9563    0.1342
           eCDF Max
distance     0.5967
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.7720        0.7758         -0.0117     1.1568    0.0499
age              27.3520       28.0303         -0.0629     0.4383    0.1091
educ             11.0117       10.2354          0.2719     0.7323    0.0566
raceblack         0.3100        0.2028          0.2667          .    0.1072
```

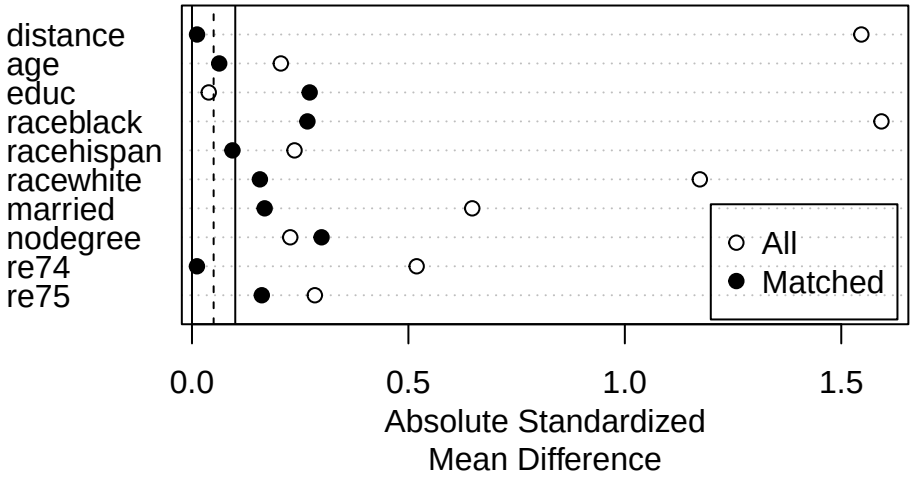
```
racehispan        0.1096        0.1422         -0.0934          .     0.0326
racewhite         0.5804        0.6550         -0.1569          .     0.0746
married           0.5967        0.5128          0.1679          .     0.0839
nodegree          0.4499        0.5967         -0.2994          .     0.1469
re74           5696.9438     5619.2365          0.0114     0.8562     0.0994
re75           1935.6090     2466.4844         -0.1613     1.5194     0.1071
           eCDF Max Std. Pair Dist.
distance     0.3543          0.0194
age          0.2191          0.8676
educ         0.2727          1.0719
raceblack    0.1072          0.3826
racehispan   0.0326          0.6407
racewhite    0.0746          0.5492
married      0.0839          0.5596
nodegree     0.1469          1.0692
re74         0.2541          0.7513
re75         0.2727          0.9781

Sample Sizes:
              Control Treated
All               429  185.
Matched (ESS)     429    6.67
Matched           429   93.
Unmatched           0   92.
Discarded           0    0.
```

```
plot(m.out3, type = "jitter", interactive = FALSE)
```



```
plot(summary(m.out3))
```




```
# Mahalanobis distance matching - no PS estimated
m.out4 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  distance = "mahalanobis")
summary(m.out4)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, distance = "mahalanobis")

Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
age          0.1577
```

```
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

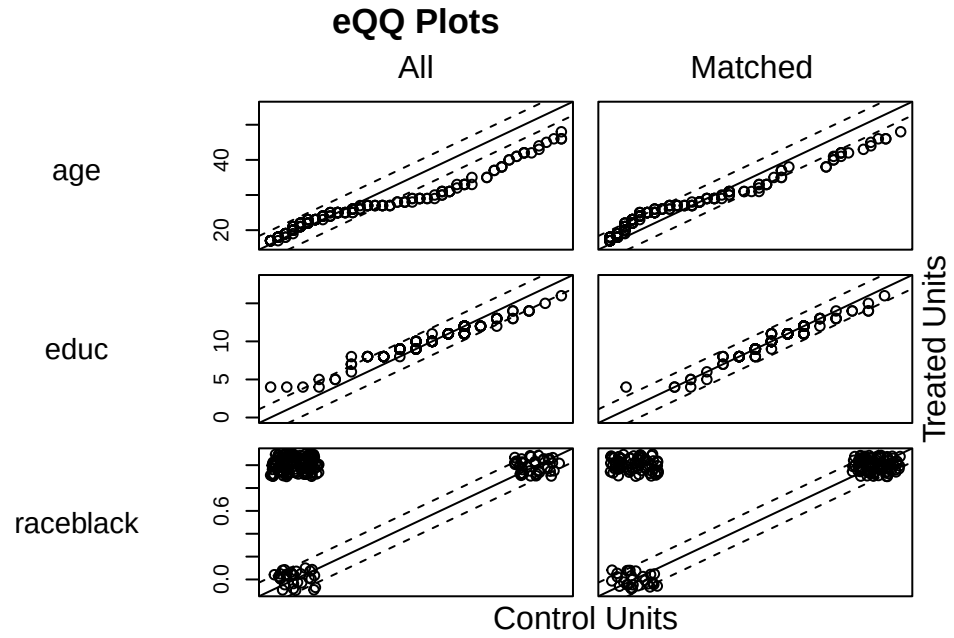
Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
age              25.8162       24.9081          0.1269     0.5327    0.0695
educ             10.3459       10.4324         -0.0430     0.8345    0.0108
raceblack         0.8432        0.4649          1.0407          .    0.3784
racehispan        0.0595        0.0595          0.0000          .    0.0000
racewhite         0.0973        0.4757         -1.2767          .    0.3784
married           0.1892        0.2486         -0.1518          .    0.0595
nodegree          0.7081        0.6595          0.1070          .    0.0486
re74           2095.5737     3305.4420         -0.2476     0.8705    0.1012
re75           1532.0553     1957.5097         -0.1322     1.0071    0.0654
           eCDF Max Std. Pair Dist.
age          0.2378          0.6195
educ         0.0486          0.2903
raceblack    0.3784          1.0407
racehispan   0.0000          0.0000
racewhite    0.3784          1.2767
married      0.0595          0.1794
nodegree     0.0486          0.1070
re74         0.3351          0.4546
re75         0.2162          0.4113

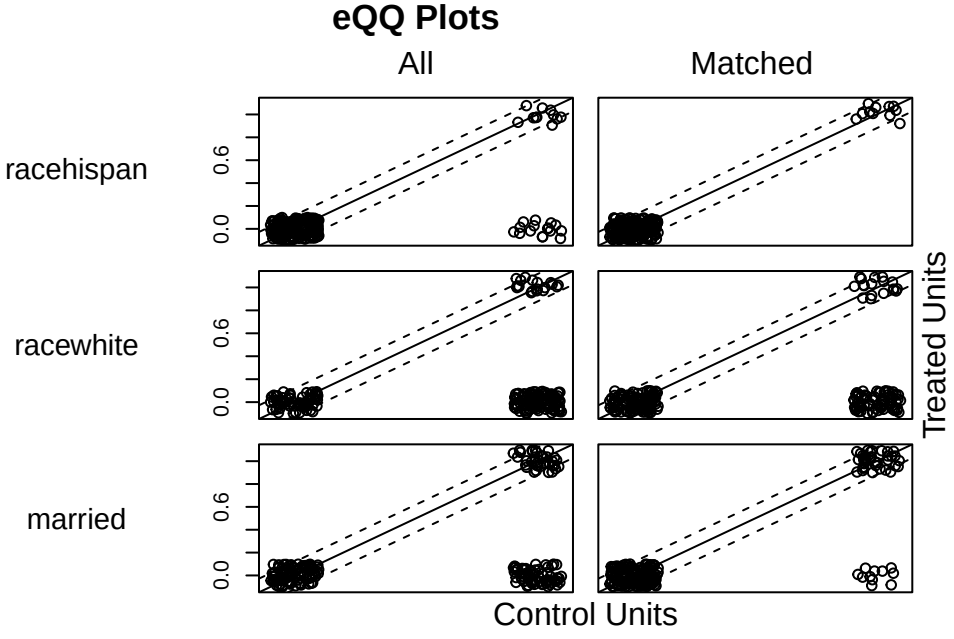
Sample Sizes:
          Control Treated
All           429     185
Matched       185     185
Unmatched     244       0
Discarded       0       0
```

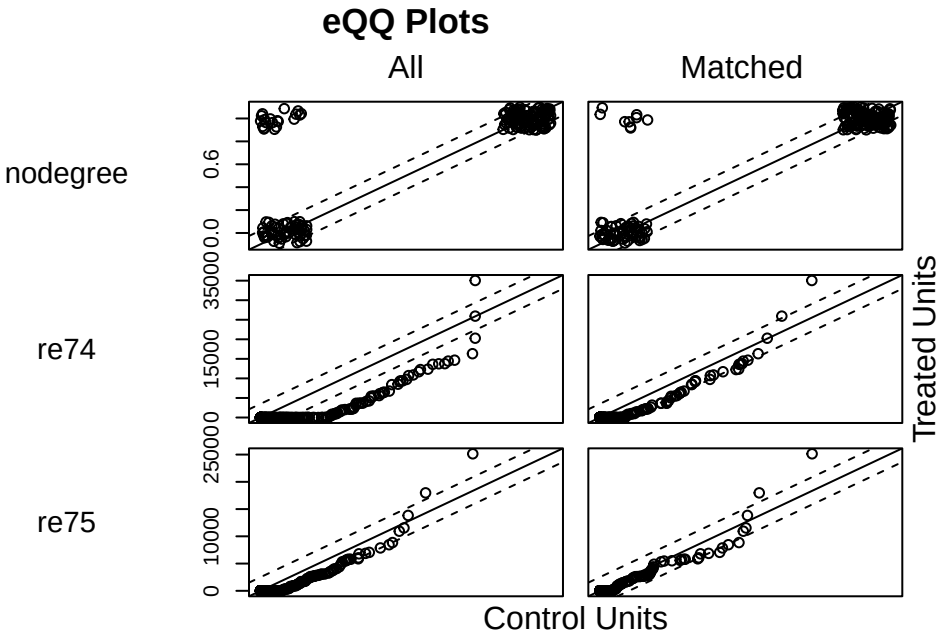
```
m.out4$distance #NULL
```

```
NULL
```

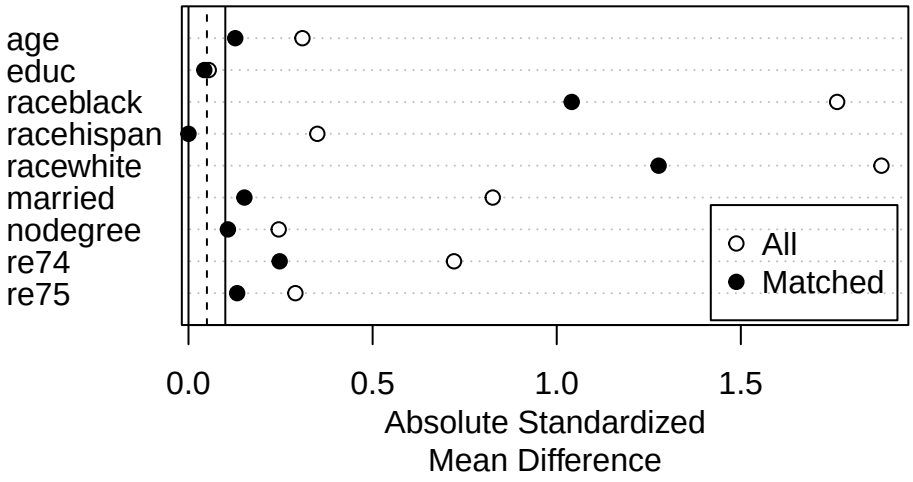
```
plot(m.out4,  interactive = FALSE)
```


eQQ Plots
All
Matched
age
educ
raceblack
20
40
0
5
10
0.0
0.6
Treated Units
Control Units


eQQ Plots
All
Matched
racehispan
racewhite
married
0.0
0.6
Treated Units
Control Units




```
plot(summary(m.out4))
```




```
# Full matching on a probit PS
m.out5 <- matchit(treat ~ age + educ + race + married +
                    nodegree + re74 + re75,
                  data = lalonde,
                  method = "full",
                  distance = "glm",
```

```
                  link = "probit")

summary(m.out5)
```

```
Call:
matchit(formula = treat ~ age + educ + race + married + nodegree +
    re74 + re75, data = lalonde, method = "full", distance = "glm",
    link = "probit")

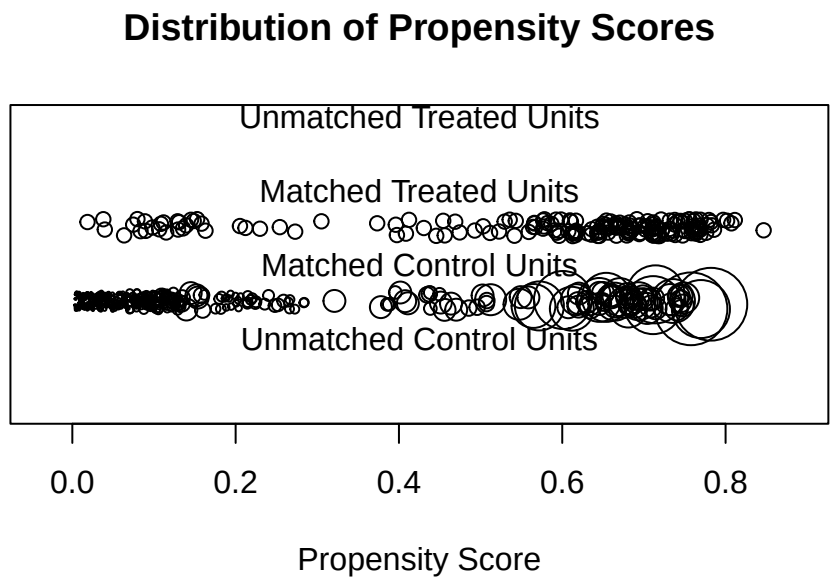
Summary of Balance for All Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5773        0.1817          1.8276     0.8777    0.3774
age              25.8162       28.0303         -0.3094     0.4400    0.0813
educ             10.3459       10.2354          0.0550     0.4959    0.0347
raceblack         0.8432        0.2028          1.7615          .    0.6404
racehispan        0.0595        0.1422         -0.3498          .    0.0827
racewhite         0.0973        0.6550         -1.8819          .    0.5577
married           0.1892        0.5128         -0.8263          .    0.3236
nodegree          0.7081        0.5967          0.2450          .    0.1114
re74           2095.5737     5619.2365         -0.7211     0.5181    0.2248
re75           1532.0553     2466.4844         -0.2903     0.9563    0.1342
           eCDF Max
distance     0.6413
age          0.1577
educ         0.1114
raceblack    0.6404
racehispan   0.0827
racewhite    0.5577
married      0.3236
nodegree     0.1114
re74         0.4470
re75         0.2876

Summary of Balance for Matched Data:
           Means Treated Means Control Std. Mean Diff. Var. Ratio eCDF Mean
distance          0.5773        0.5764          0.0045     0.9949    0.0043
age              25.8162       25.5347          0.0393     0.4790    0.0787
educ             10.3459       10.5381         -0.0956     0.6192    0.0253
raceblack         0.8432        0.8389          0.0119          .    0.0043
racehispan        0.0595        0.0492          0.0435          .    0.0103
racewhite         0.0973        0.1119         -0.0493          .    0.0146
married           0.1892        0.1633          0.0660          .    0.0259
nodegree          0.7081        0.6577          0.1110          .    0.0504
re74           2095.5737     2100.2150         -0.0009     1.3467    0.0314
re75           1532.0553     1561.4420         -0.0091     1.5906    0.0536
           eCDF Max Std. Pair Dist.
```

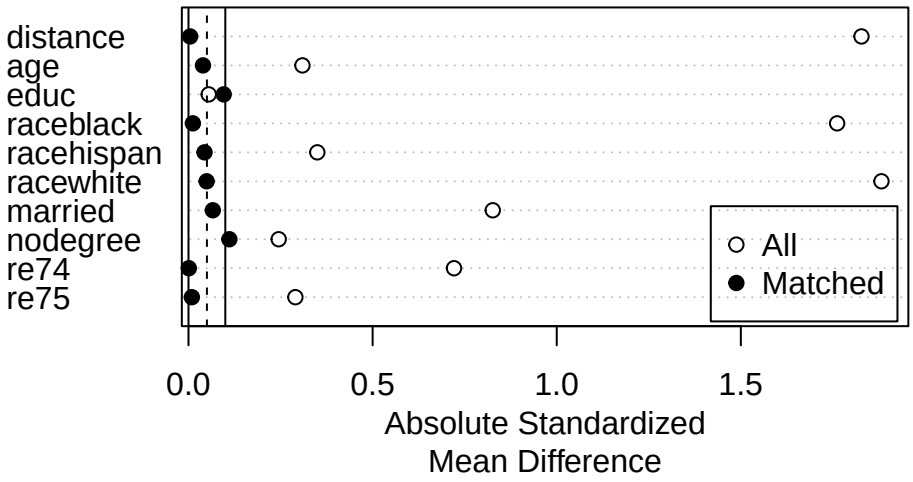
```
distance     0.0486          0.0198
age          0.2742          1.2843
educ         0.0730          1.2179
raceblack    0.0043          0.0162
racehispan   0.0103          0.4412
racewhite    0.0146          0.3454
married      0.0259          0.4473
nodegree     0.0504          0.9872
re74         0.1881          0.8387
re75         0.1984          0.8240

Sample Sizes:
              Control Treated
All            429.       185
Matched (ESS)   50.76     185
Matched        429.       185
Unmatched        0.         0
Discarded        0.         0
```

```
plot(m.out5, type = "jitter", interactive = FALSE)
```



```
plot(summary(m.out5))
```



Note that `m.out4$distance` prints `NULL`: Mahalanobis matching estimates no propensity score at all, so there is no distance column and no `distance` row in its balance table. That is the point of the method — it works on the covariate space directly rather than collapsing it to a scalar.

13.5. Estimation

After we get a matched data that we think is good enough, we can estimate the treatment effect. We use `m.out5`, the full match on a probit score.

```
m.data <- match_data(m.out5)

head(m.data)
```

```
     treat age educ   race married nodegree re74 re75       re78  distance
NSW1     1  37   11  black       1        1    0    0  9930.0460 0.6356769
NSW2     1  22    9 hispan       0        1    0    0  3595.8940 0.2298151
NSW3     1  30   12  black       0        0    0    0 24909.4500 0.6813558
NSW4     1  27   11  black       0        1    0    0  7506.1460 0.7690590
NSW5     1  33    8  black       0        1    0    0   289.7899 0.6954138
NSW6     1  22    9  black       0        1    0    0  4056.4940 0.6943658
     weights subclass
NSW1       1        1
NSW2       1       55
NSW3       1       63
```

```
NSW4       1       70
NSW5       1       79
NSW6       1       86
```

What "match_data" does is to grab the matched data with a few additional variables such as "weights", "subclass", and "distance". Then we can use it in a simple linear model with weights.

```
library("marginaleffects")

fit <- lm(re78 ~ treat * (age + educ + race + married +
                            nodegree + re74 + re75),
          data = m.data,
          weights = weights)

avg_comparisons(fit,
                variables = "treat",
                vcov = ~subclass,
                newdata = subset(m.data, treat == 1))
```

```
 Estimate Std. Error    z Pr(>|z|)   S 2.5 % 97.5 %
     1977        704 2.81  0.00501 7.6   596   3357

Term: treat
Type: response
Comparison: 1 - 0
```

The ATT is \$1,977 with a subclass-clustered standard error of \$704, $p = 0.005$, and a 95% interval of $[596, 3357]$. Training raised 1978 earnings for those who took it.

Note we allow interaction of treatment and covariates, but we do not need to demean the covariates here, since we are relying on "marginaleffects" and its "avg_comparisons" function to compute the average treatment effect. The "newdata" argument is used to specify the effect, in this case ATT. Note the weights are generated this way: since it's ATT, all treated units are weighted by 1, and control units are weighted by $p/(1-p)$, where p is proportion of treated units in the subclass. Anyway, for full matching, we get the comparison of potential outcomes for the treated, with weights, and clustered standard errors on subclasses, because that is the level at which the comparison is conducted.

13.6. Summary

Matching rests on SUTVA, unconfoundedness and overlap. It pairs treated and control units that are close in covariates. Closeness can be measured by a propensity score, estimated by logit, a GAM, CBPS or machine learning, or directly by a covariate distance such as Mahalanobis. The method then sets how units are paired: nearest neighbor, optimal pair, full and genetic matching

use a distance, while exact, coarsened exact and subclassification match within strata. The test of a match is covariate balance, checked with `cobalt`.

In the Lalonde data the raw groups are far apart, with standardized differences up to 1.88. After full matching on a probit score, a weighted regression with treatment-covariate interactions and `avg_comparisons` gives an ATT of $1,977 (704), clustered by subclass. Training raised 1978 earnings for those who took it.

Systematic treatment: R · Julia.

14. Matching and Weighting - Part 2: weighting

This chapter follows R's `WeightIt` package (Greifer, n.d.). The assumptions — SUTVA, ignorability, and overlap — are the same as in the matching chapter.

14.1. Weighting methods

Weighting creates a pseudo-population in which X is balanced across treatment and control. The most common approach is inverse probability of treatment weighting (IPTW), which uses the propensity score. Several alternatives are gaining traction.

14.1.1. Inverse probability of treatment weighting (IPTW)

IPTW is the most common weighting method. It uses the inverse of the propensity score as weights. For the **ATE**, weight each treated unit by $1/p$ and each control unit by $1/(1-p)$, where p is the propensity score. For the **ATT** — which is what the examples below request via `estimand = "ATT"` — the treated units all get weight 1 and the controls get $p/(1-p)$; you can see this in the weight summaries, where the treated range is exactly 1 to 1. Either way the point is the same: create a pseudo population in which the distribution of X is balanced between the groups.

14.1.2. Covariate balancing propensity score (CBPS)

The same idea, but with a different way to estimate the propensity score. CBPS (Imai and Ratkovic 2014) is a form of logistic regression where balance constraints are incorporated into a generalized method of moments estimation of the model coefficients.

14.1.3. Entropy balancing

Entropy balancing involves the specification of an optimization problem, the solution to which is then used to compute the weights. The constraints of the primal optimization problem correspond to covariate balance on the means (for binary and multi-category treatments) or treatment-covariate covariances (for continuous treatments), positivity of the weights, and that the weights sum to a certain value. Entropy balancing is doubly robust (for the ATT) in the sense that it is consistent either when the true propensity score model is a logistic regression of the treatment on the covariates or when the true outcome model for the control units is a linear regression of the outcome on the covariates, and it attains a semi-parametric efficiency bound when both are true. Entropy balancing will always yield exact mean balance on the included terms (not necessarily on the distribution of the covariates).

14.1.4. Energy balancing

Energy balancing is a method of estimating weights using optimization without a propensity score.

The primary benefit of energy balancing is that all features of the covariate distribution are balanced, not just means, as with other optimization-based methods like entropy balancing.

14.1.5. SuperLearner

SuperLearner works by fitting several machine learning models to the treatment and covariates and then taking a weighted combination of the generated predicted values to use as the propensity scores, which are then used to construct weights.

14.2. Example

Here is an example with Lalonde data, we are interested in the effect of treatment on "re78" (real earnings 1978).

14.2.1. IPW

```
library("WeightIt")
library(cobalt)
data(lalonde)
w.out1 <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                   data = lalonde, estimand = "ATT", method = "glm")
w.out1 #print the output
```

```
A weightit object
 - method: "glm" (propensity score weighting with GLM)
 - number of obs.: 614
 - sampling weights: none
 - treatment: 2-category
 - estimand: ATT (focal: 1)
 - covariates: age, educ, race, married, nodegree, re74, re75
```

```
summary(w.out1)
```

```
                  Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.           ||                  1.
```

```
control 0.009 |--------------------------| 3.743

- Units with the 5 most extreme weights by group:

            5     4    3     2     1
 treated    1     1    1     1     1
          597   573  381   411   303
 control 3.03 3.059 3.24 3.523 3.743

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.818 1.289   1.098       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     99.82     185
```

The treated weights are all exactly 1, as the ATT requires, and the control weights run from 0.009 to 3.743. The cost is in the effective sample size: 429 controls carry the information of 99.8 equally weighted ones, a loss of more than three quarters.

```r
library(cobalt)
bal.tab(w.out1, stats = c("m", "v"), thresholds = c(m = .05))
```

```
Balance Measures
               Type Diff.Adj         M.Threshold V.Ratio.Adj
prop.score  Distance  -0.0205     Balanced, <0.05      1.0324
age          Contin.   0.1188 Not Balanced, >0.05      0.4578
educ         Contin.  -0.0284     Balanced, <0.05      0.6636
race_black    Binary  -0.0022     Balanced, <0.05           .
race_hispan   Binary   0.0002     Balanced, <0.05           .
race_white    Binary   0.0021     Balanced, <0.05           .
married       Binary   0.0186     Balanced, <0.05           .
nodegree      Binary   0.0184     Balanced, <0.05           .
re74         Contin.  -0.0021     Balanced, <0.05      1.3206
re75         Contin.   0.0110     Balanced, <0.05      1.3938

Balance tally for mean differences
                    count
Balanced, <0.05         9
Not Balanced, >0.05     1

Variable with the greatest mean difference
```

```
 Variable Diff.Adj         M.Threshold
      age   0.1188 Not Balanced, >0.05

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     99.82     185
```

Eight of the nine covariates now pass the strict 0.05 threshold, against none before weighting (the tenth row, `prop.score`, is the distance measure itself). `age` is the exception at 0.1188. Note also the variance ratios, which IPTW does not target: `age` sits at 0.458 and `re74` at 1.321, so the *means* are balanced while the *spreads* are not.

14.2.2. Entropy balancing

```
w.out2 <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                   data = lalonde, estimand = "ATT", method = "ebal")
summary(w.out2)
```

```
                  Summary of weights

- Weight ranges:

          Min                                  Max
treated 1.           ||                     1.
control 0.008 |--------------------------| 4.062

- Units with the 5 most extreme weights by group:

             5     4    3     2     1
 treated     1     1    1     1     1
           608   381  597   303   411
 control 3.073 3.235 3.45 3.897 4.062

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       0.    0.      0.          0
control       1.834 1.287   1.101       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.       185
Weighted     98.46     185
```

```r
bal.tab(w.out2, stats = c("m", "v"), thresholds = c(m = .05))
```

```
Balance Measures
               Type Diff.Adj     M.Threshold V.Ratio.Adj
age         Contin.       -0 Balanced, <0.05      0.4096
educ        Contin.       -0 Balanced, <0.05      0.6635
race_black   Binary        0 Balanced, <0.05           .
race_hispan  Binary        0 Balanced, <0.05           .
race_white   Binary       -0 Balanced, <0.05           .
married      Binary       -0 Balanced, <0.05           .
nodegree     Binary        0 Balanced, <0.05           .
re74        Contin.       -0 Balanced, <0.05      1.3265
re75        Contin.       -0 Balanced, <0.05      1.3351

Balance tally for mean differences
                    count
Balanced, <0.05         9
Not Balanced, >0.05     0

Variable with the greatest mean difference
 Variable Diff.Adj     M.Threshold
     re74       -0 Balanced, <0.05

Effective sample sizes
           Control Treated
Unadjusted  429.       185
Adjusted     98.46     185
```

Every mean difference is now zero to three decimals — that is the constraint entropy balancing imposes directly, rather than hoping a propensity model delivers it — and all nine covariates pass. The effective sample size is 98.46, essentially the same as IPTW's 99.82, so the exact balance came at no extra cost in precision here. The variance ratios are unchanged in character (`age` 0.410, `re74` 1.327): entropy balancing constrains means, not distributions.

Suppose we are satisfied with the balance, we can use the weights to estimate the treatment effect. We can use the "marginaleffects" package to do that.

```r
# Fit outcome model
fit <- lm_weightit(re78 ~ treat * (age + educ + race + married +
                                     nodegree + re74 + re75),
                   data = lalonde, weightit = w.out2)
# G-computation for the treatment effect
library("marginaleffects")
avg_comparisons(fit, variables = "treat",
                newdata = subset(lalonde, treat == 1))
```

```
 Estimate Std. Error    z Pr(>|z|)   S 2.5 % 97.5 %
     1273        770 1.65   0.0983 3.3  -236   2783

Term: treat
Type: probs
Comparison: 1 - 0
```

The entropy-balanced ATT is \$1,273 with a standard error of \$770, $p = 0.098$, and a 95% interval of $[-236, 2783]$ that includes zero.

14.2.3. Energy balancing

```r
w.out3 <- tryCatch(
  weightit(treat ~ age + educ + race + married +
                   nodegree + re74 + re75, data = lalonde,
                 method = "energy", estimand = "ATE"),
  error = function(e) { message("Energy balancing failed: ", e$message); NULL }
)
if (!is.null(w.out3)) summary(w.out3)
```

```
                  Summary of weights

- Weight ranges:

        Min                                  Max
treated   0 |--------------------------| 14.041
control   0 |---------------------|       11.065

- Units with the 5 most extreme weights by group:

            10   179   125    137    124
 treated 5.984  6.23 6.758 13.202 14.041
           537   553   573    420    608
 control 5.499 6.404 6.599  8.274 11.065

- Weight statistics:

        Coef of Var   MAD Entropy # Zeros
treated       1.821 1.054   0.92        0
control       1.264 0.866   0.579       0

- Effective Sample Sizes:

           Control Treated
Unweighted  429.    185.
Weighted    165.42   43.04
```

```
if (!is.null(w.out3)) {
  fit <- lm_weightit(re78 ~ treat * (age + educ + race + married +
                                       nodegree + re74 + re75),
                     data = lalonde, weightit = w.out3)
  # G-computation for the treatment effect
  avg_comparisons(fit, variables = "treat")
}
```

```
 Estimate Std. Error       z Pr(>|z|)   S 2.5 % 97.5 %
      -73       1085 -0.0672    0.946 0.1 -2200   2054

Term: treat
Type: probs
Comparison: 1 - 0
```

Two things are worth noticing in that output rather than passing over it. First, the estimand changed: `w.out3` requests the **ATE**, while the entropy-balancing fit above targeted the **ATT**, so the two numbers (−73 against 1273) are not competing estimates of the same quantity. Second, look at the weight summary. Entropy balancing gives every treated unit a weight of exactly 1, as the ATT requires. Energy balancing spreads the treated weights from 0 to 14.0 — one treated unit is effectively discarded and another counts as fourteen — and the resulting standard error (1085) is much larger than entropy balancing's (770). Balancing the whole covariate *distribution* rather than just its means is a genuine advantage, but it makes the weights more variable, and that shows up directly in the precision of the estimate.

14.3. Comparing matching and weighting

Matching and weighting are popular in different fields. We used to say matching involves discarding data therefore changing the estimand, but now with full matching and other methods, matching can be done without losing data. We used to believe weighting can be unstable, as weights can be highly variable, but with CBPS, entropy balancing and energy balancing, weighting can perform much better.

So, in practice, we can try different methods, either matching or weighting, and see which one gives us better balance. Then use "marginaleffects" to get the treatment effect.

It is worth putting numbers on that, because on this data the choice changes the conclusion. Full matching in the previous chapter gave an ATT of \$1,977 with a standard error of \$704 and $p = 0.005$. Entropy balancing above gives \$1,273 with \$770 and $p = 0.098$. The two point estimates are about one standard error apart, and one interval excludes zero while the other does not.

Note which way that cuts. Entropy balancing achieves *exact* mean balance, while full matching leaves `educ` at −0.096; the better-balanced method gives the smaller and less significant estimate. Better balance on means is not the same thing as a more credible answer, and reporting whichever method happened to cross the significance line would be the wrong response to this. The honest summary reports both and says the data do not settle it.

14.4. Summary

Weighting builds a pseudo-population in which the covariates are balanced across treatment and control. IPTW weights by the inverse propensity score. CBPS estimates the score with balance constraints. Entropy balancing solves directly for weights that match covariate means exactly, and energy balancing matches the whole covariate distribution. SuperLearner estimates the score by an ensemble. The checks are balance and effective sample size.

In the Lalonde data, IPTW balances nine of ten means but leaves the variance ratios off, and 429 controls carry the information of about 100. Entropy balancing matches every mean exactly with about the same effective sample size, and gives an ATT of $1,273 (770). Energy balancing targets the ATE, with highly variable weights and a much larger standard error. Full matching gave $1,977 (704). The two ATTs are about one standard error apart, and only one excludes zero, so we report both and say the data do not settle it.

Systematic treatment: R · Julia.

15. Sensitivity analysis of OLS with sensemakr

15.1. OVB

Consider a true model,

$$Y = \delta D + X\beta + \gamma Z + \epsilon \tag{15.1}$$

If Z is unobserved we estimate

$$Y = \delta D + X\beta + u \tag{15.2}$$

and face omitted variable bias (OVB). Cinelli and Hazlett (2020) show that

$$|\widehat{\text{bias}}| = \sqrt{\frac{R^2_{Y\sim Z|D,X} R^2_{D\sim Z|X}}{1 - R^2_{D\sim Z|X}}} \left(\frac{\hat{\sigma}_{y.dx}}{\hat{\sigma}_{d.x}}\right) \tag{15.3}$$

Bias is large only when the omitted variable is strongly associated with both Y and D. The Cinelli and Hazlett framework yields three key quantities.

15.2. Partial R^2 of treatment on outcome

This measures how much variation in the outcome the treatment explains after partialling out X. In the extreme case where the confounder explains all remaining outcome variance ($R^2_{Y\sim Z|D,X} = 1$), the treatment effect goes to zero only if $R^2_{D\sim Z|X} \geq R^2_{Y\sim D|X}$. In other words, the confounder's association with treatment must exceed the treatment's partial association with the outcome.

15.3. Robust value

$RV_{q,\alpha}$ is the minimum strength of association the confounder must have with both treatment and outcome for a $(1-\alpha)$ confidence interval to include a $q\%$ change from the current estimate. With $q = 1$ this gives the threshold for the effect to include zero. Both the Z-D and Z-Y associations must exceed this threshold simultaneously.

15.4. Bounds using observed covariate(s)

The first two quantities are dimensionless but hard to interpret without context. Benchmarking against an observed covariate helps. If a covariate X_j is known to be an important predictor, we can ask how much stronger the confounder must be relative to X_j to drive the effect to zero.

$$k_D := \frac{R^2_{D\sim Z|X_{-j}}}{R^2_{D\sim X_j|X_{-j}}} \tag{15.4}$$

where X_{-j} represents all covariates other than the jth we pick as the benchmark covariate.

$$k_Y := \frac{R^2_{Y\sim Z|D,X_{-j}}}{R^2_{Y\sim X_j|D,X_{-j}}} \tag{15.5}$$

15.5. Example

Here is an example from sensemakr documentation.

```
# loads package
library(sensemakr)

# loads dataset
data("darfur")

# runs regression model
model <- lm(peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
                          pastvoted + hhsize_darfur + female + village, data = darfur)

# runs sensemakr for sensitivity analysis
sensitivity <- sensemakr(model = model,
                          treatment = "directlyharmed",
                          benchmark_covariates = "female",
                          kd = 1:3)

# short description of results
sensitivity
```

```
Sensitivity Analysis to Unobserved Confounding

Model Formula: peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
    pastvoted + hhsize_darfur + female + village

Null hypothesis: q = 1 and reduce = TRUE
```

```
Unadjusted Estimates of ' directlyharmed ':
  Coef. estimate: 0.09732
  Standard Error: 0.02326
  t-value: 4.18445

Sensitivity Statistics:
  Partial R2 of treatment with outcome: 0.02187
  Robustness Value, q = 1 : 0.13878
  Robustness Value, q = 1 alpha = 0.05 : 0.07626

For more information, check summary.
```

```r
# long description of results
summary(sensitivity)
```

```
Sensitivity Analysis to Unobserved Confounding

Model Formula: peacefactor ~ directlyharmed + age + farmer_dar + herder_dar +
    pastvoted + hhsize_darfur + female + village

Null hypothesis: q = 1 and reduce = TRUE
-- This means we are considering biases that reduce the absolute value of the current estimate.
-- The null hypothesis deemed problematic is H0:tau = 0

Unadjusted Estimates of 'directlyharmed':
  Coef. estimate: 0.0973
  Standard Error: 0.0233
  t-value (H0:tau = 0): 4.1844

Sensitivity Statistics:
  Partial R2 of treatment with outcome: 0.0219
  Robustness Value, q = 1: 0.1388
  Robustness Value, q = 1, alpha = 0.05: 0.0763

Verbal interpretation of sensitivity statistics:

-- Partial R2 of the treatment with the outcome: an extreme confounder (orthogonal to the covariates) tha

-- Robustness Value, q = 1: unobserved confounders (orthogonal to the covariates) that explain more than

-- Robustness Value, q = 1, alpha = 0.05: unobserved confounders (orthogonal to the covariates) that expl

Bounds on omitted variable bias:

--The table below shows the maximum strength of unobserved confounders with association with the treatmen

 Bound Label R2dz.x R2yz.dx      Treatment Adjusted Estimate Adjusted Se
```

```
   1x female 0.0092  0.1246 directlyharmed            0.0752      0.0219
   2x female 0.0183  0.2493 directlyharmed            0.0529      0.0204
   3x female 0.0275  0.3741 directlyharmed            0.0304      0.0187
 Adjusted T Adjusted Lower CI Adjusted Upper CI
     3.4389            0.0323            0.1182
     2.6002            0.0130            0.0929
     1.6281           -0.0063            0.0670
```

```
# minimal reporting
ovb_minimal_reporting(sensitivity,
                      format = if (knitr::is_latex_output()) "latex" else "html")
```

Outcome: *peacefactor*

Treatment:	Est.	S.E.	t-value	$R^2_{Y\sim D\|\mathbf{X}}$	$RV_{q=1}$	$RV_{q=1,\alpha=0.05}$
directlyharmed	0.097	0.023	4.184	2.2%	13.9%	7.6%
df = 783		*Bound (1x female)*: $R^2_{Y\sim Z\|\mathbf{X},D}$ = 12.5%, $R^2_{D\sim Z\|\mathbf{X}}$ = 0.9%				

Let's see how to interpret this.

$RV_{q=1}$ = 13.9%. This means that unobserved confounders that explain 13.9% of the residual variance both of the treatment and of the outcome are sufficiently strong to explain away all the observed effect. On the other hand, unobserved confounders that do not explain at least 13.9% of the residual variance both of the treatment and of the outcome are not sufficiently strong to do so.

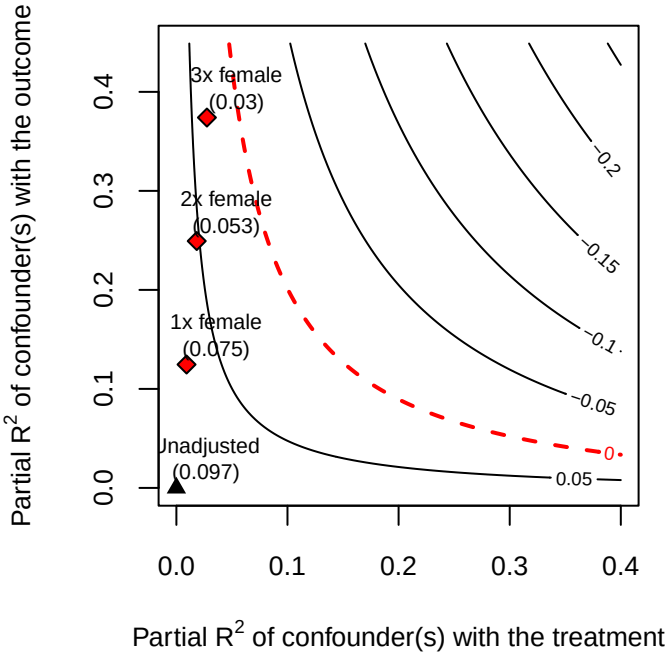
The robustness value for testing the null hypothesis that the coefficient of "directlyharmed" is zero ($RV_{q=1,\alpha=0.05}$) is 7.6%. This means that unobserved confounders that explain 7.6% of the residual variance both of the treatment and of the outcome are sufficiently strong to bring the lower bound of the confidence interval to zero (at the chosen significance level of 5%).

The partial R^2 of `directlyharmed` with `peacefactor` is 2.2%. This means that, in an extreme scenario in which we assume that unobserved confounders explain all of the left-out variance of the outcome, these unobserved confounders would need to explain at least 2.2% of the residual variance of the treatment to fully explain away the observed effect.

The above numbers are for the absolute strength the confounder needs to be to explain away the treatment effect. The authors believe in this context, "female" is an important factor. Suppose the confounder is as important as "female", what would the partial R^2 be? It is shown that $R^2_{Y\sim Z|X,D}$ = 12.5% and $R^2_{D\sim Z|X}$ = 0.9%, so a confounder as strong as "female" is not strong enough to explain away the treatment effect.

It is worth being careful about *why*, because comparing each number to $RV_{q=1}$ = 13.9% one at a time invites the wrong reading. The RV is defined for a confounder of *equal* strength on both sides, so it is not a threshold that each partial R^2 must separately clear. Here the outcome-side association (12.5%) is in fact close to the RV, which on its own might look like a near miss — but the treatment-side association is only 0.9%, more than an order of magnitude short. It is the treatment side that does the work: "female" predicts the outcome nearly well enough to matter, and barely predicts the treatment at all. The contour plot below is the honest check, since it evaluates the pair jointly rather than either coordinate alone.

```
# plot bias contour of point estimate
plot(sensitivity)
```



This plot shows that even 3 times as strong as "female", the confounder is not strong enough to explain away the treatment effect.

15.6. Sensitivity analysis for IV

Even with an IV model, we need sensitivity analysis. The worry would be there could be an unobserved confounder that affects both the instrument and the outcome.

Cinelli and Hazlett have a package "iv.sensemakr" for this. The usage is similar to "sensemakr".

The example is the returns-to-schooling data of Card (1995), 3,010 men from the 1976 NLS. The outcome is log wage, the treatment is years of education, and the instrument is `nearc4`, an indicator for growing up near a four-year college. The covariates are experience and its square, race, region, and urban indicators. The concern the sensitivity analysis addresses is a confounder affecting both proximity to a college and wages.

```
# loads package
library(iv.sensemakr)

# loads dataset
data("card")

# prepares data
y <- card$lwage  # outcome
```

```
d <- card$educ   # treatment
z <- card$nearc4 # instrument
x <- model.matrix( ~ exper + expersq + black + south + smsa + reg661 + reg662 +
                     reg663 + reg664 + reg665+ reg666 + reg667 + reg668 + smsa66,
                   data = card) # covariates
# fits IV model
card.fit <- iv_fit(y,d,z,x)

# see results
card.fit
```

```
Instrumental Variable Estimation
(Anderson-Rubin Approach)
================================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]
Note: H0 = 0, alpha = 0.05, df = 2994.
================================================
See summary for first stage and reduced form.
```

```
# runs sensitivity analysis
card.sens <- sensemakr(card.fit, benchmark_covariates = c("black", "smsa"))

# see results
card.sens
```

```
Sensitivity Analysis for Instrumental Variables
(Anderson-Rubin Approach)
================================================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2y0w.zx Lower CI Upper CI Crit. Thr.
```

```
    1x black 0.00221   0.0750  -0.0212    0.402       2.59
     1x smsa 0.00639   0.0202  -0.0192    0.396       2.57

Note: H0 = 0, q >= 1, alpha = 0.05, df = 2994.
================================================================
See summary for first stage and reduced form.
```

```
summary(card.sens)
```

```
Sensitivity Analysis for Instrumental Variables
(Anderson-Rubin Approach)
================================================================
IV Estimates:
  Coef. Estimate: 0.132
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.025, 0.285]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x r2y0w.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221   0.0750  -0.0212    0.402       2.59
     1x smsa 0.00639   0.0202  -0.0192    0.396       2.57

Note: H0 = 0, q >= 1, alpha = 0.05, df = 2994.
----------------------------------------------------------------
FS Estimates:
  Coef. Estimate: 0.32
  Standard Error: 0.0879
  t-value: 3.64
  p-value: 0.000276
  Conf. Interval: [0.148, 0.492]

Sensitivity Statistics:
  Extreme Robustness Value: 0.00313
  Robustness Value: 0.0302

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2dw.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221 0.03342    0.109    0.531       2.40
     1x smsa 0.00639 0.00498    0.120    0.520       2.27

Note: H0 = 0, q = 1, alpha = 0.05, df = 2994.
```

```
------------------------------------------------------------
RF Estimates:
  Coef. Estimate: 0.0421
  Standard Error: 0.0181
  t-value: 2.33
  p-value: 0.02
  Conf. Interval: [0.007, 0.078]

Sensitivity Statistics:
  Extreme Robustness Value: 0.000523
  Robustness Value: 0.00667

Bounds on Omitted Variable Bias:
 Bound Label  R2zw.x R2yw.zx Lower CI Upper CI Crit. Thr.
    1x black 0.00221  0.0657 -0.00418   0.0883       2.56
     1x smsa 0.00639  0.0197 -0.00429   0.0884       2.56

Note: H0 = 0, q = 1, alpha = 0.05, df = 2994.
============================================================
```

The basic idea is to run an IV regression, then apply sensemakr to that model. In this case, the authors use "black" and "smsa" as benchmark covariates. The authors believe these are important factors.

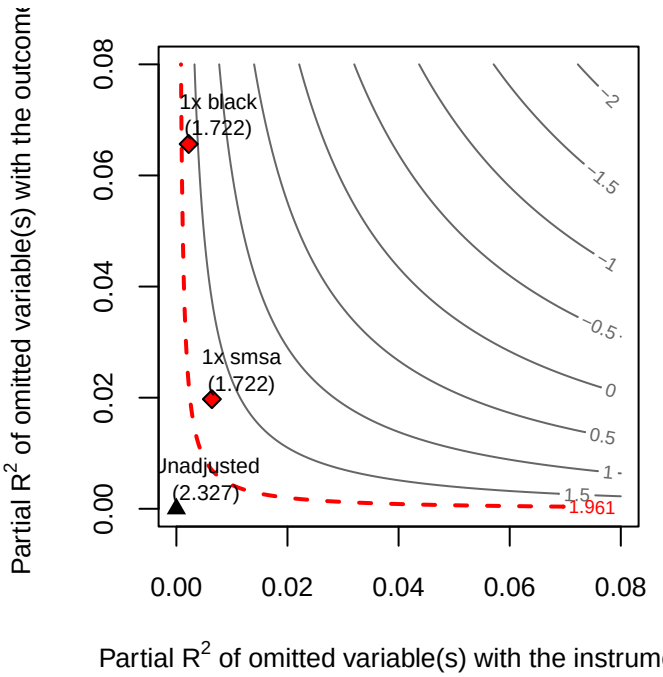
The IV estimate is 0.132 with $t = 2.33$, $p = 0.02$, and an Anderson-Rubin confidence interval of $[0.025, 0.285]$ that excludes zero. Read the sensitivity statistics next to that, and the picture is not reassuring. The Robustness Value is 0.00667 — two-thirds of one percent. Compare the OLS example earlier in this chapter, where $RV_{q=1}$ was 13.9%. This IV result is roughly twenty times more fragile than that one.

The benchmark bounds make it concrete, and they point the other way from what one might hope. A confounder as strong as `black` explains 0.22% of the residual variance of the instrument and 7.5% of the outcome, and under it the adjusted interval becomes $[-0.021, 0.402]$. For `smsa` the figures are 0.64% and 2.0%, giving $[-0.019, 0.396]$. **Both intervals contain zero.** A confounder no stronger than either benchmark covariate is enough to remove the significance of the estimate.

The `smsa` bound should not be read against the Robustness Value one coordinate at a time. The RV of 0.00667 is an equal-strength threshold: a confounder explaining 0.667% of the residual variance of *both* the instrument and the untreated potential outcome brings the interval bound to zero. The `smsa` benchmark has an instrument-side partial R^2 of 0.00639, just under the RV, but an outcome-side partial R^2 of 0.0202, three times above it. The bias depends on both jointly, so the benchmark lies well past the critical contour, which is why its adjusted interval, $[-0.019, 0.396]$, contains zero.

None of this says the Card estimate is wrong. It says the identification rests on the instrument being cleaner than `black` or `smsa` are as predictors, which is an assumption about college proximity that the data cannot check. We can also plot a contour plot to see how strong the confounder needs to be to explain away the treatment effect.

```
plot(card.sens, parm = "rf", sensitivity.of = "t-value", lim = 0.08, kz = 1, ky = 1)
```



Unfortunately the package currently only allows for one instrument. Should not be too hard to modify it to be more general.

15.7. Summary

Cinelli and Hazlett express omitted-variable bias through two partial R^2 values: the omitted confounder's association with the outcome and with the treatment. Three summaries follow. The partial R^2 of the treatment with the outcome bounds the extreme case. The robustness value is the equal strength on both sides that would change the conclusion. Benchmarks express a confounder's strength as a multiple of an observed covariate's. `sensemakr` computes all three, and `iv.sensemakr` does the same for an IV estimate.

In the Darfur example, a confounder would need to explain 13.9% of the residual variance of both treatment and outcome to remove the effect. Even one three times as strong as "female" would not. Card's IV estimate of the return to schooling is much more fragile. Its robustness value is 0.67%, and a confounder no stronger than `black` or `smsa` would put zero inside the confidence interval. The estimate rests on the instrument being cleaner than those covariates, which the data cannot check.

Systematic treatment: R · Julia.

16. Gender wage gaps: Oaxaca-Blinder and causal estimation

16.1. Oaxaca-Blinder as regression adjustment

The Oaxaca-Blinder (OB) decomposition separates the difference in mean outcomes into a part associated with observed covariates and a part associated with different regression coefficients:

$$E[Y \mid A=1] - E[Y \mid A=0] = \underbrace{(E[X \mid A=1] - E[X \mid A=0])\beta_0}_{\text{Explained (Covariate Differences)}} + \underbrace{E[X \mid A=1](\beta_1 - \beta_0)}_{\text{Unexplained (Coefficient Differences)}}$$

Here $A = 1$ denotes female and $A = 0$ denotes male. The decomposition depends on which group supplies the reference coefficients, β_0 or β_1.

Słoczyński (2013) and Słoczyński (2015) connect this decomposition to potential outcomes. If we demean the covariates at the overall sample mean, the coefficient gives the population average gender effect (τ_{PAGE}), which corresponds to the ATE. If we demean at the female-sample means, it gives the average effect for women (τ_{PATT}), which corresponds to the ATT.

Under this interpretation, the unexplained gap is the treatment effect averaged over the chosen covariate distribution. The explained part is $E[Y(0) \mid A = 1] - E[Y(0) \mid A = 0]$, the selection term in the potential-outcomes decomposition. This causal reading requires the linear model and a meaningful intervention attached to A. Without them, the explained part is only a descriptive composition difference. We can estimate the same quantities with regression adjustment or with a doubly robust estimator.

16.2. 2015 CPS data

I use a subsample from the March 2015 U.S. Current Population Survey. It contains 5,150 white, non-Hispanic, full-time workers aged 25–64. The outcome is log hourly wage (`lwage`), and `female` is the group indicator.

There are two qualifications. First, gender is not manipulable in Holland (1986)'s sense. I interpret `female` as a proxy for being perceived and treated as female by an employer, which is closer to an intervention. Second, the extract contains only full-year, full-time, non-self-employed workers. Restricting the sample conditions on employment, which can be affected by sex and by unobserved wage determinants. This opens a collider path that covariate adjustment cannot close. We would need data on non-workers instead (Lee 2009; Olivetti and Petrongolo 2008). If employed women are positively selected on wage-raising unobservables, the gaps below understate the population gap.

```r
data <- read_csv("wage2015_subsample_inference.csv") |>
  rename(socl = scl, sohs = shs, sout = so) |>
  mutate(
    exp_dm   = exp1 - mean(exp1, na.rm = TRUE),
    female   = sex,
    occ2     = factor(occ2),
    ind2     = factor(ind2),
    sohs_dm  = sohs - mean(sohs, na.rm = TRUE),
    hsg_dm   = hsg - mean(hsg, na.rm = TRUE),
    socl_dm  = socl - mean(socl, na.rm = TRUE),
    clg_dm   = clg - mean(clg, na.rm = TRUE),
    mw_dm    = mw - mean(mw, na.rm = TRUE),
    sout_dm  = sout - mean(sout, na.rm = TRUE),
    we_dm    = we - mean(we, na.rm = TRUE)
  )

# 1. Unadjusted (raw) difference
reg1 <- lm(lwage ~ female, data = data)
tidy(reg1) |> filter(term == "female")
```

```
# A tibble: 1 x 5
  term   estimate std.error statistic p.value
  <chr>     <dbl>     <dbl>     <dbl>   <dbl>
1 female  -0.0383    0.0160     -2.40  0.0165
```

```r
# 2. Regression adjustment with demeaned human capital & region controls
reg2 <- lm(lwage ~ female * (exp_dm + sohs_dm + hsg_dm + socl_dm + clg_dm + mw_dm + sout_dm + we_dm),
           data = data)
tidy(reg2) |> filter(term == "female")
```

```
# A tibble: 1 x 5
  term   estimate std.error statistic  p.value
  <chr>     <dbl>     <dbl>     <dbl>    <dbl>
1 female   -0.115    0.0147     -7.80 7.39e-15
```

16.2.1. Why adjustment widens the gap

The raw wage gap is -3.8% ($p = 0.017$). After controlling for potential experience, education, and region, the estimated gap at the sample covariate means widens to -11.5% ($p < 0.001$). Advanced degree (`ad`) is the omitted education category in this and the later regressions.

Here adjustment nearly triples the gap. We can see why by comparing the covariate means:

```
data |>
  group_by(female) |>
  summarise(
    n           = n(),
    mean_lwage  = mean(lwage),
    exp1        = mean(exp1),
    `<HS`       = mean(sohs),
    HS          = mean(hsg),
    `Some Coll` = mean(socl),
    College     = mean(clg),
    Advanced    = mean(ad)
  ) |>
  mutate(across(where(is.numeric), ~round(.x, 3))) |>
  knitr::kable(caption = "Covariate means by gender.")
```

Table 16.1.: Covariate means by gender.

female	n	mean_lwage	exp1	<HS	HS	Some Coll	College	Advanced
0	2861	2.988	13.784	0.032	0.294	0.273	0.294	0.107
1	2289	2.949	13.731	0.013	0.181	0.284	0.347	0.175

Women in this sample have more education than men: 17.5% have advanced degrees, compared with 10.7% of men, while 1.3% have less than high school, compared with 3.2% of men. Average potential experience is about 13.8 years in both groups. Because education raises wages, the raw comparison partly hides the gap. Holding education fixed gives the larger −11.3% gap within education-experience cells.

16.2.2. Regression adjustment with `marginaleffects`

Instead of manual demeaning, the same estimands can be obtained directly using `marginaleffects::avg_comparisons(`

```
reg3 <- lm(lwage ~ female * (exp1 + sohs + hsg + socl + clg + mw + sout + we), data = data)

# Population Average Gender Effect (ATE / PAGE)
avg_comparisons(reg3, variables = "female")
```

```
 Estimate Std. Error    z Pr(>|z|)    S  2.5 %  97.5 %
   -0.115     0.0147 -7.8   <0.001 47.2 -0.143 -0.0858

Term: female
Type: response
Comparison: 1 - 0
```

```
# Female Subpopulation Gender Effect (ATT / PATT)
avg_comparisons(reg3, variables = "female", newdata = subset(data, female == 1))
```

```
 Estimate Std. Error     z Pr(>|z|)    S  2.5 %  97.5 %
   -0.113     0.0148 -7.65   <0.001 45.5 -0.142 -0.0844

Term: female
Type: response
Comparison: 1 - 0
```

The ATE is −11.5%, the same as `reg2`, which centres the covariates at the sample means. The ATT, which averages over women's covariates, is −11.3%.

16.3. Should we control for occupation and industry?

Pay-gap regressions often adjust for occupation and industry. Let's see what happens here:

```
reg4 <- lm(lwage ~ female * (exp1 + sohs + hsg + socl + clg + mw + sout + we + occ2 + ind2),
           data = data)

avg_comparisons(reg4, variables = "female", newdata = subset(data, female == 1))
```

```
 Estimate Std. Error     z Pr(>|z|)    S  2.5 %  97.5 %
   -0.075     0.0162 -4.62   <0.001 18.0 -0.107 -0.0432

Term: female
Type: response
Comparison: 1 - 0
```

Adjusting for 22 occupation and industry categories narrows the gap for women (the ATT) from −11.3% to −7.5%.

16.3.1. Different controls answer different questions

If we take sex as the treatment, there is no earlier common cause to adjust for. That alone does not give us $Y(a) \perp A$. Exchangeability is a claim about the population actually analysed, and this sample holds only full-year, full-time workers — a restriction that sex and unobserved wage determinants both bear on. Conditioning on employment is the collider noted above, and it can create an association between A and $Y(a)$ inside the sample where none exists outside it. Everything below is therefore conditional on that selection, and a causal label would need an explicit intervention, a target population, and an argument about selection before it could be earned. Adding controls

changes the estimand rather than improving identification. The models below answer different questions; they are not progressively better estimates of one quantity.

Model	Held fixed	Question answered	Estimate
`reg1`	nothing	Raw gap among selected full-time workers	−3.8%
`reg3`	education, experience, region	Gap within education-experience cells (ATT)	−11.3%
`reg4`	+ occupation, industry	Gap within occupation cells too (ATT)	−7.5%

Occupation and industry are post-treatment ($A \to$ Occupation $\to Y$), and they are also a common effect of gender and of unobserved wage determinants U such as career preferences or flexibility constraints. Conditioning on them blocks the mediating path and at the same time opens $A \to$ Occupation $\leftarrow U \to Y$, which was closed before (Elwert and Winship 2014; Cinelli, Forney, and Pearl 2024).

The four-point move from −11.3% to −7.5% therefore cannot be read as occupational sorting accounting for a third of the gap. Three things are mixed into it and nothing in the output separates them: genuine mediation, collider bias from the gender-U association that conditioning induces, and ordinary confounding of the occupation-wage relationship. The direction of the second can be signed. If women who enter male-dominated, high-paying occupations are positively selected on U, then within those occupations they are better on unobservables than the men they are compared with, which pushes the within-occupation gap toward zero. Part of the attenuation may be bias rather than sorting.

Calling −7.5% a controlled direct effect is licensed only under no unmeasured mediator-outcome confounding (VanderWeele and Robinson 2014). U is exactly such a confounder, and in a wage setting it is the reason this literature exists. Report the number as the within-occupation adjusted gap and state the condition that would upgrade it to a CDE.

16.4. A nonparametric AIPW estimate

The regression adjustment still depends on its functional form. To make that part more flexible, I use augmented inverse probability weighting (AIPW) with SuperLearner through `npcausal` (Kennedy 2022). The learner library includes `earth`, `glmnet`, `ranger`, and `xgboost`.

```
library(npcausal)

# To avoid prolonged cross-fitting during interactive compilation, we load the
# pre-computed SuperLearner results (SL.earth, SL.glmnet, SL.mean, SL.ranger, SL.xgboost)
# from the book cache if present; otherwise, compute from scratch.
cache_base <- "gwg_cache/html/npcausal1_5538101f2b9ff4f8b21593068dc74ef6"
if (file.exists(paste0(cache_base, ".rdx"))) {
```

```
  cache_env <- new.env()
  lazyLoad(cache_base, envir = cache_env)
  aipw_ate <- cache_env$aipw_ate
  aipw_att <- cache_env$aipw_att
} else {
  SL.library <- c("SL.earth", "SL.glmnet", "SL.mean", "SL.ranger", "SL.xgboost")
  reg_w <- lm(lwage ~ exp_dm + sohs + hsg + socl + clg + mw + sout + we + occ2 + ind2, data = data)
  W <- as.data.frame(model.matrix(reg_w)[, -1])
  Y <- data$lwage
  A <- data$female
  set.seed(123)
  aipw_ate <- ate(y = Y, a = A, x = W, nsplits = 5, sl.lib = SL.library)
  aipw_att <- att(y = Y, a = A, x = W, nsplits = 5, sl.lib = SL.library)
}
```

```
aipw_ate$res
```

```
     parameter         est         se       ci.ll       ci.ul pval
1      E{Y(0)}  2.99695506 0.01093465  2.97552315  3.01838697    0
2      E{Y(1)}  2.93906408 0.01327342  2.91304817  2.96507999    0
3 E{Y(1)-Y(0)} -0.05789098 0.01597522 -0.08920241 -0.02657955    0
```

```
aipw_att$res
```

```
      parameter         est         se       ci.ll       ci.ul pval
1      E(Y|A=1)  2.94948490 0.01160804  2.92673314  2.97223666    0
2   E{Y(0)|A=1}  3.01689509 0.01221863  2.99294658  3.04084360    0
3 E{Y-Y(0)|A=1} -0.06741019 0.01409116 -0.09502886 -0.03979152    0
```

16.4.1. Comparing the estimates

The AIPW estimate of the ATT is -6.7% (95% CI : $[-9.5\%, -3.9\%]$), close to the interacted OLS estimate of -7.5%. This suggests that the linear model is a reasonable approximation to the same conditional mean. AIPW only relaxes the functional-form assumption. It does not solve post-treatment adjustment, selection into the sample, or the interpretation of gender as a treatment.

16.5. What the estimates mean

The raw gap is -3.8%. Holding education, experience, and region fixed gives -11.5% on average and -11.3% for women, partly because women in this sample have more education. Adding occupation and industry gives a within-occupation gap of -7.5%; it is a controlled direct effect only if there is no unmeasured occupation-wage confounding. AIPW gives -6.7%, so the within-occupation result is not driven mainly by the linear functional form. None of these methods solves selection into

full-time employment, and no single number should be called *the* gender wage gap without saying which question it answers.

Part IV.

Panel Data & DiD

17. Causal Forest in panel data

17.1. Introduction

In this simulation exercise, we use Causal Forest (Now is implemented in Generalized Random Forest) (Athey, Tibshirani, and Wager 2019, https://github.com/grf-labs/grf) to calculate the conditional average treatment effect (or heterogeneous treatment effect). We assume three different data generating processes. The first one is a linear interaction between a variable of interest and the treatment dummy. The second one assumes a nonlinear function (a step function) of a variable of interest, say X, and the treatment dummy W. The third one is also nonlinear, assuming we have variable X, but the real DGP is the interaction of log of X and W.

17.2. Linear effect

Suppose we have a panel of firms over time, treatment assignment is random. We split the data into training and test. Then we generate 10 random variables, the first two are highly correlated. Then we collapse V2 into firm means, and use that as the unit effect.

Concretely: $n = 400$ firms over $t = 10$ periods, 4,000 observations, with $p = 10$ standard normal covariates of which $V1$ and $V2$ are correlated at 0.9025. Treatment is $W \sim \text{Bernoulli}(0.5)$, assigned independently of everything, and a further coin flip splits the sample roughly in half into training and test sets (about 2,014 rows for training). The unit effect α_i is the firm mean of $V2$.

That is, we have a DGP as:

$$y_{it} = \alpha_i + V1_{it} + W_{it} + V1_{it} * W_{it} + \epsilon_{it} \tag{17.1}$$

Here α_i is not observed, and it is correlated with $V1$, so leaving it out is in principle a source of bias.

It is worth being precise about how much bias, though, because the results below are less dramatic than that setup suggests. The unit effect is the *firm mean* of $V2$ over all $t = 10$ periods, and averaging attenuates its correlation with contemporaneous $V1$ by a factor of t: with $\text{Cor}(V1_{it}, V2_{it}) = 0.9025$ we get $\text{Cov}(V1_{it}, \bar{V}2_i) = 0.9025/10$, and $\text{Cor}(V1_{it}, \alpha_i) \approx 0.29$. So the omitted-variable bias on $V1$ is only about 0.05–0.09, not a qualitative failure.

That is what the three models show: FE, RE and OLS all recover $V1 \approx 1$, $W \approx 1$ and $V1{:}W \approx 1.05$, essentially the truth. Note in particular that the *treatment* coefficients are unbiased in all three, because W is randomized and so independent of α_i — omitting a unit effect cannot bias the effect of a randomly assigned treatment. If you want a panel example where FE visibly beats OLS, the unit effect has to be correlated with $V1$ *contemporaneously* (or t has to be small), and the treatment has to be non-random.

```
# This is a simulation exercise to see how causal forest performs
# First we run a linear model, with panel setting
# Then two nonlinear situations, with panel setting.

library(lfe, quietly = TRUE)
library(lme4)
require(snowfall)
library(MASS)
library(grf)
library(tidyverse)
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)

firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)
```

```
y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ data$V1[k] + data$W[k] + data$V1[k]*data$W[k] + data$unit.effect[k]+rnorm(1,mean=0,sd=1
    }
}
data$unit=unit
data$y=y



data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
# biased.
# in general we should expect fe performs better, since we are
# omitting unit.effect in the other two models
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-3.0440 -0.6252 -0.0083  0.6221  3.6174

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1    0.98354    0.03748   26.24   <2e-16 ***
W     1.02186    0.05093   20.06   <2e-16 ***
V1:W  1.04302    0.05237   19.92   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.018 on 1612 degrees of freedom
```

```
Multiple R-squared(full model): 0.7952   Adjusted R-squared: 0.7443
Multiple R-squared(proj model): 0.7269   Adjusted R-squared: 0.659
F-statistic(full model):15.61 on 401 and 1612 DF, p-value: < 2.2e-16
F-statistic(proj model):  1430 on 3 and 1612 DF, p-value: < 2.2e-16
```

```
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 5938.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.4332 -0.6615 -0.0049  0.6640  3.8329

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.07166  0.2677
 Residual             1.04602  1.0228
Number of obs: 2014, groups:  unit, 399

Fixed effects:
            Estimate Std. Error t value
(Intercept)  1.00010    0.03539   28.26
V1           1.02961    0.03486   29.53
W            1.00333    0.04684   21.42
V1:W         1.05963    0.04814   22.01

Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.013
W    -0.656 -0.008
V1:W -0.007 -0.727  0.025
```

```
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-3.8384 -0.7101  0.0081  0.7334  4.0053
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.00363    0.03319   30.24   <2e-16 ***
V1           1.04612    0.03516   29.75   <2e-16 ***
W            0.99657    0.04712   21.15   <2e-16 ***
V1:W         1.06458    0.04840   22.00   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.057 on 2010 degrees of freedom
Multiple R-squared:  0.7249,    Adjusted R-squared:  0.7245
F-statistic:  1765 on 3 and 2010 DF,  p-value: < 2.2e-16
```

Now we run Causal Forest on this data. Two clarifications about what is and is not at stake here. First, treatment W is **randomized** (`rbinom(., 0.5)`, independent of the firm effect), so the omitted unit effect α_i does *not* confound the treatment effect — it only enters the nuisance functions (and hence precision), it does not create treatment-effect bias. The panel issue here is therefore mainly *dependence and nuisance fit*, not confounding. Second, the firm structure should be handled in two ways: we include the firm dummies in X so the forest can absorb α_i in the nuisance models, **and** we pass `clusters = firm` so that `grf`'s honest sample splitting and its variance estimates keep each firm's repeated observations together rather than treating them as independent rows. Including the dummies alone does neither of those.

```r
# ensure consistent factor levels across train/test
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)

# a is a data set of firm dummies
# Caution: this one-hot-encodes all ~400 firms into individual dummy columns
# of X. Random forests split poorly on such a high-cardinality sparse block
# of binary columns (each dummy is only informative for a tiny fraction of
# observations). A fixed-effects forest (demean y, W, and the continuous
# covariates within firm first, then run causal_forest on the demeaned
# variables) or a method that handles unit effects natively would typically
# perform better than including firm as raw dummies.
a <- as.data.frame(model.matrix(y~firm, data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W
# causal forest is run on y against V1, to X10 and firm dummies.
# mtry = ncol(X): grf's default mtry is min(ceiling(sqrt(p) + 20), p) = 41 of the
# 411 columns, so the one effect modifier V1 is a split candidate in only about
# 10% of splits and the CATE predictions shrink hard toward the mean (slope about
# 0.14 on V1 against a true 1 in this design). Letting every split consider every
# column removes that shrinkage.
# clusters = firm makes honest splitting and the variance estimates respect
```

```
# the panel structure (a firm's repeated rows are kept together, not treated
# as independent observations).
tau.forest = causal_forest(X, Y, W, clusters = as.integer(data.train$firm),
                           mtry = ncol(X))

average_treatment_effect(tau.forest, target.sample = "all")
```

```
 estimate   std.err
0.9631155 0.0556132
```

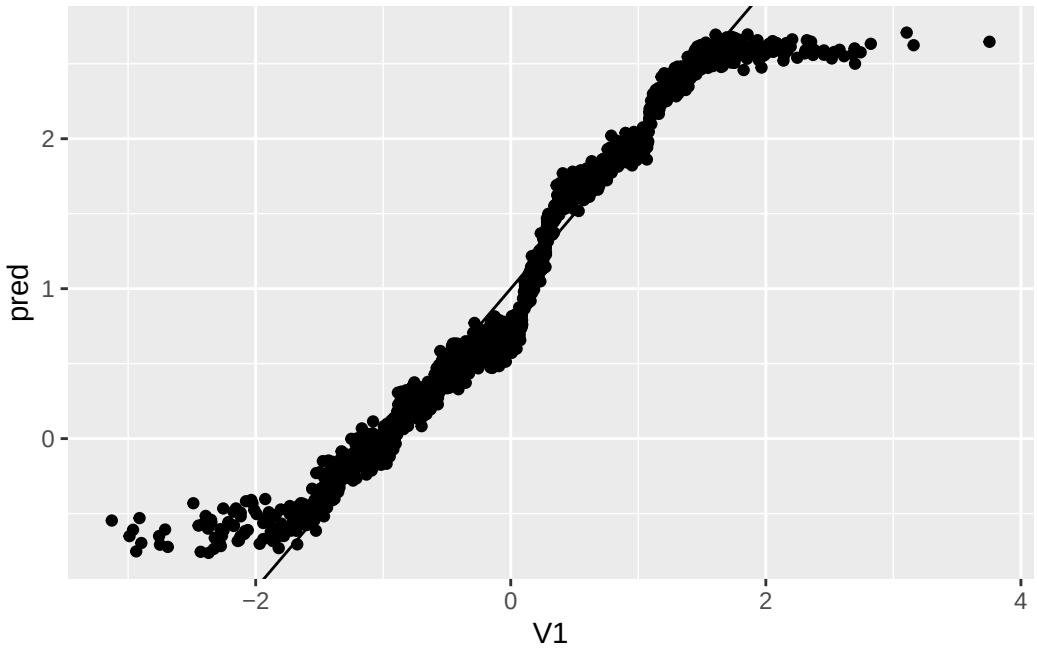
```
average_treatment_effect(tau.forest, target.sample = "treated")
```

```
 estimate   std.err
0.9498104 0.0582338
```

```
a <- as.data.frame(model.matrix(y~firm, data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.hat = predict(tau.forest, X.test, estimate.variance = TRUE)
sigma.hat = sqrt(tau.hat$variance.estimates)

data.test$pred <- tau.hat$predictions
ggplot(data.test, aes(x=V1, y=pred)) + geom_point() + geom_abline(intercept = 1, slope = 1)
```



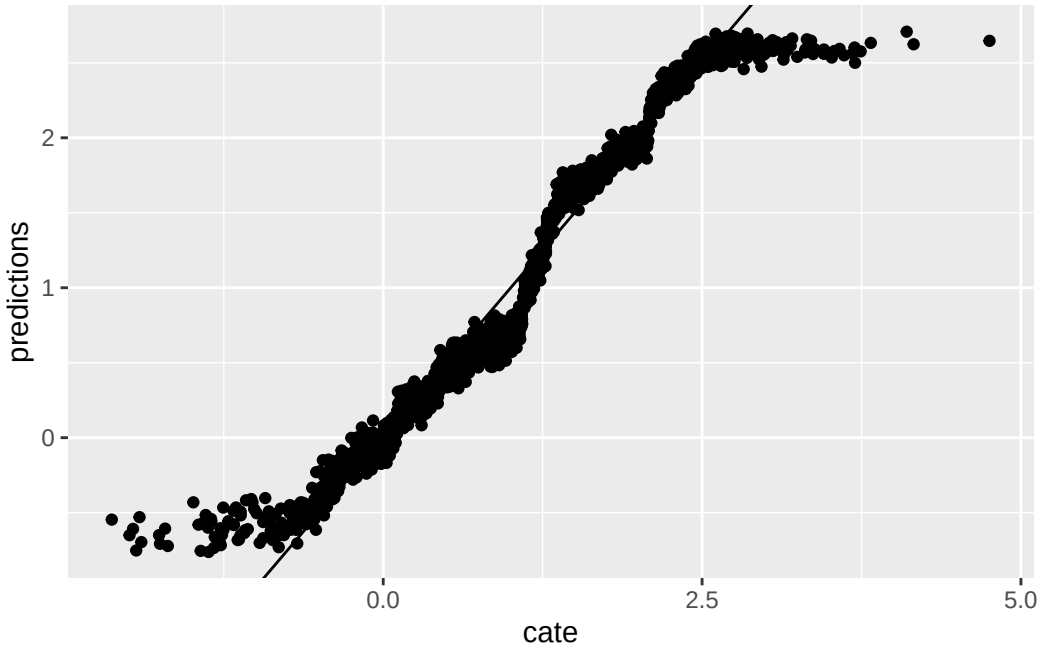
The forest's doubly-robust ATE comes out at 0.9631 with a standard error of 0.0556, and 0.9498 with 0.0582 for the second, `target.sample = "treated"`, which targets the ATT. Because W is randomized, the ATT equals the ATE here. The true ATE is $E[1+V1]=1$, so both are within one standard error of it — the forest costs a little precision against the correctly specified linear models but is not biased.

This graph is the predicted value of CATE on the test data, vs. the value of $V1$. The points should sit on the line $1+V1$. They do over most of the range and flatten in the tails, where few observations fall and a forest cannot extrapolate. This depends on `mtry = ncol(X)`. With grf's default, which offers only 41 of the 411 columns at each split, the predictions shrink to a narrow band around the mean, with a slope of about 0.14 on $V1$ instead of 1. The error bars in the later plot show how much of the remaining scatter is estimation noise.

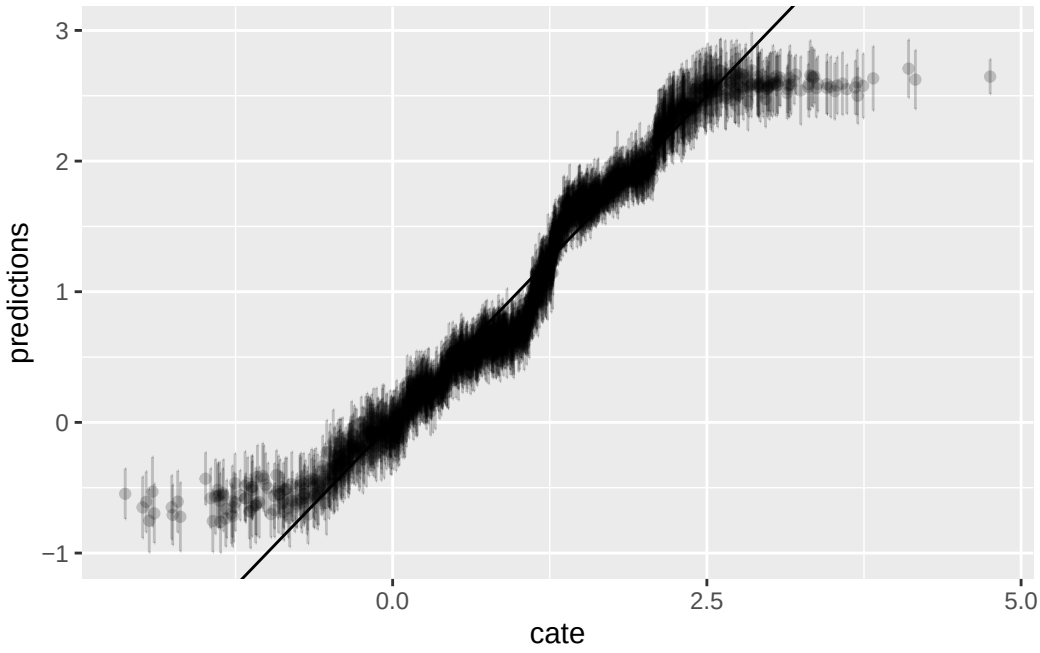
Now we graph the predicted CATE against the true CATE. The other nice feature of Causal Forest is that it returns variance for CATE for each observation. We also graph the error bar here.

```
graph.data  <- as.data.frame(tau.hat )
graph.data  <- graph.data  %>%
    mutate(cate=1+data.test$V1) %>%
    mutate(lower=predictions - sigma.hat , upper=predictions + sigma.hat )

ggplot(graph.data , aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```




```
ggplot() + geom_errorbar(graph.data , mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2) + geom_po
```



17.3. Nonlinear effect case 1

In the first situation, we assume DGP:

$$y_{it} = 1 + \alpha_i + max(V1_{it}, 0) * W_{it} + \epsilon_{it} \tag{17.2}$$

```
# if we have a nonlinear effect
# suppose we have the same X, but y is generated nonlinearly.
# in this example, y is a nonlinear function of V1 and W
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)

firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
```

```r
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)

y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ pmax(data$V1[k],0)*data$W[k]  + data$unit.effect[k]+rnorm(1,mean=0,sd=1)
    }
}
data$y=y
data$unit <- unit

data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
# biased.
# they all perform poorly
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-2.9461 -0.6225 -0.0178  0.6245  3.3537

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1   -0.02076    0.03815  -0.544    0.586
W     0.42527    0.05184   8.203 4.73e-16 ***
V1:W  0.53367    0.05331  10.012  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.036 on 1612 degrees of freedom
Multiple R-squared(full model): 0.3744   Adjusted R-squared: 0.2187
Multiple R-squared(proj model): 0.1393   Adjusted R-squared: -0.07481
F-statistic(full model):2.405 on 401 and 1612 DF, p-value: < 2.2e-16
F-statistic(proj model): 86.96 on 3 and 1612 DF, p-value: < 2.2e-16
```

```
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 6017.2

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.5905 -0.6596 -0.0031  0.6717  3.4143

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.07977  0.2824
 Residual             1.08359  1.0410
Number of obs: 2014, groups:  unit, 399

Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.99976    0.03624  27.587
V1           0.02797    0.03553   0.787
W            0.40652    0.04774   8.515
V1:W         0.54484    0.04907  11.104
```

```
Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.013
W    -0.653 -0.008
V1:W -0.007 -0.727  0.025
```

```
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0965 -0.7415  0.0095  0.7512  3.8413

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.00363    0.03386  29.643   <2e-16 ***
V1           0.04612    0.03587   1.286    0.199
W            0.39928    0.04807   8.307   <2e-16 ***
V1:W         0.54938    0.04937  11.127   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.078 on 2010 degrees of freedom
Multiple R-squared:  0.1556,    Adjusted R-squared:  0.1543
F-statistic: 123.5 on 3 and 2010 DF,  p-value: < 2.2e-16
```

These linear models miss the *shape* of the effect, but it overstates things to say they return nothing. Each recovers the best linear approximation to the true CATE $\max(V1, 0)$, which is $0.3989 + 0.5\,V1$: the fitted W coefficients are 0.40–0.43 and the $V1{:}W$ interactions are 0.53–0.55. In particular the W coefficient is essentially the true ATE, $E[\max(V1, 0)] = \phi(0) = 0.3989$. What a linear specification cannot do is see the kink at $V1 = 0$ — and that is what the forest is for.

Now we use Causal Forest. The first graph is the predicted CATE against $V1$; the second graph is the predicted CATE against true CATE. The third one adds in error bars.

The forest's ATE is 0.3695 (SE 0.0521) and 0.3682 (SE 0.0530), against the true 0.3989 — about half a standard error away, and no better than the linear model's W coefficient at recovering the average. The gain is in the shape: the fitted CATE should flatten to zero for $V1 < 0$ and rise linearly above it, which is the kink the linear interaction smooths straight through.

```
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)
```

```
# a is a data set of firm dummies
a <- as.data.frame(model.matrix(y~firm,data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W

#tau.forest = causal_forest(X, Y, W, num.trees = 4000)
a <- as.data.frame(model.matrix(y~firm,data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.forest3 = causal_forest(X, Y, W, clusters = as.integer(data.train$firm),
                            mtry = ncol(X))

average_treatment_effect(tau.forest3, target.sample = "all")
```

```
  estimate    std.err
0.36949343 0.05207733
```

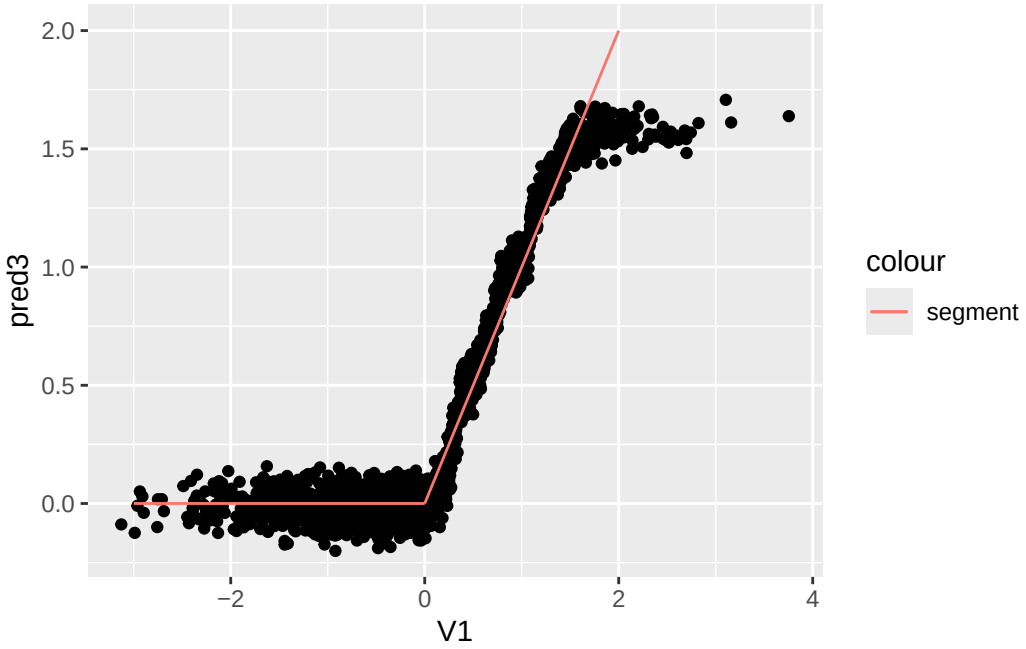
```
average_treatment_effect(tau.forest3, target.sample = "treated")
```

```
  estimate    std.err
0.36819745 0.05296265
```

```
#tau.forest3 = causal_forest(X, Y, W, num.trees = 4000)
tau.hat3 = predict(tau.forest3, X.test, estimate.variance = TRUE)
sigma.hat3 = sqrt(tau.hat3$variance.estimates)

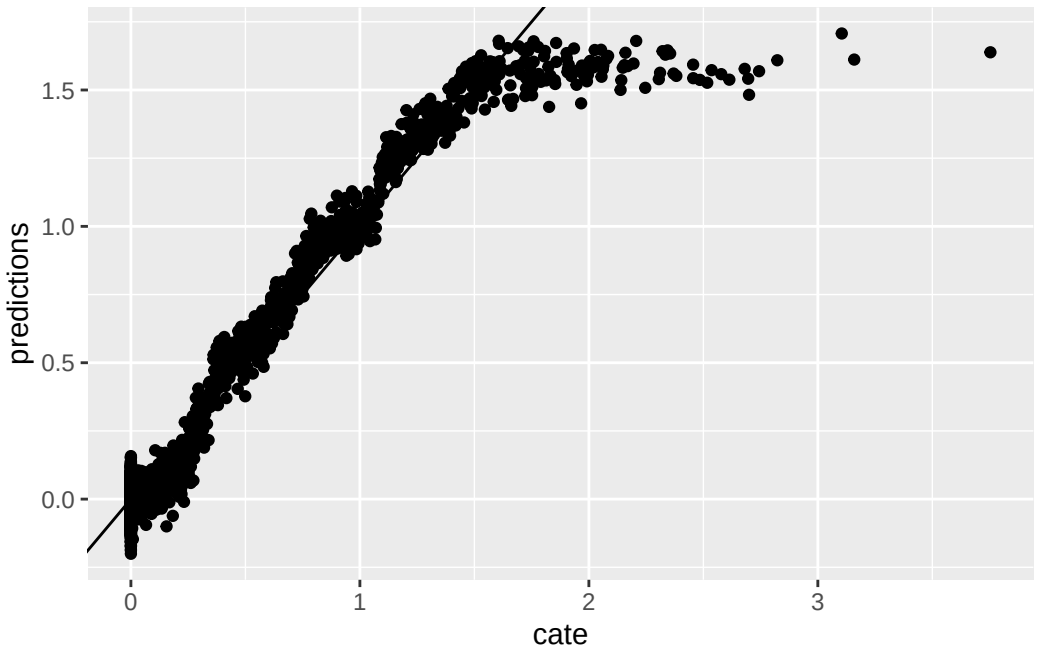
data.test$pred3 <- tau.hat3$predictions

seg1 <- data.frame(x1 = -3, x2 = 0, y1 = 0, y2 = 0)
seg2 <- data.frame(x1 = 0, x2 = 2, y1 = 0, y2 = 2)
ggplot(data.test, aes(x=V1, y=pred3)) + geom_point() +
geom_segment(aes(x = x1, y = y1, xend = x2, yend = y2, colour = "segment"), data = seg1)+
geom_segment(aes(x = x1, y = y1, xend = x2, yend = y2, colour = "segment"), data = seg2)
```

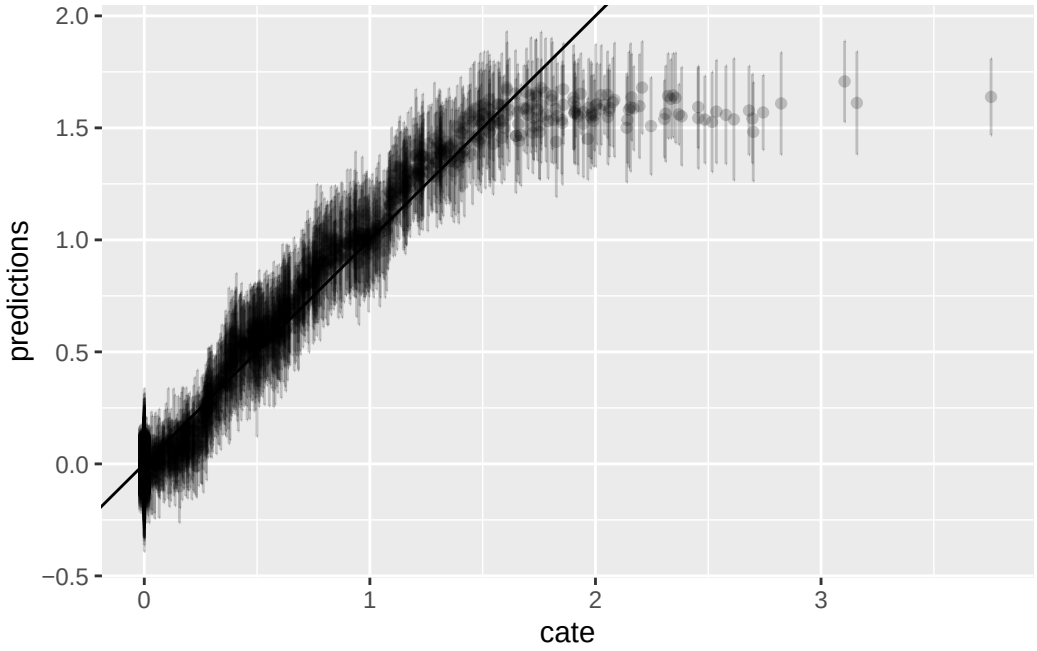



```
graph.data3 <- as.data.frame(tau.hat3)
graph.data3 <- graph.data3 %>%
    mutate(cate=pmax(data.test$V1,0)) %>%
    mutate(lower=predictions - sigma.hat3, upper=predictions + sigma.hat3)

ggplot(graph.data3, aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```



```
ggplot() + geom_errorbar(graph.data3, mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2) + geom_po
```



17.4. Nonlinear effect case 2

In the second situation, we assume DGP:

$$y_{it} = 1 + \alpha_i + \log(V1_{it}^2) * W_{it} + \min(X3_{it}, 0) + \epsilon_{it} \tag{17.3}$$

But we only observe $V1$, not the log of the squared $V1$. The extra $\min(X3_{it}, 0)$ term is a nonlinearity in the *outcome* rather than in the treatment effect, so it leaves the CATE equal to $\log(V1^2)$; it is there to give the forest's nuisance models something nonlinear to fit.

```
# if we have a nonlinear effect
# suppose we have the same X, but y is generated nonlinearly.
# another example: we observe V1, but the true treatment effect is log(V1^2)
set.seed(666)
rm(list = ls())
# t is no. of periods. p is no. of variables, n is no. of firms.
t <- 10
p <- 10
n=400
# generate p random variables
data1 = as.data.frame(matrix(rnorm(n*t*p), n*t, p))
names(data1) <- paste0("X", 1:p)
```

```
firm=seq(1,n)
time=seq(1,t)
data <- expand.grid(firm=firm,time=time)


# W is the treatment assignment
data$W <- rbinom(n*t, 1, 0.5)
data$train <- rbinom(n*t, 1, 0.5)


# generate two correlated variables
covarMat = matrix( c(1, .95^2, .95^2, 1 ) , nrow=2 , ncol=2 )
data.2  = as.data.frame(mvrnorm(n=n*t , mu=rep(0,2), Sigma=covarMat ))
names(data.2) <- c("V1", "V2")

data <- bind_cols(data,data.2) %>%
    tibble()

data <- bind_cols(data,data1)

# generate unit effect by firm means of V2, which is correlated with V1
data <- data %>%
    group_by(firm) %>%
    mutate(unit.effect=mean(V2)) %>%
    ungroup() %>%
    arrange(firm, time)

y<-c()
unit<-c()
X<-c()
k <- 0
for(i in 1:n){
    for(j in 1:t){
        k<-k+1
        unit[k]<-i
        y[k]<-1+ log(data$V1[k]^2)*data$W[k] + pmin(data$X3[k], 0) + data$unit.effect[k]+rnorm(1,mean=0,s
    }
}
data$y=y
data$unit <- unit

data.train <- data %>%
    filter(train==1)
data.test <- data %>%
    filter(train==0)

# run fixed effect, random effect, and ols to see whether they are
```

```
# biased.
# they all perform poorly
fe <- felm(y ~ V1 * W | unit, data=data.train)
re <-lmer(y~V1*W+(1 | unit), data=data.train)
ols <- lm(y ~ V1 * W , data=data.train)
summary(fe)
```

```
Call:
   felm(formula = y ~ V1 * W | unit, data = data.train)

Residuals:
     Min       1Q   Median       3Q      Max
-10.0383  -0.9478   0.0907   1.0865   5.3884

Coefficients:
     Estimate Std. Error t value Pr(>|t|)
V1   -0.01979    0.07088  -0.279    0.780
W    -1.17812    0.09632 -12.231   <2e-16 ***
V1:W  0.01450    0.09904   0.146    0.884
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.925 on 1612 degrees of freedom
Multiple R-squared(full model): 0.2876   Adjusted R-squared: 0.1104
Multiple R-squared(proj model): 0.08495   Adjusted R-squared: -0.1427
F-statistic(full model):1.623 on 401 and 1612 DF, p-value: 6.032e-11
F-statistic(proj model): 49.88 on 3 and 1612 DF, p-value: < 2.2e-16
```

```
summary(re)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ V1 * W + (1 | unit)
   Data: data.train

REML criterion at convergence: 8421.8

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.1184 -0.4977  0.0647  0.6098  2.8959

Random effects:
 Groups   Name        Variance Std.Dev.
 unit     (Intercept) 0.09312  0.3052
 Residual             3.72467  1.9299
Number of obs: 2014, groups:  unit, 399
```

```
Fixed effects:
            Estimate Std. Error t value
(Intercept)  0.57512    0.06301   9.128
V1           0.06152    0.06490   0.948
W           -1.18310    0.08706 -13.590
V1:W        -0.01892    0.08945  -0.212

Correlation of Fixed Effects:
     (Intr) V1     W
V1    0.015
W    -0.685 -0.010
V1:W -0.009 -0.727  0.028
```

```
summary(ols)
```

```
Call:
lm(formula = y ~ V1 * W, data = data.train)

Residuals:
     Min       1Q   Median       3Q      Max
-12.1215  -0.9805   0.1390   1.1999   5.7125

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.57618    0.06136   9.390   <2e-16 ***
V1           0.07086    0.06502   1.090    0.276
W           -1.18331    0.08712 -13.582   <2e-16 ***
V1:W        -0.02005    0.08949  -0.224    0.823
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.954 on 2010 degrees of freedom
Multiple R-squared:  0.08508,   Adjusted R-squared:  0.08372
F-statistic: 62.31 on 3 and 2010 DF,  p-value: < 2.2e-16
```

Here the linear models fail more completely, and instructively so. Because $\log(V1^2)$ is *even* in $V1$, its linear interaction with $V1$ is zero by construction: the fitted $V1{:}W$ terms come out at 0.015, -0.019 and -0.020, i.e. no detectable heterogeneity at all, even though the true CATE varies enormously with $V1$. The W coefficients (≈ -1.18) still approximate the true ATE, $E[\log(V1^2)] = \psi(1/2) + \log 2 = -1.2704$. So a linear model gets the average right and the heterogeneity completely wrong.

Now we run Causal Forest. Again, first graph is predicted CATE against $V1$, the second one is the predicted CATE against true CATE, the third one adds error bars.

The forest's ATE is −1.2563 (SE 0.0699) and −1.2414 (SE 0.0839) for the two variants, against the true −1.2704. Like the linear models, it gets the average about right. The difference is in the plots: the forest's fitted CATE traces the V-shape of $\log(V1^2)$ down to its minimum near $V1 = 0$, which no linear interaction can represent. It again flattens for $|V1| > 1.5$, where the data thin out.

```
all_firms <- levels(factor(data$firm))
data.train$firm <- factor(data.train$firm, levels = all_firms)
data.test$firm <- factor(data.test$firm, levels = all_firms)

# a is a data set of firm dummies
a <- as.data.frame(model.matrix(y~firm,data.train))
X <- as.matrix(bind_cols(data.train[,c(5,7:16)],a))
Y <- data.train$y
W <- data.train$W

#tau.forest = causal_forest(X, Y, W, num.trees = 4000)
a <- as.data.frame(model.matrix(y~firm,data.test))
X.test <- as.matrix(bind_cols(data.test[,c(5,7:16)],a))
Y.test <- data.test$y
W.test <- data.test$W

tau.forest4 = causal_forest(X, Y, W, clusters = as.integer(data.train$firm),
                            mtry = ncol(X))

average_treatment_effect(tau.forest4, target.sample = "all")
```

```
   estimate     std.err
-1.25633160  0.06987047
```

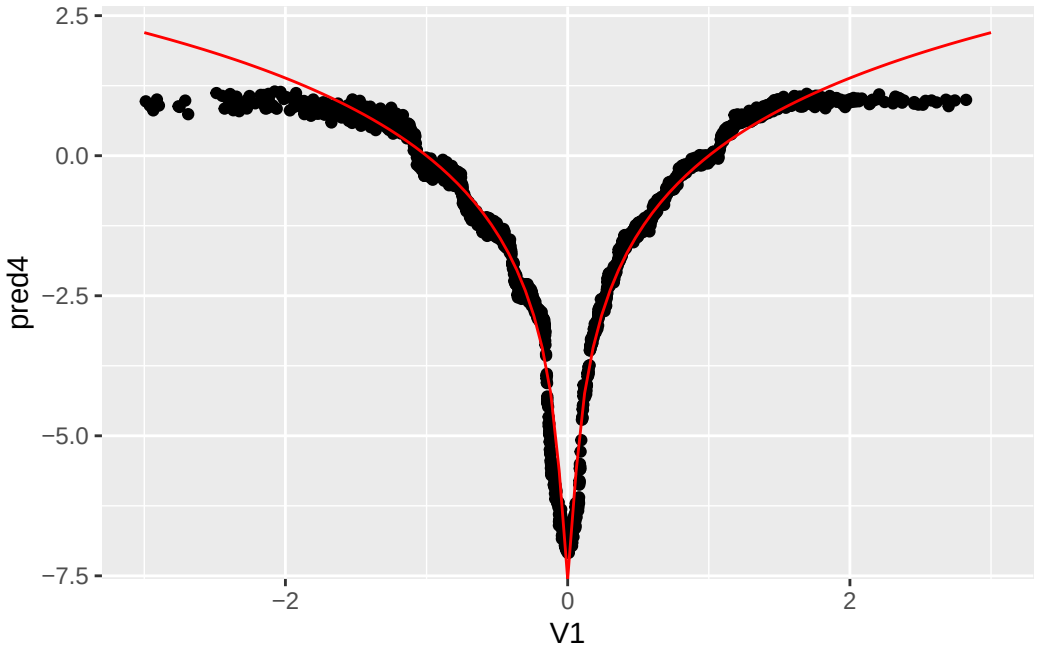
```
average_treatment_effect(tau.forest4, target.sample = "treated")
```

```
  estimate    std.err
-1.2414441  0.0838892
```

```
#tau.forest3 = causal_forest(X, Y, W, num.trees = 4000)
tau.hat4 = predict(tau.forest4, X.test, estimate.variance = TRUE)
sigma.hat4 = sqrt(tau.hat4$variance.estimates)

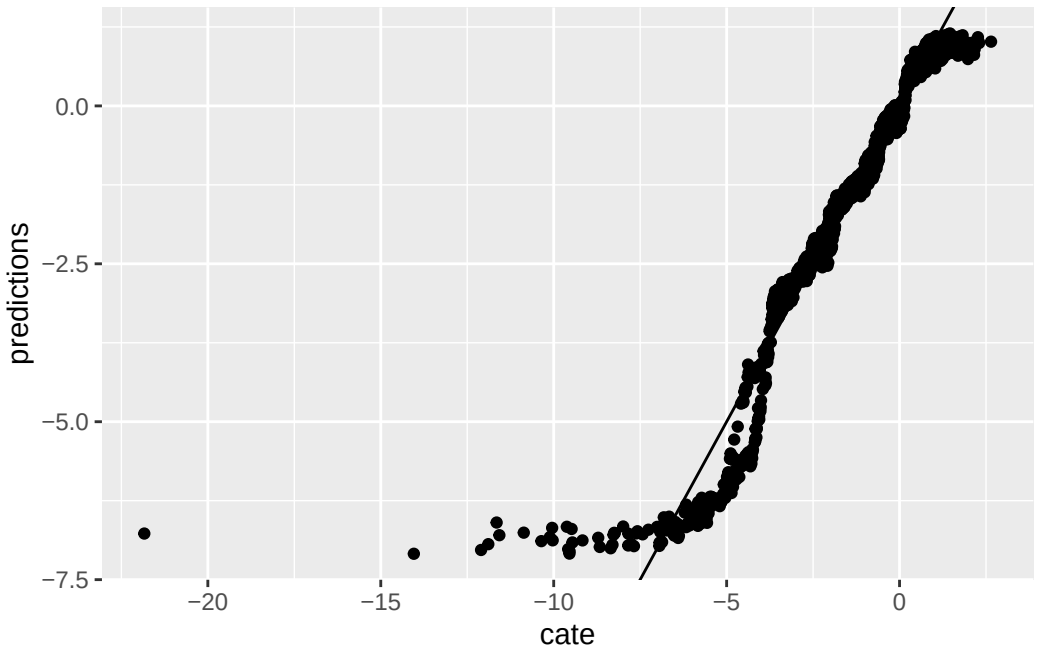
data.test$pred4 <- tau.hat4$predictions

fun.1 <- function(x) log(x^2)
ggplot(data.test, aes(x=V1, y=pred4)) + geom_point() +
stat_function(fun = fun.1, colour="red") + xlim(-3,3)
```

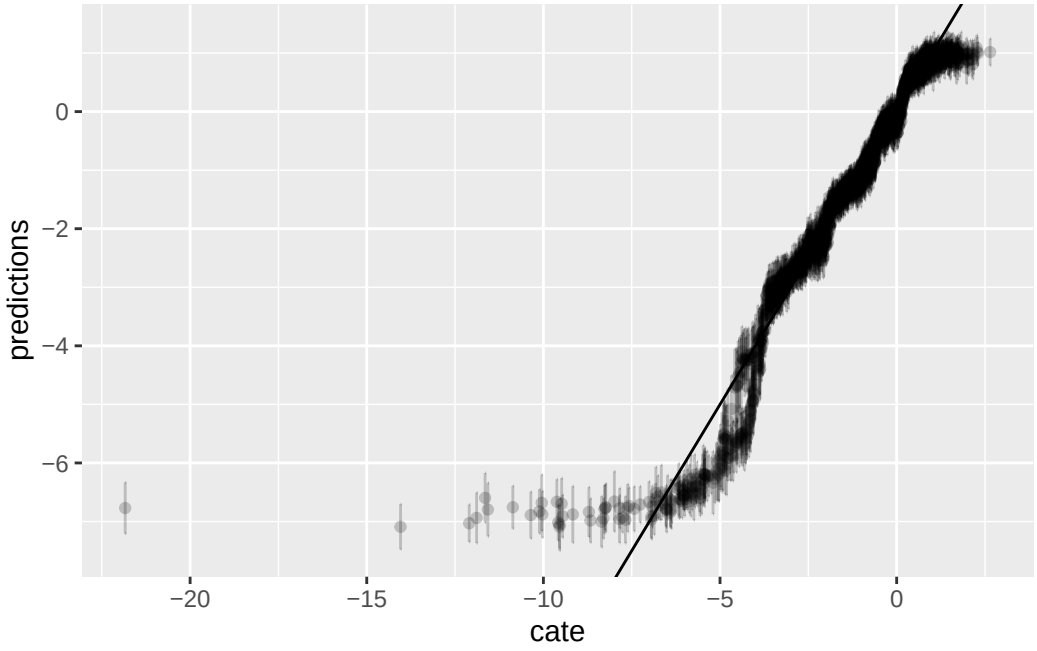



```
graph.data4 <- as.data.frame(tau.hat4)
graph.data4 <- graph.data4 %>%
    mutate(cate=log(data.test$V1^2)) %>%
    mutate(lower=predictions - sigma.hat4, upper=predictions + sigma.hat4)

ggplot(graph.data4, aes(x=cate, y=predictions)) + geom_point()  + geom_abline(slope = 1)
```



```
ggplot() + geom_errorbar(graph.data4, mapping=aes(x=cate,  ymin=lower,ymax=upper), alpha = 0.2) + geom_po
```



17.5. Summary

In all three designs the causal forest recovers the ATE within about one standard error and traces the shape of the true CATE, except in the sparse tails, where it flattens. The linear models recover the average but cannot represent the nonlinear CATEs. With 400 firm dummies in X, the forest needs `mtry = ncol(X)`; at grf's default it rarely gets to split on the one effect modifier and its CATE predictions shrink toward the mean.

Systematic treatment: R · Julia.

18. Synthetic Control in R and Stata

18.1. Synthetic control

Abadie and coauthors have a few papers on synthetic control: Abadie, Diamond, and Hainmueller (2010) and Abadie, Diamond, and Hainmueller (2015), and the survey in Abadie (2021). The key idea is for a case study situation. A single unit, say a state/firm/country is exposed to a shock, and we are interested in the effect of that shock. For example, Card and Krueger (1994) studied the effect of a minimum wage increase in New Jersey. They used diff-in-diff, with Pennsylvania as the control state. The point of synthetic control is that maybe parallel trend assumption does not hold for the control unit. Is Pennsylvania a good control for New Jersey? Does it follow the same trend as New Jersey? Maybe not. Synthetic control is to generate a more comparable control with some combination of multiple control units, which can have a much closer parallel trend as the treated unit, than using any of the control units.

Now, how to construct this synthetic control unit? Basically it is a weighted average of multiple control units. There are two sets of weights to be determined. One set for the control units, one set to determine the importance of predictors. Abadie, Diamond, and Hainmueller (2010) propose to choose a set of w_i's so that the resulting synthetic control best resembles the pre-intervention values for the treated unit of predictors of the outcome variable, subject to the constraints that the weights are non-negative and sum to 1. Another set of weights v_h for predictor importance is determined such that it minimizes the mean squared prediction error (MSPE) of this synthetic control with respect to the outcome in the pre-treatment period. The intuition of these weights are to pick w_i's to make the weighted sum of each predictor as close to the treated unit's value of predictor as close as possible, given a set of v_h's. Then v_h's are determined by how important each predictor is to predict the outcome. We don't use any information post-treatment to determine the weights. But we can use pre-treatment outcome to see the relative importance of each predictor in terms of predictor outcome. We can do this by out-of-sample validation. Basically divide the pre-treatment period into training and validation periods (that's one reason that the pre-treatment period cannot be too short). Iterate the process of choosing w_i and v_h to achieve a minimization of MSPE in the validation period.

18.2. Inference

Since synthetic control is for case study situation, we have essentially one treated unit and one synthetic control. The usual inference method would not work. Abadie et al suggested a permutation based approach. The basic idea is to permute the label of treatment; in other words, suppose one control unit is now a treated unit. Then follow the same procedure to get the synthetic control for that "treated unit"; look at the effect. Now if the treatment is truly for the treated unit only,

then there would not be any effect for all the control units. Therefore we can have a distribution of treatment effects after permutation of treatment label.

18.3. An example

The canonical synthetic control is implemented in both R and Stata (the `Synth` package, CRAN; see Abadie, Diamond and Hainmueller's *JSS* paper for the software description). We use a package called "tidysynth" here to study one of the original examples, which evaluates the impact of Proposition 99 on cigarette consumption in California.

```
library(tidysynth)
library(dplyr)
library(tidyverse)
library(ggplot2)
library(tidyr)
data("smoking")
smoking %>% dplyr::glimpse()
```

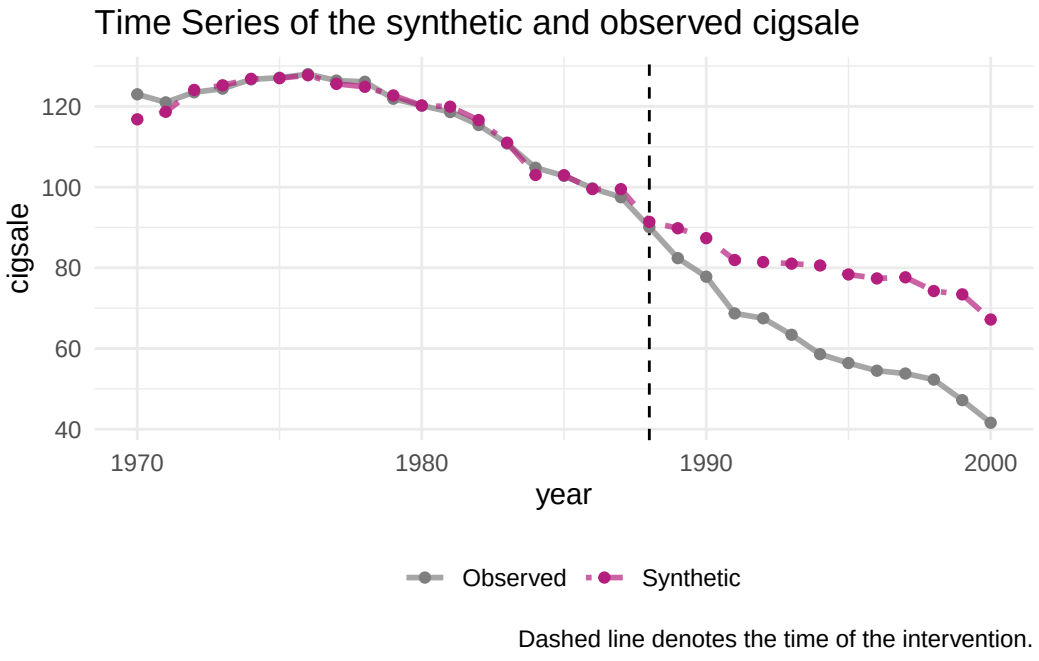
```
Rows: 1,209
Columns: 7
$ state     <chr> "Rhode Island", "Tennessee", "Indiana", "Nevada", "Louisiana~
$ year      <dbl> 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, 1970, ~
$ cigsale   <dbl> 123.9, 99.8, 134.6, 189.5, 115.9, 108.4, 265.7, 93.8, 100.3,~
$ lnincome  <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ beer      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ age15to24 <dbl> 0.1831579, 0.1780438, 0.1765159, 0.1615542, 0.1851852, 0.175~
$ retprice  <dbl> 39.3, 39.9, 30.6, 38.9, 34.3, 38.4, 31.4, 37.3, 36.7, 28.8, ~
```

The panel is 1,209 state-years: 39 states from 1970 to 2000, with per-capita cigarette sales in packs as the outcome and log income, beer sales, the share aged 15–24, and retail price as predictors. California passed Proposition 99 in 1988, so there are 19 pre-treatment years to fit the weights and 12 post-treatment years to measure the effect.

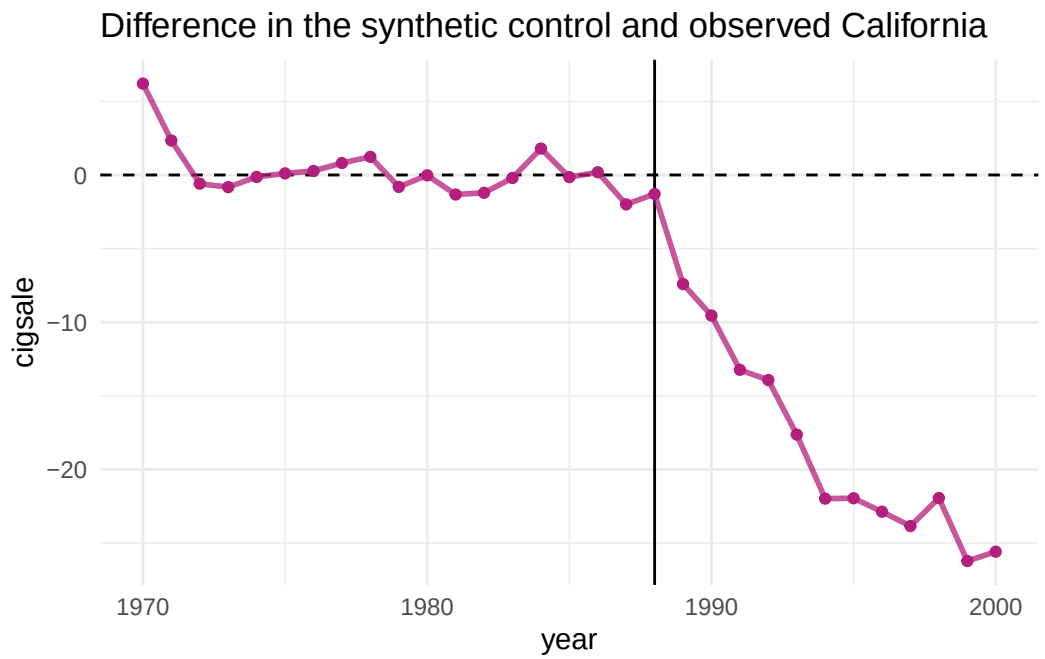
```
smoking_out <-
    smoking %>%
    # initial the synthetic control object
    synthetic_control(outcome = cigsale, # outcome
                      unit = state, # unit index in the panel data
                      time = year, # time index in the panel data
                      i_unit = "California", # unit where the intervention occurred
                      i_time = 1988, # time period when the intervention occurred
                      generate_placebos=T # generate placebo synthetic controls (for inference)
                      ) %>%
    # Generate the aggregate predictors used to fit the weights
    # average log income, retail price of cigarettes, and proportion of the
```

```
    # population between 15 and 24 years of age from 1980 - 1988
  generate_predictor(time_window = 1980:1988,
                     ln_income = mean(lnincome, na.rm = T),
                     ret_price = mean(retprice, na.rm = T),
                     youth = mean(age15to24, na.rm = T)) %>%
    # average beer consumption in the donor pool from 1984 - 1988
  generate_predictor(time_window = 1984:1988,
                     beer_sales = mean(beer, na.rm = T)) %>%
    # Lagged cigarette sales
  generate_predictor(time_window = 1975,
                     cigsale_1975 = cigsale) %>%
  generate_predictor(time_window = 1980,
                     cigsale_1980 = cigsale) %>%
  generate_predictor(time_window = 1988,
                     cigsale_1988 = cigsale) %>%
    # Generate the fitted weights for the synthetic control
  generate_weights(optimization_window = 1970:1988, # time to use in the optimization task
                   margin_ipop = .02,sigf_ipop = 7,bound_ipop = 6 # optimizer options
  ) %>%
  # Generate the synthetic control
  generate_control()

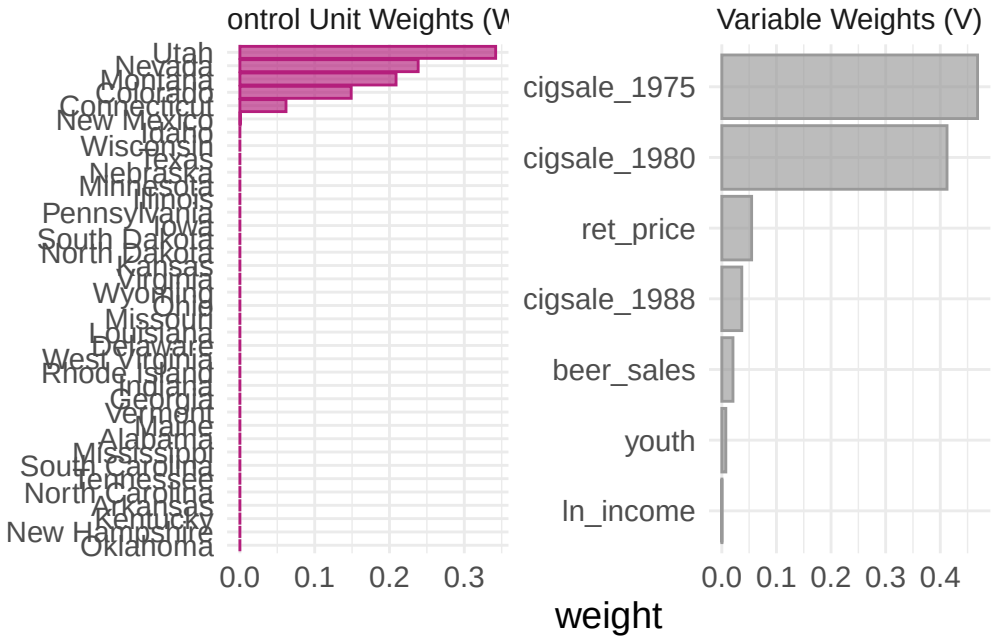
smoking_out %>% plot_trends()
```




```
smoking_out %>% plot_differences()
```




```
smoking_out %>% plot_weights()
```

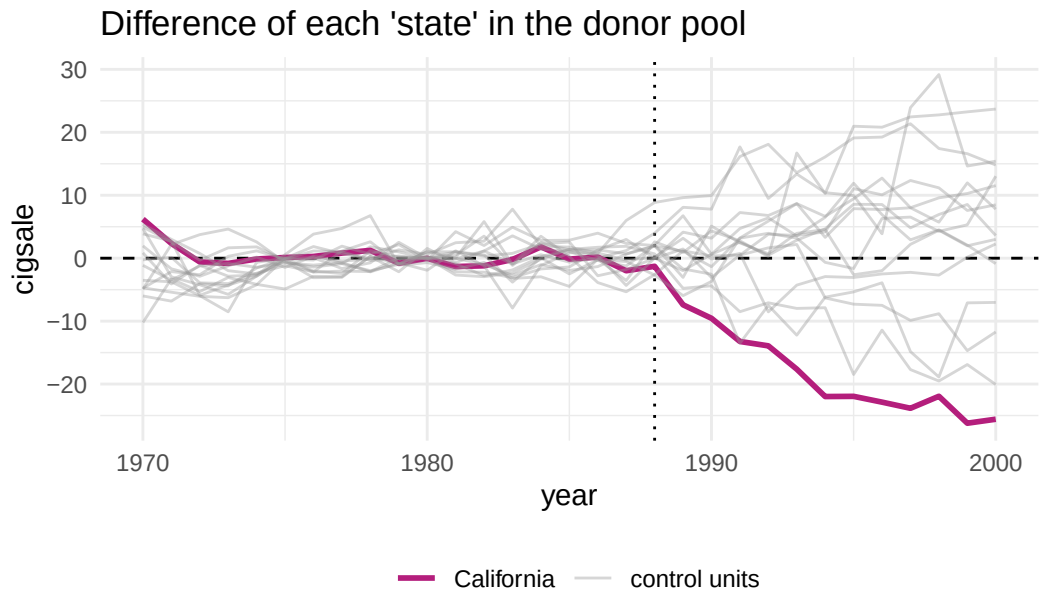



```
smoking_out %>% grab_balance_table()
```

```
# A tibble: 7 x 4
  variable     California synthetic_California donor_sample
  <chr>             <dbl>                <dbl>        <dbl>
```

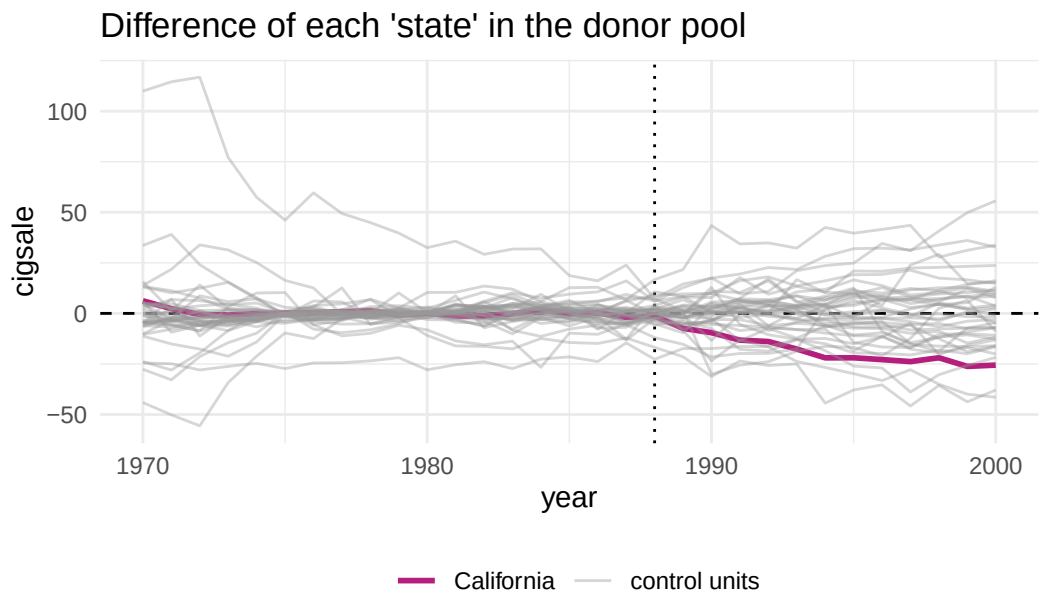
```
1 ln_income        10.1                 9.85         9.83
2 ret_price        89.4                89.4         87.3
3 youth             0.174               0.174        0.173
4 beer_sales       24.3                24.2         23.7
5 cigsale_1975    127.                127.         137.
6 cigsale_1980    120.                120.         138.
7 cigsale_1988     90.1                91.4        114.
```

```
smoking_out %>% plot_placebos()
```

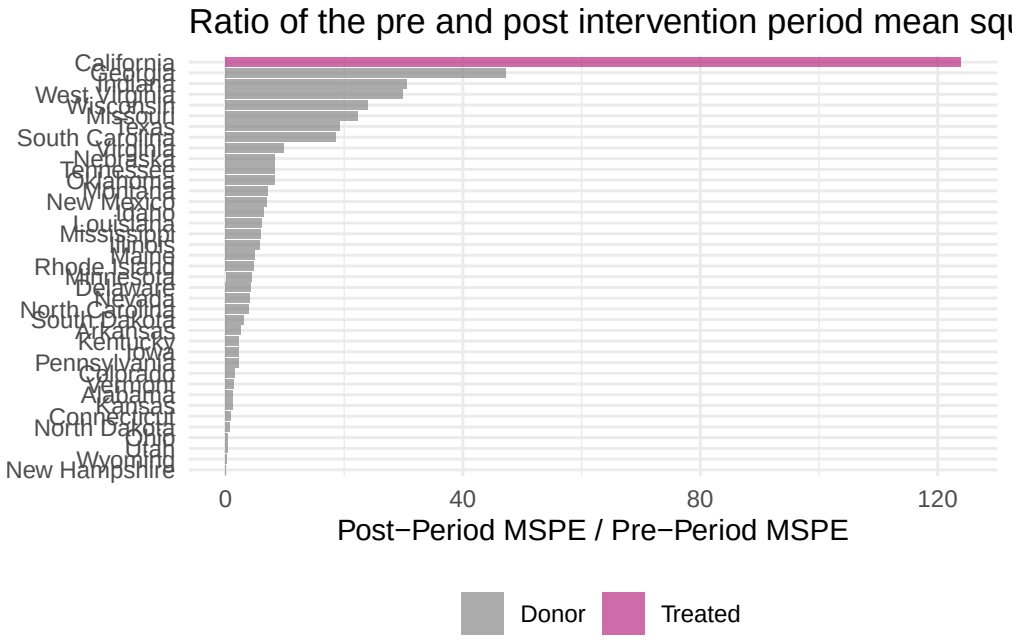


acebo cases with a pre−period RMSPE exceeding two times the treated unit's pre−period RMSPE.

```
smoking_out %>% plot_placebos(prune = FALSE)
```




```
smoking_out %>% plot_mspe_ratio()
```




```
smoking_out %>% grab_significance()
```

```
# A tibble: 39 x 8
   unit_name      type  pre_mspe post_mspe mspe_ratio  rank fishers_exact_pvalue
   <chr>          <chr>    <dbl>     <dbl>      <dbl> <int>                <dbl>
```

```
 1 California     Trea~     3.17     392.      124.       1              0.0256
 2 Georgia        Donor     3.79     179.       47.2      2              0.0513
 3 Indiana        Donor    25.2      770.       30.6      3              0.0769
 4 West Virginia  Donor     9.52     284.       29.8      4              0.103
 5 Wisconsin      Donor    11.1      268.       24.1      5              0.128
 6 Missouri       Donor     3.03      67.8      22.4      6              0.154
 7 Texas          Donor    14.4      277.       19.3      7              0.179
 8 South Carolina Donor    12.6      234.       18.6      8              0.205
 9 Virginia       Donor     9.81      96.4       9.83     9              0.231
10 Nebraska       Donor     6.30      52.9       8.40    10              0.256
# i 29 more rows
# i 1 more variable: z_score <dbl>
```

```
smoking_out %>% grab_synthetic_control()
```

```
# A tibble: 31 x 3
   time_unit real_y synth_y
       <dbl>  <dbl>   <dbl>
 1      1970   123     117.
 2      1971   121     119.
 3      1972   124.    124.
 4      1973   124.    125.
 5      1974   127.    127.
 6      1975   127.    127.
 7      1976   128     128.
 8      1977   126.    126.
 9      1978   126.    125.
10      1979   122.    123.
# i 21 more rows
```

```
smoking_out %>% grab_synthetic_control(placebo = T)
```

```
# A tibble: 1,209 x 5
   .id        .placebo time_unit real_y synth_y
   <chr>         <dbl>     <dbl>  <dbl>   <dbl>
 1 California        0      1970   123     117.
 2 California        0      1971   121     119.
 3 California        0      1972   124.    124.
 4 California        0      1973   124.    125.
 5 California        0      1974   127.    127.
 6 California        0      1975   127.    127.
 7 California        0      1976   128     128.
 8 California        0      1977   126.    126.
 9 California        0      1978   126.    125.
10 California        0      1979   122.    123.
# i 1,199 more rows
```

```
smoking_out %>%
  tidyr::unnest(cols = c(.outcome))
```

```
# A tibble: 1,482 x 50
   .id        .placebo .type   time_unit California Alabama Arkansas Colorado
   <chr>         <dbl> <chr>       <dbl>      <dbl>   <dbl>    <dbl>    <dbl>
 1 California        0 treated      1970       123       NA       NA       NA
 2 California        0 treated      1971       121       NA       NA       NA
 3 California        0 treated      1972       124.      NA       NA       NA
 4 California        0 treated      1973       124.      NA       NA       NA
 5 California        0 treated      1974       127.      NA       NA       NA
 6 California        0 treated      1975       127.      NA       NA       NA
 7 California        0 treated      1976       128       NA       NA       NA
 8 California        0 treated      1977       126.      NA       NA       NA
 9 California        0 treated      1978       126.      NA       NA       NA
10 California        0 treated      1979       122.      NA       NA       NA
# i 1,472 more rows
# i 42 more variables: Connecticut <dbl>, Delaware <dbl>, Georgia <dbl>,
#   Idaho <dbl>, Illinois <dbl>, Indiana <dbl>, Iowa <dbl>, Kansas <dbl>,
#   Kentucky <dbl>, Louisiana <dbl>, Maine <dbl>, Minnesota <dbl>,
#   Mississippi <dbl>, Missouri <dbl>, Montana <dbl>, Nebraska <dbl>,
#   Nevada <dbl>, `New Hampshire` <dbl>, `New Mexico` <dbl>,
#   `North Carolina` <dbl>, `North Dakota` <dbl>, Ohio <dbl>, ...
```

The balance table is where to judge whether the synthetic control is worth trusting, and it does well. Retail price matches to three digits (89.4 against 89.4), the youth share to 0.174 against 0.174, beer sales 24.3 against 24.2. The pre-treatment outcomes match closely too: 1975 cigarette sales 127 against 127 and 1980 sales 120 against 120. Compare those with the raw donor-pool averages in the third column — 137 and 138 — and the value of weighting is obvious. The one predictor that does not match is log income, 10.1 against 9.85, which the optimiser evidently judged less important for predicting the outcome.

The 1988 row is the one to read carefully: 90.1 against 91.4. That is the last pre-treatment year and it is fitted, not predicted, so it is a diagnostic rather than a result.

Inference comes from the placebo table. California's ratio of post- to pre-treatment MSPE is 124, the largest of all 39 units — Georgia is next at 47.2, less than half. Being ranked first of 39 gives a placebo rank of $1/39 = 0.0256$. It is an exact p-value only if the treated label could equally have fallen on any of the 39 states and the same procedure is applied under every assignment; Proposition 99 was not randomly assigned, so we read it as a placebo rank. This is a permutation statement, not a sampling one: it says no untreated state shows a post-1988 divergence from its own synthetic control as large as California's, relative to how well each was fitted beforehand.

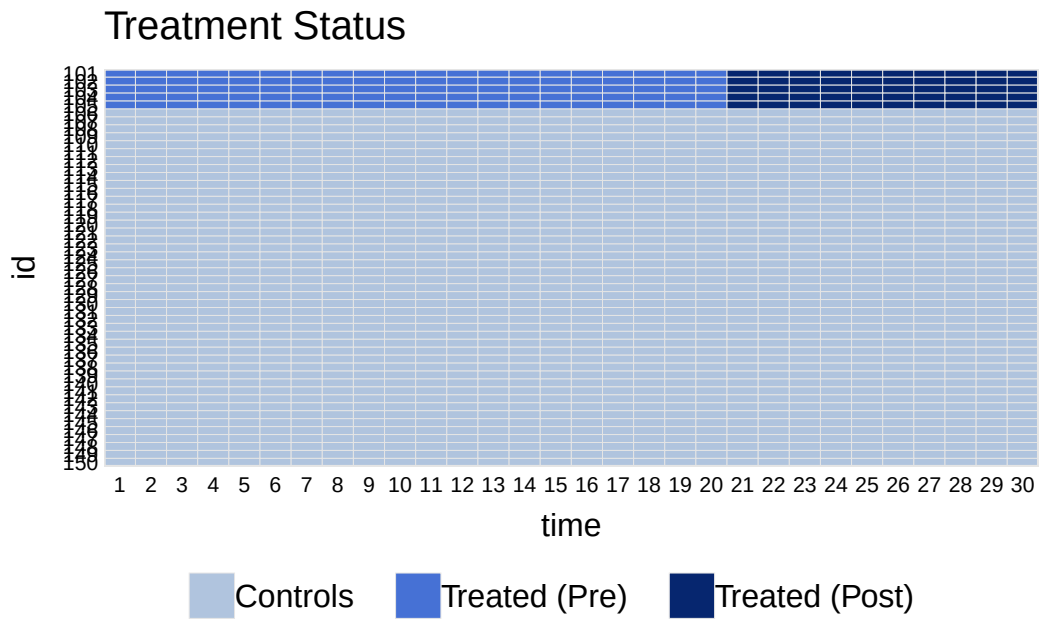
18.4. Generalized Synthetic Control

The idea of generalized SC (Xu 2017) is to combine an interactive fixed effect model and synthetic control. It is basically a panel data synthetic control. It naturally takes multiple treated units and

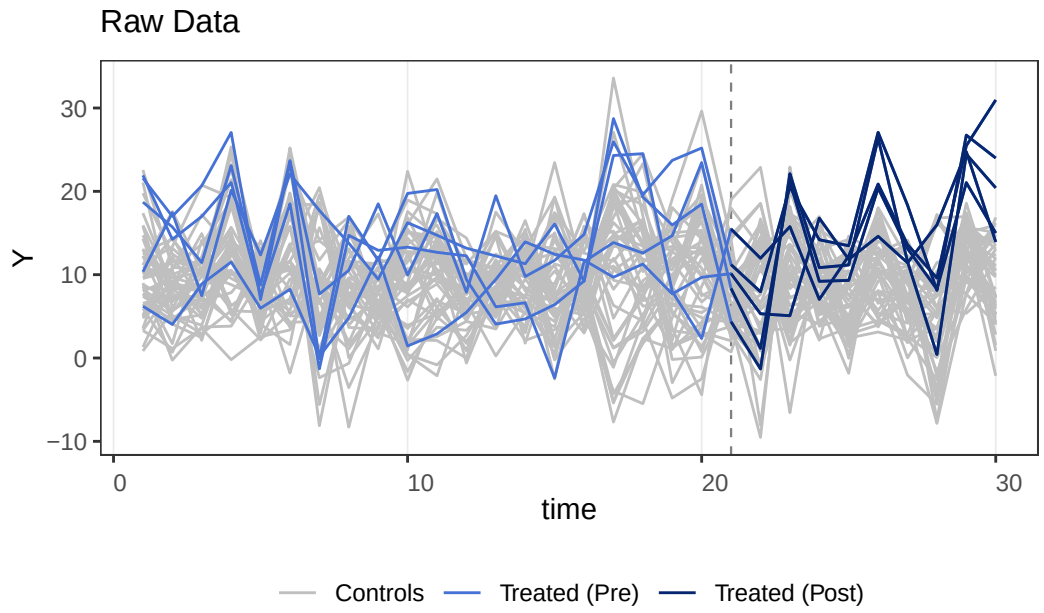
uses bootstrap to get standard errors. It generates ATT with standard errors.

Examples are here: https://yiqingxu.org/packages/gsynth/

```
## for processing C++ code
require(Rcpp)
## for plotting
require(ggplot2)
require(GGally)
## for parallel computing
require(foreach)
require(future)
require(doParallel)
require(abind)
require(lfe)
library(gsynth)
## ## Syntax has been updated since v.1.2.0.
## ## Comments and suggestions -> yiqingxu@stanford.edu.
data(gsynth)
library(panelView)
## ## See bit.ly/panelview4r for more info.
## ## Report bugs -> yiqingxu@stanford.edu.
panelview(Y ~ D, data = simdata,  index = c("id","time"), pre.post = TRUE)
```




```
panelview(Y ~ D, data = simdata,  index = c("id","time"), type = "outcome")
```




```
set.seed(2017)  # cross-validation and the parametric bootstrap are random
system.time(
    out <- gsynth(Y ~ D + X1 + X2, data = simdata, index = c("id","time"), force = "two-way", CV = TRUE,
)
```

```
   user  system elapsed
  1.915   0.003   1.919
```

```
print(out)
```

```
Call:
gsynth(formula = Y ~ D + X1 + X2, data = simdata, index = c("id",
    "time"), force = "two-way", r = c(0, 5), CV = TRUE, se = TRUE,
    nboots = 100, inference = "parametric", parallel = FALSE,
    vartype = "parametric")

Average Treatment Effect on the Treated:
     ATT.avg   S.E. CI.lower CI.upper p.value
[1,]   5.084 0.3186     4.46    5.709       0

   ~ by Period (including Pre-treatment Periods):
         ATT   S.E. CI.lower CI.upper   p.value count
-19  0.05868 0.5978  -1.1129  1.23027 9.218e-01     5
-18 -0.11741 0.5073  -1.1117  0.87688 8.170e-01     5
-17 -0.89029 0.5081  -1.8861  0.10554 7.973e-02     5
-16  1.29462 0.4659   0.3815  2.20773 5.455e-03     5
-15 -1.20350 0.6012  -2.3817 -0.02526 4.529e-02     5
```

```
-14  2.21992 0.6609   0.9245  3.51534 7.830e-04     5
-13 -0.18469 0.6951  -1.5471  1.17773 7.905e-01     5
-12  0.87413 0.4555  -0.0186  1.76687 5.497e-02     5
-11 -0.04824 0.6666  -1.3547  1.25823 9.423e-01     5
-10  0.84503 0.5220  -0.1781  1.86820 1.055e-01     5
-9   0.49754 0.5708  -0.6213  1.61636 3.834e-01     5
-8  -0.38001 0.5344  -1.4275  0.66744 4.770e-01     5
-7  -1.03493 0.4393  -1.8960 -0.17383 1.849e-02     5
-6  -0.81532 0.6829  -2.1538  0.52319 2.325e-01     5
-5  -1.13329 0.6189  -2.3464  0.07981 6.710e-02     5
-4  -0.17733 0.6236  -1.3995  1.04484 7.761e-01     5
-3   0.17671 0.9423  -1.6701  2.02350 8.512e-01     5
-2  -0.14376 0.5887  -1.2976  1.01003 8.071e-01     5
-1  -1.19697 0.5639  -2.3023 -0.09165 3.380e-02     5
0    1.35910 0.8040  -0.2166  2.93484 9.093e-02     5
1    0.34558 0.5311  -0.6953  1.38650 5.152e-01     5
2    0.15709 0.7449  -1.3030  1.61715 8.330e-01     5
3    3.64127 0.6355   2.3957  4.88687 1.007e-08     5
4    2.74975 0.7270   1.3249  4.17460 1.553e-04     5
5    4.45569 0.7928   2.9018  6.00962 1.910e-08     5
6    5.67231 0.7750   4.1534  7.19125 2.494e-13     5
7    8.17039 0.6021   6.9902  9.35055 0.000e+00     5
8    6.57085 0.9564   4.6964  8.44530 6.393e-12     5
9    9.11624 0.4644   8.2060 10.02648 0.000e+00     5
10   9.96205 0.5702   8.8444 11.07964 0.000e+00     5

Coefficients for the Covariates:
    Coef    S.E. CI.lower CI.upper p.value
X1 1.454 0.04465    1.367    1.542       0
X2 3.507 0.04711    3.415    3.600       0
```

```
out$est.att
```

```
            ATT      S.E.   CI.lower    CI.upper      p.value count
-19  0.05868370 0.5977579 -1.1129003  1.23026773 9.217949e-01     5
-18 -0.11741144 0.5073005 -1.1117022  0.87687932 8.169702e-01     5
-17 -0.89029173 0.5080842 -1.8861185  0.10553510 7.973045e-02     5
-16  1.29461792 0.4658831  0.3815038  2.20773202 5.455219e-03     5
-15 -1.20349647 0.6011524 -2.3817335 -0.02525941 4.528664e-02     5
-14  2.21991721 0.6609417  0.9244953  3.51533907 7.830470e-04     5
-13 -0.18468804 0.6951265 -1.5471109  1.17773486 7.904781e-01     5
-12  0.87413414 0.4554872 -0.0186043  1.76687258 5.496926e-02     5
-11 -0.04824208 0.6665777 -1.3547103  1.25822616 9.423052e-01     5
-10  0.84503247 0.5220314 -0.1781303  1.86819527 1.055035e-01     5
-9   0.49754000 0.5708354 -0.6212768  1.61635676 3.834267e-01     5
-8  -0.38001372 0.5344270 -1.4274714  0.66744394 4.770423e-01     5
-7  -1.03492639 0.4393428 -1.8960224 -0.17383034 1.849163e-02     5
```

```
-6  -0.81531903 0.6829265 -2.1538303  0.52319229 2.325325e-01     5
-5  -1.13328625 0.6189378 -2.3463819  0.07980945 6.709784e-02     5
-4  -0.17733004 0.6235683 -1.3995014  1.04484130 7.761196e-01     5
-3   0.17670638 0.9422582 -1.6700858  2.02349856 8.512412e-01     5
-2  -0.14376360 0.5886818 -1.2975588  1.01003159 8.070661e-01     5
-1  -1.19696676 0.5639482 -2.3022848 -0.09164869 3.379774e-02     5
0    1.35910371 0.8039598 -0.2166286  2.93483600 9.093004e-02     5
1    0.34558474 0.5310877 -0.6953281  1.38649760 5.152329e-01     5
2    0.15709192 0.7449396 -1.3029628  1.61714664 8.329819e-01     5
3    3.64127124 0.6355199  2.3956750  4.88686742 1.006713e-08     5
4    2.74975180 0.7269791  1.3248990  4.17460461 1.553011e-04     5
5    4.45569195 0.7928326  2.9017686  6.00961526 1.909953e-08     5
6    5.67231032 0.7749856  4.1533665  7.19125416 2.493561e-13     5
7    8.17038502 0.6021378  6.9902167  9.35055336 0.000000e+00     5
8    6.57084917 0.9563715  4.6963955  8.44530285 6.393108e-12     5
9    9.11624492 0.4644157  8.2060069 10.02648297 0.000000e+00     5
10   9.96204576 0.5702132  8.8444484 11.07964316 0.000000e+00     5
```

```
out$est.avg
```

```
      ATT.avg      S.E. CI.lower CI.upper p.value
[1,] 5.084123 0.3186416 4.459597 5.708649       0
```

```
out$est.beta
```

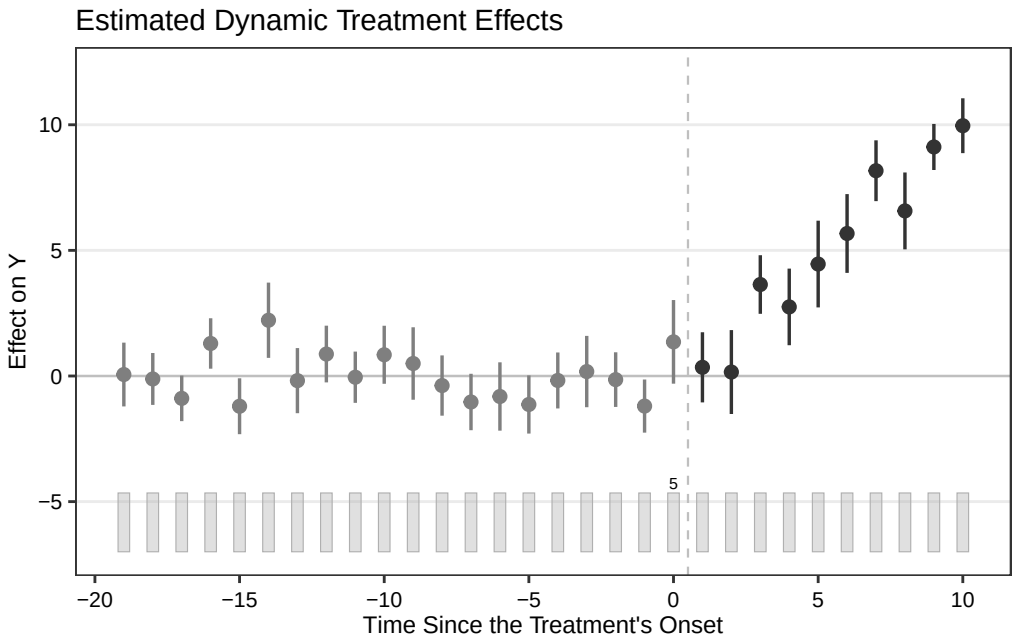
```
       Coef       S.E. CI.lower CI.upper p.value
X1 1.454447 0.04464601 1.366943 1.541952       0
X2 3.507197 0.04710639 3.414870 3.599524       0
```

```
set.seed(2017)
system.time(
out <- gsynth(Y ~ D + X1 + X2, data = simdata, index = c("id","time"),
    force = "two-way", CV = TRUE, r = c(0, 5), se = TRUE,
    inference = "parametric", nboots = 100, parallel = TRUE, cores = 4)
)
```

```
   user  system elapsed
  7.858   0.260  18.031
```

```
# cumuEff was removed in newer gsynth versions (now a wrapper for fect)
# cumu1 <- cumuEff(out, cumu = TRUE, id = NULL, period = c(0,5))
# cumu1$est.catt

plot(out)
```



`simdata` is gsynth's own simulated panel: 5 treated units and 45 controls over 30 periods, with treatment starting in period 21 for the treated group, an interactive fixed-effect structure, and two covariates.

The estimated average ATT is 5.084 with a bootstrap standard error of 0.319 and a 95% interval of $[4.460, 5.709]$. The covariate coefficients come out at 1.454 for `X1` and 3.507 for `X2`. The period-by-period table is the part worth reading, because it includes the pre-treatment periods: those run from -19 to -1 and should be centred on zero if the interactive fixed-effect model has absorbed the common factors. They mostly are — 0.059, -0.117, -0.890, 1.295, -1.203, 2.220 and so on — bouncing around zero with standard errors of about 0.5 to 0.7. Five of them (-16, -15, -14, -7 and -1) are individually significant at 5%, more than the one expected from nineteen independent tests at that level. The pre-period estimates are not independent, though, and the largest, 2.22 at -14, is small next to the post-treatment effects of 3 to 10, so this is a warning about the fit. It does not show that the factor model fails.

18.5. Summary

Synthetic control builds a comparison for a single treated unit as a weighted average of donor units. The weights are chosen to match the treated unit's predictors and pre-treatment outcomes. Inference comes from placebo runs, in which each donor is treated as if it had been treated. Generalized synthetic control combines this with an interactive fixed-effects model. It handles several treated units, bootstraps the standard errors, and reports pre-treatment estimates that should sit near zero.

For California's Proposition 99, the synthetic control matches the pre-1988 predictors and outcomes closely, except log income. California's ratio of post- to pre-treatment fit error is the largest of all 39 states, which gives a permutation p-value of $1/39 = 0.026$. On `gsynth`'s simulated panel, the

average ATT is 5.08 (0.32), and the pre-treatment estimates scatter around zero, though more of them are individually significant than chance would suggest.

Systematic treatment: Julia.

19. Same Data, Different Estimators: A Synthetic Control Comparison

19.1. One panel, four estimators

Here I use the Proposition 99 data to compare four panel estimators on the same outcome matrix: DiD, synthetic DiD, synthetic control, and TASC.

This is the standard California smoking example. There are 39 states, annual cigarette sales per capita from 1970 to 2000, and California is treated after the 1988 anti-smoking law. The useful thing about this example is that the estimators do not give exactly the same answer.

19.2. Four estimators, one panel

The panel is the California Proposition 99 data: annual per-capita cigarette sales in packs for 39 states from 1970 to 2000, arranged as a 39×31 matrix with California in the treated row, 38 donor states, and 19 pre-treatment years before the 1989 tax increase. Every estimator below sees exactly the same matrix; all that differs is how each weights the donors and the pre-periods.

```
using CSV, DataFrames, Statistics
using SynthDiD
using TASC

df    = CSV.read("california_prop99.csv", DataFrame)
# ... pivot to N×T matrix Y, N0 = 38 donors, T0 = 19 pre-treatment years
# (full setup code below)

tau_did  = did_estimate(Y, N0, T0)
tau_sdid = synthdid_estimate(Y, N0, T0)
tau_sc   = sc_estimate(Y, N0, T0)

# TASC: treated unit must be row 1
Y_tasc = vcat(Y[(N0+1):end, :], Y[1:N0, :])
model  = fit_tasc(Y_tasc; d=2, T0=T0, n_em=200, tol=1e-3)
pred   = predict_counterfactual(model, Y_tasc)
```

Estimator	ATT (packs per capita)
Difference-in-differences	−27.35

Estimator	ATT (packs per capita)
Synthetic control	−19.62
Synthetic DiD	−15.60
TASC	−17.11

(These are the values computed live in the companion Julia book's same analysis; the code in this chapter is `eval: false`, so that chapter is the authoritative source for the numbers. The DiD and SC figures also match the published `synthdid` benchmarks for this panel.)

The smallest and largest estimates differ by nearly twelve packs per capita. That is not a rounding issue. It comes from the way each estimator constructs the counterfactual.

19.3. What changes across estimators

DiD (-27.35) assigns equal weight to every control state and every pre-treatment year. California is compared to the average of all 38 donor states. In this data, that average sits well below California before treatment. A level gap alone is not a problem for DiD, which differences it away. What matters is whether the gap stays constant; if California's pre-period path diverges from the donor average, equal weights do not fit and the large DiD estimate is partly that misfit.

Synthetic control (-19.62) replaces equal weights with donor weights chosen to match California's 1970-1988 cigarette sales. Once the pre-period fit is better, the post-treatment gap shrinks. The move from about -27 to about -20 is mostly the price of using a better control group.

Synthetic DiD (-15.60) adds time weights. The estimator can put more weight on the pre-treatment years that are more useful for the post-1988 counterfactual. Here that moves the estimate a bit closer to zero.

TASC (-17.11) does something different. Instead of choosing only weights, it fits a state-space model to the pre-treatment panel:

$$x_t = A\,x_{t-1} + q_t, \quad q_t \sim \mathcal{N}(0, Q) \tag{19.1}$$

$$y_t = H\,x_t + r_t, \quad r_t \sim \mathcal{N}(0, R) \tag{19.2}$$

The matrix A describes how the latent factors move over time. After treatment, the filter continues using the donor panel, and the treated unit's row of H maps the latent state to California's counterfactual. The TASC estimate (-17.11) lands between SC and SDiD in this example.

19.4. The figure

```
# Counterfactual path for each method. omega_* / intercept_* are the donor
# weights and (for DiD/SDiD) intercepts extracted from the fitted objects
# above (e.g. tau_sdid.weights.omega for SynthDiD.jl) — this snippet is
# schematic and omits that extraction step for brevity; it is not meant to
# run as-is (hence eval: false).
cfact_did  = omega_did'  * Y[1:N0, :] .+ intercept_did   # parallel trends
cfact_sdid = omega_sdid' * Y[1:N0, :] .+ intercept_sdid
cfact_sc   = omega_sc'   * Y[1:N0, :]                    # no intercept
cfact_tasc = vec(pred.target)
tasc_se    = sqrt.(max.(vec(pred.variance), 0.0))
```

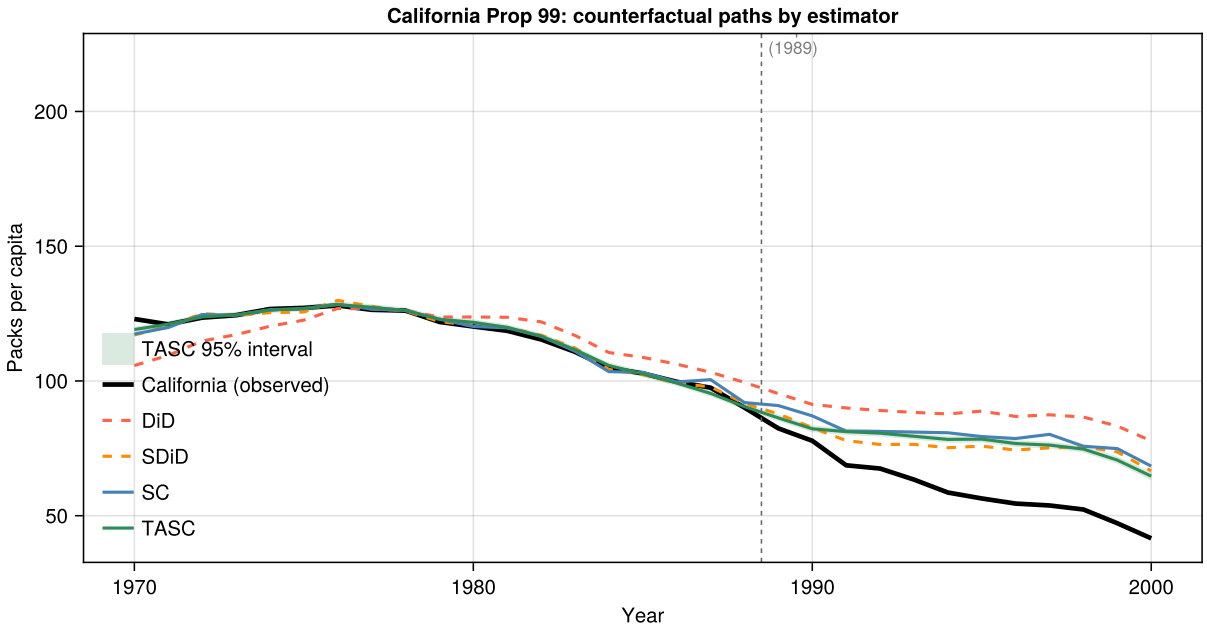


Figure 19.1.: Four counterfactual paths for California. Dashed lines (DiD, SDiD) use pre-period intercepts to enforce parallel trends; solid lines (SC, TASC) fit levels directly. The shaded band is the TASC 95% band for the latent untreated signal, $\hat{Y}_{1t}(0) \pm 1.96\,\hat{\sigma}_t$, with the model parameters fixed at their EM estimates. It leaves out observation noise and parameter uncertainty.

Two things are visible in the figure that the table cannot show.

First, the pre-treatment paths differ. The DiD control average sits well below California before treatment. SC and TASC track California more closely. This is why the DiD post-treatment gap is larger.

Second, the TASC band is narrow before treatment and wider after 1989. That is what I would expect. Before treatment, the smoother can use information from both directions in time. After treatment, the filter has to move forward without California's observed outcomes, so the uncertainty accumulates through $P_{t+1} = AP_tA^\top + Q$.

19.5. What the comparison teaches

Comparison	What mainly differs
DiD vs SC	Unit weights — equal vs optimised donor match
SC vs SDiD	Mainly time weights, but SDiD also adds an intercept and regularizes the unit weights
SC vs TASC	Model — weight fitting vs latent state-space
SC/SDiD vs TASC	Uncertainty — point estimate vs a model-based latent-signal band

Which estimator I would use depends on the setting. DiD is transparent, but the pre-period fit here is poor. SC improves the donor match. SDiD also weights time periods. TASC is useful when the outcome series has persistent dynamics and we want uncertainty for the counterfactual path, not just one average number.

For this panel, the three methods that try to match the pre-period give estimates between about -15.6 and -19.6. DiD is the outlier. I would read this as a problem with equal weights, not as evidence that the effect of Proposition 99 depends on the estimator.

19.6. Setup code

```
df     = CSV.read("california_prop99.csv", DataFrame)
states = sort(unique(df.State))
years  = sort(unique(df.Year))
N, T   = length(states), length(years)

Y_wide      = zeros(N, T)
treated_vec = zeros(Bool, N)
for row in eachrow(df)
    i = findfirst(==(row.State), states)
    t = findfirst(==(row.Year), years)
    Y_wide[i, t] = row.PacksPerCapita
    treated_vec[i] |= (row.treated == 1)
end

ctrl_idx = findall(.!treated_vec)
trt_idx  = findall(treated_vec)
Y  = Y_wide[vcat(ctrl_idx, trt_idx), :]   # donors first
N0 = length(ctrl_idx)                      # 38
T0 = findfirst(==(1989), years) - 1        # 19
```

The `SynthDiD.jl` package provides `did_estimate`, `synthdid_estimate`, and `sc_estimate`. The `TASC.jl` package provides `fit_tasc` and `predict_counterfactual`. Both packages are available on GitHub.

Systematic treatment: Julia.

20. Bartik Instrument and Rotemberg weights

20.1. Bartik Instrument

The Bartik instrument, or shift-share IV (Bartik 1991), combines initial local industry shares with national industry growth rates to construct an instrument for local economic conditions.

We follow Goldsmith-Pinkham, Sorkin, and Swift (2020) (GPSS). Consider the labor-supply elasticity:

$$y_{lt} = D_{lt}\rho + \beta_0 x_{lt} + \epsilon_{lt} \tag{20.1}$$

Here y_{lt} is wage growth in location l, x_{lt} is employment growth, and β_0 is the inverse labor-supply elasticity. The challenge is that x_{lt} is endogenous — unobserved factors may drive both wage and employment growth.

Two accounting identities underlie the instrument:

$$x_{lt} = Z_{lt}G_{lt} = \sum_{k=1}^{K} z_{lkt} g_{lkt} \tag{20.2}$$

and

$$g_{lkt} = g_{kt} + \tilde{g}_{lkt} \tag{20.3}$$

With K industries in location l, employment growth decomposes into industry-level growth rates g_{lkt} weighted by industry shares z_{lkt}. The second identity separates each location-industry growth rate into a national component and a location-specific residual. The Bartik instrument uses initial shares and national growth rates:

$$B_{lt} = Z_{l0}G_t = \sum_{k=1}^{K} z_{lk0} g_{kt} \tag{20.4}$$

The identifying argument is that initial shares are predetermined (a level, not a growth rate) and national industry growth is not driven by any single location.

20.2. Bartik Instrument in special cases

20.2.1. Two industries and one period

GPSS goes through the simplest case: two industries and one period.

$$B_l = z_{l1} g_1 + z_{l2} g_2 \tag{20.5}$$

Since there are only two industries, $z_{l1} = 1 - z_{l2}$. Then we can rewrite the Bartik instrument as

$$B_l = g_2 + (g_1 - g_2) z_{l1} \tag{20.6}$$

From this simplest case, we see that the instrument has only z_{l1}, that is related to location. If the industry share can be considered exogenous, then the instrument is valid. We have to argue that the industry share is exogenous. In this case, the instrument is a linear function of the industry share. Therefore, the Bartik instrument is the same as using the industry share as an instrument.

GPSS shows that just-identified two-stage least squares using a Bartik instrument can in fact be written as GMM with an overidentified set of instruments, where the weight matrix is the outer product of national growth rates. That is, Bartik estimator is equivalent to a GMM estimator, with each location share as one instrument, and the weight matrix is the outer product of national growth rates.

With K industries and T time periods, it's basically using $K \times T$ instruments. Still, Bartik instrument is the same as GMM with all these instruments.

20.3. Rotemberg weights

GPSS decompose the Bartik estimator:

$$\beta_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k \tag{20.7}$$

where

$$\hat{\beta}_k = (Z_k' X^\perp)^{-1} Z_k' Y^\perp \tag{20.8}$$

and

$$\hat{\alpha}_k = \frac{g_k Z_k' X^\perp}{\sum_{k'} g_{k'} Z_{k'}' X^\perp} \tag{20.9}$$

Here $X^\perp$ and $Y^\perp$ are the residualized X and Y, from the regression of X and Y on all other variables (controls).

Rotemberg weights provide a measure of how important each share is. Read the direction carefully: if industry k's own just-identified estimate $\hat{\beta}_k$ is biased by 1 percent, the overall Bartik estimate inherits α_k percent of that bias. So a large α_k means the overall answer is *sensitive* to that industry, not that the industry is itself badly biased. This is what makes the weights useful: they tell you for which instruments the exogeneity assumption has to be defended hardest. (The formal statement is GPSS Corollary D.1, given below.)

20.4. Simulations

Simulations codes are adopted from Kohei Kawaguchi's lecture notes. I modified slightly to fit different scenarios.

Here we simulate a case with 4 industries and 4 periods. We have 1000 locations, 4 periods, and 4 industries. The true β (effect of x on y) is 1.

```
library(ggplot2)
library(tidyverse)
library(modelsummary)
library(kableExtra)
library(estimatr)
set.seed(6)
L <- 1000
T <- 4
K <- 4
beta <- 1
sigma <-c(1, 1, 1)

z_lk0 <- expand_grid(
    l = 1:L,
    k = 1:K
  ) |>
mutate(z_lk0 = runif(length(l)))

g_kt <-  expand_grid(
    t = 1:T,
    k = 1:K
  ) |>
  mutate(g_kt = rnorm(n(), 0, .1))

df <- expand_grid(
    l = 1:L,
    t = 1:T,
    k = 1:K
  ) |>
  left_join(z_lk0, by = c("l", "k")) |>
  left_join(g_kt,  by = c("k", "t")) |>
  mutate(z_lkt = z_lk0 + sigma[1] * runif(length(l)),
    g_lkt = g_kt + sigma[2] * rnorm(length(l),0, .1)
  )

df <- df %>%
  group_by(t, l) |>
  mutate(
    z_lk0 = z_lk0 / sum(z_lk0),
    z_lkt = (z_lkt / sum(z_lkt)),
```

```
    x_lt = sum(z_lkt * g_lkt)
  )

df
```

```
# A tibble: 16,000 x 8
# Groups:   t, l [4,000]
       l     t     k z_lk0     g_kt z_lkt    g_lkt    x_lt
   <int> <int> <int> <dbl>    <dbl> <dbl>    <dbl>   <dbl>
 1     1     1     1 0.277  0.130   0.252  0.115    0.0400
 2     1     1     2 0.428  0.146   0.302  0.291    0.0400
 3     1     1     3 0.121 -0.165   0.321 -0.231    0.0400
 4     1     1     4 0.174 -0.110   0.126 -0.0185   0.0400
 5     1     2     1 0.277 -0.0289  0.160 -0.00619  0.0402
 6     1     2     2 0.428 -0.00130 0.356  0.0560   0.0402
 7     1     2     3 0.121  0.0433  0.272 -0.0590   0.0402
 8     1     2     4 0.174  0.0840  0.212  0.176    0.0402
 9     1     3     1 0.277 -0.101   0.237 -0.0361  -0.0738
10     1     3     2 0.428 -0.152   0.367 -0.126   -0.0738
# i 15,990 more rows
```

The simulation has $L = 1000$ locations, $T = 4$ periods and $K = 4$ industries, 16,000 location-industry-period rows. Baseline shares z_{lk0} are uniform draws normalised to sum to one within each location, national industry growth is $g_{kt} \sim N(0, 0.1^2)$, current shares add a uniform perturbation and are renormalised, and local industry growth is $g_{lkt} = g_{kt} + N(0, 0.1^2)$. Local employment growth is $x_{lt} = \sum_k z_{lkt} g_{lkt}$. The true β is 1.

This assumes we observe the industry shares at time 0 (z_{lk0}), and the industry growth rates (g_{kt}). We also observe the industry shares at time t (z_{lkt}). The employment growth rate at location l and time t (g_{lkt}) is a function of g_{kt} plus some noise. We can calculate the x_{lt}, which is the weighted sum of industry growth rates.

Next we generate the outcome at l, t level. Outcome (wage growth) is a linear function of x (employment growth) at l, t level.

```
df_lt <- df |>
  mutate(g_tilde_lkt = g_lkt - g_kt) |>
  group_by(l, t) |>
  summarise(across(c(x_lt, g_tilde_lkt),  mean),  .groups = "drop") |>
  ungroup() |>
  mutate(y_lt = beta * x_lt + sigma[3] * g_tilde_lkt)
```

The outcome is $y_{lt} = \beta x_{lt} + \sigma_3 \tilde{g}_{lt}$, where $\tilde{g}_{lt}$ is the location's mean deviation of local from national industry growth. That deviation also enters x_{lt}, which is the endogeneity: local shocks drive both employment growth and wage growth.

If we run OLS on raw data, we'll get biased results.

	(1)
(Intercept)	−0.001
	(0.001)
x_lt	1.809
	(0.006)
Num.Obs.	4000
R2	0.965
R2 Adj.	0.965
AIC	−18 854.1
BIC	−18 816.4
Log.Lik.	9433.068
F	27 701.675
RMSE	0.02

```
result_ols <-
  df_lt |>
  lm(formula = y_lt ~ x_lt + factor(t))

modelsummary(result_ols, coef_omit = "factor")
```

With period fixed effects, OLS gives 1.809 with a standard error of 0.006, against a true β of 1, far outside any confidence interval. The endogenous $\tilde{g}_{lt}$ is doing the damage, and within a period it is most of what varies in x_{lt}.

Now we generate the Bartik instrument. We calculate the Bartik instrument as the weighted sum of z_{lk0} and g_{kt}.

```
b_lt <- df |>
  group_by(l, t) |>
  summarise(b_lt = sum(z_lk0 * g_kt)) |>
  ungroup()

df_lt <- df_lt |>
  left_join(b_lt, by = c("l", "t"))


result_first_stage <-  lm(formula = x_lt ~ b_lt + factor(t), data = df_lt)

modelsummary(result_first_stage, coef_omit = "factor")
```

```
result_tsls <- df_lt |>
  estimatr::iv_robust(formula = y_lt ~ x_lt + factor(t) | b_lt + factor(t))
```

	(1)
(Intercept)	0.001
	(0.002)
b_lt	0.438
	(0.034)
Num.Obs.	4000
R2	0.595
R2 Adj.	0.595
AIC	−11 850.3
BIC	−11 812.6
Log.Lik.	5931.161
F	1470.204
RMSE	0.05

	(1)
(Intercept)	0.000
	(0.002)
x_lt	0.951
	(0.074)
Num.Obs.	4000
R2	0.812
R2 Adj.	0.811
AIC	−12 097.6
BIC	−12 059.9
RMSE	0.05

```
modelsummary::modelsummary(result_tsls, coef_omit = "factor")
```

The Bartik instrument recovers the truth: with period fixed effects, the first stage is 0.438 (0.034), and 2SLS gives 0.951 with a standard error of 0.074, against $\beta = 1$, where OLS gave 1.809. The instrument is built from baseline shares and *national* growth rates, neither of which contains the local shock $\tilde{g}_{lt}$, which is what breaks the endogeneity.

20.4.1. Calculate Rotemberg weights

Let's do IV using Bartik instruments, with period fixed effects as the only controls. We use matrix operations to recover the IV estimates, which is the same as the tsls function in R.

```
D_t = model.matrix(~ factor(t) - 1, data = df_lt)   # period dummies
B = as.matrix(cbind(D_t, df_lt$b_lt))
Y = as.matrix(df_lt$y_lt)
X = as.matrix(cbind(D_t, df_lt$x_lt))

iv = round(solve(t(B)%*%X)%*%t(B)%*%Y,3)

## Label and organize results into a data frame
beta.hat = as.data.frame(cbind(c(paste0("period ", 1:T), "x"),iv))
names(beta.hat) = c("Coeff.","Est")
beta.hat
```

```
            Coeff.    Est
factor(t)1 period 1      0
factor(t)2 period 2 -0.001
factor(t)3 period 3 -0.004
factor(t)4 period 4  0.004
                 x  0.951
```

Now we calculate the Rotemberg weights.

```
# This code is the R version of Paul Goldsmith-Pinkham's R code for Rotemberg weights on
# github; which is actually a call from a C++ code by jjchern.
# bw.R calls ComputeAlphaBeta.cpp
# dimensions:  G is KTx1, Z is LTxKT, Y is LTx1, X is LTx1
# so B=ZG is LTx1
# Here I learned the dimensions from the ADH example in Paul's github.
# in ADH example, czone(L) is 722, year(T) is 2, ind(K) is 390.
# so G is 780x1, Z is 1444x780, Y is 1444x1, X is 1444x1
y_lt <- df_lt |>
  select(l,t,y_lt)

Y <- as.matrix(y_lt[,3])

x_lt <- df_lt |>
  select(l,t,x_lt)

X <- as.matrix(x_lt[,3])
#X <- as.matrix(cbind(1,x_lt[,3]))

# global is kt level data set
# # g_data <- df |>
# #   group_by(k,t) |>
# #   summarise(g_kt=sum(g_lkt*z_lkt))
#
# G <- as.matrix(g_data[,3])
```

```
G <- as.matrix(g_kt[,3])

# generate wide Z matrix, LTxKT from df
# z_data <- df |>
#   select(l,t,k,z_lkt) |>
#   pivot_wider(names_from = k, values_from = z_lkt) |>
#   select(-l, -t)

# this makes a wide Z data, which is LTxKT, KT variables are generated by the spread function.
# basically expand the shares to KT variables, filled with 0 if there is no data.
# Use z_lk0 (the INITIAL shares), not z_lkt: the Bartik instrument is B = Z_0 G,
# so the Rotemberg decomposition has to be taken with respect to the same Z that
# builds the instrument. Using z_lkt here would decompose a different estimator.
# z_data <- df |>
#   select(l,t,k,z_lkt) |>
#   mutate(k = str_glue("t{t}_sh_ind_{k}") ) |>
#   spread(k, z_lkt, fill = 0)

z_data <- df |>
  select(l,t,k,z_lk0) |>
  mutate(k = str_glue("t{t}_sh_ind_{k}") ) |>
  pivot_wider(names_from = k, values_from = z_lk0, values_fill = 0)

Z <- as.matrix(z_data[,-c(1,2)])

# Residualize Y and X on the controls before forming the weights (GPSS). The
# controls are the period fixed effects, so the residualization demeans within
# each period. This is what makes the decomposition exact, and it keeps
# period-level shifts in X and Y out of the industry-period weights.
period <- df_lt$t
Yp <- Y - ave(as.numeric(Y), period)
Xp <- X - ave(as.numeric(X), period)
beta = (t(Z) %*% Yp)/(t(Z) %*% Xp)
alpha=(diag(as.numeric(G)) %*% t(Z)  %*% Xp) / as.numeric((t(G) %*% t(Z) %*% Xp))

bw <- tibble(g_kt, alpha=as.numeric(alpha), beta=as.numeric(beta)) |>
  mutate(product=alpha*beta)

bw
```

```
# A tibble: 16 x 6
       t     k     g_kt     alpha   beta   product
   <int> <int>    <dbl>     <dbl>  <dbl>     <dbl>
 1     1     1  0.130    0.130     0.597  0.0779
 2     1     2  0.146    0.265     1.19   0.314
 3     1     3 -0.165    0.223     0.871  0.194
```

```
 4     1     4 -0.110    0.161     1.07   0.172
 5     2     1 -0.0289   0.0121    1.17   0.0142
 6     2     2 -0.00130  0.000331  0.305  0.000101
 7     2     3  0.0433   0.00627   0.486  0.00305
 8     2     4  0.0840   0.0443    0.940  0.0417
 9     3     1 -0.101   -0.000949  2.86  -0.00271
10     3     2 -0.152    0.0885    0.920  0.0815
11     3     3 -0.0569  -0.0277    1.14  -0.0316
12     3     4 -0.0994  -0.00866  -0.514  0.00445
13     4     1  0.0266  -0.0143    0.764 -0.0109
14     4     2  0.139    0.0365   -0.555 -0.0202
15     4     3  0.0263  -0.0130    0.852 -0.0111
16     4     4  0.127    0.0980    1.27   0.124
```

```
sum(bw$product)
```

```
[1] 0.9510114
```

The sum of the product of α and β should be the same as the Bartik estimator, and it is: 0.9510 against the 2SLS estimate of 0.951. The decomposition is exact, not approximate, provided β_k is not rounded before the products are formed and X and Y are residualized on the same controls as the 2SLS regression.

Reading the table, the weights are far from uniform across the sixteen industry-period cells. Period 1 carries most of the weight — its four α values are 0.130, 0.265, 0.223 and 0.161, summing to 0.78. Period 1 also has the largest national shocks, $|g_{k1}|$ between 0.13 and 0.17, and α_k is proportional to g_k. Five cells in periods 3 and 4 have small negative weights. Because the residualization removes period means, the weights reflect share variation across locations within a period, not the period-level swings in X. The just-identified β_k estimates range from -0.56 to 2.86, but the four heavily weighted period-1 cells give 0.60, 1.19, 0.87 and 1.07; the values far from 1 all come from cells with $|\alpha| < 0.04$, where the industry-period instrument has a weak first stage.

Note there is no restriction for α, other than the sum is 1. Therefore it can be negative or positive, and individual weights can exceed 1, as long as they sum to 1. In GPSS the Rotemberg weights routinely take negative values and values greater than 1; this is a feature used to flag problematic instruments.

20.5. Why is it good to have Rotemberg weights?

GPSS Corollary D.1. shows the interpretation of the Rotemberg weights.

The percentage of bias can be written as:

$$\frac{E(\tilde{\beta})}{\beta_0} = \sum_k \alpha_k \frac{E(\tilde{\beta}_k)}{\beta_0} \tag{20.10}$$

This is to say the α_k is the share of the bias of the estimate of β that comes from the bias of the estimate of β_k. That is, how sensitive it is to the bias from industry k.

In my simulations, all the industries are random, therefore the weights are random too. But in empirical studies, they are not random, most likely dominated by some industries. This is why it is important to have Rotemberg weights. It tells us which industries are important in the estimation of the effect of interest.

20.6. What if you are using a reduced form?

If you use the reduced form, something like:

$$y_{lt} = D_{lt}\rho + \beta_0 B_{lt} + \epsilon_{lt} \tag{20.11}$$

where B_{lt} is the Bartik instrument, constructed in such a way:

$$B_{lt} = Z_{l0}G_t = \sum_{k=1}^{K} z_{lk0} g_{kt} \tag{20.12}$$

Then I think the Rotemberg weights should be calculated as:

$$\beta_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k \tag{20.13}$$

where

$$\hat{\beta}_k = (Z_k' B^\perp)^{-1} Z_k' Y^\perp \tag{20.14}$$

and

$$\hat{\alpha}_k = \frac{g_k Z_k' B^\perp}{\sum_{k'} g_{k'} Z_{k'}' B^\perp} \tag{20.15}$$

where $B^\perp$ and $Y^\perp$ are residualized B and Y.

```
result_reduced <- df_lt |>
  lm(formula = y_lt ~  b_lt + factor(t))

modelsummary::modelsummary(result_reduced, coef_omit = "factor")
```

Now we calculate the Rotemberg weights for the reduced form, for our simulated data.

```
B <- as.matrix(b_lt[,3])

# Residualize Y and B on the controls (the period fixed effects, so demean
# within period), as in the GPSS reduced-form formula above, so the
# decomposition is exact.
Yp <- Y - ave(as.numeric(Y), period)
Bp <- B - ave(as.numeric(B), period)
```

	(1)
(Intercept)	0.001
	(0.003)
b_lt	0.417
	(0.065)
Num.Obs.	4000
R2	0.290
R2 Adj.	0.289
AIC	−6790.4
BIC	−6752.6
Log.Lik.	3401.176
F	407.556
RMSE	0.10

```
beta = (t(Z) %*% Yp)/(t(Z) %*% Bp)
alpha=(diag(as.numeric(G)) %*% t(Z)  %*% Bp) / as.numeric((t(G) %*% t(Z) %*% Bp))

bw <- tibble(g_kt, alpha=as.numeric(alpha), beta=as.numeric(beta)) |>
  mutate(product = alpha * beta)

bw
```

```
# A tibble: 16 x 6
       t     k     g_kt     alpha   beta    product
   <int> <int>    <dbl>     <dbl>  <dbl>      <dbl>
 1     1     1  0.130    0.163     0.209  0.0341
 2     1     2  0.146    0.214     0.643  0.138
 3     1     3 -0.165    0.269     0.316  0.0850
 4     1     4 -0.110    0.120     0.629  0.0755
 5     2     1 -0.0289   0.0151    0.411  0.00620
 6     2     2 -0.00130  0.000335  0.132  0.0000442
 7     2     3  0.0433   0.00778   0.172  0.00134
 8     2     4  0.0840   0.0502    0.363  0.0183
 9     3     1 -0.101   -0.00182   0.654 -0.00119
10     3     2 -0.152    0.0756    0.472  0.0357
11     3     3 -0.0569  -0.0256    0.540 -0.0138
12     3     4 -0.0994  -0.00300  -0.651  0.00195
13     4     1  0.0266  -0.0142    0.337 -0.00479
14     4     2  0.139    0.0836   -0.106 -0.00887
15     4     3  0.0263  -0.0141    0.342 -0.00484
16     4     4  0.127    0.0600    0.906  0.0544
```

```
sum(bw$product)
```

```
[1] 0.4165574
```

The reduced-form decomposition sums to 0.4166 rather than 0.9510, and that is not an error: it decomposes the reduced-form coefficient (the effect of the instrument on the outcome, 0.417), not the 2SLS coefficient. The two differ by the first stage, 0.438.

20.7. Summary

The Bartik instrument combines baseline local industry shares with national industry growth. In the framework of GPSS, which this chapter follows, its validity rests on the shares being exogenous. A complementary framework treats the shocks as the source of identification: if the national growth rates g_{kt} are as good as randomly assigned across many industries, the shares need not be exogenous (Borusyak, Hull, and Jaravel 2022), and inference should then use exposure-robust standard errors that allow for correlation across locations with similar shares (Adão, Kolesár, and Morales 2019). Our simulation draws g_{kt} independently across industries and periods, which is the shock-exogeneity setup, but with only $K = 4$ industries the shock-level argument has little to work with. The Rotemberg weights are the diagnostic for the share view: they show which industry shares the estimate leans on. GPSS show that 2SLS with a Bartik instrument is GMM with each industry share as a separate instrument. The Rotemberg weights split the estimate into the just-identified estimates of the individual industries. Each weight gives the share of an industry's bias that passes into the overall estimate, so the weights show which shares the argument depends on.

In the simulation, with period fixed effects, OLS gives 1.809 against a true 1, and the Bartik 2SLS gives 0.951 (0.074). The Rotemberg decomposition reproduces the 2SLS estimate exactly. Its weights are far from uniform, and some are negative. In the reduced form the decomposition applies to the effect of the instrument on the outcome, so it sums to a different number.

Systematic treatment: R · Julia.

21. Extended TWFE and other DiD estimators

21.1. Staggered DiD without treatment reversal

This blog is adopted from the tutorial of paneltools package. https://yiqingxu.org/tutorials/panel.html#Installing_Packages

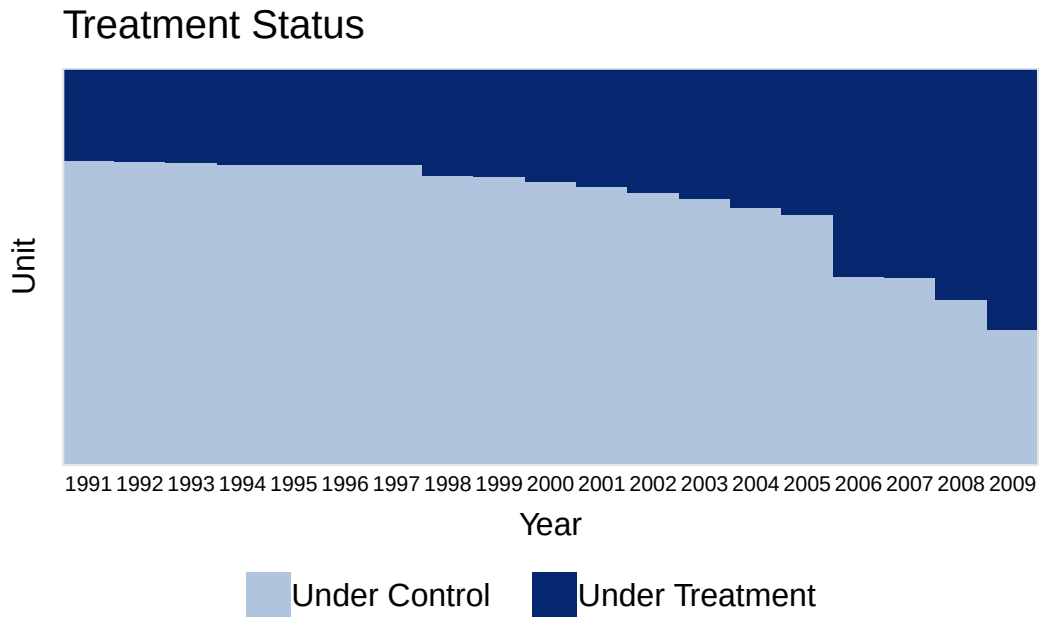
I added Wooldridge's ETWFE, which is my favorite estimator with staggered DiD. It is easy to understand and implement. It also works with nonlinear model such as logit and poisson.

```
#devtools:: install_github("xuyiqing/paneltools")
library(paneltools)
data(paneltools)
library(tidyverse)
hh2019 <- as_tibble(hh2019)
hh2019
```

```
# A tibble: 22,971 x 4
     bfs  year nat_rate_ord indirect
   <dbl> <dbl>        <dbl>    <dbl>
 1     1  1991         0           0
 2     1  1992         0           0
 3     1  1993         0           0
 4     1  1994         3.45        0
 5     1  1995         0           0
 6     1  1996         0           0
 7     1  1997         0           0
 8     1  1998         0           0
 9     1  1999         0           0
10     1  2000         0           0
# i 22,961 more rows
```

This data is from Hainmueller and Hangartner (2019), in which the authors investigate the effects of indirect democracy (versus direct democracy) on naturalization rates in Switzerland using municipality-year level panel data from 1991 to 2009. The study shows that a switch from direct to indirect democracy increased naturalization rates by an average of 1.22 percentage points (Model 1 of Table 1 in the paper).

```
#devtools:: install_github("xuyiqing/panelView")
library(panelView)
panelview(nat_rate_ord ~ indirect, data = hh2019, index = c("bfs","year"),
  xlab = "Year", ylab = "Unit", display.all = T,
  gridOff = TRUE, by.timing = TRUE)
```



The panel is 22,971 municipality-years: 1,209 Swiss municipalities observed from 1991 to 2009. The outcome `nat_rate_ord` is the ordinary naturalization rate and `indirect` marks the switch from direct to indirect democracy, adopted at different dates by different municipalities — staggered treatment timing.

21.1.1. TWFE

First we do a TWFE model on it.

```
library(fixest)
model_twfe <- feols(nat_rate_ord~indirect|bfs+year,
                      data=hh2019, cluster = "bfs")
summary( model_twfe)
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 22,971
Fixed-effects: bfs: 1,209,  year: 19
Standard-errors: Clustered (bfs)
         Estimate Std. Error t value   Pr(>|t|)
indirect  1.33932   0.186525 7.18039 1.2117e-12 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.09541     Adj. R2: 0.152719
                Within R2: 0.005173
```

TWFE gives 1.33932 with a municipality-clustered standard error of 0.1865, close to the 1.22 the original authors report. Keep that number in view: everything below is an attempt to work out whether it can be trusted.

TWFE has issues when there is staggered adoption and heterogeneous treatment effects.

Goodman-Bacon (2021) demonstrates that the two-way fixed-effects (TWFE) estimator in a staggered adoption setting can be represented as a weighted average of all possible 2x2 difference-in-differences (DID) estimates between different cohorts. However, when treatment effects change over time heterogeneously across cohorts, the "forbidden" comparisons that use post-treatment data from early adopters as controls for late adopters may introduce bias in the TWFE estimator. We employ the procedure outlined in Goodman-Bacon (2021) and decompose the TWFE estimate using the command bacon in the bacondecomp package.

"Negative weights" can be the other issue. Basically it is a weighted average of comparison between different cohorts. The weights can be negative in a TWFE model.

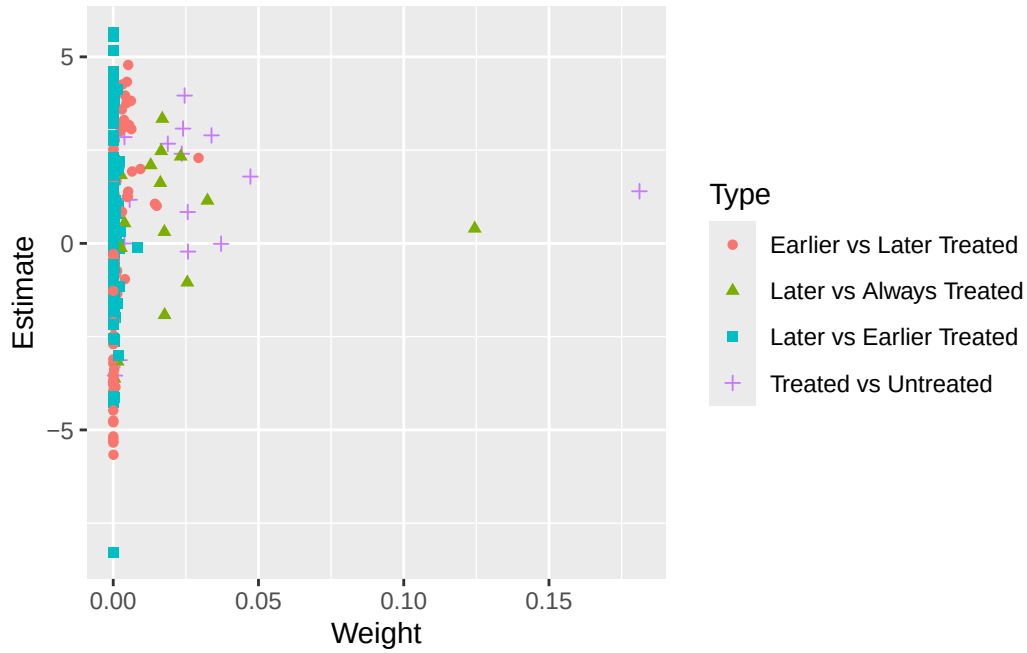
```
library(bacondecomp)
df_bacon <- bacon(nat_rate_ord~indirect,
                  data = hh2019,
                  id_var = "bfs",
                  time_var = "year")
```

```
                    type  weight avg_est
1 Earlier vs Later Treated 0.17605 1.97771
2  Later vs Always Treated 0.31446 0.75233
3 Later vs Earlier Treated 0.05170 0.32310
4     Treated vs Untreated 0.45779 1.61180
```

```
coef_bacon <- sum(df_bacon$estimate * df_bacon$weight)
coef_bacon
```

```
[1] 1.339325
```

```
ggplot(df_bacon) +
   aes(x = weight, y = estimate, shape = factor(type), color = factor(type)) +
   labs(x = "Weight", y = "Estimate", shape = "Type", color = 'Type') +
   geom_point()
```



The decomposition sums to 1.339325, reproducing the TWFE estimate exactly. It splits into four kinds of 2x2 comparison:

comparison	weight	average estimate
Treated vs Untreated	0.458	1.612
Later vs Always Treated	0.314	0.752
Earlier vs Later Treated	0.176	1.978
Later vs Earlier Treated	0.052	0.323

The clean comparisons — treated against never-treated — carry 46% of the weight and give 1.61. The forbidden ones, which use already-treated units as controls, carry 0.314 + 0.052 = 37% and give much smaller numbers, 0.75 and 0.32. That is what pulls the TWFE estimate down from 1.61 to 1.34.

The problem shown in Bacon decomposition here is the comparison between "Later" and "Earlier Treated". It's a "forbidden" comparison.

TWFE with event history type of graph.

```
df.use <- hh2019 |>
  na.omit()

df <- as.data.frame(
  hh2019 |>
    group_by(bfs) |>
    mutate(treatment_mean = mean(indirect,na.rm = TRUE))
)
df.use <- df[which(df$treatment_mean<1),]
```

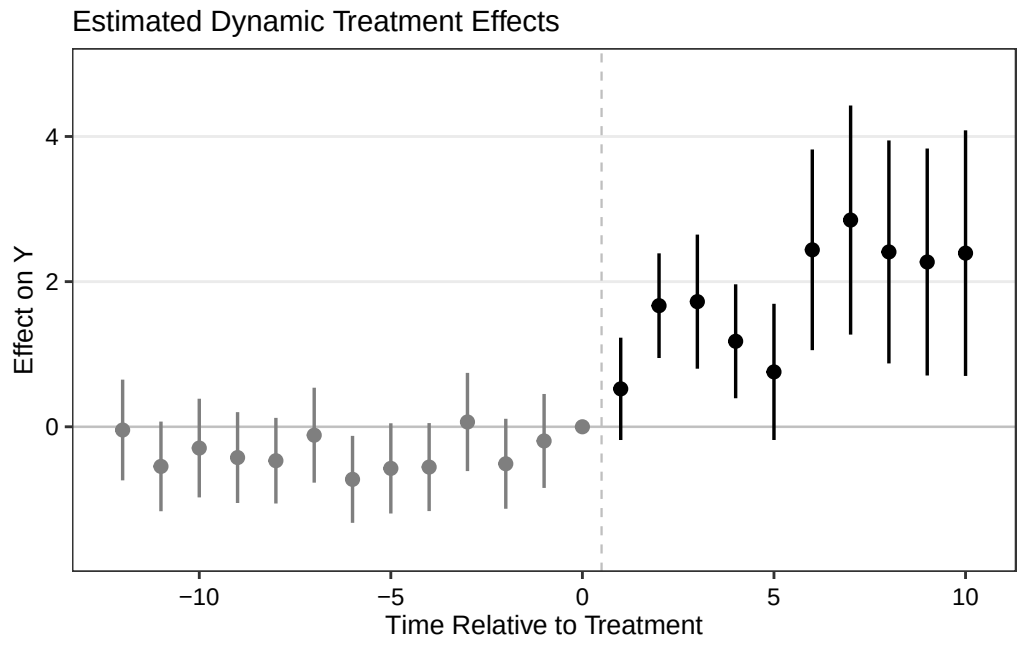
```r
df.twfe <- as_tibble(get.cohort(df.use,D="indirect",index=c("bfs","year"),start0 = TRUE))

# time_to_treatment is NA when the unit is always treated, or never treated.  It needs to be set to an a
df.twfe$treat <- as.numeric(df.twfe$treatment_mean>0) # drop always treated units
df.twfe[which(is.na(df.twfe$Time_to_Treatment)),'Time_to_Treatment'] <- 0 # can be an arbitrary value
twfe.est <- feols(nat_rate_ord ~ i(Time_to_Treatment, treat, ref = -1)| bfs + year,
                  data = df.twfe, cluster = "bfs")
summary(twfe.est)
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: bfs: 926,  year: 19
Standard-errors: Clustered (bfs)
                              Estimate Std. Error   t value   Pr(>|t|)
Time_to_Treatment::-18:treat -0.320651   0.632702 -0.506796 6.1242e-01
Time_to_Treatment::-17:treat -0.118540   0.391148 -0.303058 7.6191e-01
Time_to_Treatment::-16:treat -0.395632   0.402921 -0.981910 3.2640e-01
Time_to_Treatment::-15:treat -0.410976   0.343899 -1.195048 2.3237e-01
Time_to_Treatment::-14:treat  0.244512   0.369127  0.662406 5.0788e-01
Time_to_Treatment::-13:treat -0.044310   0.354017 -0.125164 9.0042e-01
Time_to_Treatment::-12:treat -0.545891   0.315172 -1.732042 8.3600e-02 .
Time_to_Treatment::-11:treat -0.293138   0.346835 -0.845180 3.9823e-01
Time_to_Treatment::-10:treat -0.423981   0.319375 -1.327534 1.8466e-01
Time_to_Treatment::-9:treat  -0.467114   0.301017 -1.551784 1.2106e-01
Time_to_Treatment::-8:treat  -0.115633   0.333659 -0.346561 7.2900e-01
Time_to_Treatment::-7:treat  -0.724290   0.305727 -2.369075 1.8037e-02 *
Time_to_Treatment::-6:treat  -0.572950   0.317153 -1.806540 7.1159e-02 .
Time_to_Treatment::-5:treat  -0.554993   0.309246 -1.794667 7.3033e-02 .
Time_to_Treatment::-4:treat   0.066209   0.345103  0.191853 8.4790e-01
Time_to_Treatment::-3:treat  -0.509530   0.316296 -1.610929 1.0754e-01
Time_to_Treatment::-2:treat  -0.195091   0.330220 -0.590793 5.5480e-01
Time_to_Treatment::0:treat    0.522715   0.359646  1.453415 1.4645e-01
Time_to_Treatment::1:treat    1.668834   0.367919  4.535879 6.4907e-06 ***
Time_to_Treatment::2:treat    1.724230   0.471363  3.657968 2.6858e-04 ***
Time_to_Treatment::3:treat    1.178336   0.400559  2.941732 3.3452e-03 **
Time_to_Treatment::4:treat    0.756323   0.478697  1.579962 1.1446e-01
Time_to_Treatment::5:treat    2.438229   0.705451  3.456269 5.7263e-04 ***
Time_to_Treatment::6:treat    2.848590   0.805315  3.537238 4.2442e-04 ***
Time_to_Treatment::7:treat    2.409359   0.783716  3.074275 2.1721e-03 **
Time_to_Treatment::8:treat    2.270428   0.797869  2.845617 4.5305e-03 **
Time_to_Treatment::9:treat    2.392678   0.863222  2.771799 5.6867e-03 **
Time_to_Treatment::10:treat   2.002836   0.953654  2.100170 3.5984e-02 *
Time_to_Treatment::11:treat   0.842590   0.560248  1.503961 1.3293e-01
Time_to_Treatment::12:treat   3.831918   2.300422  1.665745 9.6103e-02 .
Time_to_Treatment::13:treat   4.436437   2.322318  1.910349 5.6397e-02 .
Time_to_Treatment::14:treat   4.756977   3.485523  1.364781 1.7265e-01
```

```
Time_to_Treatment::15:treat   1.541967   1.415284  1.089511 2.7621e-01
Time_to_Treatment::16:treat   7.168943   4.844016  1.479959 1.3922e-01
Time_to_Treatment::17:treat   6.490854   2.726526  2.380632 1.7485e-02 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.42406     Adj. R2: 0.146601
                Within R2: 0.012807
```

```
twfe.output <- as.matrix(twfe.est$coeftable)
twfe.output <- as.data.frame(twfe.output)
twfe.output$Time <- c(c(-18:-2),c(0:17))+1
p.twfe <- esplot(twfe.output,Period = 'Time',Estimate = 'Estimate',
                              SE = 'Std. Error', xlim = c(-12,10))
p.twfe
```



The event study and every estimator below use `df.use`, which drops the 283 always-treated municipalities. The static TWFE above used the full panel, where "later vs always treated" comparisons carry 31% of the weight. On the same restricted sample, static TWFE gives:

```
model_twfe_use <- fixest::feols(nat_rate_ord ~ indirect | bfs + year,
                                data = df.use, cluster = "bfs")
summary(model_twfe_use)
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: bfs: 926,  year: 19
Standard-errors: Clustered (bfs)
```

```
         Estimate Std. Error t value  Pr(>|t|)
indirect  1.60858   0.195296 8.23661 6.009e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.43613     Adj. R2: 0.143692
                Within R2: 0.007415
```

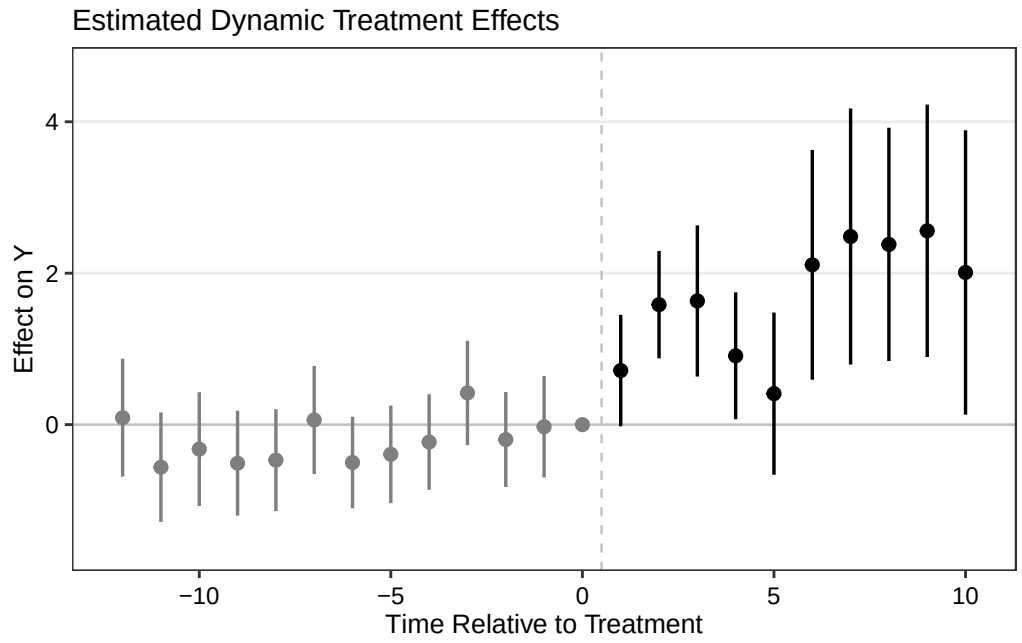
21.1.2. Sun and Abraham

The Sun and Abraham (2021) estimator is a weighted average of the average treatment effect (ATT) estimates for each cohort, obtained from a TWFE regression that includes cohort dummies fully interacted with indicators of relative time to the treatment's onset. The iw estimator is robust to heterogeneous treatment effects (HTE) and can be implemented using the command sunab in the package fixest.

```
df.sa <- df.twfe
df.sa[which(is.na(df.sa$FirstTreat)),"FirstTreat"] <- 1000 # replace NA with an arbitrary number
model.sa.1 <- feols(nat_rate_ord~sunab(FirstTreat,year)|bfs+year,
                    data = df.sa, cluster = "bfs")
summary(model.sa.1,agg = "ATT")
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: bfs: 926,  year: 19
Standard-errors: Clustered (bfs)
    Estimate Std. Error t value   Pr(>|t|)
ATT   1.3309   0.287971 4.62163 4.3474e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.34795     Adj. R2: 0.163888
                Within R2: 0.046484
```

```
sa.output <- as.matrix(model.sa.1$coeftable)
sa.output <- as.data.frame(sa.output)
sa.output$Time <- c(c(-18:-2),c(0:17))+1
p.sa <- esplot(sa.output,Period = 'Time',Estimate = 'Estimate',
                            SE = 'Std. Error', xlim = c(-12,10))
p.sa
```



21.2. Callaway and Sant'Anna

Callaway and Sant'Anna (2021) build the estimator from group-time average treatment effects $ATT(g,t)$ — the effect on cohort g at time t — each identified by comparing the treated cohort to a never-treated or not-yet-treated comparison group. The $ATT(g,t)$ are estimated by outcome regression, inverse-probability weighting, or a doubly-robust combination of the two, rather than from a single fully-interacted TWFE regression. These group-time effects are then aggregated with weights into summary parameters (overall, event-study, or by cohort). The estimator is robust to heterogeneous treatment effects (HTE) and can be implemented with the `att_gt` function in the `did` package.

```
library(did)
df.cs <- df.twfe
df.cs[which(is.na(df.cs$FirstTreat)),"FirstTreat"] <- 0 # replace NA with 0
cs.est.1 <- att_gt(yname = "nat_rate_ord",
                   gname = "FirstTreat",
                   idname = "bfs",
                   tname = "year",
                   xformla = ~1,
                   control_group = "nevertreated",
                   allow_unbalanced_panel = TRUE,
                   data = df.cs,
                   est_method = "reg")
cs.est.att.1 <- aggte(cs.est.1, type = "simple", na.rm=T, bstrap = F)
print(cs.est.att.1)
```

```
Call:
aggte(MP = cs.est.1, type = "simple", na.rm = T, bstrap = F)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multiple Time Pe
230, 2021. <https://doi.org/10.1016/j.jeconom.2020.12.001>, <https://arxiv.org/abs/1803.09015>

    ATT    Std. Error     [ 95%  Conf. Int.]
 1.3309        0.3019     0.7392      1.9226 *


---
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```

```
cs.att.1 <- aggte(cs.est.1, type = "dynamic",
                  bstrap=FALSE, cband=FALSE, na.rm=T)
print(cs.att.1)
```

```
Call:
aggte(MP = cs.est.1, type = "dynamic", na.rm = T, bstrap = FALSE,
    cband = FALSE)

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna.  "Difference-in-Differences with Multiple Time Pe
230, 2021. <https://doi.org/10.1016/j.jeconom.2020.12.001>, <https://arxiv.org/abs/1803.09015>

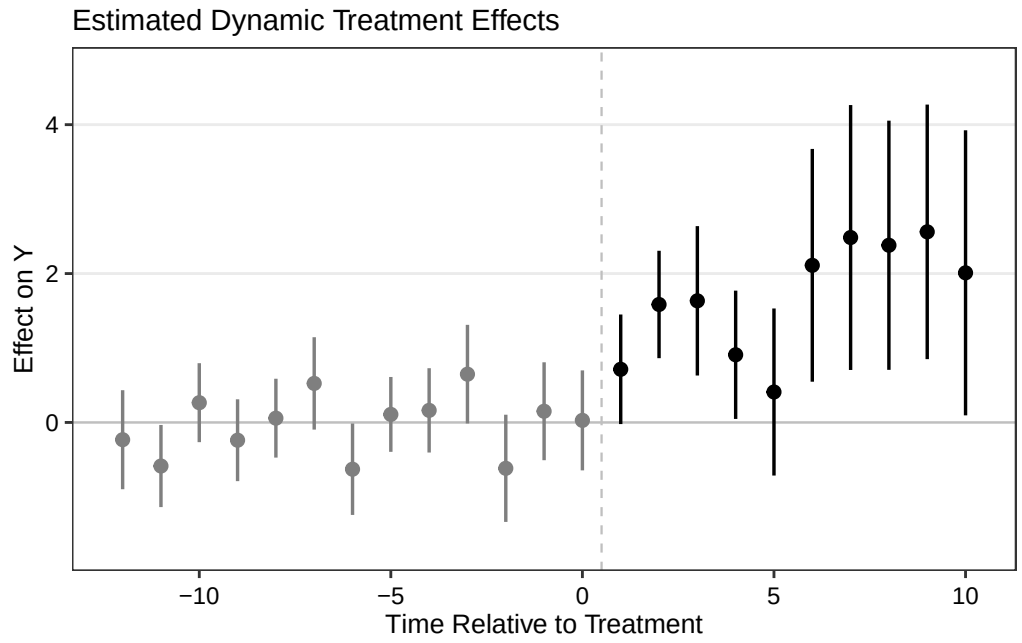

Overall summary of ATT's based on event-study/dynamic aggregation:
    ATT    Std. Error     [ 95%  Conf. Int.]
 1.2205        0.6314    -0.0171       2.458


Dynamic Effects:
 Event time Estimate Std. Error [95% Pointwise  Conf. Band]
        -17   0.1350     0.6467         -1.1324      1.4025
        -16  -0.0839     0.3583         -0.7861      0.6183
        -15   0.1467     0.3964         -0.6303      0.9236
        -14   0.7509     0.3219          0.1200      1.3818 *
        -13  -0.2318     0.3392         -0.8967      0.4331
        -12  -0.5851     0.2814         -1.1367     -0.0335 *
        -11   0.2656     0.2705         -0.2647      0.7958
        -10  -0.2386     0.2805         -0.7883      0.3111
         -9   0.0576     0.2702         -0.4721      0.5872
         -8   0.5244     0.3165         -0.0959      1.1448
```

```
         -7  -0.6282     0.3132         -1.2420     -0.0144 *
         -6   0.1081     0.2565         -0.3946      0.6108
         -5   0.1623     0.2890         -0.4041      0.7287
         -4   0.6485     0.3380         -0.0139      1.3109
         -3  -0.6161     0.3674         -1.3362      0.1040
         -2   0.1505     0.3355         -0.5071      0.8081
         -1   0.0279     0.3425         -0.6435      0.6992
          0   0.7141     0.3754         -0.0217      1.4500
          1   1.5846     0.3681          0.8632      2.3060 *
          2   1.6337     0.5118          0.6306      2.6369 *
          3   0.9090     0.4396          0.0473      1.7707 *
          4   0.4091     0.5725         -0.7130      1.5311
          5   2.1106     0.7976          0.5473      3.6739 *
          6   2.4842     0.9078          0.7051      4.2634 *
          7   2.3802     0.8538          0.7068      4.0535 *
          8   2.5602     0.8723          0.8506      4.2699 *
          9   2.0100     0.9769          0.0953      3.9246 *
         10   1.6391     1.0257         -0.3712      3.6494
         11   0.0298     0.6100         -1.1658      1.2253
         12   0.6942     2.8043         -4.8021      6.1904
         13   1.1355     2.3284         -3.4281      5.6991
         14   1.3625     3.4131         -5.3270      8.0519
         15  -2.0699     1.7160         -5.4331      1.2933
         16   2.2743     3.6182         -4.8172      9.3658
         17   0.1072     4.5338         -8.7789      8.9934
---
Signif. codes: `*' confidence band does not cover 0

Control Group:  Never Treated,  Anticipation Periods:  0
Estimation Method:  Outcome Regression
```

```
cs.output <- cbind.data.frame(Estimate = cs.att.1$att.egt,
                              SE = cs.att.1$se.egt,
                              time = cs.att.1$egt + 1)
p.cs.1 <- esplot(cs.output,Period = 'time',Estimate = 'Estimate',
                               SE = 'SE', xlim = c(-12,10))
p.cs.1
```



21.2.1. Imai, Kim, and Wang

PanelMatch is an R package implementing a set of methodological tools proposed by Imai, Kim, and Wang (2023) that enables researchers to apply matching methods for causal inference on time-series cross-sectional data with binary treatments. The package includes implementations of matching methods based on propensity scores and Mahalanobis distance, as well as weighting methods. PanelMatch enables users to easily calculate a variety of possible quantities of interest, along with standard errors. The software is flexible, allowing users to tune the matching, refinement, and estimation procedures with a large number of parameters. The package also offers a variety of visualization and diagnostic tools for researchers to better understand their data and assess their results.

```
library(PanelMatch)
df.pm <- as.data.frame(df.twfe)
# we need to convert the unit and time indicator to integer
df.pm[,"bfs"] <- as.integer(as.factor(df.pm[,"bfs"]))
df.pm[,"year"] <- as.integer(as.factor(df.pm[,"year"]))
df.pm <- df.pm[,c("bfs","year","nat_rate_ord","indirect")]

# Create PanelData object (new API)
pd <- PanelData(df.pm, unit.id = "bfs", time.id = "year",
                treatment = "indirect", outcome = "nat_rate_ord")

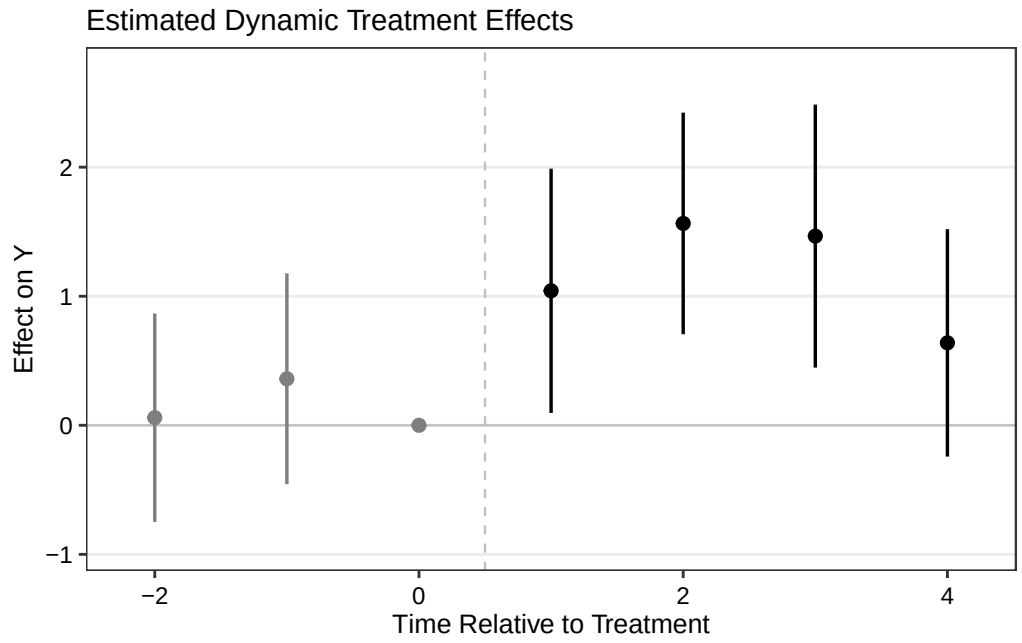
PM.results <- PanelMatch(lag=3,
                         panel.data = pd,
                         refinement.method = "none",
                         qoi = "att",
```

```
                          lead = c(0:3),
                          match.missing = TRUE)

## For pre-treatment dynamic effects
PM.results.placebo <- PanelMatch(lag=3,
                          panel.data = pd,
                          refinement.method = "none",
                          qoi = "att",
                          lead = c(0:3),
                          match.missing = TRUE,
                          placebo.test = TRUE)
PE.results.pool <- PanelEstimate(PM.results, panel.data = pd, pooled = TRUE)
summary(PE.results.pool)
```

```
     estimate std.error      2.5%    97.5%
[1,] 1.177674 0.3549721 0.4810424 1.927232
```

```
PE.results <- PanelEstimate(PM.results, panel.data = pd)
PE.results.placebo <- placebo_test(PM.results.placebo, panel.data = pd, plot = F)
est_lead <- as.vector(estimates(PE.results))
est_lag <- tryCatch(as.vector(estimates(PE.results.placebo)),
                    error = function(e) as.vector(PE.results.placebo$estimates))
sd_lead <- tryCatch(apply(PE.results$bootstrapped.estimates,2,sd),
                    error = function(e) rep(NA, length(est_lead)))
sd_lag <- tryCatch(apply(PE.results.placebo$bootstrapped.estimates,2,sd),
                   error = function(e) rep(NA, length(est_lag)))
coef <- c(est_lag, 0, est_lead)
sd <- c(sd_lag, 0, sd_lead)
pm.output <- cbind.data.frame(ATT=coef, se=sd, t=c(-2:4))
p.pm <- esplot(data = pm.output,Period = 't',
               Estimate = 'ATT',SE = 'se')
p.pm
```



21.2.2. Liu, Wang and Xu

Liu, Wang, and Xu (2024) proposes a method for estimating the fixed effect counterfactual model by utilizing the untreated observations. It is essentially an imputation method that uses the untreated observations to estimate the counterfactual outcome for the treated.

```
library(fect)
out.fect <- fect(nat_rate_ord~indirect, data = df,
                 index = c("bfs","year"),
                 method = 'fe', se = TRUE)
print(out.fect$est.avg)
```

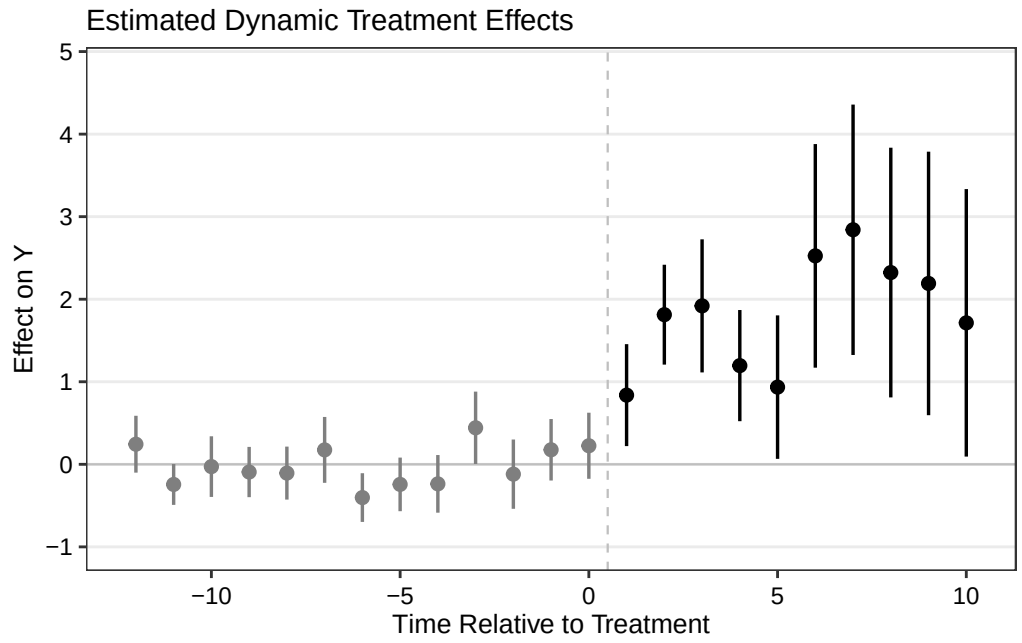
```
      ATT.avg      S.E. CI.lower CI.upper      p.value
[1,] 1.506416 0.1963763 1.121526 1.891307 1.709743e-14
```

```
fect.output <- as.matrix(out.fect$est.att)
print(fect.output)
```

```
            ATT      S.E.     CI.lower    CI.upper      p.value count
-17 -0.01985591 0.5285222 -1.055740442  1.01602863 9.700315e-01    89
-16  0.07269638 0.2486939 -0.414734614  0.56012737 7.700475e-01   157
-15 -0.21365489 0.2513343 -0.706261016  0.27895124 3.952792e-01   162
-14 -0.13153052 0.1599320 -0.444991446  0.18193041 4.108406e-01   350
-13  0.46902947 0.2181738  0.041416651  0.89664230 3.157125e-02   371
-12  0.24357664 0.1756552 -0.100701131  0.58785442 1.655408e-01   398
-11 -0.24399485 0.1261746 -0.491292431  0.00330274 5.313920e-02   417
```

```
-10 -0.02765866 0.1873913 -0.394938811  0.33962149 8.826597e-01   434
-9  -0.09347047 0.1553449 -0.397940853  0.21099991 5.473762e-01   451
-8  -0.10638473 0.1637700 -0.427367986  0.21459853 5.159516e-01   464
-7   0.17476094 0.2034793 -0.224051183  0.57357306 3.904159e-01   468
-6  -0.40297604 0.1504948 -0.697940357 -0.10801172 7.413516e-03   503
-5  -0.24326436 0.1655841 -0.567803309  0.08127460 1.417980e-01   503
-4  -0.23687833 0.1782019 -0.586147702  0.11239103 1.837592e-01   503
-3   0.44244127 0.2236361  0.004122541  0.88076000 4.788381e-02   503
-2  -0.11927690 0.2141008 -0.538906824  0.30035302 5.774549e-01   508
-1   0.17612194 0.1899998 -0.196270903  0.54851478 3.539482e-01   512
0    0.22491529 0.2042250 -0.175358266  0.62518884 2.707611e-01   514
1    0.83813813 0.3150520  0.220647498  1.45562876 7.806724e-03   514
2    1.81274486 0.3086483  1.207805220  2.41768449 4.275341e-09   425
3    1.91941052 0.4112889  1.113299078  2.72552196 3.058999e-06   357
4    1.19567464 0.3435498  0.522329444  1.86901984 5.007528e-04   352
5    0.93520961 0.4433128  0.066332488  1.80408673 3.489342e-02   164
6    2.52517848 0.6910255  1.170793357  3.87956359 2.579370e-04   143
7    2.84129736 0.7738412  1.324596459  4.35799827 2.409612e-04   116
8    2.32288547 0.7716810  0.810418523  3.83535242 2.611076e-03    97
9    2.19105962 0.8143868  0.594890868  3.78722836 7.135764e-03    80
10   1.71382906 0.8266891  0.093548199  3.33410991 3.816074e-02    63
11   1.07651548 0.9751064 -0.834658041  2.98768900 2.695940e-01    50
12  -0.30828629 0.5646155 -1.414912321  0.79833973 5.850583e-01    46
13   0.35079211 2.0383375 -3.644275938  4.34586016 8.633611e-01    11
14   0.81696815 2.4455674 -3.976255833  5.61019213 7.383337e-01    11
15   0.97427612 3.5384443 -5.960947352  7.90949960 7.830548e-01    11
16  -2.49173503 1.6630658 -5.751284139  0.76781407 1.340610e-01    11
17   1.42334105 4.9749120 -8.327307220 11.17398933 7.747986e-01     6
18  -0.01384226 4.7854270 -9.393106917  9.36542239 9.976921e-01     2
```

```r
fect.output <- as.data.frame(fect.output)
fect.output$Time <- c(-17:18)
p.fect <- esplot(fect.output,Period = 'Time',Estimate = 'ATT',
                 SE = 'S.E.',CI.lower = "CI.lower",
                 CI.upper = 'CI.upper',xlim = c(-12,10))
p.fect
```



Borusyak, Jaravel, and Spiess (2024) with package didimputation almost gives the same result as Liu, Wang, and Xu (2024).

21.2.3. Wooldridge

Wooldridge (2025), also called extended TWFE (ETWFE), is to allow more flexibility to make TWFE work properly. The problem with TWFE is that we do not specify it flexibly enough. If we allow heterogeneous treatment effect, then we cannot expect the canonical TWFE to work properly.

```
df.wool <- df.twfe
df.wool[which(is.na(df.wool$FirstTreat)),"FirstTreat"] <- 0 # replace NA with 0

library(etwfe)

model.wool = etwfe(
  fml  = nat_rate_ord ~ 1, # outcome ~ controls
  tvar = year,        # time variable
  gvar = FirstTreat, # group variable
  data = df.wool,       # dataset
  vcov = ~bfs,  # vcov adjustment (here: clustered)
  cgroup = "never"
)
model.wool
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
```

```
Fixed-effects: FirstTreat: 16,  year: 19
Standard-errors: Clustered (bfs)
                                    Estimate Std. Error  t value Pr(>|t|)
.Dtreat:FirstTreat::1992:year::1992  2.18863    1.36923  1.59844 0.110287
.Dtreat:FirstTreat::1992:year::1993 -4.18925    3.10938 -1.34730 0.178214
.Dtreat:FirstTreat::1992:year::1994 -2.50997    1.63352 -1.53654 0.124747
.Dtreat:FirstTreat::1992:year::1995 -5.23221    3.11282 -1.68086 0.093129 .
.Dtreat:FirstTreat::1992:year::1996 -5.42014    3.11207 -1.74165 0.081902 .
.Dtreat:FirstTreat::1992:year::1997 -5.15575    3.11248 -1.65648 0.097965 .
.Dtreat:FirstTreat::1992:year::1998 -5.08210    3.11345 -1.63231 0.102955
.Dtreat:FirstTreat::1992:year::1999 -5.13290    3.11447 -1.64808 0.099676 .
... 262 coefficients remaining (display them with summary() or use
argument n)
... 15 variables were removed because of collinearity
(.Dtreat:FirstTreat::1992:year::1991,
.Dtreat:FirstTreat::1993:year::1992 and 13 others [full set in
$collin.var])
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.6642     Adj. R2: 0.088474
               Within R2: 0.040642
```

```
emfx(model.wool)
```

```
 .Dtreat Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
    TRUE     1.33      0.288 4.62   <0.001 18.0 0.766    1.9

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

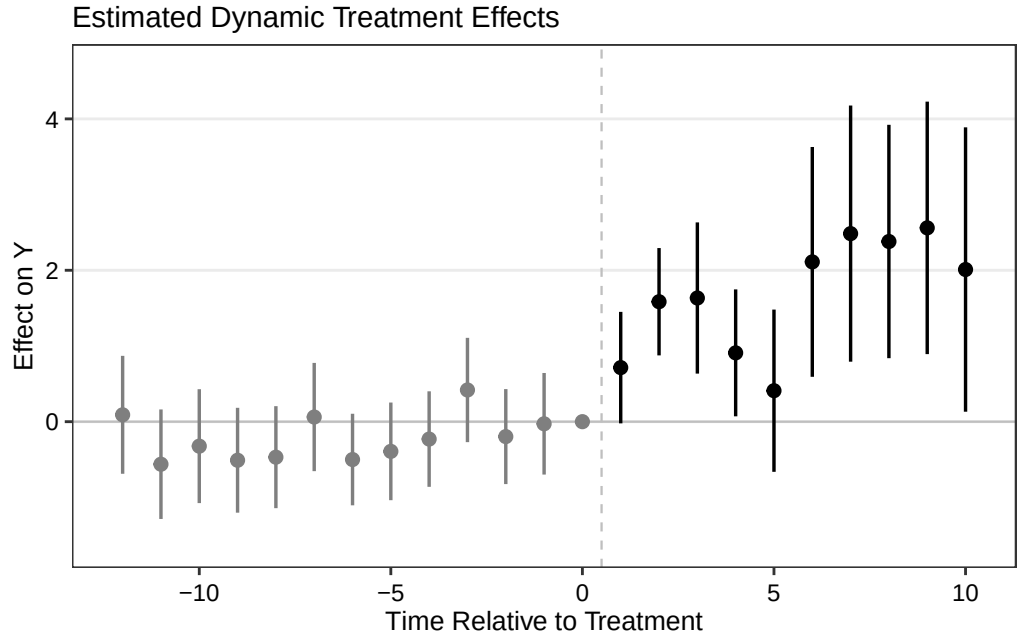
```
mod_es = emfx(model.wool, type = "event", post_only=FALSE)
mod_es
```

```
 event Estimate Std. Error       z Pr(>|z|)   S  2.5 % 97.5 %
   -18   -0.266      0.789 -0.3371    0.736 0.4 -1.812  1.280
   -17    0.178      0.503  0.3538    0.723 0.5 -0.808  1.164
   -16    0.114      0.519  0.2198    0.826 0.3 -0.903  1.131
   -15   -0.104      0.408 -0.2550    0.799 0.3 -0.903  0.695
   -14    0.410      0.413  0.9927    0.321 1.6 -0.400  1.220
--- 26 rows omitted. See ?print.marginaleffects ---
    13    1.135      1.998  0.5683    0.570 0.8 -2.781  5.052
    14    1.362      3.129  0.4355    0.663 0.6 -4.770  7.495
    15   -2.070      1.439 -1.4387    0.150 2.7 -4.890  0.750
```

```
    16    2.274      3.080  0.7385    0.460 1.1 -3.762  8.311
    17    0.107      4.576  0.0234    0.981 0.0 -8.861  9.076
Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```
wool.output <- mod_es |>
  filter(event != -1) |>
  mutate(Time = event + 1) |>
  as.data.frame()

p.wool <- esplot(wool.output,Period = 'Time',Estimate = 'estimate',
                 SE = 'std.error', xlim = c(-12,10))
p.wool
```



Note that in order to get pre-event estimates, we need to set "cgroup=never" in the etwfe function. Otherwise, ETWFE will only give us post-event estimates. Under the default setting, the control group comprises the "not-yet" treated units, therefore all pre-treatment effects are mechanistically set to zero. By setting "cgroup=never", we only use never treated units as the control group. This way we can get pre-treatment estimates, but we are not using all available information for the control group, therefore the estimates may not be as efficient as the default setting. That cost is visible in the two runs below: the never-treated version gives an ATT of 1.33 with a standard error of 0.288, while the default not-yet-treated version gives 1.51 with a standard error of 0.188 — a little over a third narrower.

Also we don't have control variable here. If we have, it is very easy to incorporate into ETWFE. Wooldridge used the Mundlak device to incorporate control variables, which is a very simple and intuitive way to do it.

If we do the default setting and plot only post-event estimates:

```
model.wool2 = etwfe(
  fml  = nat_rate_ord ~ 1, # outcome ~ controls
  tvar = year,        # time variable
  gvar = FirstTreat, # group variable
  data = df.wool,       # dataset
  vcov = ~bfs  # vcov adjustment (here: clustered)
)
model.wool2
```

```
OLS estimation, Dep. Var.: nat_rate_ord
Observations: 17,594
Fixed-effects: FirstTreat: 16,  year: 19
Standard-errors: Clustered (bfs)
                                    Estimate Std. Error  t value Pr(>|t|)
.Dtreat:FirstTreat::1992:year::1992  1.99670    1.35883  1.46943 0.142057
.Dtreat:FirstTreat::1992:year::1993 -4.52291    3.09436 -1.46166 0.144174
.Dtreat:FirstTreat::1992:year::1994 -2.62350    1.61688 -1.62257 0.105022
.Dtreat:FirstTreat::1992:year::1995 -5.17241    3.09454 -1.67146 0.094969 .
.Dtreat:FirstTreat::1992:year::1996 -5.08905    3.09404 -1.64479 0.100353
.Dtreat:FirstTreat::1992:year::1997 -5.17232    3.09482 -1.67128 0.095004 .
.Dtreat:FirstTreat::1992:year::1998 -5.39134    3.09606 -1.74135 0.081954 .
.Dtreat:FirstTreat::1992:year::1999 -5.09733    3.09495 -1.64698 0.099901 .
... 121 coefficients remaining (display them with summary() or use
argument n)
... 141 variables were removed because of collinearity
(.Dtreat:FirstTreat::1993:year::1992,
.Dtreat:FirstTreat::1994:year::1992 and 139 others [full set in
$collin.var])
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 4.6999     Adj. R2: 0.081954
               Within R2: 0.0259
```

```
emfx(model.wool2)
```

```
 .Dtreat Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
    TRUE     1.51      0.188 8.02   <0.001 49.7  1.14   1.87

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

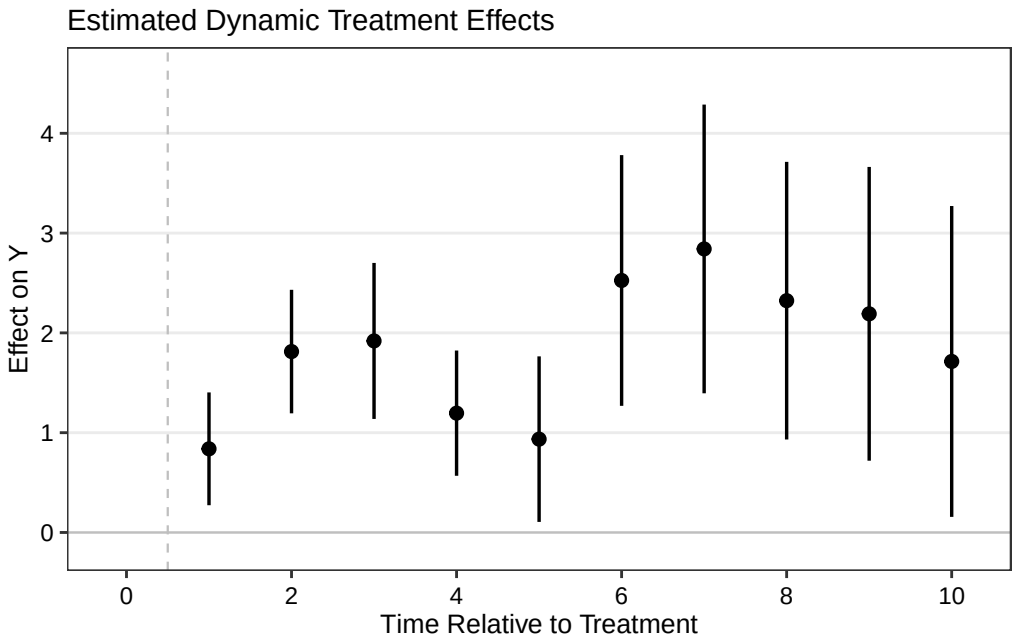
```
mod_es2 = emfx(model.wool2, type = "event")
mod_es2
```

```
 event Estimate Std. Error        z Pr(>|z|)    S  2.5 % 97.5 %
     0   0.8381      0.288  2.90643  0.00366  8.1  0.273  1.403
     1   1.8127      0.316  5.74107  < 0.001 26.7  1.194  2.432
     2   1.9194      0.399  4.81257  < 0.001 19.4  1.138  2.701
     3   1.1957      0.320  3.73394  < 0.001 12.4  0.568  1.823
     4   0.9352      0.423  2.20947  0.02714  5.2  0.106  1.765
     5   2.5252      0.641  3.93965  < 0.001 13.6  1.269  3.781
     6   2.8413      0.738  3.84931  < 0.001 13.0  1.395  4.288
     7   2.3229      0.710  3.27278  0.00106  9.9  0.932  3.714
     8   2.1911      0.751  2.91737  0.00353  8.1  0.719  3.663
     9   1.7138      0.795  2.15674  0.03103  5.0  0.156  3.271
    10   1.0765      0.941  1.14366  0.25277  2.0 -0.768  2.921
    11  -0.3083      0.498 -0.61885  0.53602  0.9 -1.285  0.668
    12   0.3508      1.892  0.18538  0.85293  0.2 -3.358  4.060
    13   0.8170      2.012  0.40611  0.68466  0.5 -3.126  4.760
    14   0.9743      3.351  0.29076  0.77123  0.4 -5.593  7.542
    15  -2.4917      1.413 -1.76355  0.07781  3.7 -5.261  0.278
    16   1.4233      4.430  0.32126  0.74801  0.4 -7.260 10.107
    17  -0.0138      4.556 -0.00304  0.99758  0.0 -8.944  8.916

Term: .Dtreat
Type: response
Comparison: TRUE - FALSE
```

```
wool.output2 <- mod_es2 |>
  filter(event != -1) |>
  mutate(Time = event + 1) |>
  as.data.frame()

p.wool2 <- esplot(wool.output2, Period = 'Time',Estimate = 'estimate',
                  SE = 'std.error', xlim = c(0,10))
p.wool2
```



21.3. Comparing the estimators

It is worth collecting what all of these produced on the same data:

estimator	ATT	s.e.
TWFE (full panel)	1.339	0.187
TWFE (without always-treated)	1.609	0.195
Sun and Abraham	1.331	0.288
Callaway and Sant'Anna (simple)	1.331	0.302
Callaway and Sant'Anna (dynamic)	1.221	0.631
Imai, Kim and Wang	1.178	0.335
Liu, Wang and Xu	1.506	0.204
Wooldridge ETWFE	1.330	0.288

Three things stand out. Sun-Abraham, Callaway-Sant'Anna and ETWFE agree to three digits at 1.33 — they are different routes to the same never-treated-control estimand, and they land on the same number. The spread across the six heterogeneity-robust estimators, 1.18 to 1.51, is smaller than their standard errors, so on this data the choice among them does not change the conclusion.

TWFE needs more care, because the two TWFE rows use different samples. On the full panel it gives 1.339, close to the robust estimators. But those estimators all use `df.use`, which drops the 283 always-treated municipalities. On that same sample TWFE gives 1.609, about 0.28 above Sun-Abraham, Callaway-Sant'Anna and ETWFE, or about 1.4 of its standard errors. The full-panel TWFE lands near 1.33 because its "later vs always treated" comparisons, 31% of the weight with an average of 0.75, pull it down. So the closeness says nothing about how small the contamination is. Compared on the same sample, TWFE is the outlier.

21.4. Staggered DiD with treatment reversal

With treatment reversal (ie., the treated units go back to control), Wooldridge's method still works. However, I don't think package "etwfe" has it implemented.

Several packages handle treatment reversal. Besides `fect` and `PanelMatch`, the estimator of Chaisemartin and D'Haultfœuille (2020), implemented in `DIDmultiplegt` and its successor `did_multiplegt_dyn` (R and Stata), was designed for treatments that switch on and off. It compares units whose treatment changes between two periods with units whose treatment does not, so it never uses a switching unit as a control.

We can probably manually do ETWFE with treatment reversal. See Wooldridge's dropbox for the code in Stata. The basic idea is to treat cohorts as different treatment beginning time and exit time combinations. Then do the same regression with interactions of cohort dummies with the time dummies. It is very flexible. The tricky part is to get the marginal effect and its standard error right.

21.5. Summary

With staggered adoption and effects that vary across cohorts, a single TWFE coefficient mixes clean comparisons with forbidden ones that use already-treated units as controls. The Goodman-Bacon decomposition shows the mix. Sun-Abraham, Callaway-Sant'Anna, imputation (Borusyak et al., Liu-Wang-Xu), PanelMatch and Wooldridge's ETWFE all avoid the forbidden comparisons in different ways. ETWFE is the simplest: a TWFE regression flexible enough to give each cohort its own effect in each period.

On the Swiss naturalization data, TWFE on the full panel gives 1.339. The decomposition shows 37% of the weight on forbidden comparisons, which pull the estimate down from the clean comparisons' 1.61. Sun-Abraham, Callaway-Sant'Anna and ETWFE agree at 1.33, and the six robust estimators fall between 1.18 and 1.51. They drop the always-treated municipalities, and on that sample TWFE gives 1.609, so the full-panel agreement is a coincidence of samples. Treatment reversal needs `fect`, PanelMatch or `did_multiplegt_dyn`, or ETWFE built by hand with exit cohorts.

Systematic treatment: R · Julia.

22. Causal Panel (why DDDiD)

22.1. Why DDDiD

Guido Imbens has been saying DDDiD (Don't Do Diff in Diff). But why?

To see that, DiD objective function (in the form of TWFE) is:

$$(\hat{\tau}^{did}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \tag{22.1}$$

We can see that in this objective function, we are allowing only α_i and β_t to capture the difference between units and time periods. The obvious drawback is that control units that are less similar to a treated unit is given the same weight as those that are more similar. A time period that is far from the beginning of the treatment period is given the same weight as those that are closer. Say the treatment period is 2000, we have observations from 1988 to 2005, then 1988 observations are the same as 1999 observations. This is not ideal. The same logic applies to the units.

22.2. Synthetic Control

SC objective function:

$$(\hat{\tau}^{sc}, \hat{\mu}, \hat{\beta}) = \underset{\tau,\mu,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} (Y_{it} - \mu - \beta_t - W_{it}\tau)^2 \hat{\omega}_i^{sc} \tag{22.2}$$

It's a weighted version of DiD, but note carefully that it is **not** a nesting of it: setting every $\hat{\omega}_i^{sc}$ to 1 here does not recover the DiD objective above, because SC has no α_i term — it would give a pooled regression with time effects only. Dropping the unit fixed effects is a substantive restriction, not a re-weighting. (DiD *is* nested in synthetic DiD below, which keeps α_i and sets both weights to be constant; that is the right way to see DiD as a special case.) The weights are set to optimally match donor units to the treated unit so that they are as close as possible at each time point. There is no time weight.

22.3. Synthetic DiD

Arkhangelsky et al. (2021) tries to combine the idea of DiD and SC. SC assigns different weights to different control units. The standard DiD is a TWFE, assigning equal weights to all time periods and units.

$$(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{argmin} \Sigma_{i=1}^{N} \Sigma_{t=1}^{T} \hat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \tag{22.3}$$

SDiD sets another weight in addition to SC weights, which changes over time. The SC weights are trying to construct a control unit that is close to the treated unit; the SDiD weights are trying to put more weights on pre-treatment periods that are more similar to post-treatment periods.

22.4. Example from synthdid

The data are the California Proposition 99 panel: annual per-capita cigarette sales in packs for 39 states from 1970 to 2000, with California the single treated unit, 38 donors, 19 pre-treatment years and 12 post-treatment years.

```
library(synthdid)
data('california_prop99')
setup = panel.matrices(california_prop99)
tau.hat = synthdid_estimate(setup$Y, setup$N0, setup$T0)
summary(tau.hat)
```

```
$estimate
[1] -15.60383

$se
     [,1]
[1,]   NA

$controls
               estimate 1
Nevada              0.124
New Hampshire       0.105
Connecticut         0.078
Delaware            0.070
Colorado            0.058
Illinois            0.053
Nebraska            0.048
Montana             0.045
Utah                0.042
New Mexico          0.041
Minnesota           0.039
Wisconsin           0.037
West Virginia       0.034
North Carolina      0.033
Idaho               0.031
Ohio                0.031
Maine               0.028
```

```
Iowa                0.026

$periods
     estimate 1
1988      0.427
1986      0.366
1987      0.206

$dimensions
          N1           N0 N0.effective           T1           T0 T0.effective
       1.000       38.000       16.388       12.000       19.000        2.783
```

The estimate is −15.60 packs per capita, and the summary shows how it gets there. The unit weights are concentrated — Nevada 0.124, New Hampshire 0.105, Connecticut 0.078 — giving an effective 16.4 donors out of 38. The time weights are more concentrated still: 0.427 on 1988, 0.366 on 1986 and 0.206 on 1987, which is an effective 2.8 pre-periods out of 19. That is the "time-aware" part of synthetic DiD in numbers, and it is the mechanism behind the estimator's advantage over plain DiD.

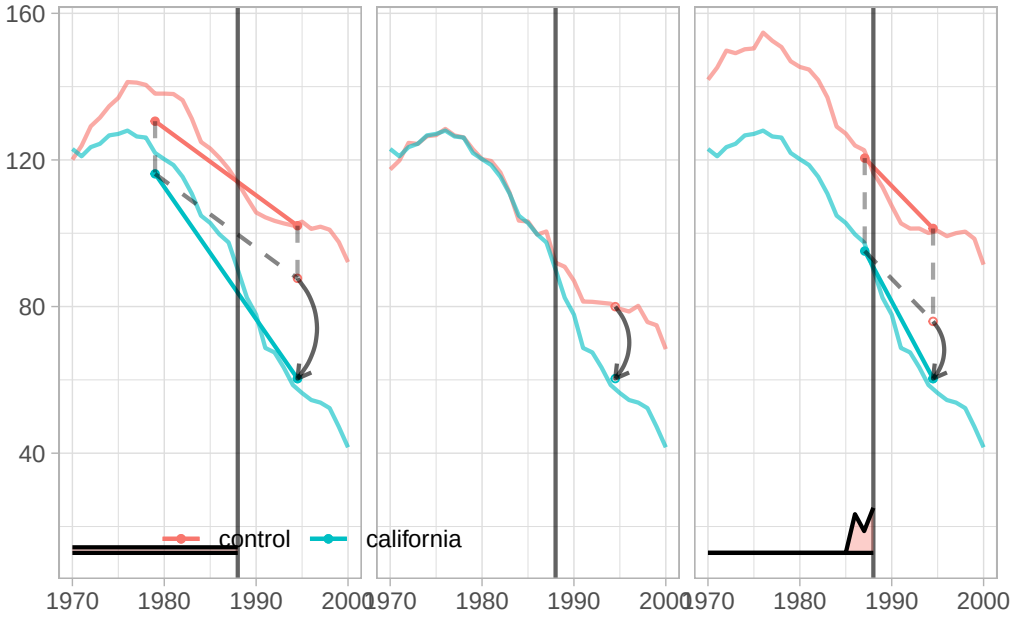
Note also that `se` comes back `NA`. That is not a glitch: the default is a jackknife standard error, which deletes one treated unit at a time, and California is the only treated unit. The placebo method is the usable option here.

```
tau.sc   = sc_estimate(setup$Y, setup$N0, setup$T0)
tau.did  = did_estimate(setup$Y, setup$N0, setup$T0)
estimates = list(tau.did, tau.sc, tau.hat)
names(estimates) = c('Diff-in-Diff', 'Synthetic Control', 'Synthetic Diff-in-Diff')
print(unlist(estimates))
```

```
          Diff-in-Diff      Synthetic Control Synthetic Diff-in-Diff
             -27.34911              -19.61966              -15.60383
```

Plain DiD gives −27.35, synthetic control −19.62, and synthetic DiD −15.60. The spread across the three is nearly twelve packs per capita, on the same data. Not, though, on the same identification assumptions: the three differ in what they require of the donor pool and the pre-period, which is part of why they disagree. DiD moves furthest because it weights all 38 donors and all 19 pre-periods equally: states unlike California, and years unlike the late 1980s, all count in full.

```
plot <- synthdid_plot(estimates, facet.vertical=FALSE,
              control.name='control', treated.name='california',
              lambda.comparable=TRUE, se.method = 'none',
              trajectory.linetype = 1, line.width=.75, effect.curvature=-.4,
              trajectory.alpha=.7, effect.alpha=.7,
              diagram.alpha=1, onset.alpha=.7) +
    theme(legend.position=c(.26,.07), legend.direction='horizontal',
          legend.key=element_blank(), legend.background=element_blank(),
          strip.background=element_blank(), strip.text.x = element_blank())
plot
```



Each panel shows California's path against its weighted control, with the estimated effect as the post-1989 gap. Note: λ_t is plotted at the bottom, so the pre-periods that actually enter each comparison are visible — for synthetic DiD they pile up on 1986–1988, while for DiD the weight is flat across all nineteen years.

22.5. Summary

To me, "DDDiD" is saying DiD is often too simple. That is not the same as saying a weighted estimator dominates. If parallel trends and no anticipation genuinely hold, DiD is identified, and synthetic DiD is not then a strictly better estimator – it is robust to a different failure mode, at the price of its own assumptions: a donor pool whose weighted pre-period path tracks the treated unit, and the low-rank or fixed-effects structure that justifies regularized unit and time weights. Nor does a closer pre-treatment fit by itself establish smaller post-treatment bias. The useful reading is that the three estimators lean on different structures, so agreement between them is informative and disagreement tells you which structure is doing the work.

Systematic treatment: R · Julia.

Part V.

Count Data & Specialized Models

23. Which count data model to use

23.1. A comparison of various count data models with extra zeros

In empirical studies, data sets with a lot of zeros are often hard to model. There are various models to deal with it: zero-inflated Poisson model, Negative Binomial (NB) model, hurdle model, etc.

Here we are following a zero-inflated model's thinking: model the data with two processes. One is a Bernoulli process, the other one is a count data process (Poisson or NB).

We'd like to see, in this simulation exercise, how different models perform with changes of sample size and percentage of zeros (we expect the less zero, the better a plain Poisson model would perform). Therefore we vary sample size n and an indicator of how much percentage of zeros in the data θ.

For the count data process (y_c):

$$y_c = \lfloor \exp(2x + u) \rfloor, \qquad u \sim N(0, \sigma^2) \tag{23.1}$$

Note what that is and is not. It is a floored lognormal variable, not a draw from a Poisson distribution, and not normal either. The flooring matters: it makes the conditional mean only approximately $\exp(2x + \sigma^2/2)$, so the log-linear index is not exactly $2x$ and "bias relative to 2" is not a clean common target across the models compared below.

For the Bernoulli process (y_b):

$$z_1 = 4z + \theta \tag{23.2}$$

$$logit(y_b) = z_1 \tag{23.3}$$

$$p_y = \frac{e^{z_1}}{1 + e^{z_1}} \tag{23.4}$$

where $y_b \sim Bernoulli(p_y)$ is the realized 0/1 draw. Combining these two processes:

$$y = y_c \cdot y_b \tag{23.5}$$

so $y = y_c$ when $y_b = 1$ and $y = 0$ when $y_b = 0$.

23.1.1. Zero-inflated Poisson models

A zero-inflated Poisson needs specifying both the binary process and the count process correctly. More often than not, we don't have a model for the binary process. Many people simply use the same explanatory variables for both processes. We simulate both situations. Case 1: suppose we observe z, and case 2: suppose we don't observe z. In the graph below, they are labeled zip1 and zip2.

23.1.2. Poisson model

A plain Poisson model returns a consistent estimator for the coefficients, with or without Poisson-distributed data. We expect Poisson model's performance improve with sample size. Note that the standard errors from a Poisson model needs adjustment, which we do not discuss in this post.

23.1.3. NB model

NB model is used widely to handle "overdispersion" problem. That is, the variance far exceeds the mean, therefore the Poisson model is considered inappropriate. NB model addresses that by allowing an extra parameter. However, many people also use it to model "extra zero" situation, we'll see in our simulation it may not be better than a plain Poisson model.

23.1.4. Log-linear model

What about an OLS model with $log(y + 1)$?

23.1.5. Hurdle model

A hurdle model models the zero's and other values separately; that is, the zero's are from a binomial process only, the other positive values are from a truncated count data process. We assume here, in the simulation, we don't observe z. Therefore, x is determining both binary and count processes. In the graph below, it's labeled hurdle.

```
library(MASS)
library(pscl)
library(parallel)
set.seed(666)

gen.sim <- function(df) {
  nobs <- df["nobs"]; th <- df["th"]
  z    <- rnorm(nobs, 0, 1)
  x    <- rnorm(nobs, 0, 1)
  u    <- rnorm(nobs, 0, 1)
  y.count  <- floor(exp(2 * x + u))
  z1       <- 4 * z + th
  prob     <- plogis(z1)
  y.logit  <- rbinom(nobs, size = 1, prob = prob)
  y        <- ifelse(y.logit == 1, y.count, y.logit)

  m1 <- zeroinfl(y ~ x | z)
  m4 <- zeroinfl(y ~ x | x)
  m2 <- glm(y ~ x, family = "poisson")
  m3 <- lm(log(y + 1) ~ x)
  m5 <- glm.nb(y ~ x)
```

```
  m6 <- hurdle(y ~ x)

  c(zip1       = coef(m1, "count")["x"] - 2,
    poisson    = coef(m2)["x"] - 2,
    log.linear = coef(m3)["x"] - 2,
    zip2       = coef(m4, "count")["x"] - 2,
    nb         = coef(m5)["x"] - 2,
    hurdle     = coef(m6, "count")["x"] - 2)
}

data.grid <- expand.grid(
  nobs = ceiling(exp(seq(4, 9, 1)) / 100) * 100,
  nsim = seq(1, 100, 1),
  th   = seq(-4, 4, 2)
)

cl      <- makeCluster(detectCores() - 1)
clusterEvalQ(cl, { library(MASS); library(pscl) })
clusterExport(cl, "gen.sim")
results <- parApply(cl, data.grid, 1, gen.sim)
stopCluster(cl)

forshiny <- cbind(data.grid, t(results))
write.csv(forshiny, "results.csv", row.names = FALSE)
```

Count data models can be used even if data is not "counts"; for example, some non-negative non-integer numbers. In fact, Poisson model is consistent even if data is not Poisson-distributed, if the model specification is correct on modeling the log of expected counts. We simulate both scenarios: Case 1, data is generated from a Poisson process. Case 2, data is a floored lognormal – $\lfloor \exp(2x + u) \rfloor$ with u normal – so it is non-negative and integer-valued without being a count from any count distribution, and we use count data models on it anyway. (Earlier versions of this text called case 2 "Normal"; the variable being modelled is not normal, only the u inside the exponential is.) The above code is for case 2.

We simulate 100 times with θ ranging from -4 to 4, lower number means higher percentage of zeros; number of observations from e^4 to e^9 (roughly from 50 to 8000).

Since there are many simulations, we use base R's `parallel` library to speed things up.

For raw code, please visit case1: poisson and case2: normal. The code above updates the original `snowfall`-based version to use base R `parallel`.

Status of these simulation findings. The grid above is retained for its code rather than as current evidence, and the ranking it once carried should not be read as verified. Two things undercut it. The metric for the log-linear row was wrong in the original run – see the note below – and the case-2 data-generating process is a floored lognormal whose log-linear index is not exactly $2x$, so subtracting 2 from each model's slope does not compare like with like. The chunk is marked `eval: false`, so nothing here regenerates it.

What the run suggested, and what remains plausible on grounds independent of it: plain Poisson QMLE holds up well in large samples even with unmodelled excess zeros *when the conditional mean is correctly specified*; zero-inflated Poisson does better when the zero process is known; and log-linear regression on $\log(y+1)$ targets a different quantity altogether. For evidence rather than recollection, the worked example in the next section is the part of this chapter that actually runs.

(Note: the `log.linear` bias line above previously read `exp(coef(m3)["x"]) - 2` instead of `coef(m3)["x"] - 2`, which — since the log-linear coefficient is already on the log scale — manufactured a spurious ≈5.4 apparent bias regardless of the model's actual performance. That has been corrected here to match the other rows. Log-linear regression on `log(y+1)` is still expected to be biased relative to Poisson-family models in a zero-inflated count setting, for real reasons (the log transform's nonlinearity near zero, and the fact that $E[\log(y+1)] \neq \log(E[y]+1)$), but the *magnitude* of "consistently the worst" above reflects the original (buggy) run and has not been re-verified against a corrected simulation.)

23.2. Quick worked example

All five models on a single small dataset (n = 1000, ~40% zeros).

The DGP is a zero-inflated Poisson built in two independent pieces. We draw $x \sim N(0,1)$ and $z \sim N(0,1)$, independent of each other. The count part is $y_c \sim \text{Poisson}(\exp(1.5 + 0.8x))$ and the zero part is $y_b \sim \text{Bernoulli}(\Lambda(1+2z))$, with the observed outcome $y = y_c \cdot y_b$. So the true slope on x is 0.8, and — this is the feature that drives the results — the zero process depends on z alone, never on x.

```
library(MASS)
library(pscl)
set.seed(42)

n  <- 1000
x  <- rnorm(n)
z  <- rnorm(n)                          # zero-inflation driver
mu <- exp(1.5 + 0.8 * x)               # count mean
y_c <- rpois(n, lambda = mu)
y_b <- rbinom(n, 1, plogis(1 + 2 * z)) # 1 = non-zero
y   <- y_c * y_b                        # zero-inflated count

cat("Zero fraction:", round(mean(y == 0), 2), "\n")
```

```
Zero fraction: 0.39
```

```
cat("Mean (non-zero):", round(mean(y[y > 0]), 2), "\n\n")
```

```
Mean (non-zero): 6.79
```

```
# 1. Poisson
m_pois <- glm(y ~ x, family = poisson)

# 2. Negative binomial
m_nb <- glm.nb(y ~ x)

# 3. Zero-inflated Poisson (ZIP) — correct zero model
m_zip <- zeroinfl(y ~ x | z)

# 4. ZIP — wrong zero model (only x)
m_zip2 <- zeroinfl(y ~ x | x)

# 5. Hurdle
m_hurdle <- hurdle(y ~ x)

# 6. Log-linear OLS on log(y + 1)
m_loglin <- lm(log(y + 1) ~ x)

# Compare x coefficient (true = 0.8)
coefs <- c(
  Poisson    = coef(m_pois)["x"],
  NB         = coef(m_nb)["x"],
  `ZIP (correct z)` = coef(m_zip, "count")["x"],
  `ZIP (wrong)` = coef(m_zip2, "count")["x"],
  Hurdle     = coef(m_hurdle, "count")["x"],
  `Log-linear log(y+1)` = coef(m_loglin)["x"]
)

knitr::kable(
  data.frame(Model = names(coefs), `x coefficient` = round(coefs, 3),
             Bias = round(coefs - 0.8, 3)),
  caption = "True x coefficient = 0.8"
)
```

Table 23.1.: True x coefficient = 0.8

	Model	x.coefficient	Bias
Poisson.x	Poisson.x	0.850	0.050
NB.x	NB.x	0.824	0.024
ZIP (correct z).x	ZIP (correct z).x	0.803	0.003
ZIP (wrong).x	ZIP (wrong).x	0.802	0.002
Hurdle.x	Hurdle.x	0.803	0.003
Log-linear log(y+1).x	Log-linear log(y+1).x	0.458	-0.342

The sample comes out 39% zeros with a non-zero mean of 6.79. The estimated slopes are 0.850 for Poisson, 0.824 for negative binomial, 0.803 for ZIP with the correct zero model, 0.802 for ZIP with

the wrong one, 0.803 for the hurdle model, and 0.458 for log-linear OLS.

The first five models recover the true `x` slope of 0.8 reasonably well; the log-linear row is included to make the earlier claim checkable, and needs reading with care — see the note after the table. This is the key point of *this* DGP: the zero-inflation process depends on `z`, which is generated **independently of `x`**, so omitting the zero model distorts the zero mass and the intercept, not the slope on `x`. The hurdle model and even the ZIP with a *misspecified* (x-only) zero model still recover the slope closely; the plain Poisson shows a small upward bias *in this particular sample* (n = 1000), but that is finite-sample noise, not a systematic property of the model. Since the zero-inflation driver `z` is independent of `x`, the conditional mean here is correctly specified for Poisson QMLE — increasing `n` from 1,000 to 1,000,000 shrinks the Poisson `x`-coefficient bias from about +0.04 to effectively 0, confirming consistency. The unmodeled excess zeros create overdispersion, but overdispersion under a correctly-specified conditional mean affects only the standard errors (which should be corrected with a robust/sandwich estimator), not the consistency of the slope.

About the log-linear row: its coefficient is **not** estimating the same thing as the others, which is the real reason to avoid it rather than any particular bias number. The count models target $\log E[y \mid x]$, whereas OLS on $\log(y+1)$ targets $E[\log(y+1) \mid x]$, and $E[\log(y+1)] \neq \log(E[y]+1)$ by Jensen's inequality. The +1 offset makes the discrepancy worst exactly where the mass of zeros is. So comparing its coefficient to 0.8 is an apples-to-oranges comparison, and the honest statement is that log-linear answers a different question — not that it answers this one badly by some measured amount. A misspecified zero process biases the count *slope* only when the zero process is correlated with the count covariates — which is not the case here. (Change `z` to depend on `x` and the omitted-zero-process slope bias appears.)

23.3. Mixed count models with glmmTMB

When data has both overdispersion *and* random effects (clustered counts, panel data, repeated measures), `glmmTMB` handles the full range in one package:

```
library(glmmTMB)

# Zero-inflated negative binomial with random intercept per subject
m_glmmtmb <- glmmTMB(
  count ~ treatment + time + (1 | subject),   # count model
  ziformula = ~ treatment,                     # zero-inflation model
  family = nbinom2,
  data = your_data
)

summary(m_glmmtmb)

# Conditional vs marginal predictions
predict(m_glmmtmb, type = "conditional")  # excluding zero-inflation
predict(m_glmmtmb, type = "response")     # including zero-inflation
```

`glmmTMB` supports: Poisson, NB1, NB2, zero-inflated variants of all of the above, hurdle models, beta-binomial, tweedie, and more. The `ziformula` argument takes a one-sided formula for the zero-inflation component — use `~ 1` for a constant zero-inflation rate, or covariates to model what drives the structural zeros.

23.4. Summary

Count data with many zeros invite several models: Poisson, negative binomial, zero-inflated, hurdle and log-linear OLS on $\log(y + 1)$. Poisson is a quasi-MLE and is consistent for the slope whenever the conditional mean is correctly specified. That holds even with excess zeros and overdispersion, which affect only the standard errors, so robust standard errors are needed. Zero-inflated and hurdle models help when the zero process is known. Log-linear OLS targets a different quantity altogether, by Jensen's inequality.

In the worked example the zero process depends on a variable unrelated to x. Every count model recovers the true slope of 0.8 closely: Poisson 0.850, negative binomial 0.824, and the zero-inflated and hurdle models about 0.80, even with a misspecified zero model. Log-linear OLS gives 0.458, an answer to a different question. `glmmTMB` fits all of these with random effects for clustered counts.

24. What model to use for rare events

24.1. Introduction

In empirical studies, people are worried about rare event situation. That is, when you have, for example, lots of 0's and only a few 1's, or vice versa. Do you run a logit model, or do you use a "rare event logit"? When should you use either approach? Or there is a third approach?

Paul Allison said in his blog (Allison 2012):

"Prompted by a 2001 article by King and Zeng, many researchers worry about whether they can legitimately use conventional logistic regression for data in which events are rare. Although King and Zeng accurately described the problem and proposed an appropriate solution, there are still a lot of misconceptions about this issue.

The problem is not specifically the rarity of events, but rather the possibility of a small number of cases on the rarer of the two outcomes. If you have a sample size of 1000 but only 20 events, you have a problem. If you have a sample size of 10,000 with 200 events, you may be OK. If your sample has 100,000 cases with 2000 events, you're golden."

In general I agree with him. However, when exactly should we use the "relogit" of King and Zeng (2001)?

Allison also mentioned two other methods. One is called the Firth method (Firth 1993), a penalized likelihood approach. The other one is the exact logistic regression, which is for small samples.

In this simulation exercise, I compare logit against bias-reduced logistic regression (`brglm2` with `method="brglmFit"`, `type="AS_mean"`) and the Firth model (`logistf`).

One thing to be clear about before reading any comparison of the latter two: for a logistic regression **they are the same estimator.** Firth's penalized likelihood (penalizing by the Jeffreys prior) coincides with mean bias reduction for GLMs with a canonical link, and logit is canonical for the binomial. So `brglm2`'s `AS_mean` and `logistf` solve the same estimating equations and differ only in numerical tolerance and in how each package builds standard errors and intervals (`logistf` uses penalized profile likelihood). A demonstration, on a deliberately rare-event DGP: $n = 500$, $x \sim N(0, 1)$, and $y \sim \text{Bernoulli}(\Lambda(-3.5 + 0.8x))$, which puts the event rate around 3% — roughly fifteen events in the whole sample.

```
library(brglm2); library(logistf)
set.seed(11)
n <- 500; x <- rnorm(n); y <- rbinom(n, 1, plogis(-3.5 + 0.8 * x))
d <- data.frame(y, x)

rbind(
```

```
  `plain logit`     = coef(glm(y ~ x, data = d, family = binomial)),
  `brglm2 AS_mean`  = coef(glm(y ~ x, data = d, family = binomial,
                               method = "brglmFit", type = "AS_mean")),
  `logistf (Firth)` = coef(logistf(y ~ x, data = d))
)
```

```
                (Intercept)         x
plain logit       -3.176125 0.7378324
brglm2 AS_mean    -3.134453 0.7277674
logistf (Firth)   -3.134453 0.7277675
```

The two bias-reduced fits give −3.134453 and 0.7277674 against −3.134453 and 0.7277675: they agree to about 10^{-7}, which is numerical tolerance rather than any statistical difference. Plain logit gives −3.176 and 0.738, so on this draw the bias correction moves the slope by about 0.010 and the intercept by 0.042 — both estimators sitting below the true 0.8 and above the true −3.5, as expected with only fifteen events. This matters for reading the findings below: what the simulation can tell us is how **logit** compares with **Firth-type bias reduction**, not which of two implementations of the same estimator is better. If a genuinely distinct third method is wanted, `brglm2`'s `type = "AS_median"` (median bias reduction) is one, and King and Zeng's `relogit` correction is another.

24.2. Simulation

Here I have some code for using multiple cores to run these three models. The bias-reduced logistic regression is implemented in R via the `brglm2` package, the Firth method is implemented in `logistf`.

```
library(brglm2)
library(logistf)
require(snowfall)
set.seed(666)

# initialize parallel cores.
sfInit(parallel=TRUE, cpus=6)

gen.sim <- function(df){
  x <- rnorm(df['nobs'], 0, 1)

  # generate binary data
  p <- df['p']
  # NB: alpha = qlogis(p) makes p the event probability at x = 0, not the
  # marginal event probability. Because the slope on x is 2 and x ~ N(0,1),
  # the marginal rate is much higher than p: integrating plogis(alpha + 2x)
  # over x gives 0.044, 0.134 and 0.203 for p = 0.01, 0.05 and 0.10.
  alpha <- -log((1-p)/p)
  z = alpha + 2*x
```

```
  pr = 1/(1+exp(-z))
  y = rbinom(df['nobs'], 1, pr)
  df = data.frame(y=y, x=x)

  # With small nobs and small p (e.g. nobs=10, p=.01), y can easily come out
  # all-zero (about 64% chance in that cell, since the marginal event rate is
  # 0.044, not 0.01), which makes glm/logistf unable to
  # estimate a coefficient on x. Guard against that rather than letting a
  # single degenerate draw crash the whole parallel simulation loop.
  if (var(y) == 0) {
    return(c(logit=NA, brglm2=NA, logistf=NA))
  }

  # logit
  m1 <- tryCatch(glm(y ~ x, family='binomial'), error = function(e) NULL)
  m1.x <- if (!is.null(m1)) summary(m1)$coefficients['x','Estimate'] - 2 else NA

  # bias-reduced logistic regression (brglm2)
  m2 <- tryCatch(
    glm(y ~ x, family = binomial(link = "logit"), data = df,
        method = "brglmFit", type = "AS_mean"),
    error = function(e) NULL
  )
  m2.x <- if (!is.null(m2)) coef(m2)['x'] - 2 else NA

  # logistf (Firth)
  m3 <- tryCatch(logistf(y ~ x, data=df), error = function(e) NULL)
  m3.x <- if (!is.null(m3)) coef(m3)['x'] - 2 else NA

  return(c(logit=m1.x, brglm2=m2.x, logistf=m3.x))
}

# set parameter space
sim.grid = seq(1, 100, 1)
p.grid = c(.01, .05, .1)
nobs.grid = c(10, 30, 50, 100, 200, 500, 1000, 10000)

data.grid <- expand.grid(nobs.grid, sim.grid, p.grid)
names(data.grid) <- c('nobs', 'nsim', 'p')

# export functions and libraries to parallel workers
sfExport(list=list("gen.sim"))
sfLibrary(brglm2)
sfLibrary(logistf)

results <- data.frame(t(sfApply(data.grid, 1, gen.sim)))
```

```
# stop the cluster
sfStop()

forshiny <- cbind(data.grid, results)
write.csv(forshiny, 'results.csv')
```

We simulate 100 times with sample size from 10 to 10000, event probability .01, .05, and .1.

Since there are many simulations, we used the `snowfall` library to speed things up.

(The original post plotted bias and MSE by sample size and event probability from the `results.csv` output above; those figures are not reproduced here — the findings are summarized in prose below.)

A caveat on how to read the grid before the findings. The parameter `p` is the event probability at $x = 0$, not the marginal event probability: with a slope of 2 on $x \sim N(0, 1)$, the marginal rate is roughly 0.044, 0.134 and 0.203 for `p` = 0.01, 0.05 and 0.10. So the expected number of events in a cell is closer to n times those numbers than to np, by a factor of about four at the rare end. What follows is stated in terms of the realized event count, which is the quantity that actually governs the behaviour.

Many samples in the rare cells cannot be fit at all. Because observations are independent, the chance that a sample has no events is $(1 - \bar{p})^n$, where $\bar{p}$ is the marginal event rate. At $p = 0.01$ this is 0.64 for $n = 10$, 0.26 for $n = 30$ and 0.10 for $n = 50$; at $p = 0.05$ and $n = 10$ it is 0.24. The code returns `NA` for those samples and for fits that fail, so bias and MSE are averages over the samples where estimation succeeded. They describe conditional performance and should be read alongside the failure rate, which the original summaries did not report. The thresholds of five and fifty events below describe this DGP and design, not a general rule.

When the rarer of the two outcomes has only a handful of observations – fewer than about five events in the sample – none of these three models performs well. This is understandable: there is almost nothing in the data to identify the slope. Once the rarer group has more than about fifty events, there is not much difference between the three. In between, we found that Firth-type bias reduction — whether computed by `brglm2` or by `logistf` — performs better than plain logit. Any apparent ranking *between* those two should be discounted, since as shown above they are the same estimator; differences at that level reflect numerical tolerance or differing standard-error conventions, not a substantive advantage.

In the small sample situation, maybe it's better to use the exact logistic regression.

24.3. Summary

Rare events are a problem when the rarer outcome has few cases in absolute terms, more than when it is a small share of the sample. Firth's penalized likelihood and mean bias reduction are the same estimator for logit, so `logistf` and `brglm2`'s `AS_mean` agree to numerical tolerance. The real comparison is between plain logit and Firth-type bias reduction.

In the simulation, with fewer than about five events no model does well, and with more than about fifty the three agree. In between, bias reduction beats plain logit. For very small samples exact

logistic regression is an option. King and Zeng's `relogit` and median bias reduction are genuinely different alternatives.

25. Poisson model with exposure

25.1. Poisson with exposure or Poisson rate

Poisson models for count data often arise when units are observed for different lengths of time. A hospital admission count for a patient followed for 30 days is not directly comparable to a count for a patient followed for 365 days. The natural quantity of interest is the *rate* — admissions per unit time — not the raw count.

There are three related but distinct specifications, each making a different assumption:

Model 1 — Poisson with offset (rate model):

$$\log\big(E[Y]/t\big) = \beta_0 + \beta_1 X \tag{25.1}$$

Equivalently, $\log E[Y] = \beta_0 + \beta_1 X + \log t$, where $\log t$ is included as a *fixed* offset (coefficient constrained to 1). This is the standard exposure model: the expected count is proportional to exposure time.

Model 2 — Poisson on the ratio Y/t:

$$\log\big(E[Y/t]\big) = \beta_0 + \beta_1 X \tag{25.2}$$

This looks similar, but it is not equivalent to the offset model, and strictly speaking Y/t is **not a Poisson outcome at all**: a Poisson likelihood is defined for non-negative integer counts, and the continuous ratio Y/t is not one. Because t is a fixed constant for each unit, $E[Y/t] = E[Y]/t$ exactly (linearity of expectation), so the *conditional mean* is still log-linear in X; that is why the point estimates are often similar. But the Poisson variance assumption fails: $\mathrm{Var}(Y/t) = \lambda/t^2$, whereas the mean is $E[Y/t] = \lambda/t$, so the variance is $1/t$ times the mean rather than equal to it. (Writing it as λ/t would make variance equal mean and quietly restore equidispersion, which is precisely what fails here.) So fitting `glm(rate ~ x, family = poisson)` is a quasi-likelihood *mean model* rather than a correct Poisson likelihood. If you use it, treat it as a workaround: weight by exposure (`weights = t`) so the implied variance is right, and/or use robust standard errors. Write X_i for unit i's covariates, including the constant, and β for their coefficients. With these weights the rate model's score is $\sum_i t_i X_i\{Y_i/t_i - \exp(X_i'\beta)\} = \sum_i X_i\{Y_i - t_i \exp(X_i'\beta)\}$, the offset model's score, so the two give identical coefficients. Robust standard errors do not reproduce this; they change the variance, not the estimating equations. The offset model (Model 1) avoids the issue entirely by keeping the integer count as the outcome.

Model 3 — Unconstrained log-exposure:

$$\log E[Y] = \beta_0 + \beta_1 X + \gamma \log t \tag{25.3}$$

Here γ is estimated from the data. If $\hat{\gamma} \approx 1$, the proportionality assumption is supported. If $\hat{\gamma} < 1$, counts grow sub-linearly with exposure — a testable, substantive claim.

25.2. Simulation

```r
library(tidyverse)
set.seed(42)

n <- 800

# Simulate a health study: count of adverse events
# Covariate: age group (0 = younger, 1 = older)
age_old <- rbinom(n, 1, 0.45)

# Exposure: days under observation (varies 7–365)
exposure <- sample(seq(7, 365, by = 7), n, replace = TRUE)

# True model: rate = exp(-3 + 1.2 * age_old)
# Count ~ Poisson(rate * exposure)
true_rate <- exp(-3 + 1.2 * age_old)
count <- rpois(n, lambda = true_rate * exposure)

df <- data.frame(count, age_old, exposure, log_exposure = log(exposure))
summary(df[, c("count", "exposure")])
```

```
     count          exposure
 Min.   : 0.00   Min.   :  7.0
 1st Qu.: 7.00   1st Qu.:112.0
 Median :13.00   Median :196.0
 Mean   :18.95   Mean   :192.9
 3rd Qu.:27.00   3rd Qu.:281.8
 Max.   :71.00   Max.   :364.0
```

We simulate $n = 800$ patients. Age group is age_old $\sim$ Bernoulli(0.45), and exposure is days under observation, drawn uniformly from the weekly grid 7 to 364 and independently of age. The event rate is $\exp(-3 + 1.2\,\text{age_old})$ per day and the count is Poisson(rate $\times$ exposure), so the true intercept is -3, the true age effect is 1.2, and the true exposure elasticity is $\gamma = 1$ by construction.

The data has a wide range of exposure times — 7 to 364 days, median 196, with counts from 0 to 71 and a mean of 19. Exposure is drawn independently of age, so $E[Y \mid \text{age_old}] = E[t]\exp(\beta_0 + \beta_1 \text{age_old})$: ignoring exposure would change the intercept and the variance, but it would not bias the age coefficient here. We use this design to compare the estimating equations and their precision. If older patients were followed longer, omitting exposure would change the age coefficient.

25.3. Fitting the three models

```
# Model 1: offset — coefficient on log(exposure) constrained to 1
m1 <- glm(count ~ age_old + offset(log(exposure)),
          family = poisson, data = df)

# Model 2: Poisson on the rate Y/t (different response)
df$rate <- df$count / df$exposure
m2 <- glm(rate ~ age_old,
          family = poisson, data = df)

# Model 3: log(exposure) as a free covariate
m3 <- glm(count ~ age_old + log_exposure,
          family = poisson, data = df)

# Collect coefficients
results <- data.frame(
  Model = c("1. Offset (constrained γ=1)",
            "2. Rate Y/t as response",
            "3. Unconstrained log(t)"),
  Intercept = c(coef(m1)[1], coef(m2)[1], coef(m3)[1]),
  age_old   = c(coef(m1)[2], coef(m2)[2], coef(m3)[2]),
  gamma_log_t = c(1, NA, coef(m3)[3])
)

knitr::kable(results, digits = 3,
  caption = "Coefficient estimates: true intercept = -3, true age effect = 1.2, true γ = 1")
```

Table 25.1.: Coefficient estimates: true intercept $= -3$, true age effect $= 1.2$, true $\boldsymbol{\gamma} = 1$

Model	Intercept	age_old	gamma_log_t
1. Offset (constrained $\boldsymbol{\gamma}$=1)	-3.009	1.212	1.000
2. Rate Y/t as response	-2.980	1.176	NA
3. Unconstrained log(t)	-2.801	1.212	0.962

All three recover the truth. The offset model gives an intercept of -3.009 and an age effect of 1.212 against true values of -3 and 1.2; the rate model gives -2.980 and 1.176; the unconstrained model gives -2.801, 1.212 and $\hat{\gamma} = 0.962$ against a true γ of 1.

Reading the table: Model 1 and Model 3 give nearly identical age-group coefficients because the true $\gamma = 1$ (counts are indeed proportional to exposure time). Model 2 differs slightly in the intercept because the response variable changes. The estimated $\hat{\gamma}$ from Model 3 should be close to 1.

25.4. When $\gamma \neq 1$

To see why the unconstrained model matters, repeat the simulation with a sub-linear count process ($\gamma = 0.6$: an initial burst of events that doesn't grow proportionally with long follow-up):

```
count2 <- rpois(n, lambda = true_rate * exposure^0.6)
df2 <- data.frame(count2, age_old, exposure, log_exposure = log(exposure))

m1b <- glm(count2 ~ age_old + offset(log(exposure)),
           family = poisson, data = df2)
m3b <- glm(count2 ~ age_old + log_exposure,
           family = poisson, data = df2)

cat("Offset model age coefficient:      ", round(coef(m1b)[2], 3), "\n")
```

```
Offset model age coefficient:       1.171
```

```
cat("Unconstrained model age coefficient:", round(coef(m3b)[2], 3), "\n")
```

```
Unconstrained model age coefficient: 1.168
```

```
cat("Estimated gamma (log-exposure):    ", round(coef(m3b)[3], 3),
    " (true = 0.6)\n")
```

```
Estimated gamma (log-exposure):     0.566  (true = 0.6)
```

Two things to read off this, and the first is not what one might expect.

The offset model's age coefficient (1.171) and the unconstrained model's (1.168) are essentially the same, and both sit just below the true 1.2. Forcing $\gamma = 1$ when the truth is 0.6 does **not** bias the treatment coefficient here — and the reason is structural: `exposure` is drawn by `sample(...)` independently of `age_old`, so the misspecified exposure elasticity is absorbed by the intercept and cannot contaminate a coefficient on a covariate orthogonal to it. The constraint would bite if exposure were *correlated* with the covariate of interest — for instance if older patients were systematically followed longer, which is common in real cohort data and is the case worth worrying about.

What the unconstrained model does give, unambiguously, is γ itself: it recovers $\hat{\gamma} = 0.566$ against a truth of 0.6, whereas the offset model cannot report it at all because it fixes it at 1. So Model 3's role is diagnostic — it tells you whether the proportionality assumption holds — rather than protective of the treatment estimate.

25.5. Stata equivalent

```
* Model 1: offset
poisson count age_old, exposure(exposure)

* Model 2: rate as response
gen rate = count / exposure
* Stata will warn that the dependent variable is non-integer (poisson expects
* integer counts by default) -- this is expected and can be ignored here,
* since Poisson QMLE is valid for continuous non-negative outcomes too.
poisson rate age_old

* Model 3: unconstrained log(t)
gen log_exposure = log(exposure)
poisson count age_old log_exposure
```

25.6. Which model to use?

Situation	Recommended model
Counts grow proportionally with time	Model 1 (offset)
Uncertain about proportionality	Model 3 (unconstrained), test $\hat{\gamma} = 1$
Rate is the natural estimand	Model 1 (offset) — it estimates a rate directly; Model 2 only as a weighted/robust quasi-model
Long follow-up with declining event rate	Model 3, or a flexible function of follow-up time; a negative binomial with offset still assumes counts proportional to t
Overdispersion	Negative binomial with offset, or Model 1 with robust standard errors

In most health and social-science applications, Model 1 (Poisson with offset) is the default. Model 3 is the diagnostic check: fit it, look at $\hat{\gamma}$, and return to Model 1 if the constraint is supported by the data.

Systematic treatment: R · Julia.

26. Instrumental variables in fixed-effects Poisson models

26.1. Why a control function is needed

Suppose a panel count-data model has the conditional mean

$$E[y_{it} \mid x_{it}, c_i] = c_i \exp(x_{it}\beta)$$

If a variable in x_{it} is endogenous, linear two-stage least squares does not fit this exponential conditional mean. Nonlinear IV commands such as Stata's `ivpoisson` also cannot absorb high-dimensional individual and time fixed effects.

Wei Lin and Wooldridge (2019) propose a two-step control-function estimator for this setting.

26.2. The two-step estimator

Let w_{it} be a continuous endogenous regressor and z_{it} an excluded instrument. The first stage regresses w_{it} on the instrument and exogenous controls while absorbing all fixed effects:

$$w_{it} = z_{it}\pi + x_{it1}\gamma + c_i + f_t + v_{it}$$

We save the OLS residual $\hat{v}_{it}$ and include it in a fixed-effects Poisson model:

$$E[y_{it} \mid w_{it}, x_{it1}, c_i, f_t, \hat{v}_{it}] = c_i f_t \exp(w_{it}\beta_1 + x_{it1}\beta_2 + \rho\hat{v}_{it})$$

In Stata, the two commands are `reghdfe ..., resid` and `ppmlhdfe`. In R, we can use `fixest::feols()` and `fixest::fepois()`. Testing $H_0 : \rho = 0$ gives a test of exogeneity. Two inference questions sit here and they have different answers. The test of $\rho = 0$ is fine as it stands: under that null the first stage contributes nothing to the asymptotic variance, so the usual t-statistic on $\hat{v}_{it}$ is valid. Inference on the *corrected* coefficient β_1 is the problem, because $\hat{v}_{it}$ is estimated and under the alternative that estimation enters the variance. Either an analytic two-step correction or a bootstrap repeating both stages is needed there. I use a cluster bootstrap over units.

26.3. A continuous endogenous variable in Stata

I use Stata's `website.dta`, which has 500 observations. `visits` is the count outcome, `time` is the endogenous regressor, `phone` is the instrument, and `frfam` is an exogenous control. The model absorbs fixed effects for the advertising campaigns (12 in the estimation sample, after a singleton campaign is dropped) and for `female`.

```
webuse website, clear

* Step 1: Linear first stage absorbing both fixed effects
reghdfe time phone frfam, absorb(ad female) resid
predict double u2h_fe, resid

* Step 2: FE Poisson with control function residual
ppmlhdfe visits time u2h_fe frfam, absorb(ad female)
```

```
(Visits to website)

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   2,    484) =      29.02
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.3137
                                                  Adj R-squared   =     0.2938
                                                  Within R-sq.    =     0.1071
                                                  Root MSE        =     2.2239

------------------------------------------------------------------------------
        time | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       phone |   .2705429   .0626834     4.32   0.000     .1473778     .393708
       frfam |   .2539231   .0619373     4.10   0.000     .1322239    .3756222
       _cons |   1.406026   .2549245     5.52   0.000     .9051305    1.906921
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(1 missing value generated)
```

```
Iteration 1:   deviance = 3.7303e+02  eps = .         iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.57  P
Iteration 2:   deviance = 3.5849e+02  eps = 4.06e-02  iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.59
Iteration 3:   deviance = 3.5840e+02  eps = 2.46e-04  iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.60
Iteration 4:   deviance = 3.5840e+02  eps = 3.17e-08  iters = 2    tol = 1.0e-0
> 4
>    min(eta) =  -0.60
Iteration 5:   deviance = 3.5840e+02  eps = 9.72e-16  iters = 2    tol = 1.0e-0
> 5
>    min(eta) =  -0.60   S
Iteration 6:   deviance = 3.5840e+02  eps = 1.62e-16  iters = 1    tol = 1.0e-0
> 9
>    min(eta) =  -0.60   S O
------------------------------------------------------------------------------
> ----------------------------
(legend: p: exact partial-out   s: exact solver   h: step-halving   o: epsilon
> below tolerance)
Converged in 6 iterations and 14 HDFE sub-iterations (tol = 1.0e-08)

HDFE PPML regression                              No. of obs      =        499
Absorbing 2 HDFE groups                           Residual df     =        483
                                                  Wald chi2(3)    =     169.93
Deviance             =  358.4027719               Prob > chi2     =     0.0000
Log pseudolikelihood = -993.8547331               Pseudo R2       =     0.2976
------------------------------------------------------------------------------
             |               Robust
      visits | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .0448103   .0448665     1.00   0.318    -.0431265    .1327471
      u2h_fe |   .0765144   .0459404     1.67   0.096    -.0135272     .166556
       frfam |   .0004108   .0205161     0.02   0.984       -.0398    .0406215
       _cons |   1.510566   .1033122    14.62   0.000     1.308078    1.713054
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+
```

The coefficient on `time` is 0.0448 ($p = 0.318$). The residual coefficient is $\hat{\rho} = 0.0765$ ($p = 0.096$):

positive, but imprecisely estimated. The p value is not a measure of how large the selection is.

26.3.1. A comparison with linear models

In a linear model, the control-function estimator gives the same coefficient as 2SLS. I fit both as a check on the two-step code:

```
clear all
webuse website, clear

* 1. Linear 2SLS
ivreghdfe visits frfam (time=phone), absorb(ad female)

* 2. Linear control function
reghdfe time phone frfam, absorb(ad female) resid
predict double u2h_fe, resid
reghdfe visits time u2h_fe frfam, absorb(ad female)

* 3. Naive models (ignoring endogeneity)
reghdfe visits time frfam, absorb(ad female)
ppmlhdfe visits time frfam, absorb(ad female)
```

```
(Visits to website)

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

IV (2SLS) estimation
--------------------

Estimates efficient for homoskedasticity only
Statistics consistent for homoskedasticity only

                                                      Number of obs =      499
                                                      F(  2,   484) =     8.31
                                                      Prob > F      =   0.0003
Total (centered) SS     =  3335.918099                Centered R2   =   0.3971
Total (uncentered) SS   =  3335.918099                Uncentered R2 =   0.3971
Residual SS             =  2011.119506                Root MSE      =    2.038

------------------------------------------------------------------------------
      visits | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .5642802   .2123745     2.66   0.008     .1469904      .98157
       frfam |    -.04045   .0922873    -0.44   0.661    -.2217832    .1408833
------------------------------------------------------------------------------
```

```
Underidentification test (Anderson canon. corr. LM statistic):          18.494
                                                    Chi-sq(1) P-val =    0.0000
------------------------------------------------------------------------------
Weak identification test (Cragg-Donald Wald F statistic):               18.628
Stock-Yogo weak ID test critical values: 10% maximal IV size             16.38
                                         15% maximal IV size              8.96
                                         20% maximal IV size              6.66
                                         25% maximal IV size              5.53
Source: Stock-Yogo (2005).  Reproduced by permission.
------------------------------------------------------------------------------
Sargan statistic (overidentification test of all instruments):           0.000
                                                  (equation exactly identified)
------------------------------------------------------------------------------
Instrumented:         time
Included instruments: frfam
Excluded instruments: phone
Partialled-out:       _cons
                      nb: total SS, model F and R2s are after partialling-out;
                          any small-sample adjustments include partialled-out
                          variables in regressor count K
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   2,    484) =      29.02
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.3137
                                                  Adj R-squared   =     0.2938
                                                  Within R-sq.    =     0.1071
                                                  Root MSE        =     2.2239

------------------------------------------------------------------------------
        time | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       phone |   .2705429   .0626834     4.32   0.000     .1473778     .393708
       frfam |   .2539231   .0619373     4.10   0.000     .1322239    .3756222
       _cons |   1.406026   .2549245     5.52   0.000     .9051305    1.906921
------------------------------------------------------------------------------
```

```
Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(1 missing value generated)

(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   3,    483) =     117.11
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.7677
                                                  Adj R-squared   =     0.7605
                                                  Within R-sq.    =     0.4211
                                                  Root MSE        =     1.9996

------------------------------------------------------------------------------
      visits | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .5642802   .2083276     2.71   0.007       .15494    .9736205
      u2h_fe |   .1827169   .2122987     0.86   0.390    -.2344262    .5998601
       frfam |    -.04045   .0905287    -0.45   0.655    -.2183288    .1374288
       _cons |   3.357879   .4427337     7.58   0.000     2.487957    4.227801
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(dropped 1 singleton observations)
(MWFE estimator converged in 3 iterations)

HDFE Linear regression                            Number of obs   =        499
Absorbing 2 HDFE groups                           F(   2,    484) =     175.38
                                                  Prob > F        =     0.0000
                                                  R-squared       =     0.7673
                                                  Adj R-squared   =     0.7606
                                                  Within R-sq.    =     0.4202
                                                  Root MSE        =     1.9991
```

```
------------------------------------------------------------------------------
      visits | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .7402254   .0400951    18.46   0.000     .6614435    .8190073
       frfam |  -.1034523    .053248    -1.94   0.053     -.208078    .0011735
       _cons |   3.027348   .2202211    13.75   0.000     2.594641    3.460055
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+

(dropped 1 observations that are either singletons or separated by a fixed effe
> ct)
Iteration 1:   deviance = 3.7573e+02  eps = .         iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.64  P
Iteration 2:   deviance = 3.6107e+02  eps = 4.06e-02  iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.64
Iteration 3:   deviance = 3.6098e+02  eps = 2.57e-04  iters = 3    tol = 1.0e-0
> 4
>    min(eta) =  -0.64
Iteration 4:   deviance = 3.6098e+02  eps = 3.76e-08  iters = 2    tol = 1.0e-0
> 4
>    min(eta) =  -0.64
Iteration 5:   deviance = 3.6098e+02  eps = 1.45e-15  iters = 2    tol = 1.0e-0
> 5
>    min(eta) =  -0.64   S
Iteration 6:   deviance = 3.6098e+02  eps = 0.00e+00  iters = 1    tol = 1.0e-0
> 9
>    min(eta) =  -0.64   S O
------------------------------------------------------------------------------
> ------------------------------
(legend: p: exact partial-out   s: exact solver   h: step-halving   o: epsilon
> below tolerance)
Converged in 6 iterations and 14 HDFE sub-iterations (tol = 1.0e-08)

HDFE PPML regression                              No. of obs      =        499
Absorbing 2 HDFE groups                           Residual df     =        484
                                                  Wald chi2(2)    =     164.67
Deviance             =  360.9797175               Prob > chi2     =     0.0000
Log pseudolikelihood = -995.1432059               Pseudo R2       =     0.2967
```

```
------------------------------------------------------------------------------
             |               Robust
      visits | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        time |   .1187612   .0095895    12.38   0.000     .0999662    .1375563
       frfam |  -.0281572   .0099614    -2.83   0.005    -.0476812   -.0086332
       _cons |   1.345121   .0497098    27.06   0.000     1.247692    1.442551
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
          ad |        12           0          12     |
      female |         2           1           1     |
-----------------------------------------------------+
```

Model	Estimator	`time` Coef.	Std. Err.	p-value
Naive Linear FE	`reghdfe`	0.7402	0.0401	< 0.001
Linear 2SLS	`ivreghdfe`	0.5643	0.2124	0.008
Linear Control Function	Manual two-step	0.5643	0.2083	0.007
Naive FE Poisson	`ppmlhdfe` (no residual)	0.1188	0.0096	< 0.001
Control-Function FE Poisson	`ppmlhdfe` (with $\hat{v}$)	0.0448	0.0449	0.318

The manual linear control function reproduces the 2SLS coefficient exactly. In the Poisson model, adding the control-function residual reduces the `time` coefficient from 0.1188 to 0.0448.

26.3.2. Cluster bootstrap in Stata

The bootstrap can draw the same cluster more than once. Each copy must get a new identifier through `idcluster()`; otherwise Stata pools the copies into one fixed effect.

```
clear all
capture program drop ppmlhdfe_cf
program ppmlhdfe_cf, rclass
  reghdfe time phone frfam, absorb(newid female) resid
  predict double u2h_fe, resid
  ppmlhdfe visits time u2h_fe frfam, absorb(newid female)
  return scalar b_time = _b[time]
  return scalar b_u2h  = _b[u2h_fe]
```

```
  drop u2h_fe
  xtset, clear
end

webuse website, clear
xtset, clear
bootstrap r(b_time) r(b_u2h), reps(1000) seed(123) cluster(ad) idcluster(newid) nodots: ppmlhdfe_cf
```

```
(Visits to website)



Bootstrap results                                       Number of obs =   499
                                                        Replications  = 1,000

      Command: ppmlhdfe_cf
        _bs_1: r(b_time)
        _bs_2: r(b_u2h)

                                    (Replications based on 12 clusters in ad)
------------------------------------------------------------------------------
             |   Observed   Bootstrap                         Normal-based
             | coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _bs_1 |   .0448103   .0477457     0.94   0.348    -.0487697    .1383902
       _bs_2 |   .0765144   .0486568     1.57   0.116    -.0188511    .1718799
------------------------------------------------------------------------------
```

The bootstrap standard error on `time` is 0.0477 (compared to the unadjusted analytic standard error of 0.0449).

26.4. The same estimator in R

In R, I use `fixest::feols()` and `fixest::fepois()`. `feols()` drops a singleton cluster here (`ad == 10`), so we use `obs()` to align the residuals with the rows kept in the first stage:

```
library(fixest)
library(haven)

df <- as.data.frame(read_dta("https://www.stata-press.com/data/r18/website.dta"))
df$ad     <- as.factor(df$ad)
df$female <- as.factor(df$female)

# Step 1: Linear first stage
fs <- feols(time ~ phone + frfam | ad + female, data = df)
```

```r
# Align rows (feols drops singletons)
d <- df[obs(fs), ]
d$u2h_fe <- residuals(fs)

# Step 2: FE Poisson with and without control function
m_naive <- fepois(visits ~ time + frfam          | ad + female, data = d, notes = FALSE,
                  vcov = "hetero")  # robust, as ppmlhdfe reports by default
m_cf    <- fepois(visits ~ time + u2h_fe + frfam | ad + female, data = d, notes = FALSE,
                  vcov = "hetero")

etable(m_naive, m_cf, headers = c("Naive FE Poisson", "Control Function"), se.below = TRUE)
```

```
                        m_naive            m_cf
                Naive FE Poisson Control Function
Dependent Var.:           visits           visits

time                   0.1188***         0.0448
                      (0.0097)          (0.0456)
frfam                 -0.0282**          0.0004
                      (0.0101)          (0.0208)
u2h_fe                                   0.0765
                                        (0.0466)
Fixed-Effects:        ----------        --------
ad                           Yes               Yes
female                       Yes               Yes
_________________     __________        ________
S.E. type             Hete.-rob.        Het.-rob.
Observations                 499               499
Squared Cor.             0.75130           0.75528
Pseudo R2                0.29670           0.29761
BIC                      2,083.5           2,087.1
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The point estimates match Stata to seven digits: 0.1187612 (naive) and 0.0448103 (control function). With `vcov = "hetero"` the standard errors are heteroskedasticity-robust, as `ppmlhdfe` reports by default: 0.0097 and 0.0456 on `time`, against Stata's 0.0096 and 0.0449. The small gap is `fixest`'s default small-sample correction.

26.4.1. Cluster bootstrap in R

```r
set.seed(123)
B <- 1000
clusters   <- unique(d$ad)
```

```
boot_coefs <- matrix(NA_real_, nrow = B, ncol = 2, dimnames = list(NULL, c("time", "resid")))

for (b in seq_len(B)) {
  drawn <- sample(clusters, length(clusters), replace = TRUE)
  boot_data <- do.call(rbind, lapply(seq_along(drawn), function(k) {
    sub <- d[d$ad == drawn[k], ]
    sub$boot_id <- k
    sub
  }))
  boot_data$boot_id <- as.factor(boot_data$boot_id)

  fs_b <- feols(time ~ phone + frfam | boot_id + female, data = boot_data, notes = FALSE)
  bb   <- boot_data[obs(fs_b), ]
  bb$u_b <- residuals(fs_b)

  m_b <- try(fepois(visits ~ time + u_b + frfam | boot_id + female, data = bb, notes = FALSE), silent = T
  if (!inherits(m_b, "try-error")) boot_coefs[b, ] <- coef(m_b)[c("time", "u_b")]
}

cat("Completed replications :", sum(complete.cases(boot_coefs)), "\n")
```

```
Completed replications : 1000
```

```
cat("Bootstrap SE, time     :", round(sd(boot_coefs[, "time"],  na.rm = TRUE), 4), "\n")
```

```
Bootstrap SE, time     : 0.045
```

```
cat("Bootstrap SE, residual :", round(sd(boot_coefs[, "resid"], na.rm = TRUE), 4), "\n")
```

```
Bootstrap SE, residual : 0.0457
```

The R bootstrap yields SE = 0.0450. With only 12 clusters in `ad`, small differences across software packages (0.0477 in Stata vs. 0.0450 in R) reflect finite-sample sensitivity in how rare degenerate resamples and singletons are handled.

26.5. A binary endogenous regressor

When D_{it} is binary, as with program participation, a linear first stage is misspecified because its error cannot satisfy the required conditional-independence restriction.

26.5.1. A CRE logit first stage

A first-stage logit with a dummy for every unit has an incidental-parameters problem when T is small. For example, its slopes can be about twice as large as they should be when $T = 2$. I instead use pooled logit with a Mundlak adjustment, adding the unit means of the time-varying covariates:

$$P(D_{it} = 1 \mid z_{it}, \bar{z}_i) = \Lambda(z_{it}\gamma + \bar{z}_i\pi)$$

For logit, the generalized residual is simply the response residual,

$$\hat{r}_{it} = D_{it} - \hat{p}_{it}.$$

We then include $\hat{r}_{it}$ in the fixed-effects Poisson model.

26.5.2. A simulation

Let's simulate 2,000 units with unbalanced panels, where T_i ranges from 2 to 6. The treatment D is binary and endogenous, and its true coefficient is $\beta = 0.30$.

```
preserve
clear
set seed 20260731
set obs 2000
gen i = _n
gen double ci = rnormal()*0.5
gen T = 2 + int(runiform()*5)        /* 2-6 observations per unit */
expand T
bysort i: gen t = _n

gen double z = rnormal()             /* instrument */
gen double e = rnormal()
gen double u = 0.8*e + rnormal()*0.6 /* correlated with e -> endogeneity */
gen byte   d = (-0.2 + 0.9*z + ci + e > 0)
gen double mu = exp(0.3*d + ci + 0.5*u)
gen y = rpoisson(mu)                 /* true beta = 0.3 */

* 1. Naive FE Poisson (ignores endogeneity)
quietly ppmlhdfe y d, absorb(i)
scalar b_naive = _b[d]

* 2. CF with CRE (Mundlak) logit first stage
bysort i: egen double z_i = mean(z)
quietly logit d z z_i
predict double phat, pr
gen double gr = d - phat
quietly ppmlhdfe y d gr, absorb(i)
```

```
scalar b_cre = _b[d]

* 3. CF with linear first stage
quietly reghdfe d z, absorb(i) resid
predict double v_lin, resid
quietly ppmlhdfe y d v_lin, absorb(i)
scalar b_lin = _b[d]

di "True coefficient on d       = 0.300000"
di "Naive FE Poisson            = " %8.6f b_naive
di "CF, CRE (Mundlak) Logit     = " %8.6f b_cre
di "CF, Linear First Stage      = " %8.6f b_lin
restore
```

```
Number of observations (_N) was 0, now 2,000.
```

```
(5,969 observations created)
```

```
True coefficient on d       = 0.300000

Naive FE Poisson            = 0.747206

CF, CRE (Mundlak) Logit     = 0.293829

CF, Linear First Stage      = 0.277644
```

The naive fixed-effects Poisson estimate is 0.747, more than 2.4 times the true value. The CRE logit control function gives 0.294, while the linear first-stage control function gives 0.278.

These two are not specifications of equal standing, and reporting them side by side as coequal sensitivity checks would misrepresent the second. The CRE logit control function is the better of the two here, but it is an approximation too. The DGP draws treatment from a normal latent error, so the first stage is a probit, not a logit, and with normal errors the selection factor, the mean of $\exp(0.5u)$ given treatment and the instrument (with u the outcome error in the simulation code), is a nonlinear function of the treatment index. A linear residual term in the exponential mean captures it only approximately, and 0.294 against 0.30 in one draw does not establish consistency. The linear first stage is a misspecified benchmark: as noted above, a linear reduced form for a binary D_{it} has no valid independence assumption behind it, so it is inconsistent whatever the sample size. Agreement between the two says the misspecification happens not to bite much in this sample. It is not evidence that the linear version is an alternative estimator. When the approximation is in doubt, the comparison to reach for is with the exact correction of Terza (1998) or with multiplicative-moment GMM. Both carry stated assumptions, and both are set out in the Poisson IV chapter of the causal econometrics guide.

26.6. What to remember

The control-function approach adds the first-stage residual to the exponential outcome model. For a continuous endogenous variable, we can combine `reghdfe` with `ppmlhdfe` in Stata, or `feols()` with `fepois()` in R. Inference should bootstrap both stages by cluster. For a binary endogenous variable, a CRE logit first stage avoids the incidental-parameters problem from unit dummies, and its residual is $d - \hat{p}$.

Part VI.

Causal Inference Methods

27. Using machine learning for causal effect in observational study

27.1. A simulation for an OLS model

In an observational study, we need to assume we have the functional form to get causal effect estimated correctly, in addition to the assumption of treatment being exogenous.

The simulation makes that concrete. We draw $n = 2000$ observations of three correlated standard normals (x, z, w), with covariances 0.8^2 between x and w, 0.5^2 between x and z, and 0.6^2 between z and w. The DGP then runs on the *logs of their squares*, not on the variables themselves: treatment is $A \sim \text{Bernoulli}(1/(1+e^g))$ with $g = \log(w^2 + 0.01) + N(0,1)$, the untreated outcome is $y_0 = \log(x^2 + 0.01) + N(0,1)$, and the individual treatment effect is $\tau_i = 2 + N(0,1)$, so the true ATE is exactly 2.

The point is what an analyst sees. Treatment depends on $\log w^2$ and the outcome on $\log x^2$, but only x, z and w are observed. This is the common situation: we have the right variables and the wrong functional form.

```
library(MASS)
library(ggplot2)
library(dplyr)
library(tmle)
library(glmnet)
set.seed(366)

nobs <- 2000
xw <- .8
xz <- .5
zw <- .6
nrow <- 3
ncol <- 3
covarMat = matrix( c(1^2, xz^2, xw^2, xz^2, 1^2, zw^2,  xw^2, zw^2, 1^2 ) , nrow=ncol , ncol=ncol )

mu <- rep(0,3)
rawvars <- mvrnorm(n=nobs, mu=mu, Sigma=covarMat)
df <- as_tibble(rawvars, .name_repair = "minimal")
names(df) <- c('x','z','w')
df <- df %>%
    # A small constant inside each log() keeps log(v^2) from diverging to
    # large negative values when v is close to 0 (e.g. log(w^2) alone ranges
```

```
    # down to about -15 here), which otherwise pushes the propensity score
    # P(A=1|W) toward 0/1 for those units and violates positivity.
    mutate(log.x=log(x^2 + 0.01), log.z=log(z^2 + 0.01), log.w=log(w^2 + 0.01), z.sqr=z^2, w.sqr=w^2) %>%
    mutate(g.var= log.w  + rnorm(nobs)) %>%
    mutate(A = rbinom(nobs, 1, 1/(1+exp((g.var))))) %>%
    mutate(y0=rnorm(nobs) + log.x) %>%
    mutate(tau.true = 2  + rnorm(nobs), y1=y0+tau.true, treat=A, y = treat*y1 + (1-treat)*y0)
lm1 <- lm(y ~ A + log.w + log.x , data=df)
summary(lm1)
```

```
Call:
lm(formula = y ~ A + log.w + log.x, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5207 -0.8338 -0.0089  0.8386  4.1534

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.01311    0.04774   0.275    0.784
A            1.95487    0.06703  29.166   <2e-16 ***
log.w        0.02835    0.01945   1.458    0.145
log.x        1.00090    0.01748  57.264   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.253 on 1996 degrees of freedom
Multiple R-squared:  0.6746,    Adjusted R-squared:  0.6741
F-statistic:  1380 on 3 and 1996 DF,  p-value: < 2.2e-16
```

```
lm2 <- lm(y ~ A , data=df)
summary(lm2)
```

```
Call:
lm(formula = y ~ A, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-6.1667 -1.4416  0.1372  1.4760  6.1625

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75733    0.07539  -10.04   <2e-16 ***
A            1.47007    0.09571   15.36   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.077 on 1998 degrees of freedom
Multiple R-squared:  0.1056,    Adjusted R-squared:  0.1052
F-statistic: 235.9 on 1 and 1998 DF,  p-value: < 2.2e-16
```

```
lm3 <- lm(y ~ A + w, data=df)
summary(lm3)
```

```

Call:
lm(formula = y ~ A + w, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-6.1657 -1.4531  0.1338  1.4780  6.0957

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75631    0.07541 -10.030   <2e-16 ***
A            1.46876    0.09573  15.343   <2e-16 ***
w           -0.03874    0.04464  -0.868    0.386
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.077 on 1997 degrees of freedom
Multiple R-squared:  0.1059,    Adjusted R-squared:  0.105
F-statistic: 118.3 on 2 and 1997 DF,  p-value: < 2.2e-16
```

```
lm4 <- lm(y ~ A + w + x, data=df)
summary(lm4)
```

```

Call:
lm(formula = y ~ A + w + x, data = df)

Residuals:
   Min     1Q Median     3Q    Max
-6.167 -1.445  0.147  1.473  6.205

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.75360    0.07539  -9.996   <2e-16 ***
A            1.46387    0.09573  15.292   <2e-16 ***
w           -0.09981    0.05764  -1.732   0.0835 .
```

```
x            0.10281    0.06141   1.674   0.0943 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.076 on 1996 degrees of freedom
Multiple R-squared:  0.1072,    Adjusted R-squared:  0.1059
F-statistic: 79.88 on 3 and 1996 DF,  p-value: < 2.2e-16
```

In this example, treatment assignment process is determined by logged w, and outcome is determined by logged x and treatment. However, what we observe is w and x. In observational studies, this happens all the time. In fact, this is an ideal situation, that we observe variables that are determinants of outcome, although we are not sure about the functional form that determines the outcome. However, this example shows that unless we have observed exactly the factors themselves (in this case logged x, w, which determines the DGP), we have biased estimates of the true treatment effect.

The four estimates are 1.955, 1.470, 1.469 and 1.464 against a true 2.

Model 1 is the only model with reasonable estimate of treatment effect (which is 2 in this case), at 1.955 with a standard error of 0.067 — and it is also the only one given the correct functional form, `log.x` and `log.w`. Its `log.x` coefficient of 1.001 recovers the DGP's coefficient of 1 almost exactly. Model 2 is a model with endogeneity: A is correlated with the missing variable logged x. Model 3 and 4 we have x and w, but not logged, therefore still biased.

The lesson here is the functional form does matter. However, we have no way of knowing the functional form. What can we do here?

```
# Algorithm set trimmed to 7 fast learners for render speed.
# A production analysis would also include SL.randomForest, SL.gbm, SL.gam.
Q.SL.library <- c("SL.glmnet","SL.glm","SL.glm.interaction", "SL.rpart","SL.bayesglm","SL.step","SL.mean"
g.SL.library <- c("SL.glmnet","SL.glm","SL.glm.interaction", "SL.rpart","SL.bayesglm","SL.step","SL.mean"

# tmle1 uses x and w; tmle2 additionally includes z.  Note z is not in the true
# treatment or outcome DGP -- it is only correlated with x and w -- so adding it
# should not change the estimate much.
set.seed(367)  # SuperLearner's cross-validation folds are random
tmle1 <- tmle(Y = df$y, A = df$treat, W = df[,c('x','w')], g.SL.library = g.SL.library , Q.SL.library = Q
tmle1
```

```
Marginal mean under treatment (EY1)
  Parameter Estimate:  0.91815
  Estimated Variance:  0.0033321
             p-value:  <2e-16
   95% Conf Interval:  (0.80501, 1.0313)

Marginal mean under comparator (EY0)
  Parameter Estimate:  -1.0301
  Estimated Variance:  0.0037223
```

```
            p-value:  <2e-16
   95% Conf Interval:  (-1.1496, -0.91049)

Additive Effect
  Parameter Estimate:  1.9482
  Estimated Variance:  0.004408
            p-value:  <2e-16
   95% Conf Interval:  (1.8181, 2.0783)

Additive Effect among the Treated
  Parameter Estimate:  1.9154
  Estimated Variance:  0.0056111
            p-value:  <2e-16
   95% Conf Interval:  (1.7686, 2.0622)

Additive Effect among the Controls
  Parameter Estimate:  2.0147
  Estimated Variance:  0.0051554
            p-value:  <2e-16
   95% Conf Interval:  (1.874, 2.1554)
```

```
set.seed(368)
tmle2 <- tmle(Y = df$y, A = df$treat, W = df[,c('x','w', 'z')], g.SL.library = g.SL.library , Q.SL.libra
tmle2
```

```
Marginal mean under treatment (EY1)
  Parameter Estimate:  0.91164
  Estimated Variance:  0.0033311
            p-value:  <2e-16
   95% Conf Interval:  (0.79852, 1.0248)

Marginal mean under comparator (EY0)
  Parameter Estimate:  -1.0283
  Estimated Variance:  0.0036383
            p-value:  <2e-16
   95% Conf Interval:  (-1.1465, -0.91003)

Additive Effect
  Parameter Estimate:  1.9399
  Estimated Variance:  0.0043666
            p-value:  <2e-16
   95% Conf Interval:  (1.8104, 2.0694)

Additive Effect among the Treated
  Parameter Estimate:  1.9056
  Estimated Variance:  0.0055007
            p-value:  <2e-16
```

```
    95% Conf Interval:  (1.7603, 2.051)

 Additive Effect among the Controls
   Parameter Estimate:  1.9933
   Estimated Variance:  0.0052122
              p-value:  <2e-16
    95% Conf Interval:  (1.8518, 2.1348)
```

We use the targeted maximum likelihood method of Laan and Rubin (2006; see also Laan and Rose 2011). It uses SuperLearner as the initial estimator. It's an ensemble of multiple machine learning algorithms. Therefore it does not need to assume the functional form of the DGP. Even if we don't have the variables that determines the DGP of outcome, if we observe some functions (even nonlinear functions) of these variables, we can still get reasonable estimates of the treatment effect.

In this example, we used multiple popular machine learning algorithms in modeling both treatment assignment process and the outcome process. The first TMLE model is with x and w (note not the logged x and w which are in the true DGP), the second one with an additional variable z.

TMLE gives an additive effect of 1.9482 with x and w, and 1.9399 when z is added — against a true 2, and against 1.464 to 1.470 for the linear models given the same untransformed variables. The 95% intervals, $[1.818, 2.078]$ and $[1.810, 2.069]$, both cover 2. Adding z changes almost nothing, which is what should happen: z is in neither the treatment nor the outcome equation and is merely correlated with the variables that are.

It seems that TMLE results are less biased than the linear models with x and w. That is putting it mildly: the linear models miss by about 0.53 and TMLE by about 0.04, an order of magnitude better, purely from not having to guess the functional form. It may not be better than the linear model with logged x and w, but in empirical studies, we often cannot assume we have the variables in the DGP, but only some proxy of the variables in the DGP. I'll do more simulations to see whether TMLE does perform better in the situation that we are not sure about the functional form. We should expect that is the case.

The simple `tmle` package example used here is for a binary treatment. TMLE itself is not limited to binary treatments: there are TMLE estimators for continuous, multivalued, longitudinal, stochastic, and survival settings (e.g. the `ltmle` and `lmtp` packages), though which estimand and package you use differs by setting.

It's about time we embrace machine learning techniques into studies of causal effect in observational studies.

27.2. Summary

Regression gives a causal effect only if it has both the right variables and the right functional form. In the simulation the outcome and treatment depend on the logs of the squares of x and w. With those transformations, OLS gives 1.955 against a true effect of 2. With x and w entered directly, it gives about 1.47, biased by half a unit even though the right variables are in the model.

TMLE with a SuperLearner ensemble for both the treatment and the outcome does not need the functional form. Given only x and w it gives 1.948, and 1.940 with the irrelevant z added. Both intervals cover 2. That is an order of magnitude closer than the linear models with the same variables.

Systematic treatment: R · Julia.

28. Mediation analysis in R and Stata

28.1. Mediation analysis

The traditional mediation model has two equations:

$$Y = aX + bM + \epsilon_1 \tag{28.1}$$

$$M = cX + \epsilon_2 \tag{28.2}$$

The effect of X on Y decomposes into a direct effect and an indirect effect through M. The classical approach is Baron and Kenny's four-step procedure. Modern implementations use SEM to estimate both equations jointly, as in R's `lavaan` and Stata's `sem`.

```
library(lavaan)
set.seed(1234)
n <- 10000
X <- rnorm(n)
M <- 0.5*X + rnorm(n)
Y <- 0.7*M + 0.3*X + rnorm(n)
Data <- data.frame(X = X, Y = Y, M = M)
model <- '   Y ~ a*X + b*M
             M ~ c*X
           # direct effect (a)
           # indirect effect (b*c)
             bc := b*c
           # total effect
             total := a + (b*c)
         '
fit <- sem(model, data = Data)
summary(fit)
```

```
lavaan 0.7-2 ended normally after 1 iteration

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                         5

  Number of observations                         10000

Model Test User Model:
```

```
  Test statistic                                0.000
  Degrees of freedom                                0

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Expected
  Information saturated (h1) model          Structured

Regressions:
                   Estimate  Std.Err  z-value  P(>|z|)
  Y ~
    X          (a)    0.286    0.011   25.233    0.000
    M          (b)    0.699    0.010   70.384    0.000
  M ~
    X          (c)    0.511    0.010   50.161    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Y                 0.998    0.014   70.711    0.000
   .M                 1.012    0.014   70.711    0.000

Defined Parameters:
                   Estimate  Std.Err  z-value  P(>|z|)
    bc                 0.357    0.009   40.849    0.000
    total              0.643    0.012   51.956    0.000
```

We simulate $n = 10{,}000$ observations with $X \sim N(0,1)$, $M = 0.5X + \varepsilon_2$ and $Y = 0.7M + 0.3X + \varepsilon_1$, both errors standard normal. So the true direct effect is $a = 0.3$, the mediator coefficient is $b = 0.7$, the treatment-to-mediator path is $c = 0.5$, the indirect effect is $bc = 0.35$ and the total is 0.65. There is no confounding of any kind.

`lavaan` recovers all of it: $a = 0.286$, $b = 0.699$, $c = 0.511$, giving $bc = 0.357$ and a total of 0.643. The model is saturated (0 degrees of freedom), so the perfect fit statistic carries no information.

The same analysis in Stata:

```
set obs 10000
gen x= rnormal()
gen m= .5*x + rnormal()
gen y= .7*m + .3*x +rnormal()
sem (y <- x m) (m <- x)
estat teffects
```

```
Number of observations (_N) was 0, now 10,000.
```

```
Endogenous variables
  Observed: y m

Exogenous variables
  Observed: x

Fitting target model:
Iteration 0:  Log likelihood = -42567.918
Iteration 1:  Log likelihood = -42567.918

Structural equation model                               Number of obs = 10,000
Estimation method: ml

Log likelihood = -42567.918

------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .2971221   .0113774    26.12   0.000     .2748228    .3194215
       _cons |  -.0000391   .0101082    -0.00   0.997    -.0198508    .0197725
  -----------+----------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
       _cons |   .0187204   .0099478     1.88   0.060    -.0007769    .0382177
-------------+----------------------------------------------------------------
     var(e.y)|   1.021391   .0144446                      .9934684    1.050098
     var(e.m)|   .9895863   .0139949                      .9625336    1.017399
------------------------------------------------------------------------------
LR test of model vs. saturated: chi2(0) = 0.00                Prob > chi2 = .



Direct effects
------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .2971221   .0113774    26.12   0.000     .2748228    .3194215
```

```
  -----------+------------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
-------------------------------------------------------------------------------


Indirect effects
-------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+-----------------------------------------------------------------
Structural   |
  y          |
           m |          0  (no path)
           x |   .3595823   .0087834    40.94   0.000     .3423672    .3767974
  -----------+-----------------------------------------------------------------
  m          |
           x |          0  (no path)
-------------------------------------------------------------------------------


Total effects
-------------------------------------------------------------------------------
             |                 OIM
             | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+-----------------------------------------------------------------
Structural   |
  y          |
           m |   .7136545   .0101594    70.25   0.000     .6937424    .7335666
           x |   .6567044   .0124172    52.89   0.000     .6323672    .6810417
  -----------+-----------------------------------------------------------------
  m          |
           x |   .5038605   .0100014    50.38   0.000     .4842581    .5234628
-------------------------------------------------------------------------------
```

The above examples should have direct effect of .3 and indirect effect of .35, and total effect of .65.

28.2. Causal Mediation analysis

The traditional mediation analysis has been criticized for the lack of causal interpretation. Without manipulation of the mediator, it is hard to interpret the effects causally, because even if the treatment is from random experiments, the mediator is often not. Therefore there could be an unmeasured confounder that is causing both M and Y.

R's "mediation" package is for causal mediation analysis. It uses simulation to estimate the causal effects of treatment, under assumptions of sequential ignorability.

Identifying assumptions for natural (in)direct effects

Before reading the `mediate()` output causally, the following must hold (this is "sequential ignorability" plus the standard support and link conditions):

1. **Treatment ignorability** — no unmeasured treatment–outcome confounding given covariates W.
2. **Mediator ignorability** — no unmeasured mediator–outcome confounding given W and treatment.
3. **No treatment-induced confounding** — no variable affected by the treatment confounds the mediator–outcome relationship (this is the strong, often-violated condition).
4. **Positivity** — both treatment and mediator have positive probability across W.
5. **Consistency** — observed outcomes equal the corresponding potential outcomes.
6. **Cross-world independence** — the condition underlying natural effects, which no single experiment can test.

Mediator ignorability and the no-treatment-induced-confounding condition are the ones that usually break, because the mediator is rarely randomized.

It estimates the following quantities:

$$\tau_i = Y_i(1, M_i(1)) - Y_i(0, M_i(0)) \tag{28.3}$$

This is the total treatment effect, which is to say, what's the change in Y if we change each unit from control to treated, hypothetically?

Then this can be decomposed into the causal mediation effects:

$$\delta_i(t) = Y_i(t, M_i(1)) - Y_i(t, M_i(0)) \tag{28.4}$$

for treatment status $t = 0, 1$. This is to say, given treatment status, what's the mediation effect?

$$\eta_i(t) = Y_i(1, M_i(t)) - Y_i(0, M_i(t)) \tag{28.5}$$

for treatment status $t = 0, 1$. This is to say, given mediator status for each treatment status, what's the direct effect?

Therefore there are four quantities estimated, direct and mediation effect for treated and control.

R's "mediation" needs users to feed two models, outcome model and mediation model.

If we study the same data, we would expect it returns the same estimates as the traditional methods. However, the causal mediation models can be much more flexible in outcome and mediation models.

```
library(mediation)
med.fit <- lm(M ~ X, data=Data)
summary(med.fit)
```

```
Call:
lm(formula = M ~ X, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.1134 -0.6972  0.0086  0.6837  3.7309

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.007814   0.010060  -0.777    0.437
X            0.510954   0.010187  50.156   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.006 on 9998 degrees of freedom
Multiple R-squared:  0.201, Adjusted R-squared:  0.2009
F-statistic:  2516 on 1 and 9998 DF,  p-value: < 2.2e-16
```

```
out.fit <- lm(Y ~ X + M,data=Data)
summary(out.fit)
```

```
Call:
lm(formula = Y ~ X + M, data = Data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.0680 -0.6771 -0.0081  0.6643  3.8304

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.004170   0.009992   0.417    0.676
X           0.285582   0.011319  25.229   <2e-16 ***
M           0.699006   0.009933  70.373   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9991 on 9997 degrees of freedom
Multiple R-squared:  0.4734,    Adjusted R-squared:  0.4733
F-statistic:  4494 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```
set.seed(123)
med.out <- mediate(med.fit, out.fit, treat = "X", mediator = "M", sims = 100)
summary(med.out)
```

```
Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

               Estimate 95% CI Lower 95% CI Upper   p-value
ACME            0.35684      0.34208      0.37741 < 2.2e-16 ***
ADE             0.28540      0.26522      0.30604 < 2.2e-16 ***
Total Effect    0.64224      0.62091      0.66752 < 2.2e-16 ***
Prop. Mediated  0.55530      0.53571      0.57897 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 10000


Simulations: 100
```

And it does. The two component regressions give $c = 0.510954$ and, in the outcome equation, $a = 0.285582$ and $b = 0.699006$ — the same numbers `lavaan` produced. `mediate()` then reports ACME 0.35684 (against a true 0.35), ADE 0.28540 (against 0.30) and a total effect of 0.64224 (against 0.65), with the proportion mediated at 0.555. The simulation-based intervals all exclude zero.

The point is that on this data the causal machinery adds nothing — which is the right outcome for a DGP with no confounding, a linear outcome model and no treatment-mediator interaction. Its value appears when those conditions fail.

28.3. Binary outcome

For example, in the case of binary outcome, the traditional approach will have difficulties. We can estimate the outcome model and mediator model jointly, but the total effects are not easy to decompose into direct and indirect effect (see Imai, Keele, and Tingley 2010, 320).

The causal mediation analysis framework is much more general.

```
library(mediation)
data(framing)
med.fit <- lm(emo ~ treat + age + educ + gender + income, data = framing)
out.fit <- glm(cong_mesg ~ emo + treat + age + educ + gender + income,
               data = framing, family = binomial("probit"))
set.seed(123)
med.out <- mediate(med.fit, out.fit, treat = "treat", mediator = "emo",
                   robustSE = TRUE, sims = 100)
summary(med.out)
```

```
Causal Mediation Analysis

Quasi-Bayesian Confidence Intervals

                           Estimate 95% CI Lower 95% CI Upper p-value
ACME (control)             0.081341     0.037680     0.122101  <2e-16 ***
ACME (treated)             0.082417     0.036606     0.126460  <2e-16 ***
ADE (control)              0.018299    -0.105898     0.158469    0.72
ADE (treated)              0.019375    -0.115315     0.172002    0.72
Total Effect               0.100716    -0.028036     0.259054    0.24
Prop. Mediated (control)   0.632609   -13.705651     4.305615    0.24
Prop. Mediated (treated)   0.661761   -12.422330     4.036601    0.24
ACME (average)             0.081879     0.037143     0.124080  <2e-16 ***
ADE (average)              0.018837    -0.110606     0.165198    0.72
Prop. Mediated (average)   0.647185   -13.063991     4.171108    0.24
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 265


Simulations: 100
```

Here the decomposition does something the linear example could not. The `framing` data has a binary outcome (`cong_mesg`, whether the subject sent a message to Congress) modelled by probit, with anxiety (`emo`) as the mediator. ACME comes out at 0.0813 under control and 0.0824 under treatment, both significant, while the direct effects are 0.0183 and 0.0194 with $p = 0.72$ and the total effect is 0.1007 with $p = 0.24$.

Read that pattern carefully: the *total* effect is not distinguishable from zero, and neither is the direct effect, yet the mediated effect clearly is. A study reporting only the total effect would have concluded the framing treatment did nothing. The proportion mediated is reported as 0.63, but note its interval runs from -13.7 to 4.3 — a ratio whose denominator is indistinguishable from zero is not a usable quantity, and this is the standard reason not to lean on "proportion mediated".

"mediation" package has more functionalities, such as multilevel, interaction of treatment and mediator, etc.

Stata's sem and gsem commands can model different situations, but the direct effect and indirect effects are not easy to compute, especially when you have binary outcome, or other non-continuous outcome situations. They are not designed for causal mediation analysis.

28.4. Going further: formal causal mediation

The `mediation` package above implements sequential ignorability — a specific set of identifying assumptions. The causal mediation chapter covers the assumptions more formally, introduces the controlled direct effect (CDE) and natural direct/indirect effects (NDE/NIE) using potential-outcome

notation, and discusses identification with unmeasured mediator–outcome confounding. If your setting involves a non-binary outcome, treatment–mediator interaction, or you need a doubly-robust estimator, start there.

28.5. Summary

Classical mediation splits the effect of a treatment into a direct path and an indirect path through a mediator, estimated by two regressions or by SEM. Causal mediation defines the indirect and direct effects with potential outcomes. It needs sequential ignorability, meaning no unmeasured confounding of treatment or mediator, and no confounder of the mediator and outcome that treatment affects. The `mediation` package estimates these effects by simulation from an outcome model and a mediator model.

In the linear simulation, `lavaan` and `mediation` agree: the indirect effect is about 0.357 against a true 0.35, and the direct effect 0.286 against 0.3. With no confounding and a linear model, the causal machinery adds nothing. With the binary outcome in the `framing` data, the mediated effect through anxiety is about 0.08 and significant, while the total effect is not. A study that reported only the total effect would have missed it. The proportion mediated is unusable there, because its denominator cannot be told apart from zero.

Systematic treatment: R · Julia.

29. Endogeneity with censored variable

29.1. Bunching for the outcome variable

Bunching is a mass point in the distribution of some variable — for example, the excess density of earned income at a kink in the tax schedule (Saez 2010). The idea is to compare the observed distribution with a counterfactual "manipulation-free" distribution and use the excess mass to estimate behavioral elasticities.

Bunching is closely related to regression discontinuity. RD assumes no manipulation at the threshold; bunching estimators relax that assumption and estimate the fraction of manipulators directly (Kleven (2016)).

C. Caetano, Caetano, and Nielsen (2024) extend bunching to the treatment variable, which is the focus of this chapter.

29.2. Bunching for the treatment variable

29.2.1. A test of endogeneity

C. Caetano (2015) showed that, if the distribution of T_i has bunching at $T_i = \bar{T}$, it is possible to test the exogeneity of T_i . When one compares the outcome of observations at the bunching point and those around it, the treatment itself is very similar. Therefore, there cannot be more than a marginal difference in the outcome that is due to treatment variation, since the treatment hardly varies.

C. Caetano (2015)'s estimation can be done in a two step process:

(1) estimate $E[Y_i|T_i = \bar{T}, X_i]$ non-parametrically

(2) do a local linear regression $E[Y_i|T_i = \bar{T}, X_i] - Y_i$ onto T_i at $\bar{T}$, using only observations such that $T_i > \bar{T}$.

The approach is known as the Discontinuity Test.

29.2.2. Dummy test

G. Caetano and Maheshri (2018) proposed a simpler test, which is simply regressing Y_i on T_i , X_i and a dummy variable for $T_i > \bar{T}$. The coefficient of the dummy variable is the test statistic. If the coefficient is significant, which is to say there is a discontinuity on outcome, then T_i is endogenous.

This is simpler, but with linear functional form assumption.

The logic of the endogeneity test is that around the bunching point, the treatment is very similar, so the outcome should be similar. If there is a discontinuity in the outcome, then it has to be one of two reasons. One is that treatment has discontinuous effect on outcome around the bunching point. Usually we can reasonably rule out that possibility. Second is that there is an unobserved variable that is causing both the treatment and the outcome. That unobserved variable has a discontinuous effect on the outcome. If that's the case, then the treatment is endogenous.

29.2.3. Treatment effect estimation

C. Caetano, Caetano, and Nielsen (2024) proposed a method to estimate the treatment effect when there is bunching. It seems to me there is no reason that we cannot use it for the censoring situation. There are a lot of situations that we have censored treatment variable, when we study continuous treatment variable.

Suppose we have

$$Y = \beta T + Z\gamma + \delta\eta + \epsilon \tag{29.1}$$

where T is the treatment variable, η is the unobserved variable that is causing both the treatment and the outcome, Z is the control variables, and ϵ is the error term. We only observe Y, T, Z.

Suppose T^* is the latent variable that is censored at $T^* = 0$. We observe $T = max(T^*, 0)$. We assume that T^* is continuous over its support.

$$T^* = Z\pi + \eta \tag{29.2}$$

From the two equations, we can derive:

$$E[Y|T, Z] = (\beta + \delta)T + Z(\gamma - \pi\delta) + \delta E[T^*|T^* \leq 0, Z]1(T = 0) \tag{29.3}$$

or

$$E[Y|T, Z] = \beta T + Z(\gamma - \pi\delta) + \delta(T + E[T^*|T^* \leq 0, Z]1(T = 0)) \tag{29.4}$$

Therefore we can "correct" for endogeneity by simply including an additional term $\delta(T + E[T^*|T^* \leq 0, Z]1(T = 0))$ in the regression.

However, the prediction of $\delta(T + E[T^*|T^* \leq 0, Z]1(T = 0))$ is out of sample. We have to make some additional assumptions for the distribution of T^*. For example, we can assume that T^* is normally distributed. Or, we can assume it is symmetric. Then we can use the right tail to estimate the left tail.

29.3. Simulation

Let's do a simulation to see how it works.

We draw $n = 1000$ observations. Two controls (Z_1, Z_2) are joint normal with unit variances and correlation 0.5, and the unobserved confounder is $\eta \sim N(1,1)$. The latent treatment is $T^* = Z_1 + 2Z_2 + 1.5\eta$, censored at zero so that $T = \max(T^*, 0)$, and the outcome is $Y = 2T + 0.5Z_1 + 0.5Z_2 + 2\eta + N(0,1)$.

So the true treatment effect is $\beta = 2$, and η enters both the treatment and the outcome with a positive sign — the endogeneity. Note Y depends on the *observed* censored T, not on T^*, which is what makes the mass point at $T = 0$ informative: among those units the treatment is held fixed while η still varies.

```
library(MASS)
library(ggplot2)
library(dplyr)
# set seed
set.seed(123)
# number of observations
n=1000
# generate Z and make it dataframe
Z=as.data.frame(mvrnorm(n=n, mu=c(0,0), Sigma=matrix(c(1,0.5,0.5,1),2,2)))
data <- Z
names(data) <- c("Z1", "Z2")
# generate eta
# eta is normally distributed
data$eta=rnorm(n, 1,1)
# generate T*
# T* is normally distributed
# Y depends on the OBSERVED (censored) treatment T, not the latent T_star --
# matching the structural equation Y = beta*T + Z*gamma + delta*eta + epsilon
# above. If Y depended on T_star directly, T=0 would not fully "block" the
# latent variable's effect on Y, and the treatment effect of T would not even
# be well defined.
data <- data %>%
  mutate(T_star=Z1 + 2*Z2 + 1.5*eta) %>%
  mutate(T=ifelse(T_star>0, T_star, 0)) %>%
  mutate(Y=2*T + 0.5*Z1 + 0.5*Z2 + 2*eta + rnorm(n, 0, 1))

# regress Y on T, Z1, Z2
lm1 <- lm(Y ~ T + Z1 + Z2, data=data)
summary(lm1)
```

```
Call:
lm(formula = Y ~ T + Z1 + Z2, data = data)
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-8.2706 -0.8700  0.1009  1.0955  4.0476

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.42394    0.09588  -4.422 1.09e-05 ***
T            3.14972    0.03906  80.631  < 2e-16 ***
Z1          -0.29574    0.06667  -4.436 1.02e-05 ***
Z2          -1.14069    0.08060 -14.153  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.639 on 996 degrees of freedom
Multiple R-squared:  0.9343,    Adjusted R-squared:  0.9341
F-statistic:  4719 on 3 and 996 DF,  p-value: < 2.2e-16
```

Without including *eta*, the effect estimate is biased: OLS gives 3.150 against a true 2, an upward bias of 57%, and the standard error of 0.039 makes it look extremely precise. Let's do a dummy test.

```
lm2 <- lm(Y ~ T + Z1 + Z2 + I(T>0), data=data)
summary(lm2)
```

```
Call:
lm(formula = Y ~ T + Z1 + Z2 + I(T > 0), data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-8.0768 -0.8363  0.0406  0.9126  4.8213

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.68077    0.11961 -14.052   <2e-16 ***
T             3.06030    0.03570  85.728   <2e-16 ***
Z1           -0.52092    0.06189  -8.416   <2e-16 ***
Z2           -1.48038    0.07601 -19.476   <2e-16 ***
I(T > 0)TRUE  2.10995    0.13884  15.197   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.477 on 995 degrees of freedom
Multiple R-squared:  0.9466,    Adjusted R-squared:  0.9464
F-statistic:  4414 on 4 and 995 DF,  p-value: < 2.2e-16
```

It shows that the dummy variable is significant: the coefficient on $1(T > 0)$ is 2.110 with a standard error of 0.139, $t = 15.2$. It means that T is endogenous. The logic is the one set out above — at the

bunching point the treatment barely varies, so a jump of this size in the outcome has to come from something else moving discontinuously, and η is the candidate.

29.3.1. Correcting for endogeneity

The correction term is $E[T^* \mid T^* \leq 0, Z]$, on the scale of the *treatment* T^* — so we need to model T^* given Z, not Y given T and Z. Let's assume T^* is normally distributed given Z: $T^* \mid Z \sim N(Z\pi, \sigma^2)$. Since we observe $T = T^*$ whenever $T^* > 0$ and $T = 0$ otherwise, this is exactly a (left-)censored (Tobit) regression of T on Z, using **all** observations (not just the right tail): the censored observations still inform π and σ through the likelihood, they just don't reveal T^*'s value directly. Given $\hat{\pi}$ and $\hat{\sigma}$, the standard truncated-normal mean formula gives us $E[T^* \mid T^* \leq 0, Z]$ directly, with no separate mirroring step needed:

$$E[T^* \mid T^* \leq 0, Z] = Z\hat{\pi} - \hat{\sigma}\,\frac{\phi(-Z\hat{\pi}/\hat{\sigma})}{\Phi(-Z\hat{\pi}/\hat{\sigma})} \tag{29.5}$$

```r
library(AER)

tob <- tobit(T ~ Z1 + Z2, left = 0, data = data)
lin_pred  <- coef(tob)["(Intercept)"] + coef(tob)["Z1"] * data$Z1 + coef(tob)["Z2"] * data$Z2
sigma_hat <- tob$scale
alpha     <- -lin_pred / sigma_hat
E_Tstar_below0 <- lin_pred - sigma_hat * dnorm(alpha) / pnorm(alpha)

# here I create an extra term for the correction:
# when T > 0, it is T; when T = 0, it is the truncated-normal conditional
# mean of T* given T* <= 0 and Z, computed above.
data <- data %>%
  mutate(extra = if_else(T > 0, T, E_Tstar_below0))

lm4 <- lm(Y ~ T + Z1 + Z2 + extra, data=data)
summary(lm4)
```

```
Call:
lm(formula = Y ~ T + Z1 + Z2 + extra, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-5.8902 -0.7686  0.0480  0.7935  5.4430

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.03693    0.07979   0.463    0.644
T            1.98735    0.05917  33.585   <2e-16 ***
Z1          -0.82513    0.05838 -14.133   <2e-16 ***
```

```
Z2          -2.31062    0.08224 -28.097   <2e-16 ***
extra        1.36256    0.05873  23.200   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.321 on 995 degrees of freedom
Multiple R-squared:  0.9573,    Adjusted R-squared:  0.9572
F-statistic:  5582 on 4 and 995 DF,  p-value: < 2.2e-16
```

Under the normality assumption, this recovers β (the true coefficient on T is 2) very closely — 1.987 with a standard error of 0.059, against OLS's 3.150. The correction term itself enters with coefficient 1.363, which is the estimate of δ, the effect of η on Y; the DGP sets that path at $2/1.5 = 1.33$ once expressed per unit of T^*. Note the control coefficients also move, from -0.296 and -1.141 to -0.825 and -2.311. These are not estimates of the structural $\gamma = (0.5, 0.5)$. By Equation 29.4, the coefficient on Z is the composite $\gamma - \pi\delta$, which in this DGP is $(0.5 - 1 \times 4/3,\ 0.5 - 2 \times 4/3) = (-0.833, -2.167)$, and the estimates are close to it. To recover γ itself, enter the generalized residual, `extra` minus the linear prediction $Z\hat{\pi}$, in place of `extra`. The parametric correction does much better than a naive symmetric-mirroring heuristic that estimates the right-tail conditional mean by an unweighted linear regression of T on Z restricted to a somewhat arbitrary right-tail cutoff: that heuristic's accuracy turns out to be quite sensitive to the cutoff and sample used, because the true conditional mean of T^* given T^* beyond a cutoff is a nonlinear (inverse-Mills-ratio) function of Z, not linear — so a plain linear regression on the selected right-tail subsample is itself misspecified. The parametric (Tobit) route sidesteps that by modeling the untruncated latent distribution directly and using the exact conditional-mean formula.

29.4. Conclusion

We are making assumptions for the distribution of the continuous treatment variable. Symmetry is the minimum assumption. We can make other parametric assumptions on the distribution of T^*. This is a trade-off between making distribution assumptions and ignoring endogeneity. The good news is that we don't need an instrument. We are using the censoring part of the treatment variable to deal with endogeneity.

30. G-computation (g-formula) with time varying covariates

30.1. The problem

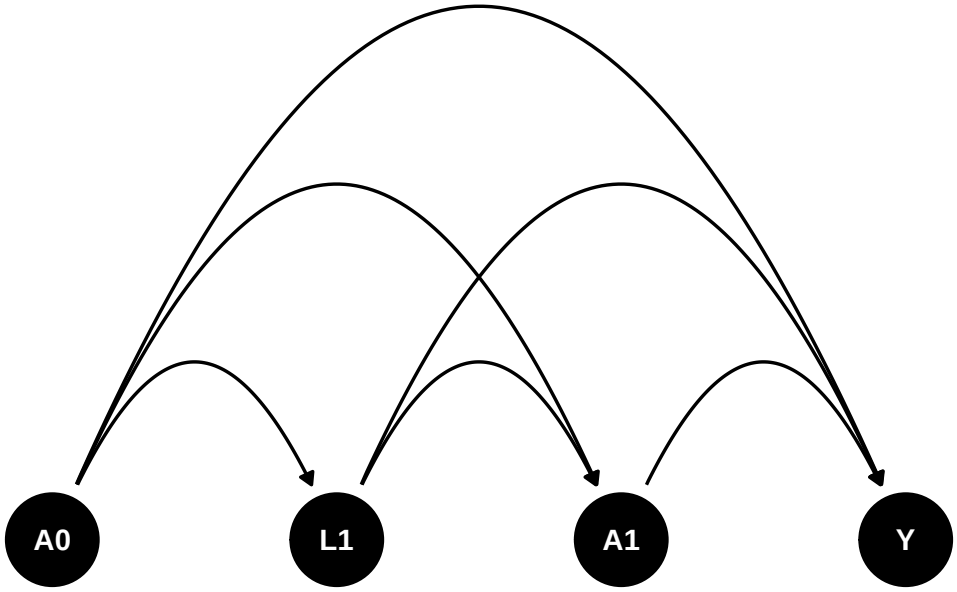
The g-formula (g-computation algorithm) is a method to estimate the causal effect of a time varying treatment or covariates, developed by Robins (1986). Note this is distinct from **g-estimation** of structural nested models (Robins and Rotnitzky 1992), a different g-methods estimator that uses instrumental-variable-like estimating equations and does not require modeling the joint distribution of all time-varying confounders. This chapter — and the `gfoRmula`/`gmethods` packages used below — implements the g-formula, not g-estimation.

We can imagine such a scenario. Suppose $A0$ and $A1$ are time 0 and time 1 treatment, can be binary or continuous. $L1$ is a time 1 covariate. It can be some intermediate outcome. Say HbA1c. HbA1c can be a result of treatment $A0$. Then it can be a cause for the next period treatment $A1$. Y is the outcome. U is the unmeasured confounder. The causal diagram is shown below. How do we estimate the effect of treatment?

```
library(dagitty)
library(ggdag)
library(ggplot2)
library(gfoRmula)
library(gmethods)
library(tidyverse)

dag <- dagify(
  Y ~ A1 + A0 + L1 ,
  A1 ~ A0 + L1,
  L1 ~ A0 ,
  coords = time_ordered_coords()
)

dag %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_dag_point() +
  geom_dag_edges_arc() +
  geom_dag_text() +
  theme_dag()
```



30.2. Assumptions

There are four potential outcomes, with A_0 and A_1 can be 0 or 1. The causal estimand is

$$\tau_{a_0,a_1,a_0',a_1'} = E[Y^{a_0,a_1} - Y^{a_0',a_1'}] \tag{30.1}$$

To identify the effects, we need assumptions.

1. Sequential ignorability

$$Y^{a_0,a_1} \perp A_0 \tag{30.2}$$

$$Y^{a_0,a_1} \perp A_1^{a_0} | A_0, L_1^{a_0} \tag{30.3}$$

In general, treatment at time t is randomized with probabilities depending on the observed past (covariates and intermediate outcomes): at any time t, for every regime $\bar{a}$,

$$Y^{\bar{a}} \perp A_t \mid H_t \tag{30.4}$$

where H_t is the **observed** history of treatment and covariates up to time t,

$$H_t = (A_0, A_1, ..., A_{t-1}; L_1, ..., L_t). \tag{30.5}$$

Read this in terms of the *observed* treatment A_t and the *observed* history H_t: conditional on what has actually happened up to time t, the next treatment is independent of the future potential outcome. The superscripted objects in the two-period statement above (e.g. $A_1^{a_0}$, $L_1^{a_0}$) denote the values that would arise under setting $A_0 = a_0$; by consistency these equal the observed A_1, L_1 on the subset of units with $A_0 = a_0$, which is what lets us write the conditioning event in terms of observed variables. The statement itself still involves the potential outcome $Y^{\bar{a}}$, which we do not observe for units that followed a different regime. Sequential ignorability is an identifying assumption. We use it to derive an observed-data functional, and observing that functional does not test it.

2. Positivity

$$0 < P(A_0 = a_0) < 1 \quad \text{for each } a_0 \in \{0, 1\} \tag{30.6}$$

$$0 < P(A_1 = a_1 \mid A_0 = a_0, L_1 = l) < 1 \quad \text{for each } (a_0, a_1) \text{ and each } l \text{ with } P(A_0 = a_0, L_1 = l) > 0 \tag{30.7}$$

In general, for a discrete treatment,

$$0 < P(A_t = a_t \mid H_t = h_t) < 1 \quad \text{for every } a_t \text{ and every history } h_t \text{ of positive density} \tag{30.8}$$

For a continuous treatment the point probabilities are zero. Let $f(a_t \mid h_t)$ be the conditional density of A_t given $H_t = h_t$. Positivity then requires $f(a_t \mid h_t) > 0$ for every dose a_t in the regime and every history h_t the data can produce.

Positivity has to hold at *every* treatment value and *every* history the data can produce, not only at the value and history that happen to appear in a worked example. A regime with no one following it at some history is a regime the data cannot speak to.

3. Consistency

$$\text{if } (A_0, A_1) = (a_0, a_1) \text{ then } Y = Y^{a_0, a_1} \tag{30.9}$$

This is an implication, not a conditional expectation. It says the potential outcome under the regime a unit actually followed *is* that unit's observed outcome. Written as a conditional equality it would look like a statement about a subpopulation's average, which is not what consistency asserts.

30.3. Identifiability

$$\begin{aligned} E[Y^{a_0,a_1}] &= E[Y^{a_0,a_1} | A_0 = a_0] \\ &= E[Y^{A_0,a_1} | A_0 = a_0] \\ &= \sum_{l=0,1} E[Y^{A_0,a_1} | A_0 = a_0, L = l] P(L = l | A_0 = a_0) \\ &= \sum_{l=0,1} E[Y^{A_0,a_1} | A_0 = a_0, L = l, A_1 = a_1] P(L = l | A_0 = a_0) \\ &= \sum_{l=0,1} E[Y^{A_0,A_1} | A_0 = a_0, L = l, A_1 = a_1] P(L = l | A_0 = a_0) \end{aligned} \tag{30.10}$$

Each equality earns its place from a named assumption. The first adds the conditioning event $A_0 = a_0$, which is sequential ignorability at $t = 0$, Equation 30.2. The second replaces a_0 by A_0 inside the potential outcome, legitimate because we are already on the event where the two agree. The third is the law of total expectation over L. The fourth adds $A_1 = a_1$ to the conditioning set, and that is sequential ignorability at $t = 1$, Equation 30.3, applied conditional on (A_0, L). The fifth replaces a_1 by A_1 on the same grounds as the second, after which consistency, Equation 30.9, turns the potential outcome into the observed Y. Positivity, Equation 30.8, is what keeps every conditional expectation in the chain well defined.

The general case is built the same way, one period at a time: condition on the history so far, apply ignorability at that period, average over the next covariate, and repeat.

30.4. Estimation

The parametric g-formula needs two kinds of component. The first is the outcome model; the second is a set of models for the time-varying covariates, or intermediate outcomes (confounders).

30.5. Parametric g-formula

In this simple case, if Y is continuous, we can use a linear model. If Y is binary, we can use a logistic model.

$$E[Y|A_0, A_1, L_1] = \beta_0 + \beta_1 A_0 + \beta_2 L_1 + \beta_3 A_1 \tag{30.11}$$

Suppose L_1 is binary,

$$P(L_1 = 1|A_0) = logit^{-1}(\gamma_0 + \gamma_1 A_0) \tag{30.12}$$

$$L_1|A_0 \sim Bernoulli(logit^{-1}(\gamma_0 + \gamma_1 A_0)) \tag{30.13}$$

We may need to simulate the distribution of the time-varying covariates for the expected value of the potential outcomes.

30.6. An example

The simulated two-period example below is adapted from Christopher Boyer's epidemiology lab materials (christopherbboyer.com). I keep his data-generating process and parameter values, and rewrite the estimation to compare the parametric g-formula against the naive regression.

First we set up a simulated data set.

```
# setting up the data -----------------------------------------------------

set.seed(1986)  # the noise added to Y below is random

# The code below creates the frequency table, and then turns it into long format
# data. Read through the code carefully and run it.

# re-create the frequency table for homework 3
hw3_freq <- tribble(
  ~A_0, ~L_1, ~A_1, ~N, ~Y_1,
  0, 0, 0, 6000, 60,
  0, 0, 1, 2000, 60,
```

```
  0, 1, 0, 2000, 210,
  0, 1, 1, 6000, 210,
  1, 0, 0, 3000, 240,
  1, 0, 1, 1000, 240,
  1, 1, 0, 3000, 120,
  1, 1, 1, 9000, 120
)

# expand the frequency table to 1 row per person
# (first just assign everyone the average Y value)
dat <- uncount(hw3_freq, N) %>%
  # give those rows an id number
  rowid_to_column(var = "id") %>%
  # for each of the variables except for id, split it into two
  # rows, one for each time point
  pivot_longer(-id,
               names_to = c(".value", "time"),
               names_sep = "_"
  ) %>%
  # make sure that time is read as a number
  mutate(time = parse_number(time),
         # add some random error to Y
         # (nrow(.) means a different value for each row of the dataset in use
         Y = Y + rnorm(nrow(.), 0, 10))


# create lagged variables
# group_by(id) makes this robust to row order -- lag() would otherwise
# silently lag across a different person's rows if dat were ever re-sorted
# (e.g. by time first). Currently harmless here since dat is in id-major
# order with exactly one row per id per time, but safer to be explicit.
t1 <- dat %>%
  group_by(id) %>%
  mutate(lag_A = lag(A)) %>%
  ungroup() %>%
  filter(time == 1)
```

Then calculate the g-formula by hand.

The recipe is the same in both cases: fit a model for the time-1 covariate L given past treatment, fit a model for Y given treatment, past treatment and L, then simulate 10,000 people forward with the treatments *fixed* at the regime's values, drawing L from its fitted conditional distribution rather than conditioning on the observed one.

First for $E[Y^{1,1}]$.

```
# fit models for covariate and outcome given past on t1
L_mod <- glm(L ~ lag_A, data = t1, family = binomial())
```

```r
Y_mod <- glm(Y ~ A * lag_A * L, data = t1)

# create new data set with space for 10,000 simulated entries
set.seed(1987)  # the simulated covariates below are random draws
nsim <- 10000
baseline_pop <- tibble(id = 1:nsim)

# fix intervention values in the new data set to be consistent with the regime
new_dat_11 <- baseline_pop %>%
  mutate(lag_A = 1, A = 1)

# predict covariate means at time 1
new_dat_11$pL <- predict(L_mod, newdata = new_dat_11, type = "response")

# simulate covariate values at time 1
new_dat_11$L <- rbinom(nsim, 1, new_dat_11$pL)

# predict outcome at time 1 based on simulated covariates and fixed treatments
new_dat_11$Ey <- predict(Y_mod, newdata = new_dat_11, type = "response")

# take mean across all simulations to get g-formula estimate!
mean(new_dat_11$Ey)
```

```
[1] 150.6583
```

That gives $E[Y^{1,1}] = 150.66$.

Then for $E[Y^{0,0}]$.

```r
# for 0, 0
new_dat_00 <- baseline_pop %>%
  mutate(lag_A = 0, A = 0)

# predict covariate means at time 1
new_dat_00$pL <- predict(L_mod, newdata = new_dat_00, type = "response")

# simulate covariate values at time 1
new_dat_00$L <- rbinom(nsim, 1, new_dat_00$pL)

# predict outcome at time 1 based on simulated covariates and fixed treatments
new_dat_00$Ey <- predict(Y_mod, newdata = new_dat_00, type = "response")

# take mean across all simulations to get g-formula estimate!
mean(new_dat_00$Ey)
```

```
[1] 134.1713
```

And $E[Y^{0,0}] = 134.17$, so always-treat versus never-treat is worth $150.66 - 134.17 = 16.49$ by hand.

Now with the gfoRmula package.

```
# running the g-formula ------------------------------------------------

# add your comments to the lines below
id <- "id"
time_name <- "time"
covnames <- c("L", "A")
outcome_name <- "Y"
#
outcome_type <- "continuous_eof"
#
histories <- c(lagged)
#
histvars <- list(c("A"))
#
covtypes <- c("binary", "binary")
covparams <- list(covmodels = c(
  L ~ lag1_A,
  A ~ lag1_A * L
))
ymodel <- Y ~ A * L * lag1_A
#
intvars <- list("A", "A", "A", "A")
#
interventions <- list(
  list(c(static, c(0, 0))),
  list(c(static, c(0, 1))),
  list(c(static, c(1, 0))),
  list(c(static, c(1, 1)))
)
int_descript <- c(
  "0, 0",
  "0, 1",
  "1, 0",
  "1, 1"
)

# put it all together
# (the warning "obs_data was coerced to a data table" is fine!)
gform_res <- gfoRmula::gformula(
  obs_data = dat,
  id = id,
  time_name = time_name,
  covnames = covnames,
```

```
  outcome_name = outcome_name,
  outcome_type = outcome_type,
  covtypes = covtypes,
  covparams = covparams,
  ymodel = ymodel,
  intvars = intvars,
  interventions = interventions,
  int_descript = int_descript,
  ref_int = 1,
  histories = histories,
  histvars = histvars,
  sim_data_b = TRUE,
  seed = 345636
)
# Use the following code to explore the results. (You can ignore Intervention 0,
# the natural course intervention, for now.) How many models were fit? How can
# you tell which of the data in the `sim_data` object was simulated or predicted
# from a model? Does these results match your answers to the earlier questions,
gform_res
```

```
PREDICTED RISK UNDER MULTIPLE INTERVENTIONS

Intervention     Description
0        Natural course
1        0, 0
2        0, 1
3        1, 0
4        1, 1

Sample size = 32000, Monte Carlo sample size = 32000
Number of bootstrap samples = 0
Reference intervention = 1



     k Interv.  NP mean g-form mean Mean ratio Mean difference % Intervened On
 <num>   <num>    <num>       <num>      <num>           <num>           <num>
     1       0 142.6015    141.7317  1.0512988      6.91588694         0.00000
     1       1       NA    134.8158  1.0000000      0.00000000        74.87812
     1       2       NA    134.7618  0.9995993     -0.05401877        75.12188
     1       3       NA    150.2145  1.1142202     15.39869407        80.85625
     1       4       NA    150.1904  1.1140412     15.37455992        69.14375
 Aver % Intervened On
                <num>
              0.00000
             49.74531
             50.25469
```

```
            55.99375
            44.00625
```

```
gform_res$coeffs
```

```
$L
 (Intercept)       lag1_A
8.985178e-14 1.098612e+00

$A
  (Intercept)        lag1_A             L      lag1_A:L
-1.098612e+00 -2.273084e-13  2.197225e+00  2.983558e-14

$Y
  (Intercept)             A             L        lag1_A           A:L
  60.12623682    0.08403462  150.10153695  180.02242928   -0.27744195
     A:lag1_A      L:lag1_A    A:L:lag1_A
  -0.17630509 -270.17383143    0.36841169
```

```
gform_res$sim_data
```

```
$`Natural course`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>      <num>  <num>    <num>
    1:     1     0    NA     0        TRUE          0      0       NA
    2:     1     1     1     1        TRUE          0      0 210.0344
    3:     2     0    NA     0        TRUE          0      0       NA
    4:     2     1     1     1        TRUE          0      0 210.0344
    5:     3     0    NA     0        TRUE          0      0       NA
   ---
63996: 31998     1     1     1        TRUE          0      1 120.0751
63997: 31999     0    NA     1        TRUE          0      0       NA
63998: 31999     1     1     1        TRUE          0      1 120.0751
63999: 32000     0    NA     1        TRUE          0      0       NA
64000: 32000     1     1     1        TRUE          0      1 120.0751

$`0, 0`
          id  time     L     A eligible_pt intervened lag1_A        Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>     <num>
    1:     1     0    NA     0        TRUE      FALSE      0        NA
    2:     1     1     1     0        TRUE       TRUE      0 210.22777
    3:     2     0    NA     0        TRUE      FALSE      0        NA
    4:     2     1     1     0        TRUE       TRUE      0 210.22777
    5:     3     0    NA     0        TRUE      FALSE      0        NA
   ---
63996: 31998     1     0     0        TRUE      FALSE      0  60.12624
```

```
63997: 31999     0    NA     0        TRUE       TRUE      0        NA
63998: 31999     1     0     0        TRUE      FALSE      0  60.12624
63999: 32000     0    NA     0        TRUE       TRUE      0        NA
64000: 32000     1     1     0        TRUE       TRUE      0 210.22777

$`0, 1`
          id  time     L     A eligible_pt intervened lag1_A        Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>     <num>
    1:     1     0    NA     0        TRUE      FALSE      0        NA
    2:     1     1     1     1        TRUE      FALSE      0 210.03437
    3:     2     0    NA     0        TRUE      FALSE      0        NA
    4:     2     1     1     1        TRUE      FALSE      0 210.03437
    5:     3     0    NA     0        TRUE      FALSE      0        NA
   ---
63996: 31998     1     0     1        TRUE       TRUE      0  60.21027
63997: 31999     0    NA     0        TRUE       TRUE      0        NA
63998: 31999     1     0     1        TRUE       TRUE      0  60.21027
63999: 32000     0    NA     0        TRUE       TRUE      0        NA
64000: 32000     1     1     1        TRUE      FALSE      0 210.03437

$`1, 0`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>    <num>
    1:     1     0    NA     1        TRUE       TRUE      0       NA
    2:     1     1     1     0        TRUE       TRUE      1 120.0764
    3:     2     0    NA     1        TRUE       TRUE      0       NA
    4:     2     1     1     0        TRUE       TRUE      1 120.0764
    5:     3     0    NA     1        TRUE       TRUE      0       NA
   ---
63996: 31998     1     1     0        TRUE       TRUE      1 120.0764
63997: 31999     0    NA     1        TRUE      FALSE      0       NA
63998: 31999     1     1     0        TRUE       TRUE      1 120.0764
63999: 32000     0    NA     1        TRUE      FALSE      0       NA
64000: 32000     1     1     0        TRUE       TRUE      1 120.0764

$`1, 1`
          id  time     L     A eligible_pt intervened lag1_A       Ey
       <int> <num> <num> <num>      <lgcl>     <lgcl>  <num>    <num>
    1:     1     0    NA     1        TRUE       TRUE      0       NA
    2:     1     1     1     1        TRUE      FALSE      1 120.0751
    3:     2     0    NA     1        TRUE       TRUE      0       NA
    4:     2     1     1     1        TRUE      FALSE      1 120.0751
    5:     3     0    NA     1        TRUE       TRUE      0       NA
   ---
63996: 31998     1     1     1        TRUE      FALSE      1 120.0751
63997: 31999     0    NA     1        TRUE      FALSE      0       NA
63998: 31999     1     1     1        TRUE      FALSE      1 120.0751
63999: 32000     0    NA     1        TRUE      FALSE      0       NA
```

```
64000: 32000     1     1     1        TRUE      FALSE      1 120.0751
```

The package agrees with the hand calculation. Intervention 4 ("1, 1", always treat) gives a g-formula mean of 150.19 against the 150.66 computed by hand, and intervention 1 ("0, 0", never treat) gives 134.82 against 134.17. The gaps are Monte Carlo error: both routes simulate the covariate, and the hand calculation uses only 10,000 draws, which puts a standard error of about 0.5 on the always-treat mean and 0.75 on the never-treat mean, or 0.9 on their difference. The table's own contrast of the two, 15.37, is 1.1 from the 16.49 of the hand calculation, about one such standard error.

The natural course is 141.73, sitting between the two regimes, and the non-parametric mean of the observed data is 142.60, which is the check that the model-based natural course is not badly misspecified.

The other two regimes are worth a glance. Treating only at time 0 gives 150.21 and treating only at time 1 gives 134.76 — so essentially all of the effect comes from the first period's treatment, and the second period adds nothing.

Reading `gform_res$coeffs` shows why this needs the g-formula rather than a regression. The L model has a coefficient of 1.0986 on lagged treatment, so past treatment moves the time-1 covariate; and the Y model has 150.10 on L and -270.17 on the $L \times \text{lag}_1 A$ interaction, so that covariate in turn moves the outcome. L is a time-varying confounder affected by past treatment, which is exactly the configuration where conditioning on it blocks part of the effect and omitting it leaves confounding.

30.7. A second example

Christopher Boyer has a package "gmethods" that can be used to estimate the g-formula. To me it's a bit more intuitive than the gfoRmula package. Here is an example in the "gmethods" package, with the same model in the "gfoRmula" package.

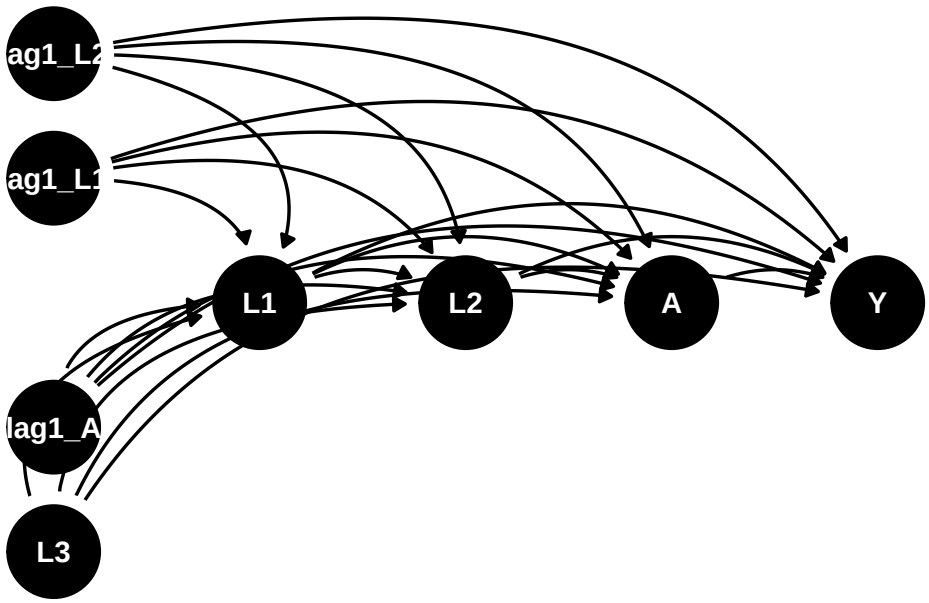
The example dataset continuous_eofdata again consists of 7,500 observations on 2,500 individuals with a maximum of 7 follow-up times, where the outcome corresponds to a characteristic only in the last interval (e.g., systolic blood pressure in interval 7). The variables in the dataset are: t0: The time index id: A unique identifier for each individual L1: A time-varying covariate; categorical L2: A time-varying covariate; continuous L3: A baseline covariate; continuous A: The treatment variable; binary Y: The outcome of interest; continuous

The goal is to estimate the mean outcome at time $K + 1 = 7$ under "never treat" versus "always treat."

The DAG would be something like:

```
dag <- dagify(
  Y ~ A + L1 + L2 + L3 + lag1_A + lag1_L1 + lag1_L2 ,
  L1 ~ lag1_A + lag1_L1 + lag1_L2 + L3 ,
  L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3,
  A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3,
  coords = time_ordered_coords()
)
```

```
dag %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_dag_point() +
  geom_dag_edges_arc() +
  geom_dag_text() +
  theme_dag()
```



30.7.1. Results from gfoRmula

First use the gFormula package.

```
id <- 'id'
time_name <- 't0'
covnames <- c('L1', 'L2', 'A')
outcome_name <- 'Y'
covtypes <- c('categorical', 'normal', 'binary')
histories <- c(lagged)
histvars <- list(c('A', 'L1', 'L2'))
covparams <- list(covmodels = c(L1 ~ lag1_A + lag1_L1 + L3 + t0 +
                                  lag1_L2,
                                L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3 + t0,
                                A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3 + t0))
ymodel <- Y ~ A + L1 + L2 + lag1_A + lag1_L1 + lag1_L2 + L3
intvars <- list('A', 'A')
interventions <- list(list(c(static, rep(0, 7))),
                      list(c(static, rep(1, 7))))
```

```
int_descript <- c('Never treat', 'Always treat')
nsimul <- 50000

gform_cont_eof <- gformula_continuous_eof(obs_data = continuous_eofdata,
                                          id = id,
                                          time_name = time_name,
                                          covnames = covnames,
                                          outcome_name = outcome_name,
                                          covtypes = covtypes,
                                          covparams = covparams, ymodel = ymodel,
                                          intvars = intvars,
                                          interventions = interventions,
                                          int_descript = int_descript,
                                          histories = histories,
                                          histvars = histvars,
                                          model_fits = TRUE,
                                          basecovs = c("L3"),
                                          nsimul = nsimul, seed = 1234)
gform_cont_eof
```

```
PREDICTED RISK UNDER MULTIPLE INTERVENTIONS

Intervention     Description
0        Natural course
1        Never treat
2        Always treat

Sample size = 2500, Monte Carlo sample size = 50000
Number of bootstrap samples = 0
Reference intervention = natural course (0)


     k Interv.   NP mean g-form mean Mean ratio Mean difference % Intervened On
 <num>   <num>     <num>       <num>      <num>           <num>           <num>
     6       0 -4.414543   -4.363027  1.0000000       0.0000000           0.000
     6       1        NA   -3.133081  0.7180981       1.2299454         100.000
     6       2        NA   -4.602952  1.0549904      -0.2399248          74.038
 Aver % Intervened On
                <num>
              0.00000
             82.35200
             17.49571
```

30.7.2. Results from gmethods

Now use the gmethods package.

```
gf <- gmethods::gformula(
  outcome_model = list(
    formula = Y ~ A + L1 + L2 + lag1_A + lag1_L1 + lag1_L2 + L3,
    link = "identity",
    family = "normal"
  ),
  covariate_model = list(
    "L1" = list(
      formula = L1 ~ lag1_A + lag1_L1 + L3 + t0 + lag1_L2,
      family = "categorical"
    ),
    "L2" = list(
      formula = L2 ~ lag1_A + L1 + lag1_L1 + lag1_L2 + L3 + t0,
      link = "identity",
      family = "normal"
    ),
    "A" = list(
      formula = A ~ lag1_A + L1 + L2 + lag1_L1 + lag1_L2 + L3 + t0,
      link = "logit",
      family = "binomial"
    )
  ),
  data = continuous_eofdata,
  survival = FALSE,
  id = 'id',
  time = 't0'
)

s <- simulate(
  gf,
  interventions = list(
    "Never treat" = function(data, time) set(data, j = "A", value = 0),
    "Always treat" = function(data, time) set(data, j = "A", value = 1)
  ),
  n_samples = 50000
)

s$results
```

```
$means
  intervention  estimate conf.low conf.high
1            0 -4.360358       NA        NA
2            1 -3.139260       NA        NA
3            2 -4.602994       NA        NA

$ratios
  intervention  estimate conf.low conf.high
```

```
1            0        NA       NA        NA
2            1 0.7199547       NA        NA
3            2 1.0556459       NA        NA

$diffs
  intervention  estimate conf.low conf.high
1            0        NA       NA        NA
2            1  1.221098       NA        NA
3            2 -0.242636       NA        NA
```

Both packages agree closely. `gfoRmula` gives a natural course of −4.363, never-treat −3.133 and a mean difference of 1.230; `gmethods` gives −4.360, −3.139 and 1.221, with always-treat at −4.603 and a difference of −0.243. The two implementations differ in the third decimal, which is Monte Carlo noise from the 50,000-draw simulation, not a disagreement about the estimand.

Note the results are very close. Also note that in the covariates and treatment models, there is a time variable t0. This is because these models are pooled regression with all time points. Therefore a time fixed effect is included. In the outcome model, there is only the last period data, so the time variable is not included.

Standard errors can be calculated with bootstrapping.

30.8. Summary

With treatment over time, a covariate can be both a confounder of later treatment and a result of earlier treatment. Conditioning on it blocks part of the effect, and leaving it out leaves confounding. The g-formula handles this under sequential ignorability, positivity and consistency. It models the time-varying covariate given past treatment and the outcome given the full history. Then it simulates the population forward with the treatments fixed at the regime of interest.

In the two-period example the hand calculation gives 150.66 under always-treat and 134.17 under never-treat, a difference of 16.49. `gfoRmula` gives 15.37, with the gap due to Monte Carlo error. Almost all of the effect comes from the first period's treatment. In the seven-period example, `gfoRmula` and `gmethods` agree to the third decimal. Standard errors come from the bootstrap.

Systematic treatment: R · Julia.

31. Policy learning by policytree

31.1. Optimal policy

I have a few posts about estimating ATE or CATE. The target has been getting the treatment effect or conditional treatment effect accurately. However, sometimes after estimating the treatment effects, we may be interested in the optimal treatment assignment. For example, a doctor may be interested in how to assign treatment based on patient characteristics. OSHA might be interested in which firms to send inspector to, based on firm characteristics, or past history. We may think we can just assign the treatment to those who benefited most (largest CATE), or any patient who has a positive CATE. However, the objective function for policy is different from the CATE target.

Suppose we have {Y(0), Y(1), X} and we have estimated the CATE as $\tau(x) = E[Y(1)-Y(0)|X = x]$. The policy learning problem is to find a policy function that assigns treatment to maximize some utility function.

Suppose that policy is

$$\pi \ : \ X \to \{0, 1\} \tag{31.1}$$

This policy is to assign treatment or control based on X.

The natural objective function is a value function $V(\pi) = E[Y_i(\pi(X_i))]$, which is the expected value of outcome with this policy. We'd like to maximize this function.

$$V(\pi) = E[Y_i(\pi(X_i))] = E[Y_i(0)] + E[\tau(X)\pi(X)] \tag{31.2}$$

Since π only enters through $E[\tau(X)\pi(X)]$, the maximizer is exactly $\pi(X) = 1\{\tau(X) > 0\}$ — treat everyone with a positive effect. A threshold $c \neq 0$ is not a free choice here: it appears once treatment is costly, where maximizing $E[(\tau(X) - c)\pi(X)]$ gives the rule $\tau(X) > c$ with c the per-unit cost. However, estimating $\tau(X)$ is different from maximizing V.

First, we don't have the true τ. The loss function in estimating $\tau(X)$ is different from the loss function in maximizing V. The loss function in estimating $\tau(X)$ is the squared error loss. We'll see maximizing V is a different problem.

Second, the X we use to estimate τ is different from the ones we can use for policy. For example, we have some variables used in CATE estimation that we cannot use for policy, like gender, or age, for discrimination reasons. Or there are variables we have after the experiment that we cannot use for CATE estimation, but we'd like to use for policy learning.

31.2. IPW loss

Kitagawa and Tetenov (2018) proposed a loss function for policy learning. The loss function is based on the inverse propensity weighting.

$$\hat{\pi} = argmax\{\hat{V}(\pi) : \pi \in \Pi\} \tag{31.3}$$

$$\hat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n} \frac{1(W_i = \pi(X_i))}{P[W_i = \pi(X_i)|X_i]} Y_i \tag{31.4}$$

This is to say when we have $W_i = \pi(X_i)$, we can use the outcome Y_i to estimate the value function. That is, when we have a policy, we then select the observations that matches that policy then weight the outcome by the inverse propensity score. This is to adjust for the fact that there is selection bias for that policy. Therefore we re-weight it by IPW.

Obviously we don't have the true propensity score of $W_i = \pi(X_i)$, so we need to estimate it. We can estimate it by machine learning.

31.3. AIPW loss

The IPW loss can be not robust to model misspecification. Athey and Wager (2021) proposed a loss function that is robust to model misspecification. The loss function is based on the doubly robust estimator, AIPW.

$$\hat{\pi} = argmax\{\hat{V}(\pi) : \pi \in \Pi\} \tag{31.5}$$

$$\hat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n} (2\pi(X_i) - 1)\hat{\Gamma}_i \tag{31.6}$$

$$\Gamma_i = \hat{\mu}(1)(X_i) - \hat{\mu}(0)(X_i) + \frac{W_i}{\hat{e}(X_i)}(Y_i - \hat{\mu}(1)(X_i)) - \frac{1 - W_i}{1 - \hat{e}(X_i)}(Y_i - \hat{\mu}(0)(X_i)) \tag{31.7}$$

Note the Γ_i is the score from the AIPW estimator. It's basically the efficient influence function we learned from the nonparametric estimation of ATE.

One clarification about Equation 31.6, because the name is easy to over-read. With scalar doubly robust effect scores $\hat{\Gamma}_i$, that quantity is a *centered empirical welfare objective*, not an estimate of the outcome-level policy value $E[Y(\pi(X))]$. It has the same maximizer, up to an additive constant and a factor of two, which is all the optimization needs. An estimator of the value itself uses action-specific scores, or adds a control-outcome score to $\pi(X_i)\hat{\Gamma}_i$. So maximize Equation 31.6, but do not report its value as the welfare of the learned rule.

We'd like to maximize this objective.

The basic idea is for each policy, calculate the score from AIPW estimator. Then calculate the value function. Then search over the policy space for the policy that maximizes the value function.

This is very computationally intensive task. Right now seems it would work for "shallow" trees. That is, we allow 2 or 3 levels of trees. Otherwise, the search space could be too large to search.

31.4. Policytree

The simulation is built so that the optimal policy is known. We draw $n = 1000$ observations with ten independent standard normal covariates. The propensity is $e(X) = 1/(1 + e^{X_3})$, so treatment depends on X_3 alone. The true CATE is

$$\tau(X) = \frac{1}{1 + e^{(X_1 + X_2)/2}} - \tfrac{1}{2}, \tag{31.8}$$

and the outcome is $Y = X_3 + W\tau(X) + N(0, 1)$.

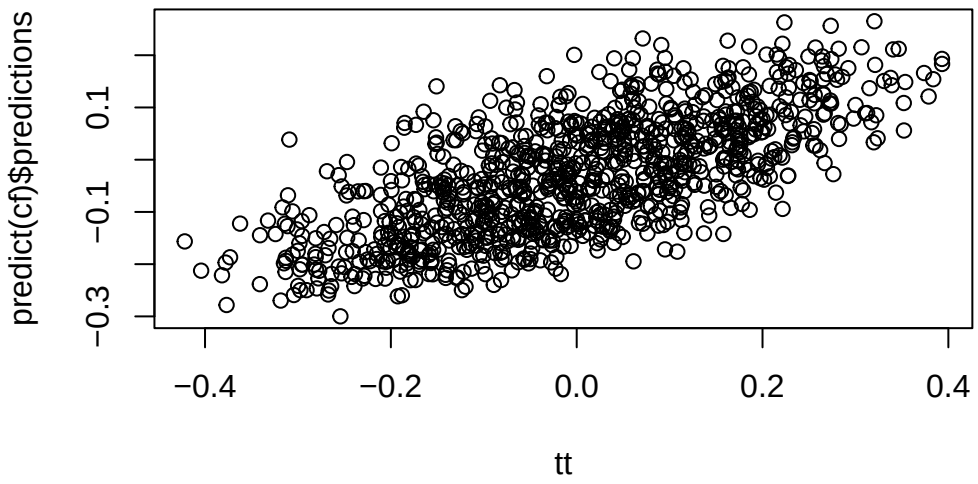
Because $\tau(X) > 0$ exactly when $X_1 + X_2 < 0$, the optimal policy is the half-plane below the line $X_1 + X_2 = 0$ — which is the diagonal drawn on the last plot. Note that only X_1 and X_2 matter for the policy, X_3 matters only for selection, and X_4 through X_{10} are pure noise.

```
library(grf)
library(policytree)
library(DiagrammeR)
set.seed(31)
n <- 1000
p <- 10

X <- matrix(rnorm(n * p), n, p)
ee <- 1 / (1 + exp(X[, 3]))
tt <- 1 / (1 + exp((X[, 1] + X[, 2]) / 2)) - 0.5
W <- rbinom(n, 1, ee)
Y <- X[, 3] + W * tt + rnorm(n)

cf <- causal_forest(X, Y, W)

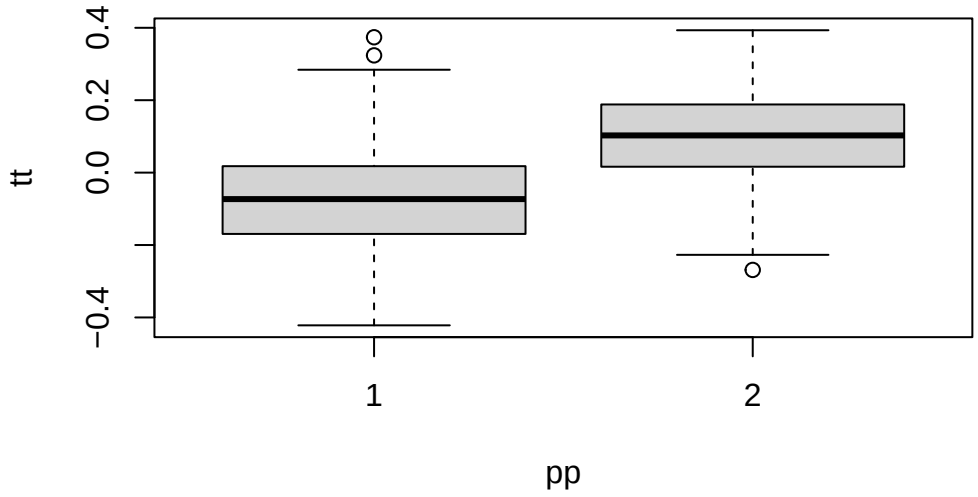
plot(tt, predict(cf)$predictions)
```



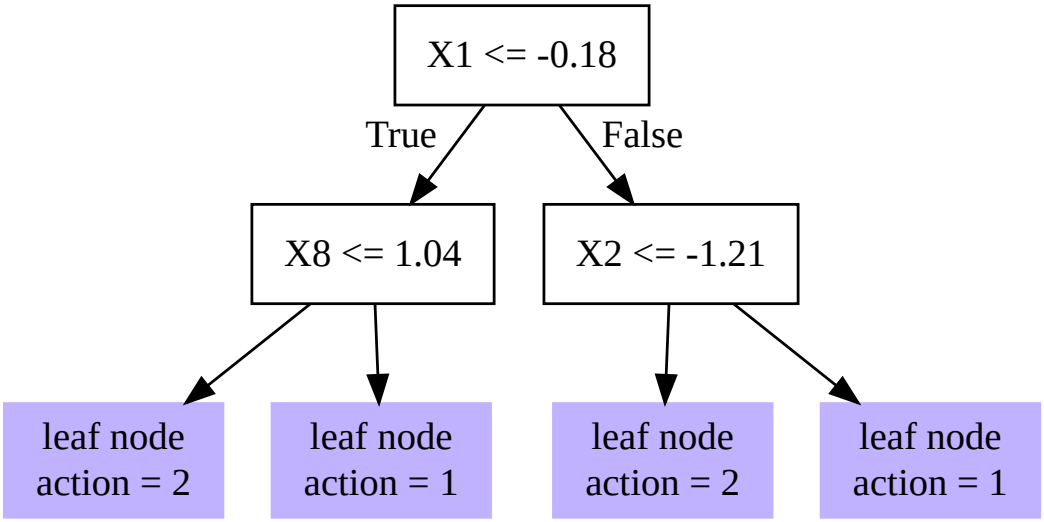

```
dr <- double_robust_scores(cf)
tree <- policy_tree(X, dr, 2)
tree
```

```
policy_tree object
Tree depth:  2
Actions:  1: control 2: treated
Variable splits:
(1) split_variable: X1  split_value: -0.179677
  (2) split_variable: X8  split_value: 1.0422
    (4) * action: 2
    (5) * action: 1
  (3) split_variable: X2  split_value: -1.20838
    (6) * action: 2
    (7) * action: 1
```

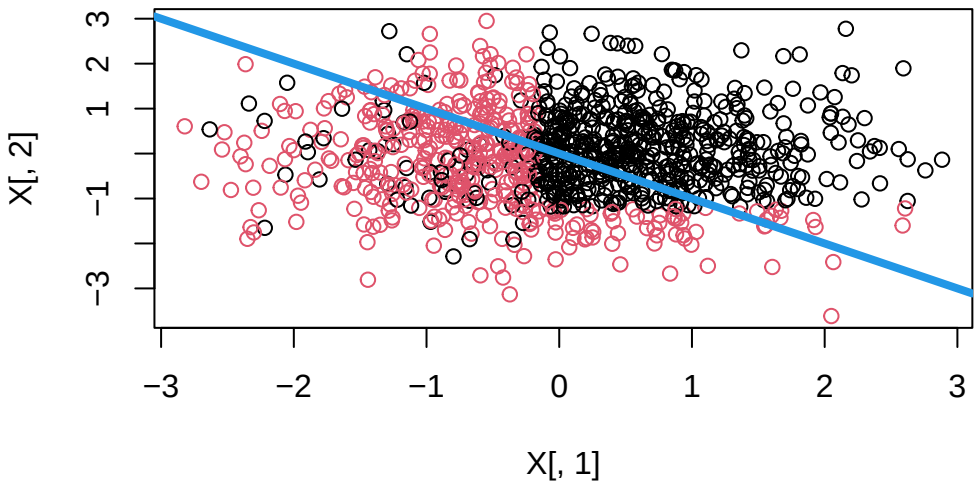
```
pp <- predict(tree, X)
boxplot(tt ~ pp)
```




```
plot(tree)
```


X1 <= -0.18
True
False
X8 <= 1.04
X2 <= -1.21
leaf node
action = 2
leaf node
action = 1
leaf node
action = 2
leaf node
action = 1

```
plot(X[, 1], X[, 2], col = pp)
abline(0, -1, lwd = 4, col = 4)
```



In the example, first "causal_forest" is used to estimate the CATE. "double_robust_scores" is used to calculate the AIPW score. Then "policy_tree" is used to estimate the policy tree. The level of tree is set to 2.

Read the fitted tree against the truth rather than on its own terms. The optimal rule is a *diagonal* boundary, $X_1 + X_2 < 0$, and a depth-2 axis-aligned tree cannot represent it exactly. That is approximation error. It need not be large: a depth-2 tree that splits on X_1 and X_2 can follow the diagonal with a coarse staircase. Runs of this chunk often split instead on noise covariates such as X_8 and X_{10}. That is a different error, estimation error: the doubly robust scores are noisy, and searching over every covariate and threshold in one sample can pick splits that fit the noise. The last plot shows both: the coloured regions are rectangles, the true boundary is the drawn line, and the gap between them mixes the class's approximation error with the estimated rule's mistakes. Comparing the estimated rule with the best depth-2 tree in population would separate the two.

The `boxplot(tt ~ pp)` is the honest summary: it shows the distribution of the *true* effect within each recommended action. A good rule puts most of the positive-τ mass in the "treat" box.

In this draw (seed 31) the tree splits first on X_1 at -0.18. Below it, it treats unless the noise covariate X_8 exceeds 1.04; above it, it treats only when $X_2 < -1.21$. That is a coarse staircase along the true boundary $X_1 + X_2 < 0$, plus one split on noise. The boxplot passes the check: the median true effect is about 0.10 among units the rule treats and about -0.07 among those it does not, though both boxes contain units on the wrong side of zero. With another seed the splits, and how well the rule does, can change, because the signal is weak: the true effect has a standard deviation of about 0.16, against X_3 plus noise with a standard deviation of about 1.4.

31.5. Summary

Policy learning chooses whom to treat so as to maximize the expected outcome. With no cost the best rule treats everyone with a positive conditional effect, and with a cost c everyone whose effect exceeds c. Estimating the effects well is a different problem from choosing the rule, and the variables a policy may use can differ from those used to estimate effects. Kitagawa and Tetenov maximize an inverse-propensity-weighted estimate of the policy's value. Athey and Wager use doubly robust AIPW scores, and `policytree` searches over shallow trees with those scores.

In the simulation the optimal rule is a diagonal boundary. A depth-2 tree can only make perpendicular cuts, so its splits often land on noise variables and say little about which variables matter. The useful check is the distribution of the true effect within each recommended action. The simulation sets no seed, so the particular splits change from run to run.

Systematic treatment: R · Julia.

32. Recent causal inference tools

In this post, I'll use simulated data to see how a few recently developed causal inference packages work.

In other posts, I have discussed some methods involving machine learning, such as TMLE, or causal forest. This time we'll talk about nonparametric methods and double machine learning.

32.1. Identification

First suppose we have a "target" parameter in mind, which is some function of the some unknown distribution P^*, called a functional, say $\psi^*(P^*)$. For example ATE, $E(Y^1 - Y^0)$.

The goal is to have $\psi^*(P^*)$ estimated by $\psi(\hat{P})$, where P is some observational distribution, and ψ is some known function. This is called identification.

For example, suppose we are interested in the ATE, $E[Y^1 - Y^0]$. The identification is done by deriving from $E(Y^a|X) = E(Y|X, A = a)$; basically from the target parameter, which is counterfactual, to something estimable from the observed variables in the sample. This identification can only be done with assumptions such as ignorability, positivity and consistency.

In this case, $\psi^* = E(Y^a)$, and $\psi = E\{E(Y|X)\}$.

32.2. Nonparametric estimation in a nutshell

After identification, we then need to find a way to construct a good estimator $\hat{\psi}$ for $\psi(P)$.

Suppose we have a sample Z, which can be A, W, Y (treatment, covariates, and outcome) in causal inference situation.

We can use a parametric model, assuming we know what kind of distribution P is. We assume $P = P_\theta$. Then we can use MLE to get $\psi(P_\theta)$. This is what we do in parametric modeling. But this can be too restrictive, we usually have no way of knowing how good P_θ is; in other words, the model can be misspecified thus the target parameter is estimated with bias.

If we don't make any assumptions about the distribution, then we are in nonparametric world. The natural way to approximate P is to use the empirical distribution $\hat{P}$. Then $\psi^*(\hat{P})$ is the "plug-in" estimator. For example we can use sample mean to estimate the population mean, etc. The plug-in is often not the best choice *when the nuisance functions are estimated flexibly* (e.g. with machine learning): the plug-in then inherits the slow convergence rate of those nuisance estimates and carries a first-order regularization ("plug-in") bias, so it is generally not $\sqrt{n}$ consistent in that regime. (This is not a universal indictment of plug-in estimators: with a correctly specified, smooth parametric

model a plug-in / g-computation estimator can itself be $\sqrt{n}$ efficient. The problem is specifically the flexible / high-dimensional nuisance case.)

Another choice is nonparametric estimator based on influence function. The nice thing about this type of estimators based on efficient influence function is that they are often $\sqrt{n}$ consistent, meaning they converge to the truth as fast as $\sqrt{n}$ rate. This means $\hat{\psi} - \psi$ not only goes to 0 with n goes to infinity, but also as fast as $1/\sqrt{n}$ goes to 0. Why does this matter? Because we have limited sample size, when we are based on asymptotics, the faster the bias goes to 0 the smaller sample size we need.

If we have an estimator such that

$$\sqrt{n}(\hat{\psi} - \psi) = \sqrt{n}P_n(\phi) + o_P(1) \to N(0, var(\phi)) \tag{32.1}$$

Then this estimator is $\sqrt{n}$ consistent and asymptotically normal; and ϕ is the influence function. The efficient influence function is the one with variance at the lower bound (similar to parametric case for Cramer-Rao lower bound).

Consider the mean of the treated potential outcome, $\psi = E\{E(Y|X, A = 1)\} = E[Y^1]$. Its efficient influence function is

$$\phi_1(Z; P) = \frac{A}{\pi(X)}\{Y - \mu_1(X)\} + \mu_1(X) - \psi \tag{32.2}$$

where $\pi(X) = P(A = 1|X)$ the propensity score model, $\mu_1(X) = E(Y|X, A = 1)$, the outcome model. The ATE $E[Y^1 - Y^0]$ has efficient influence function $\phi_1 - \phi_0$, where ϕ_0 is the analogous control-arm term $\frac{1-A}{1-\pi(X)}\{Y - \mu_0(X)\} + \mu_0(X) - E[Y^0]$ with $\mu_0(X) = E(Y|X, A = 0)$.

The idea of IF-based estimator is to use so called "one-step correction" to estimate the "plug-in" bias. The bias is approximated by first deriving the influence function; then based on the influence function, the bias can be estimated by using the influence function as a slope to the path from "plug-in" estimate to true parameter. This path can be non-linear; therefore this one-step correction may not be perfect, but it should be closer to the truth than the original estimate.

For example, an IF-based bias-corrected estimator

$$\hat{\psi} = P_n[\frac{A}{\hat{\pi}(X)}\{Y - \hat{\mu}(X)\} + \hat{\mu}(X)] \tag{32.3}$$

where P_n means sample mean. This is the familiar AIPW estimator. It is known to be doubly robust. Since we know the influence function, we can calculate its variance. Under asymptotic linearity, $\hat{\psi} - \psi \approx \frac{1}{n}\sum_i \phi_1(Z_i)$, so the variance of the estimator is approximately $\text{Var}(\phi_1)/n$; the influence function and the estimator do not have the same variance. We estimate the nuisance parameters, form $\hat{\psi}$, and divide the sample variance of the estimated influence values by n. This estimator is the treated-arm mean $E[Y^1]$; the ATE subtracts the analogous control-arm estimator, and its influence function is $\phi_1 - \phi_0$.

In this estimator, both $\hat{\mu}$ and $\hat{\pi}$ can be done using any estimator, such as machine learning.

The IF based estimator only works if either $\phi(P)$ is not too complex (empirical processes theory involved, a lot of difficult situations won't apply), or we separate $\hat{P}$ and P_n to prevent overfitting. The latter is by sample splitting. Sample splitting is usually preferred because it needs less assumption than empirical processes. The idea is to split the sample, do estimation of nuisance parameters (such as μ and π) in one sample, and construct estimator in the other. Cross-fitting controls the

empirical-process term. It does not remove the other requirements: the nuisance errors must shrink fast enough that their product is $o_p(n^{-1/2})$, and overlap must hold. Under those conditions the EIF-based estimator is $\sqrt{n}$ consistent and asymptotically normal, with variance $\mathrm{Var}(\phi)/n$.

Unfortunately influence function is hard to derive for different situations. Ed Kennedy has done a lot of work on using IF on causal inference http://www.ehkennedy.com/. Here we use his "npcausal" package.

32.3. Simulation with a known treatment effect

The simulated data is from https://migariane.github.io/TMLE.nb.html

We draw $n = 1000$ observations with four covariates: $w_1 \sim \text{Bernoulli}(0.5)$, $w_2 \sim \text{Bernoulli}(0.65)$, $w_3 \sim U(0,4)$ and $w_4 \sim U(0,5)$. Treatment is $A \sim \text{Bernoulli}(\Lambda(-0.4 + 0.2w_2 + 0.15w_3 + 0.2w_4 + 0.15w_2w_4))$, and both potential outcomes are Bernoulli with the same logit index shifted by the treatment, $Y(d) \sim \text{Bernoulli}(\Lambda(-1+d-0.1w_1+0.3w_2+0.25w_3+0.2w_4+0.15w_2w_4))$. Both nuisance functions are nonlinear in the covariates, which is what makes flexible estimation worthwhile. Since both potential outcomes are simulated, the realized-sample effect is known: 0.203.

```
library(npcausal)
library(boot)
library(MASS)
library(SuperLearner)
library(tmle)
library(tidyverse)
library(ranger)
```

```
set.seed(123)
n=1000
w1 <- rbinom(n, size=1, prob=0.5)
w2 <- rbinom(n, size=1, prob=0.65)
w3 <- round(runif(n, min=0, max=4), digits=3)
w4 <- round(runif(n, min=0, max=5), digits=3)
A  <- rbinom(n, size=1, prob= plogis(-0.4 + 0.2*w2 + 0.15*w3 + 0.2*w4 + 0.15*w2*w4))
Y.1 <- rbinom(n, size=1, prob= plogis(-1 + 1 -0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4))
Y.0 <- rbinom(n, size=1, prob= plogis(-1 + 0 -0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4))
Y <- Y.1*A + Y.0*(1 - A)
W<-data.frame(cbind(w1,w2,w3,w4))
data <-  data.frame(w1, w2, w3, w4, A, Y, Y.1, Y.0)
True_Psi <- mean(data$Y.1-data$Y.0);
cat(" True_Psi:", True_Psi)
```

```
 True_Psi: 0.203
```

We first include a few learners. "npcausal" has a set of default learners.

```
SL.library <- c("SL.earth","SL.gam","SL.glm","SL.glmnet","SL.glm.interaction", "SL.mean","SL.ranger", "SL
```

32.3.1. Plug-in estimator from outcome regression

```
SL.outcome<- SuperLearner(Y=data$Y, X=data %>% select(w1,w2,w3,w4,A),
                                    SL.library=SL.library, family="binomial")
SL.outcome.obs<- predict(SL.outcome, X=data %>% select(w1,w2,w3,w4,A))$pred
# predict the PO Y^1
SL.outcome.1<- predict(SL.outcome, newdata=data %>% select(w1,w2,w3,w4,A) %>% mutate(A=1))$pred
# predict the PO Y^0
SL.outcome.0<- predict(SL.outcome, newdata=data %>% select(w1,w2,w3,w4,A) %>% mutate(A=0))$pred
```

This is to do machine learning on the outcome equation; then get prediction on treated vs control, to get $\hat{Y}^1$ and $\hat{Y}^0$. The sample average of the difference is our plug-in estimator.

```
SL.plugin<-mean(SL.outcome.1-SL.outcome.0)
SL.plugin
```

```
[1] 0.2050486
```

The plug-in estimate is 0.2050 against the realized 0.203. It's not too far from the truth in this case — but it comes with no standard error, and the regularisation bias of the SuperLearner passes straight into the average with nothing to correct it.

```
Q=data.frame(QAW=SL.outcome.obs, Q0W=SL.outcome.0,Q1W=SL.outcome.1)
```

32.3.2. AIPW with machine learning

We can use "npcausal" to do AIPW, or we can manually calculate it.

First we get propensity score. Then, based on both $\hat{\mu}$ and $\hat{\pi}$, we can get the AIPW estimator.

```
set.seed(123)
SL.g<- SuperLearner(Y=data$A, X=data %>% select(w1,w2,w3,w4),
                    SL.library=SL.library, family="binomial")
g1W <- SL.g$SL.predict
g0W<- 1- g1W
IF.1<-((data$A/g1W)*(data$Y-Q$Q1W)+Q$Q1W)
IF.0<-(((1-data$A)/g0W)*(data$Y-Q$Q0W)+Q$Q0W)
IF<-IF.1-IF.0
aipw.1<-mean(IF.1);aipw.0<-mean(IF.0)
aipw.manual<-aipw.1-aipw.0
ci.lb<-mean(IF)-qnorm(.975)*sd(IF)/sqrt(length(IF))
ci.ub<-mean(IF)+qnorm(.975)*sd(IF)/sqrt(length(IF))
```

```
res.manual.aipw<-c(aipw.manual,ci.lb, ci.ub)
res.manual.aipw
```

```
[1] 0.2123436 0.1511466 0.2735405
```

Now we use "npcausal".

```
library(npcausal)
aipw<- ate(y=Y, a=A, x=W, nsplits=1, sl.lib=SL.library)
```

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     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5719531 0.02594164 0.5211075 0.6227987    0
2      E{Y(1)} 0.7860664 0.01580281 0.7550929 0.8170399    0
3 E{Y(1)-Y(0)} 0.2141133 0.03004191 0.1552311 0.2729955    0
```

```
aipw$res
```

```
     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5719531 0.02594164 0.5211075 0.6227987    0
2      E{Y(1)} 0.7860664 0.01580281 0.7550929 0.8170399    0
3 E{Y(1)-Y(0)} 0.2141133 0.03004191 0.1552311 0.2729955    0
```

The manual AIPW gives 0.2123 with a 95% interval of $[0.1511, 0.2735]$, and `npcausal` with `nsplits=1` gives 0.2141 with a standard error of 0.0300 and an interval of $[0.1552, 0.2730]$. It also reports the two arm means separately, $E[Y(0)] = 0.5720$ and $E[Y(1)] = 0.7861$.

The results from "npcausal" are almost the same as the manual results. Two clarifications. First, the manual estimator above *is* bias-corrected in the relevant sense: AIPW is exactly the one-step correction of the plug-in estimator using the efficient influence function. What we did *not* do is **cross-fitting** (sample splitting), which is a separate device. Second, that matters for inference: because the manual example fits the SuperLearner nuisances on the *full* sample and then uses `sd(IF)/sqrt(n)`, the resulting confidence interval is only valid under empirical-process (Donsker) conditions that flexible ML learners typically violate. With cross-fitting, the influence-function variance is valid without those conditions. "npcausal" does the cross-fitting for you when you set `nsplits > 1` (here `nsplits = 1` turns it off, for comparison with the manual full-sample version).

```
aipw2<- ate(y=Y, a=A, x=W, nsplits=10, sl.lib=SL.library)
```

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     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5725400 0.02933282 0.5150477 0.6300324    0
2      E{Y(1)} 0.7871840 0.01627910 0.7552770 0.8190910    0
3 E{Y(1)-Y(0)} 0.2146439 0.03325488 0.1494644 0.2798235    0
```

With `nsplits=10` the cross-fitted estimate is essentially unchanged, 0.2146 against 0.2141. That is

the reassuring case: cross-fitting is insurance against overfitting bias, and when the nuisance models are not badly overfitted it costs nothing.

32.3.3. TMLE

We can use TMLE for ATE.

```
library(tmle)
TMLE<- tmle(Y=data$Y,A=data$A,W=W, family="binomial", Q.SL.library=SL.library, g.SL.library=SL.library)

TMLE$estimates$ATE
```

```
$psi
[1] 0.2117905

$var.psi
[1] 0.001019485

$CI
[1] 0.1492101 0.2743709

$pvalue
[1] 3.287231e-11

$bs.var
[1] NA

$bs.CI.twosided
[1] NA NA

$bs.CI.onesided.lower
[1] -Inf   NA

$bs.CI.onesided.upper
[1]  NA Inf
```

TMLE gives 0.2118 with a 95% interval of $[0.1492, 0.2744]$ — within 0.003 of both AIPW versions, as it should be, since all three solve the same efficient influence-function equation and differ only in how.

32.3.4. DoubleML

The R package DoubleML implements the double/debiased machine learning framework of Chernozhukov et al. (2018). It provides functionalities to estimate parameters in causal models based on machine learning methods. The double machine learning framework consists of three key ingredients: Neyman orthogonality, High-quality machine learning estimation and Sample splitting.

They consider a partially linear model:

$$y_i = \theta d_i + g_0(x_i) + \eta_i \tag{32.4}$$

$$d_i = m_0(x_i) + v_i \tag{32.5}$$

This model is quite general, except it does not allow interaction of d and x; therefore no heterogeneous treatment effect across x. But "DoubleML" implements more than partially linear model, it actually allows for heterogeneous treatment effects, in models such as interactive regression model.

The basic idea of doubleML is to use any machine learning algorithm to estimate outcome equation ($l_0(X) = E(Y|X)$) and treatment equation ($m_0(X) = E(D|X)$). Then get the residuals, namely $\hat{W} = Y - \hat{l}_0(X)$ and $\hat{V} = D - \hat{m}_0(X)$.

Then regress $\hat{W}$ on $\hat{V}$. Based on FWL theorem, you get $\hat{\theta}$.

An important component here is to specify a Neyman-orthogonal score function. For the regression of $\hat{W}$ on $\hat{V}$ just described, it is the partialling-out score of Robinson (1988), the product of the outcome residual net of θ times the treatment residual and the treatment residual:

$$\psi(W; \theta, l, m) = ((Y - l(X)) - \theta(D - m(X)))(D - m(X)) \tag{32.6}$$

This is `DoubleMLPLR`'s default, `score = "partialling out"`. The alternative `"IV-type"` score, $(Y - D\theta - g(X))(D - m(X))$, uses the nuisance $g_0(X) = E[Y - D\theta_0 \mid X]$ in place of $l_0(X)$.

The estimator $\hat{\theta}$ is obtained by solving the equation that sets the sample mean of this score function to 0.

And the variance of this score function is used as the variance of the doubleML estimator's variance.

```
library(DoubleML)
dml_data = DoubleMLData$new(data,
                            y_col = "Y",
                            d_cols = "A",
                            x_cols = c("w1","w2","w3","w4"))
print(dml_data)
```

```
================= DoubleMLData Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: w1, w2, w3, w4
Instrument(s):
Selection variable:
No. Observations: 1000
```

```
library(mlr3)
library(mlr3learners)
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")

learner = lrn("regr.ranger", num.trees=500,  max.depth=5, min.node.size=2)
ml_g = learner$clone()
ml_m = learner$clone()

learner = lrn("regr.glmnet")
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()

set.seed(3141)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_g=ml_g, ml_m=ml_m)
obj_dml_plr$fit()
print(obj_dml_plr)
```

```
================== DoubleMLPLR Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: w1, w2, w3, w4
Instrument(s):
Selection variable:
No. Observations: 1000

------------------ Score & algorithm ------------------
Score function: partialling out
DML algorithm: dml2

------------------ Machine learner   ------------------
ml_l: regr.ranger
ml_m: regr.ranger

------------------ Resampling        ------------------
No. folds: 5
No. repeated sample splits: 1
Apply cross-fitting: TRUE

------------------ Fit summary       ------------------
 Estimates and significance testing of the effect of target variables
  Estimate. Std. Error t value Pr(>|t|)
A   0.23271    0.03224   7.219 5.25e-13 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: this example (and the EAGeR example below) both have a **binary** treatment `A` and outcome `Y`, but `DoubleMLPLR` assumes a partially linear model with a single constant effect `theta` and continuous-style regression learners (`ml_g`, `ml_m`) for both nuisance functions. The more appropriate DoubleML estimator for a binary treatment/outcome is the Interactive Regression Model, `DoubleMLIRM`, which uses classification learners for the propensity score and allows heterogeneous treatment effects. `DoubleMLPLR` will still run and often gives a reasonable ATE-like summary, but `DoubleMLIRM` is the estimator actually designed for this setting.

32.4. EAGeR trial simulation (AIPW package)

This simulation is based on a package "AIPW", which has a simulation data "eager_sim_rct". A denotes the binary treatment assignment, Y is the binary outcome, and W_g represents the covariate that affects the treatment assignment, which in our case was deemed to be the eligibility stratum indicator, sampled with replacement from the EAGeR Trial. Similarly, W_Q is a set of baseline prognostic covariates, which were also sampled with replacement from the EAGeR Trial, and includes the number of prior pregnancy losses, age, number of months of trying to conceive prior to randomization, body mass index (weight (kg)/height (m)2), and mean arterial blood pressure (denoted W1…5 , respectively).

Data and the true risk difference. Unlike the simulation above, this is a resampled trial: 1,228 observations with a binary treatment and outcome, and the package records the risk difference used to generate it, so we again have a target to check against.

```
library(AIPW)
data("eager_sim_rct")
glimpse(eager_sim_rct)
```

```
Rows: 1,228
Columns: 8
$ sim_Y             <int> 0, 1, 1, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0~
$ sim_Tx            <int> 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0~
$ eligibility       <dbl> 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0~
$ loss_num          <dbl> 2, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 1, 1, 2, 2, 2, 1~
$ age               <dbl> 21.93, 30.65, 30.96, 32.24, 31.49, 27.99, 31.60, 25.~
$ time_try_pregnant <dbl> 2, 1, 0, 9, 1, 10, 8, 6, 1, 0, 1, 1, 3, 3, 0, 10, 10~
$ BMI               <dbl> 19.51963, 25.56474, 20.75446, 21.65728, 24.27480, 19~
$ meanAP            <dbl> 73.33333, 79.00000, 79.33333, 63.83333, 97.44444, 85~
```

```
attributes(eager_sim_rct)$true_rd
```

```
[1] 0.1324569
```

```
data <- eager_sim_rct %>%
  rename(Y=sim_Y, A=sim_Tx) %>%
  tibble()

W <- data %>%
  select(-Y,-A)
```

32.4.1. Plug-in estimator from outcome regression

```
SL.outcome<- SuperLearner(Y=data$Y, X=data %>% select(-Y),
                                    SL.library=SL.library, family="binomial")
SL.outcome.obs<- predict(SL.outcome, X=data %>% select(-Y))$pred
# predict the PO Y^1
SL.outcome.1<- predict(SL.outcome, newdata=data %>% select(-Y) %>% mutate(A=1))$pred
# predict the PO Y^0
SL.outcome.0<- predict(SL.outcome, newdata=data %>% select(-Y) %>% mutate(A=0))$pred
```

This is to do machine learning on the outcome equation; then get prediction on treated vs control, to get $\hat{Y}^1$ and $\hat{Y}^0$. The sample average of the difference is our plug-in estimator.

```
SL.plugin<-mean(SL.outcome.1-SL.outcome.0)
SL.plugin
```

```
[1] 0.1185897
```

```
Q=data.frame(QAW=SL.outcome.obs, Q0W=SL.outcome.0,Q1W=SL.outcome.1)
```

The plug-in gives 0.1186 against the recorded true risk difference of 0.1325 — about 0.014 low.

32.4.2. AIPW with machine learning

```
set.seed(123)
SL.g<- SuperLearner(Y=data$A, X=data %>% select(-Y, -A),
                    SL.library=SL.library, family="binomial")
g1W <- SL.g$SL.predict
g0W<- 1- g1W
IF.1<-((data$A/g1W)*(data$Y-Q$Q1W)+Q$Q1W)
IF.0<-(((1-data$A)/g0W)*(data$Y-Q$Q0W)+Q$Q0W)
IF<-IF.1-IF.0
aipw.1<-mean(IF.1);aipw.0<-mean(IF.0)
aipw.manual<-aipw.1-aipw.0
ci.lb<-mean(IF)-qnorm(.975)*sd(IF)/sqrt(length(IF))
```

```
ci.ub<-mean(IF)+qnorm(.975)*sd(IF)/sqrt(length(IF))
res.manual.aipw<-c(aipw.manual,ci.lb, ci.ub)
res.manual.aipw
```

```
[1] 0.13103847 0.06938821 0.19268873
```

Now we use "npcausal".

```
library(npcausal)
aipw<- ate(y=data$Y, a=data$A, x=W, nsplits=1, sl.lib=SL.library)
```

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     parameter       est         se      ci.ll     ci.ul pval
1      E{Y(0)} 0.4455985 0.02240994 0.40167501 0.4895220    0
2      E{Y(1)} 0.5753483 0.02105948 0.53407176 0.6166249    0
3 E{Y(1)-Y(0)} 0.1297499 0.03059579 0.06978212 0.1897176    0
```

The results from "npcausal" are almost the same as the manual results. Two clarifications. First, the manual estimator above *is* bias-corrected in the relevant sense: AIPW is exactly the one-step correction of the plug-in estimator using the efficient influence function. What we did *not* do is **cross-fitting** (sample splitting), which is a separate device. Second, that matters for inference: because the manual example fits the SuperLearner nuisances on the *full* sample and then uses `sd(IF)/sqrt(n)`, the resulting confidence interval is only valid under empirical-process (Donsker) conditions that flexible ML learners typically violate. With cross-fitting, the influence-function variance is valid without those conditions. "npcausal" does the cross-fitting for you when you set `nsplits > 1` (here `nsplits = 1` turns it off, for comparison with the manual full-sample version).

```
aipw2<- ate(y=data$Y, a=data$A, x=W, nsplits=10, sl.lib=SL.library)
```

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     parameter       est         se      ci.ll     ci.ul pval
1      E{Y(0)} 0.4454472 0.02499646 0.39645419 0.4944403    0
2      E{Y(1)} 0.5750179 0.02214021 0.53162307 0.6184127    0
3 E{Y(1)-Y(0)} 0.1295707 0.03321609 0.06446711 0.1946742    0
```

32.4.3. DoubleML

```
library(DoubleML)
dml_data = DoubleMLData$new(as.data.frame(data),
                            y_col = "Y",
                            d_cols = "A",
                            x_cols = c("eligibility" , "loss_num","age", "time_try_pregnant","BMI","mean
print(dml_data)
```

```
================= DoubleMLData Object ==================
```

```
------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: eligibility, loss_num, age, time_try_pregnant, BMI, meanAP
Instrument(s):
Selection variable:
No. Observations: 1228
```

```r
library(mlr3)
library(mlr3learners)
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")

learner = lrn("regr.ranger", num.trees=500,  max.depth=5, min.node.size=2)
ml_g = learner$clone()
ml_m = learner$clone()

learner = lrn("regr.glmnet")
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()

set.seed(3141)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_g=ml_g, ml_m=ml_m)
obj_dml_plr$fit()
print(obj_dml_plr)
```

```
================== DoubleMLPLR Object ==================


------------------ Data summary      ------------------
Outcome variable: Y
Treatment variable(s): A
Covariates: eligibility, loss_num, age, time_try_pregnant, BMI, meanAP
Instrument(s):
Selection variable:
No. Observations: 1228

------------------ Score & algorithm ------------------
Score function: partialling out
DML algorithm: dml2

------------------ Machine learner   ------------------
ml_l: regr.ranger
ml_m: regr.ranger

------------------ Resampling        ------------------
```

```
No. folds: 5
No. repeated sample splits: 1
Apply cross-fitting: TRUE

------------------ Fit summary       ------------------
 Estimates and significance testing of the effect of target variables
  Estimate. Std. Error t value Pr(>|t|)
A   0.12505    0.03292   3.798 0.000146 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note: as with the earlier PLR example, `A` and `Y` here are **binary** (this is the EAGeR trial data), so `DoubleMLIRM` (which uses classification learners for the propensity score and allows heterogeneous effects) would be the more appropriate DoubleML estimator than `DoubleMLPLR`'s partially linear, constant-effect model — see the note above.

32.5. Summary

Identification turns a causal target, such as $E(Y^1 - Y^0)$, into a functional of the observed distribution. A plug-in estimate of that functional from machine-learning nuisance fits inherits their regularization bias and has no standard error. Influence-function estimators correct the bias with a one-step term. AIPW from `npcausal`, TMLE and DoubleML are all of this kind, and sample splitting keeps overfitting out of the correction. DoubleML's partially linear model assumes a constant effect, so with a binary treatment and outcome its interactive model is the better fit.

In the simulation with a realized effect of 0.203, the plug-in gives 0.2050, AIPW 0.2123 by hand and 0.2141 from `npcausal`, and TMLE 0.2118, all within 0.003 of each other. Cross-fitting barely changes the estimate here. In the EAGeR trial data the plug-in gives 0.1186 against a true risk difference of 0.1325, and the one-step corrections move it toward the truth: 0.1310 by hand, 0.1297 from `npcausal` and 0.1296 with 10-fold cross-fitting.

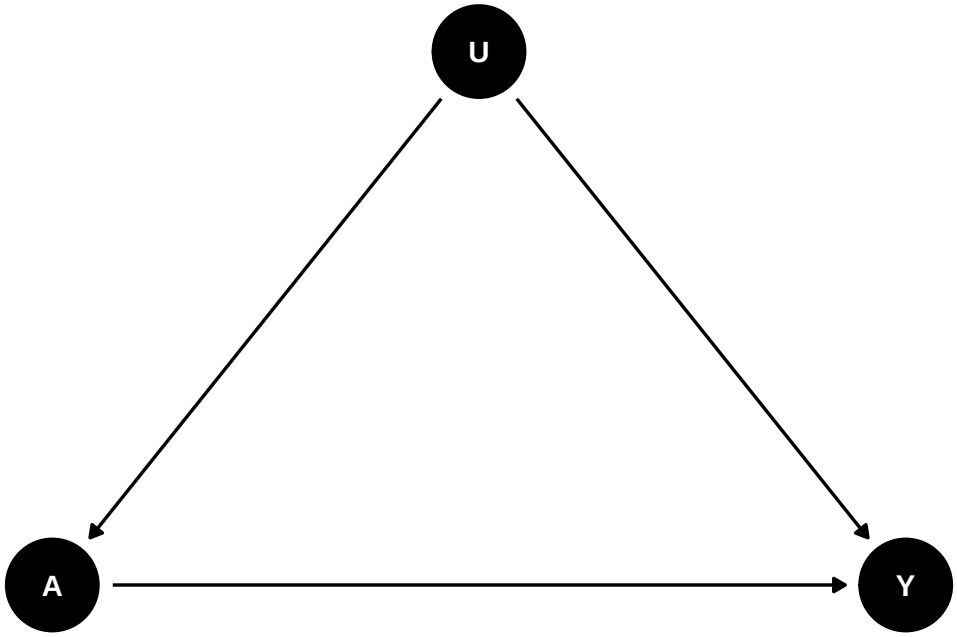
Systematic treatment: R · Julia.

33. Intro to proximal causal inference

This chapter follows Tchetgen Tchetgen and co-authors (Miao, Geng, and Tchetgen Tchetgen 2018; Tchetgen Tchetgen et al. 2024), drawing primarily on Xu Shi's talk on proximal causal inference.

33.1. Unobserved confounders

The central problem for observational studies is unobserved confounding. Even after conditioning on observed covariates X, an unobserved confounder U may affect both treatment A and outcome Y:

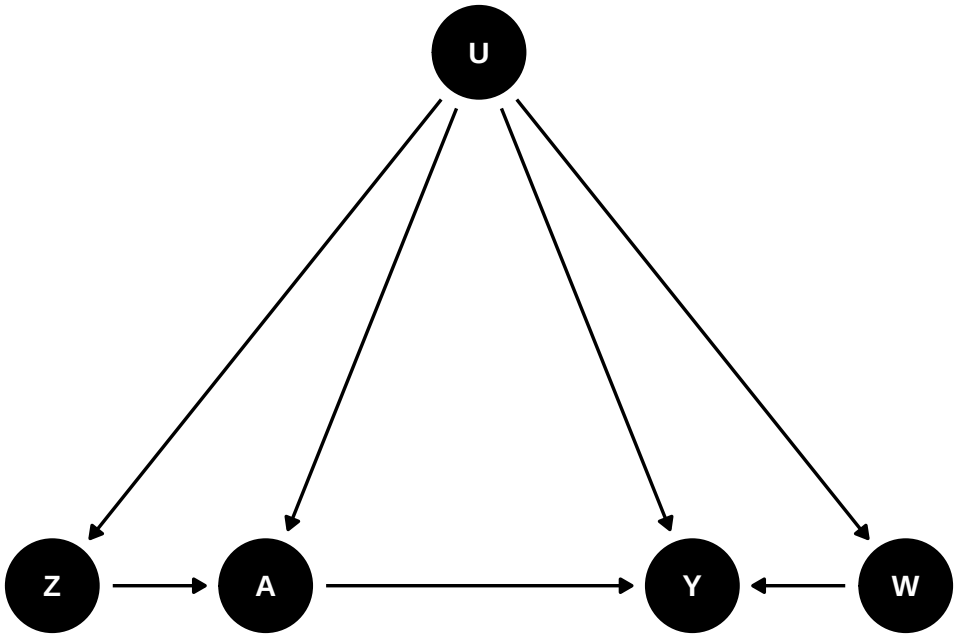


In this case, we have biased estimates of the treatment effect, regressing Y on A controlling for X.

33.2. Proximal causal inference

Proximal causal inference (also called negative controls) is a framework that allows us to estimate the causal effect of treatment A on outcome Y while accounting for unobserved confounders.

We'll need a DAG like this:

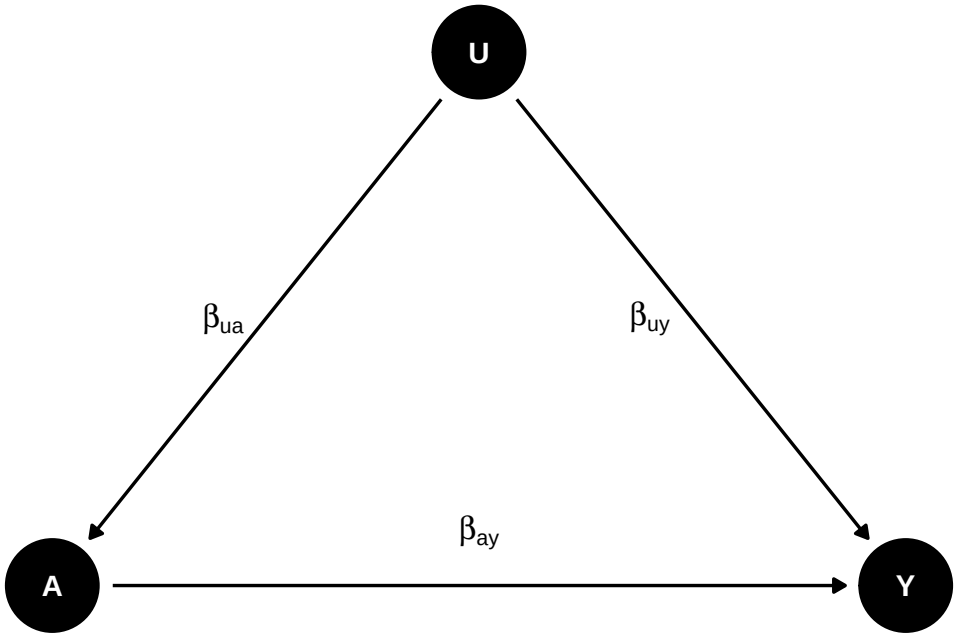


Basically we need two proxies for U to control for confounding. Z is called Negative Control for Exposure (NCE) and W is called Negative Control for Outcome (NCO). There is no link from Z to Y, and no link from W to A or Z.

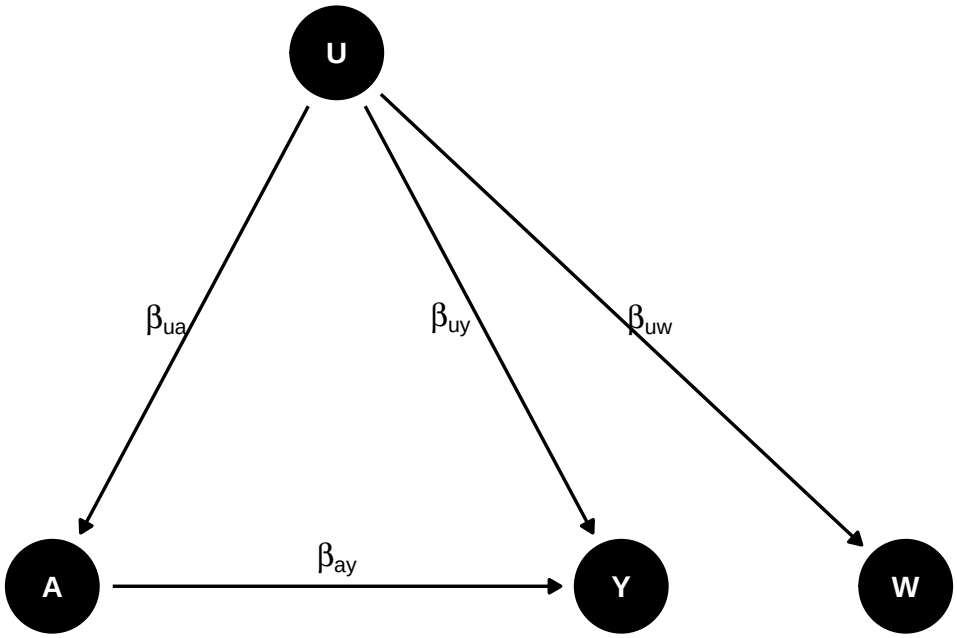
Z is an NCE if $Y(a, z) = Y(a)$ and $Z \perp Y(a)|U$, meaning that Z satisfies the exclusion restriction that there is no direct link from Z to Y, and that Z is independent of Y given U.

W is an NCO if $W(a, z) = W$ and $W \perp (A, Z)|U$, meaning that A and Z have no direct link to W.

33.2.1. Why this works

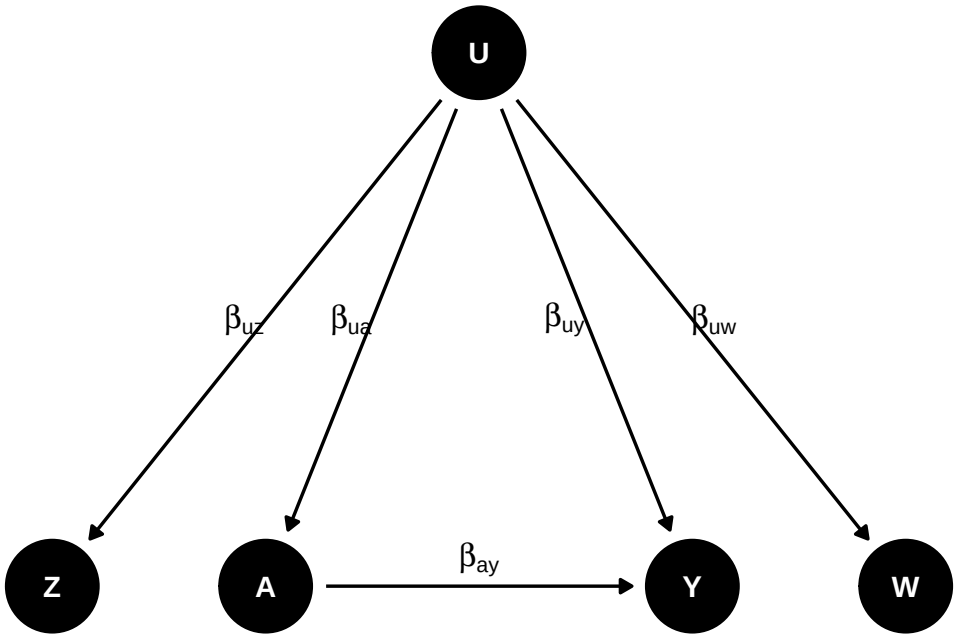


If we regress Y on A, then we get $\beta_{ay} + \beta_{uy}\beta_{ua}$, which is biased. How do we recover β_{ay}?



Suppose we have W as proxy of U and $\beta_{uw} = \beta_{uy}$, then we can get the true effect. We can do regression of Y on A, and then W on A. The first regression gives us $\beta_{ay} + \beta_{uy}\beta_{ua}$, and the second regression gives us $\beta_{uw}\beta_{ua}$. We can then get β_{ay} from the difference of the two coefficients. This is similar to a DiD setting. Say W is Y_{t-1}, and assume the effect of U is the same on Y and Y_{t-1}, then effect of A on Y by the difference of regressing Y on A and Y_{t-1} on A.

However, most likely $\beta_{uw} \neq \beta_{uy}$. Then we need another proxy Z of U to recover the true effect. This is also called "double proxy", or "double negative control".



33.2.2. A linear model

Suppose we have the following linear model:

$$E[Y|A, Z, U] = \beta_{y0} + \beta_{ay}A + \beta_{uy}U \tag{33.1}$$

$$E[W|A, Z, U] = \beta_{w0} + \beta_{uw}U \tag{33.2}$$

$$E[U|A, Z] = \beta_{u0} + \beta_{zu}Z + \beta_{au}A \tag{33.3}$$

β_{ay} is the true effect of A on Y.

$$E[Y|A, Z] = \beta_{y0} + \beta_{ay}A + \beta_{uy}(\beta_{u0} + \beta_{zu}Z + \beta_{au}A) \tag{33.4}$$

$$E[W|A, Z] = \beta_{w0} + \beta_{uw}(\beta_{u0} + \beta_{zu}Z + \beta_{au}A) \tag{33.5}$$

From the last two regressions we can recover β_{ay}:

If we define $\delta_a^y = \beta_{ay} + \beta_{uy}\beta_{au}$ (regression coefficient of A on the Y regression) and $\delta_a^w = \beta_{uw}\beta_{au}$, (regression coefficient of A on the W regression), similarly for δ_z^y and δ_z^w, then we have

$$\beta_{ay} = \delta_a^y - \delta_a^w \frac{\delta_z^y}{\delta_z^w} \tag{33.6}$$

In this **linear-Gaussian special case only**, we can look at this as a two-stage-least-squares-style regression and replace U with the predicted value of W:

$$E[Y|A,Z] = \left(\beta_{y0} - \frac{\beta_{uy}}{\beta_{uw}}\beta_{w0}\right) + \beta_{ay}A + \frac{\beta_{uy}}{\beta_{uw}}E[W|A,Z] \tag{33.7}$$

since $E[U \mid A,Z] = (E[W \mid A,Z] - \beta_{w0})/\beta_{uw}$ from the W equation above. The coefficient on the predicted value of W is β_{uy}/β_{uw}, not β_{uy} alone — it only reduces to β_{uy} when $\beta_{uw} = 1$. So we first regress W on A and Z, then regress Y on A and the predicted value of W.

This "predict W from A and Z, then plug it in" shortcut is **not** the general proximal identification argument, and it is not the same as a standard IV/2SLS specification (we are not instrumenting an endogenous regressor with `ivreg`). In general proximal causal inference, the outcome bridge function $h(A,W)$ is identified by solving the conditional moment restriction $E[Y \mid A,Z] = E[h(A,W) \mid A,Z]$, which requires completeness (relevance) conditions linking the treatment-inducing proxy Z and the outcome-inducing proxy W to the hidden confounder U. The generated regressor $S = \widehat{E}[W \mid A,Z]$ is only a valid substitute under the linear-Gaussian structure above; even then the two-stage standard errors are wrong as written, because S is generated in the first stage (bootstrap the whole procedure).

33.2.3. A nonparametric model

We don't have to have a linear model, this can be done with a nonparametric model as well.

$$E[Y(a)] = E[h(a,W)] \tag{33.8}$$

where $E[Y|A,Z] = E[h(A,W)|A,Z]$.

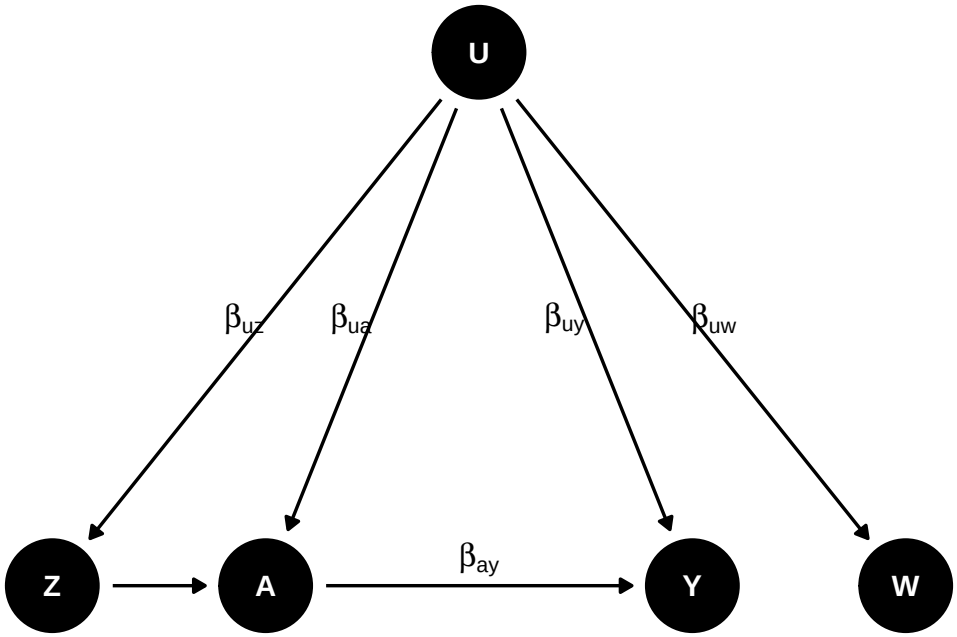
That is, we can estimate h as a function of A and W, identified using Z, nonparametrically. This h function is called outcome bridge function. In the linear case, it is $h(A,W) = \beta_0 + \beta_a A + \beta_w W$.

33.2.4. How to find proximal variables

Kummerfeld, Lim, and Shi (2024) propose the Data-driven Automated Negative Control Estimation (DANCE) algorithm to find negative controls. It tests candidate variables for a special kind of negative control, the disconnected negative control, applies a negative control estimator to each pair of validated variables, and combines the results into one estimate and interval. Like a search for instruments, it can only screen candidates against testable implications; the untestable exclusions still have to be argued.

33.3. Example

Let's generate data based on this DAG:




```
set.seed(123)
n <- 5000
# Set coefficients
b_uw <- 0.5 # effect of U on W
b_uy <- 0.5 # effect of U on Y
b_ay <- 0.5 # effect of A on Y
b_uz <- 0.5 # effect of U on Z
b_ua <- 0.5 # effect of U on A
b_za <- 0.5 # effect of Z on A

U <- rnorm(n)
W <- b_uw * U + rnorm(n) # W is a function of U
Z <- b_uz * U + rnorm(n) # Z is a function of U
A <- b_ua * U + b_za * Z + rnorm(n) # A is a function of U and Z
Y <- b_uy * U + b_ay * A + rnorm(n) # Y is a function of U and A
# A <- rbinom(n, 1, plogis(-1 + U / 2)) # A is a binary treatment


# Create a data frame with the generated variables

data <- data.frame(W = W, Z = Z, A = A, Y = Y)
```

The simulation sets $n = 5000$ and every coefficient to 0.5. The hidden confounder is $U \sim N(0, 1)$; the two proxies are $W = 0.5U + \varepsilon_W$ and $Z = 0.5U + \varepsilon_Z$; treatment is $A = 0.5U + 0.5Z + \varepsilon_A$; and

the outcome is $Y = 0.5U + 0.5A + \varepsilon_Y$, all errors standard normal. The true effect of A on Y is therefore 0.5.

Note the roles: Z affects A but not Y directly, W is affected by U but affects nothing, and neither is a valid control for U on its own.

If we run regression of Y on A, Z, W,

```
m1 <- lm(Y ~ A + Z + W, data = data)
summary(m1)
```

```
Call:
lm(formula = Y ~ A + Z + W, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5836 -0.7167  0.0005  0.7305  4.4866

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.006347   0.015279  -0.415    0.678
A            0.660021   0.014123  46.734  < 2e-16 ***
Z            0.085004   0.016650   5.105 3.43e-07 ***
W            0.132798   0.014165   9.375  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.08 on 4996 degrees of freedom
Multiple R-squared:  0.4556,    Adjusted R-squared:  0.4553
F-statistic:  1394 on 3 and 4996 DF,  p-value: < 2.2e-16
```

We don't get the true effect of A on Y: the coefficient on A is 0.660 against a true 0.5, a bias of 32%. Conditioning on both proxies does not remove the confounding, because neither W nor Z *is* U — each is a noisy reflection of it, and controlling for a noisy proxy leaves residual confounding. The proxies do carry signal (both are significant, at 0.085 and 0.133), which is what makes this failure mode deceptive: the regression looks like it is adjusting for the confounder.

If we do the 2sls using W and Z, then we get the true effect.

```
# Linear-Gaussian (Y, W) toy case ONLY. This two-stage plug-in identifies
# the effect under the linear structure above; it is not the general
# proximal estimator, and the reported SEs below are NOT valid because S is
# a generated regressor (bootstrap the whole two-step procedure for inference).
# Perform the first-stage linear regression W ~ A + Z
m1 <- lm(W ~ A + Z, data = data)
# Compute the estimated E[W|A,Z]
S <- m1$fitted.values
```

```r
# Perform the second-stage linear regression Y ~ A + E[W|A,Z]
m2 <- lm(Y ~ A + S, data = data)
summary(m2)   # point estimate OK under the linear model; SEs need bootstrap
```

```
Call:
lm(formula = Y ~ A + S, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-4.5477 -0.7309  0.0059  0.7395  4.3715

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.005049   0.015408  -0.328    0.743
A            0.489342   0.043210  11.325  < 2e-16 ***
S            1.143480   0.199171   5.741 9.96e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.089 on 4997 degrees of freedom
Multiple R-squared:  0.446, Adjusted R-squared:  0.4458
F-statistic:  2012 on 2 and 4997 DF,  p-value: < 2.2e-16
```

The two-stage estimate is 0.489, against the true 0.5 and the naive 0.660. Using the two proxies in the right way — one to build a predicted $E[W \mid A, Z]$ and that prediction as the control — recovers the effect, where using both directly as regressors did not.

The reported standard error of 0.043 should be ignored, for the reason flagged in the code comment: S is a generated regressor, so the second-stage standard errors are wrong as written and need a bootstrap over both stages.

33.4. Summary

When an unobserved confounder affects both treatment and outcome, regression on observed covariates stays biased. Proximal causal inference uses two proxies of the confounder. A negative control exposure affects the outcome only through the confounder. A negative control outcome is affected by the confounder but not by the treatment. One proxy is enough only in the special case where it loads on the confounder exactly as the outcome does, so in general two are needed. The effect is identified through an outcome bridge function that solves a conditional moment restriction, and that requires completeness conditions linking the proxies to the confounder. In the linear-Gaussian case this reduces to a two-stage regression on the predicted negative control outcome.

In the simulation the true effect is 0.5. Regression on the treatment and both proxies gives 0.660, still confounded because each proxy is a noisy reflection of the confounder. The two-stage estimate

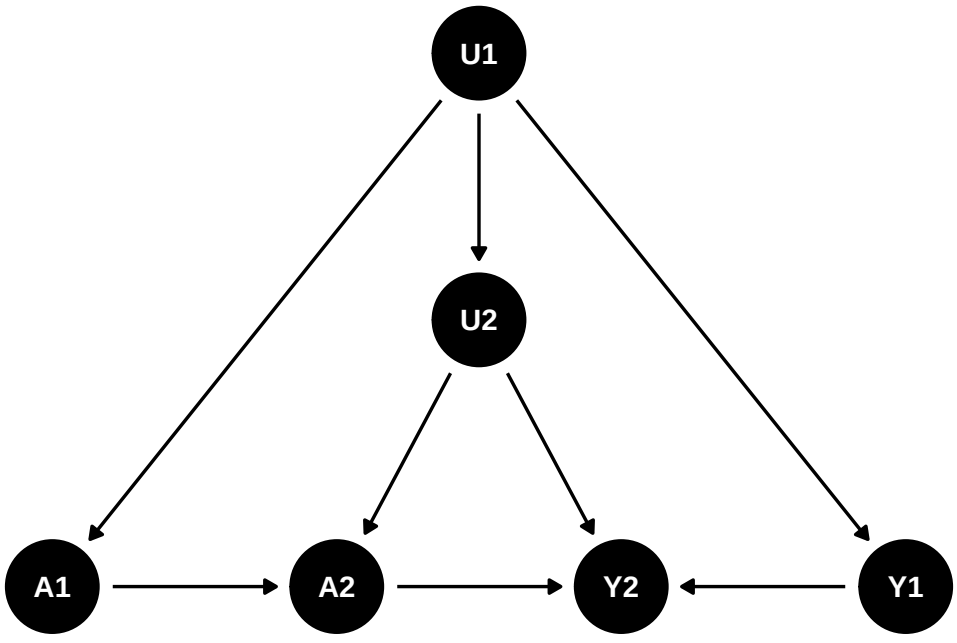
gives 0.489. Its printed standard error is wrong, since the first stage is a generated regressor. The whole procedure has to be bootstrapped.

Systematic treatment: R · Julia.

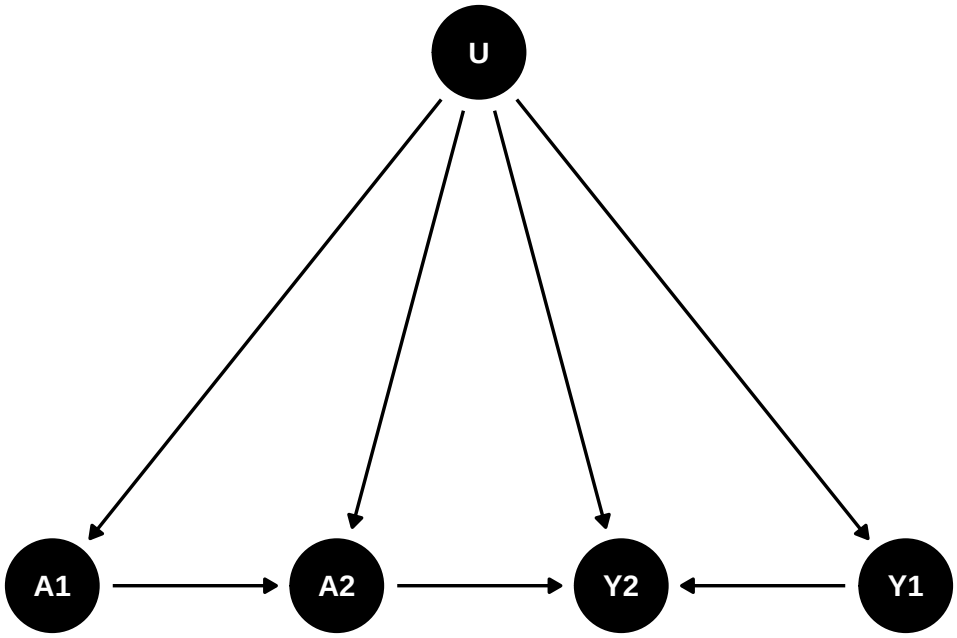
34. Simulation on proximal causal inference with panel data

This chapter extends proximal causal inference to a panel-data setting. The question is whether lagged treatment and lagged outcome can serve as negative control exposure (NCE) and negative control outcome (NCO), respectively, to account for unobserved confounders.

Consider a two-period DAG:



Suppose that's the case for the causal structure, then I can reasonably combine U1 and U2 into a single unobserved confounder U:



That would fit into the double negative control framework. The problem is that we have to assume there is no link from A1 to Y1, which is odd. Can we assume in period 1 there is no effect, but in period 2, there is an effect?

34.1. A simulation with panel data

I generate two period data based on the DAG above.

```
library(dplyr)
library(rootSolve)
library(tidyverse)

gen_data <- function(seed, n){
  set.seed(seed)
  u <- rnorm(n, mean = 0, sd = 1)  # unobserved confounder
  x1 <- rnorm(n, mean = 0, sd = 1)  # observed confounder
  x2 <- rnorm(n, mean = 0, sd = 1)  # irrelevant covariate: enters no equation below
  a1 <- rbinom(n, 1, plogis(2*u + 1.5*x1))  # treatment at time 1
  a2 <- rbinom(n, 1, plogis(u + a1 +  x1))  # treatment at time 2
  y1 <-  1.5*u + .5*x1  +  rnorm(n, mean = 0, sd = 1)  # outcome at time 1
  y2 <-  y1 + 2.5*u + 2.5*x1 + 3*a2 + rnorm(n, mean = 0, sd = 1)  # outcome at time 2
  data <- data.frame(u=u, x1=x1, x2=x2, a1=a1, a2=a2, y1=y1, y2=y2)
  return(data)
}
data <- gen_data(333, 100000)
```

```
data_sim <- data |>
  select(a1, a2, y2,  x1, x2, y1) |>
  rename(a = a2, y = y2,  x = x1, w = y1, z=a1)
```

The simulation draws $n = 100{,}000$ two-period observations. There is an unobserved confounder $u \sim N(0,1)$, an observed confounder $x_1 \sim N(0,1)$, and a second covariate $x_2 \sim N(0,1)$ that enters neither treatment nor outcome. Treatment at time 1 is $a_1 \sim \text{Bernoulli}(\Lambda(2u + 1.5x_1))$ and at time 2 is $a_2 \sim \text{Bernoulli}(\Lambda(u + a_1 + x_1))$. The outcomes are $y_1 = 1.5u + 0.5x_1 + \varepsilon_1$ and $y_2 = y_1 + 2.5u + 2.5x_1 + 3a_2 + \varepsilon_2$.

So the effect of interest is the 3 on a_2. The construction then relabels a_1 as the negative control exposure z and y_1 as the negative control outcome w — note that y_1 does not depend on a_1, which is exactly the assumption discussed at the end of this chapter.

We can use the simple 2sls:

```
lm1 <- lm(w ~ a + x + z, data = data_sim)
data_sim$what <- lm1$fitted.values
lm2 <- lm(y ~ a + x +  what, data = data_sim)
summary(lm2)
```

```
Call:
lm(formula = y ~ a + x + what, data = data_sim)

Residuals:
     Min       1Q   Median       3Q      Max
-16.4284  -2.3083   0.0071   2.3020  20.7590

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.01293    0.02315  -0.558    0.577
a            3.02113    0.03528  85.625   <2e-16 ***
x            1.67203    0.01245 134.327   <2e-16 ***
what         2.66578    0.01747 152.631   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.456 on 99996 degrees of freedom
Multiple R-squared:  0.6822,    Adjusted R-squared:  0.6822
F-statistic: 7.155e+04 on 3 and 99996 DF,  p-value: < 2.2e-16
```

The 2SLS coefficient on a is 3.021 with a standard error of 0.035, against the true 3. The coefficient on the fitted proxy `what` is 2.666, which is the ratio of the confounder's loading on the outcome to its loading on the proxy, not an effect of interest.

Or we can use more sophisticated methods. Here I use the m-estimation code by Paul Zivich: https://github.com/pzivich/publications-code/blob/master/ProximalCI/proximal_sims_r.R. This code is to construct estimating equations, then use root finding to find the parameter estimates.

```
pcl_m2 <- function(theta, mydata, out = "ef"){
  # read in variables
  a <- mydata$a
  x <- mydata$x
  w <- mydata$w
  z <- mydata$z
  y <- mydata$y
  # model for what
  what <- theta[1] + theta[2] * a + theta[3] * x + theta[4] * z

  # matrix of covars in model for what
  wmat <- as.matrix(cbind(rep(1, nrow(mydata)), a, x, z), ncol = 4,
                    nrow = nrow(mydata))

  # estimating function for what
  ef1 <- (t(wmat)  %*% (w - what))            # gives px1 vector (summed over units)
  ef1_b <- wmat * (w - what)                  # gives nxp matrix (rows are units)

  # model for yhat
  yhat <- theta[5] + theta[6]*a + theta[7]*x + theta[8] * what

  # matrix of covars in model for yhat
  ymat <- as.matrix(cbind(rep(1, nrow(mydata)), a, x, what), ncol = 4,
                    nrow = nrow(mydata))

  #estimating function for yhat
  ef2 <- t(ymat) %*% (y - yhat)               # px1 vector, summed over units
  ef2_b <- (ymat) * (y - yhat)                # nxp matrix (rows are units)
  f <- c(ef1, ef2)                            # p-dimensional vector
  if(out == "full") return(cbind(ef1_b, ef2_b)) else return(f) # full used in var calculation only
}

# call m-estimator for PCL
mpcl_mest <- function(dat){
  theta_start <- rep(.05, 8)
  results <- mestimator(pcl_m2, init = theta_start, dat, mydata = dat)
  return(results)
}

##' M-estimation engine
mestimator <- function(ef, init, dat, ...){
  # get estimated coefficients
  fit <- multiroot(ef, start = init, out = "ef",
                   rtol = 1e-9, atol = 1e-12, ctol = 1e-12, ...  )
  betahat <- fit$root
  n <- nrow(dat)
```

```
  # bread
  # derivative of estimating function at betahat
  pd <- gradient(ef, betahat, out = "ef", ...)
  pd <- as.matrix(pd)
  bread <- -pd/n
  se_mod <- sqrt(abs(diag(-solve(pd))))

  # meat1
  efhat <- ef(betahat, out = "full", ...)
  meat <- list()
  meat1 <- (t(efhat) %*% efhat)/n

  # sandwich
  sandwich1 <- (solve(bread) %*% meat1 %*% t(solve(bread)))/n
  se_sandwich1 <- sqrt(diag(sandwich1))

  results <- data.frame(betahat, se_sandwich1, se_mod)
  return(results)
}
```

```
rmpcl <- unname(mpcl_mest(data_sim)[6, c(1:2)])
rmpcl
```

```
6 3.021126 0.0198433
```

```
#mpcl_mest(data_sim)
```

The m-estimator gives 3.0211 with a standard error of 0.0198.

Both the 2sls and the m-estimation give us the same estimate of the causal effect of treatment on outcome, which is 3 — identical to four decimal places, 3.021126 either way. The standard errors differ, 0.0198 against 0.0353, and the m-estimator's is the one to use: it accounts for the estimation of the first stage, which the plug-in 2SLS standard error does not. The m estimator can be more general. 2sls is good for the linear case.

However, the main issue is the assumption that there is no link from A1 to Y1.

This resembles the no-anticipation assumption of DiD, but it is a different restriction. No anticipation says treatment has no effect before it starts. Here A_1 is a treatment that has started, and we require its effect on Y_1 to be zero. We cannot meet that by making A_1 zero for everyone: A_1 is the negative control exposure, and its variation is what the bridge function uses to learn about U. A constant A_1 removes the information the method needs.

This is a restrictive identifying assumption, not a technical footnote. Using A_{t-1} as the negative control exposure and Y_{t-1} as the negative control outcome only identifies the effect if two separate exclusions hold: A_1 must have no direct effect on Y_1, so that Y_1 is a clean proxy for U rather than a partly post-treatment quantity; and A_1 must have no direct effect on Y_2 except through U,

which is the exclusion restriction that makes A_1 admissible as a negative control exposure in the first place. If either link is actually present — for example, if treatment has an immediate effect in the same period it starts, which is the empirically common case — this proximal construction is misspecified and the estimate above should not be trusted. Before applying this approach, researchers should have a substantive reason to believe treatment effects only appear with a lag. Two further conditions are separate from the exclusions. The proxies must be relevant, moving with U strongly enough to pin down the bridge function, and a bridge function must exist and be unique. The two-stage calculation can return a number when any of these fails, so its numerical success is not evidence for them.

This two-period design is a special case of a broader literature. Deaner (2021) uses observations from other periods as proxies for unobserved heterogeneity in panel models, and Ying et al. (2023) extend proximal inference to time-varying treatments with time-varying proxies under a marginal structural model. They are the place to start when the exclusions needed here are implausible.

34.2. Summary

In a two-period panel, last period's treatment and outcome can act as the two negative controls of proximal inference for this period's effect. This works only if the lagged treatment has no effect on the lagged outcome, and if the confounding acts through one shared unobserved factor in both periods. The first condition is close to the no-anticipation assumption of DiD. It is restrictive, and it has to be argued case by case.

In the simulation with $n = 100{,}000$ and a true effect of 3, both the two-stage regression and Zivich's M-estimator give 3.021. The M-estimator's standard error, 0.0198, accounts for the estimated first stage, unlike the two-stage regression's 0.035.

Systematic treatment: R · Julia.

35. Longitudinal modified treatment policy (LMTP)

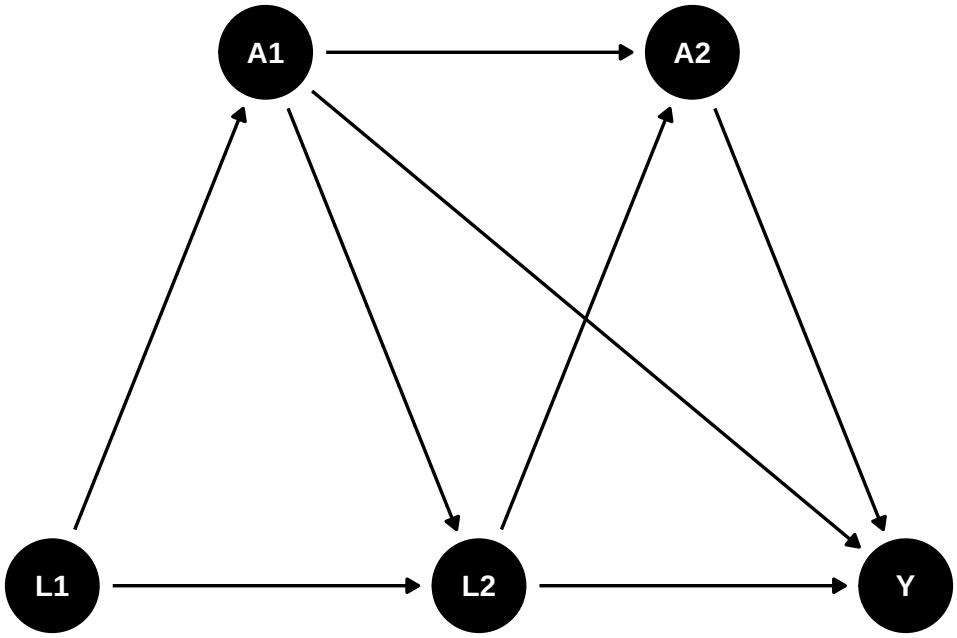
The longitudinal modified treatment policy (LMTP) framework, developed in epidemiology and biostatistics, handles settings where the standard ATE is poorly defined: continuous or multivalued treatments, time-varying exposures, or hypothetical interventions that would violate positivity. This chapter follows Nicholas Williams.

Suppose we have such a DAG:

```
library(ggplot2)
library(ggdag)

g <- dagify(
  A1 ~ L1,
  L2 ~ L1,
  L2 ~ A1,
  A2 ~ A1,
  A2 ~ L2,
  Y ~ A2,
  Y ~ L2,
  Y ~ A1,
  exposure = "A2",
  outcome = "Y",
  coords = list(x = c(L1 = 1,  L2 = 2, Y = 3, A1 = 1.5, A2 = 2.5),
                y = c(L1 = 1,  L2 = 1, Y = 1, A1 = 1.5, A2 = 1.5))
)

ggdag(g) +
  theme_dag()
```



We have multiple time points, and the treatment A is time-varying. We are interested in not only the ATE of A_2 on Y, but also some other hypothetical interventions.

35.1. Definitions and assumptions

35.1.1. Assumptions

1. Consistency. The observed outcome under the treatment actually received equals the corresponding potential outcome, and the shifted intervention is well defined — no hidden versions of it.
2. Sequential unconfoundedness. There are no unmeasured confounders of treatment at any time point given the history. The stronger *sequential randomization* condition conditions on the full history including past natural values of treatment; some estimators need that version and others do not, so it is worth knowing which one a given result assumes.
3. Positivity, in the form the shift requires. Roughly: if a unit with history h_t can receive a_t, then the shifted value $d(a_t, h_t)$ must also have support given h_t. This is where modified treatment policies earn their keep. A static always-treat or never-treat rule can demand support the data does not have, while a bounded shift of the observed treatment stays inside it.
4. For continuous treatments, a smoothness condition. The influence-function estimators below assume the shift function is piecewise smooth and invertible in a_t, which is what lets the density of the shifted treatment be written in terms of the observed one.

The always-treat and never-treat demonstration below is, on that reading, an unusual lead example for this framework: it is the easiest case to follow, and also the case whose positivity is most likely to fail. A bounded continuous shift is the canonical LMTP application. Díaz et al. (2023) state the assumptions precisely.

35.1.2. Dynamic treatment regime

Some notations commonly used in this literature: We observe $Z = (L_0, A_0, L_1, A_1, Y)$, where L is the observed confounder, A is the treatment, and Y is the outcome. The treatment A can be time-varying. We can also have multiple treatments at different time points. For example, we can have A_2, A_3, etc. In this graph, baseline L_0 affects A_0, which in turn affects L_1, which then affects A_1, and finally A_1 affects the outcome Y.

History H_t is the history of data up to time t, right before A_t. for example, $H_1 = (L_0, A_0, L_1)$, and $H_2 = (L_0, A_0, L_1, A_1)$. d is the hypothetical intervention function, or shift function, which is a function of the history H_t and the treatment A_t. For example, $d_0(a_0, h_0, \epsilon_0)$ is a user-given function to map a_0, h_0, and ϵ_0 to a potential treatment value. The function d can be deterministic, or it can be stochastic. Then we can replace A_0 with $A_0^d = d_0(A_0, H_0, \epsilon_0)$. Then after that $A_1(A_0^d)$ is called the natural value of treatment.

This is very general, comparing to the static treatment regime.

Suppose treatment A is a function of the history of treatment and confounders, for example, $A = d(A_1, L_1, A_2, L_2)$. A can be set to a fixed value, say 1 or 0, or some value A^d. This function d can be anything, it can be taking a deterministic value, or it can be a function that takes the natural treatment value A as input. In the package "lmtp", this is called a shift function, or hypothetical intervention. For example, d can be set to 1 if $age < 30$, or d can be set to double the natural value of A. Many possibilities.

In comparison, for ATE, we only need to set A to 0 or 1.

Under this LMTP, the causal parameter is

$$\theta = E[Y^{\bar{A}^d}] \tag{35.1}$$

$Y^{\bar{A}^d}$ is the potential outcome under the hypothetical intervention $\bar{A}^d$. At time 1, $A_1^d = d(A_1, H_1)$.

35.1.3. Modified treatment policy

A simulated data set makes the estimation concrete.

This simulation is from Susmann et al. (2024). I modified slightly to fit the DAG above.

```
library(tidyverse)
library(tidyr)

mtp <- function(data, trt) {
  a <- data[[trt]]
  a * 0 + 1
}
simulate_data <- function(seed, N, tau, sigma = 0.5) {
  set.seed(seed)

  data <- tibble(id = 1:N)
```

```
  for(t in 1:tau) {
    Lt <- paste0("L_", t)
    Ltd <- paste0("L_", t, "d")

    At <- paste0("A_", t)
    Atd <- paste0("A_", t, "d")

    Lt1 <- paste0("L_", t - 1)
    Lt1d <- paste0("L_", t - 1, "d")

    At1 <- paste0("A_", t - 1)
    At1d <- paste0("A_", t - 1, "d")

    if(t == 1) {
      data[[Lt]] <- runif(N, 0, 1)
      data[[Ltd]] <- data[[Lt]]

      # A_1 depends on L_1 (the DAG edge L1 -> A1).
      data[[At]] <- rbinom(N, size = 1, prob = plogis(-0.5 + 0.5 * data[[Lt]]))
    }
    else {
      # L_t responds to the previous treatment A_{t-1} (the DAG edge A1 -> L2), and
      # A_t responds to L_t (L2 -> A2). L_2 is affected by A_1 and confounds A_2:
      # the treatment-confounder feedback this chapter is about.
      data[[Lt]] <- rnorm(N, mean = 0.25 * data[[Lt1]] + 0.4 * data[[At1]], 0.5)
      data[[Ltd]] <- rnorm(N, mean = 0.25 * data[[Lt1d]] + 0.4 * data[[At1d]], 0.5)

      data[[At]] <- rbinom(N, size = 1, prob = plogis(0.5 - 0.2 * data[[At1]] + 0.5 * data[[Lt]]))
    }

    data[[Atd]] <- mtp(data, At)
  }
  # Y depends on the final treatment A_2 and confounder L_2, and also directly
  # on the first-period treatment A_1 (the DAG edge A1 -> Y).
  data$Y  <- rnorm(N, data[[At]] + data[[Lt]] + 0.5 * data[["A_1"]], sigma)
  data$Yd <- rnorm(N, data[[Atd]] + data[[Ltd]] + 0.5 * data[["A_1d"]], sigma)
  data
}

simulated_data1 <- simulate_data(seed = 123, N = 10000, tau = 2)

mean(simulated_data1$Yd)
```

```
[1] 2.028593
```

```
mtp <- function(data, trt) {
  a <- data[[trt]]
  a * 0 + 0
}

simulated_data2 <- simulate_data(seed = 123, N = 10000, tau = 2)

mean(simulated_data2$Yd)
```

```
[1] 0.1285934
```

```
mean(simulated_data1$Yd) - mean(simulated_data2$Yd)
```

```
[1] 1.9
```

Note in this simulation the variables ending with "d" are the variables under hypothetical intervention, or modified treatment policy. L_1 is from Uniform$(0, 1)$, and L_2 is normal with mean $0.25L_1 + 0.4A_1$ and standard deviation 0.5 – so the time-varying confounder L_2 depends on the earlier treatment A_1, which is the time-varying confounding this chapter is about. The treatment A_1 is from a Bernoulli distribution with probability plogis$(-0.5+0.5L_1)$, and the treatment A_2 is from a Bernoulli distribution with probability plogis$(0.5-0.2A_1+0.5L_2)$. So L_2 is moved by the earlier treatment and in turn moves the later one. The outcome Y is from a normal distribution with mean $A_2+L_2+0.5A_1$, so Y depends on both the current treatment A_2 and the lagged treatment A_1. In the code, the first modified treatment policy sets treatment to 1 in every period (`simulated_data1`) and the second sets it to 0 in every period (`simulated_data2`); the reported difference $E[Y^d_{\text{always 1}}] - E[Y^d_{\text{always 0}}]$ is the effect of the always-treat versus never-treat static regimes.

The always-treat regime gives $E[Y^d] = 2.029$, never-treat gives 0.129, and their difference is 1.900 — and that 1.9 decomposes exactly into the three routes the DGP contains: 1 from A_2 direct, 0.5 from A_1 direct, and 0.4 through the $A_1 \to L_2 \to Y$ path, since L_2 shifts by 0.4 and enters Y with coefficient 1.

In this case, we can just do a linear regression to get the effect of A2 on Y, knowing the exact DAG.

```
m1 <- glm(Y ~ L_2 + A_1   + A_2, data = simulated_data1)
summary(m1)
```

```
Call:
glm(formula = Y ~ L_2 + A_1 + A_2, data = simulated_data1)

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.001540   0.009296  -0.166    0.868
L_2          0.999305   0.009953 100.400   <2e-16 ***
```

```
A_1          0.489431   0.010905  44.882   <2e-16 ***
A_2          1.005200   0.010432  96.358   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.2500106)

    Null deviance: 9986.8  on 9999  degrees of freedom
Residual deviance: 2499.1  on 9996  degrees of freedom
AIC: 14522

Number of Fisher Scoring iterations: 2
```

Knowing the DAG, the regression recovers every structural coefficient: L_2 at 0.9993, A_1 at 0.4894 and A_2 at 1.0052, against true values of 1, 0.5 and 1. Note this identifies the *direct* effect of A_1 only; its path through L_2 is blocked by conditioning on L_2, which is why the total of 1.9 above cannot be read off this table.

Now consider a different MTP: set half of the time to 0 and leave the other half unchanged.

$$d(a_t, \epsilon_t) = \begin{cases} 0, & \text{if } \epsilon_t < .5 \text{ and } a_t = 1 \\ a_t, & \text{otherwise} \end{cases} \tag{35.2}$$

This is, say, to set half of smokers to non-smokers, and the other half remain smokers.

```r
# mtp <- function(data, trt) {
#   a <- data[[trt]]
#   epsilon <- rbinom(nrow(data), size = 1, prob = 0.5)
#   ifelse(epsilon <.5 & a == 1, 0, a)
# }
d <- function(a) {
  epsilon <- runif(length(a))
  ifelse(epsilon < 0.5 & a == 1, 0, a)
}
simulated_data1$m3_d <- simulated_data1$Y

m2 <- glm(m3_d ~ L_1 + A_1 + L_2 + A_2, data = simulated_data1)
summary(m2)
```

```
Call:
glm(formula = m3_d ~ L_1 + A_1 + L_2 + A_2, data = simulated_data1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.00775    0.01249  -0.620    0.535
L_1          0.01312    0.01764   0.744    0.457
```

```
A_1          0.48944    0.01090  44.882   <2e-16 ***
L_2          0.99828    0.01005  99.342   <2e-16 ***
A_2          1.00517    0.01043  96.351   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.2500218)

    Null deviance: 9986.8  on 9999  degrees of freedom
Residual deviance: 2499.0  on 9995  degrees of freedom
AIC: 14524

Number of Fisher Scoring iterations: 2
```

```
simulated_data1$m2_d <- predict(m2, mutate(simulated_data1, A_2 = d(A_2)))

m1 <- glm(m2_d ~ L_1 + A_1, data = simulated_data1)
summary(m1)
```

```

Call:
glm(formula = m2_d ~ L_1 + A_1, data = simulated_data1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.30919    0.01522   20.32   <2e-16 ***
L_1          0.25652    0.02476   10.36   <2e-16 ***
A_1          0.89371    0.01434   62.32   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.5024226)

    Null deviance: 7070.0  on 9999  degrees of freedom
Residual deviance: 5022.7  on 9997  degrees of freedom
AIC: 21501

Number of Fisher Scoring iterations: 2
```

```
m1_d <- predict(m1, mutate(simulated_data1, A_1 = d(A_1)))
mean(m1_d)
```

```
[1] 0.6273608
```

The truth for this regime is not the same simulation with `mtp` swapped in. The shift acts on the *natural* value of each treatment, the value it would take after the earlier shift has already moved L_2. So we simulate that sequence directly, on a large sample:

```r
set.seed(2026)
n_big <- 1e6
L1  <- runif(n_big)
A1d <- d(rbinom(n_big, 1, plogis(-0.5 + 0.5 * L1)))            # shift the natural A_1
L2d <- rnorm(n_big, 0.25 * L1 + 0.4 * A1d, 0.5)                # L_2 under the shifted A_1
A2d <- d(rbinom(n_big, 1, plogis(0.5 - 0.2 * A1d + 0.5 * L2d))) # shift the natural A_2
truth_half <- mean(A2d + L2d + 0.5 * A1d)                       # E[Y] under the regime
truth_half
```

```
[1] 0.6396128
```

The recursion makes the blocked path visible. The first regression, on the full history, again gives $A_1 = 0.4894$ — the direct effect. The second, after substituting the shifted A_2 and regressing on L_1 and A_1 only, gives $A_1 = 0.8937$, close to the $0.5 + 0.4 = 0.9$ of the direct path and the path through L_2: dropping L_2 from the conditioning set lets A_1's indirect effect back in. The final average is 0.627, against a true 0.640 under the stochastic half-off regime. The gap is sampling noise, not misspecification. In this DGP the logit index of A_2 is $0.5 - 0.2A_1 + 0.5L_2$, and substituting $L_2 = 0.25L_1 + 0.4A_1 + \nu_2$ gives $0.5 + 0.125L_1 + 0.5\nu_2$, where ν_2 is the normal error of L_2: the direct path and the path through L_2 cancel. So the mean of the shifted A_2 given (L_1, A_1) does not depend on A_1 and is almost exactly linear in L_1 over $[0, 1]$, and the second regression is close to correctly specified. What is left is the variation of one sample of 10,000 and of the random draws in `d()`. Both numbers sit between the always-treat 2.029 and never-treat 0.129.

Note this recursive process is based on Díaz et al. (2023).

Here is how exactly we estimate it: we start with the last time point. Regress it on previous treatment and confounders, and then get the predicted value with A changed based on the MTP. Regress that predicted value on the previous treatment and confounders, get predicted values with A changed based on MTP. Repeat until the first time point. The average of the predicted value at time 1 is the expected value under this MTP.

The shift function `d()` above is a *stochastic* intervention: at each call it draws a fresh $\epsilon \sim \text{Uniform}(0, 1)$ and, with probability 0.5, switches a treated unit ($a = 1$) to 0, otherwise leaves a unchanged. The regime is "at each time point, draw independently conditional on history and apply this rule"; because the draw is independent across time points and units, calling `d()` separately for A_2 and A_1 is what implements that regime. (By contrast, the `d1`/`d2` functions used with `lmtp_tmle` below are *deterministic* static regimes – always 1 and always 0.)

This is basically g-formula extended to longitudinal data.

In a general case, this is the generalized g-formula to estimate θ:

Set $m_{\tau+1} = Y$, let $A_t^d = d(A_t, H_t)$. For $t = \tau, \ldots, 1$, recursively define:

$$m_t(a_t, h_t) = E[m_{t+1}(A_{t+1}^d, H_{t+1}) | A_t = a_t, H_t = h_t] \tag{35.3}$$

Then $\theta = E[m_1(A_1^d, L_1)]$.

We start from the last period. Regress $m_{\tau+1}$, which is Y on A_τ and H_τ. Then get the predicted value with A_τ changed based on the MTP. Then regress that predicted value on $A_{\tau-1}$ and $H_{\tau-1}$.

Repeat until the first time point. The average of the predicted value at time 1 is the expected value under this MTP.

35.1.4. Estimators

The authors advocate two estimators, TMLE and SDR (sequentially doubly robust estimator). The procedures are the same, starting from the last time point, then apply TMLE or SDR, iterate to the first time point.

35.2. Using lmtp package

```
library(tidyverse)
library(lmtp)
d1 <- function(data, trt) {
  rep(1, nrow(data))
}
A <- "A_2"
Y <- "Y"
W <- c("L_1", "L_2", "A_1")

set.seed(34465)

treat <- lmtp_tmle(
  data = simulated_data1,
  trt = "A_2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d1,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

show_lmtp(treat)
```

```
     Estimate: 1.505
Std. error: 0.009
```

```
d2 <- function(data, trt) {
  rep(0, nrow(data))
}
A <- "A_2"
```

```
Y <- "Y"
W <- c("L_1", "L_2", "A_1")

set.seed(34465)

control <- lmtp_tmle(
  data = simulated_data1,
  trt = "A_2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d2,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

show_lmtp(control)
```

```
     Estimate: 0.499
Std. error: 0.011
```

```
show_lmtp(lmtp_contrast(treat, ref = control))
```

```
  shift   ref estimate std.error conf.low conf.high p.value
1   1.5 0.499     1.01    0.0104    0.986      1.03  <0.001
```

Note that `folds = 1` turns off cross-fitting (sample splitting). That is fine for a small, fast demonstration with `SL.glm`, but in applied work with flexible learners you should use several folds so the nuisance fits are cross-fitted; otherwise the standard errors can be anti-conservative.

Note on the estimand: this `lmtp_tmle` example only shifts A_2 (`trt = "A_2"`), treating A_1 as an observed baseline covariate (via `baseline = W`) rather than also intervening on it. That is a different, single-period policy from the always-1-vs-always-0 *both-period* regime computed manually via `simulated_data1$Yd`/`simulated_data2$Yd` in the `simulate_data()` function above — the two numbers are not expected to match, and should not be used as a sanity check against each other. To replicate the both-period always-treat/never-treat contrast with `lmtp`, shift both time points by passing `trt = c("A_1", "A_2")` and supplying `time_vary` (the time-varying covariates measured before each treatment time, e.g. `list(character(0), "L_2")`), with `shift` functions that map each treatment in the vector to 1 (or 0).

35.3. Another example

The data set bmi, from the DynTxRegime package, are simulated to reflect a two-stage RCT {A1, A2} that studied the effect of meal replacement (MR) shakes versus a calorie deficit (CD) diet on adolescent obesity.

35.3.1. Shift function 1

Consider a shift function that assigns meal replacement to all observations at time 1, but only meal replacement at time 2 to those observations whose 4-month BMI is greater than 30.

```
library(DynTxRegime)
data(bmiData)
bmi <- bmiData
glimpse(bmi)
```

```
Rows: 210
Columns: 8
$ gender      <int> 0, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1~
$ race        <int> 1, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1, 0~
$ parentBMI   <dbl> 31.59683, 30.17564, 30.27918, 27.49256, 26.42350, 29.30970~
$ baselineBMI <dbl> 35.84005, 37.30396, 36.83889, 36.70679, 34.84207, 36.68640~
$ month4BMI   <dbl> 34.22717, 36.38014, 34.42168, 32.52011, 33.72922, 32.06622~
$ month12BMI  <dbl> 34.27263, 36.38401, 34.41447, 32.52397, 33.73546, 32.15977~
$ A1          <chr> "CD", "CD", "MR", "CD", "CD", "MR", "CD", "CD", "CD", "CD"~
$ A2          <chr> "MR", "MR", "CD", "CD", "CD", "MR", "MR", "CD", "MR", "MR"~
```

```
d_dtr <- function(data, trt) {
  if (trt == "A1") return(rep("MR", nrow(data)))

  ifelse(data$month4BMI > 30, "MR", "CD")
}

fit_dtr <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = d_dtr,
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
)
```

```
show_lmtp(fit_dtr)
```

```
     Estimate: 35.854
Std. error: 0.362
```

35.3.2. Shift function 2

Suppose we are interested in comparing the dynamic treatment regime to a static treatment regime where all patients receive meal replacement at both time points. Using the SDR estimator, estimate the effect of this static intervention.

```
fit_MR <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = \(data, trt) rep("MR", nrow(data)),
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
)

show_lmtp(fit_MR)
```

```
     Estimate: 35.831
Std. error: 0.361
```

Let's also estimate the effect of an intervention where all patients receive a calorie deficit diet at both time points.

```
fit_CD <- lmtp_sdr(
  data = bmi,
  trt = c("A1", "A2"),
  outcome = "month12BMI",
  baseline = c("gender", "race", "parentBMI"),
  time_vary = list("baselineBMI", "month4BMI"),
  shift = \(data, trt) rep("CD", nrow(data)),
  outcome_type = "continuous",
  folds = 1,
  learners_trt = "SL.glm",
```

```
  learners_outcome = c("SL.mean", "SL.glm", "SL.gam")
)

show_lmtp(fit_CD)
```

```
     Estimate: 35.053
Std. error: 0.301
```

Finally, we can compare the three treatment regimes using the `lmtp_contrast` function.

```
show_lmtp(lmtp_contrast(fit_dtr, fit_MR, ref = fit_CD))
```

```
  shift  ref estimate std.error conf.low conf.high p.value
1  35.9 35.1    0.801     0.335    0.144      1.46  0.0169
2  35.8 35.1    0.778     0.335    0.122      1.43  0.0201
```

35.4. Summary

A modified treatment policy sets each unit's treatment to a function of its observed value and history: a shift, a cap, or a random switch. That makes it possible to ask about realistic interventions with continuous or time-varying treatments, where static always-treat or never-treat contrasts may violate positivity. The estimand is identified by a generalized g-formula. Starting at the last period, we regress the outcome on the history, predict under the shifted treatment, and repeat backwards. The `lmtp` package estimates it with TMLE or the sequentially doubly robust estimator. Both need consistency, sequential unconfoundedness and positivity for the shifted treatment.

In the simulation, always-treat gives 2.029 and never-treat 0.129. The difference of 1.900 splits into the direct effects of the two treatments and the path through the time-varying confounder. The policy that switches half of the treated units off has a true mean of 0.640; the linear recursion estimates 0.627. The recursion shows the indirect path returning once the confounder is dropped from the conditioning set.

Systematic treatment: R · Julia.

36. Incremental effects for continuous exposures

The LMTP chapter answered questions like "what if every unit's dose were one unit lower?" That shift needs positivity at the shifted dose. A unit whose dose sits near the edge of the support, or just beside a gap in it, is moved to a dose that no comparable unit ever received. Schindl, Shen, and Kennedy (2026) study an intervention that avoids this. Instead of moving each unit's dose, it reweights the distribution each unit's dose is drawn from, so that higher (or lower) doses become more likely. Doses the unit could never receive stay impossible.

For a binary treatment this is the incremental propensity score intervention of Kennedy (2019), which multiplies every unit's odds of treatment by a factor. `npcausal::ipsi()` estimates it, and `lmtp::ipsi()` gives a shift function for binary exposures that scales the probability of treatment by a risk ratio. This chapter is about the continuous version, for which the authors provide replication code but no package. We write the estimator, check it in a simulation, and apply it to the authors' example, political ads and campaign contributions.

36.1. The exponential tilt

We observe covariates X, a dose A and an outcome Y. Let $\pi(a \mid x)$ be the conditional distribution of the dose and $\mu(x, a) = E(Y \mid X = x, A = a)$ the outcome regression. For a number δ, the tilted distribution of the dose is (Schindl, Shen, and Kennedy 2026, eq. 1; the construction is from Díaz and Hejazi 2020)

$$q_\delta(a \mid x) = \frac{e^{\delta a}\, \pi(a \mid x)}{\int e^{\delta t}\, \pi(t \mid x)\, dt}. \tag{36.1}$$

The denominator only makes the result integrate to one. On the log scale, $\log q_\delta(a \mid x) = \log \pi(a \mid x) + \delta a - \text{const}$: we add a line of slope δ to the log density, which tips it toward high doses when $\delta > 0$ and toward low doses when $\delta < 0$. That is why it is called a tilt. It goes back to Esscher (1932) in actuarial work and is the standard device in importance sampling. A normal density with mean m and variance s^2, tilted by δ, becomes a normal with mean $m + \delta s^2$ and the same variance: adding δa to the exponent $-(a-m)^2/(2s^2)$ and completing the square moves the centre to $m + \delta s^2$ and leaves the coefficient on a^2 alone. On a bounded support it cannot shift forever, so the mass piles up against the edge, and as $\delta \to \infty$ all of it ends up at the top dose.

Two features matter for what follows. First, δ is not a dose. It is the slope of the log likelihood ratio between q_δ and π, measured in one over the units of A. The dose change it implies is the change in the conditional mean. The derivative of the tilted mean with respect to δ is the tilted variance, so for small δ the change is about $\delta \operatorname{var}(A \mid X)$. Second, where $\pi(a \mid x) = 0$, so is $q_\delta(a \mid x)$. The tilt never sends anyone to a dose they could not have received.

The estimand is the mean outcome when X keeps its distribution and each unit's dose is drawn from q_δ (Schindl, Shen, and Kennedy 2026, eq. 2):

$$\psi(\delta) = E\left[\int \mu(X, a)\, q_\delta(a \mid X)\, da\right]. \tag{36.2}$$

Under consistency and conditional exchangeability, $\psi(\delta)$ is identified by Equation 36.2, and it needs no positivity, because the integral only uses μ where π is positive. At $\delta = 0$ the tilt does nothing and $\psi(0) = E(Y)$. That gives an exact check on any estimator.

36.2. Tilt or shift

The two interventions differ in what they act on. A shift, the modified treatment policy of Díaz Muñoz and Laan (2012) and Haneuse and Rotnitzky (2013) that `lmtp` estimates, acts on each unit's own dose: it replaces the observed A with $A + s$, or with $A + s$ only when that dose is still feasible. A tilt acts on the distribution the dose is drawn from. Each unit keeps its covariates and receives a new dose drawn from $q_\delta(\cdot \mid X)$, and its observed dose plays no part.

Four differences follow. A shift needs positivity at the shifted dose, $\pi(a + s \mid x) > 0$ wherever $\pi(a \mid x) > 0$; a tilt needs none. A shift moves everyone by s in dose units; a tilt moves the conditional mean by about $\delta \operatorname{var}(A \mid X)$, so units whose dose is more uncertain given their covariates move further. The efficient influence function of a shift weights residuals by the density ratio $\pi(A - s \mid X)/\pi(A \mid X)$ (Díaz et al. 2023), which `lmtp` estimates with a classifier; the tilt weights them by $e^{\delta A}/E(e^{\delta A} \mid X)$. And when the dose is normal given X with variance $\sigma^2(X)$, the tilt by δ is exactly a shift by $\delta\sigma^2(X)$, by the completing-the-square argument above, so with constant variance the two are the same intervention. They part ways when the support is bounded, lumpy or has holes, which is the setting where the shift's positivity condition bites.

36.3. Efficiency, and why the tilt size matters

Write $E_Q(\mu \mid X) = \int \mu(X, a)\, q_\delta(a \mid X)\, da$ for the regression averaged over the tilted distribution. For bounded δ, the efficient influence function is the following. It is Proposition 1 of Schindl, Shen, and Kennedy (2026), who write it as three terms that add up to this form, and it was first derived by Díaz and Hejazi (2020).

$$\varphi(Z; \delta) = \frac{q_\delta(A \mid X)}{\pi(A \mid X)}\{Y - E_Q(\mu \mid X)\} + E_Q(\mu \mid X) - \psi(\delta), \tag{36.3}$$

and the one-step estimator replaces the nuisances by cross-fitted estimates and averages $\hat{\varphi} + \hat{\psi}$ over the sample. The likelihood ratio in Equation 36.3 has a simple form, $q_\delta(A \mid X)/\pi(A \mid X) = e^{\delta A}/\int e^{\delta t}\pi(t \mid X)\, dt$, by Equation 36.1, so we never need the density at the observed dose, only integrals against it.

That ratio grows with δ. If π is uniform on $[0, 1]$, it equals $\delta e^{\delta a}/(e^\delta - 1)$, which is at least δ at $a = 1$ (Schindl, Shen, and Kennedy 2026, sec. 4.1). Schindl, Shen, and Kennedy (2026) show that, when the dose density is bounded below on an interval at the top of the support (they call this

weak positivity), the variance of φ is bounded above and below by multiples of δ (their Lemma 1). A two-point argument then gives a minimax lower bound of order $\sqrt{\delta/n}$ for any estimator (their Theorem 2), and the cross-fitted one-step estimator attains it (their Theorems 3 and 4). The effective sample size is n/δ. Without weak positivity, for example when the density goes to zero at the top edge, the variance can grow like any power of δ (their Section 10). As $\delta \to \infty$ in both directions around an interior dose a', a "reflected" tilt also gives an estimate of the dose-response curve $E(Y^{a'})$, with δ playing the role of an inverse bandwidth. We come back to that at the end.

36.4. Implementing it

The paper estimates $\pi(a \mid x)$ on a grid of design points a_j by regressing the kernel-transformed outcome $K_h(A - a_j)$ on X (their Section 5.1), since $E\{K_h(A-a) \mid X = x\}$ is close to $\pi(a \mid x)$ for small h. The integrals in Equation 36.1 and $E_Q(\mu \mid X)$ are then weighted sums over the grid. We keep that approach and change four things.

The first is the bandwidth. The authors' code chooses h by the cross-validated RMSE of the regression of $K_h(A - a_j)$ on X. The variance of $K_h(A-a)$ is of order $1/h$, so the criterion is not comparable across bandwidths. Run on a subsample of the application data, it picks an end of its grid at every design point: the largest bandwidth wherever doses are nearby, and the smallest wherever the design point falls between the discrete dose values, since there a narrow kernel sees no data and its target is close to zero. We fix h by Silverman's rule. The second is normalisation. Their $\hat{\pi}$ for $\psi(\delta)$ is not rescaled to integrate to one, so $\hat{q}_\delta/\hat{\pi}$ is off by a factor $1/\int \hat{\pi}$, and $\hat{\psi}(0)$ need not equal $\bar{Y}$. We renormalise each unit's density. The third is the edge of the support. A Gaussian kernel loses about half its mass at the boundary, which is exactly where the tilt puts its weight, so we reflect the kernel at both ends. The fourth is specific to the application: 12% of the ZIP codes there have no ads at all, so many units share the dose $A = 0$ exactly and no smooth density describes them. The tilt in Equation 36.1 still works: the zero dose keeps weight $e^0 = 1$, so $q_\delta/\pi = e^{\delta A}/\nu(X)$ with $\nu(x) = E(e^{\delta A} \mid X = x)$, also at $A = 0$. We estimate $P(A = 0 \mid X)$ with a classification forest and fit the kernel to the positive doses only. The influence function keeps the form of Equation 36.3, since the derivation behind Proposition 1 uses only that q_δ/π is a likelihood ratio, which it still is when some units have dose exactly zero. The simulation below supports this: with the true dose distribution the estimator is unbiased with 95% coverage. The paper instead smooths the zeros into the density.

The paper also mentions a version with no density at all (their Remark 3). Since $E_Q(\mu \mid X) = \eta(X)/\nu(X)$ with $\eta(x) = E\{e^{\delta A}\mu(X, A) \mid X = x\}$, both pieces are regressions on X. The estimated influence value is $e^{\delta A}(Y - \hat{\eta}/\hat{\nu})/\hat{\nu} + \hat{\eta}/\hat{\nu}$. Its expectation given X is $\eta/\hat{\nu} - \nu\hat{\eta}/\hat{\nu}^2 + \hat{\eta}/\hat{\nu}$, and subtracting η/ν leaves a bias of $(\hat{\nu} - \nu)(\hat{\eta}/\hat{\nu} - \eta/\nu)/\hat{\nu}$, a product of errors, so it is as doubly robust as the density version. The authors report that it is unstable, and we include it in the simulation to see.

All four estimators below share the outcome forest and the cross-fitting. `tilt_grid()` takes a function that returns the conditional dose distribution on a grid, so the same code runs with ours, with the paper's, and with the true distribution in the simulation.

```
library(ranger)

# ---- Our estimator: tilts through regressions on X -------------------------
# With nu(x) = E[exp(delta A) | X = x] and eta(x) = E[exp(delta A) mu(X, A) | X = x],
```

```
# the tilted mean of mu is eta / nu and q / pi at the observed dose is
# exp(delta A) / nu. Both are regressions on X, so no conditional density is
# needed and an atom in the dose distribution (here, A = 0) is handled as it is.
# The influence value is (q / pi)(Y - eta / nu) + eta / nu. eta is fitted to
# exp(delta A) mu_hat(X, A), using out-of-bag predictions of mu on the training
# fold. Exponentials are scaled by exp(-delta c) with c = max(A) for delta > 0,
# which cancels in every ratio. With shift = TRUE we also return the implied
# change in the mean dose, E_Q(A | X) - E(A | X), averaged over units.
tilt_regression <- function(y, a, x, deltas, folds, shift = FALSE, cluster = NULL,
                            num.trees = 200, threads = 1) {
  x  <- as.data.frame(x)
  n  <- length(y)
  phi <- matrix(NA_real_, n, length(deltas))
  dA  <- matrix(NA_real_, n, length(deltas))
  reg <- function(target, tr, va) {
    f <- ranger(t ~ ., data = data.frame(x[tr, , drop = FALSE], t = target),
                num.trees = num.trees, num.threads = threads)
    predict(f, data = x[va, , drop = FALSE], num.threads = threads)$predictions
  }
  for (k in sort(unique(folds))) {
    va <- which(folds == k); tr <- which(folds != k)
    fmu <- ranger(y ~ ., data = data.frame(x[tr, , drop = FALSE], a = a[tr], y = y[tr]),
                  num.trees = 500, num.threads = threads)
    mu_oob <- fmu$predictions
    if (shift) ea0 <- reg(a[tr], tr, va)
    for (j in seq_along(deltas)) {
      d   <- deltas[j]
      c0  <- if (d > 0) max(a) else 0
      t   <- exp(d * (a[tr] - c0))
      nu  <- reg(t, tr, va)
      eq  <- reg(t * mu_oob, tr, va) / nu
      qp  <- exp(d * (a[va] - c0)) / nu
      phi[va, j] <- qp * (y[va] - eq) + eq
      if (shift) dA[va, j] <- reg(t * a[tr], tr, va) / nu - ea0
    }
  }
  out <- data.frame(delta = deltas, est = colMeans(phi), se = if_se(phi))
  if (!is.null(cluster)) out$se_cluster <- if_se(phi, cluster)
  if (shift) out$mean_shift <- colMeans(dA)
  out
}

# ---- Conditional law of the dose on a grid ----------------------------------
# pi(a | x) at each of D design points a_j is the regression on X of the
# kernel-transformed outcome K_h(A - a_j), as in Section 5.1 of the paper, with
# two changes to the authors' code: h is fixed by Silverman's rule (their CV on
# the RMSE of K_h always picks the largest h) and each unit's density is
```

```
# renormalised to integrate to one. With atom = TRUE (our default) the zeros
# get their own probability p0(x) from a classification forest and the kernel
# is fitted to the positive doses only; with reflect = TRUE the kernel is
# reflected at the ends of the support so it does not lose mass there. The
# paper's version is atom = FALSE, reflect = FALSE.
fit_kernel_law <- function(a_tr, x_tr, x_va, atom = TRUE, reflect = TRUE, D = 50,
                           num.trees = 100, threads = 1) {
  p0 <- rep(0, nrow(x_va))
  if (atom) {
    d0 <- data.frame(x_tr, zero = factor(a_tr == 0, levels = c(FALSE, TRUE)))
    f0 <- ranger(zero ~ ., data = d0, probability = TRUE,
                 num.trees = 300, num.threads = threads)
    p0 <- predict(f0, data = x_va, num.threads = threads)$predictions[, "TRUE"]
    x_tr <- x_tr[a_tr > 0, , drop = FALSE]; a_tr <- a_tr[a_tr > 0]
  }
  L <- min(a_tr); U <- max(a_tr); h <- bw.nrd0(a_tr)
  ag <- L + (U - L) * (seq_len(D) - 0.5) / D
  Fh <- sapply(ag, function(g) {
    k <- dnorm((a_tr - g) / h)
    if (reflect) k <- k + dnorm((2 * L - a_tr - g) / h) + dnorm((2 * U - a_tr - g) / h)
    f <- ranger(k ~ ., data = data.frame(x_tr, k = k / h),
                num.trees = num.trees, num.threads = threads)
    predict(f, data = x_va, num.threads = threads)$predictions
  })
  list(p0 = p0, agrid = matrix(ag, nrow(x_va), D, byrow = TRUE),
       W = Fh / rowSums(Fh), da = (U - L) / D)
}

mu_on_grid <- function(fmu, x_va, agrid, threads = 1) {
  G  <- ncol(agrid)
  xx <- x_va[rep(seq_len(nrow(x_va)), times = G), , drop = FALSE]
  pr <- predict(fmu, data = data.frame(xx, a = as.vector(agrid)),
                num.threads = threads)$predictions
  matrix(pr, nrow(x_va), G)
}

# Grid estimator for a given conditional law: law(a_tr, x_tr, x_va) returns an
# atom probability p0 at A = 0, a dose grid, weights summing to one per unit and
# the cell width da. Used with fit_kernel_law and, in the simulation, the truth.
tilt_grid <- function(y, a, x, deltas, folds, law, cluster = NULL, threads = 1) {
  x <- as.data.frame(x)
  n <- length(y)
  phi <- matrix(NA_real_, n, length(deltas))
  dA  <- matrix(NA_real_, n, length(deltas))
  for (k in sort(unique(folds))) {
    va <- which(folds == k); tr <- which(folds != k)
    fmu <- ranger(y ~ ., data = data.frame(x[tr, , drop = FALSE], a = a[tr], y = y[tr]),
```

```
                  num.trees = 500, num.threads = threads)
    f    <- law(a[tr], x[tr, , drop = FALSE], x[va, , drop = FALSE])
    ea0  <- (1 - f$p0) * rowSums(f$W * f$agrid)              # E(A | X)
    mu_g <- mu_on_grid(fmu, x[va, , drop = FALSE], f$agrid, threads)
    mu_0 <- predict(fmu, data = data.frame(x[va, , drop = FALSE], a = 0),
                    num.threads = threads)$predictions
    for (j in seq_along(deltas)) {
      d  <- deltas[j]
      c0 <- if (d > 0) max(a) else 0
      # exact integral of exp(delta a) over each grid cell of width da
      cf <- if (d == 0) 1 else sinh(d * f$da / 2) / (d * f$da / 2)
      ew <- cf * f$W * exp(d * (f$agrid - c0))
      nu <- f$p0 * exp(-d * c0) + (1 - f$p0) * rowSums(ew)
      eq <- (f$p0 * exp(-d * c0) * mu_0 + (1 - f$p0) * rowSums(ew * mu_g)) / nu
      qp <- exp(d * (a[va] - c0)) / nu
      phi[va, j] <- qp * (y[va] - eq) + eq
      dA[va, j]  <- (1 - f$p0) * rowSums(ew * f$agrid) / nu - ea0
    }
  }
  out <- data.frame(delta = deltas, est = colMeans(phi), se = if_se(phi),
                    mean_shift = colMeans(dA))
  if (!is.null(cluster)) out$se_cluster <- if_se(phi, cluster)
  out
}

# Standard error of a mean of influence values, one column per estimand:
# iid by default, or clustered when units share a cluster (here, a market).
# With a cluster, the estimators return both.
if_se <- function(phi, cluster = NULL) {
  n <- nrow(phi)
  if (is.null(cluster)) return(apply(phi, 2, sd) / sqrt(n))
  ctr <- sweep(phi, 2, colMeans(phi))
  sqrt(colSums(rowsum(ctr, cluster)^2)) / n
}

# Reflected tilt at interior doses a1 (dose-response): tilt up below a1 and
# down above it, with side weights omega = P(A <= a1 | X) and 1 - omega. The
# atom at 0 lies below every a1 > 0. The influence value is (r / pi)(Y - U) + U,
# where U is the tilted mean of mu on the side of a1 where the unit's dose lies.
reflect_grid <- function(y, a, x, a1s, delta, folds, law, cluster = NULL,
                         threads = 1) {
  x <- as.data.frame(x)
  n <- length(y)
  phi <- matrix(NA_real_, n, length(a1s))
  cf  <- function(da) sinh(delta * da / 2) / (delta * da / 2)
  for (k in sort(unique(folds))) {
    va <- which(folds == k); tr <- which(folds != k)
```

```
    fmu <- ranger(y ~ ., data = data.frame(x[tr, , drop = FALSE], a = a[tr], y = y[tr]),
                  num.trees = 500, num.threads = threads)
    f    <- law(a[tr], x[tr, , drop = FALSE], x[va, , drop = FALSE])
    mu_g <- mu_on_grid(fmu, x[va, , drop = FALSE], f$agrid, threads)
    mu_0 <- predict(fmu, data = data.frame(x[va, , drop = FALSE], a = 0),
                    num.threads = threads)$predictions
    for (j in seq_along(a1s)) {
      a1  <- a1s[j]
      lo  <- f$agrid <= a1
      elo <- cf(f$da) * f$W * exp(delta * (f$agrid - a1)) * lo     # at most 1
      ehi <- cf(f$da) * f$W * exp(-delta * (f$agrid - a1)) * !lo   # at most 1
      p0s <- f$p0 * exp(-delta * a1)
      D1  <- p0s + (1 - f$p0) * rowSums(elo)
      D2  <- (1 - f$p0) * rowSums(ehi)
      om  <- f$p0 + (1 - f$p0) * rowSums(f$W * lo)
      m1  <- (p0s * mu_0 + (1 - f$p0) * rowSums(elo * mu_g)) / D1
      m2  <- ((1 - f$p0) * rowSums(ehi * mu_g)) / D2
      low <- a[va] <= a1
      rp  <- ifelse(low, om * exp(delta * (a[va] - a1)) / D1,
                         (1 - om) * exp(-delta * (a[va] - a1)) / D2)
      U   <- ifelse(low, m1, m2)
      phi[va, j] <- rp * (y[va] - U) + U
    }
  }
  out <- data.frame(a1 = a1s, delta = delta, est = colMeans(phi), se = if_se(phi))
  if (!is.null(cluster)) out$se_cluster <- if_se(phi, cluster)
  out
}
```

36.5. Simulation

We check two things: whether the one-step estimator is right when the dose distribution is known, and what estimating it costs. The covariates X_1 and X_2 are independent uniforms on $(0, 1)$. With probability $\text{expit}(-1.5 + X_1)$ the dose is zero. Otherwise it is drawn from a density on $(0, 1)$ that depends on X_2 and is at least 0.5 everywhere, so weak positivity holds at both edges. The outcome is $Y = 1 + X_1 + X_2 + 2A + \sin(3A) + \varepsilon$ with standard normal ε, and the true $\psi(\delta)$ comes from numerical integration.

All estimators are the cross-fitted one-step estimator of Equation 36.3 with the same outcome forest. They differ only in the dose distribution plugged into it: the known one that generated the data (the oracle), which no one has in practice; ours, which models the zeros separately; and the paper's, which smooths the zeros into the density. We use $n = 1000$ with 200 replications and $n = 4000$ with 100, and tilts from -10 to 25. The run takes about two hours on six cores, so the code reads saved results.

```r
mu_true <- function(x1, x2, a) 1 + x1 + x2 + 2 * a + sin(3 * a)
p0_true <- function(x1) plogis(-1.5 + x1)
# density of A given A > 0: half Uniform(0, 1), half Beta(1 + 2 x2, 1), so it is
# at least 0.5 on all of [0, 1] (weak positivity at both edges)
f_pos <- function(a, x2) 0.5 + 0.5 * dbeta(a, 1 + 2 * x2, 1)

sim_data <- function(n) {
  x1 <- runif(n); x2 <- runif(n)
  zero <- runif(n) < p0_true(x1)
  pos <- ifelse(runif(n) < 0.5, runif(n), rbeta(n, 1 + 2 * x2, 1))
  a <- ifelse(zero, 0, pos)
  y <- mu_true(x1, x2, a) + rnorm(n)
  list(y = y, a = a, x = data.frame(x1 = x1, x2 = x2))
}

# True psi(delta) by quadrature: midpoint grids over (x1, x2) and over a.
true_psi <- function(deltas, nx = 100, na = 2000) {
  g  <- (seq_len(nx) - 0.5) / nx
  ag <- (seq_len(na) - 0.5) / na
  X  <- expand.grid(x1 = g, x2 = g)
  sapply(deltas, function(d) {
    p0 <- p0_true(X$x1)
    num <- den <- numeric(nrow(X))
    c0 <- if (d > 0) 1 else 0                          # scale exp(delta a) by exp(-delta c0)
    for (j in seq_len(nrow(X))) {
      f  <- f_pos(ag, X$x2[j]) / na
      ew <- f * exp(d * (ag - c0))
      den[j] <- sum(ew)
      num[j] <- sum(ew * mu_true(X$x1[j], X$x2[j], ag))
    }
    s  <- exp(-d * c0)                                 # same scaling for the atom
    mean((p0 * s * mu_true(X$x1, X$x2, 0) + (1 - p0) * num) /
         (p0 * s + (1 - p0) * den))
  })
}

oracle_law <- function(a_tr, x_tr, x_va, G = 200) {
  ag <- (seq_len(G) - 0.5) / G
  W  <- t(sapply(x_va$x2, function(v) { f <- f_pos(ag, v); f / sum(f) }))
  list(p0 = p0_true(x_va$x1), agrid = matrix(ag, nrow(x_va), G, byrow = TRUE), W = W,
       da = 1 / G)
}

one_rep <- function(n, deltas) {
  d <- sim_data(n)
  folds <- sample(rep(1:5, length.out = n))
  out <- rbind(
```

```
    data.frame(estimator = "regression",
               tilt_regression(d$y, d$a, d$x, deltas, folds)),
    data.frame(estimator = "mixed",
               tilt_grid(d$y, d$a, d$x, deltas, folds, law = fit_kernel_law)),
    data.frame(estimator = "paper",
               tilt_grid(d$y, d$a, d$x, deltas, folds, law = function(...)
                 fit_kernel_law(..., atom = FALSE, reflect = FALSE))),
    data.frame(estimator = "oracle",
               tilt_grid(d$y, d$a, d$x, deltas, folds, law = oracle_law)))
  stopifnot(all(abs(out$est[out$delta == 0] - mean(d$y)) < 1e-8))
  out$n <- n
  out
}
```

```
library(parallel)
# About 40 minutes for n = 1000 and 95 for n = 4000 on six cores, so the chunk
# reads the saved results when they exist.
if (!file.exists("incremental-effects-sim.rds")) {
  RNGkind("L'Ecuyer-CMRG"); set.seed(20261005)
  deltas <- c(-10, -5, -2, 0, 2, 5, 10, 25)
  truth  <- data.frame(delta = deltas, truth = true_psi(deltas))
  res <- list()
  for (n in c(1000, 4000)) {
    reps <- if (n == 1000) 200 else 100
    t0 <- Sys.time()
    out <- mclapply(seq_len(reps), function(r) {
      o <- try(one_rep(n, deltas)); if (!inherits(o, "try-error")) o$rep <- r; o
    }, mc.cores = 6, mc.set.seed = TRUE)
    bad <- vapply(out, inherits, logical(1), "try-error")
    if (any(bad)) message("n=", n, ": ", sum(bad), " failed reps: ", out[bad][[1]])
    res[[as.character(n)]] <- do.call(rbind, out[!bad])
    message("n=", n, " done in ", format(Sys.time() - t0))
    saveRDS(list(truth = truth, sims = do.call(rbind, res)), "incremental-effects-sim.rds")
  }
}
sim <- readRDS("incremental-effects-sim.rds")
```

Table 36.1.: Bias and coverage of nominal 95% intervals for $\psi(\delta)$. All three estimators use Equation 36.3 with the same outcome forest and differ only in the dose distribution π: the known one used to generate the data (the oracle), ours (zeros modelled separately), or the paper's (zeros smoothed into the density).

n	δ	Known π: bias	Known π: cover	Ours: bias	Ours: cover	Paper's: bias	Paper's: cover
1000	-5	-0.004	0.96	-0.116	0.68	-0.514	0.00
1000	-2	-0.002	0.93	-0.070	0.76	-0.158	0.34

Table 36.1.: Bias and coverage of nominal 95% intervals for $\psi(\delta)$. All three estimators use Equation 36.3 with the same outcome forest and differ only in the dose distribution π: the known one used to generate the data (the oracle), ours (zeros modelled separately), or the paper's (zeros smoothed into the density).

n	δ	Known π: bias	Known π: cover	Ours: bias	Ours: cover	Paper's: bias	Paper's: cover
1000	2	0.000	0.96	0.055	0.84	0.038	0.90
1000	10	0.002	0.96	0.046	0.96	0.053	0.96
4000	-5	0.000	0.94	-0.088	0.30	-0.423	0.00
4000	-2	0.001	0.95	-0.057	0.49	-0.141	0.00
4000	2	-0.001	0.94	0.044	0.59	0.032	0.65
4000	10	-0.003	0.94	0.004	0.96	0.010	0.94

With the true dose distribution the estimator is unbiased and covers at about 95% at every tilt, so Equation 36.3 and the one-step estimator are right. With our estimated distribution the bias at $\delta = -5$ falls only from -0.116 to -0.088 when n quadruples, while the standard deviation halves, so coverage falls from 0.68 to 0.3. The reason is that the remainder contains the squared error of the density estimate (Schindl, Shen, and Kennedy 2026, Theorem 3), so a good outcome model does not protect us and the density has to be estimated well at every dose. Smoothing the zeros into the density, as the paper's code does, is worse: at negative tilts its coverage is close to zero at both sample sizes. The density-free regression version of the previous section, which we also ran, was unstable at large positive tilts, as the authors reported.

36.6. Political ads and campaign contributions

The application follows Urban and Niebler (2014), with data from Fong, Hazlett, and Imai (2018) as distributed in the authors' repository. The units are 16,265 ZIP codes in states that were not competitive in the 2008 presidential election. Media markets cross state lines, so some of these ZIP codes received ads aimed at a neighbouring competitive state. The dose is $\log(\text{ads} + 1)$. The outcome is the ZIP code's total contributions to the two presidential campaigns, C, in thousands of dollars, entered as $\log(C + 1)$. Thirty-six percent of ZIP codes gave nothing and the median gave $750. It is a total, not an amount per person, which is one reason log population is among the eight covariates, which are those of the paper. Two facts about the dose shape the analysis. Twelve percent of ZIP codes have no ads, and none has between about 3.6 and 5.0 log ads. And ads are bought by media market, so every ZIP code in one of the 138 markets has the same dose. The dose takes only 73 distinct values. The paper treats the ZIP codes as independent. We cross-fit with folds made of whole markets and report standard errors clustered by market next to the iid ones.

```
e <- new.env(); load("FECForCEM.RData", envir = e); fec <- e$x
y <- log(fec$Cont + 1)
a <- log(fec$TotAds + 1)
X <- model.matrix(~ -1 + log(TotalPop) + PercentOver65 + log(Inc + 1) + PercentHispanic +
                    PercentBlack + density + per_collegegrads + CanCommute, data = fec)
colnames(X) <- make.names(colnames(X))
```

```
market <- fec$DMACode
set.seed(20261005)
mk <- unique(market)
folds <- sample(rep(1:5, length.out = length(mk)))[match(market, mk)]
if (!file.exists("incremental-effects-app.rds")) {
  deltas <- seq(-2.5, 5, by = 0.05)
  mixed <- tilt_grid(y, a, X, deltas, folds, cluster = market, threads = 6,
                     law = function(...) fit_kernel_law(..., D = 100, threads = 6))
  paper <- tilt_grid(y, a, X, deltas, folds, cluster = market, threads = 6,
                     law = function(...) fit_kernel_law(..., atom = FALSE, reflect = FALSE,
                                                        D = 100, threads = 6))
  stopifnot(abs(mixed$est[mixed$delta == 0] - mean(y)) < 1e-8,
            abs(paper$est[paper$delta == 0] - mean(y)) < 1e-8)
  apos <- a[a > 0]
  delta_dr <- length(y)^(1/3) / diff(range(apos))
  a1s <- c(1, 2, 3, 5.5, 6.5, 7.5, 8.5, 9.5)
  dr <- reflect_grid(y, a, X, a1s, delta_dr, folds, cluster = market, threads = 6,
                     law = function(...) fit_kernel_law(..., D = 100, threads = 6))
  saveRDS(list(mixed = mixed, paper = paper, dr = dr, delta_dr = delta_dr, ybar = mean(y)),
          "incremental-effects-app.rds")
}
app <- readRDS("incremental-effects-app.rds")
```

The check at zero holds: $\hat{\psi}(0) = 1.2431$, the sample mean of log contributions. The first thing the bottom panel shows is how strong these tilts are. The dose has variance 10.8 across ZIP codes, so $\delta = 0.05$ already moves the mean dose by 0.48 log ads, and the paper's largest tilt, $\delta = 5$, moves it by 5.51, which puts nearly every ZIP code at the largest ad counts. The range of δ the paper plots, from -2.5 to 5, runs from almost no ads to the maximum.

The estimated curve rises slightly from 1.243 to a peak of 1.25 near $\delta = 0.15$, then falls to 0.745 at $\delta = 5$. Negative tilts lower it to 1.045 at $\delta = -2.5$. For positive tilts the two versions agree, because the zeros get little weight there. For negative tilts the paper's version gives lower values, 0.758 at $\delta = -2.5$, which is the same direction as its bias in the simulation.

The standard errors clustered by market are about ten times the iid ones: 0.12 against 0.012 at $\delta = 0$. Ads vary across 138 markets, and the iid errors treat 16,265 ZIP codes as independent doses. With intervals of about ± 0.24, moderate tilts cannot be told apart from no tilt, and the simulation warns that even these intervals are too narrow at moderate δ. We read the curve as descriptive.

This does not match the paper's application. It describes its results as average individual contributions rising from about \$2.00 at $\delta = 0$ to \$2.20 at $\delta = 5$, with most of the gain below $\delta = 1$. The outcome is neither individual nor in dollars: it is the log of ZIP-level totals in thousands of dollars. Undoing the log at $\delta = 0$ gives $e^{\hat{\psi}(0)} - 1 = 2.47$ thousand dollars per ZIP code, and that is a geometric-type average of totals, not a mean contribution. The missing normalisation of $\hat{\pi}$ in their code does not account for the difference either: run on a subsample of the data, their code puts $\hat{\psi}(0)$ within 0.01 of the sample mean. We could not find how the dollar figures were computed.

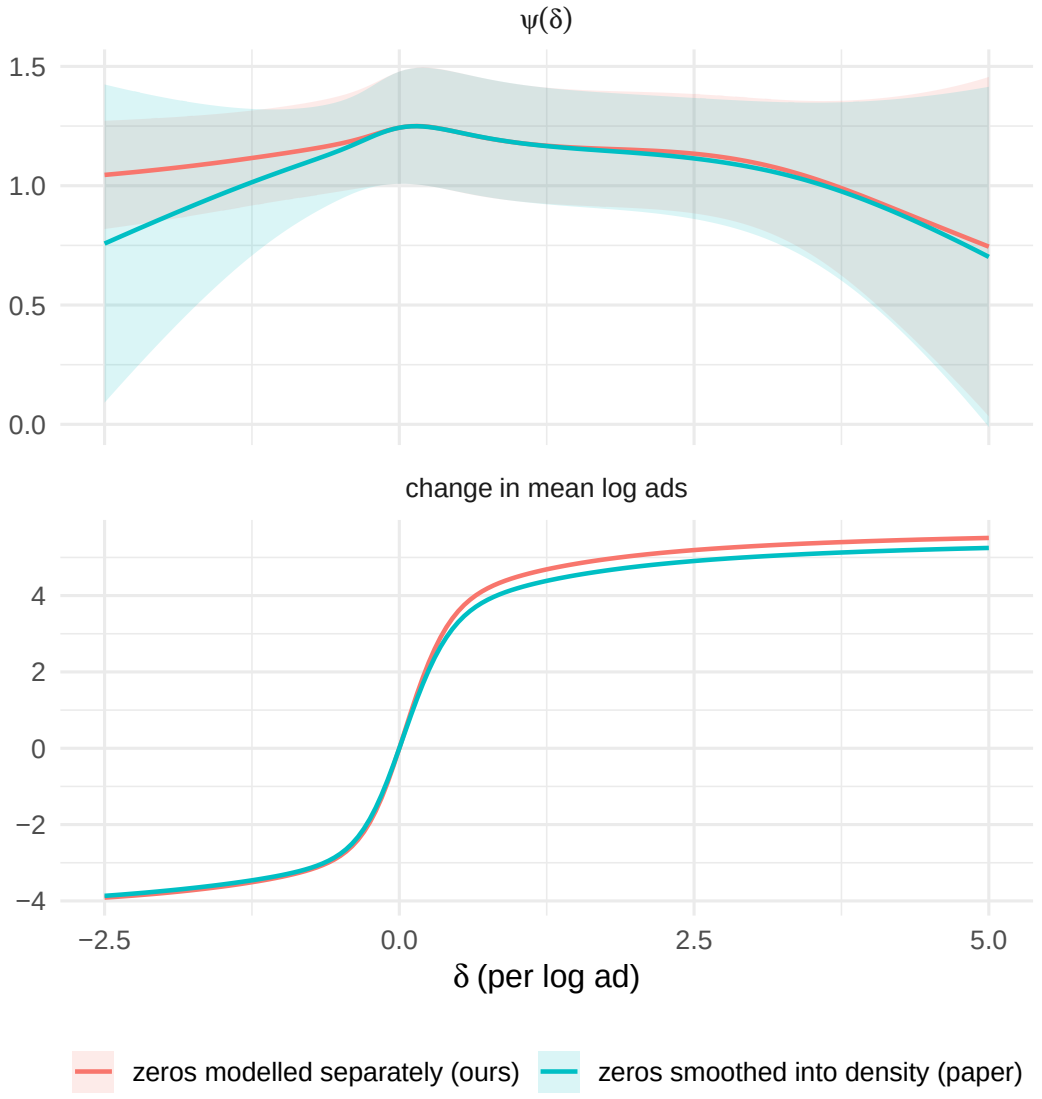


Figure 36.1.: Estimated incremental effect of political ads on log contributions (top) and the implied change in the mean dose (bottom), with the zeros modelled separately (ours) and smoothed into the density (the paper's). Bands are 95% intervals clustered by media market.

36.7. Shift and tilt on the same ZIP codes

We now estimate shifts with `lmtp` on the same data: the same covariates, a ranger learner for both nuisances, cross-fitting and standard errors by market. Each unit's dose moves by $s = \pm 0.5$ log ads when the shifted dose stays inside the observed range and is left alone otherwise. For each shift we take the tilt whose implied change in the mean dose is the same, read off the fine grid of δ above, so the two interventions move the average dose by the same amount.

```
# Shift interventions on the same ZIP codes with lmtp: move each unit's dose by
# s log ads when the shifted dose stays inside the observed range, and leave it
# unchanged otherwise. Same covariates, a ranger learner for both nuisances,
# cross-fitting and standard errors by media market.
library(lmtp); library(SuperLearner)
SL.ranger6 <- function(...) SL.ranger(..., num.trees = 500, num.threads = 6)
feasible_shift <- function(s, lo, hi) function(data, trt) {
  a <- data[[trt]]
  ifelse(a + s >= lo & a + s <= hi, a + s, a)
}
fec_df <- data.frame(X, a = a, y = y, market = market)
if (!file.exists("incremental-effects-lmtp.rds")) {
  shifts <- c(-0.5, 0.5)
  lmtp_fits <- lapply(shifts, function(s) {
    f <- lmtp_sdr(fec_df, trt = "a", outcome = "y", baseline = colnames(X),
                  shift = feasible_shift(s, min(a), max(a)), mtp = TRUE,
                  outcome_type = "continuous", id = "market",
                  learners_outcome = "SL.ranger6", learners_trt = "SL.ranger6", folds = 5)
    d <- feasible_shift(s, min(a), max(a))(fec_df, "a")
    data.frame(s = s, est = f$estimate@x, se = f$estimate@std_error,
               mean_shift = mean(d - a), share_moved = mean(d != a))
  })
  lmtp_res <- do.call(rbind, lmtp_fits)
  saveRDS(lmtp_res, "incremental-effects-lmtp.rds")
}
lmtp_res <- readRDS("incremental-effects-lmtp.rds")
```

Table 36.2.: Shift (lmtp) and tilt with the same change in the mean dose. Standard errors clustered by market; the tilt estimate is interpolated on the grid of δ.

s	Share moved	Mean change in dose	Shift estimate	SE	Matched δ	Tilt estimate	SE
-0.5	0.879	-0.439	1.241	0.202	-0.047	1.238	0.118
0.5	0.975	0.488	1.251	0.207	0.051	1.247	0.122

For a change of half a log ad in the average dose, the two estimates are almost the same: 1.251 for the shift and 1.247 for the tilt when the mean dose rises, and 1.241 against 1.238 when it falls. Both

are within a few thousandths of $\bar{Y} = 1.243$, and the differences are far inside the standard errors. Small changes in ads move contributions very little under either intervention.

They differ in who moves. The shift moves everyone it can by the same amount: 97.5% of ZIP codes for the upward shift, and 87.9% for the downward one, since the ZIP codes with no ads cannot go lower. The tilt with the same mean moves units in proportion to the spread of their dose given covariates and never sends anyone to a dose they could not receive. The shift also sends every ZIP code it moves to an ad count that no market had. The dose takes only 73 values, and none of the shifted doses, 14,289 downward and 15,859 upward, equals one of them. Staying inside the observed range therefore does not make a shifted dose supported. The data say nothing directly about outcomes at those doses, and `lmtp`'s outcome model and density-ratio classifier fill the gaps by smoothing between the observed doses. The tilt, by contrast, only reweights doses that occur. Its standard errors here are larger, 0.207 against 0.122.

So the two estimates are close, but they do not rest on the same evidence. The tilt estimate is identified from the doses the markets received. The shift estimate depends on how the models extrapolate to doses no market received, so the comparison is a check of that extrapolation and not a second identified estimate that agrees with the first. Under both, a small change in ads moves contributions very little. The choice of intervention matters for what the estimate means and for what the data must support. A shift asks what would happen if every ZIP code saw half a log ad more. It is stated in dose units, and it needs the shifted doses to be ones that comparable ZIP codes actually received. A tilt asks what would happen if higher ad counts became more likely. It never needs an impossible dose, but its δ has to be translated into a change in the dose, as the bottom panel of Figure 36.1 does. We would use a shift when the shifted doses are plausible, and a tilt when the edges of the support, gaps in it, or a pile of zeros rule shifts out. Either way the intervals depend on estimating the dose distribution well. Our simulation shows this for the tilt; we have not checked it for `lmtp`.

36.8. Dose-response by a reflected tilt

To reach an interior dose a', tilt upward below a' and downward above it. With $\omega(x) = P(A \le a' \mid X = x)$, the reflected tilt is the following [Schindl, Shen, and Kennedy (2026), Section 6; Theorem 5 gives its influence function]

$$r_\delta(a \mid x) = \omega(x)\,\frac{e^{\delta a}\pi(a \mid x)\,1(a \le a')}{\int_{t \le a'} e^{\delta t}\pi(t \mid x)\,dt} + \{1 - \omega(x)\}\,\frac{e^{-\delta a}\pi(a \mid x)\,1(a > a')}{\int_{t > a'} e^{-\delta t}\pi(t \mid x)\,dt}, \tag{36.4}$$

and $\psi_R(\delta, a')$ averages μ over it. As $\delta \to \infty$ it converges to $E(Y^{a'})$. If μ is Lipschitz in the dose, the estimate is within $O_P(1/\delta + \sqrt{\delta/n})$ of $E(Y^{a'})$ (their Theorem 6), so δ of order $n^{1/3}$ gives the usual $n^{-1/3}$ rate. The two sides of a' get the weights ω and $1 - \omega$, which makes the tilt an asymmetric exponential kernel. Its bias is first order in $1/\delta$, which is why the result stops at Lipschitz smoothness. The rate is stated for a dose on $[0, 1]$, so we set $\delta = n^{1/3}$ on that scale, which is $\delta = 2.72$ per log ad here, and evaluate the curve away from the hole, where weak positivity fails.

Two qualifications apply to the figure below. First, the dose takes only 73 values, and none of the eight targets a' is one of them; the nearest observed doses are 0.008 to 0.099 log ads away. Theorem 6 assumes a positive conditional density of the dose near a', which a discrete dose does not have. As

δ grows, the tilt applied to the true dose distribution piles onto the observed doses next to a', not onto a' itself. Our estimated dose density is smooth, so it puts mass between the observed doses, but the data cannot tell us the outcome there. Second, the $n^{1/3}$ rule is derived for independent units. We use $n = 16{,}265$ ZIP codes, but the dose is set by 138 markets, and with $n = 138$ the same rule gives a δ about five times smaller. We therefore read the curve as a smoothed, model-dependent description of how contributions vary with ads, not as estimates of $E(Y^{a'})$.

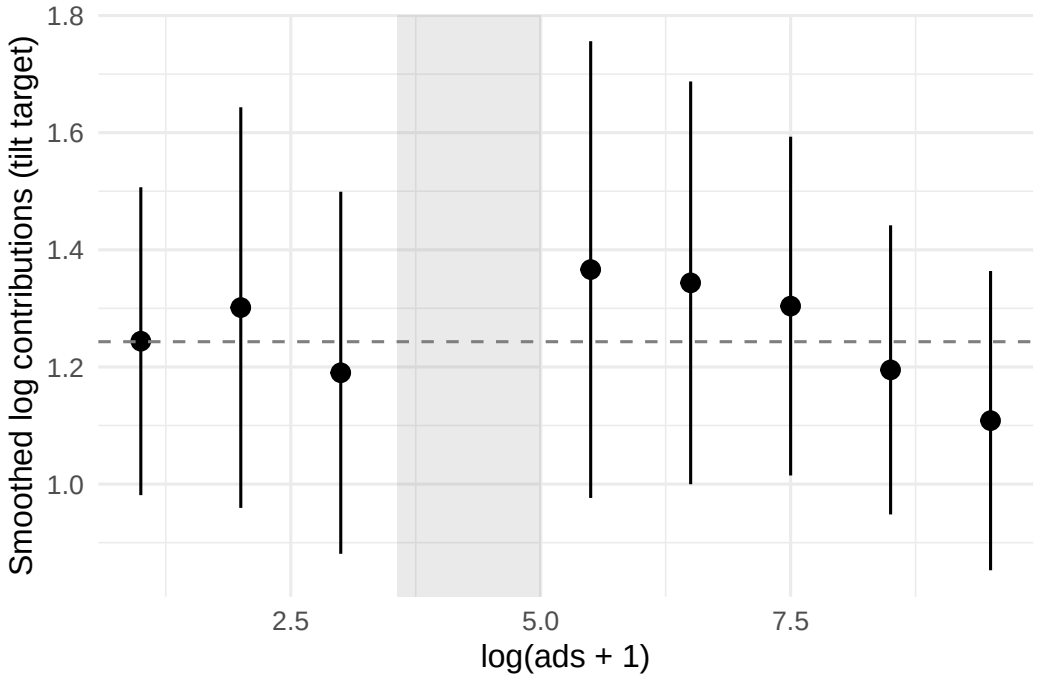


Figure 36.2.: Smoothed, model-dependent description of log contributions against log ads by the reflected tilt, with 95% intervals clustered by market. The curve is not evaluated in the hole of the dose distribution between about 3.6 and 5.0.

The estimated curve is flat within its noise. It rises from 1.244 at one log ad to 1.366 at 5.5 and falls to 1.108 at 9.5, with clustered standard errors between 0.13 and 0.2. The dashed line is the sample mean. The same caveats apply as before: intervals that understate uncertainty when the density is estimated, a bias of order $1/\delta$, and targets that lie between the observed doses.

36.9. Summary

An incremental effect tilts each unit's dose distribution toward higher or lower doses, $q_\delta(a \mid x) \propto e^{\delta a}\pi(a \mid x)$, and asks for the mean outcome under the tilted distribution. Doses a unit could never receive stay impossible, so the effect is identified without positivity, which a shift of the dose needs. The price is that δ is not a dose: it changes the mean dose by about $\delta \operatorname{var}(A \mid X)$. Its efficient influence function weights residuals by $e^{\delta A}/E(e^{\delta A} \mid X)$. Under weak positivity at the top of the support, the best attainable error is of order $\sqrt{\delta/n}$, and the cross-fitted one-step estimator attains it when the nuisances are good enough.

"Good enough" is the catch. In our simulation the one-step estimator is unbiased and covers at

95% when the dose distribution is known. With the distribution estimated by the paper's kernel regression, made to integrate to one, reflected at the edges and given a separate probability for zero doses, the bias falls more slowly than the standard error, and coverage at moderate tilts gets worse as n grows. The remainder contains the squared error of the density estimate, so the conditional density has to be estimated well, uniformly over doses. Smoothing the zeros into the density biases negative tilts badly, and the regression version without a density is unstable.

In the political-ads data the tilts are strong, standard errors must be clustered by the 138 media markets that set the dose, and the curve is flat within those errors. Shifting the dose by half a log ad with `lmtp` gives almost the same estimate as a tilt with the same change in the mean dose. The shifted doses are values no market received, so the shift estimate rests on the models' smoothing between observed doses. The reflected-tilt dose-response curve rests on the same smoothing, because its targets also lie between the 73 observed doses.

Systematic treatment: R · Julia.

Part VII.

Advanced Topics

37. Doing the same multi-level model with R and Stata

This post is a demo for doing the same multi-level models with R and stata. R's multilevel models are mostly done with lme4 and nlme. Stata's multilevel models are mostly done with mixed and menl. The two packages are not exactly the same. `mixed` does in fact support several residual covariance structures via its `residuals()` option (e.g. `residuals(exponential, t(year))`, as well as `ar`, `ma`, `toeplitz`, etc.) — so it's not that `mixed` cannot handle a structured residual at all, but `menl`'s nonlinear-mixed-model framework gives more flexibility to combine that with custom nonlinear mean/random-effect specifications (like the CEM model below), which is why we reach for it here. Likewise, in R lme4's "lmer" does not allow this specification. We have to use nlme. But it's good to know how to do the same thing in both packages.

For example, we want to do a consensus emergence model (CEM) as described in Lang, Bliese, and Voogt (2018). The CEM model is basically a growth model with the residual being a exponential function of the time variable. We want to do the same thing in both R and stata.

We start with simpler models.

37.1. Stata and R syntax comparison for the same three level model

We run a three level model specified in stata with mixed and menl. In R with nlme and lme4. This is a random intercept model. Only intercept is varying for each unit at each level. In this data set supposedly state is nested within region. There are situations that levels are not nested, then we need to specify cross effect models. We do not discuss that here. One point to watch: `nlme`'s `lme` function expects the grouping factors in the `random` specification ordered from the *highest* level down. Here `state` is nested within `region`, so `region` must be listed before `state`; reversing the order leads `lme` to treat regions as nested within states, which is not the intended structure.

There are different ways to specify this model in nlme. I show two ways here (including lme2 which is commented out).

37.1.1. Stata

```
* let's see how to use menl
webuse productivity
* to replicate mixed with menl
mixed gsp || region: || state:
menl gsp = {b1:},  define(b1: U1[region] UU1[region>state])
```

```
(Public capital productivity)


Performing EM optimization ...

Performing gradient-based optimization:
Iteration 0:  Log likelihood =  200.86162
Iteration 1:  Log likelihood =  200.86162

Computing standard errors ...

Mixed-effects ML regression                                   Number of obs = 816

        Grouping information
        -------------------------------------------------------------
                        |     No. of       Observations per group
         Group variable |     groups    Minimum    Average    Maximum
        ----------------+--------------------------------------------
                 region |          9         51       90.7        136
                  state |         48         17       17.0         17
        -------------------------------------------------------------

                                                              Wald chi2(0)  =   .
Log likelihood =  200.86162                                   Prob > chi2   =   .

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _cons |   10.65961   .2503806    42.57   0.000     10.16887    11.15035
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Identity             |
                  var(_cons) |   .4376123   .2697616        .13073    1.464887
-----------------------------+------------------------------------------------
state: Identity              |
                  var(_cons) |   .6080626   .1382733      .3893904    .9495357
-----------------------------+------------------------------------------------
               var(Residual) |   .0246633   .0012586      .0223159    .0272577
------------------------------------------------------------------------------
LR test vs. linear model: chi2(2) = 2750.56                   Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.


Obtaining starting values by EM:
```

```
Alternating PNLS/LME algorithm:

Iteration 1:  Linearization log likelihood =  200.86162
Iteration 2:  Linearization log likelihood =  200.86162

Computing standard errors:

Mixed-effects ML nonlinear regression           Number of obs     =        816

      Grouping information
      ------------------------------------------------------------------
                           |     No. of        Observations per group
                      Path |     groups    Minimum    Average    Maximum
      ---------------------+--------------------------------------------
                    region |          9         51       90.7        136
             region>state |         48         17       17.0         17
      ------------------------------------------------------------------


Linearization log likelihood =  200.86162

          b1: U1[region] UU1[region>state]

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
b1           |
       _cons |   10.65961   .2505341    42.55   0.000     10.16857    11.15065
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Identity             |
                     var(U1) |   .4376119   .2686479      .1313834    1.457598
-----------------------------+------------------------------------------------
region>state: Identity       |
                    var(UU1) |   .6080625   .1381527      .3895418    .9491666
-----------------------------+------------------------------------------------
               var(Residual) |   .0246633   .0012586      .0223159    .0272577
------------------------------------------------------------------------------
```

37.1.2. R

The data are `productivity`: 816 observations of gross state product, with 48 states nested in 9 regions, 17 years each. The model is intercept-only with random intercepts at both levels.

These two R packages return same results. Note that lmer reports variance, while lme reports standard deviation.

And they do: `lmer` reports variances of 0.60775 for state-within-region and 0.50977 for region, with a residual of 0.02466; `lme` reports the corresponding standard deviations 0.77958, 0.71398 and 0.15705. Squaring the `lme` figures returns the `lmer` ones ($0.77958^2 = 0.6077$), so the two agree exactly and differ only in what they choose to print. The fixed intercept is 10.665 either way.

Note where the variance sits: between regions and between states, with almost none left within a state over time — the residual standard deviation of 0.157 is about a fifth of the state-level one. Most of the variation in this outcome is cross-sectional, not longitudinal.

```r
library(tidyverse)
library(haven)
library(dplyr)
library(readxl)
library(nlme)
library(lme4)
data <- read_dta('https://www.stata-press.com/data/r18/productivity.dta')
# lme4 for three level model
# (1|region) + (1|state) specifies CROSSED random effects; it numerically
# coincides with the intended nested structure here only because each
# state's label happens to be unique across the whole dataset (never reused
# across regions) -- verified: identical logLik to the explicitly nested
# (1|region/state) below. The nested form is the conceptually correct one
# and is robust even if state labels were ever reused across regions.
lmer1 <- lmer(gsp ~ 1 + (1|region/state), data=data)
summary(lmer1)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: gsp ~ 1 + (1 | region/state)
   Data: data

REML criterion at convergence: -400.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8328 -0.6757  0.0474  0.6382  3.3928

Random effects:
 Groups       Name        Variance Std.Dev.
 state:region (Intercept) 0.60775  0.7796
 region       (Intercept) 0.50977  0.7140
 Residual                 0.02466  0.1570
Number of obs: 816, groups:  state:region, 48; region, 9

Fixed effects:
            Estimate Std. Error t value
(Intercept)   10.665      0.266    40.1
```

```
lme1 <- lme(gsp ~ 1, random= list( region=~1, state=~1), data=data)
#or
#lme1 <- lme(gsp ~ 1, random= list(region = pdSymm(~1), state=pdSymm(~1)), data=data)
#but
#lme1 <- lme(gsp ~ 1, random= list( state=~1, region=~1), data=data)
#would be a different result!
#The other way to specify:
#lme2 <- lme(gsp ~ 1, random= ~ 1 | region/state, data=data)
summary(lme1)
```

```
Linear mixed-effects model fit by REML
  Data: data
        AIC       BIC   logLik
  -392.8509 -374.0381 200.4254

Random effects:
 Formula: ~1 | region
        (Intercept)
StdDev:   0.7139792

 Formula: ~1 | state %in% region
        (Intercept)  Residual
StdDev:   0.7795814 0.1570457

Fixed effects:  gsp ~ 1
               Value Std.Error  DF  t-value p-value
(Intercept) 10.66483 0.2659842 768 40.09574       0

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.83282326 -0.67572390  0.04744387  0.63817953  3.39281358

Number of Observations: 816
Number of Groups:
           region state %in% region
                9                48
```

37.2. A growth model in R and stata

We implement a growth model in R and Stata, with time centred at 1970 and a correlated random intercept and slope for region. All four fits agree: `mixed`, `menl`, `lmer` and `lme` reach the same log likelihood, 888.824. The random slope on time is small (its variance is 0.000056), but it is estimated.

37.2.1. Stata

```
* same growth model with mixed and menl
* Stata's "mixed ... || region: year" defaults to an INDEPENDENT covariance
* structure for region's random intercept and slope (var(year), var(_cons),
* cov=0 by construction) -- verified via `estat recovariance`. R's lmer
* "(year|region)" and lme's "region=~year" both default to an UNSTRUCTURED
* (correlated) covariance. Add covariance(unstructured) to match R, in both
* mixed and menl (menl's random effects are also independent by default).
* Time is centred at the first year, t = year - 1970, so the random intercept
* is the region's level in 1970 rather than an extrapolation to year 0.
webuse productivity, clear
gen t = year - 1970
mixed gsp t || region: t, covariance(unstructured) || state:
menl gsp =  {b2:} + {U1[region]}*t, define(b2: t U0[region]  U11[region>state]) covariance(U0 U1, unstruc
```

```
(Public capital productivity)



Performing EM optimization ...

Performing gradient-based optimization:
Iteration 0:  Log likelihood =  888.82273
Iteration 1:  Log likelihood =  888.82404
Iteration 2:  Log likelihood =  888.82404

Computing standard errors ...

Mixed-effects ML regression                             Number of obs =    816

        Grouping information
        -------------------------------------------------------------
                        |     No. of       Observations per group
         Group variable |     groups    Minimum    Average    Maximum
        ----------------+--------------------------------------------
                 region |          9         51       90.7        136
                  state |         48         17       17.0         17
        -------------------------------------------------------------

                                                        Wald chi2(1)  = 107.46
Log likelihood =  888.82404                             Prob > chi2   = 0.0000

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

```
           t |   .0263275   .0025397    10.37   0.000     .0213497    .0313053
       _cons |   10.43297   .2626235    39.73   0.000     9.918239     10.9477
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Unstructured         |
                     var(t) |    .000056   .0000274      .0000215     .000146
                  var(_cons) |   .4974817   .2974088      .1541361    1.605646
                cov(t,_cons) |  -.0045713   .0025218      -.009514    .0003714
-----------------------------+------------------------------------------------
state: Identity              |
                  var(_cons) |   .6121894   .1393574      .3918504     .956426
-----------------------------+------------------------------------------------
               var(Residual) |   .0039862   .0002046      .0036046    .0044081
------------------------------------------------------------------------------
LR test vs. linear model: chi2(4) = 4112.15               Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.


Obtaining starting values by EM:

Alternating PNLS/LME algorithm:

Iteration 1:  Linearization log likelihood =  888.82404
Iteration 2:  Linearization log likelihood =  888.82404

Computing standard errors:

Mixed-effects ML nonlinear regression           Number of obs     =        816

      Grouping information
      ------------------------------------------------------------------
                       |     No. of        Observations per group
                  Path |     groups    Minimum    Average    Maximum
      -----------------+------------------------------------------------
                region |          9         51       90.7        136
        region>state |         48         17       17.0         17
      ------------------------------------------------------------------

                                                Wald chi2(1)      =     107.20
Linearization log likelihood =  888.82404       Prob > chi2       =     0.0000

          b2: t U0[region] U11[region>state]

------------------------------------------------------------------------------
```

```
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
b2           |
           t |   .0263275   .0025428    10.35   0.000     .0213436    .0313114
       _cons |   10.43297   .2629456    39.68   0.000     9.917608    10.94834
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Unstructured         |
                     var(U0) |     .49748   .2952832      .1554312    1.592257
                     var(U1) |    .000056   .0000274      .0000215    .0001459
                   cov(U0,U1) |  -.0045713   .0025207     -.0095118    .0003693
-----------------------------+------------------------------------------------
region>state: Identity       |
                    var(U11) |   .6121883   .1389459      .3923659     .955166
-----------------------------+------------------------------------------------
               var(Residual) |   .0039862   .0002046      .0036046    .0044081
------------------------------------------------------------------------------
```

37.2.2. R

```
# lme4 for three level model, growth model
data$t <- data$year - 1970   # centre time at the first year, as in Stata
# state codes are unique across regions, so (1|state) is the nested state effect
lmer_growth1 <- lmer(gsp ~ 1 + t + (t|region) +  (1|state), data=data,  REML=FALSE,
                     control = lmerControl(optimizer = "bobyqa"))
summary(lmer_growth1)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: gsp ~ 1 + t + (t | region) + (1 | state)
   Data: data
Control: lmerControl(optimizer = "bobyqa")

      AIC       BIC    logLik -2*log(L)  df.resid
  -1763.6   -1730.7     888.8   -1777.6       809

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.9078 -0.5767  0.0122  0.5985  4.5440

Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 state    (Intercept) 6.122e-01 0.782425
```

```
 region   (Intercept) 4.975e-01 0.705317
          t           5.598e-05 0.007482 -0.87
 Residual             3.986e-03 0.063136
Number of obs: 816, groups:  state, 48; region, 9

Fixed effects:
            Estimate Std. Error t value
(Intercept) 10.43297    0.26262   39.73
t            0.02633    0.00254   10.37

Correlation of Fixed Effects:
  (Intr)
t -0.766
```

```
# nlme for three level model, growth model
#lme_growth1<-lme(gsp ~ year, random = list(region=pdSymm(~year), state=pdIdent(~1)),data=data, method="M
lme_growth1<-lme(gsp ~ t, random = list(region=~t, state=~1),data=data, method="ML")
summary(lme_growth1)
```

```
Linear mixed-effects model fit by maximum likelihood
  Data: data
        AIC       BIC  logLik
  -1763.648 -1730.717 888.824

Random effects:
 Formula: ~t | region
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev      Corr
(Intercept) 0.705322591 (Intr)
t           0.007482037 -0.866

 Formula: ~1 | state %in% region
        (Intercept)   Residual
StdDev:   0.7824247 0.06313614

Fixed effects:  gsp ~ t
                Value  Std.Error  DF  t-value p-value
(Intercept) 10.432972 0.26294558 767 39.67730       0
t            0.026327 0.00254284 767 10.35357       0
 Correlation:
  (Intr)
t -0.766

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-3.90782851 -0.57667741  0.01216096  0.59852643  4.54401910
```

```
Number of Observations: 816
Number of Groups:
           region state %in% region
                9                48
```

37.3. CEM in R and Stata

"CEM" is a growth model with the residual being a exponential function of the time variable.

37.3.1. Stata

With raw calendar years (1970–1986) in the variance function, $\exp(\gamma \cdot \text{year})$ is evaluated near year 2000 and the model does not converge. Centring time at 1970 fixes that.

```
webuse productivity, clear
gen t = year - 1970

* residual variance sigma2 * exp(2 * gamma * t), with t centred at 1970
menl gsp =  {b2:} + {U1[region]}*t, define(b2: t U0[region]  U11[region>state]) covariance(U0 U1, unstru
```

```
(Public capital productivity)



Mixed-effects ML nonlinear regression           Number of obs     =        816

      Grouping information
      ------------------------------------------------------------------
                       |     No. of        Observations per group
                  Path |     groups    Minimum    Average    Maximum
      -----------------+------------------------------------------------
                region |          9         51       90.7        136
          region>state |         48         17       17.0         17
      ------------------------------------------------------------------

                                                Wald chi2(1)      =      96.58
Linearization log likelihood =  908.40814       Prob > chi2       =     0.0000

          b2: t U0[region] U11[region>state]

------------------------------------------------------------------------------
         gsp | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
b2           |
           t |   .0273381   .0027818     9.83   0.000     .0218859    .0327904
```

```
       _cons |   10.43041   .2648885    39.38   0.000     9.911242    10.94959
------------------------------------------------------------------------------

------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
region: Unstructured         |
                     var(U0) |   .5064182   .3001106      .1585167    1.617869
                     var(U1) |   .0000675   .0000329       .000026    .0001753
                  cov(U0,U1) |  -.0047686   .0027319      -.010123    .0005858
-----------------------------+------------------------------------------------
region>state: Identity       |
                    var(U11) |   .6088354   .1384617      .3898694    .9507812
-----------------------------+------------------------------------------------
Residual variance:           |
  Exponential t              |
                     sigma2 |   .0015338     .00022      .0011579    .0020318
                      gamma |   .0556869   .0081777      .0396589    .0717148
------------------------------------------------------------------------------
```

37.3.2. R

With centred time, R and Stata agree.

```
lme_growth1<-lme(gsp ~ t, random = list(region=pdSymm(~t), state=pdIdent(~1)),data=data, method="ML")
lme_growth2<-update(lme_growth1,weights=varExp( form = ~ t), method="ML", control = lmeControl(opt = "op
summary(lme_growth2)
```

```
Linear mixed-effects model fit by maximum likelihood
  Data: data
        AIC       BIC   logLik
  -1800.816 -1763.181 908.4081

Random effects:
 Formula: ~t | region
 Structure: General positive-definite
            StdDev      Corr
(Intercept) 0.711318393 (Intr)
t           0.008217065 -0.816

 Formula: ~1 | state %in% region
        (Intercept)   Residual
StdDev:   0.7802004 0.03916322

Variance function:
 Structure: Exponential of variance covariate
```

```
 Formula: ~t
 Parameter estimates:
     expon
0.05569042
Fixed effects:  gsp ~ t
                Value  Std.Error  DF  t-value p-value
(Intercept) 10.430356 0.26478882 767 39.39123       0
t            0.027338 0.00278269 767  9.82436       0
 Correlation:
  (Intr)
t -0.724

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-3.63657023 -0.60788373  0.03464838  0.62215692  3.89628444

Number of Observations: 816
Number of Groups:
           region state %in% region
                9                48
```

```
delta1<-lme_growth2$modelStruct$varStruct
delta1
```

```
Variance function structure of class varExp representing
     expon
0.05569042
```

37.4. Summary

Stata's `mixed` and `menl` and R's `lme4` and `nlme` fit the same multilevel models with different syntax and different printed quantities. In the three-level random-intercept model for state productivity, `lmer` and `lme` agree exactly. One reports variances and the other standard deviations, and squaring the latter returns the former. Most of the variation lies between regions and between states, with little within a state over time.

The growth model and the consensus emergence model, a growth model whose residual variance changes exponentially over time, give the same answers in R and Stata once two things match: time is centred at the first year, and the random intercept and slope are allowed to correlate in every package (`menl` and `mixed` default to independent random effects, `lmer` and `lme` to correlated ones). Then all four growth fits reach a log likelihood of 888.824, and both emergence fits reach 908.408 with the same variance-function parameter, 0.0557. With raw calendar years the emergence model did not converge, and the packages disagreed.

38. Modeling variation in Likert-scale data

38.1. Why the mean constrains the variance

Suppose we have a 1–7 satisfaction or agreement scale and want to know whether one group shows more disagreement than another. A natural approach is to calculate a standard deviation for each unit and regress it on group indicators or other covariates.

The problem is that the variance of a bounded outcome depends mechanically on its mean. The Bhatia and Davis (2000) inequality is the population version of that statement: for a *distribution* on $[a, b]$ with mean μ,

$$\mathrm{Var}(X) \leq (\mu - a)(b - \mu) \tag{38.1}$$

The bound is sharp over distributions, attained by the two-point distribution putting mass $(\mu - a)/(b - a)$ at b and the rest at a. Whether a given *sample* can reach it is a different question, and with few raters it usually cannot, because the counts at each endpoint have to be whole numbers. That matters below.

For a 1–7 item, the maximum possible standard deviation over distributions is therefore

$$\mathrm{SD}_{\max}(\mu) = \sqrt{(\mu - 1)(7 - \mu)}$$

```
mu_grid <- seq(1, 7, length.out = 200)
sd_max  <- sqrt(pmax((mu_grid - 1) * (7 - mu_grid), 0))

ggplot(data.frame(mu = mu_grid, sd_max = sd_max), aes(mu, sd_max)) +
  geom_line(linewidth = 1, color = "#2d5fa3") +
  labs(x = expression("Mean Rating (" * mu * ")"), y = expression(SD[max](mu)),
       title = "Maximum Possible SD as a Function of the Mean") +
  theme_minimal(base_size = 12)
```

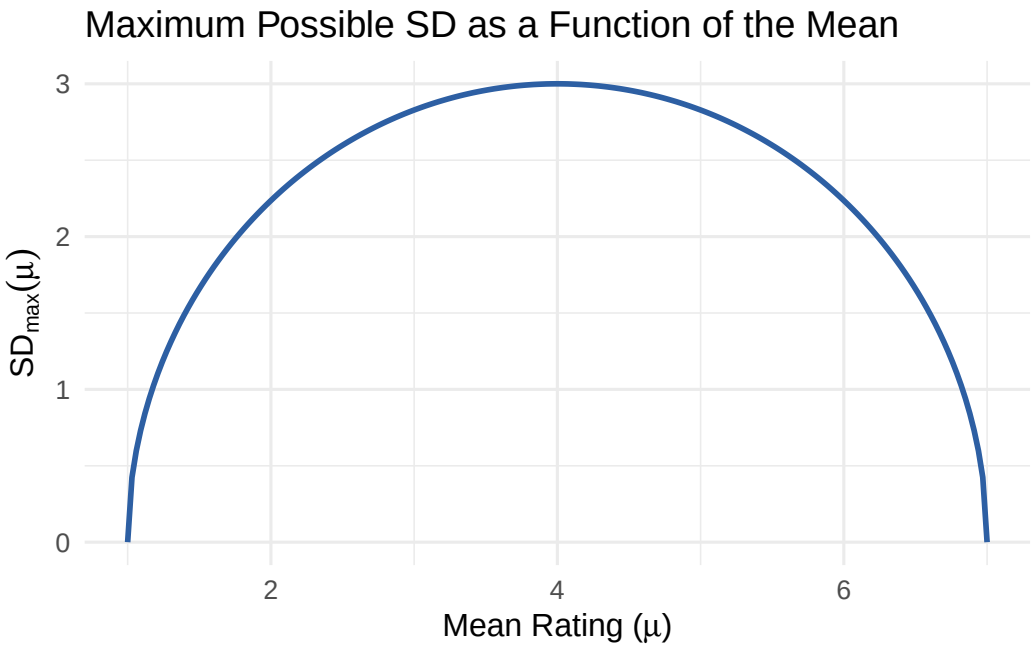


Figure 38.1.: The mechanical ceiling on SD as a function of the mean on a 1–7 scale.

The maximum is 3 at the middle of the scale, where $\mu = 4$, and falls to zero at either endpoint. Satisfaction and performance ratings are often close to the upper endpoint. A group with a higher mean can then have a smaller observed standard deviation even when its underlying dispersion is the same.

38.1.1. Why simple adjustments do not solve it

Dividing the standard deviation by the mean does not remove the dependence. That adjustment assumes that variance grows proportionally with the mean, while variance on a bounded scale is compressed at both endpoints. In this example, dividing by μ only moves the peak from $\mu = 4$ to $\mu = 1.75$.

A Gaussian location-scale model has a related problem. A model such as `sigma ~ group` treats the response as unbounded, so it can interpret compression near an endpoint as a real difference in residual variation.

38.2. An ordinal location-scale model

One way to handle the bounds is to treat the Likert response as a discretized latent variable Y^*:

$$Y_i^* = \eta_i + \sigma_i \epsilon_i, \qquad \epsilon_i \sim N(0, 1)$$

where η_i is the location predictor and σ_i is the latent residual standard deviation. The observed rating $Y_i \in \{1, \dots, K\}$ is determined by ordered cutpoints $\theta_1 < \cdots < \theta_{K-1}$:

$$Y_i = k \iff \theta_{k-1} < Y_i^* \leq \theta_k$$

Under a probit link, the cumulative probability is:

$$P(Y_i \leq k) = \Phi\left(\frac{\theta_k - \eta_i}{\sigma_i}\right) = \Phi(\text{disc}_i(\theta_k - \eta_i))$$

where discrimination is defined as the inverse of latent dispersion: $\text{disc}_i \equiv 1/\sigma_i$.

In `brms`, we model $\log(\text{disc}_i)$ directly. Higher discrimination means smaller latent dispersion, or more agreement. Lower discrimination means greater spread. The cutpoints are estimated on an unbounded latent scale, so an observed response near 7 does not by itself imply smaller latent dispersion. This interpretation assumes that the groups use the response categories in the same way.

38.3. A simulation

Let's simulate 100 subjects in each group, with six ratings for each subject. Both groups have latent standard deviation $\sigma = 1$. Group A has a latent mean near the ceiling ($\mu_A = 2.4$), while Group B is closer to the middle ($\mu_B = 0.6$).

```
set.seed(42)

n_per_group <- 100
n_raters    <- 6
sigma_true  <- 1.0   # Identical latent dispersion across groups
mu_A <- 2.4          # Group A: high mean (near ceiling)
mu_B <- 0.6          # Group B: mid-scale mean

subjects <- data.frame(
  subject = 1:(2 * n_per_group),
  group   = rep(c("A", "B"), each = n_per_group),
  mu      = rep(c(mu_A, mu_B), each = n_per_group)
)

df <- subjects[rep(seq_len(nrow(subjects)), each = n_raters), ]
df$latent <- rnorm(nrow(df), mean = df$mu, sd = sigma_true)

# Discretize using 6 fixed cutpoints into categories 1 through 7
cuts <- c(-2, -1.2, -0.5, 0.2, 1.0, 1.8)
df$rating     <- as.integer(cut(df$latent, breaks = c(-Inf, cuts, Inf), labels = 1:7))
df$rating_ord <- factor(df$rating, ordered = TRUE, levels = 1:7)
```

38.3.1. The observed ratings

```
subj_summary <- df |>
  group_by(subject, group) |>
  summarise(mean_rating = mean(rating), sd_rating = sd(rating), .groups = "drop")

subj_summary |>
  group_by(group) |>
  summarise(mean_rating = round(mean(mean_rating), 2),
            mean_sd     = round(mean(sd_rating), 2)) |>
  knitr::kable(caption = "Observed summary statistics by group.")
```

Table 38.1.: Observed summary statistics by group.

group	mean_rating	mean_sd
A	6.64	0.58
B	4.89	1.22

```
# sd() uses the n - 1 denominator; the Bhatia-Davis bound is for the population
# (divide-by-n) variance. With six ratings the sample-SD ceiling is sqrt(6/5) times it.
sd_max_sample <- sqrt(6 / 5) * sd_max
ggplot(subj_summary, aes(mean_rating, sd_rating, color = group)) +
  geom_jitter(width = 0.05, height = 0, alpha = 0.5) +
  geom_line(data = data.frame(mean_rating = mu_grid, sd_rating = sd_max_sample),
            aes(mean_rating, sd_rating), color = "grey40", inherit.aes = FALSE) +
  labs(x = "Subject Mean Rating", y = "Subject SD",
       title = "Group A's SD is compressed by the boundary") +
  theme_minimal(base_size = 12)
```

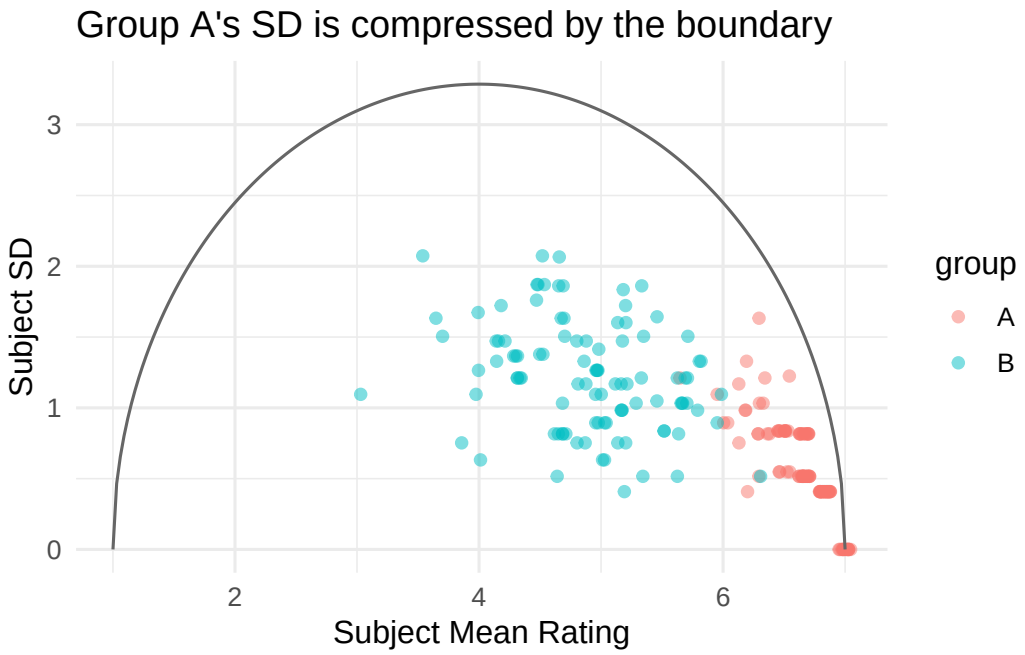


Figure 38.2.: Subject-level mean vs. SD against the Bhatia-Davis bound, rescaled by $\sqrt{6/5}$ because `sd()` divides by $n-1=5$.

Each subject's SD uses the $n-1$ denominator, so its ceiling is $\sqrt{6/5}$ times the population bound: six ratings split between 1 and 7 give a sample SD of 3.29, not 3. The observed standard deviation averages 0.58 in Group A and 1.22 in Group B. If we looked only at these numbers, we would conclude that raters agree much more about Group A, even though the two latent standard deviations are identical by construction.

38.4. Comparing Gaussian and ordinal models

I fit two models in `brms`. The Gaussian model makes $\log(\sigma)$ a function of group. The ordinal cumulative-probit model instead makes log(disc) a function of group, with Group A used as the reference ($\text{disc}_A = 1$).

```
# 1. Gaussian location-scale model
fit_gaussian <- brm(
  bf(rating ~ group + (1 | subject), sigma ~ group),
  data = df, family = gaussian(),
  chains = 2, iter = 2000, warmup = 1000, cores = 2, seed = 1,
  refresh = 0, silent = 2
)

# 2. Ordinal cumulative-probit location-scale model
fit_ordinal <- brm(
  bf(rating_ord ~ group + (1 | subject), disc ~ 0 + group),
```

```
  data = df, family = cumulative("probit"), init = 0,
  prior = c(
    prior(constant(0), class = "b", coef = "groupA", dpar = "disc"),
    prior(normal(0, 1), class = "b", coef = "groupB", dpar = "disc")
  ),
  chains = 2, iter = 2000, warmup = 1000, cores = 2, seed = 1,
  refresh = 0, silent = 2
)
```

```
knitr::kable(round(fixef(fit_gaussian), 2), caption = "Gaussian MELSM: Incorrectly finds group difference
```

Table 38.2.: Gaussian MELSM: Incorrectly finds group differences in residual SD.

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	6.64	0.03	6.58	6.69
sigma_Intercept	-0.41	0.03	-0.46	-0.35
groupB	-1.75	0.06	-1.86	-1.63
sigma_groupB	0.68	0.04	0.60	0.76

```
knitr::kable(round(fixef(fit_ordinal), 2), caption = "Ordinal MELSM: Correctly recovers equal latent disp
```

Table 38.3.: Ordinal MELSM: Correctly recovers equal latent dispersion.

	Estimate	Est.Error	Q2.5	Q97.5
Intercept[1]	-4.39	0.32	-5.06	-3.81
Intercept[2]	-3.65	0.23	-4.12	-3.22
Intercept[3]	-2.98	0.17	-3.34	-2.66
Intercept[4]	-2.27	0.12	-2.52	-2.04
Intercept[5]	-1.44	0.07	-1.59	-1.30
Intercept[6]	-0.63	0.06	-0.74	-0.52
groupB	-1.89	0.10	-2.11	-1.69
disc_groupA	0.00	0.00	0.00	0.00
disc_groupB	-0.01	0.08	-0.17	0.15

38.4.1. Results

The Gaussian coefficient on `sigma_groupB` is 0.68. It estimates that Group B's residual standard deviation is $\exp(0.68) \approx 1.97$ times Group A's (95% CI : $[1.82, 2.14]$). This apparent difference comes from the scale ceiling.

The ordinal coefficient on `disc_groupB` is -0.01 (95% CI : $[-0.17, 0.15]$), which implies a latent standard-deviation ratio of $\exp(0.01) \approx 1.01$. The ordinal model recovers the equal dispersion used to generate the data.

38.4.2. An important assumption

The data were generated from the ordinal model's own functional form: a latent normal variable, a constant σ, and one `cuts` vector shared by both groups. It is therefore not surprising that the ordinal model fits well. What the example shows is narrower: boundary compression by itself does not create a difference in the latent dispersion parameter, while it does create one in the Gaussian model.

The example does not show that the model can distinguish dispersion from threshold differences, because I did not simulate any threshold difference. Common thresholds across groups, or a model that allows them to vary, are needed for identification. If groups use the categories differently, threshold shifts and `disc` can trade off: the cutpoints can absorb a real dispersion difference, or a threshold difference can appear as a change in `disc`. On real data, we need information outside the basic model, such as anchoring vignettes or a partial-invariance specification, to check this assumption.

38.5. A descriptive alternative

A simpler descriptive measure is the ratio of the observed variance to the maximum possible variance at the same mean:

$$R = \frac{\mathrm{Var}(Y)}{\mathrm{Var}_{\max}(\mu)} \in [0, 1]$$

where $R = 0$ indicates perfect agreement and $R = 1$ indicates maximum polarization.

The top of that range is not generally reachable at the subject level. Enumerate every set of six ratings on a 1–7 item and only 7 of the 37 achievable means admit a configuration that attains the Bhatia–Davis bound, namely the integer means where an all-endpoint split is possible. Elsewhere the ceiling is lower: the largest attainable R is 0.83 at a mean of 3.5, 0.45 at 1.5, and 0.14 at 7/6. A subject-level R is therefore measured against a denominator the ratings could not have produced, and the shortfall itself varies with the mean, which is the dependence R was meant to remove.

Either divide by the sharp finite-sample maximum at the observed mean and rater count, obtained by the enumeration just described, or keep the Bhatia–Davis denominator and say plainly that $R \leq 1$ is measured against a possibly unattainable ceiling. The pooled calculation below is the mild case: with a whole group's ratings the achievable means are dense enough that the distributional bound is close to sharp.

With only six ratings per subject, R is noisy and can be undefined when every rater chooses 7, giving 0/0. We can instead calculate it after pooling the ratings within each group:

```
df |>
  group_by(group) |>
  summarise(
    mean_rating = mean(rating),
    var_obs     = mean((rating - mean(rating))^2),
    var_max     = (mean(rating) - 1) * (7 - mean(rating)),
```

```
    R_pooled    = round(var_obs / var_max, 3)
  ) |>
  knitr::kable(caption = "Group-level pooled R metric.")
```

Table 38.4.: Group-level pooled R metric.

group	mean_rating	var_obs	var_max	R_pooled
A	6.638333	0.4475306	2.039197	0.219
B	4.893333	1.7286222	8.201956	0.211

Both groups give $R \approx 0.21$–0.22. Dividing by the maximum attainable variance makes the descriptive measure more comparable across groups with different means. It is still only a normalization, not a model-based correction, and it has no latent dispersion parameter to interpret.

38.6. Which approach to use

Raw standard deviations are hard to compare when group means differ on a bounded scale. If the goal is a model of underlying response dispersion, I would use the ordinal location-scale model and examine whether common thresholds are plausible. If the goal is only a descriptive comparison, pooled $R = \mathrm{Var}/\mathrm{Var}_{\max}$ is easier to explain, as long as it is not interpreted as a latent parameter.

39. Conjoint design and analysis using R

This post is to illustrate how to use R to do basic conjoint analysis.

39.1. Conjoint Analysis

Marketing people used to use Sawtooth to do conjoint analysis.

What is conjoint analysis? Here is the intro from Sawtooth:

"Conjoint analysis methods use statistical analysis to compute mathematical representations of survey respondents' preferences for product features, simulate how attribute changes affect demand, and even predict market acceptance of new products before they launch.

This powerful research methodology has become very popular with market researchers and economists for the valuable insight that few other research methods can provide."

"Consider the task of purchasing a new vehicle.

When searching for the "right" vehicle, you typically consider a multitude of variables: type, make (brand), model, year, mileage, price, color, accessories packages, etc. You can see how this is a complex choice to make.

How does anyone possibly make such a complicated decision?

You think about what features matter most to you and then make tradeoffs.

For example, one vehicle could be an acceptable make and model, but at a high price point, whereas a second option is not the ideal make and model but has a more palatable price.

The tradeoff in this simple example is whether the make and model is more preferable to a reasonable price, or vice versa.

A conjoint survey captures the relationship between different attributes of a product and how preferable they are both comparatively and when combined.

Because of this, conjoint analysis is the gold standard solution for testing multiple features simultaneously, when limited A/B testing doesn't go far enough."

"In simple terms, the conjoint survey design algorithm generates a balanced survey design with product profiles, also known as concepts, formulated to have ideal statistical properties (such as level balance and independence of attributes).

Each profile is made up of multiple conjoined product features (hence, conjoint analysis), such as brand, type, engine, and color, each with systematically varied levels.

These product concepts are then included in a series of questions, usually 8-20, called choice tasks, that make up the conjoint analysis portion of the survey."

39.2. Example

The example uses the tea data that ships with the `conjoint` package (Bąk and Bartłomowicz 2012). A hundred respondents rate 13 tea profiles on a scale from 0 to 10. Each profile combines four attributes: price (low, medium, high), variety (black, green, red), kind (bags, granulated, leafy) and aroma (yes, no).

```
library(tidyverse)
library(conjoint)

data(tea)   # tprof: the 13 profiles; tprefm: 100 x 13 ratings; tlevn: level names

levels_list <- list(price   = c("low", "medium", "high"),
                    variety = c("black", "green", "red"),
                    kind    = c("bags", "granulated", "leafy"),
                    aroma   = c("yes", "no"))

profiles <- tprof
profiles[] <- Map(function(x, l) factor(l[x], levels = l), tprof, levels_list)

tea_long <- tibble(respondent = rep(1:100, each = 13),
                   profile    = rep(1:13, times = 100),
                   rating     = as.vector(t(tprefm))) |>
  bind_cols(profiles[rep(1:13, times = 100), ])

tea_long
```

```
# A tibble: 1,300 x 7
   respondent profile rating price  variety kind       aroma
 *      <int>   <int>  <int> <fct>  <fct>   <fct>      <fct>
 1          1       1      8 high   black   bags       yes
 2          1       2      1 low    green   bags       yes
 3          1       3      1 medium green   granulated yes
 4          1       4      3 medium black   leafy      yes
 5          1       5      9 high   red     leafy      yes
 6          1       6      2 medium black   bags       no
 7          1       7      7 high   green   bags       no
 8          1       8      2 medium red     bags       no
 9          1       9      2 high   black   granulated no
10          1      10      2 low    red     granulated no
# i 1,290 more rows
```

The data are 1,300 rows, 100 respondents times 13 profiles.

39.3. Conjoint design of experiment

With three, three, three and two levels, the full factorial has $3 \times 3 \times 3 \times 2 = 54$ profiles. Asking every respondent to rate 54 teas is too much, so a conjoint study shows a fraction of them. The question is which fraction. A good design lets us estimate the effect of every attribute level, with little correlation between the attributes.

`AlgDesign` searches for such designs. `optFederov()` picks the subset of the full factorial that maximizes the determinant of the information matrix for a main-effects model, the D-optimality criterion. A main-effects model here has 8 parameters, an intercept plus $2+2+2+1$ level contrasts, so a design needs at least 8 profiles. We search for designs of several sizes and report their D-efficiency (Kuhfeld, Tobias, and Garratt 1994) and whether every level appears equally often.

```
library(AlgDesign)

full <- expand.grid(lapply(levels_list, factor))

set.seed(123)
designs <- map(c(9, 12, 13, 18, 27, 54),
               \(n) optFederov(~ ., data = full, nTrials = n, criterion = "D"))

tibble(trials       = c(9, 12, 13, 18, 27, 54),
       D_efficiency = map_dbl(designs, "Dea"),
       balanced     = map_lgl(designs, \(d) all(map_lgl(d$design,
                                    \(f) length(unique(table(f))) == 1))))
```

```
# A tibble: 6 x 3
  trials D_efficiency balanced
   <dbl>        <dbl> <lgl>
1      9        0.146 FALSE
2     12        0.779 TRUE
3     13        0.67  FALSE
4     18        1     TRUE
5     27        0.969 FALSE
6     54        1     TRUE
```

D-efficiency measures how much information a design carries per profile, relative to an ideal orthogonal design. A design with efficiency 0.8 needs about 25% more observations to give the same standard errors as one with efficiency 1. Balance requires a number of profiles divisible by 3 and by 2. Among these sizes, only 12, 18 and 54 are balanced. The 18-profile design reaches an efficiency of 1: it is an orthogonal main-effects design, a third of the full factorial with nothing lost for a main-effects model.

The tea study used 13 profiles. That number cannot be balanced, and the design shows it.

```
map(profiles, table)
```

```
$price

   low medium   high
     3      5      5

$variety

black green   red
    5     5     3

$kind

      bags granulated      leafy
         5          3          5

$aroma

yes  no
  5   8
```

```
X <- model.matrix(~ price + variety + kind + aroma, profiles)[, -1]
round(cor(X), 2)
```

```
               pricemedium pricehigh varietygreen varietyred kindgranulated
pricemedium           1.00     -0.62         0.02      -0.06           -0.06
pricehigh            -0.62      1.00         0.02      -0.06           -0.06
varietygreen          0.02      0.02         1.00      -0.43           -0.06
varietyred           -0.06     -0.06        -0.43       1.00            0.13
kindgranulated       -0.06     -0.06        -0.06       0.13            1.00
kindleafy             0.02      0.02         0.02      -0.06           -0.43
aromano              -0.02     -0.02        -0.02       0.06            0.06
               kindleafy aromano
pricemedium         0.02   -0.02
pricehigh           0.02   -0.02
varietygreen        0.02   -0.02
varietyred         -0.06    0.06
kindgranulated     -0.43    0.06
kindleafy           1.00   -0.02
aromano            -0.02    1.00
```

Low price and red variety appear 3 times, while other levels appear 5 times, and aroma is "no" in 8 of the 13 profiles. The correlations between dummies of the *same* attribute, such as -0.62 for medium and high price, come from the coding and are expected. What matters is the correlation *between* attributes, and it stays small: the largest is 0.13, between red variety and granulated kind. The attributes are close to independent, so each attribute's effect can be estimated without much contamination from the others.

39.4. Conjoint analysis

The part-worths are the coefficients of a regression of the rating on attribute dummies. Each respondent rates 13 profiles, so the ratings are clustered by respondent, and the standard errors have to allow for that.

```
library(sandwich)
library(lmtest)

partworths_lm <- lm(rating ~ price + variety + kind + aroma, data = tea_long)
coeftest(partworths_lm, vcov = vcovCL(partworths_lm, cluster = ~ respondent))
```

```
t test of coefficients:

               Estimate Std. Error t value  Pr(>|t|)
(Intercept)     4.95614    0.25090 19.7538 < 2.2e-16 ***
pricemedium    -0.38334    0.18135 -2.1139  0.034717 *
pricehigh      -0.33734    0.23621 -1.4282  0.153483
varietygreen   -0.58000    0.25521 -2.2727  0.023210 *
varietyred     -1.26466    0.25189 -5.0206 5.870e-07 ***
kindgranulated -1.02666    0.10561 -9.7216 < 2.2e-16 ***
kindleafy       0.61600    0.22559  2.7306  0.006407 **
aromano        -0.82155    0.17027 -4.8250 1.567e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The baseline profile is low-priced black tea in bags with aroma. Relative to it, red tea lowers the rating by 1.26 points and green tea by 0.58. Granulated tea lowers it by 1.03, while leafy tea raises it by 0.62. Tea without aroma is rated 0.82 lower. Price matters least: medium and high prices lower the rating by about 0.35, and only the medium-price contrast is significant at 5%.

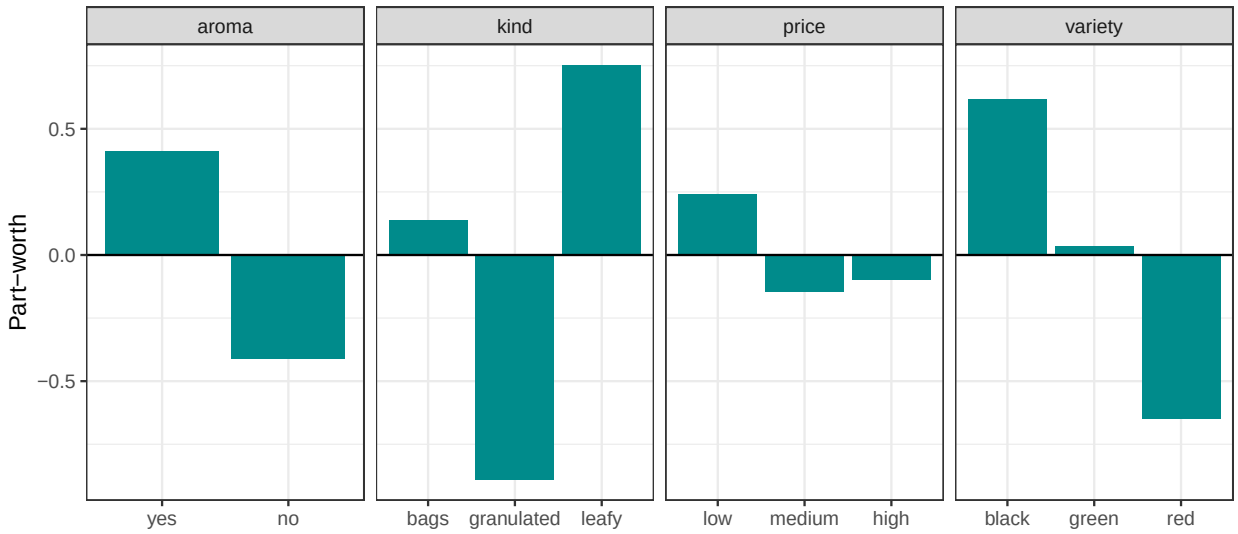
Conjoint software usually reports part-worths in effect coding, where the part-worths of each attribute sum to zero. They are the same coefficients, re-centred within each attribute. The importance of an attribute is the range of its part-worths as a share of the sum of the ranges.

```
part_worths <- imap_dfr(levels_list, \(lv, attr) {
  b <- c(0, coef(partworths_lm)[paste0(attr, lv[-1])])
  tibble(attribute = attr, level = factor(lv, levels = lv), part_worth = b - mean(b))
})

importance <- part_worths |>
  group_by(attribute) |>
  summarise(range = diff(range(part_worth))) |>
  mutate(importance = round(100 * range / sum(range), 1))
importance
```

```
# A tibble: 4 x 3
  attribute range importance
  <chr>     <dbl>      <dbl>
1 aroma     0.822       20
2 kind      1.64        39.9
3 price     0.383        9.3
4 variety   1.26        30.8
```

```
ggplot(part_worths, aes(level, part_worth)) +
  geom_col(fill = "darkcyan") +
  geom_hline(yintercept = 0) +
  facet_wrap(~ attribute, scales = "free_x", nrow = 1) +
  labs(x = NULL, y = "Part-worth") +
  theme_bw()
```



Kind is the most important attribute, with 39.9% of the total range, followed by variety with 30.8%, aroma with 20.0% and price with 9.3%.

The `conjoint` package computes the same quantities. Its pooled part-worths match ours exactly. Its importance figures do not, and the reason is worth knowing. `caImportance()` computes each respondent's importance from that respondent's own part-worths, and then averages across respondents. Individual part-worths from 13 ratings are noisy, and noise widens every attribute's range, so the averaged importance spreads more evenly across attributes than the pooled figure.

```
# conjoint's functions switch R to sum-to-zero contrasts globally and do not
# switch back, which would break every later regression; save and restore.
op <- options("contrasts")
invisible(capture.output(pooled <- caUtilities(y = tprefm, x = tprof, z = tlevn)))
invisible(capture.output(imp <- caImportance(y = tprefm, x = tprof)))
options(op)

round(pooled, 3)                      # intercept, then effect-coded part-worths
```

```
 [1]  3.553  0.240 -0.143 -0.097  0.615  0.035 -0.650  0.137 -0.890  0.753
[11]  0.411 -0.411
```

```
setNames(imp, names(levels_list))     # averaged over respondents
```

```
  price variety    kind   aroma
  24.76   32.22   27.15   15.88
```

The averaged importances are 24.8 for price, 32.2 for variety, 27.2 for kind and 15.9 for aroma. Price moves from least important to second. The two figures answer different questions. The pooled range describes the average respondent. The averaged importance describes how much each attribute matters within a respondent, and it is inflated by sampling noise in the individual fits.

39.5. Causal effect in conjoint analysis

Hainmueller, Hopkins, and Yamamoto (2014) define the average marginal component effect (AMCE): the average change in the outcome when one attribute moves from one level to another, averaged over the distribution of the other attributes. It is not to be confused with the ACME, the average causal mediation effect of Imai, Keele, and Tingley (2010). When the attribute levels are randomized independently of each other and of the respondents, the AMCE is identified, and it is estimated by the coefficients of a regression of the outcome on the attribute dummies. Standard errors are clustered by respondent. The `cjoint` package implements it.

```
library(cjoint)

amce_fit <- amce(rating ~ price + variety + kind + aroma, data = as.data.frame(tea_long),
                 cluster = TRUE, respondent.id = "respondent")

imap_dfr(amce_fit$estimates, \(m, attr)
  tibble(attribute = attr, level = colnames(m),
         amce = round(m["AMCE", ], 3), se = round(m["Std. Error", ], 3)))
```

```
# A tibble: 7 x 4
  attribute level            amce    se
  <chr>     <chr>           <dbl> <dbl>
1 aroma     aromano        -0.822 0.17
2 kind      kindgranulated -1.03  0.106
3 kind      kindleafy       0.616 0.226
4 price     pricemedium    -0.383 0.181
5 price     pricehigh      -0.337 0.236
6 variety   varietygreen   -0.58  0.255
7 variety   varietyred     -1.26  0.252
```

The AMCEs are exactly the part-worth coefficients above, with the same clustered standard errors. The tea profiles were not randomized; they are a fixed 13-profile design. The attributes in it are close to independent, which is why the regression coefficients still read as average effects of each attribute. The causal reading also needs the ratings to reflect the attributes shown and nothing else about the profile, and it holds only for the population of profiles the design represents.

39.6. Summary

A conjoint study asks respondents to rate or choose among product profiles whose attributes vary by design. The full factorial is usually too long, so we use a fraction chosen for D-efficiency and balance. For the tea attributes, an 18-profile design is fully efficient and balanced. The study's own 13 profiles cannot be balanced, but the correlations between attributes stay below 0.13.

The analysis is a regression of the ratings on attribute dummies, with standard errors clustered by respondent. The coefficients are the part-worths. Kind and variety matter most, and price least. Importance from the pooled part-worths differs from `conjoint`'s average of individual importances, because individual fits are noisy. When attribute levels are independent, the same coefficients are Hainmueller et al.'s average marginal component effects, which `cjoint` reproduces.

40. Stata 19 CATE command

40.1. Stata 19 CATE command

Stata 19 introduces a `cate` command for estimating conditional average treatment effects (CATE). Fernando Rios-Avila provides an overview.

> ⚠ If a Stata chunk here renders with no output
>
> The Stata chunks below are cached (`cache: true`) because the `cate` command does lasso cross-fitting plus a random forest, and a full re-execution of this chapter takes over five minutes. The cache is safe for prose edits, which leave every chunk served uniformly from cache. It is **not** safe if you edit the *code* of one Stata chunk: the edited chunk re-executes while the earlier `collectcode: true` chunks are still served from cache, their code never enters Statamarkdown's accumulated buffer, every command runs against missing data, and the chunk renders silently blank – and that empty result is then cached in turn.
> If that happens, delete `stata-cate_cache/` **entirely** (not just the offending chunk's entry) along with `_freeze/stata-cate/`, then re-render so all chunks run in order. Clearing `_freeze/` alone is not enough. See the note in this book's `CLAUDE.md`.

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)
cate po (assets $catecovars) (e401k), group(incomecat) nolog
```

```
(Excerpt from Chernozhukov and Hansen (2004))
```

```
Conditional average treatment effects     Number of observations       = 9,913
Estimator:       Partialing out           Number of folds in cross-fit =    10
Outcome model:   Linear lasso             Number of outcome controls   =    17
Treatment model: Logit lasso              Number of treatment controls =    17
CATE model:      Random forest            Number of CATE variables     =    17

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
GATE         |
```

```
   incomecat |
           0 |   3999.888   989.8742     4.04   0.000      2059.77    5940.006
           1 |   1424.931   1668.081     0.85   0.393    -1844.447     4694.31
           2 |   5092.801   1344.769     3.79   0.000     2457.102    7728.499
           3 |   8700.512   2272.126     3.83   0.000     4247.226     13153.8
           4 |   20350.97   4717.093     4.31   0.000     11105.64     29596.3
-------------+------------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |    7914.24    1150.79     6.88   0.000     5658.734    10169.75
-------------+------------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14012.54   834.0963    16.80   0.000     12377.74    15647.34
------------------------------------------------------------------------------
```

The data are `assets3`, from Chernozhukov and Hansen (2004): 9,913 households, with net financial assets as the outcome and 401(k) eligibility as the treatment. The group ATEs by income category show strong heterogeneity — $4,000 in the lowest category, $1,425 in the second (not significant), then $5,093, $8,701 and $20,350 as income rises. A single ATE would hide a fivefold spread.

He said "We can achieve comparable results using standard regression with full interactions:"

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)

* you do not want to see all interactions.
qui:reg assets i.incomecat##i.e401k##c.($catecovars), robust

* Average Treatment Effect (ATE)
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r))

* Group Average Treatment Effects (GATEs)
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r)) over(incomecat)
```

```
(Excerpt from Chernozhukov and Hansen (2004))
```

```
Contrasts of predictive margins                         Number of obs = 9,913
Model VCE: Robust

Expression: Linear prediction, predict()
```

```
1._at: e401k = 0
2._at: e401k = 1

------------------------------------------------
             |         df           F        P>F
-------------+----------------------------------
         _at |          1       44.72     0.0000
             |
 Denominator |       9833
------------------------------------------------

------------------------------------------------------------------
             |            Delta-method
             |   Contrast   std. err.     [95% conf. interval]
-------------+----------------------------------------------------
         _at |
   (2 vs 1)  |   7642.407   1142.797      5402.291    9882.524
------------------------------------------------------------------


Contrasts of predictive margins                         Number of obs = 9,913
Model VCE: Robust

Expression: Linear prediction, predict()
Over:       incomecat
1._at: 0.incomecat
           e401k = 0
       1.incomecat
           e401k = 0
       2.incomecat
           e401k = 0
       3.incomecat
           e401k = 0
       4.incomecat
           e401k = 0
2._at: 0.incomecat
           e401k = 1
       1.incomecat
           e401k = 1
       2.incomecat
           e401k = 1
       3.incomecat
           e401k = 1
       4.incomecat
           e401k = 1

------------------------------------------------
              |         df           F        P>F
```

```
--------------+----------------------------------
_at@incomecat |
  (2 vs 1) 0  |          1       16.08     0.0001
  (2 vs 1) 1  |          1        0.67     0.4123
  (2 vs 1) 2  |          1       16.03     0.0001
  (2 vs 1) 3  |          1       13.77     0.0002
  (2 vs 1) 4  |          1       17.26     0.0000
       Joint  |          5       12.76     0.0000
              |
  Denominator |       9833
-------------------------------------------------

------------------------------------------------------------------
              |            Delta-method
              |   Contrast   std. err.     [95% conf. interval]
--------------+---------------------------------------------------
_at@incomecat |
  (2 vs 1) 0  |   3594.182   896.3298      1837.191    5351.172
  (2 vs 1) 1  |     1283.3   1565.151     -1784.718    4351.317
  (2 vs 1) 2  |   5056.144   1262.728      2580.938     7531.35
  (2 vs 1) 3  |   8610.622   2320.156      4062.641     13158.6
  (2 vs 1) 4  |   19665.61   4733.551      10386.88    28944.34
------------------------------------------------------------------
```

Numerically these two sets of estimations may be close, but I don't think they are from the same model, even the same idea. The CATE command with "PO" option is to estimate the CATE with "partialling out" approach.

What Fernando did is a "regression adjustment" type of model, which is modeling outcome model and allow interaction of treatment and covariates. The "partialling out" approach is sometimes called double ML.

Let's see how to replicate "teffects ra" with "reg":

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)

teffects ra (assets age educ i.(incomecat pension married twoearn ira ownhome))(e401k)

qui: reg assets i.e401k##c.(age educ) i.e401k##i.(incomecat pension married twoearn ira ownhome)
margins, at(e401k=(0 1)) noestimcheck contrast(atcontrast(r))
```

```
(Excerpt from Chernozhukov and Hansen (2004))
```

```
Iteration 0:  EE criterion = 1.291e-21
```

```
Iteration 1:  EE criterion = 1.046e-23

Treatment-effects estimation                    Number of obs     =      9,913
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |   7929.242   1216.273     6.52   0.000      5545.39    10313.09
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14094.12   863.7641    16.32   0.000     12401.18    15787.07
------------------------------------------------------------------------------



Contrasts of predictive margins                         Number of obs = 9,913
Model VCE: OLS

Expression: Linear prediction, predict()
1._at: e401k = 0
2._at: e401k = 1

-------------------------------------------------
             |         df           F        P>F
-------------+-----------------------------------
         _at |          1       34.39     0.0000
             |
 Denominator |       9889
-------------------------------------------------

------------------------------------------------------------------
             |            Delta-method
             |   Contrast   std. err.     [95% conf. interval]
-------------+----------------------------------------------------
         _at |
   (2 vs 1)  |   7929.242   1352.153      5278.746    10579.74
------------------------------------------------------------------
```

40.1.1. DoubleML

The R package DoubleML implements the double/debiased machine learning framework of Chernozhukov et al. (2018). It provides functionalities to estimate parameters in causal models based on machine learning methods. The double machine learning framework consists of three key ingredients: Neyman orthogonality, High-quality machine learning estimation and Sample splitting.

They consider a partially linear model:

$$y_i = \theta d_i + g_0(x_i) + \eta_i \tag{40.1}$$

$$d_i = m_0(x_i) + v_i \tag{40.2}$$

This model is quite general, except it does not allow interaction of d and x; therefore no heterogeneous treatment effect across x. But "DoubleML" implements more than partially linear model, it actually allows for heterogeneous treatment effects, in models such as interactive regression model.

The basic idea of doubleML is to use any machine learning algorithm to estimate outcome equation ($l_0(X) = E(Y|X)$) and treatment equation ($m_0(X) = E(D|X)$). Then get the residuals, namely $\hat{W} = Y - \hat{l}_0(X)$ and $\hat{V} = D - \hat{m}_0(X)$.

Then regress $\hat{W}$ on $\hat{V}$. Based on FWL theorem, you get $\hat{\theta}$.

An important component here is to specify a Neyman-orthogonal score function. In the case of PLR, it can be the product of the two residuals:

$$\psi(W; \theta, \eta) = (Y - D\theta - g(X))(D - m(X)) \tag{40.3}$$

The estimators $\hat{\theta}$ is to solve the equation that the sample mean of this score function being 0.

The variance of the estimator comes from this score, scaled. Under suitable regularity conditions, $\mathrm{Var}(\hat{\theta}) \approx E[\psi^2]/(nJ^2)$, where $J = E[\partial\psi/\partial\theta] = -E[(D - m(X))^2]$ is the derivative of the score in θ and n the sample size. Cross-fitting makes this valid with flexible learners, provided the nuisance estimates converge fast enough and overlap holds.

Now we try to replicate the CATE estimation in Stata using the R package DoubleML. I have not found a way to include factor variables in the lasso regression in R, so I included them as numeric variables.

The example in stata's cate command is to use lasso on both outcome and treatment regression, then random forest on the individual treatment effect to get ATE. Here I try to replicate it in R.

```
library(haven)
library(DoubleML)
library(mlr3)
library(mlr3learners)
library(data.table)
library(dplyr)
# get rid of labels because doubleML does not like them
```

```
data <- read_dta("https://www.stata-press.com/data/r19/assets3.dta") |>
  zap_labels()


  # pension, married, twoearn, ira, ownhome are already binary 0/1, so
  # passing them as numeric is fine. incomecat, however, has 5 unordered
  # categories (0-4) -- passing it as a single numeric column would force
  # the ML learners (especially the linear regr.glmnet below) to treat
  # category 2 as twice category 1, which is not a meaningful assumption
  # for a nominal variable. One-hot encode it into dummy columns instead.
incomecat_dummies <- model.matrix(~ as.factor(incomecat) - 1, data = data)
colnames(incomecat_dummies) <- paste0("incomecat_", 0:(ncol(incomecat_dummies) - 1))
data <- cbind(data, incomecat_dummies)

assets3 <- data.table::setDT(data)
dml_data = DoubleMLData$new(data = assets3,
                            y_col = "assets",
                            d_cols = "e401k",
                            x_cols = c("age","educ", colnames(incomecat_dummies),"pension", "married","
print(dml_data)
```

```
================= DoubleMLData Object ==================


------------------ Data summary      ------------------
Outcome variable: assets
Treatment variable(s): e401k
Covariates: age, educ, incomecat_0, incomecat_1, incomecat_2, incomecat_3, incomecat_4, pension, married
Instrument(s):
Selection variable:
No. Observations: 9913
```

```
lgr::get_logger("mlr3")$set_threshold("warn")
learner = lrn("regr.glmnet", lambda=1)
ml_g_sim = learner$clone()
ml_m_sim = learner$clone()
set.seed(123)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_l=ml_g_sim, ml_m=ml_m_sim, n_folds=10)
obj_dml_plr$fit()
obj_dml_plr$summary()
```

```
Estimates and significance testing of the effect of target variables
      Estimate. Std. Error t value Pr(>|t|)
e401k      7553       1281   5.897 3.69e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

DoubleML with lasso learners gives an ATE of $7,553 with a standard error of $1,281.

If we use random forest on the two models:

```
# suppress messages from mlr3 package during fitting
lgr::get_logger("mlr3")$set_threshold("warn")
learner = lrn("regr.ranger", num.trees=2000,  max.depth=5, min.node.size=2,
              num.threads = 1)  # single thread so set.seed() fixes the forest
ml_l = learner$clone()
ml_m = learner$clone()
# learner = lrn("regr.glmnet")
# ml_g_sim = learner$clone()
# ml_m_sim = learner$clone()
set.seed(123)
obj_dml_plr = DoubleMLPLR$new(dml_data, ml_l=ml_l, ml_m=ml_m, n_folds=5)
obj_dml_plr$fit()
obj_dml_plr$summary()
```

```
Estimates and significance testing of the effect of target variables
      Estimate. Std. Error t value Pr(>|t|)
e401k      9114       1375   6.627 3.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can do the same in Stata:

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)
cate po (assets $catecovars) (e401k),  omethod(rforest) tmethod(rforest) nolog  xfolds(5)
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Conditional average treatment effects     Number of observations       = 9,913
Estimator:       Partialing out           Number of folds in cross-fit =     5
Outcome model:   Random forest            Number of outcome controls   =    17
Treatment model: Random forest            Number of treatment controls =    17
CATE model:      Random forest            Number of CATE variables     =    17

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
```

```
  (Eligible  |
         vs  |
Not elig..)  |   8282.334   1181.772     7.01   0.000     5966.103    10598.57
-------------+------------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   14002.82   855.4206    16.37   0.000     12326.23    15679.42
------------------------------------------------------------------------------
```

With random forests DoubleML gives $9,114 (SE $1,375), while Stata's `cate` with `rforest` for both models gives $8,282 (SE $1,182). The gap, about $830, is smaller than either standard error. The two packages tune and cross-fit their forests differently, so the numbers need not match exactly.

40.2. AIPW

Stata's cate also has "aipw" option.

```
clear all
webuse assets3
global catecovars age educ i.(incomecat pension married twoearn ira ownhome)
cate aipw (assets $catecovars) (e401k),  nolog
```

```
(Excerpt from Chernozhukov and Hansen (2004))



Conditional average treatment effects     Number of observations       = 9,913
Estimator:       Augmented IPW            Number of folds in cross-fit =    10
Outcome model:   Linear lasso             Number of outcome controls   =    17
Treatment model: Logit lasso              Number of treatment controls =    17
CATE model:      Random forest            Number of CATE variables     =    17

------------------------------------------------------------------------------
             |               Robust
      assets | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
ATE          |
       e401k |
  (Eligible  |
         vs  |
Not elig..)  |   8185.616   1162.852     7.04   0.000     5906.468    10464.76
-------------+----------------------------------------------------------------
POmean       |
       e401k |
Not eligi..  |   13943.48    836.799    16.66   0.000     12303.39    15583.58
------------------------------------------------------------------------------
```

Stata's AIPW gives an ATE of $8,186 with a standard error of $1,163, on 9,913 households.

I'll try to do this with "npcausal":

```
library(npcausal)
#SL.library <- c("SL.earth","SL.gam","SL.glmnet","SL.glm.interaction", "SL.mean","SL.ranger", "SL.xgboost
#SL.library <- c("SL.glmnet")
SL.lasso = function(...) {
  SL.glmnet(..., alpha=1)
}
SL.library <- c("SL.lasso")

Y <- data$assets
A <- data$e401k
# incomecat is nominal (5 categories); use the one-hot dummies created above
# rather than the raw numeric column, for the same reason as in the
# DoubleML example.
# Two coercions, both for npcausal/SuperLearner:
#   * as.data.frame(): `data` became a data.table in the DoubleML chunk
#     (setDT() works by reference), and data.table evaluates a character
#     vector in j instead of selecting columns.
#   * as.numeric on every column: read_dta() returns haven_labelled columns,
#     which SuperLearner rejects.
W <- as.data.frame(data)[, c("age", "educ", grep("^incomecat_", names(data), value = TRUE), "pension", "n
W <- as.data.frame(lapply(W, as.numeric))
# Wrapped so a failure inside npcausal is reported rather than stopping the render.
aipw <- tryCatch(
  ate(y = Y, a = A, x = W, nsplits = 10, sl.lib = SL.library),
  error = function(e) {
    message("npcausal::ate() failed: ", conditionMessage(e))
    NULL
  }
)
```

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     parameter       est        se    ci.ll    ci.ul pval
1      E{Y(0)} 13968.168  828.5825 12344.15 15592.19    0
2      E{Y(1)} 22032.806  890.5290 20287.37 23778.24    0
3 E{Y(1)-Y(0)}  8064.638 1157.6473  5795.65 10333.63    0
```

```
if (!is.null(aipw)) aipw$res
```

```
     parameter       est        se    ci.ll    ci.ul pval
1      E{Y(0)} 13968.168  828.5825 12344.15 15592.19    0
2      E{Y(1)} 22032.806  890.5290 20287.37 23778.24    0
3 E{Y(1)-Y(0)}  8064.638 1157.6473  5795.65 10333.63    0
```

Note here to match the Stata results, I included only lasso (`glmnet`) in the SuperLearner library. `npcausal` gives an ATE of $8,065 with a standard error of $1,158, against $8,186 from Stata's AIPW.

Or we can use a similar package with more functionalities: "AIPW".

```
library(AIPW)
library(SuperLearner)

#SuperLearner libraries for outcome (Q) and exposure models (g)
#sl.lib <- c("SL.glmnet")

AIPW_sl <- AIPW$new(Y= Y,
                    A= A,
```

```
                    W= W,
                    Q.SL.library = SL.library,
                    g.SL.library = SL.library,
                    k_split = 10,
                    verbose=FALSE)

AIPW_sl$fit()
AIPW_sl$summary()
AIPW_sl$result
```

```
                  Estimate        SE   95% LCL  95% UCL    N
Mean of Exposure 22148.599  880.3483 20423.117 23874.08 3682
Mean of Control  13907.211  826.3618 12287.541 15526.88 6231
Mean Difference   8241.389 1149.5322  5988.306 10494.47 9913
```

```
# library(ggplot2)
# AIPW_sl$plot.p_score()
# AIPW_sl$plot.ip_weights()
```

`AIPW` gives $8,241 with a standard error of $1,150, close to `npcausal` and to Stata's AIPW.

40.3. Summary

Stata 19's `cate` command estimates conditional average treatment effects with partialling out, which is double machine learning, or with AIPW. It uses lasso or random forests for the nuisance models. A fully interacted regression gives similar numbers, but it is regression adjustment, a different idea. DoubleML, `npcausal` and `AIPW` in R implement the same estimators.

In the 401(k) data, the group ATEs rise from about $4,000 in the lowest income category to $20,350 in the highest. A single ATE would hide that spread. Overall ATEs depend on the learners and the software: $7,553 from DoubleML with lasso, $9,114 with random forests in R against $8,282 in Stata, and AIPW gives $8,186 in Stata, $8,065 from `npcausal` and $8,241 from `AIPW`. The estimates span $7,553 to $9,114 with standard errors between $1,150 and $1,375, so the choice of learner and software moves the ATE by up to about one standard error.

Systematic treatment: R · Julia.

41. Spatial econometrics introduction

41.1. Why do we need spatial econometrics?

'Everything is related to everything else, but near things are more related than distant things' (Tobler 1970). This is the first law of geography. This is the reason why we need spatial econometrics. The spatial econometrics is to account for the spatial dependence in the data. The spatial dependence is the correlation between the observations that are close to each other. This is a common feature in many datasets. For example, in the housing market, the price of a house is related to the prices of the houses nearby. In the crime data, the crime rate in one neighborhood is related to the crime rate in the neighboring neighborhoods. In the health data, the health outcomes of people in one area are related to the health outcomes of people in the neighboring areas. In the environmental data, the pollution level in one area is related to the pollution level in the neighboring areas. In the economic data, the economic performance of one region is related to the economic performance of the neighboring regions. In the social network data, the behavior of one person is related to the behavior of the people in the neighboring areas.

Econometricians love OLS. It is worth being precise about *what* spatial dependence breaks. Spatially correlated **errors** alone (the term $u = \rho W u + \epsilon$ below, with exogenous regressors) do not make the OLS coefficients inconsistent — they make the conventional standard errors wrong, so inference is invalid even though the point estimates are fine. OLS coefficients become **biased/inconsistent** in the genuinely simultaneous or omitted-variable cases: a spatially lagged *outcome* Wy on the right-hand side (endogenous by construction), omitted spatially correlated confounders, or endogenous spillovers. The sections below treat these cases separately: S2SLS for the spatial-lag-of-outcome model, and spatial-error models for the correlated-error case.

41.2. A general linear spatial model

$$y = \lambda W y + X\beta + WX\gamma + u \tag{41.1}$$

$$u = \rho W u + \epsilon \tag{41.2}$$

where Y is the dependent variable, X is the matrix of independent variables, β is the vector of coefficients, and u is the error term. The error term u is spatially correlated.

We can see the major difference is W. The term Wy is called "spatially lagged y", which in my opinion is misleading. This term is likely from time-series counter-part of lagged y, but there is nothing about lags in the spatial data. W is the spatial weights matrix. It is a matrix that describes the spatial structure of the data. The simplest example of W is the case that we define an element of W, $w_{i,j}$ to be 1 if unit i and j are neighbors, 0 if not.

In this simplest case, our model is saying we have a pattern we need to take account of: neighboring units are related, they seem to display similar pattern in terms of our study. This commonality of neighbors can be causal, for example, if we study the effect of minimum wage effect on employment, then if the neighboring state's wage increase might negatively impact the focal state's employment, because people go for higher wages. This is the spatial spillover effect. This can also be non-causal, for example, if we study the effect of temperature on crop yield, then the neighboring state's temperature might be a good predictor of the focal state's temperature. This is the spatial autocorrelation. In either case, we need to take account of the spatial structure in the data.

The reason behind the spatial pattern can be all kinds. It can be x_i causing y_j, or y_i causing y_j, or some unobserved factors causing both y_i and y_j, or some other reasons. Ignoring this spatial structure could make OLS biased. Betz, Cook, and Hollenbach (2020) show that ignoring spatial interdependence in the outcome results in asymptotically biased estimates even when instruments are randomly assigned.

41.3. Estimation with S2SLS for a less general model

Because we have the term Wy, we introduced endogeneity. Wy is necessarily correlated with u, even if u is not spatially correlated. Suppose u is not spatially correlated.

The spatial two stage least squares (S2SLS) estimator is one solution. Usually we use X and spatial lags of X as instruments. Typically instruments H is (X, WX, W^2X).

S2SLS is similar to 2SLS. The first stage is

$$\hat{Z} = PZ = [X, WX, \hat{Wy}] \tag{41.3}$$

where

$$\hat{Wy} = PWy \tag{41.4}$$

and

$$P = H(H'H)^{-1}H' \tag{41.5}$$

Then

$$\hat{\theta}_{S2SLS} = (\hat{Z}'Z)^{-1}\hat{Z}'y \tag{41.6}$$

where $\hat{\theta}_{S2SLS}$ is the full coefficient vector (β, γ, λ).

The first stage is to regress WY on H . The second stage is to regress Y on the predicted WY from the first stage and X and WX. The coefficient of X is the S2SLS estimator.

41.4. Estimation with a more general model

Suppose instead the error term is spatially correlated

$$u = \rho W u + \epsilon \tag{41.7}$$

Then the estimation is more complicated. The generalized spatial two stage least squares (GS2SLS) estimator is needed. See Kelejian and Prucha (1998) for details.

41.5. An example

This example is from R package "sphet".

The Boston data contain information on property values and characteristics in the area of Boston, Massachusetts and has been widely used for illustrating spatial models. Specifically, there is a total of 506 units of observation for each of which a variety of attributes are avail- able, such as: (corrected) median values of owner-occupied homes (CMEDV); per-capita crime rate (CRIM); nitric oxides concentration (NOX); average number of rooms (RM); proportions of residential land zoned for lots over 25,000 sq. ft (ZN); proportions of non-retail business acres per town (INDUS); Charles River dummy variable (CHAS); proportions of units built prior to 1940 (AGE); weighted distances to five Boston employment centers (DIS); index of accessibil- ity to highways (RAD); property-tax rate (TAX); pupil-teacher ratios (PTRATIO); proportion of blacks (B); and % of the lower status of the population (LSTAT).

The dataset with Pace's tract coordinates is available to the R community as part of "spdep".

The spatial weights matrix is a sphere of influence neighbors list also available from spdep once the Boston data are loaded. Here the weight matrix comes with the data. Usually it's the most heavy lifting part of the study.

```
library(sphet)
library(spdep)
data("boston", package = "spData")
listw <- nb2listw(boston.soi)
```

The function read.gwt2dist reads a '.GWT' file (e.g., generated using the function distance). In this example we are using the matrix of tract point coordinates projected to UTM zone 19 boston.utm available from spdep to generate a '.GWT' file of the 10 nearest neighbors.

```
id1 <- seq(1, nrow(boston.utm))
tmp <- distance(boston.utm, region.id = id1, output = TRUE,
                type = "NN", nn = 10, shape.name = "shapefile",
                region.id.name = "id1", firstline = TRUE,
                file.name = "boston_nn_10.GWT")
coldist <- read.gwt2dist(file = "boston_nn_10.GWT", region.id = id1, skip = 1)
class(coldist)
```

```
[1] "sphet"    "distance" "nb"       "GWT"
```

```
summary(coldist)
```

```
 Number of observations:
 n: 506

 Distance summary:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.5441  0.9588  1.5843  2.0848  2.6389 11.6388

 Neighbors summary:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     10      10      10      10      10      10
```

The `boston` data are 506 census tracts. The 10-nearest-neighbour weights give every tract exactly 10 neighbours, with distances running from 0.54 to 11.64 and a median of 1.58.

Spatial models in sphet are fitted by the functions stslshac and gstslshet. "stslshac" is Spatial two stages least square with HAC standard errors. "gstslshet" is GM estimation of a Cliff-Ord type model with Heteroskedastic Innovations, which is the general model we discussed before, with spatially correlated errors.

The theoretical model above is the Spatial Durbin Model (SDM), which includes WX as regressors (with their own coefficient γ), not just as instruments for Wy. If we only pass X (no WX) in the formula below, `stslshac`'s `W2X` argument still uses spatial lags of X internally, but only as *instruments* for the endogenous Wy – γ is never estimated, so the fitted model is actually the simpler Spatial Autoregressive (SAR) model ($\gamma = 0$), not SDM. To match the SDM specification, include WX explicitly as regressors:

```
# Build X with the transformations used in the model, then lag those same
# columns, so that WX is W times the X that enters the equation.
boston.c2 <- boston.c
boston.c2$CHAS    <- as.numeric(as.character(boston.c$CHAS))
boston.c2$NOX2    <- boston.c$NOX^2
boston.c2$RM2     <- boston.c$RM^2
boston.c2$logDIS  <- log(boston.c$DIS)
boston.c2$logRAD  <- log(boston.c$RAD)
boston.c2$logLSTAT <- log(boston.c$LSTAT)
boston_x_vars <- c("CRIM","ZN","INDUS","CHAS","NOX2","RM2","AGE","logDIS","logRAD",
                   "TAX","PTRATIO","B","logLSTAT")
for (v in boston_x_vars) {
  boston.c2[[paste0("W_", v)]] <- lag.listw(listw, boston.c2[[v]])
}

sdm_formula <- as.formula(paste(
```

```
  "log(CMEDV) ~", paste(boston_x_vars, collapse = " + "), "+",
  paste0("W_", boston_x_vars, collapse = " + ")
))

res <- stslshac(sdm_formula,
                data = boston.c2, listw,
                distance = coldist, type = "Triangular", HAC = TRUE)
summary(res)
```

```
=======================================================
=======================================================
                  Spatial Lag Model
                 HAC standard errors
=======================================================
=======================================================

Call:
stslshac(formula = sdm_formula, data = boston.c2, listw = listw,
    HAC = TRUE, distance = coldist, type = "Triangular")

Residuals:
    Min.  1st Qu.   Median  3rd Qu.     Max.
-0.65686 -0.05883 -0.00552  0.06881  0.63021

Coefficients:
               Estimate SHAC St.Er. t-value  Pr(>|t|)
Wy           0.88470068  0.09510328  9.3025 < 2.2e-16 ***
(Intercept)  0.39658921  0.47701731  0.8314 0.4057512
CRIM        -0.00456290  0.00143993 -3.1688 0.0015305 **
ZN           0.00074768  0.00048502  1.5416 0.1231800
INDUS       -0.00079659  0.00294332 -0.2706 0.7866658
CHAS        -0.05704354  0.03840546 -1.4853 0.1374650
NOX2        -0.02036599  0.21906066 -0.0930 0.9259277
RM2          0.00819110  0.00286987  2.8542 0.0043149 **
AGE         -0.00137929  0.00047500 -2.9038 0.0036871 **
logDIS      -0.12367413  0.09563235 -1.2932 0.1959333
logRAD       0.06301327  0.02182438  2.8873 0.0038858 **
TAX         -0.00050391  0.00013156 -3.8302 0.0001281 ***
PTRATIO     -0.01262399  0.00596866 -2.1150 0.0344262 *
B            0.00056687  0.00012017  4.7170 2.393e-06 ***
logLSTAT    -0.23596749  0.04002901 -5.8949 3.749e-09 ***
W_CRIM       0.00013364  0.00286797  0.0466 0.9628336
W_ZN        -0.00071594  0.00054990 -1.3019 0.1929355
W_INDUS      0.00046770  0.00358762  0.1304 0.8962768
W_CHAS       0.08705789  0.06268800  1.3887 0.1649091
W_NOX2      -0.11358804  0.23988785 -0.4735 0.6358531
W_RM2       -0.00499204  0.00404315 -1.2347 0.2169461
```

```
W_AGE        0.00121748  0.00060442  2.0143 0.0439774 *
W_logDIS     0.07529641  0.09535585  0.7896 0.4297404
W_logRAD    -0.05066086  0.02863080 -1.7695 0.0768182 .
W_TAX        0.00052235  0.00017520  2.9815 0.0028688 **
W_PTRATIO    0.00947813  0.00769857  1.2312 0.2182651
W_B         -0.00052453  0.00014948 -3.5089 0.0004500 ***
W_logLSTAT   0.22161764  0.05307447  4.1756 2.972e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The spatial lag coefficient `Wy` is 0.885 with a HAC standard error of 0.095, $t = 9.3$ — strong spatial dependence in house values, which is what motivates the whole exercise. Among the own-tract regressors, log(LSTAT) (`logLSTAT`) enters at -0.236 and RM^2 (`RM2`) at 0.0082. The lagged regressors `W_` are W applied to these same transformed columns.

The function gstslshet is used to estimate the GS2SLS model. Unlike `stslshac`, `gstslshet`'s GMM machinery hits an exactly-singular matrix on this dataset once the WX regressors are added (likely due to near-collinearity introduced by the additional lagged terms combined with its own internal instrument construction), so we keep the original (SAR, no WX) specification here — this estimates the spatial-error model with $\gamma = 0$, not the full Durbin extension shown above for `stslshac`.

```
res <- gstslshet(log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2) + I(RM^2)
                 + AGE + log(DIS) + log(RAD) + TAX + PTRATIO + B + log(LSTAT),
                 data = boston.c, listw = listw, initial.value = 0.2)
summary(res)
```

```
Call:
gstslshet(formula = log(CMEDV) ~ CRIM + ZN + INDUS + CHAS + I(NOX^2) +
    I(RM^2) + AGE + log(DIS) + log(RAD) + TAX + PTRATIO + B +
    log(LSTAT), data = boston.c, listw = listw, initial.value = 0.2)

Residuals:
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-0.569385 -0.073165 -0.001675  0.000531  0.071500  0.740309

Coefficients:
               Estimate  Std. Error t-value  Pr(>|t|)
(Intercept)  2.51316605  0.26749367  9.3952 < 2.2e-16 ***
CRIM        -0.00662744  0.00144522 -4.5858 4.523e-06 ***
ZN           0.00038299  0.00036563  1.0475 0.2948733
INDUS        0.00159352  0.00179772  0.8864 0.3753967
CHAS1       -0.00447974  0.03689065 -0.1214 0.9033480
I(NOX^2)    -0.27295899  0.11561412 -2.3609 0.0182283 *
I(RM^2)      0.00744059  0.00199637  3.7270 0.0001937 ***
AGE         -0.00045400  0.00045572 -0.9962 0.3191333
log(DIS)    -0.16517174  0.03484858 -4.7397 2.140e-06 ***
```

```
log(RAD)     0.07453521  0.01752830  4.2523 2.116e-05 ***
TAX         -0.00041956  0.00010763 -3.8981 9.697e-05 ***
PTRATIO     -0.01412661  0.00410143 -3.4443 0.0005725 ***
B            0.00035970  0.00011182  3.2168 0.0012964 **
log(LSTAT)  -0.24593826  0.03213364 -7.6536 1.954e-14 ***
lambda       0.42407826  0.04463747  9.5005 < 2.2e-16 ***
rho          0.29587455  0.08614291  3.4347 0.0005932 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This specification separates the two channels the previous one merged. In the notation of Equation 41.1 and Equation 41.2, which `gstslshet` also uses, λ, the spatial lag on y, is 0.424 with a standard error of 0.045, and ρ, the spatial correlation in the errors, is 0.296 with 0.086. Both are significant. Note $\lambda = 0.424$ here against `Wy` $= 0.885$ above. The two fits differ in two ways at once: this one allows spatially correlated *errors*, and it drops the spatially lagged covariates WX. So the fall in λ cannot be attributed to the error structure alone. Part of what the lag model attributed to neighbours' outcomes may be correlated unobservables, and part may be the WX terms. Separating them needs a comparison that holds the mean specification fixed. The coefficients are not spillover effects either: with a spatial lag, the effect of a regressor runs through the multiplier $(I - \lambda W)^{-1}$, so direct and indirect impacts have to be computed from it.

In stata, you would use "spivregress" which replaced user-written "spivreg". Stata has a set of commands for spatial econometrics. Check out stata's "sp" manual.

41.6. Summary

Spatial models let a unit's outcome depend on its neighbours through a weights matrix W. Spatially correlated errors alone leave OLS consistent but invalidate its standard errors. A spatial lag of the outcome, Wy, is endogenous, so OLS is inconsistent. Spatial two-stage least squares instruments Wy with spatial lags of the regressors. GS2SLS also allows spatially correlated errors. To fit the spatial Durbin model, WX must enter as regressors. Using it only as instruments fits the simpler spatial autoregressive model.

In the Boston house-price data, the spatial Durbin lag model gives a spatial coefficient of 0.885. Once correlated errors are allowed for, the lag falls to 0.424 and the error correlation is 0.296. Much of what looked like neighbours' outcomes affecting each other is reassigned to correlated unobservables. That is the identification problem of spatial econometrics, visible in two numbers.

42. Causal simulation

Evans and Didelez (2024) address the inverse problem: given a causal model, what joint distribution does it imply? The usual direction is observational data to causal structure to parameter estimates; this paper starts from the structure and asks what data it would generate.

Consider the joint distribution $p(x, a, y)$, where x is covariates, a is treatment, and y is outcome.

$$\begin{aligned} p(x,a,y) &= p(x,a)p(y|x,a) \\ &= p(x,a)p_a(y|x) \\ &= p(x,a)\frac{p_a(x,y)}{p_a(x)} \\ &= p(x,a)\frac{p_a(x)p_a(y)c(x,y|a)}{p_a(x)} \\ &= p(x,a)p_a(y)c(x,y|a) \end{aligned} \tag{42.1}$$

The joint distribution factorizes into the joint of (x, a), the interventional marginal $p_a(y) = p(y \mid do(a))$, and the copula $c(x, y \mid a)$. The second line is where the causal content enters, and it is not an algebraic identity: replacing the observational conditional $p(y \mid x, a)$ by the interventional $p_a(y \mid x)$ requires the causal model to make them agree – consistency together with exchangeability given x, or the equivalent structural condition. For an arbitrary observed joint distribution the two differ, and the rest of the factorization inherits that assumption. The derivation starts from $p(y \mid x, a)$ and absorbs the x-dependence into the copula, leaving only the marginal $p_a(y)$. Each piece can be specified separately: (x, a) as the "past," $p_a(y)$ as the marginal structural model, and the copula as the dependence structure between x and y given a.

A copula links a joint distribution to its marginals:

$$p(x, y) = f(x)g(y)c(F(x), G(y)) \tag{42.2}$$

where $F(x)$ and $G(y)$ are the marginal distribution functions of x and y. Keep the two objects apart: the copula C is a distribution function on $[0, 1]^2$ with uniform marginals, and c in Equation 42.2 is its *density*, which is what multiplies $f(x)g(y)$. It can be used to generate joint distributions from marginal distributions.

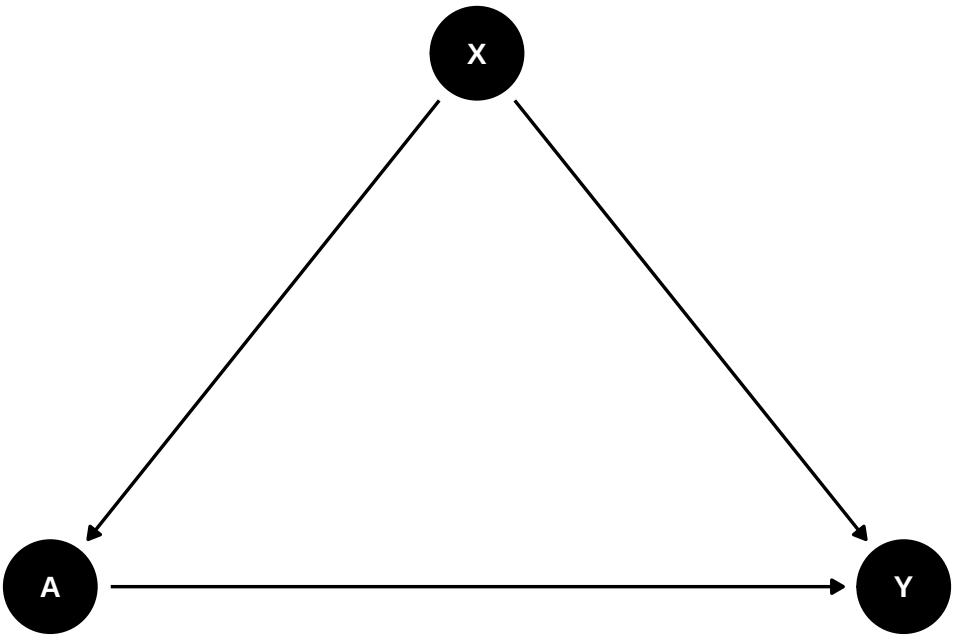
Here x, a and y can all be vectors.

42.1. Examples

What do we use this for? We can use this to simulate data from a causal model, and then fit different models to the simulated data. This allows to compare the performance of different models.

42.1.1. Example 1

Suppose we have a causal graph like this:



We need to specify $P(y|do(a)) = \sum_x P(x)P(y|x,a)$, where $P(x)$ is the marginal distribution of x, and $P(y|x,a)$ is the conditional distribution of y given x and a. This is the "marginal structural model" (Robins, Hernán, and Brumback 2000). Also the "g-formula".

We also need to specify $P(x,a)$, the "past".

Finally we need to specify the copula $c(x,y|a)$, which captures the dependence between x and y given a. Depending on different situations, different copula can be used.

In the "causl" package example, we specify:

$$X \sim N(0,1) \tag{42.3}$$

$$A|X = x \sim N(x/2,1) \tag{42.4}$$

$$Y|do(A = a) \sim N((a-1)/2,1) \tag{42.5}$$

and Gaussian copula with correlation $\rho = 2expit(1) - 1$.

```
library(causl)
# formulae corresponding to covariates, treatments, outcomes and the dependence
forms <- list(X ~ 1, A ~ X, Y ~ A, ~ 1)
# vector of model families (3=gamma/exponential, 1=normal/Gaussian)
fam <- c(1, 1, 1, 1)
# list of parameters, including 'beta' (regression params) and 'phi' dispersion
pars <- list(X = list(beta=0, phi=1),
             A = list(beta=c(0,0.5), phi=1),
             Y = list(beta=c(-0.5,0.5), phi=1),
```

```
             cop = list(beta=1))

## now create a `causl_model` object
cm <- causl_model(formulas=forms, family=fam, pars=pars,method="inversion")

# now simulate 1000 observations
set.seed(123456)
data <- rfrugal(n=1000, causl_model=cm)
head(data)
```

```
            X           A          Y
1  0.83373317  1.10709111  1.1416713
2 -0.27604777 -0.27218853 -1.0495996
3 -0.35500184  0.77575209  1.3179678
4  0.08748742 -0.05959823 -0.6106023
5  2.25225573  0.61909216  2.0789188
6  0.83446013  0.52019822  0.3229069
```

In this example, first we specify the structure of the model using a list of formulas. The first formula is for the covariates, the second for the treatments, the third for the outcomes, and the fourth for the dependence structure (copula). Note that outcome Y only depends on A in this interventional distribution.

Then we specify the families of the random variables. Here we use normal distribution for X, normal distribution for A and Y, and Gaussian copula for the dependence structure.

Finally we specify the parameters of the model, including regression coefficients and dispersion parameters. For example, $Y \sim A$ has two coefficients, intercept -.5 and slope .5.

The `causl_model` object is then created using these components.

In this simple example, if we run a regression with A, X in the model, on the simulated data, we would get the correct effect back:

```
lm1 <- lm(Y ~ A*X, data=data)
summary(lm1)
```

```

Call:
lm(formula = Y ~ A * X, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-2.6326 -0.6212 -0.0105  0.6051  3.3132

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.491922   0.030413 -16.175   <2e-16 ***
```

```
A            0.493471   0.027820  17.738   <2e-16 ***
X            0.468038   0.031878  14.682   <2e-16 ***
A:X         -0.009247   0.022360  -0.414    0.679
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8953 on 996 degrees of freedom
Multiple R-squared:  0.4875,    Adjusted R-squared:  0.4859
F-statistic: 315.8 on 3 and 996 DF,  p-value: < 2.2e-16
```

The true interventional model is $Y \mid do(A = a) \sim N((a-1)/2, 1)$, so the true intercept is -0.5 and the true coefficient on A is 0.5. The naive regression of Y on A and X gives $A = 0.493$ and an intercept of -0.492, with the interaction $A{:}X$ at -0.009 and insignificant ($p = 0.68$), which is correct — the DGP has no such interaction.

Note this regression *also* includes X, which is the adjustment the g-formula calls for. Its coefficient of 0.468 is not a causal quantity; it is there to close the backdoor.

We can also use MLE:

```
out <- fit_causl(data, formulas = list(X ~ 1, Y ~ A, ~ 1), family = c(1, 1, 1))
out
```

```
log-likelihood:  -2716.696
X ~ 1
              est.   s.e. sandwich
(intercept) 0.0108 0.0314   0.0314
  residual s.e.:  0.983 0.044 0.0447

Y ~ A
              est.   s.e. sandwich
(intercept) -0.491 0.0319   0.0319
A            0.494 0.0278   0.0295
  residual s.e.:  1.01 0.0472 0.0475

copula parameters:
cop ~ 1
             est.  s.e. sandwich
(intercept) 0.998 0.069   0.0734
```

`fit_causl` recovers the same structure directly from the frugal parameterisation: intercept -0.491 and A at 0.494, against the true -0.5 and 0.5, with a copula parameter of 0.998 against the specified 1. The sandwich standard errors barely differ from the model-based ones, which is what should happen when the model is correctly specified.

42.1.2. Plasmode simulation

In reality, we have a data set and we have a causal model. We want to simulate data from the causal model, but we want the simulated data to have the same distribution as the original data. This is called "plasmode simulation". For example, we don't want to use the real outcome variable. Instead, we simulate X's from the original data, Then simulate Y and A from the causal model. We can then test which method works better in estimating the causal effect of A on Y.

```
# A tibble: 4,802 x 58
     x_1 x_2     x_3   x_4   x_5   x_6   x_7   x_8   x_9  x_10  x_11  x_12  x_13
   <int> <fct> <dbl> <dbl> <int> <int> <int> <int> <int> <int> <int> <int> <int>
 1    29 C         1     7    60    85     0     0     1     0     0     1     0
 2    27 C         0     0    64   178     0     0     0     0     0     2     1
 3    27 C         0     0    60   102     0     0     0     0     0     1     0
 4    37 C         0     0    65   174     0     0     0     0     0     1     0
 5    24 C        20    14    63   129     0     0     0     0     0     1     0
 6    27 C        40    15    63   135     0     0     0     0     0     1     1
 7    26 C        20     8    69   140     0     0     0     1     0     0     0
 8    33 C         0     0    60   110     0     0     0     1     0     0     1
 9    28 C         3     5    61   160     0     0     0     0     0     3     1
10    31 C         0     0    63   114     0     1     0     0     0     0     0
# i 4,792 more rows
# i 45 more variables: x_14 <int>, x_15 <int>, x_16 <int>, x_17 <int>,
#   x_18 <int>, x_19 <int>, x_20 <int>, x_21 <fct>, x_22 <int>, x_23 <int>,
#   x_24 <fct>, x_25 <int>, x_26 <int>, x_27 <int>, x_28 <int>, x_29 <int>,
#   x_30 <int>, x_31 <int>, x_32 <int>, x_33 <int>, x_34 <int>, x_35 <int>,
#   x_36 <int>, x_37 <int>, x_38 <int>, x_39 <int>, x_40 <int>, x_41 <int>,
#   x_42 <int>, x_43 <int>, x_44 <int>, x_45 <int>, x_46 <int>, x_47 <int>, ...
```

```r
# Model for the causal effect of smoking on birthweight
forms <- list(list(),
              list(A ~ x_1 + x_3 + x_4),
              list(Y ~ A),
              list(~ 1))
# fams <- list(integer(0), 5, 1, 1)
fams <- list(integer(0), "binomial", "gaussian", 1)
pars <- list(A = list(beta=c(-1.5,0.03,0.02,0.05)),
             Y = list(beta=c(3200, -500), phi=400^2),
             cop = list(beta=-1))

cm2 <- causl_model(formulas=forms, family=fams, pars=pars, dat = dat)

set.seed(123456)
data2 <- rfrugal(causl_model=cm2)
head(data2)
```

```
  x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_10 x_11 x_12 x_13 x_14 x_15 x_16 x_17
```

```
1  29   C   1   7  60  85   0   0   1    0    0    1    0    0    2    0    0
2  27   C   0   0  64 178   0   0   0    0    0    2    1    0    2    0    0
3  27   C   0   0  60 102   0   0   0    0    0    1    0    0    0    0    0
4  37   C   0   0  65 174   0   0   0    0    0    1    0    0    1    0    0
5  24   C  20  14  63 129   0   0   0    0    0    1    0    0    0    0    0
6  27   C  40  15  63 135   0   0   0    0    0    1    1    2    2    0    0
  x_18 x_19 x_20 x_21 x_22 x_23 x_24 x_25 x_26 x_27 x_28 x_29 x_30 x_31 x_32
1   12   35   10    J    1   43    B   17   64  105   30   11    0    0    0
2   12   75   12    J    1   50    E   12   90  183   40    8    0    0    0
3   12   35   10    J    1   57    E   14   70  121   27   12    0    0    0
4   11   35   12    J    1   43    E   10   80  185   39   10    0    0    0
5    9   35   15    J    1   33    E   14   70  160   28    7    0    0    0
6    9   45   10    J    1   40    E   15   64  160   31    9    0    0    0
  x_33 x_34 x_35 x_36 x_37 x_38 x_39 x_40 x_41 x_42 x_43 x_44 x_45 x_46 x_47
1    0   79   28  340   48    1    9    9    3    0    3   54   16    0    0
2    0   69   30  430   73    0    9    9    3    0    5   51   16    0    0
3    0   84   29  450   76    0    3    6    2    0    4   41   13    0    0
4    0   82   30  440   78    1    8    9    2    0    8   70   16    0    0
5    1   85   35  360   71    1    5    8    0    2    3   65   18    0    0
6    1   81   28  515   56    1    7    9    2    0    1   63   20    0    0
  x_48 x_49 x_50 x_51 x_52 x_53 x_54 x_55 x_56 x_57 x_58 A        Y
1    0    0    0    0    0    0    0    0    0   45   39 1 2125.677
2    0    0    0    0    0    0    0    0    0   46   42 1 2689.114
3    0    1    0    0    0    0    0    0    0   45   40 0 3442.800
4    0    0    0    0    0    0    0    0    0   47   40 0 3212.584
5    0    2    0    0    0    0    0    0    0   47   43 0 2992.775
6    0    0    0    0    0    0    0    0    0   45   44 0 2738.893
```

```
# we can also use rfrugalParam(), which is older version of rfrugal().
#datAY <- rfrugalParam(formulas=forms, family=fams, pars=pars, dat=dat)
```

In this specification, we have a binary treatment A and a continuous outcome Y. The covariates are x_1, x_3 and x_4. As in the first example above, the copula's `beta` parameter maps to the actual Gaussian correlation via $\rho = 2\,\mathrm{expit}(\beta) - 1$; with `beta=-1` this gives $\rho = 2\,\mathrm{expit}(-1) - 1 \approx -0.46$, not -1 (a correlation of exactly -1 would require $\beta \to -\infty$ and would make the joint distribution singular). We can then simulate the data.

42.2. A more complicated model

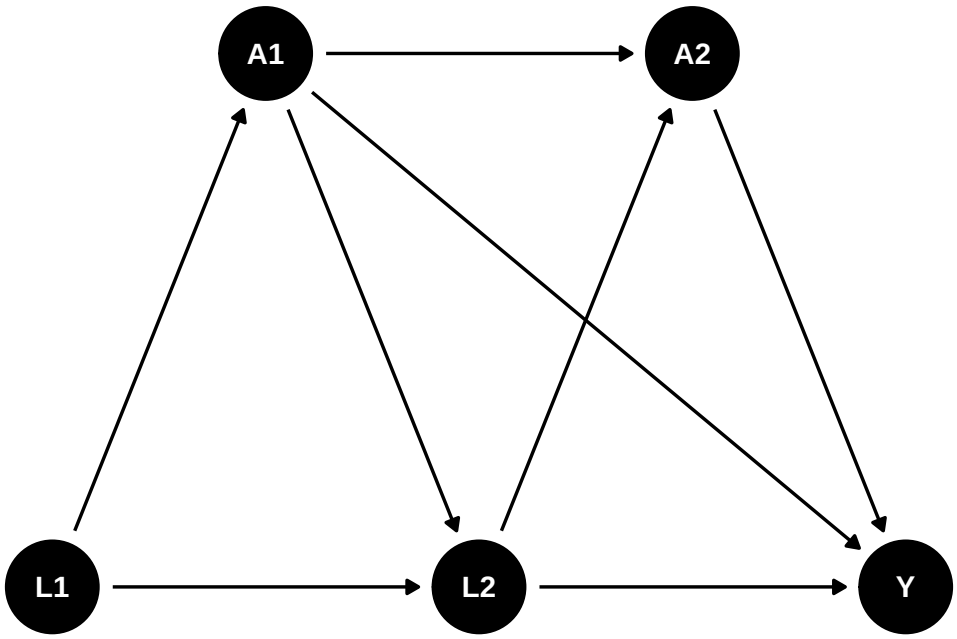
Let's simulate a more complicated model:

```
library(ggplot2)
library(ggdag)

g <- dagify(
  A1 ~ L1,
```

```
  L2 ~ L1,
  L2 ~ A1,
  A2 ~ A1,
  A2 ~ L2,
  Y ~ A2,
  Y ~ L2,
  Y ~ A1,
  exposure = "A2",
  outcome = "Y",
  coords = list(x = c(L1 = 1,  L2 = 2, Y = 3, A1 = 1.5, A2 = 2.5),
                y = c(L1 = 1,  L2 = 1, Y = 1, A1 = 1.5, A2 = 1.5))
)

ggdag(g) +
  theme_dag()
```



Let's see how to write a causal model:

```
# formulae corresponding to covariates, treatments, outcomes and the dependence
forms <- list(list(L1  ~ 1, L2 ~ L1*A1),   # covariates
              list(A1 ~ L1, A2 ~ L2*A1), # treatments
              Y ~ A1*A2*L2,             # outcome
              ~ 1)

# vector of model families (3=gamma/exponential, 1=normal/Gaussian)

fams <- list(c(1, 1), c(5,5), 1, 1)
pars <- list(L1 = list(beta=0, phi=1),
```

```
             L2 = list(beta=c(0.3,0.5,-0.2,-0.1), phi=1),
             A1 = list(beta=c(-0.3,0.4), phi=1),
             A2 = list(beta=c(0.5,0.3,0.1,0), phi=1),
             Y = list(beta=c(0,1,2,0.5,1,0.2,0,0.8), phi=1),
             cop = list(beta = 0.5))

# we can use rfrugalParam() to simulate data from the model, or rfrugal().
#dat <- rfrugalParam(n=1e4, formulas=forms, family = fams, pars=pars)
#head(dat)

cm3 <- causl_model(formulas=forms, family=fams, pars=pars)

set.seed(123456)
data3 <- rfrugal(n=1e4,causl_model=cm3)
head(data3)
```

```
           L1          L2 A1 A2         Y
1  0.83373317  1.91691936  1  1 8.1680612
2 -0.27604777  0.17008294  1  1 5.2276976
3 -0.35500184  0.16907708  1  1 2.8194835
4  0.08748742  0.19688985  0  1 2.6789661
5  2.25225573 -0.29257431  0  0 0.3180188
6  0.83446013  0.02661196  0  1 1.5694426
```

The outcome formula `Y ~ A1*A2*L2` specifies the causal margin, the distribution of Y under interventions on A_1 and A_2 given L_2. The observational regression of Y on the same terms recovers the ATE of A_2, but not every coefficient:

```
m1 <- glm(Y ~ A1*A2*L2, data = data3)
summary(m1)
```

```
Call:
glm(formula = Y ~ A1 * A2 * L2, data = data3)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.11524    0.02110  -5.461 4.85e-08 ***
A1           1.11589    0.03245  34.391  < 2e-16 ***
A2           2.02332    0.02682  75.454  < 2e-16 ***
L2           0.80424    0.01929  41.682  < 2e-16 ***
A1:A2        0.98345    0.04090  24.044  < 2e-16 ***
A1:L2        0.16028    0.02976   5.386 7.37e-08 ***
A2:L2       -0.03446    0.02401  -1.435    0.151
A1:A2:L2     0.83541    0.03693  22.620  < 2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 0.9100255)

    Null deviance: 53902  on 9999  degrees of freedom
Residual deviance:  9093  on 9992  degrees of freedom
AIC: 27446

Number of Fisher Scoring iterations: 2
```

```
library(marginaleffects)
avg_comparisons(
    model = m1,
    variables = "A2")
```

```
 Estimate Std. Error   z Pr(>|z|)   S 2.5 % 97.5 %
     2.51     0.0202 124   <0.001 Inf  2.47   2.55

Term: A2
Type: response
Comparison: 1 - 0
```

The fitted model has $A_2 = 2.023$, $A_1{:}A_2 = 0.983$ and $A_1{:}A_2{:}L_2 = 0.835$, so the effect of A_2 depends on both A_1 and L_2 and cannot be read off any single coefficient. `avg_comparisons` averages over the observed distribution and returns 2.51 with a standard error of 0.020, against a true ATE of 2.5.

The individual coefficients are another matter. The intercept is -0.115, A_1 is 1.116 and L_2 is 0.804, against 0, 1 and 0.5 in `pars$Y$beta`. The Gaussian copula with parameter 0.5 ties Y to the covariates L_1 and L_2, so the observed conditional mean $E[Y \mid A_1, A_2, L_2]$ also depends on L_1, which the regression omits, and the coefficients on terms correlated with L_1 absorb that dependence. The ATE of A_2 survives because A_2 is independent of L_1 given (A_1, L_2) in this DAG.

Let's use "lmtp" package to estimate the treatment effect of A_2 on Y.

```
library(lmtp)

d1 <- function(data, trt) {
  rep(1, nrow(data))
}
A <- "A2"
Y <- "Y"
W <- c("L1", "L2", "A1")

set.seed(34465)

treat <- lmtp_tmle(
```

```
  data = data3,
  trt = "A2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d1,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

show_lmtp(treat)
```

```
     Estimate: 3.046
Std. error: 0.022
```

```
d2 <- function(data, trt) {
  rep(0, nrow(data))
}
control <- lmtp_tmle(
  data = data3,
  trt = "A2",
  outcome = "Y",
  baseline = W,
  outcome_type = "continuous",
  shift = d2,
  folds = 1,
  learners_trt = "SL.glm",
  learners_outcome = "SL.glm"
)

show_lmtp(control)
```

```
     Estimate: 0.542
Std. error: 0.021
```

```
show_lmtp(lmtp_contrast(treat, ref = control))
```

```
  shift   ref estimate std.error conf.low conf.high p.value
1  3.05 0.542      2.5    0.0227     2.46      2.55  <0.001
```

42.3. Summary

Evans and Didelez's frugal parameterization builds a joint distribution from three pieces: the distribution of covariates and treatment, the interventional distribution of the outcome, and a copula linking covariates and outcome given treatment. Because the causal effect is a parameter of the model, we can simulate data with exactly the effect we want, and then check whether an estimator recovers it. Plasmode simulation does the same while keeping the covariate distribution of a real data set.

In the first example the true effect is 0.5. Regression on treatment and the covariate gives 0.493, and `fit_causl` gives 0.494 with the copula parameter near its true value. In the time-varying example, the outcome model has interactions, so no single coefficient is the effect. Averaging over the observed covariates gives an ATE of 2.51 against a true 2.5.

43. Frengression and Engression

In the causal simulation chapter, I used Evans and Didelez (2024)'s decomposition of the joint distribution $p(x, a, y)$ into three pieces:

$$p(x, a, y) = p(x, a) \cdot p_a(y) \cdot c(x, y|a) \tag{43.1}$$

where $p(x, a)$ is the "past" distribution of covariates and treatment, $p_a(y)$ is the causal margin, and $c(x, y|a)$ is the copula linking covariates and outcome given treatment. The `causl` package implements this with parametric families. We specify Gaussian, binomial, and so on for each piece.

But real data can be hard to specify parametrically. We may not know the right copula family. The marginal distributions may have mass points, skewness, or other features that are annoying to write down.

Frengression (Yang et al. 2025) keeps the same decomposition but learns the pieces with neural networks.

There are two uses: causal simulation with a learned generative model, and interventional prediction using learned confounding structure. The second part uses engression, so I start there.

43.1. Engression

Engression (Shen and Meinshausen 2025) stands for energy regression. Standard regression estimates $E[Y|X]$. Engression tries to learn the whole conditional distribution $P(Y|X)$ using neural networks trained with the energy score.

43.1.1. The Energy Score

The energy score is a proper scoring rule for distributions. Given observations Y and samples $\hat{Y}, \hat{Y}'$ from the model:

$$S(Y, \hat{Y}, \hat{Y}') = E\|Y - \hat{Y}\| - \frac{1}{2}E\|\hat{Y} - \hat{Y}'\| \tag{43.2}$$

The first term rewards samples close to the truth. The second term keeps the simulated distribution from collapsing to one point. This is why the method targets the full distribution rather than only the mean.

43.1.2. The Stochastic Network (StoNet)

The building block is a `StoNet`, a neural network that adds random noise $\epsilon \sim N(0, I)$ to the input at each layer:

$$\text{StoNet}(x) = f(x, \epsilon), \quad \epsilon \sim N(0, I) \tag{43.3}$$

Different noise draws produce different outputs. Each output is a sample from the learned conditional distribution. So we can:

- **Predict the mean**: average over many noise draws $\to E[Y|X = x]$
- **Predict quantiles**: take quantiles of the noise draws $\to$ conditional quantile regression
- **Sample**: each forward pass with fresh noise is a sample from $P(Y|X = x)$

43.1.3. Why Engression?

Compared with quantile regression or mixture density networks, engression is useful because:

1. it handles multivariate responses directly;
2. it does not require us to specify a Gaussian mixture or a list of quantiles;
3. it gives samples from the conditional distribution, which is convenient for simulation.

43.1.4. Quick Example

The `engression` R package, separate from `Rfrengression`, provides the standalone estimator. The code below shows the API. The `Rfrengression` examples later are the ones I actually run here.

```
library(engression)

# Nonlinear DGP with heteroskedastic noise
n <- 1000
X <- matrix(rnorm(n * 2), ncol = 2)
Y <- (X[,1])^2 + X[,2] + rnorm(n) * (0.5 + abs(X[,1]))

# Fit engression; it learns the full conditional distribution
engr <- engression(X, Y)

# Predict conditional mean (like lm, but nonlinear)
Yhat_mean <- predict(engr, X, type = "mean")

# Predict conditional quantiles (like quantile regression, but joint)
Yhat_quant <- predict(engr, X, type = "quantiles")

# Sample from the learned distribution (generative)
Ysample <- predict(engr, X, type = "sample", nsample = 1)
```

The point is that engression gives a distributional model of $Y|X$, not just a fitted mean.

43.2. From Engression to Frengression

Frengression is frugal simulation plus engression. It uses three engression models to represent the decomposition of $P(X, A, Y)$. Each sub-network is trained with the energy score.

43.3. The Architecture

Frengression has three sub-networks, matching the three pieces of the decomposition:

Decomposition	causl (parametric)	Frengression (neural net)
$P(X, A)$ — the past	Specified by family + params	`model_xz`: noise-only generator
$P_a(Y)$ — causal margin	Specified by family + params	`model_y`: takes (x, eta), outputs y
$c(X, Y \mid A)$ — copula	Gaussian/Frank/etc. copula	`model_eta`: takes (x, z), outputs eta

The important part is `model_eta`. It takes (x, z) and outputs a latent variable `eta` that carries the dependence between confounders and outcome. During training, a Z-permutation trick forces `eta` to be marginally $N(0, I)$. This means:

- When `eta` comes from `model_eta(x,z)`, it carries the observational confounding information.
- When `eta` is sampled independently from $N(0, 1)$, it gives the interventional prediction.

Same network, different source of `eta`. That is the trick.

43.4. Using Rfrengression

The `Rfrengression` package is a native R implementation using `torch`.

```
library(Rfrengression)
```

43.4.1. Example: Binary Treatment DGP

Use a DGP with binary treatment, binary outcome, and four confounders. We draw $n = 2000$ observations with $w_1, w_2 \sim \text{Bernoulli}(0.5)$, $w_3 \sim U(0, 4)$ and $w_4 \sim U(0, 5)$; treatment is $A \sim \text{Bernoulli}(\Lambda(-0.4 + 0.2w_2 + 0.15w_3 + 0.2w_4 + 0.15w_2w_4))$; and both potential outcomes are Bernoulli with the same logit index shifted by the treatment. Since both are simulated, the realized ATE is known:

```r
set.seed(42)
n <- 2000

w1 <- rbinom(n, 1, 0.5)
w2 <- rbinom(n, 1, 0.5)
w3 <- round(runif(n, 0, 4), 3)
w4 <- round(runif(n, 0, 5), 3)

A <- rbinom(n, 1, plogis(-0.4 + 0.2*w2 + 0.15*w3 + 0.2*w4 + 0.15*w2*w4))
lin <- -1 - 0.1*w1 + 0.3*w2 + 0.25*w3 + 0.2*w4 + 0.15*w2*w4
p1  <- plogis(lin + 1)
p0  <- plogis(lin)
Y.1 <- rbinom(n, 1, p1)
Y.0 <- rbinom(n, 1, p0)
Y <- Y.1 * A + Y.0 * (1 - A)

# Two benchmarks, and they are not the same quantity. The first is the realized
# finite-sample ATE in this particular draw of the Bernoulli outcomes; the second
# averages the true individual probabilities over the sampled covariates, so it
# carries no outcome sampling noise.
true_ate  <- mean(Y.1 - Y.0)
bench_ate <- mean(p1 - p0)
cat("Realized ATE (this draw):", round(true_ate, 4), "\n")
```

```
Realized ATE (this draw): 0.1935
```

```r
cat("Covariate-averaged ATE:  ", round(bench_ate, 4), "\n")
```

```
Covariate-averaged ATE:   0.2024
```

43.4.2. Train Frengression

Training has two stages: first the outcome model (`model_y` + `model_eta`), then the marginal model (`model_xz`).

```r
# torch keeps its own generator; R's set.seed() does not reach it.
torch::torch_manual_seed(1)

x_mat <- matrix(A, ncol = 1)
z_mat <- as.matrix(data.frame(w1, w2, w3, w4))
y_mat <- matrix(Y, ncol = 1)

model <- frengression(x_dim = 1, y_dim = 1, z_dim = 4,
                      x_binary = TRUE, y_binary = TRUE,
                      z_binary_dims = 2, noise_dim = 10)
```

```
# tol = 0 turns off train_y's default early stopping (tol = 0.01), which
# otherwise halts as soon as the two energy-loss terms move by less than 0.01
# between iterations -- after 67 iterations in this draw.
model <- train_y(model, x_mat, z_mat, y_mat,
                 num_iters = 1000, lr = 1e-3, print_every = 500, tol = 0)
```

```
Epoch 1: loss 1.2606, loss_y 0.4787 (0.4977, 0.0380), loss_eta 0.7818 (0.8583, 0.1528)
Epoch 500: loss 0.7616, loss_y 0.1932 (0.3678, 0.3492), loss_eta 0.5683 (1.1332, 1.1297)
Epoch 1000: loss 0.7550, loss_y 0.1892 (0.3752, 0.3720), loss_eta 0.5658 (1.1018, 1.0719)
```

```
model <- train_xz(model, x_mat, z_mat,
                  num_iters = 1000, lr = 1e-4, print_every = 500)
```

```
Epoch 1: loss 3.4499, loss1 3.6071, loss2 0.3143
Epoch 500: loss 1.3388, loss1 2.6926, loss2 2.7078
Epoch 1000: loss 1.3458, loss1 2.7336, loss2 2.7756
```

43.4.3. Estimate ATE

For interventional prediction, `eta` is sampled from $N(0,1)$ instead of being generated from the observed confounding structure:

```
torch::torch_manual_seed(1)

y1 <- predict(model, matrix(1, ncol = 1), type = "mean", nsample = 2000, trim = 0)
y0 <- predict(model, matrix(0, ncol = 1), type = "mean", nsample = 2000, trim = 0)

cat("Frengression ATE:", round(y1 - y0, 4), "\n")
```

```
Frengression ATE: 0.1895
```

```
cat("True ATE:", round(true_ate, 4), "\n")
```

```
True ATE: 0.1935
```

The realized ATE in this draw is 0.1935, and averaging the true probabilities over the sampled covariates gives 0.2024. The gap between those two is outcome sampling noise, not model error, and it is worth keeping them apart: the realized figure is the ATE of the potential outcomes actually drawn, while the covariate-averaged figure is the smoother target the estimator is really chasing. With early stopping turned off, frengression's 0.1895 lands 0.004 below the realized ATE and 0.013 below the covariate-averaged one.

The training chunks carry `cache: false`, so they re-run on every render, and torch keeps a generator of its own that R's `set.seed()` does not reach. Seeding it with `torch_manual_seed()` fixes the

weight initialisation, the training draws, and the sampling inside `predict()`, so the figure above is reproducible from one render to the next. On a CUDA device some nondeterminism can survive the seed, from cuDNN algorithm selection and atomic accumulation in the backward pass.

43.5. Frengression as DGP

Frengression can estimate an ATE, but that is not its main advantage. If I only want an ATE, I would usually use `npcausal`, TMLE, or DoubleML. The more interesting use is to train a data-generating process for benchmarking.

The idea is:

1. Train frengression on observational data, so it learns the confounding structure.
2. Replace the causal margin with a known function via `specify_causal()`.
3. Generate synthetic data with realistic confounding and known ground truth.

```
# Specify a known causal effect: Y = sigmoid(0.2*X + eta)
model_bench <- specify_causal(model, function(x, eta) {
  torch::torch_sigmoid(0.2 * x + eta)
})

# Y is binary: Y = 1{sigmoid(0.2*x + eta) > 0.5} = 1{0.2*x + eta > 0}, eta ~ N(0,1),
# so P(Y = 1 | do(x)) = pnorm(0.2 * x) and the true ATE is pnorm(0.2) - 0.5.
true_ate_spec <- pnorm(0.2) - 0.5
y1_true <- predict(model_bench, matrix(1, ncol = 1), type = "mean", nsample = 5000, trim = 0)
y0_true <- predict(model_bench, matrix(0, ncol = 1), type = "mean", nsample = 5000, trim = 0)
cat("True ATE (specified, exact):", round(true_ate_spec, 4), "\n")
```

```
True ATE (specified, exact): 0.0793
```

```
cat("Monte Carlo check from predict():", round(as.numeric(y1_true - y0_true), 4), "\n")
```

```
Monte Carlo check from predict(): 0.0748
```

```
# The generator draws eta from the learned model_eta, whose marginal is trained
# toward N(0,1). The ATE in the generated population is P(eta > -0.2) - P(eta > 0)
# under that learned marginal; check it on a large draw.
eta_gen <- torch::with_no_grad({
  xz  <- model_bench$model_xz$forward(n = 100000)
  x_g <- (xz[, 1, drop = FALSE] > 0)$to(dtype = torch::torch_float())
  z_g <- xz[, 2:(1 + model_bench$z_dim), drop = FALSE]
  zb  <- model_bench$z_binary_dims
  if (zb > 0) z_g[, 1:zb] <- (z_g[, 1:zb] > 0)$to(dtype = torch::torch_float())
  as.numeric(as.array(model_bench$model_eta$forward(torch::torch_cat(list(x_g, z_g), dim = 2))))
})
cat(sprintf("Generator eta: mean %.3f, sd %.3f; implied ATE %.4f\n",
            mean(eta_gen), sd(eta_gen), mean(eta_gen > -0.2) - mean(eta_gen > 0)))
```

```
Generator eta: mean 0.044, sd 0.946; implied ATE 0.0785
```

The specified margin has an exact ATE of $\Phi(0.2) - 0.5 = 0.0793$, where Φ is the standard normal distribution function, and `predict()` with `trim = 0` reproduces it to Monte Carlo error. The population the generator actually draws uses the learned η, whose mean is 0.044 and standard deviation 0.946 rather than 0 and 1, and its ATE is 0.0785. We score the benchmark against the exact 0.0793. The 0.0008 gap is the generator's approximation error, small next to the biases below.

Now we can generate synthetic data and benchmark any method:

```r
syn <- sample_joint(model_bench, sample_size = 1000)
cat("Synthetic data: n =", nrow(syn$x), "\n")
```

```
Synthetic data: n = 1000
```

```r
cat("Treatment prevalence:", mean(syn$x), "\n")
```

```
Treatment prevalence: 0.904
```

```r
cat("Outcome prevalence:", mean(syn$y), "\n")
```

```
Outcome prevalence: 0.586
```

43.6. Real Data: LaLonde

This is most useful with real data. The LaLonde job-training data have many of the problems that make simulations hard to write by hand: zeros in prior earnings, uneven covariate distributions, and poor overlap.

```r
library(MatchIt)
data("lalonde", package = "MatchIt")
lalonde$black <- as.integer(lalonde$race == "black")
lalonde$hispan <- as.integer(lalonde$race == "hispan")

x_lal <- matrix(lalonde$treat, ncol = 1)
y_lal <- matrix(lalonde$re78, ncol = 1)
# frengression's z_binary_dims assumes the FIRST z_binary_dims columns of z
# are the binary ones -- so the binary covariates (black, hispan, married,
# nodegree) must come first, followed by the continuous ones (age, educ,
# re74, re75). The original column order (age, educ first) put two
# continuous variables where binary ones were expected, and vice versa.
z_lal <- as.matrix(lalonde[, c("black", "hispan", "married", "nodegree",
                               "age", "educ", "re74", "re75")])
```

```
# Standardize for training
# Standardize only the continuous columns; the first four must stay 0/1.
z_sc <- z_lal
z_sc[, 5:8] <- scale(z_lal[, 5:8])
y_sc <- (y_lal - mean(y_lal)) / sd(y_lal)

model_lal <- frengression(x_dim = 1, y_dim = 1, z_dim = 8,
                          x_binary = TRUE, y_binary = FALSE,
                          z_binary_dims = 4, noise_dim = 10)

model_lal <- train_y(model_lal, x_lal, z_sc, y_sc,
                     num_iters = 1000, lr = 1e-3, print_every = 500)
```

```
Epoch 1: loss 1.4288, loss_y 0.6976 (0.7627, 0.1303), loss_eta 0.7313 (0.8021, 0.1417)
Stopping at iter 96
```

```
model_lal <- train_xz(model_lal, x_lal, z_sc,
                      num_iters = 1000, lr = 1e-4, print_every = 500)
```

```
Epoch 1: loss 1.9727, loss1 2.1577, loss2 0.3699
Epoch 500: loss 1.4583, loss1 2.8513, loss2 2.7859
Epoch 1000: loss 1.4970, loss1 2.9272, loss2 2.8603
```

Now specify a known effect and generate data:

```
model_lal_bench <- specify_causal(model_lal, function(x, eta) {
  0.3 * x + eta
})

# Generate synthetic LaLonde-like data with known ATE = 0.3 (standardized)
syn_lal <- sample_joint(model_lal_bench, sample_size = 2000)
cat("Synthetic LaLonde: n =", nrow(syn_lal$x), "\n")
```

```
Synthetic LaLonde: n = 2000
```

```
cat("Treatment prevalence:", round(mean(syn_lal$x), 3), "\n")
```

```
Treatment prevalence: 0.032
```

```
cat("True ATE:", 0.3, "(standardized) =", round(0.3 * sd(y_lal), 0), "dollars\n")
```

```
True ATE: 0.3 (standardized) = 2241 dollars
```

The synthetic data inherits LaLonde's confounding structure, but now the true effect is known. This is the part that is hard to do with a hand-written DGP.

43.7. Comparison: causl vs frengression

Both packages use the same decomposition:

Feature	causl	Frengression
Marginal $P(X, A)$	Parametric families	Learned by neural net
Causal margin $P_a(Y)$	Parametric families	Learned (or specified)
Copula	Gaussian, Frank, etc.	Learned by `model_eta`
Requires specification	Yes — families + params	No — learns from data
Can learn from real data	Limited (plasmode)	Yes — full joint
Distributional flexibility	Limited by family choice	Arbitrary

The `causl` approach is more transparent because we know exactly what we specified. Frengression is more flexible because it can learn structure we did not write down. I would treat them as complements. Frengression is not the first neural generator for benchmarking: RealCause (Neal, Huang, and Raghupathi 2020) fits flexible generative models to real data and then sets the treatment effect, to build benchmarks with known ground truth that look like real data. What the frugal parameterization adds is that the causal margin is specified directly and separately from the confounding, so the true effect is exact by construction.

43.8. Benchmarking: Frengression vs Semiparametric Methods

Now compare frengression against standard semiparametric causal inference tools on the same data. The three comparison methods are:

- `npcausal`: AIPW estimator with SuperLearner nuisance models
- `tmle`: targeted minimum loss-based estimation with SuperLearner
- `DoubleML`: the interactive regression model (`DoubleMLIRM`) with `ranger`, the doubly robust ATE estimator for a binary treatment

These methods are built for ATE estimation. They should usually be better for this specific estimand. The comparison is still useful because frengression is trying to learn much more than the ATE.

```
library(npcausal)
library(tmle)
library(SuperLearner)
library(DoubleML)
library(mlr3)
library(mlr3learners)
library(tidyverse)
library(knitr)
```

43.8.1. Single-Dataset Comparison

First, run all four methods on the binary DGP data created above.

```
W <- data.frame(w1, w2, w3, w4)
SL.library <- c("SL.earth", "SL.glm.interaction", "SL.mean",
                "SL.ranger", "SL.glmnet")

set.seed(123)
aipw_fit <- ate(y = Y, a = A, x = W, nsplits = 2, sl.lib = SL.library)
```

```
  |
  |                                                                      |   0%
  |
  |=========                                                             |  12%
  |
  |==================                                                    |  25%
  |
  |==========================                                            |  38%
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  |==================================                                    |  50%
  |
  |===========================================                           |  62%
  |
  |====================================================                  |  75%
  |
  |============================================================          |  88%
  |
  |======================================================================| 100%
     parameter       est         se     ci.ll     ci.ul pval
1      E{Y(0)} 0.5697312 0.02038159 0.5297832 0.6096791    0
2      E{Y(1)} 0.7608322 0.01207030 0.7371744 0.7844900    0
3 E{Y(1)-Y(0)} 0.1911010 0.02335102 0.1453330 0.2368690    0
```

```
ate_npcausal <- aipw_fit$res$est[3]
se_npcausal  <- aipw_fit$res$se[3]
ci_npcausal  <- c(aipw_fit$res$ci.ll[3], aipw_fit$res$ci.ul[3])

cat("npcausal ATE:", round(ate_npcausal, 4), "\n")
```

```
npcausal ATE: 0.1911
```

```
cat("SE:", round(se_npcausal, 4), "\n")
```

```
SE: 0.0234
```

```
set.seed(123)
tmle_fit <- tmle(Y = Y, A = A, W = W, family = "binomial",
                 Q.SL.library = SL.library, g.SL.library = SL.library)

ate_tmle <- tmle_fit$estimates$ATE$psi
se_tmle  <- sqrt(tmle_fit$estimates$ATE$var.psi)
ci_tmle  <- tmle_fit$estimates$ATE$CI

cat("TMLE ATE:", round(ate_tmle, 4), "\n")
```

```
TMLE ATE: 0.1916
```

```
cat("SE:", round(se_tmle, 4), "\n")
```

```
SE: 0.0209
```

```
lgr::get_logger("mlr3")$set_threshold("warn")
data_obs <- data.frame(w1, w2, w3, w4, A, Y)

dml_data <- DoubleMLData$new(data_obs,
                             y_col = "Y", d_cols = "A",
                             x_cols = c("w1", "w2", "w3", "w4"))

# Binary treatment with heterogeneous effects: the interactive regression model
# (IRM) targets the ATE; the partially linear model would target a
# variance-weighted average of the effects instead.
learner_g <- lrn("regr.ranger", num.trees = 500, max.depth = 5, min.node.size = 2)
learner_m <- lrn("classif.ranger", num.trees = 500, max.depth = 5, min.node.size = 2,
                 predict_type = "prob")

set.seed(123)
dml_plr <- DoubleMLIRM$new(dml_data, ml_g = learner_g, ml_m = learner_m)
dml_plr$fit()

ate_dml <- dml_plr$coef
se_dml  <- dml_plr$se
ci_dml  <- c(dml_plr$confint()[1], dml_plr$confint()[2])

cat("DoubleML ATE:", round(ate_dml, 4), "\n")
```

```
DoubleML ATE: 0.1806
```

```
cat("SE:", round(se_dml, 4), "\n")
```

```
SE: 0.0231
```

```r
# Frengression ATE already computed above (y1 - y0)
ate_freng <- as.numeric(y1 - y0)

# Repeated prediction from the one fitted network. This measures Monte Carlo
# integration error in predict(), not sampling or training uncertainty: the data
# are not resampled and the network is not refitted. It is not a standard error.
set.seed(123)
ate_mc_draws <- replicate(200, {
  predict(model, matrix(1, ncol = 1), type = "mean", nsample = 500, trim = 0) -
    predict(model, matrix(0, ncol = 1), type = "mean", nsample = 500, trim = 0)
})
mc_error_freng <- sd(ate_mc_draws)
cat("Monte Carlo integration error of the frengression ATE:", round(mc_error_freng, 4), "\n")
```

```
Monte Carlo integration error of the frengression ATE: 0.0278
```

```r
results <- data.frame(
  Method = c("True ATE", "Frengression", "npcausal (AIPW)", "TMLE", "DoubleML (IRM)"),
  ATE = c(true_ate, ate_freng, ate_npcausal, ate_tmle, ate_dml),
  SE = c(NA, NA, se_npcausal, se_tmle, se_dml),       # no SE for frengression
  CI_lower = c(NA, NA, ci_npcausal[1], ci_tmle[1], ci_dml[1]),
  CI_upper = c(NA, NA, ci_npcausal[2], ci_tmle[2], ci_dml[2])
)

results$Bias <- results$ATE - true_ate
results$Covers <- ifelse(is.na(results$CI_lower), NA,
                         results$CI_lower <= true_ate & true_ate <= results$CI_upper)

kable(results, digits = 4,
      caption = "ATE Estimation: Frengression vs Semiparametric Methods")
```

Table 43.3.: ATE Estimation: Frengression vs Semiparametric Methods

Method	ATE	SE	CI_lower	CI_upper	Bias	Covers
True ATE	0.1935	NA	NA	NA	0.0000	NA
Frengression	0.1895	NA	NA	NA	-0.0040	NA
npcausal (AIPW)	0.1911	0.0234	0.1453	0.2369	-0.0024	TRUE
TMLE	0.1916	0.0209	0.1506	0.2326	-0.0019	TRUE
DoubleML (IRM)	0.1806	0.0231	0.1353	0.2259	-0.0129	TRUE

On this single dataset all four are close. TMLE gives 0.1916 and npcausal 0.1911, within 0.003 of the true 0.1935; DoubleML's interactive model gives 0.1806, 0.013 below; frengression gives 0.1895, 0.004 below. We report no standard error for frengression. The spread of repeated predictions from the one fitted network, 0.0278, measures only the Monte Carlo error of the integration in `predict()`: it does not resample the data or refit the network. The three reported intervals all cover the truth.

The semiparametric estimators target the ATE directly. Frengression learns the joint distribution and then computes the ATE as one functional of that distribution. So I would not expect it to be the most precise ATE estimator in this table.

43.8.2. Monte Carlo Benchmarking (Simulated DGP)

This is where frengression is more useful. Use the model trained above, plug in a known causal effect with `specify_causal()`, generate 10 synthetic datasets, and benchmark all four estimators.

```
B <- 10
n_syn <- 2000

methods <- c("Frengression", "npcausal", "TMLE", "DoubleML")
ate_mat <- matrix(NA, nrow = B, ncol = length(methods),
                  dimnames = list(NULL, methods))
se_mat  <- ate_mat
cover_mat <- ate_mat

SL.lib_fast <- c("SL.glm.interaction", "SL.ranger", "SL.mean")

# True ATE of the specified causal margin (exact, derived above)
true_ate_bench <- true_ate_spec

set.seed(2024)
for (b in seq_len(B)) {
  syn <- sample_joint(model_bench, sample_size = n_syn)

  x_syn <- as.numeric(syn$x)
  y_syn <- as.numeric(syn$y)
  z_syn <- as.data.frame(syn$z)
  colnames(z_syn) <- paste0("z", 1:ncol(z_syn))
  df_syn <- cbind(data.frame(A = x_syn, Y = y_syn), z_syn)

  # --- Frengression ---
  tryCatch({
    torch::torch_manual_seed(1000 + b)  # torch's own generator; set.seed() does not reach it
    mod_f <- frengression(x_dim = 1, y_dim = 1, z_dim = ncol(z_syn),
                          x_binary = TRUE, y_binary = TRUE,
                          z_binary_dims = model_bench$z_binary_dims,
                          noise_dim = 10, hidden_dim = 100, num_layer = 3)
    mod_f <- train_y(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                     matrix(y_syn, ncol=1),
                     num_iters = 500, lr = 1e-3, silent = TRUE)
    mod_f <- train_xz(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                      num_iters = 500, lr = 1e-4, silent = TRUE)

    y1_f <- predict(mod_f, matrix(1, ncol=1), type="mean", nsample=1000, trim=0)
```

```
    y0_f <- predict(mod_f, matrix(0, ncol=1), type="mean", nsample=1000, trim=0)
    ate_mat[b, "Frengression"] <- y1_f - y0_f
    # No standard error or interval for frengression: SE and coverage stay NA.
  }, error = function(e) message("Frengression failed on rep ", b, ": ", e$message))

  # --- npcausal ---
  tryCatch({
    fit_np <- ate(y = y_syn, a = x_syn, x = z_syn,
                  nsplits = 2, sl.lib = SL.lib_fast)
    ate_mat[b, "npcausal"] <- fit_np$res$est[3]
    se_mat[b, "npcausal"]  <- fit_np$res$se[3]
    ci <- c(fit_np$res$ci.ll[3], fit_np$res$ci.ul[3])
    cover_mat[b, "npcausal"] <- (ci[1] <= true_ate_bench & true_ate_bench <= ci[2])
  }, error = function(e) message("npcausal failed on rep ", b, ": ", e$message))

  # --- TMLE ---
  tryCatch({
    fit_tmle <- tmle(Y = y_syn, A = x_syn, W = z_syn, family = "binomial",
                     Q.SL.library = SL.lib_fast, g.SL.library = SL.lib_fast)
    ate_mat[b, "TMLE"] <- fit_tmle$estimates$ATE$psi
    se_mat[b, "TMLE"]  <- sqrt(fit_tmle$estimates$ATE$var.psi)
    ci_t <- fit_tmle$estimates$ATE$CI
    cover_mat[b, "TMLE"] <- (ci_t[1] <= true_ate_bench & true_ate_bench <= ci_t[2])
  }, error = function(e) message("TMLE failed on rep ", b, ": ", e$message))

  # --- DoubleML ---
  tryCatch({
    dml_d <- DoubleMLData$new(df_syn, y_col = "Y", d_cols = "A",
                              x_cols = colnames(z_syn))
    lr_g <- lrn("regr.ranger", num.trees = 300, max.depth = 5, min.node.size = 2)
    lr_m <- lrn("classif.ranger", num.trees = 300, max.depth = 5, min.node.size = 2,
                predict_type = "prob")
    dml_obj <- DoubleMLIRM$new(dml_d, ml_g = lr_g, ml_m = lr_m)
    dml_obj$fit()
    ate_mat[b, "DoubleML"] <- dml_obj$coef
    se_mat[b, "DoubleML"]  <- dml_obj$se
    ci_d <- c(dml_obj$confint()[1], dml_obj$confint()[2])
    cover_mat[b, "DoubleML"] <- (ci_d[1] <= true_ate_bench & true_ate_bench <= ci_d[2])
  }, error = function(e) message("DoubleML failed on rep ", b, ": ", e$message))

  if (b %% 5 == 0) cat("Completed replication", b, "of", B, "\n")
}
```

```
Completed replication 5 of 10
```

```
Completed replication 10 of 10
```

```
bench_summary <- data.frame(
  Method = colnames(ate_mat),
  Mean_ATE = colMeans(ate_mat, na.rm = TRUE),
  Bias = colMeans(ate_mat, na.rm = TRUE) - true_ate_bench,
  SD = apply(ate_mat, 2, sd, na.rm = TRUE),
  RMSE = sqrt(colMeans((ate_mat - true_ate_bench)^2, na.rm = TRUE)),
  Mean_SE = colMeans(se_mat, na.rm = TRUE),
  Coverage_95 = colMeans(cover_mat, na.rm = TRUE),
  N_valid = colSums(!is.na(ate_mat))
)
bench_summary[, c("Mean_SE", "Coverage_95")][is.na(bench_summary[, c("Mean_SE", "Coverage_95")])] <- NA

kable(bench_summary, digits = 4,
      caption = paste0("Monte Carlo Benchmarking (B=", B,
                       ", n=", n_syn, ", True ATE=",
                       round(true_ate_bench, 4), ")"))
```

Table 43.4.: Monte Carlo Benchmarking (B=10, n=2000, True ATE=0.0793)

	Method	Mean_ATE	Bias	SD	RMSE	Mean_SE	Coverage_95	N_valid
Frengression	Frengression	0.0820	0.0027	0.0389	0.0370	NA	NA	10
npcausal	npcausal	0.2797	0.2004	0.1552	0.2487	0.0733	0.2	10
TMLE	TMLE	0.1226	0.0434	0.0848	0.0914	0.0273	0.5	10
DoubleML	DoubleML	0.1714	0.0922	0.0411	0.1001	0.0407	0.4	10

```
ate_long <- as.data.frame(ate_mat) |>
  pivot_longer(everything(), names_to = "Method", values_to = "ATE") |>
  filter(!is.na(ATE))

# Trim extreme outliers for readable plot
q_bounds <- quantile(ate_long$ATE, c(0.02, 0.98), na.rm = TRUE)
ate_long_trim <- ate_long |> filter(ATE >= q_bounds[1] & ATE <= q_bounds[2])

ggplot(ate_long_trim, aes(x = Method, y = ATE, fill = Method)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = true_ate_bench, linetype = "dashed", color = "red",
             linewidth = 1) +
  annotate("text", x = 0.5, y = true_ate_bench, label = "True ATE",
           hjust = 0, vjust = -0.5, color = "red") +
  labs(title = "ATE Estimates Across Frengression-Generated Datasets",
       subtitle = paste0("B=", B, " replications, n=", n_syn, " per dataset"),
       y = "Estimated ATE") +
  theme_minimal() +
  theme(legend.position = "none")
```

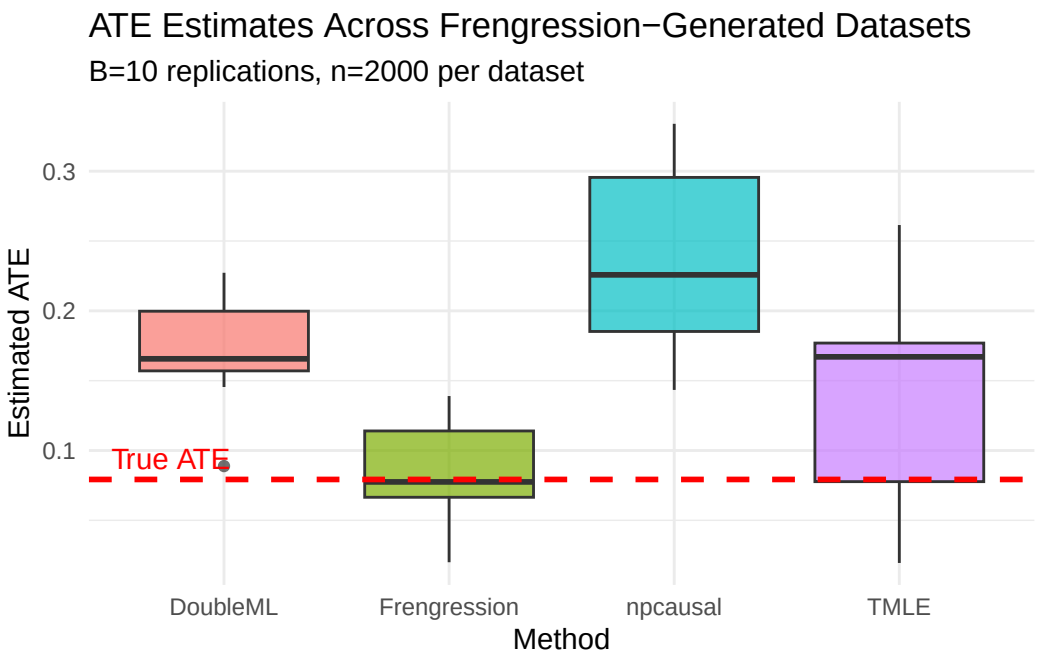


Figure 43.1.: ATE Estimates Across Synthetic Datasets

43.8.3. Monte Carlo Benchmarking (LaLonde)

Now do the same exercise with the LaLonde-trained model. Here the confounding structure comes from the real data rather than from my hand-written simulation.

```
# Y = 0.3*x + eta, so Y(1) - Y(0) = 0.3 for every unit: the true ATE is exactly 0.3.
true_ate_lal <- 0.3
y1_lal_true <- predict(model_lal_bench, matrix(1, ncol=1), type="mean", nsample=5000, trim=0)
y0_lal_true <- predict(model_lal_bench, matrix(0, ncol=1), type="mean", nsample=5000, trim=0)
cat("Monte Carlo check from predict():", round(as.numeric(y1_lal_true - y0_lal_true), 4), "\n")
```

```
Monte Carlo check from predict(): 0.2986
```

```
cat("True ATE (standardized):", round(true_ate_lal, 4), "\n")
```

```
True ATE (standardized): 0.3
```

```
cat("True ATE (dollars):", round(true_ate_lal * sd(y_lal), 0), "\n")
```

```
True ATE (dollars): 2241
```

```r
B_lal <- 10
n_syn_lal <- 2000

methods <- c("Frengression", "npcausal", "TMLE", "DoubleML")
ate_mat_lal <- matrix(NA, nrow = B_lal, ncol = length(methods),
                      dimnames = list(NULL, methods))
se_mat_lal  <- ate_mat_lal
cover_mat_lal <- ate_mat_lal

SL.lib_fast <- c("SL.glm.interaction", "SL.ranger", "SL.mean")

set.seed(2024)
for (b in seq_len(B_lal)) {
  syn <- sample_joint(model_lal_bench, sample_size = n_syn_lal)

  x_syn <- as.numeric(syn$x)
  y_syn <- as.numeric(syn$y)
  z_syn <- as.data.frame(syn$z)
  colnames(z_syn) <- paste0("z", 1:ncol(z_syn))
  df_syn <- cbind(data.frame(A = x_syn, Y = y_syn), z_syn)

  # --- Frengression ---
  tryCatch({
    torch::torch_manual_seed(2000 + b)
    mod_f <- frengression(x_dim = 1, y_dim = 1, z_dim = ncol(z_syn),
                          x_binary = TRUE, y_binary = FALSE,
                          z_binary_dims = 4,
                          noise_dim = 10, hidden_dim = 100, num_layer = 3)
    mod_f <- train_y(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                     matrix(y_syn, ncol=1),
                     num_iters = 500, lr = 1e-3, silent = TRUE)
    mod_f <- train_xz(mod_f, matrix(x_syn, ncol=1), as.matrix(z_syn),
                      num_iters = 500, lr = 1e-4, silent = TRUE)

    y1_f <- predict(mod_f, matrix(1, ncol=1), type="mean", nsample=1000, trim=0)
    y0_f <- predict(mod_f, matrix(0, ncol=1), type="mean", nsample=1000, trim=0)
    ate_mat_lal[b, "Frengression"] <- y1_f - y0_f

    # No standard error or interval for frengression: SE and coverage stay NA.
  }, error = function(e) message("Frengression failed on rep ", b, ": ", e$message))

  # --- npcausal ---
  tryCatch({
    fit_np <- ate(y = y_syn, a = x_syn, x = z_syn,
                  nsplits = 2, sl.lib = SL.lib_fast)
    ate_mat_lal[b, "npcausal"] <- fit_np$res$est[3]
    se_mat_lal[b, "npcausal"]  <- fit_np$res$se[3]
```

```
    ci <- c(fit_np$res$ci.ll[3], fit_np$res$ci.ul[3])
    cover_mat_lal[b, "npcausal"] <- (ci[1] <= true_ate_lal & true_ate_lal <= ci[2])
  }, error = function(e) message("npcausal failed on rep ", b, ": ", e$message))

  # --- TMLE ---
  tryCatch({
    fit_tmle <- tmle(Y = y_syn, A = x_syn, W = z_syn, family = "gaussian",
                     Q.SL.library = SL.lib_fast, g.SL.library = SL.lib_fast)
    ate_mat_lal[b, "TMLE"] <- fit_tmle$estimates$ATE$psi
    se_mat_lal[b, "TMLE"]  <- sqrt(fit_tmle$estimates$ATE$var.psi)
    ci_t <- fit_tmle$estimates$ATE$CI
    cover_mat_lal[b, "TMLE"] <- (ci_t[1] <= true_ate_lal & true_ate_lal <= ci_t[2])
  }, error = function(e) message("TMLE failed on rep ", b, ": ", e$message))

  # --- DoubleML ---
  tryCatch({
    dml_d <- DoubleMLData$new(df_syn, y_col = "Y", d_cols = "A",
                              x_cols = colnames(z_syn))
    lr_g <- lrn("regr.ranger", num.trees = 300, max.depth = 5, min.node.size = 2)
    lr_m <- lrn("classif.ranger", num.trees = 300, max.depth = 5, min.node.size = 2,
                predict_type = "prob")
    dml_obj <- DoubleMLIRM$new(dml_d, ml_g = lr_g, ml_m = lr_m)
    dml_obj$fit()
    ate_mat_lal[b, "DoubleML"] <- dml_obj$coef
    se_mat_lal[b, "DoubleML"]  <- dml_obj$se
    ci_d <- c(dml_obj$confint()[1], dml_obj$confint()[2])
    cover_mat_lal[b, "DoubleML"] <- (ci_d[1] <= true_ate_lal & true_ate_lal <= ci_d[2])
  }, error = function(e) message("DoubleML failed on rep ", b, ": ", e$message))

  if (b %% 5 == 0) cat("Completed replication", b, "of", B_lal, "\n")
}
```

```
Completed replication 5 of 10
```

```
Completed replication 10 of 10
```

```
bench_lal <- data.frame(
  Method = colnames(ate_mat_lal),
  Mean_ATE = colMeans(ate_mat_lal, na.rm = TRUE),
  Bias = colMeans(ate_mat_lal, na.rm = TRUE) - true_ate_lal,
  SD = apply(ate_mat_lal, 2, sd, na.rm = TRUE),
  RMSE = sqrt(colMeans((ate_mat_lal - true_ate_lal)^2, na.rm = TRUE)),
  Mean_SE = colMeans(se_mat_lal, na.rm = TRUE),
  Coverage_95 = colMeans(cover_mat_lal, na.rm = TRUE),
  N_valid = colSums(!is.na(ate_mat_lal))
)
```

```
bench_lal[, c("Mean_SE", "Coverage_95")][is.na(bench_lal[, c("Mean_SE", "Coverage_95")])] <- NA   # no SE

kable(bench_lal, digits = 4,
      caption = paste0("LaLonde-Based Benchmarking (B=", B_lal,
                       ", n=", n_syn_lal,
                       ", True ATE=", round(true_ate_lal, 4),
                       " [", round(true_ate_lal * sd(y_lal), 0), " dollars])"))
```

Table 43.5.: LaLonde-Based Benchmarking (B=10, n=2000, True ATE=0.3 [2241 dollars])

	Method	Mean_ATE	Bias	SD	RMSE	Mean_SE	Coverage_95	N_valid
Frengression	Frengression	-0.0487	-0.3487	0.1881	0.3917	NA	NA	10
npcausal	npcausal	-0.4664	-0.7664	0.4418	0.8736	0.0780	0.1	10
TMLE	TMLE	0.5203	0.2203	0.6856	0.6867	0.0206	0.0	10
DoubleML	DoubleML	-0.5070	-0.8070	0.2630	0.8447	0.0815	0.1	10

```
ate_long_lal <- as.data.frame(ate_mat_lal) |>
  pivot_longer(everything(), names_to = "Method", values_to = "ATE") |>
  filter(!is.na(ATE))

q_bounds_lal <- quantile(ate_long_lal$ATE, c(0.02, 0.98), na.rm = TRUE)
ate_long_lal_trim <- ate_long_lal |>
  filter(ATE >= q_bounds_lal[1] & ATE <= q_bounds_lal[2])

ggplot(ate_long_lal_trim, aes(x = Method, y = ATE, fill = Method)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = true_ate_lal, linetype = "dashed", color = "red",
             linewidth = 1) +
  annotate("text", x = 0.5, y = true_ate_lal, label = "True ATE",
           hjust = 0, vjust = -0.5, color = "red") +
  labs(title = "ATE Estimates: LaLonde-Based Frengression DGP",
       subtitle = paste0("B=", B_lal, " replications, n=", n_syn_lal,
                         " per dataset\n(confounding learned from real data)"),
       y = "Estimated ATE (standardized)") +
  theme_minimal() +
  theme(legend.position = "none")
```

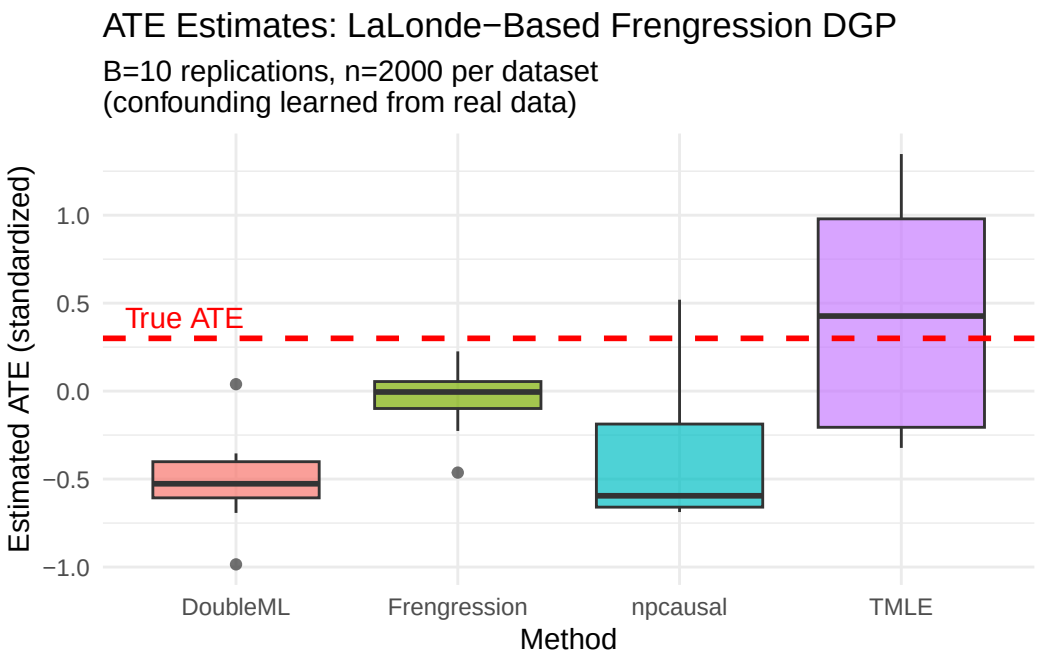


Figure 43.2.: ATE Estimates from LaLonde-Based Synthetic Data

On the hand-written DGP (true ATE 0.0793), frengression does best, with a bias of 0.0027 and an RMSE of 0.0370. TMLE has a bias of 0.0434 and an RMSE of 0.0914, DoubleML's interactive model 0.0922 and 0.1001, and npcausal 0.2004 and 0.2487, with coverage of 50%, 40% and 20%. The generated data treat 90% of units, so some propensity scores are close to one and the inverse-propensity terms in the three doubly robust estimators are unstable. On the LaLonde-generated data (true ATE 0.3, or \$2,241), only 3.2% of units are treated, about 64 of 2,000, against 30% in the real LaLonde sample: the learned treatment model badly understates treatment, and every method struggles. Frengression's bias is −0.3487 with an RMSE of 0.3917; TMLE's is 0.2203 with an RMSE of 0.6867 and 0% coverage; npcausal and DoubleML are off by −0.7664 and −0.8070, with 10% coverage. Frengression has no coverage entry because we have no valid standard error for it.

The coverage column shows where reported standard errors fail. A method whose interval covers the truth in 0 or 1 of 10 replications is not merely imprecise; its reported standard error is wrong. TMLE's mean SE on the simulated benchmark is 0.0273 against an actual sampling SD of 0.0848, understating the uncertainty by a factor of about three.

One caveat before drawing conclusions: $B = 10$ replications is a very small Monte Carlo, so these coverage rates carry a standard error of about 0.15 themselves, and the ranking of the methods can change from one run of ten to the next. An earlier run of this chapter, which used the partially linear DoubleML model, scaled the binary covariates and stopped training early, ranked DoubleML first on the hand-written DGP. What is stable across runs is that weak overlap, 90% treated in one design and 3% in the other, makes the reported standard errors of the doubly robust estimators far too small.

All methods show more bias on the LaLonde-generated data than on the hand-written simulation. That is the point of this exercise. The synthetic data has the real data's skewed prior-earnings distribution, and, because the learned treatment model understates treatment, far worse overlap

than the real data. Before using such a generator as a benchmark, its treatment and covariate margins should be checked against the real data; here the treatment margin fails that check.

43.9. Summary

Frengression uses the Evans and Didelez simulation decomposition, but learns the pieces nonparametrically. The practical use is not that it beats TMLE or AIPW for a simple ATE. The practical use is that it can learn a realistic observational distribution, let us impose a known causal effect, and then generate synthetic data where the truth is known.

That makes it useful for benchmarking. The simulation is still artificial, but it is less artificial than writing down a clean logistic model and pretending that it looks like real observational data.

Systematic treatment: Julia.

44. Partial Interference

Most causal inference methods assume SUTVA: one unit's treatment does not affect another unit's outcome. This is often not true. Vaccinating your neighbor can reduce your risk of infection. A job training program for one worker may affect another worker's job prospect. A student's test score may depend on whether classmates received tutoring.

Partial interference (Sobel 2006; Hudgens and Halloran 2008) is one way to make this manageable. Units are organized into clusters. Interference can happen within a cluster, but not between clusters. This is still a strong assumption, but it is weaker than no interference.

Qu et al. (2024) develop clustered IPW and AIPW estimators for this setting, and the `CausalModel` R package implements them. As the number of clusters grows, their AIPW estimator is asymptotically normal and attains a semiparametric efficiency bound that allows outcomes in a cluster to be correlated (their Theorem 2). Here I just simulate data and see how the functions work.

44.1. The Setup

44.1.1. Clusters and Groups

Each unit i belongs to a cluster c, such as a household, classroom, or village. Within each cluster, units are also organized into groups indexed by $j = 0, 1, \dots, J-1$. The group structure $\mathbf{s} = (s_0, s_1, \dots, s_{J-1})$ gives the number of units per group.

For example, with `group_struct = c(2, 3)`:

- Group 0 has 2 units
- Group 1 has 3 units
- Each cluster has 5 units total

The groups can be real roles, such as parents and children, or they can just be a modeling device.

44.1.2. The Neighborhood Configuration G

The key object is the neighborhood configuration $G_i = (g_0, g_1, \dots, g_{J-1})$. Here g_j counts how many other units in group j within unit i's cluster are treated. "Other" means excluding unit i itself.

With `group_struct = c(2, 3)`:

- If unit i is in group 0, then $g_0 \in \{0, 1\}$ (the other unit in group 0 is treated or not) and $g_1 \in \{0, 1, 2, 3\}$ (0 to 3 units in group 1 are treated)
- G records the treatment environment that unit i experiences

44.1.3. Heterogeneous Treatment Effects $\beta(G)$

Under partial interference, the treatment effect for unit i depends on its neighborhood G_i:

$$\beta(G) = E[Y_i(1, G) - Y_i(0, G)] \tag{44.1}$$

where $Y_i(z, G)$ is unit i's potential outcome when its own treatment is z and its neighborhood configuration is G.

A simple model is to make the treatment effect linear in the neighborhood:

$$\beta(G) = \tau + \sum_{j=0}^{J-1} \gamma_j \cdot g_j \tag{44.2}$$

where:

- τ is the direct effect when no neighbors are treated
- γ_j is how much each treated neighbor in group j changes the direct effect

$\beta(G)$ is a direct effect at a fixed neighborhood: own treatment changes and G is held fixed. A spillover effect is the other contrast. It holds own treatment z fixed and changes the neighborhood, $E[Y_i(z, G) - Y_i(z, G')]$. So γ_j is not by itself the spillover of a treated neighbor. It is the spillover for treated units minus the spillover for untreated units, per treated neighbor in group j. In the simulation below, untreated outcomes do not depend on G, so untreated units get no spillover and treated units get γ_j per treated neighbor in group j. The package estimates β separately for each group of the focal unit, so it is really $\beta_j(G)$ for a unit in group j. In the simulation all groups share one $\beta(G)$.

44.1.3.1. Example

With `group_struct = c(2, 3)`, $\tau = 1$, $\gamma_0 = 0.5$, $\gamma_1 = 0.1$:

Neighborhood G	Formula	$\beta(G)$
$(0, 0)$	$1 + 0 \times 0.5 + 0 \times 0.1$	1.0
$(1, 0)$	$1 + 1 \times 0.5 + 0 \times 0.1$	1.5
$(0, 1)$	$1 + 0 \times 0.5 + 1 \times 0.1$	1.1
$(1, 1)$	$1 + 1 \times 0.5 + 1 \times 0.1$	1.6

Having one treated neighbor in group 0 adds 0.5 to the treatment effect; having one in group 1 adds only 0.1.

44.2. Estimation

44.2.1. Two Propensity Models

Standard IPW uses one propensity score, $P(Z_i = 1|X_i)$. Under interference, we need two:

1. **Individual propensity**: $P(Z_i = 1|X_i)$ — probability of being treated given covariates (logistic regression)
2. **Neighborhood propensity**: $P(G_i = g|Z_i, X_i)$ — probability of neighborhood configuration g given **own treatment** and covariates (multinomial logistic regression)

The second piece is the new part. It models how likely a particular treatment environment is, conditional on covariates. Here X_i includes cluster-level covariates, such as the mean of the neighbors' covariates, and not only unit i's own. The neighbors' treatments depend on their covariates, so the focal unit's covariates alone do not determine the distribution of G_i. The `clustered()` constructor below adds the cluster means.

Note the conditioning on Z_i in the second piece. Own treatment Z_i and the neighborhood exposure G_i are usually **dependent by design** — for a network or cluster exposure mapping, G_i is built from the treatments of i's neighbours, which are correlated with i's own assignment mechanism. The joint exposure probability must therefore be modeled as a joint distribution $P(Z_i = z, G_i = g \mid X_i)$, or via the valid factorization $P(Z_i = z \mid X_i)\, P(G_i = g \mid Z_i = z, X_i)$. A product of the two *marginals* $P(Z_i \mid X_i)P(G_i \mid X_i)$ is only correct when own and neighborhood treatment are conditionally independent given X_i, which is not the typical case.

44.2.2. Clustered IPW

The clustered IPW estimator for $\beta(g)$ reweights outcomes by the joint exposure probability, written here as the conditional factorization:

$$\hat{\beta}_{\text{IPW}}(g) = \frac{1}{N}\sum_i \left[\frac{\mathbf{1}(Z_i = 1, G_i = g)}{P(Z_i = 1|X_i) \cdot P(G_i = g|Z_i = 1, X_i)} - \frac{\mathbf{1}(Z_i = 0, G_i = g)}{P(Z_i = 0|X_i) \cdot P(G_i = g|Z_i = 0, X_i)}\right] Y_i \tag{44.3}$$

This gives a separate treatment-effect estimate at each neighborhood configuration g.

44.2.3. Clustered AIPW (Doubly Robust)

The AIPW version adds outcome models:

- Fit separate outcome models $E[Y|X, Z, G = g]$ for each configuration g
- Combine with propensity weighting so the estimator is consistent if either the outcome model or the (joint) exposure-propensity model is correct. Here the propensity part must correctly model the *joint* exposure mechanism $P(Z_i, G_i \mid X_i)$, not two independent marginals.

This is the estimator I would usually try first, for the same reason standard AIPW is attractive.

44.2.4. Variance Estimation

Standard errors are computed by the matching-based variance estimator of Qu et al. (2024, Appendix A.5), which they propose for testing which interference structure holds and show to be consistent. Within each group and neighborhood configuration, units are matched on standardized covariates, and the variance is estimated from the matched pairs. This avoids a cluster bootstrap, which can be expensive here.

44.3. Using CausalModel

```
library(CausalModel)
library(knitr)
set.seed(42)
```

44.3.1. Generate Clustered Data

The `generate_fixed_cluster()` function creates synthetic data where the interference structure is known. We build 200 clusters of 5 units each — 2 in group 0 and 3 in group 1, 1,000 units in all. The direct effect is $\tau = 1$ when no neighbor is treated. Each treated neighbour in group 0 raises it by 0.5, and each in group 1 by 0.1. The outcome is $Y_i = (1 + Z_i)X_{ci}'b + Z_i\beta(G_i) + \varepsilon_i$, where X_{ci} stacks unit i's two covariates and their cluster means, b is a fixed coefficient vector and ε_i is standard normal noise. Untreated outcomes therefore do not depend on the neighbours. Treatment is drawn from a logistic model in the same X_{ci}, so own treatment and the neighbours' treatments both depend on the cluster means. Treatment prevalence comes out at 0.477.

```
cdata <- generate_fixed_cluster(
  clusters = 200,
  group_struct = c(2, 3),   # 2 units in group 0, 3 in group 1
  tau = 1,                   # direct effect
  gamma = c(0.5, 0.1)        # spillover: 0.5 per treated in group 0, 0.1 in group 1
)

cat("Total units:", length(cdata$Y), "\n")
```

```
Total units: 1000
```

```
cat("Units per cluster:", sum(c(2, 3)), "\n")
```

```
Units per cluster: 5
```

```
cat("Treatment prevalence:", round(mean(cdata$Z), 3), "\n")
```

```
Treatment prevalence: 0.477
```

Each unit has:

- `Y` — observed outcome
- `Z` — treatment indicator (0/1)
- `X` — individual covariates
- `cluster_labels` — which cluster the unit belongs to
- `group_labels` — which group within the cluster (0-indexed)
- `ingroup_labels` — position within the group (0-indexed)

44.3.2. Construct the Clustered Object

```
cl_obj <- clustered(
  Y = cdata$Y,
  Z = cdata$Z,
  X = cdata$X,
  cluster_labels = cdata$cluster_labels,
  group_labels = cdata$group_labels,
  ingroup_labels = cdata$ingroup_labels,
  n_matches = 50
)
```

The `clustered()` constructor does several things:

1. Augments individual covariates with cluster-level aggregates (mean of neighbors' covariates)
2. Fits the individual propensity model (logistic regression)
3. Fits the neighborhood propensity model (multinomial logistic regression)
4. Computes G for each unit — the number of treated neighbors per group

44.3.3. Estimate Treatment Effects

44.3.3.1. Clustered AIPW

```
result_aipw <- est_via_aipw(cl_obj)
```

The result is a list with one element per group. Each element contains `beta_g`, the treatment effects at each neighborhood configuration, and `se`, the standard errors:

```
for (j in seq_along(result_aipw)) {
  cat(sprintf("Group %d:\n", j - 1))
  valid <- !is.nan(result_aipw[[j]]$beta_g) & result_aipw[[j]]$se > 0
  df <- data.frame(
    g_index = which(valid) - 1,
    beta_g = round(result_aipw[[j]]$beta_g[valid], 3),
    se = round(result_aipw[[j]]$se[valid], 3)
```

```
  )
  print(df, row.names = FALSE)
  cat("\n")
}
```

```
Group 0:
 g_index beta_g    se
       0  0.677 0.364
       1  1.572 0.347
       3  1.447 0.315
       4  1.870 0.257
       6  1.141 0.276
       7  2.003 0.276
       9  1.015 0.428
      10  2.674 0.384

Group 1:
 g_index beta_g    se
       0  1.528 0.311
       1  1.634 0.228
       2  2.126 0.331
       3  1.280 0.266
       4  1.799 0.188
       5  2.180 0.269
       6  1.272 0.419
       7  2.137 0.247
       8  2.850 0.389
```

44.3.3.2. Clustered IPW

```
result_ipw <- est_via_ipw(cl_obj)
```

```
cat("Clustered IPW beta_g (group 0):\n")
```

```
Clustered IPW beta_g (group 0):
```

```
valid <- !is.nan(result_ipw[[1]]$beta_g) & result_ipw[[1]]$se > 0
round(result_ipw[[1]]$beta_g[valid], 3)
```

```
[1] 0.491 1.591 1.404 1.865 0.984 2.002 1.182 2.327
```

44.3.4. Interpreting the Results

The neighborhood configurations are encoded as integers. For `group_struct = c(2, 3)`, the encoding for group 0 is:

$$G_{\text{encoded}} = g_0 + g_1 \times (s_0 + 1) \tag{44.4}$$

where g_0 and g_1 are the numbers of *other* treated units in group 0 and group 1, and $s_0 = 2$ and $s_1 = 3$ are the group sizes. The stride for the second dimension is $s_0 + 1 = 3$, so the grid has $(s_0 + 1)(s_1 + 1) = 12$ slots, numbered 0 to 11, for every focal unit. Which slots can be filled depends on the focal unit's own group. For a unit in group 0, at most $s_0 - 1 = 1$ other group-0 unit can be treated, so $g_0 = 2$ is impossible and slots 2, 5, 8 and 11 stay empty. For a unit in group 1, both group-0 units are others, so $g_0 \in \{0, 1, 2\}$, but at most $s_1 - 1 = 2$ other group-1 units can be treated, so $g_1 = 3$ is impossible and slots 9, 10 and 11 stay empty. These are structural zeros, not missing data.

g_0	g_1	Encoded	True β
0	0	0	1.0
1	0	1	1.5
0	1	3	1.1
1	1	4	1.6
0	2	6	1.2
1	2	7	1.7

Some configurations do not have enough observations. The `se > 0` filter removes degenerate cases.

Reading the estimates against that table: configuration 0 (no treated neighbours) gives 0.677 against a true 1.0, configuration 1 gives 1.572 against 1.5, configuration 3 gives 1.447 against 1.1, and configuration 4 gives 1.870 against 1.6. Single-dataset estimates for individual configurations are noisy, which is what the Monte Carlo below is for.

Clustered IPW gives 0.491, 1.591, 1.404, 1.865 and so on across the same configurations — close to the AIPW numbers but not identical, as expected from two different estimators of the same quantities.

44.4. Monte Carlo Validation

Now run a small Monte Carlo. Generate data 100 times, estimate $\beta(G)$ each time, and check bias, SE calibration, and coverage.

```
n_clusters <- 200
group_struct <- c(2, 3)
tau_int <- 1
gamma_int <- c(0.5, 0.1)
n_reps <- 100
```

```
# True beta(g) function
true_beta <- function(g) tau_int + sum(g * gamma_int)

# Track results for group 0 at three neighborhood configurations
g_indices <- list(c(0, 0), c(1, 0), c(0, 1))
g_encoded <- sapply(g_indices, function(g) {
  g[1] + g[2] * (group_struct[1] + 1) + 1  # 1-indexed; stride = max_g0 + 1
})
true_values <- sapply(g_indices, true_beta)

int_results <- array(NA, dim = c(n_reps, length(g_indices), 2),
                     dimnames = list(NULL, NULL, c("beta", "se")))
```

```
set.seed(2024)
for (i in seq_len(n_reps)) {
  cdata <- generate_fixed_cluster(
    clusters = n_clusters, group_struct = group_struct,
    tau = tau_int, gamma = gamma_int
  )
  cl_obj <- clustered(
    Y = cdata$Y, Z = cdata$Z, X = cdata$X,
    cluster_labels = cdata$cluster_labels,
    group_labels = cdata$group_labels,
    ingroup_labels = cdata$ingroup_labels,
    n_matches = 50
  )
  res <- est_via_aipw(cl_obj)

  for (j in seq_along(g_indices)) {
    idx <- g_encoded[j]
    int_results[i, j, "beta"] <- res[[1]]$beta_g[idx]
    int_results[i, j, "se"] <- res[[1]]$se[idx]
  }
}
```

```
int_rows <- list()
for (j in seq_along(g_indices)) {
  betas <- int_results[, j, "beta"]
  ses <- int_results[, j, "se"]

  valid <- !is.nan(betas) & !is.nan(ses) & ses > 0
  betas <- betas[valid]
  ses <- ses[valid]
  tv <- true_values[j]

  ci_lo <- betas - 1.96 * ses
  ci_hi <- betas + 1.96 * ses
```

```
  int_rows[[j]] <- data.frame(
    g = paste0("(", paste(g_indices[[j]], collapse = ","), ")"),
    True_beta = tv,
    Mean_est = mean(betas),
    Bias = mean(betas) - tv,
    Emp_SE = sd(betas),
    Mean_SE = mean(ses),
    Coverage = mean(ci_lo <= tv & tv <= ci_hi),
    N_valid = length(betas)
  )
}
tbl_int <- do.call(rbind, int_rows)
kable(tbl_int, digits = 3, row.names = FALSE,
      caption = sprintf("Clustered AIPW (group 0, %d clusters, %d reps)",
                        n_clusters, n_reps))
```

Table 44.3.: Clustered AIPW (group 0, 200 clusters, 100 reps)

g	True_beta	Mean_est	Bias	Emp_SE	Mean_SE	Coverage	N_valid
(0,0)	1.0	2.107	1.107	8.760	0.448	0.820	100
(1,0)	1.5	1.569	0.069	0.549	0.415	0.899	99
(0,1)	1.1	1.083	-0.017	0.248	0.246	0.930	100

Over 100 replications the picture is mixed, and the failure is informative. Configuration $(1,0)$ recovers its truth well — mean 1.569 against 1.5, bias 0.069, coverage 89.9% — and $(0,1)$ better still, 1.083 against 1.1 with 93.0% coverage. But $(0,0)$ is badly behaved: mean 2.107 against a true 1.0, and an empirical standard deviation of 8.760 against a mean reported standard error of 0.448. The estimator is not merely biased there; its own standard error understates the sampling variability by a factor of nearly twenty, and coverage is 82% only because a few enormous outliers drag the mean while most draws sit near the truth.

The reason is positivity. Configuration $(0,0)$ requires a unit with *no* treated neighbours in either group, and with treatment prevalence near 0.5 in clusters of five that configuration is rare, so the inverse-probability weights attached to it are extreme.

The paper's own simulation points the same way on a smaller scale (Qu et al. 2024, Table 2). With 5,000 clusters of size four, AIPW's coverage is 94.88% to 95.60%, close to nominal. But its mean squared error, 0.0038 to 0.0039, is larger than a random forest's, 0.0027, for every outcome model, and larger than OLS's, 0.0003, when the outcome model is linear. The text there says AIPW has much smaller MSE. What the table shows is that AIPW gives valid intervals, not that it is the most precise.

```
par(mfrow = c(1, length(g_indices)), mar = c(3, 3, 2, 1))
for (j in seq_along(g_indices)) {
  betas <- int_results[, j, "beta"]
  ses <- int_results[, j, "se"]
```

```
  valid <- !is.nan(betas) & !is.nan(ses) & ses > 0
  t_stats <- (betas[valid] - true_values[j]) / ses[valid]
  g_label <- paste0("(", paste(g_indices[[j]], collapse = ","), ")")
  hist(t_stats, breaks = 20, freq = FALSE, col = "lightblue",
       main = sprintf("g = %s", g_label),
       xlab = "", xlim = c(-4, 4))
  curve(dnorm(x), add = TRUE, col = "red", lwd = 2)
  abline(v = 0, lty = 2)
}
```

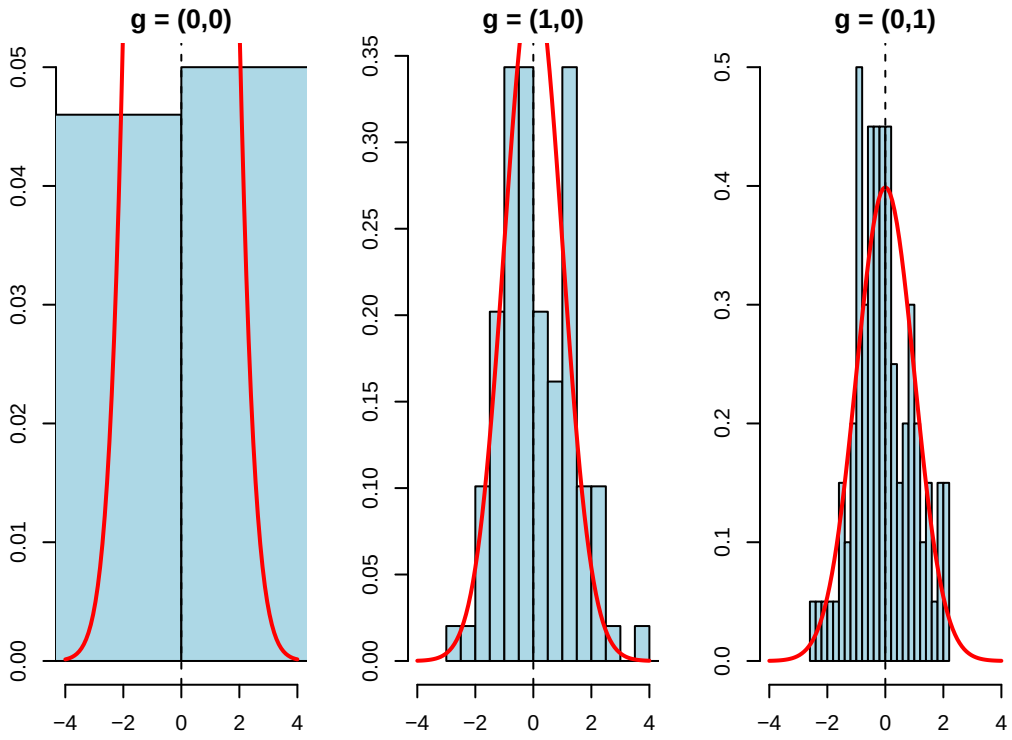


Figure 44.1.: Studentized statistics under clustered AIPW

If both the estimator and SE work well, the studentized statistics should look roughly normal.

44.5. What If You Ignore Interference?

What happens if we ignore interference and use a standard AIPW estimator? We get one number that averages over neighborhood configurations. That number is hard to interpret.

```
# Generate one dataset with interference
set.seed(42)
cdata <- generate_fixed_cluster(
  clusters = 200,
```

```
  group_struct = c(2, 3),
  tau = 1,
  gamma = c(0.5, 0.1)
)

# Naive: ignore clusters, treat as standard observational data
obs_naive <- observational(cdata$Y, cdata$Z, cdata$X)
naive_ate <- est_via_aipw(obs_naive)

cat("Naive AIPW (ignoring interference):", round(naive_ate$ate, 3), "\n")
```

```
Naive AIPW (ignoring interference): 1.611
```

```
cat("SE:", round(naive_ate$se, 3), "\n\n")
```

```
SE: 0.068
```

```
# Correct: use clustered estimator
cl_obj <- clustered(
  Y = cdata$Y, Z = cdata$Z, X = cdata$X,
  cluster_labels = cdata$cluster_labels,
  group_labels = cdata$group_labels,
  ingroup_labels = cdata$ingroup_labels,
  n_matches = 50
)
result <- est_via_aipw(cl_obj)

cat("Clustered AIPW beta_g (group 0):\n")
```

```
Clustered AIPW beta_g (group 0):
```

```
valid <- !is.nan(result[[1]]$beta_g) & result[[1]]$se > 0
df <- data.frame(
  g_index = which(valid) - 1,
  beta_g = round(result[[1]]$beta_g[valid], 3),
  se = round(result[[1]]$se[valid], 3)
)
print(df, row.names = FALSE)
```

```
 g_index beta_g    se
       0  0.677 0.364
       1  1.572 0.347
       3  1.447 0.315
       4  1.870 0.257
```

```
 6  1.141 0.276
 7  2.003 0.276
 9  1.015 0.428
10  2.674 0.384
```

The naive estimate compares treated units, whose neighborhoods follow the distribution of G given $Z = 1$, with untreated units, whose neighborhoods follow the distribution given $Z = 0$. These differ here, because own and neighbors' treatments both depend on the cluster means. Write $m_0(g) = E[Y_i(0, g)]$ for the mean untreated outcome at neighborhood g. If treatment were as good as random given G, the naive contrast would be an average of $\beta(G)$ over the treated units' neighborhoods, plus the difference in $m_0(G)$ between the two neighborhood distributions. That second term is a spillover on untreated units; it is zero here because untreated outcomes do not depend on G. But treatment is not random given G. It depends on the cluster covariate means, which also move untreated outcomes, while the naive AIPW adjusts only for each unit's own covariates. So the selected means $E[Y_i(0, g) \mid Z_i, G_i = g]$ differ from $m_0(g)$, and the naive contrast also carries residual selection. Ignoring interference can mix direct effects, differences in neighborhood exposure and residual selection; a decomposition in population means needs exchangeability for the joint exposure (Z_i, G_i). The clustered estimator adds the cluster covariate means, models the joint exposure, and reports the direct effect separately at each neighborhood configuration. It reports the direct effect separately at each neighborhood configuration.

44.6. Summary

Partial interference is a compromise. We do not assume away spillovers, but we do assume they stay inside clusters. Once we make that assumption, the neighborhood configuration G becomes part of the estimand. The effect is no longer just treatment versus control; it is treatment versus control under a particular neighborhood treatment environment.

The `CausalModel` package estimates these effects with clustered IPW and AIPW. The important practical lesson is that a standard AIPW estimate is not a good substitute. It collapses direct effects and spillovers into one number.

45. Power analysis with list experiment

This chapter follows Ulrich et al. (2012).

45.1. List experiment

The list experiment (also called the item count technique, Miller (1984)) elicits the prevalence of a sensitive attribute without asking respondents to reveal their own status. A control group of n_c respondents receives k neutral items and reports the total count of "yes" answers. An experimental group of n_e respondents receives the same k items plus the sensitive item. The expected counts are $\mu_c = \sum_{i=1}^{k} \pi_i$ and $\mu_e = \sum_{i=1}^{k} \pi_i + \pi_S$, so the difference in means estimates π_S:

$$\hat{\pi}_S = \hat{\mu}_e - \hat{\mu}_c. \tag{45.1}$$

45.2. Power analysis

45.2.1. Define power

Suppose π_s is the proportion of the population that has the sensitive attribute, or the proportion of the population that would answer the sensitive question with "yes", such as the proportion of people who have committed a crime. The null hypothesis $H0$ is that $\pi_s = 0$, and the alternative hypothesis $H1$ is that $\pi_s > 0$.

The expected mean and the sampling variance of $\hat{\pi}_s$ under the null hypothesis H0 are denoted μ_0 and σ_0^2, respectively. Likewise, let μ_1 and σ_1^2 be the mean and the variance under the alternative hypothesis H1 that π_s has a specific value that is greater than zero, for instance, $H1 : \pi_s = 0.05$.

The statistical power $P(\text{“}H_1\text{”} \mid H_1)$ of a test represents the probability of rejecting $H0$ in favor of $H1$ given that $H1$ is true.

Write $z_\alpha = \Phi^{-1}(\alpha)$ for the lower α quantile of the standard normal, which is negative ($z_{0.05} = -1.64$),

so the one-sided critical value is $c_{1-\alpha} = \mu_0 - z_\alpha \sigma_0$. On a general level, the power is computed as,

$$\begin{aligned} P(\text{“}H_1\text{”}|H_1) &= P(\hat{\pi}_s > c_{1-\alpha}) \\ &= P(\frac{\hat{\pi}_s - \mu_1}{\sigma_1} > \frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= P(Z > \frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= 1 - \Phi(\frac{c_{1-\alpha} - \mu_1}{\sigma_1}) \\ &= \Phi(\frac{\mu_1 - c_{1-\alpha}}{\sigma_1}) \\ &= \Phi(\frac{\mu_1 - \mu_0 + z_\alpha \sigma_0}{\sigma_1}) \end{aligned} \tag{45.2}$$

Consider a special case, $\mu_0 = 0$, $\mu_1 = 1$, $\sigma_0 = \sigma_1 = 1$, and $z_\alpha = -1.64$, then the power is 0.26. This is the one side power. If we consider two sided power, then the power is .168, since $z_{\alpha/2} = -1.96$.

45.2.2. Power for list experiment

Under H1, the sampling variance of $\hat{\pi}_s$ is

$$\sigma_1^2 = \frac{\pi_s(1-\pi_s)}{n_e} + (\frac{1}{n_e} + \frac{1}{n_c}) \sum_{i=1}^{k} \pi_i(1-\pi_i) \tag{45.3}$$

where π_i is the probability of answering the neutral question N_i with “yes”. These formulas assume that a respondent's answers to the items are independent of each other and of the sensitive item. In general the neutral count has variance $\sum_j \mathrm{Var}(N_j) + 2\sum_{j<k} \mathrm{Cov}(N_j, N_k)$, and the treatment-list count also includes the covariances with the sensitive item. Random assignment of lists does not make answers within a list independent. The prevalence contrast itself rests on two further assumptions: adding the sensitive item does not change answers to the neutral items, and respondents report the sensitive item truthfully.

Under H0, the sampling variance of $\hat{\pi}_s$ is

$$\sigma_0^2 = (\frac{1}{n_e} + \frac{1}{n_c}) \sum_{i=1}^{k} \pi_i(1-\pi_i) \tag{45.4}$$

Assuming $n_e = n_c = n$, then the power is

$$P(\text{"}H_1\text{"}|H_1) = \Phi(\frac{\pi_s + z_\alpha \sqrt{\frac{2}{n}\sum_{i=1}^{k} \pi_i(1-\pi_i)}}{\sqrt{\frac{\pi_s(1-\pi_s)}{n} + \frac{2}{n}\sum_{i=1}^{k} \pi_i(1-\pi_i)}}) \tag{45.5}$$

For example, when you have $\pi_s = .1$, $n = 500$, $k = 4$, $\pi_i = .1$, then the power is

```
num=.1+qnorm(.05)*sqrt((1/250)*(4*.1*.9))
denom=sqrt((.1*.9/500+(1/250)*4*.1*.9))
pnorm(num/denom)
```

```
[1] 0.8247802
```

With 500 per arm, 4 neutral questions, a sensitive-item prevalence of 0.1 and neutral-item probabilities of 0.1, the power comes out at 0.825.

45.2.3. Sample size for list experiment

Assuming $n_e = n_c$,

$$a_1 = \sqrt{2\pi_s(1-\pi_s) + 4\sum_{i=1}^{k}\pi_i(1-\pi_i)} \tag{45.6}$$

$$a_0 = 2\sqrt{\sum_{i=1}^{k}\pi_i(1-\pi_i)} \tag{45.7}$$

Sample size needed for a power of $1-\beta$ is

$$n = (\frac{(z_{1-\beta})a_1 - z_\alpha a_0}{\mu_1 - \mu_0})^2 \tag{45.8}$$

where β is type 2 error, α is type 1 error. Here n is the **total** sample size; each arm receives $n/2$. The factor $\sqrt{2}$ inside $a_1 = \sqrt{2\pi_s(1-\pi_s) + 4\sum\pi_i(1-\pi_i)}$ and $a_0 = 2\sqrt{\sum\pi_i(1-\pi_i)}$ converts the per-arm variances σ_1^2, σ_0^2 above into a total-sample formula, so this n is twice the per-group size that the power section uses.

Let's look at an example: suppose the proportion of the population that would answer the sensitive question with "yes" is 0.1. The probability of answering the neutral question N_i with "yes" is 0.1. The type 1 error is 0.05, and the type 2 error is 0.2. Suppose we have 4 neutral questions. What is the sample size needed?

```
a1=sqrt(2*.1*.9+4*4*.1*.9)
a0=2*sqrt(4*.1*.9)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.1)^2
n
```

```
[1] 927.2228
```

The required total sample is 927, so about 464 per arm.

Suppose probability of answering the neutral question N_i with "yes" is .3.

```
a1=sqrt(2*.1*.9+4*4*.3*.7)
a0=2*sqrt(4*.3*.7)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.1)^2
n
```

```
[1] 2114.682
```

Raising the neutral-item probability from 0.1 to 0.3 pushes the requirement to 2,115 — more than double. The neutral questions are pure noise from the estimand's point of view, and their variance $\pi_i(1-\pi_i)$ is maximised at 0.5, so choosing neutral items with probabilities near 0 or 1 is what keeps the design efficient.

Suppose the probability of answering the sensitive question with "yes" is .05.

```
a1=sqrt(2*.05*.95+4*4*.3*.7)
a0=2*sqrt(4*.3*.7)
n=((qnorm(.8)*a1-qnorm(.05)*a0)/.05)^2
n
```

```
[1] 8388.512
```

Halving the sensitive-item prevalence from 0.1 to 0.05 raises the requirement to 8,389, a fourfold increase. The effect enters the denominator of the sample-size formula squared, so rare sensitive behaviours are expensive to measure this way: the list experiment gets more honest answers but needs a much larger n.

45.3. Summary

A list experiment estimates the prevalence of a sensitive attribute as the difference in mean item counts between respondents shown the list with the sensitive item and respondents shown the list without it. Its power depends on the prevalence, the sample size, and the variance of the neutral items, which adds noise without information. Ulrich et al. give closed forms for the power and for the total sample size needed.

With 500 respondents per arm, four neutral items answered yes with probability 0.1 and a prevalence of 0.1, the power is 0.825. Reaching 80% power with those values needs a total of 927 respondents. Raising the neutral-item probability to 0.3 pushes that to 2,115. Halving the prevalence to 0.05 pushes it to 8,389, because the effect enters the formula squared. Neutral items with probabilities near 0 or 1 keep the design efficient.

46. Causal mediation analysis

46.1. Traditional mediation analysis

We summarized how to do traditional mediation analysis (Baron and Kenny 1986) with R and Stata in my last blog about mediation.

Traditionally mediation model can be represented in the following equations:

$$Y = aA + bM + \epsilon_1 \tag{46.1}$$

$$M = cA + \epsilon_2 \tag{46.2}$$

There are three main variables, A (treatment, or exposure), M (mediator) and Y (outcome). There are two pathways from A to Y, a direct effect, and an indirect effect through A -> M -> Y.

The traditional approach is to model these two equations directly, and get the estimates of direct and indirect effects.

46.2. Causal mediation analysis

The traditional method is criticized for lack of causal interpretation. The causal mediation analysis makes assumptions up front. If those assumptions are met, then you have a causal interpretation.

In a random experiment, usually A is randomly assigned, but M is not. If there is a confounder on the M to Y pathway, then we don't have a causal effect, or say we cannot estimate causal effect without confounder bias.

Causal mediation analysis needs that too; it's not like if you use a causal mediation package then you get the causal effect.

The other shortcoming of the traditional mediation method is that it assumes a homogeneous treatment indirect effect; that is, there is no A - M interaction. With causal mediation, this interaction can be allowed.

46.2.1. Controlled direct effect

Suppose we can set A and M at will to any (a, m), then we have the potential outcome $Y(a, m)$.

Controlled direct effect (CDE) is defined as

$$CDE(m) = E(Y(1, m) - Y(0, m)) \tag{46.3}$$

That is, setting M to m, what is the effect of A on Y?

46.2.1.1. Assumptions

What are the assumptions for CDE to be identified (estimable)?

1. Conditional treatment randomization. Suppose we observe confounders W, which can be a joint set of confounders for the A to Y pathway, or the M to Y pathway.

   $$Y(a, m) \perp A|W \tag{46.4}$$

   This is the usual conditional independence (unconfoundedness, ignorability, etc.) assumption. This is saying the assignment of treatment, given covariates W, has nothing to do with potential outcomes.

2. Conditional mediator randomization.

   $$Y(a, m) \perp M|W, A = a \tag{46.5}$$

   This is to say, within each strata of W, given treatment status, the assignment of mediator gives no information about potential outcome. This randomization is usually not implemented during many experiments (random trials). This is the assumption that makes a lot of mediation hard to make a causal claim.

3. Positivity.

   There are two positivity assumptions:

   $$P(A = a|W = w) > 0 \tag{46.6}$$

   for all w.

   This is the usual positivity assumption.

   $$P(M = m|W = w, A = a) > 0 \tag{46.7}$$

   for all w. This is the mediator positivity.

If these assumptions can be met, then we can do causal mediation analysis. In reality, the ignorability assumptions are untestable. Positivity is not testable either, though it is partly diagnosable: estimated propensity distributions and observed cell counts reveal practical violations, and a fitted propensity close to zero or one in some region is a warning. What cannot be established from a finite sample is the population condition itself. Consistency is the third requirement, alongside exchangeability and positivity.

46.2.1.2. Estimation

CDE can be estimated using G-computation method, or IPW, or the doubly-robust methods, one of them is AIPW (augmented IPW).

G-computation is to model the outcome equation:

$$CDE_G(m) = E_W[Q(1,m,W) - Q(0,m,W)], \qquad Q(a,m,w) = E[Y \mid A=a, M=m, W=w] \tag{46.8}$$

$$CDE_G(m) = \sum_w [Q(1,m,w) - Q(0,m,w)]P(W=w) \tag{46.9}$$

Writing the inner conditional mean as Q keeps the two operations apart: Q is a regression on observed variables, and the outer expectation is over the covariate distribution. Both equations are the same statement, the second spelling out the outer expectation for discrete W.

IPW is to model the treatment assignment equation and the mediator assignment equation:

$$E[Y(a,m)] = E[\frac{I(A=a, M=m)}{P(A=a, M=m|W)}Y] \tag{46.10}$$

Therefore,

$$CDE_{ipw}(m) = E[\frac{I(A=1, M=m)}{g_m(m|1,W)g_A(1|W)} - \frac{I(A=0, M=m)}{g_m(m|0,W)g_A(0|W)}Y] \tag{46.11}$$

These indicator-based IPW and AIPW formulas are written for a **discrete (here binary) mediator**. For a continuous mediator the indicator $I(M=m)$ and the probability g_m must be replaced by a conditional density (or a kernel / stochastic-intervention formulation); the formulas do not apply as written.

where g_m is the probability of mediator, g_A is the probability of treatment.

For the AIPW estimator:

$$CDE_{AIPW} = CDE_G(m) + B(\bar{Q}, g_M, g_A) \tag{46.12}$$

where $\bar{Q}$ is the mean outcome function.

$$B(\bar{Q}, g_M, g_A) = \frac{1}{n}\sum_n \frac{I(M=m, A=1)}{g_m(m|1,W)g_A(1|W)}[Y - \bar{Q}(m,1,W)] - \frac{1}{n}\sum_n \frac{I(M=m, A=0)}{g_m(m|0,W)g_A(0|W)}[Y - \bar{Q}(m,0,W)] \tag{46.13}$$

46.2.1.3. simulation example

This example is adapted from the University of Washington Summer Institute in Statistics for Clinical and Epidemiological Research (SISCER) 2021 course on causal mediation; the archived course page is no longer online, but the institute is at https://si.biostat.washington.edu/.

We simulate $n = 5000$ observations. There are two independent standard normal covariates, W_1 confounding the treatment-outcome relationship and W_2 confounding the mediator-outcome relationship. Treatment is $A \sim \text{Bernoulli}(\Lambda(-1 + W_1/2))$, the mediator is $M \sim \text{Bernoulli}(\Lambda(-2 + A/2 + W_2/3))$, and the outcome is $Y \sim \text{Bernoulli}(\Lambda(-1 + A - M/2 + W_1/3 + W_2/3))$. All three are binary.

The target is $CDE(0)$, the effect of treatment holding the mediator at 0 for everyone. Because the DGP is fully specified we can integrate it out rather than simulate: $CDE(0) = E_W[\Lambda(W_1/3 + W_2/3) - \Lambda(-1 + W_1/3 + W_2/3)] = 0.222$.

First, G-computation:

```
set.seed(1234)
n <- 5000
# confounder of A/Y
W1 <- rnorm(n)
# confounder of M/Y
W2 <- rnorm(n)
# treatment
A <- rbinom(n, 1, plogis(-1 + W1 / 2))
# binary mediator
M <- rbinom(n, 1, plogis(-2 + A / 2 + W2 / 3))
# binary outcome
Y <- rbinom(n, 1, plogis(-1 + A - M / 2 + W1 / 3 + W2 / 3))
full_data <- data.frame(W1 = W1, W2 = W2, A = A, M = M, Y = Y)

# fit outcome regression
or_fit <- glm(Y ~ A + M + W1 + W2, family = binomial(), data = full_data)
# new data setting A and M
data_A1_M0 <- data_A0_M0 <- full_data
data_A1_M0$A <- 1; data_A1_M0$M <- 0
data_A0_M0$A <- 0; data_A0_M0$M <- 0
# predict on new data
Qbar_A1_M0 <- predict(or_fit, newdata = data_A1_M0, type = "response")
Qbar_A0_M0 <- predict(or_fit, newdata = data_A0_M0, type = "response")
# gcomp estimate of CDE(0)
mean(Qbar_A1_M0 - Qbar_A0_M0)
```

```
[1] 0.2515527
```

G-computation gives 0.2516.

Second, IPW:

```
# model for P(A = 1 | W)
ps_fit1 <- glm(A ~ W1 + W2, family = binomial(), data = full_data)
P_A1_W <- predict(ps_fit1, type = "response")
P_A0_W <- 1 - P_A1_W
# model for P(M = 0 | A, W)
ps_fit2 <- glm(M ~ A + W1 + W2, family = binomial(), data = full_data)
# P(M = 0 | A = 1, W)
data_A1 <- full_data; data_A1$A <- 1
P_M0_A1_W <- 1 - predict(ps_fit2, newdata = data_A1, type = "response")
# P(M = 0 | A = 0, W)
data_A0 <- full_data; data_A0$A <- 0
P_M0_A0_W <- 1 - predict(ps_fit2, newdata = data_A0, type = "response")
# ipw estimate of CDE(0)
mean( (A == 1) / P_A1_W * (M == 0) / P_M0_A1_W * Y ) -
mean( (A == 0) / P_A0_W * (M == 0) / P_M0_A0_W * Y )
```

```
[1] 0.2553091
```

IPW gives 0.2553.

AIPW:

```
# aipw estimate of E[Y(1,0)]
aiptw_EY_A1_M0 <- mean(Qbar_A1_M0) +
mean( (A == 1) / P_A1_W * (M == 0) / P_M0_A1_W * (Y - Qbar_A1_M0) )
# aipw estimate of E[Y(0,0)]
aiptw_EY_A0_M0 <- mean(Qbar_A0_M0) +
mean( (A == 0) / P_A0_W * (M == 0) / P_M0_A0_W * (Y - Qbar_A0_M0) )
# aipw estimate of CDE(0)
aiptw_EY_A1_M0 - aiptw_EY_A0_M0
```

```
[1] 0.2554265
```

AIPW gives 0.2554.

The three estimates cluster at 0.25 against a true 0.222, all about 0.03 high. That is one unlucky draw rather than bias: every model here is correctly specified, and over 400 replications of this design the g-computation estimator averages 0.2206 with a sampling standard deviation of 0.0159, so this seed sits about 1.9 standard deviations above the truth. Note that the three agree with each other far more closely than any of them agrees with the truth — they share the same data and the same fitted nuisance models, so their agreement checks the implementation, not the sampling error.

46.2.2. Natural direct and indirect effect

CDE is to study the effect of treatment, given the level of mediator. Instead, the Natural Effect is to set mediator to its natural value with the value of treatment, that is, $M = M(a)$.

$$ATE = NIE + NDE = (E[Y(1, M(1))] - E[Y(1, M(0))]) + (E[Y(1, M(0))] - E[Y(0, M(0))]) \tag{46.14}$$

There are two different ways to decompose the ATE in terms of NIE and NDE. Above is one of them.

The advantage of NDE and NIE comparing to CDE is that it's more "natural"; that is, you don't set the level of mediator deterministically. And it can decompose the ATE into direct and indirect effects.

However, there is an additional assumption for NDE and NIE identified.

Assumption 4:

$$Y(a, m) \perp M(a^*)|W \tag{46.15}$$

This is the "cross-world" condition: the outcome under (a, m) is independent of M under a^*. These two situations cannot happen in the same world; you cannot set A to both a and a^*. No experiment can implement it.

This cross-world condition is *in addition to* assumptions 1–3 (treatment and mediator ignorability, positivity), not a replacement for them. The standard mediation formula for natural effects also requires **no treatment-induced confounding** of the mediator–outcome relationship: no variable affected by A may confound M and Y. When such a confounder exists, natural effects are not identified by this formula and interventional direct/indirect effects are used instead.

Simulation example from the same website materials:

```
# fit outcome regression (include interaction because we can)
or_fit <- glm(Y ~ A + M + W1 + W2 + A*M + M*W1,
family = binomial(), data = full_data)
# need E(Y | A = 0/1, M = 0/1, W1 = W1i, W2 = W2i)
get_EY_a_m_Wi <- function(full_data, or_fit, a, m){
data_Aa_Mm_Wi <- full_data
data_Aa_Mm_Wi$A <- a; data_Aa_Mm_Wi$M <- m
predict(or_fit, newdata = data_Aa_Mm_Wi, type = "response")
}
EY_A0_M0_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 0, m = 0)
EY_A0_M1_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 0, m = 1)
EY_A1_M0_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 1, m = 0)
EY_A1_M1_Wi <- get_EY_a_m_Wi(full_data, or_fit, a = 1, m = 1)

# include interactions -- why not?
med_fit <- glm(M ~ A*W1 + W1*W2, family = binomial(), data = full_data)
```

```
# estimates of P(M = m | A = a, W = W_i)
get_Pm_a_Wi <- function(full_data, med_fit, a, m){
data_Aa_Wi <- full_data; data_Aa_Wi$A <- a
p <- predict(med_fit, newdata = data_Aa_Wi, type = "response")
if(m == 1){
p
}else{
1 - p
}
}
PM0_A0_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 0, m = 0)
PM1_A0_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 0, m = 1)
PM0_A1_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 1, m = 0)
PM1_A1_Wi <- get_Pm_a_Wi(full_data, med_fit, a = 1, m = 1)

# E(E(Y | A = 1, M, W) | A = 1, W)
EY1M1_Wi <- EY_A1_M1_Wi * PM1_A1_Wi + EY_A1_M0_Wi * PM0_A1_Wi
# E(E(Y | A = 0, M, W) | A = 1, W)
EY0M1_Wi <- EY_A0_M1_Wi * PM1_A1_Wi + EY_A0_M0_Wi * PM0_A1_Wi
# E(E(Y | A = 1, M, W) | A = 0, W)
EY1M0_Wi <- EY_A1_M1_Wi * PM1_A0_Wi + EY_A1_M0_Wi * PM0_A0_Wi
# E(E(Y | A = 0, M, W) | A = 0, W)
EY0M0_Wi <- EY_A0_M1_Wi * PM1_A0_Wi + EY_A0_M0_Wi * PM0_A0_Wi

# estimate of E[Y(1, M(1))]
E_Y1M1 <- mean(EY1M1_Wi)
# estimate of E[Y(0, M(1))]
E_Y0M1 <- mean(EY0M1_Wi)
# estimate of E[Y(1, M(0))]
E_Y1M0 <- mean(EY1M0_Wi)
# estimate of E[Y(0, M(0))]
E_Y0M0 <- mean(EY0M0_Wi)

# natural direct and indirect effects (the total effect is their sum)
NDE <- E_Y1M0 - E_Y0M0
NIE <- E_Y1M1 - E_Y1M0
round(c(E_Y1M1 = E_Y1M1, E_Y1M0 = E_Y1M0, E_Y0M0 = E_Y0M0,
        NDE = NDE, NIE = NIE, TE = E_Y1M1 - E_Y0M0), 4)
```

```
 E_Y1M1  E_Y1M0  E_Y0M0     NDE     NIE      TE
 0.5002  0.5099  0.2625  0.2475 -0.0098  0.2377
```

The natural direct effect is $0.5099 - 0.2625 = 0.2475$ and the natural indirect effect is $0.5002 - 0.5099 = -0.0098$; they sum to the total effect, 0.2377. The signs match the DGP: treatment raises the outcome directly, and it raises the probability of the mediator, which lowers the outcome, so the indirect effect is small and negative.

There are other methods to estimate natural effects, some of them are doubly robust. The "mediation" package can be used to do it.

46.2.3. When there is an exposure-induced confounder

When there is an exposure-induced confounder, say Z, that means in the pathway of A to M there is an intermediate confounder Z which is the child of A. When that happens and we observe Z, CDE can still be estimated, but NIE and NDE cannot. The reason is that the presence of Z implies that the cross-world assumption cannot hold.

46.2.4. Interventional direct and indirect effects

People are not happy with the cross-world assumption in general. Interventional direct and indirect effects are introduced to avoid this assumption and still be able to decompose the ATE.

$$IIE + IDE = (E[Y(1, G_1)] - E[Y(1, G_0)]) + (E[Y(1, G_0)] - E[Y(0, G_0)]) = E[Y(1, G_1)] - E[Y(0, G_0)] \tag{46.16}$$

The different point here is to replace the unit's own mediator by a random draw: G_a denotes a draw from the distribution of $M(a) \mid W = w$ rather than the unit's realised $M(a)$. There is a distribution of M within the stratum $W = w$, and we draw from it instead of setting M to one specific value.

Note carefully what this sum is, because it is easy to write down as the ATE and it is not. $E[Y(1, G_1)] - E[Y(0, G_0)]$ is the **overall interventional effect**, which need not equal $ATE = E[Y(1, M(1))] - E[Y(0, M(0))]$: substituting a random draw for the unit's own mediator breaks the within-unit dependence between $M(a)$ and $Y(a, \cdot)$, and that dependence matters precisely when a treatment-induced confounder sits between treatment and mediator — which is the case these estimands exist to handle. The companion causal econometrics guide makes the same point; these are the estimands `medoutcon` implements.

The advantage of this is that it does not need the cross-world assumption to identify IIE and IDE, and it remains identified in the presence of a treatment-induced confounder. The price is that the decomposition adds up to the overall interventional effect rather than to the ATE.

46.2.4.1. an example

Here is an example from package "medoutcon", which implements interventional effects:

```
library(data.table)
library(tidyverse)
library(medoutcon)
set.seed(1584)

# produces a simple data set based on a causal model with mediation
make_example_data <- function(n_obs = 1000) {
  ## baseline covariates
```

```
  w_1 <- rbinom(n_obs, 1, prob = 0.6)
  w_2 <- rbinom(n_obs, 1, prob = 0.3)
  w_3 <- rbinom(n_obs, 1, prob = pmin(0.2 + (w_1 + w_2) / 3, 1))
  w <- cbind(w_1, w_2, w_3)
  w_names <- paste("W", seq_len(ncol(w)), sep = "_")

  ## exposure
  a <- as.numeric(rbinom(n_obs, 1, plogis(rowSums(w) - 2)))

  ## mediator-outcome confounder affected by treatment
  z <- rbinom(n_obs, 1, plogis(rowMeans(-log(2) + w - a) + 0.2))

  ## mediator -- could be multivariate
  m <- rbinom(n_obs, 1, plogis(rowSums(log(3) * w[, -3] + a - z)))
  m_names <- "M"

  ## outcome
  y <- rbinom(n_obs, 1, plogis(1 / (rowSums(w) - z + a + m)))

  ## construct output
  dat <- as.data.table(cbind(w = w, a = a, z = z, m = m, y = y))
  setnames(dat, c(w_names, "A", "Z", m_names, "Y"))
  return(dat)
}

# set seed and simulate example data
example_data <- make_example_data()
w_names <- str_subset(colnames(example_data), "W")
m_names <- str_subset(colnames(example_data), "M")

# quick look at the data
head(example_data)
```

```
     W_1   W_2   W_3     A     Z     M     Y
   <num> <num> <num> <num> <num> <num> <num>
1:     1     0     1     0     0     0     1
2:     0     1     0     0     0     1     0
3:     1     1     1     1     0     1     1
4:     0     1     1     0     0     1     0
5:     0     0     0     0     0     1     1
6:     1     0     1     1     0     1     0
```

```
# compute one-step estimate of the interventional direct effect
os_de <- medoutcon(W = example_data[, ..w_names],
                   A = example_data$A,
                   Z = example_data$Z,
                   M = example_data[, ..m_names],
```

```
                  Y = example_data$Y,
                  effect = "direct",
                  estimator = "onestep")
# print() reports through message(), which this chunk suppresses
summary(os_de)
```

```
# A tibble: 1 x 7
  lwr_ci param_est upr_ci var_est eif_mean estimator param
   <dbl>     <dbl>  <dbl>   <dbl>    <dbl> <chr>     <chr>
1 -0.170   -0.0708 0.0281 0.00254 1.12e-16 onestep   direct_interventional
```

```
# compute targeted minimum loss estimate of the interventional direct effect
tmle_de <- medoutcon(W = example_data[, ..w_names],
                     A = example_data$A,
                     Z = example_data$Z,
                     M = example_data[, ..m_names],
                     Y = example_data$Y,
                     effect = "direct",
                     estimator = "tmle")
# print() reports through message(), which this chunk suppresses
summary(tmle_de)
```

```
# A tibble: 1 x 7
  lwr_ci param_est upr_ci var_est eif_mean estimator param
   <dbl>     <dbl>  <dbl>   <dbl>    <dbl> <chr>     <chr>
1 -0.169   -0.0679 0.0332 0.00266   0.0210 tmle      direct_interventional
```

The one-step estimate of the interventional direct effect is -0.071 with a 95% interval of $[-0.170, 0.028]$, and the TMLE estimate is -0.068 with $[-0.169, 0.033]$. The two estimators agree closely, and both intervals include zero at this sample size.

46.2.4.2. Potential problem

However, Miles (2023) points out that the interventional effects do not satisfy "sharp mediational null". "Intuitively, an indirect effect measure should be null whenever there is no individual-level effect for anyone in the population." He found a counter example that the interventional effects cannot pass this "sharp mediational null" criterion.

More recently, Ivan Diaz and coauthors have proposed alternative mediation estimands (discussed under the name "causal influence" in some of that line of work) aimed at satisfying the sharp mediational null where interventional effects fail to. We have not worked through the details of that proposal here, and won't attempt to summarize its identification conditions or estimator without first working through the derivation carefully – readers interested in this specific gap should go directly to Miles's counterexample paper and the more recent Diaz et al. work rather than relying on a secondhand summary.

46.3. Summary

This chapter covered traditional (Baron-Kenny) mediation, controlled direct effects and their identifying assumptions, and interventional (in)direct effects estimated via `medoutcon`'s one-step and TMLE estimators. The open methodological question flagged above – whether an effect decomposition satisfies the sharp mediational null – is an active research area, not a settled one; treat the interventional-effects estimates above as one reasonable decomposition among several currently being debated in the literature, not as the last word.

Systematic treatment: R · Julia.

47. Uplift Modeling and CATE

Uplift modeling is about finding whom treatment changes. In a marketing example, the motivating idea is to separate people who buy because of the campaign from people who would have bought anyway. The data cannot do that person by person: each person shows only one potential outcome. What they can identify is the expected incremental response for people with given characteristics, the conditional average treatment effect (CATE),

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]. \tag{47.1}$$

For a binary purchase, $\tau(x)$ is the share of persuadable people at x minus the share the campaign puts off. A group with a large $\tau(x)$ has a large net share of persuadables, but its members' individual response types remain unknown. A targeting rule learned from $\hat{\tau}(x)$ should be evaluated on independent data. The word "uplift" is more common in industry. The word "CATE" is more common in the causal inference literature.

47.1. Python: causalml uplift trees

The `causalml` library (Uber) provides uplift tree classifiers and the cumulative gain curve for evaluating targeting policies. The code below follows the library's tutorial with synthetic data.

I am copying this code from https://github.com/uber/causalml/blob/master/docs/examples/uplift_trees_with_synthetic_data.ipynb.

> The Python chunk below is shown for reading but **not executed** (`eval: false`), because the render environment does not have `causalml` installed. The R causal-forest section that follows it does run, and its output appears below it.

```
import numpy as np
import pandas as pd

from causalml.dataset import make_uplift_classification
from causalml.inference.tree import UpliftRandomForestClassifier
from causalml.metrics import plot_gain

from sklearn.model_selection import train_test_split
import importlib
print(importlib.metadata.version('causalml') )

df, x_names = make_uplift_classification()
df.head()
```

```
# Look at the conversion rate and sample size in each group
df.pivot_table(values='conversion',
               index='treatment_group_key',
               aggfunc=[np.mean, np.size],
               margins=True)

# Split data to training and testing samples for model validation (next section)
df_train, df_test = train_test_split(df, test_size=0.2, random_state=111)

from causalml.inference.tree import UpliftTreeClassifier

clf = UpliftTreeClassifier(control_name='control')
clf.fit(df_train[x_names].values,
         treatment=df_train['treatment_group_key'].values,
         y=df_train['conversion'].values)
p = clf.predict(df_test[x_names].values)

df_res = pd.DataFrame(p, columns=clf.classes_)
df_res.head()

uplift_model = UpliftRandomForestClassifier(control_name='control')

uplift_model.fit(df_train[x_names].values,
                 treatment=df_train['treatment_group_key'].values,
                 y=df_train['conversion'].values)

df_res = uplift_model.predict(df_test[x_names].values, full_output=True)
print(df_res.shape)
df_res.head()

y_pred = uplift_model.predict(df_test[x_names].values)

# Put the predictions to a DataFrame for a neater presentation
# The output of `predict()` is a numpy array with the shape of [n_sample, n_treatment] excluding the
# predictions for the control group.
result = pd.DataFrame(y_pred,
                      columns=uplift_model.classes_[1:])
result.head()

# If all deltas are negative, assign to control; otherwise assign to the treatment
# with the highest delta
best_treatment = np.where((result < 0).all(axis=1),
                          'control',
                          result.idxmax(axis=1))

# Create indicator variables for whether a unit happened to have the
# recommended treatment or was in the control group
```

```
actual_is_best = np.where(df_test['treatment_group_key'] == best_treatment, 1, 0)
actual_is_control = np.where(df_test['treatment_group_key'] == 'control', 1, 0)

synthetic = (actual_is_best == 1) | (actual_is_control == 1)
synth = result[synthetic]

auuc_metrics = (synth.assign(is_treated = 1 - actual_is_control[synthetic],
                             conversion = df_test.loc[synthetic, 'conversion'].values,
                             uplift_tree = synth.max(axis=1))
                     .drop(columns=list(uplift_model.classes_[1:])))

plot_gain(auuc_metrics, outcome_col='conversion', treatment_col='is_treated')
```

A caution on this evaluation. The gain curve above keeps only the units whose *observed* treatment happened to match the recommended treatment (or who were controls), and then compares outcomes. This is a common tutorial shortcut, but it estimates the value of the recommended policy **only if treatment was randomly assigned** (so that "units who happened to get the recommended arm" are exchangeable with those who did not). With observational treatment assignment, this selection is itself confounded, and a valid off-policy evaluation needs an inverse-probability / policy-value estimator that reweights by the propensity of the observed treatment. Read the gain curve here as a diagnostic under randomized assignment, not as a general off-policy evaluation.

47.2. R: causal forest for CATE

In R, the `grf` package implements causal forests (Wager and Athey 2018; Athey, Tibshirani, and Wager 2019). The setup is simpler than the uplift tree example: provide outcomes, treatments, and covariates, and the forest estimates CATEs using honest splitting.

We simulate $n = 5000$ users with five independent standard normal covariates. Treatment is randomly assigned, $W \sim \text{Bernoulli}(0.5)$, so there is no confounding. The true CATE is $\tau(X) = 0.3 + 0.5X_1$ — the intervention helps high-x_1 users more — and the outcome is $Y = \tau(X)W + X_2 + \varepsilon$ with $\varepsilon \sim N(0, 0.5^2)$. The average true effect is therefore 0.3, and only x_1 modifies it while x_2 affects the outcome level and x_3 to x_5 do nothing.

```
library(grf)
library(ggplot2)

# Simulate a marketing intervention
# True CATE: treatment helps high-engagement users more
set.seed(123)
n  <- 5000
X  <- matrix(rnorm(n * 5), n, 5)
colnames(X) <- paste0("x", 1:5)
W  <- rbinom(n, 1, 0.5)                          # random treatment
tau_true <- 0.3 + 0.5 * X[, 1]                   # CATE depends on x1
Y  <- tau_true * W + X[, 2] + rnorm(n, sd = 0.5) # observed outcome
```

```
# Fit causal forest
cf <- causal_forest(X, Y, W, num.trees = 2000, seed = 42)

# Estimate CATE for each unit
tau_hat <- predict(cf, estimate.variance = TRUE)
cate    <- tau_hat$predictions
cate_se <- sqrt(tau_hat$variance.estimates)

cat("Mean estimated CATE:", round(mean(cate), 3), "\n")
```

```
Mean estimated CATE: 0.307
```

```
cat("True mean CATE:     ", round(mean(tau_true), 3), "\n")
```

```
True mean CATE:      0.3
```

```
# Who benefits? Targeting policy
# Sort by predicted CATE, target top 30%
threshold <- quantile(cate, 0.70)
targeted  <- cate >= threshold

# Report the estimated and the simulated-true subgroup averages side by side.
# The first pair is what the forest predicts; the second is the truth under the
# same targeting rule, available here only because this is a simulation.
cat("Estimated CATE, targeted group:", round(mean(cate[targeted]), 3), "\n")
```

```
Estimated CATE, targeted group: 0.858
```

```
cat("Estimated CATE, non-targeted:  ", round(mean(cate[!targeted]), 3), "\n")
```

```
Estimated CATE, non-targeted:   0.071
```

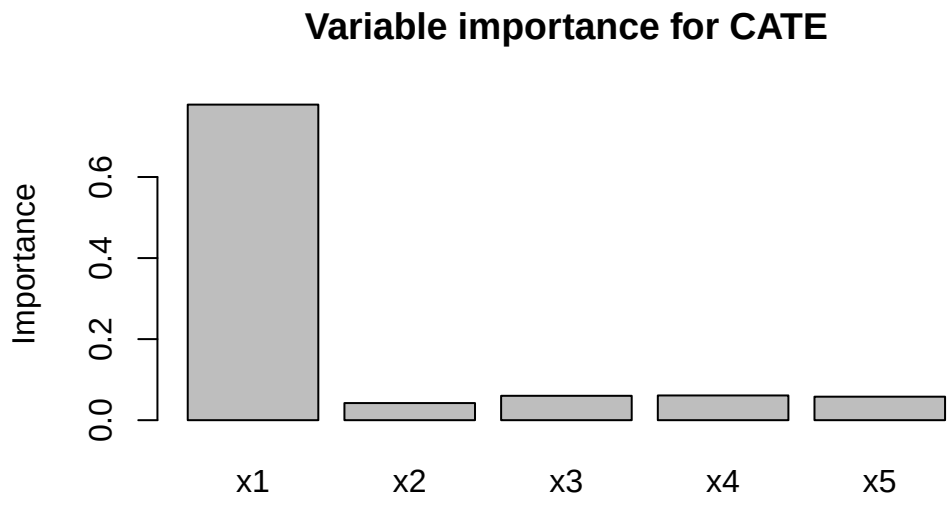
```
cat("True CATE,      targeted group:", round(mean(tau_true[targeted]), 3), "\n")
```

```
True CATE,      targeted group: 0.872
```

```
cat("True CATE,      non-targeted:  ", round(mean(tau_true[!targeted]), 3), "\n")
```

```
True CATE,      non-targeted:   0.054
```

```
# Variable importance: which covariates drive heterogeneity?
# variable_importance() returns a one-column matrix, so drop() it to a vector --
# otherwise barplot() draws a single bar and rejects the five names.
imp <- drop(variable_importance(cf))
barplot(imp, names.arg = colnames(X),
        main = "Variable importance for CATE",
        ylab = "Importance")
```




```
# Omnibus test: is there significant CATE heterogeneity?
test_calibration(cf)
```

```
Best linear fit using forest predictions (on held-out data)
as well as the mean forest prediction as regressors, along
with one-sided heteroskedasticity-robust (HC3) SEs:

                              Estimate Std. Error t value    Pr(>t)
mean.forest.prediction        0.999033   0.046573  21.451 < 2.2e-16 ***
differential.forest.prediction 1.041590   0.031644  32.916 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The forest gives a mean estimated CATE of 0.307 against the true 0.300. Sorting on the estimate, the targeted group averages an *estimated* CATE of 0.858 against 0.071 for the rest. Because this is a simulation we can also look at the truth under that same rule: 0.872 and 0.054, a roughly sixteenfold gap where the estimates implied twelve. Either way the separation is wide, which is

what makes the targeting worth doing rather than treating everyone. The estimated pair is what a practitioner would actually have; the true pair is available here only to audit the exercise.

The `test_calibration()` output is a useful diagnostic. It checks whether units with higher predicted CATEs also appear to have higher treatment effects.

47.3. Uplift vs CATE: the same thing with different vocabulary

Uplift term	Causal inference term
Uplift score	$\hat{\tau}(x)$ — estimated CATE
Persuadables	Covariate profiles with $\tau(x) > 0$
Do-not-disturbs	Covariate profiles with $\tau(x) < 0$
Cumulative gain curve	Policy value curve / AUTOC
Uplift tree	Causal tree / honest tree
Uplift forest	Causal forest (grf)

The two vocabularies are not quite identical. "Persuadable" names an individual; $\tau(x)$ is an average over everyone sharing the profile x. A positive $\tau(x)$ says that subgroup gains on average. It does not say each member of it gains. Sorting individuals into persuadables and do-not-disturbs needs assumptions about individual potential outcomes, and a CATE does not supply them.

The Python `causalml` package comes from the marketing/tech vocabulary. `grf` comes from the econometrics vocabulary. They are trying to estimate the same quantity. For publication, I would usually want the inference and diagnostics from `grf`. For a targeting exercise, the gain-curve tools in `causalml` are also useful.

47.4. Summary

Uplift modelling and CATE estimation target the same quantity, the expected effect of treatment for units with given covariates. Industry calls it uplift and the causal-inference literature calls it the CATE. `causalml`'s uplift trees and gain curves come from the first tradition, and `grf`'s causal forests, with honest splitting and calibration tests, from the second. The gain-curve shortcut that keeps only units whose observed treatment matches the recommendation is valid only under random assignment. With observational data, policy evaluation needs propensity reweighting.

In the simulation with a true average effect of 0.3, the causal forest's mean CATE is 0.307. Targeting the units with the highest estimated effects separates them sharply: their estimated average effect is 0.858 against 0.071 for the rest, and the true values are 0.872 and 0.054. A positive CATE says a subgroup gains on average. It does not say every member does.

Systematic treatment: R · Julia.

48. Using Numpyro

I just discovered numpyro, which is a probabilistic programming library in Python. It can be coupled with Jax and is very powerful.

Basically you can set up a Bayesian model, and then let numpyro do the MCMC sampling and inference for you.

Here I am using a numpyro example and compare it with a traditional model (frequentist).

48.1. Example 1

I am using the numpyro example here: https://num.pyro.ai/en/stable/tutorials/bayesian_hierarchical_linear_regression.html

The data set is from https://www.kaggle.com/c/osic-pulmonary-fibrosis-progression

"Pulmonary fibrosis is a disorder with no known cause and no known cure, created by scarring of the lungs. In this competition, we were asked to predict a patient's severity of decline in lung function. Lung function is assessed based on output from a spirometer, which measures the forced vital capacity (FVC), i.e. the volume of air exhaled.

In medical applications, it is useful to evaluate a model's confidence in its decisions. Accordingly, the metric used to rank the teams was designed to reflect both the accuracy and certainty of each prediction."

I read it in R first.

```
library(tidyverse)
# read in csv file
library(readr)

data <- read_csv("osic_pulmonary_fibrosis.csv") |>
  arrange(Patient, Weeks)

data
```

```
# A tibble: 1,549 x 7
   Patient                   Weeks   FVC Percent   Age Sex   SmokingStatus
   <chr>                     <dbl> <dbl>   <dbl> <dbl> <chr> <chr>
 1 ID00007637202177411956430    -4  2315    58.3    79 Male  Ex-smoker
 2 ID00007637202177411956430     5  2214    55.7    79 Male  Ex-smoker
 3 ID00007637202177411956430     7  2061    51.9    79 Male  Ex-smoker
```

```
 4 ID00007637202177411956430     9  2144    54.0    79 Male  Ex-smoker
 5 ID00007637202177411956430    11  2069    52.1    79 Male  Ex-smoker
 6 ID00007637202177411956430    17  2101    52.9    79 Male  Ex-smoker
 7 ID00007637202177411956430    29  2000    50.3    79 Male  Ex-smoker
 8 ID00007637202177411956430    41  2064    51.9    79 Male  Ex-smoker
 9 ID00007637202177411956430    57  2057    51.8    79 Male  Ex-smoker
10 ID00009637202177434476278     8  3660    85.3    69 Male  Ex-smoker
# i 1,539 more rows
```

48.1.1. A random effect model

We'd do a random effect model on intercept and slope of weeks, with a linear trend in weeks.

```r
# a random effect model with FVC as DV, and a linear time trend, with random effect on Patient
library(lme4)
reg1 <- lmer(FVC ~ Weeks + (1 + Weeks | Patient), data = data)
summary(reg1)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: FVC ~ Weeks + (1 + Weeks | Patient)
   Data: data

REML criterion at convergence: 20891.4

Scaled residuals:
    Min      1Q  Median      3Q     Max
-9.3209 -0.4257  0.0063  0.4458  5.6990

Random effects:
 Groups   Name        Variance  Std.Dev. Corr
 Patient  (Intercept) 686691.11 828.668
          Weeks           25.88   5.087  -0.14
 Residual              18592.11 136.353
Number of obs: 1549, groups:  Patient, 176

Fixed effects:
             Estimate Std. Error t value
(Intercept) 2810.2843    62.9497  44.643
Weeks         -4.2629     0.4377  -9.739

Correlation of Fixed Effects:
      (Intr)
Weeks -0.174
```

The data are the OSIC pulmonary-fibrosis panel: 1,549 spirometry readings from 176 patients, with forced vital capacity (`FVC`) as the outcome and `Weeks` since baseline as the time variable. The model gives each patient a random intercept and slope.

The fixed effects are an intercept of 2810.3 (SE 62.9) and a slope of −4.263 (SE 0.438): average FVC declines by about 4.3 units per week, and the t of −9.7 leaves little doubt about the direction. The variance components show where the heterogeneity lives — a patient-intercept standard deviation of 828.7 against a residual of 136.4, so patients differ far more in level than any single measurement deviates from that patient's own line. The slope SD of 5.09 against a mean slope of −4.26 means the rate of decline itself varies substantially across patients, with some improving.

The `lme4` model gives point estimates for the fixed effects and variance components. Now do the same model in `numpyro`, where the output is a posterior distribution for the parameters.

48.2. Example 2: Numpyro hierarchical model

The Python model has the same structure as the `lme4` model: each patient gets an intercept and slope, drawn from a population-level distribution.

> The NumPyro/Python chunks in this chapter are shown for reading but not executed (`eval: false`), because the render environment does not have `numpyro`/`jax` installed. Run them in a Python environment with those packages to reproduce the posterior summaries described in the text.

```
import pandas as pd
import numpy as np
import jax
import jax.numpy as jnp
import numpyro
import numpyro.distributions as dist
from numpyro.infer import MCMC, NUTS

# Load the same data
data = pd.read_csv("osic_pulmonary_fibrosis.csv").sort_values(["Patient", "Weeks"])

# Encode patient IDs as integers
patients, patient_ids = pd.factorize(data["Patient"])
n_patients = len(patient_ids)
weeks = jnp.array(data["Weeks"].values, dtype=float)
fvc   = jnp.array(data["FVC"].values,   dtype=float)
pat   = jnp.array(patients,              dtype=int)
```

Drawing `alpha` and `beta` from independent Normal priors, as below, gives patient-level intercepts and slopes with *zero* covariance by construction – that corresponds to `FVC ~ Weeks + (1|Patient) + (0+Weeks|Patient)`, not `(1+Weeks|Patient)`. `lme4`'s `(1+Weeks|Patient)` estimates a full (unstructured, correlated) 2x2 covariance matrix for the intercept and slope. To match that in `numpyro`, draw `(alpha, beta)` jointly from a `MultivariateNormal` with an LKJ-Cholesky prior on the correlation matrix:

```
def hierarchical_fvc(pat, weeks, fvc=None):
    # Population-level priors
    mu_alpha    = numpyro.sample("mu_alpha",    dist.Normal(2500., 500.))
    mu_beta     = numpyro.sample("mu_beta",     dist.Normal(0., 5.))
    sigma_obs   = numpyro.sample("sigma_obs",   dist.HalfNormal(300.))

    # Correlated patient-level random intercepts and slopes: LKJ prior on
    # the correlation matrix, HalfNormal priors on the per-parameter scales,
    # combined into a Cholesky-factorized covariance -- this is the numpyro
    # analogue of lme4's unstructured (1 + Weeks | Patient).
    sigma_re = numpyro.sample("sigma_re", dist.HalfNormal(jnp.array([300., 3.])))
    L_corr   = numpyro.sample("L_corr", dist.LKJCholesky(2, concentration=2.0))
    L_cov    = jnp.diag(sigma_re) @ L_corr

    mu_re = jnp.stack([mu_alpha, mu_beta])
    with numpyro.plate("patients", n_patients):
        re = numpyro.sample("re", dist.MultivariateNormal(mu_re, scale_tril=L_cov))
    alpha, beta = re[:, 0], re[:, 1]
    # Record these as sites, or they are only local variables and
    # mcmc.get_samples() will not contain "alpha" or "beta".
    numpyro.deterministic("alpha", alpha)
    numpyro.deterministic("beta", beta)

    # Likelihood
    mu = alpha[pat] + beta[pat] * weeks
    numpyro.sample("fvc", dist.Normal(mu, sigma_obs), obs=fvc)

# Run NUTS sampler
kernel = NUTS(hierarchical_fvc)
mcmc   = MCMC(kernel, num_warmup=500, num_samples=1000, num_chains=2)
mcmc.run(jax.random.PRNGKey(0), pat, weeks, fvc)
mcmc.print_summary()
```

`mcmc.print_summary()` reports posterior means, standard deviations and credible intervals for each site: `mu_alpha`, `mu_beta`, the two-vector `sigma_re`, `L_corr`, and `sigma_obs`.

No output is shown here, and that is deliberate. This chapter's Python is not executed during the render, so any table printed at this point would be written by hand rather than produced by the model. An earlier version of this section did exactly that, and the invented numbers drifted away from the model in two visible ways: they reported `sigma_alpha` and `sigma_beta`, which this parameterization does not create – it has the vector site `sigma_re` – and they disagreed with the `lme4` fixed effects fitted above. Treat the code below as a sketch of the comparison, not as a record of one that was run.

```
# Compare population-level estimates to lme4
samples = mcmc.get_samples()
```

```
print("Numpyro posterior mean mu_alpha:", jnp.mean(samples["mu_alpha"]).item())
print("Numpyro posterior mean mu_beta: ", jnp.mean(samples["mu_beta"]).item())
print()
print("lme4 fixed effects (from summary(reg1) above):")
print("  Intercept:", 2810.3)
print("  Weeks:    ", -4.263)

# Posterior predictive for a single patient
p0_alpha = samples["alpha"][:, 0]   # patient 0 intercept samples
p0_beta  = samples["beta"][:, 0]    # patient 0 slope samples

# Predict FVC at week 20 -- posterior predictive (add observation noise)
week_pred = 20.
mu_pred   = p0_alpha + p0_beta * week_pred
fvc_pred  = dist.Normal(mu_pred, samples["sigma_obs"]).sample(jax.random.PRNGKey(1))
print(f"\nPatient 0 FVC at week 20:")
print(f"  Mean: {jnp.mean(fvc_pred):.1f}")
print(f"  90% CI: [{jnp.quantile(fvc_pred, 0.05):.1f}, {jnp.quantile(fvc_pred, 0.95):.1f}]")
```

48.3. Comparing the two approaches

	lme4	numpyro
Estimation	REML (restricted maximum likelihood)	MCMC (full posterior)
Output	Point estimates + SE	Full posterior distributions
Uncertainty	Approximate (Wald-type CI)	Simulation-based posterior credible intervals
Computation	Fast (seconds)	Slower (minutes for 1000 samples)
Prediction	`predict()` gives point estimates	Posterior predictive — full distribution

The main advantage of `numpyro` shows up in prediction. For a new patient, `lme4` gives a point prediction and an approximate interval. `numpyro` gives a posterior predictive distribution. Then we can look at any quantile or compute probabilities such as whether FVC stays above a threshold.

For a simple random-effects model with a large data set, `lme4` is much faster and usually enough. `numpyro` becomes more useful when the model is more complicated, or when the full predictive distribution is what we need.

48.4. Summary

`numpyro` is a probabilistic programming library on JAX that fits Bayesian models by MCMC. A hierarchical model with patient-level intercepts and slopes can be written in it to mirror an `lme4`

random-effects model. Independent priors on the intercept and slope match uncorrelated random effects. Matching `lme4`'s correlated specification needs a joint multivariate normal draw with an LKJ prior on the correlation.

On the pulmonary-fibrosis data, `lme4` gives an average decline in FVC of about 4.3 units per week. Patients differ far more in level than within their own trajectories, and the rate of decline varies a lot across patients. The `numpyro` code is shown but not run in this render, so no posterior numbers are reported. Its advantage is the full posterior predictive distribution for a new patient. For a simple random-effects model, `lme4` is faster and usually enough.

Systematic treatment: R · Julia.

49. Equivalence testing and two one-sided tests (TOST)

49.1. Why non-significance is not evidence of no effect

Suppose we use the usual null hypothesis significance test:

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_1 : \theta \neq 0$$

Failing to reject H_0 ($p > 0.05$) does not show that $\theta = 0$. We can get a large p-value because the true effect is small, but we can also get one because the sample is small or the outcome is noisy.

The usual test cannot tell these cases apart. If we want evidence that an effect is small enough to ignore, we need an equivalence test.

49.2. Two one-sided tests

We first choose a smallest effect size of interest. I write the resulting equivalence interval as $[-\Delta, +\Delta]$. An effect inside this interval is small enough to be negligible for the application.

The two one-sided tests (TOST) procedure (Schuirmann 1987; Lakens 2017) uses two null hypotheses:

$$H_{01} : \theta \leq -\Delta \quad \text{and} \quad H_{02} : \theta \geq +\Delta$$

If we reject both null hypotheses at level α, we declare equivalence. The claim is exactly that: both nonequivalence nulls are rejected at level α, so the procedure falsely declares equivalence at most α of the time for any θ outside the interval. It is not a statement that θ lies inside $[-\Delta, +\Delta]$ with probability $1 - 2\alpha$. A frequentist interval does not carry that reading.

An equivalent way to do the test is to check whether the $(1 - 2\alpha)$ confidence interval falls entirely inside $[-\Delta, +\Delta]$. When $\alpha = 0.05$, we use the 90% confidence interval.

49.3. Using TOSTER in R

Let's use the 1974 Motor Trend data in `mtcars`. We compare quarter-mile acceleration time (`qsec`) between automatic ($n = 19$) and manual ($n = 13$) transmissions.

```
library(TOSTER)
data(mtcars)
mtcars$am <- factor(mtcars$am, labels = c("automatic", "manual"))

# Standard two-sample t-test
t.test(qsec ~ am, data = mtcars)
```

```
    Welch Two Sample t-test

data:  qsec by am
t = 1.2878, df = 25.534, p-value = 0.2093
alternative hypothesis: true difference in means between group automatic and group manual is not equal to
95 percent confidence interval:
 -0.4918522  2.1381679
sample estimates:
mean in group automatic    mean in group manual
               18.18316                17.36000
```

The usual two-sample t-test gives $p = 0.21$. This tells us that the data do not clearly distinguish the groups, but it is not evidence that they are equivalent.

49.3.1. TOST with $\Delta = 1$ second

Suppose we pre-specify that a difference of less than 1 second ($\Delta = 1$) is negligible:

```
t_TOST(formula = qsec ~ am, data = mtcars, eqb = 1, var.equal = FALSE)
```

```
Welch Two Sample t-test

The equivalence test was non-significant, t(25.53) = -0.28, p = 0.39
The null hypothesis test was non-significant, t(25.53) = 1.288, p = 0.21
NHST: don't reject null significance hypothesis that the effect is equal to zero
TOST: don't reject null equivalence hypothesis

TOST Results
                 t    df p.value
t-test      1.2878 25.53   0.209
TOST Lower  2.8524 25.53   0.004
```

```
TOST Upper -0.2767 25.53   0.392

Effect Sizes
               Estimate     SE              C.I. Conf. Level
Raw              0.8232 0.6392 [-0.2678, 1.9141]         0.9
Hedges's g(av)   0.4522 0.3783 [-0.1375, 1.0342]         0.9
Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

The 90% confidence interval is $[-0.27, 1.91]$, so it extends beyond $+1.0$. We fail to reject H_{02} and cannot claim equivalence within ± 1 second.

49.3.2. TOST with $\Delta = 2$ seconds

If the substantive tolerance is instead $\Delta = 2$ seconds:

```
t_TOST(formula = qsec ~ am, data = mtcars, eqb = 2, var.equal = FALSE)
```

```
Welch Two Sample t-test

The equivalence test was significant, t(25.53) = -1.8, p = 0.04
The null hypothesis test was non-significant, t(25.53) = 1.288, p = 0.21
NHST: don't reject null significance hypothesis that the effect is equal to zero
TOST: reject null equivalence hypothesis

TOST Results
                t    df p.value
t-test      1.288 25.53   0.209
TOST Lower  4.417 25.53 < 0.001
TOST Upper -1.841 25.53   0.039

Effect Sizes
               Estimate     SE              C.I. Conf. Level
Raw              0.8232 0.6392 [-0.2678, 1.9141]         0.9
Hedges's g(av)   0.4522 0.3783 [-0.1375, 1.0342]         0.9
Note: SMD confidence intervals are an approximation. See vignette("SMD_calcs").
```

Now the 90% confidence interval $[-0.27, 1.91]$ lies completely inside $[-2, 2]$. Both one-sided tests reject ($p_1 < 0.001$, $p_2 = 0.039$), so the groups are statistically equivalent within ± 2 seconds at $\alpha = 0.05$.

49.4. Bayesian equivalence testing and units

We can also evaluate a Bayes factor for an interval null:

$$\mathrm{BF}_{\text{in/out}} = \frac{p(\text{data} \mid \theta \in [-\Delta, +\Delta])}{p(\text{data} \mid \theta \notin [-\Delta, +\Delta])}$$

Both terms are marginal likelihoods, so each one integrates the likelihood over a *prior* for θ conditional on its region. `BayesFactor` puts a Cauchy prior on Cohen's d and truncates it to the interval in the numerator and to the complement in the denominator. The number that comes out depends on that prior and on Δ, not on the data alone.

The two R packages use different units. `TOSTER::t_TOST(eqb = ...)` takes the bound in the original outcome units, which are seconds here. `BayesFactor::ttestBF(nullInterval = ...)` expects a standardized effect size in Cohen's d.

We therefore need to divide the raw bound by the pooled standard deviation before passing it to `ttestBF`:

```
library(BayesFactor)

# Compute pooled SD to standardize the 2-second bound
grp <- split(mtcars$qsec, mtcars$am)
n1 <- length(grp[[1]])
n2 <- length(grp[[2]])
sd_pooled <- sqrt(((n1 - 1) * var(grp[[1]]) + (n2 - 1) * var(grp[[2]])) / (n1 + n2 - 2))

delta_d <- 2 / sd_pooled  # convert +/- 2 seconds to Cohen's d
c(sd_pooled = sd_pooled, delta_d = delta_d)
```

```
sd_pooled   delta_d
 1.767842  1.131323
```

With $s_{\text{pooled}} \approx 1.77$, $\Delta = 2$ corresponds to $d \approx 1.13$. We can now test the interval null:

```
ttestBF(formula = qsec ~ am, data = mtcars, nullInterval = c(-delta_d, delta_d))
```

```
Bayes factor analysis
--------------
[1] Alt., r=0.707 -1.1313225176573<d<1.1313225176573    : 0.9813314  ±0%
[2] Alt., r=0.707 !(-1.1313225176573<d<1.1313225176573) : 0.02203644 ±0.03%

Against denominator:
  Null, mu1-mu2 = 0
---
Bayes factor type: BFindepSample, JZS
```

The Bayes factor is 0.98 for an effect inside $[-\Delta, +\Delta]$ relative to the point null and 0.022 for an effect outside it. Their ratio is about $45 : 1$ ($0.98/0.022 \approx 44.5$) in favor of the equivalence interval. The two analyses point the same way for this bound and this prior. They are not the same test and they answer different questions: TOST controls the rate at which equivalence is wrongly declared, while the Bayes factor weighs evidence for two regions under a prior we chose. Agreement here is a fact about this example, not a general correspondence.

49.5. Choosing the equivalence bound

The data cannot choose Δ for us. It has to come from the application. We might use the smallest effect with a meaningful practical or economic consequence, or the smallest effect that would justify the cost of an intervention. Another option is Simonsohn (2015)'s "small telescopes" benchmark: use the effect size that an earlier study had only 33% power to detect, denoted by $d_{33\%}$. The 33% refers to power, not to 33% of the original estimate; $d_{33\%}$ is a little more than half of $d_{80\%}$.

49.6. What to report

A large p-value is not evidence of no effect. For TOST, report the point estimate, the pre-specified equivalence bound, the $(1-2\alpha)$ confidence interval, and the two test decisions. This gives readers enough information to see whether another reasonable choice of Δ would change the conclusion. For a Bayesian interval test, also check whether the software expects the bound in the original units or in standardized units.

References

Each chapter lists the works it cites at the foot of its own page. This is the combined list, and it prints only works actually cited somewhere in the book, so an entry here means some chapter rests on it. The bibliography source is `references.bib` in the book repository.

Many entries were carried over unchanged from the bibliographies of the two causal econometrics guides, keeping the same citation keys, so a passage can move between the three books without a key being renamed.

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